

MAIN DIESEL ENGINE

△ 2	FOR FINAL	10.11.10	T.H.KIM	J.T.SEO	T.B.KIM
△ 1	FOR WORKING	09.12.21	S.M.LEE	J.T.SEO	T.B.KIM
△ 0	FOR APPROVAL	09.02.13	S.M.LEE	J.T.SEO	T.B.KIM
REVISION	DESCRIPTION	DATE	DWG	CHECK	APPD
SHIPYARD / OWNER	STX / NNC	APPROVED BY		T.B.KIM	
HULL NO.	S-4012	CHECKED BY		J.T.SEO	
MODEL	7S50MC-C7	CHECKED BY			
KW x RPM	11,060 kW x 127rpm	DRAWN BY		S.M.LEE	
DWG NO.	7S50MC-C7-H10B591	ISSUE DATE		2009.02.13	

stx Heavy Industries Co., Ltd.

DRAWING LIST

HULL NO. : S-4012

ENGINE TYPE : 7S50MC-C7

SHIPYARD / OWNER : STX SHIPBUILDING / NNC

NO	DESCRIPTION	DWG. NO.	REVISION		REMARK
			DWG. NO.	DATE	
01	SPECIFICATION OF DIESEL ENGINE	GS-H10B591		09.12.08	REV. 
1.	GENERAL SPECIFICATION	-			
2.	DIESEL ENGINE PARTICULARS	-			
3.	SCOPE OF SUPPLY	-			
4.	SHOP TEST AND OVERHAUL INSP.	-			
5.	TORSIONAL VIBRATION CALCULATION SHEET	-		09.12.07	ADD. 
	AXIAL VIBRATION CALCULATION SHEET	-		09.12.07	ADD. 
6.	SHOP TEST REPORT	-		10.11.10	ADD. 
02	GENERAL ARRANGEMENT				
1.	ENGINE OUTLINE	2805007560			
2.	ENGINE SPACE REQUIREMENT	2805007570			
3.	ENGINE GALLERY OUTLINE	2805029420	2805029420.1	09.09.10	REV. 
4.	TURNING GEAR ARRANGEMENT	2805014880			
5.	TURNING GEAR ASS'Y	YRTG20-27			
6.	MIST CATCHER DRAIN ARRANGEMENT	280500146A			
6-1.	VARIOUS PIPE - EXH. SIDE	2805009260			
7.	JACK-UP BRACKET ARRANGEMENT	2805002400			
8.	ARR. OF FLYWHEEL COVER	280101732A	2801024530	09.09.10	CHG. 
8-1.	FLYWHEEL COVER		2801024540	09.09.10	ADD. 
03	INSTALLATION AND COUPLING				
1.	ARRANGEMENT OF ENGINE SEATING	2805002410			
2.	ENGINE SEATING WITH EPOXY CHOCK	2805004390			
3.	ASS'Y OF SIDE CHOCK	2805000390			
3-1.	SIDE CHOCK	2805000400			
3-2.	LINER FOR SIDE CHOCK ASS'Y(PORT)	2805000410			
3-3.	LINER FOR SIDE CHOCK ASS'Y(STAR BOARD)	2805000420			
3-4.	LINER FOR SIDE CHOCK(PORT)	2805000430			
3-5.	LINER FOR SIDE CHOCK(STAR BOARD)	2805000440			
3-6.	LOCK PLATE, SIDE CHOCK LINER	2805000450			
4.	ASS'Y OF END CHOCK	2805000460			

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NO	DESCRIPTION	DWG. NO.	REVISION		REMARK
			DWG. NO.	DATE	
4-1.	END CHOCK	2805000470			
4-2.	LINER FOR END CHOCK	2805000480			
4-3.	ASS`Y OF END CHOCK BOLT	2805000490			
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5.	ASS`Y OF HOLDING DOWN BOLT	2805000550			
5-1.	HOLDING DOWN BOLT	2805000560			
5-2.	ROUND NUT	2805000570			
5-3.	DISTANCE TUBE	2805000580			
5-4.	SPHERICAL WASHER	2805000590			
5-5.	SPHERICAL NUT	2805000600			
5-6.	PROTECTING CAP	2805000610			
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7.	TOP BRACING ARRANGEMENT	2805029440			
7-1.	PREPARATION FOR MECH. TOP BRACING	2805000640			
7-2.	TOP BRACING FORCES	2805000650			
7-3.	DATA FOR TOP BRACING	2805000660			
7-4.	TOP BRACING INSTRUCTIONS	2805000670			
7-5.	TOP BRACING OUTLINE	280500208B	2801023690	09.09.10	CHG. 
7-5-1.	PLATE		2802042420	09.09.10	ADD. 
04	GENERAL OF PIPING SYSTEM				
1.	ENGINE PIPE CONNECTIONS	2805030080	2805030080.1A	09.07.30	REV. 
1-1.	AIR REDUCING UNIT SUPPORT ASS`Y	280100930A	2801023640	09.09.10	CHG. 
2.	LIST OF FLANGES	2805029460	2805033090	09.12.22	CHG. 
3.	ARR. OF L.O OUTLET(VERTICAL)	2805010090	2805010091	09.09.10	REV. 
4.	BASIC SYMBOL	1105000110			
5.	IDENTIFICATION CODES	9005000240	900500024A	10.06.22	REV. 

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NO	DESCRIPTION	DWG. NO.	REVISION		REMARK
			DWG. NO.	DATE	
05	FUEL OIL SYSTEM				
1.	FUEL OIL PIPES	2805029470			
2.	FUEL OIL DRAIN LINE	2805006450			
3.	FUEL OIL PIPES, HEATING	2805000890			
4.	FUEL OIL PIPES-INSULATION	2805001410			
5.	RECOMMENDED FUEL OIL SPECIFICATION	9005000290			
06	LUBRICATING OIL SYSTEM				
1.	LUB. & COOLING OIL PIPES	2805029480			
2.	T/C L.O PIPES	2805020250			
3.	CYLINDER LUB. OIL PIPES	2805006750			
4.	STUFFING BOX DRAIN PIPES	2805029700			
5.	CONSUMPTION & L.O CHART FOR M/E	290500101A	290500101B	09.09.10	REV. 
07	COOLING WATER SYSTEM				
1.	L.T F.W COOLING PIPES	2805029490			
2.	H.T F.W COOLING PIPES	2805019210			
3.	RECOMMENDED INHIBITORS FOR COOLING	2705001000			
	WATER TREATMENT				
4.	METALLIC FLEXIBLE HOSE	2802014240			
08	STARTING AIR SYSTEM				
1.	STARTING AIR PIPES	2805015960			
2.	CONTROL & SAFETY AIR PIPES	2805019470			
3.	CYL. LUBRICATOR CONTROL AIR PIPE	2805012490			
09	COMBUSTION(SCAVENGE) AIR SYSTEM				
1.	SCAVENGING AIR PIPES	2805019220			
2.	SCAV. AIR SPACE-DRAIN PIPES	2805029710			
3.	FIRE EXTINGUISH IN SCAV. BOX	2805020260			
4.	AUXILIARY BLOWER	HAA-334/80N			
5.	SCAVENGING AIR COOLER	LKM-(Y)JT			

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NO	DESCRIPTION	DWG. NO.	REVISION		REMARK
			DWG. NO.	DATE	
10	EXHAUST GAS SYSTEM				
1.	EXHAUST GAS PIPES	2805019900			
2.	TURBOCHARGER	TCA 66-20			
3.	EXH. GAS T/C OUTLET	2801022480			
11	OTHER PIPING SYSTEM				
1.	SEALING AIR PIPES FOR EXH. V/V	2805029520			
2.	A/C CLEANING PIPES	2805015640			
3.	T/C CLEANING PIPE	2805030000			
4.	BEDPLATE DRAIN PIPES	2805029720			
5.	U-TUBE MANOMETER PIPES	2805007760			
6.	OIL MIST DETECTOR PIPES	2805030010			
12	ELEXTRIC AND CONTROL SYSTEM				
13	GEAR BOX		NOT APPLIED		
14	ALTERNATOR		NOT APPLIED		
15	SPARE PARTS AND TOOLS				
1.	MAIN ENGINE SPARE PARTS	2805030850	2805030851	09.12.18	REV. 
2.	LIST OF LARGE SPARE PARTS	2805018180			
3.	MAIN ENGINE TOOLS	2805030860	2805030861	10.11.10	REV. 
4.	LIST OF LARGE TOOL PARTS	2805014190			
5.	TOOL PANELS		2805028540	09.09.10	ADD. 
6.	SPARE & TOOL PARTS FOR ACCESORY PARTS	2805030090	2805030091	10.06.22	REV. 
	-END-				

SPECIFICATION OF DIESEL ENGINE

△ 2	FOR FINAL	2010.11.10	T.H.KIM	J.T.SEO	T.B.KIM
△ 1	FOR WORKING	2009.12.08	S.M.LEE	J.T.SEO	T.B.KIM
△ 0	FOR APPROVAL	2009.02.13	S.M.LEE	J.T.SEO	T.B.KIM
REVISION	DESCRIPTION	DATE	DWG	CHECK	APPD
SHIPYARD / OWNER	STX OFFSHORE & SHIPBUILDING /NNC	APPROVED BY		T.B.KIM	
HULL NO.	S-4012	CHECKED BY		-	
MODEL	7S50MC-C7	CHECKED BY		J.T.SEO	
kW × RPM	11,060 × 127 × 1set / Ship	DRAWN BY		S.M.LEE	
DWG NO.	GS-H10B591	ISSUE DATE		2008.02.13	



STX Heavy Industries., Ltd

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Chapter 1. General specification

Chapter 2. Diesel engine particulars

Chapter 3. Scope of supply

Chapter 4. Shop test & overhaul inspection

ABBREVIATION

STD : Standard parts

OPT : Optional parts with extra price

B : Built - on engine

S : Equipments separately delivered
and installed on hull side

To be used for filling in :

☒ : Included in scope of supply

Chapter 1. General Specification

STD	Q'ty OPT /ship	Description	Remarks
☒	☐	Rules and Regulations	
☒	☐	Classification society	LR
☒	☐	Unattended engine room	UMS
☒	☐	Standards : KS, JIS and maker's standard	
☒	☐	To comply with IMO NOx emission limitations	
☒	☐	To be submitted Nox EIAPP Certificate with class : LR	
☒	☐	Unit	
		- Engine output	☒ kW
		- Pressure	☒ kg/cm2 ☒ Mpa or Pa
		- Temperature	°C
☒	☐	Final paint color	
		- Diesel engine	Munsell No.: 7.5 BG 7/2
		- Electrical equipment	Munsell No.: 7.5 BG 7/2
		- Loosely supply items	According to maker's standard
☒	☐	Calculation of torsional and axial vibrations	
☒	☐	Standard calculation of torsional vibration for direct coupled engine	
☒	☐	Calculation of axial vibrations required by classification society(LR, BV, RINA, CCS)	
☒	☐	Barred speed range accepted by owner	
☒	☐	Approval of the calculation results to classification society to be carried out by engine maker	Exclusive of special calculations(*)
☒	☐	Measurement of torsional vibration during sea trial	by engine maker (1st vessel only)
☒	☐	Measurement of axial vibration during sea trial (if required by classification society)	by engine maker (1st vessel only)
☒	☐	Documentation	
☒	☐	Name plates on engine in English	
☒	☐	Caution plates on engine in English & Arabic	
☒	15	Sets of Approval drawings in English with 1 CD	for project
☒	15	Sets of Working drawings in English	for project
☒	4	Sets of Final drawings in English with 5 CD	for vessel
☒	4	Sets of Instruction manuals in English with 5 CD	for vessel
☒	2	Certificate of classification society including original	

STD	Q'ty OPT /ship	Description	Remarks
☒	<u>4</u>	Copies of shop test record	
☒		Testing of diesel engine with water brake Official shop test on Marine Diesel Oil with water brake according to class requirements	<u>refer to the chapter 4 of details</u>
☒		Buyer will participate. Notice to be given to buyer <u>21</u> days before testing	
☒		Inspection after shop trial of components from one cylinder unit only	
☒		Voltage on board for electrical consumers Electrical power available - AC 440V, 3Φ, 60Hz - AC 220V, 1Φ, 60Hz - DC 24V - Motor starting method : Y-Δ starting <u>60</u> kW above	<u>Normally above 10% of Generator capacity</u>
		Dismantling, packing and shipping of engine Dismantling and packing of engine Available lifting capacity and number of crane hooks required must be informed by buyer : _____ tons _____ hooks	_____ _____ _____
		Dispatch pattern A (for short distance transportation, max. 8weeks from delivery to installation) Dispatch pattern B (for overseas or long distance transportation, max. 6months)	
☒		Dispatch pattern A1 or B1 Engine complete Spare parts & tools Mass of heaviest item : _____ tons	<u>Drying machine to be provide for transportation/storage period, to be returned to seller on Buyer's account in case of dispatch pattern B</u>
☒		Lending of lifting tool for complete engine for installation purpose (hardware) exclusive of lifting wire/rope/shackles	<u>lifting tools to be returned to engine builder on buyer's account after the installation of the engine</u>

Chapter 2. Diesel Engine Particulars

1. Engine type : 2-stroke, single acting, direct reversible, cross head diesel engine with exhasut gas turbocharger and air cooler
2. Engine model : 7S50MC-C7
3. Cylinder bore x Stroke : 500 mm x 2,000 mm
4. Engine rating & revolution

- Nominal max. continuous rating	: 11,060 kW on flywheel at 127 rpm
- Specified max. continuous rating	: 11,060 kW on flywheel at 127 rpm
- Continuous service rating (CSR)	: 9,954 kW on flywheel at 123 rpm
- Optimsed rating	: 11,060 kW on flywheel at 127 rpm
- Auxiliary machinery and shaft line for main engine to be dimensioned for on engine output of	: 11,060 kW on flywheel at 127 rpm
- Overload capacity	: 12,166 kW on flywheel at 131 rpm
	110% one (1) hour in every twelve (12) hours

At the based condition of

- Altitude	: Sea level (1000 mbar)	
- Ambient temperature	: Max 45 °C	
- Relative humidity	: 60 %	
- Sea water temperature	: 32 °C	- Fresh water temperature : 36 °C
5. Piston speed at MCR : 8.47 m/s
6. Mean effective pressure at MCR : 19.0 bar
7. Max. combustion pressure (Pmax) : 150 bar
8. Direction of engine rotation (ahead) : Clockwise (viewing from stern side to engine)
9. Firing order (ahead) : 1-7-2-5-4-3-6
10. Starting : Compressed air Max. 30 kg/cm²
11. Cooling system : Cylinder liner, cylinder cover & T/C : by F.W
Scavenging air cooler : F.W for central cooling
12. Using fuel oil : HFO 380 cSt / 50 °C (up to 700 cSt / 50 °C)
13. Specific fuel oil consumption at SMCR : 171 g/kW hr + 5 % tolerance
This data is given in accordance with ISO 3046/1 conditions and using diesel oil with lower calorific value of 42,700 kJ/kg (10,200 kcal/kg)

- Ambient air temperature	: 25 °C
- Ambient air pressure	: 1000 mbar
- Charge air coolant temperature	: 25 °C
- Exh. Gas back pressure after T/C	: Max. 300 mm Aq
14. Lub oil consumption : Approximately 4 ~ 5 kg/cyl./24 h
15. Cylinder Lub oil consumption : 0.7 ~ 1.5 g/ kWh

Chapter 3. Scope of Supply

STD	Q'ty OPT /ship	Inst	Description	Remarks
Engine basic				
☒	___	B	Basic engine of reversible type	_____
☒	___	B	With built-in thrust bearing	For direct coupled with propeller
☒	___	B	Starboard design engine With exhaust pipes on starboard side	_____
☒	___	B	Camshaft part with indicator cam	_____
☒	___	B	Indicator drive including planimeter	planimeter type : Digital type
☒	___	B	Exhaust valves with Nimonic	_____
☒	___	B	Electronic governor system	model : AC-C20
☒	___	B	Turning gear with el. Motor (Insulation class : B or F)	maker : Kongsberg power source : AC 440V, 3Ph, 60Hz motor output : 2.2 kW Starting method : D.O.L ☒ with space heater
Axial vibration				
☒	___	B	Axial vibration damper	_____
☒	___	B	Mechanical measuring device for condition check of axial vibration check	_____
Torsional vibration				
☒	___	B	Standard size Turning wheel complete(fly wheel)	J=3,539.5kgm2, Weight : 2,977kg
2nd order moment compensators				
☒	___	B	Without 2nd order moment compensator on aft end	On 4, 5, and 6-cylinder engines only
☒	___	B	Not prepared for 2nd order moment compensator on fore end	_____
Crankshaft & coupling bolts				
☒	___	B	Crankshaft of forged or casted semi-built type	Coupling hole : Final size
Engine seating & gallery				
☒	___	B	Engine prepared for mounting at epoxy chocks	_____
☒	___	B	Engine alignment method : Jacking bracket type	_____
☒	___	B	Galleries complete including gratings, stanchions & railings (excluding ladders)	_____
☒	___	B	Cables & lightings at the engine side under the gallery	Q'ty and position : As maker standard

STD	Q'ty OPT /ship	Inst	Description	Remarks
☒	___	B	Preparation for top bracing on exhaust side	_____
☒	___	S	Mechanical top bracing:friction shims, distance plate retaining plates and bolts	Double bar type _____
Micellaneous				
☒	___	S	Drilled counter flanges, gaskets & bolts/nuts only for non-JIS connections	_____
☒	___	S	Touch up paint	quantity : 18L x 1 can _____
Fuel Oil System				
☒	___	B	Fuel pump without VIT-type	The part load optimising is not valid. _____
☒	___	B	Insulated fuel oil pipes and main return pipes	_____
☒	___	B	Drain box with level alarm for fuel oil leakage	refer to sensor list _____
☒	___	B	Steam tracing in copper pipes for heating of fuel oil pipes	_____
☒	___	B	Safety cover for F.O pipe connection	according to the SOLAS rule _____
L.O System				
☒	___	B	Oil pan with vertical outlets	L.O drain pipe with 3 sheets rubber plate _____
☒	___	B	Uni-lubricating system for bearing, chain drive piston cooling and camshaft	_____
☒	___	B	Safety cover for L.O pipe connection	according to the SOLAS rule _____
Cylinder L.O system				
☒	___	B	Engine internal cylinder L.O piping	_____
☒	___	B	Mechanical cyl. Lubricators, speed & load change dependent type (Maker : ☒ Korean maker)	with heating element _____
Jacket cooling fresh water system				
☒	___	B	Jacket cooling water piping on engine for freshwater cooling of cylinders	_____
☒	___	S	Flexibile joint for J.C.F.W outlet	_____
Low temperature cooling water system				
☒	___	B	Central cooling with freshwater at steel pipes	_____
Starting air system				
☒	___	B	Without slow turning before starting	_____
☒	___	S	Double pressure reducing unit for safety device & control air (☒ Engine mounting type)	press. reducing : 30 → 7 kg/cm ² _____

Q'ty		Inst	Description	Remarks
STD	OPT /ship			
Scavenging air system				
☒	___	B	Engine internal scavenging air piping	_____
☒	___		Designed for 4.5 bar working pressure of cooling water system	<u>Recommended for central cooling</u> _____
☒	___	B	Air cooler with coated cast iron covers	_____
☒	___	B	Float alarm for excessive drain from water mist catcher	_____
☒	___	B	Two(2) auxiliary blowers with electric motors per engine (Insulation class : B or F)	<u>power source : AC 440V, 3Ph, 60Hz</u> <u>motor output : 55kW</u> <u>Starting method : D.O.L</u> <u>☒ with space heater</u>
☒	___	B	Steam fire extinguishing for scavenge air spaces	_____
Exhaust gas system				
☒	___	B	Turbochargers, type TCA	<u>TCA 66-20</u> _____
☒	___	B	Dry cleaning of turbine side	<u>To be excepted water washing for MET</u> _____
☒	___	B	Common L.O system with main L.O for turbocharger	_____
☒	___	B	T/C located on exhaust side of engine	_____
☒	___	B	Gas outlet 15' from vertical, away from engine 	_____
☒	___	B	Exhaust gas receiver without by-pass flange	_____
☒	___	B	Exhaust gas T/C outlet pipe(Transition piece)	_____
☒	___	B	T/C tacho pick-up & converter	_____
☒	___	S	T/C tachometer for ECR	_____
Manoeuvring system				
☒	___	B	Manoeuvring system including various pneumatic & solenoid valves & piping etc., for local & remote control Emergency console on engine	☒ <u>without slow turning</u> _____
☒	___	B	Tag for pneumatic parts	_____
☒	___	B	Line filter for control air & safety air inlet	_____ _____
☒	___	S	Repair kit for pneumatic parts	_____
Spare parts & tools				
☒	___	S	Standard spare parts in accordance with class requirement & maker's standard	_____
☒	___	S	Standard tools for maintenance of main engine for normal lifting procedure	_____

STD	Q'ty OPT /ship	Inst	Description	Remarks
☒	___	S	Grinding machine for exh. valve(electric type)	power source : AC 440V, 3Ph, 60Hz
☒	___	S	Hydraulic jack for holding down bolts and hydraulic jack for end chock bolts(epoxy mounting)	maker : Maker's standard
☒	___	S	Fuel valve test rig with a bench	maker : Maker's standard
☒	___	S	Work table for exhaust valve housing	
☒	___	S	The tool panels and tools should be completely assembled and packed per panel.	
Pressure switch arrangement				
☒	___	B	Switch board arrangement (refer to sensor lists)	
Group for thermometer (dial type)				
☒	___	B	Standard thermometers	range boss (°C) size
			- F.O inlet	0~200 PF3/4
			- Main L.O & P.C.O inlet	0~100 PF3/4
			- Piston cooling oil outlet(1/cyl.)	0~100 PF3/4
			- T/C L.O outlet(TCA,TPL,MET type)	0~200 PF3/4
			- Jacket Cooling F.W inlet	0~100 PF3/4
			- Jacket Cooling F.W outlet(1/cyl.)	0~150 PF3/4
			- C.W inlet to air cooler	0~100 PF3/4
			- C.W outlet from air cooler	0~100 PF3/4
			- Scavenge air inlet of air cooler	0~250 PF3/4
			- Scavenge air outlet of air cooler	0~100 PF3/4
			- Scavenge air receiver	0~120 PF3/4
			- Exh. gas each cyl. outlet (1/cyl.)	50~650 PF3/4
			- Exh. gas T/C inlet	50~650 PF3/4
			- Exh. gas T/C outlet	50~650 PF3/4
			- Thrust bearing	0~120 PF1/2
☒	1	B	Extra thermometer(s)	
			☒ 1 Cooling F. W outlet-common	0~100 PF3/4
Engine pressure gauge arrangement(Oil filled type)				
☒	___	B	Standard Press. Gauges	unit ☒ kg/cm2 ☒ Mpa
				Type Range
			- F.O inlet	bourdon type, 0 ~ 15
			- Scavenge air receiver	bourdon type, 0 ~ 4
			- MLO & PCO inlet	bourdon type, 0 ~ 6
			- Starting air inlet	bourdon type, 0 ~ 50
			- Jacket C.F.W inlet	bourdon type, 0 ~ 6
			- C.W inlet of Air cooler	bourdon type, 0 ~ 6
			- Control air inlet(separate panel)	bourdon type, 0 ~ 10
			- Safety air inlet(separate panel)	bourdon type, 0 ~ 10
			- T/C cleaning line(separately mounted)	bourdon type, 0 ~ 4
			- Exh. Valve spring air inlet(separate panel)	bourdon type, 0 ~ 10
			- T/C L. O inlet(TCA,MET,NA , TPL type)	bourdon type, 0 ~ 6
☒	___	B	U-type manometer(water) for suction air silencer(VTR, TCA and TPL type only)	
☒	___	B	U-type manometer(water) for differential pressure of air cooler inlet/outlet	

STD	Q'ty OPT /ship	Inst	Description	Remarks
☒	___	B	Pressure gauge for differential pressure between exh. gas & scavenge air	
☒	8	S	Pressure gauges for ECR unit ☒ kg/cm2 ☒ Mpa	
			- MLO & PCO inlet(6k)	Electric type
			- Jacket C.F.W inlet(6k)	Electric type
			- C.W inlet of Air cooler(6k)	Electric type
			- F.O inlet(16k)	Electric type
			- Starting air inlet(40k)	Electric type
			- Scavenge air inlet(4k)	Electric type
			- Control air inlet(10k)	Electric type
			- T/C L.O inlet(6k)	Electric type
			Oil mist detector(refer to sensor list)	
☒	___	B	Oil mist detector	☒ 2 Maker: Graviner MK6
			Bridge manoeuvring system	
☒	___	S	Bridge manoeuvring system	maker : Kongsberg model : AC-C20 details: refer to separate EOD
			Aux. Blower starter	
☒	___	S	Starter for aux. blower	
			- Power supply	AC 440V, 3φ, 60Hz
			- Composition	1Panel / 2 aux. blower
			- Control mode	manual + auto
			- Starting method	D.O.L
			- Protection grade	IP 44
			- Mounting type	wall mounting
			- Running hour meter	
			- Others	maker's standard
			T/G motor starter	
☒	___	S	Starter for turning gear motor	
			- Power supply	AC 440V, 3φ, 60Hz
			- Composition	1 panel / 1 eng
			- Control mode	manual
			- Starting method	D.O.L
			- Protection grade	IP 44
			- Mounting type	wall mounting
			- Others	maker's standard
☒	___	S	Remote push button box with 20m cables, 1 plug and 2 receptacles for starter	
			Connection between main engine & shipyard's equipment	
☒	___	B	Main Junction Box	
			Components for additional function	
☒	___	B	Bearing conditioning monitoring system	Maker : AMOT

Chapter 4. Shop test and overhaul inspection

4-1. Inspection

All inspections of the engine are to be carried out in accordance with the Rules and Regulations of the classification society and engine-maker's inspection standard.

Material tests are to be carried out on items which are required by the Rules and Regulations of the classification society

4-2. Shop test and overhauling

The following tests are to be carried out at engine-maker's test bench, using engine-maker's facility, lub. Oil and diesel oil in the presence of the surveyor(s) from classification society and owner.

Lub. Oil and fuel oil are to be selected by the engine-maker.

	Testing item	Running hours
1.	25 % load	30 Min
2.	50 % load	30 Min
3.	75 % load	30 Min
4.	Service load (90% load)	60 Min
5.	100 % load	60 Min
6.	110 % load	30 Min
7.	Governor test	
8.	Safety device test (over speed, main lub oil low pressure of engine inlet, thrust bearing temperature high, turning gear interlock, emergency stop)	
9.	Starting and reversing(for reversible engine only) test	
10.	Minimum revolution running test	
11.	Fuel oil consumption test	
12.	Astern running(for reversible engine only) test at no load	
13.	Main engine bearing clearance must be carried out and recorded at shop by MAN B&W standard dial gauge.	
14.	Nox measurement at shop test (1st engine only in series engines, if required)	

After shop test, appearance inspections are to be carried out on the following parts in the presence of the surveyor(s) from the classification society and owner.

1. Crankshaft without lifting up from bedplate
(1 set of main and crankpin bearing to be dismantled.)
2. 1 set of main bearing
(shells to be dismantled from engine)
3. 1 set of crankpin bearing
(with dismantling from engine)
4. 1 set of crosshead bearing & guide shoe
(without dismantling from engine)
5. 1 set of crosshead pin and guide shoe
(without dismantling from engine)

6. 1 set of cylinder liner
(liner surface, without dismantling from engine)
7. 1 set of cylinder cover
(combustion surface, dismantled from engine)
8. 1 set of piston complete
(dismantled from engine)
9. Camshaft driving chain and chain wheel
(without dismantling from engine)

Remark

The purchaser has the right to send its representative(s) to the engine-maker's plant for the purpose of witnessing inspections and test.

Such representative(s) is (are) not to disturb the manufacturing and testing of the engine

Failure to send such representative(s) is deemed a waiver of the said right and the purchaser will be obliged to accept the result of the said inspections or tests which are accepted by the classification society.

PLAN HISTORY					
REV.	DATE	DESCRIPTION	DRAWN	CHECKED	APPROVED
0	2009. 10. 01.	ISSUED & SUBMITTED TO CLASS	M.S. KIM	H.J. KIM	G.J. KOO
NO. OF PAGES (INCL. COVER) : 19					
YARD	STX OFFSHORE & SHIPBUILDING		HULL NO.	S-4006/12/14/29	
VESSEL TYPE	81K DWT B/C		CLASS	LR	
PLANT	7S50MC-C MAIN PROPULSION				
APPRD : Geum-jae Koo		TITLE : CALCULATION OF TORSIONAL VIBRATION			
CHKED : Hyung-jin Kim					
DRAWN : Min-seok Kim					
DATE : 2009. 10. 01.					
STX Heavy Industries Co., Ltd. Startower 10F, 73-37, Yongho-dong, Changwon Gyeongsangnam-do, Korea Post. 641-842 Tel: +82-55-278-9676 Fax: +82-55-284-2190			DOCUMENT NO. TV-0911-NB733		

PLANT INFORMATION

1. ENGINE

MODEL : 7S50MC-C
POWER AT MCR (KW) : 11060.00
SPEED AT MCR (RPM) : 127.00

1.1 CRANK SHAFT : FSB 242457 / SOLID CRANKPIN
1.2 VIBRATION DAMPER : NONE
1.3 TUNING WHEEL : NONE
1.4 TURNING WHEEL : M.O.I. = 3442.3 KGM2

2. SHAFTING & PROPELLER

MANUFACTURER :
GENERAL ARRANGEMENT DRAWING :

2.1 INTER. SHAFT : D 410.0 / 0.0, UTS = MIN. 600 N/MM2
2.2 PROPE. SHAFT : D 500.0 / 0.0, UTS = MIN. 600 N/MM2
2.3 PROPELLER : 4-BLADE F.P.PROPELLER
: M.O.I. (IN WATER) = 38685.0 KGM2

OPERATING CONDITIONS

1. NORMAL FIRING : PAGE 1 - 11
2. MISFIRING CYL. NO. 7 : PAGE 12 - 18

CONCLUSION

THERE HAS BEEN FOUND SATISFACTORY TORSIONAL VIBRATION BEHAVIOUR IN BOTH
NORMAL AND ONE CYLINDER MISFIRING CONDITION.

MAXIMUM VIBRATORY STRESSES IN THE CRANKSHAFT NEVER EXCEEDS MAN DIESEL
CRANKSHAFT LIMITS.

MAXIMUM VIBRATORY STRESSES IN THE INTERMEDIATE AND PROPELLER SHAFT ARE BELOW
THE RULE GUIDE LINES.

RESTRICTIONS :

BARRED SPEED RANGE (41 ~ 51 RPM) SHOULD BE PROVIDED AGAINST 1 - 7.0 ORDER
UNDER NORMAL AND ONE CYLINDER MISFIRING OPERATION.

IN CASE OF ONE CYLINDER MISFIRING OPERATION THE ENGINE SPEED SHOULD NOT EXCEED
102 RPM (80% OF MCR) IN ORDER TO AVOID THE 3.0 ORDER RESONANCE AND THERMAL
OVERLOADING OF THE WORKING CYLINDERS.

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PARTICULARS OF SYSTEM

DESCRIPTION ON SYSTEM

ENGINE : 7S50MC-C : TURNWHL (J=3442.3 KGM2) :
PROPELLER : 4BLADE-FPP : : J(WTR) = 38685.0KGM2

ENGINE

MODEL	:	7S50MC-C
2 OR 4 STROKE	:	2.00
ANGLE BETWEEN BANKS	(DEG) :	0.00
OUTPUT AT MCR	(KW) :	11060.00
SPEED AT MCR	(RPM) :	127.00
BORE	(MM) :	500.00
STROKE	(MM) :	2000.00
NO. OF CYLINDER	:	7.00
RECIPROCATING WEIGHT	(KG) :	2721.00
CON-ROD LENGTH	(MM) :	2050.00

DATA FOR MASS ELEMENTS :

MASS NO	MASS NO	DESCRIPTION	RPM RATIO	MASS MOMENT OF INERTIA (KGM2)	DAMPING COEFF (NMS/RAD)	FIRING ANGLE (DEG)	CODE
1	1	FL+CW	1.000	230.000	0.000		0
2	2	CYL 1	1.000	4482.000	-58.800	0.0	1
3	3	CYL 2	1.000	4482.000	-58.800	102.9	1
4	4	CYL 3	1.000	4482.000	-58.800	257.1	1
5	5	CYL 4	1.000	4482.000	-58.800	205.7	1
6	6	CYL 5	1.000	4482.000	-58.800	154.3	1
7	7	CYL 6	1.000	4482.000	-58.800	308.6	1
8	8	CYL 7	1.000	4482.000	-58.800	51.4	1
9	9	CHMDRV+THRU	1.000	1358.000	0.000		0
10	10	TURNWHL	1.000	3558.800	0.000		0
11	11	FLANGE	1.000	260.600	0.000		0
12	12	PROPELLER	1.000	38842.301	0.000		4

DATA FOR SHAFT ELEMENTS :

SHAFT NO	SHAFT NO	DESCRIPTION	RPM RATIO	TORSIONAL STIFFNESS (MMN/RAD)	DAMPING COEFF (NMS/RAD)	SHAFT DIA (MM)	CODE
1	1	2 CRANK S1	1.000	848.176	-100.000	599.9	2
2	2	3 CRANK S2	1.000	813.008	-50.000	599.9	1
3	3	4 CRANK S3	1.000	844.595	-50.000	599.9	1
4	4	5 CRANK S4	1.000	782.473	-50.000	599.9	1
5	5	6 CRANK S5	1.000	783.085	-50.000	599.9	1
6	6	7 CRANK S6	1.000	843.170	-50.000	599.9	1
7	7	8 CRANK S7	1.000	867.303	-50.000	599.9	1
8	8	9 CRANK S8	1.000	1100.110	-100.000	599.9	1
9	9	10 CRANK S9	1.000	1626.016	0.000	599.9	2
10	10	11 INTER S10	1.000	33.540	0.000	410.0	4
11	11	12 PROP S11	1.000	74.699	0.000	500.0	5

NOTE : DAMPINGS WITH NEGATIVE SIGN ARE DYNAMIC MAGNIFIERS.
 THE MASS DAMPING FOR THE PROPELLER IS DEPENDENT ON
 THE PROPELLER REVOLUTION.

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NATURAL FREQUENCIES & MODE SHAPES

MODE NO	1	2	3
FREQUENCY (CPM)	323.9	1595.7	3113.9

MASS NO	MODE SHAPES		
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1	1.0000	1.0000	0.9317
2	0.9997	0.9925	0.9054
3	0.9930	0.8318	0.3466
4	0.9805	0.5539	-0.3869
5	0.9606	0.1653	-0.9430
6	0.9344	-0.2494	-0.9247
7	0.9043	-0.5975	-0.3851
8	0.8697	-0.8497	0.3511
9	0.8383	-0.9519	0.7794
10	0.8163	-0.9988	1.0000
11	-0.3513	-0.3146	0.4095
12	-0.8741	0.0233	-0.0075

MODE NO	1	2	3
FREQUENCY (CPM)	323.9	1595.7	3113.9

SHAFT NO	INERTIA TORQUE (KNM)		
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1	0.267E+03	0.639E+04	0.223E+05
2	0.542E+04	0.131E+06	0.454E+06
3	0.105E+05	0.235E+06	0.620E+06
4	0.156E+05	0.304E+06	0.435E+06
5	0.205E+05	0.325E+06	-0.143E+05
6	0.254E+05	0.294E+06	-0.455E+06
7	0.300E+05	0.219E+06	-0.639E+06
8	0.345E+05	0.112E+06	-0.471E+06
9	0.358E+05	0.763E+05	-0.359E+06
10	0.392E+05	-0.229E+05	0.198E+05
11	0.391E+05	-0.252E+05	0.312E+05

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RESONANCE ENGINE SPEED & VECTOR SUMMATION

	NATURAL FREQUENCY (CPM)		
ORDER *	323.9	1595.7	3113.9

1.0 *	324/ 0.049	1596/ 0.180	3114/ 3.059
2.0 *	162/ 0.022	798/ 0.045	1557/ 0.468
3.0 *	108/ 0.219	532/ 3.288	1038/ 1.430
4.0 *	81/ 0.212	399/ 3.285	778/ 1.428
5.0 *	65/ 0.019	319/ 0.044	623/ 0.469
6.0 *	54/ 0.044	266/ 0.172	519/ 3.059
7.0 *	46/ 6.644	228/ 0.853	445/ 1.099
8.0 *	40/ 0.055	199/ 0.189	389/ 3.060
9.0 *	36/ 0.031	177/ 0.046	346/ 0.466
10.0 *	32/ 0.227	160/ 3.292	311/ 1.432
11.0 *	29/ 0.205	145/ 3.281	283/ 1.425
12.0 *	27/ 0.023	133/ 0.043	259/ 0.470
13.0 *	25/ 0.039	123/ 0.164	240/ 3.059
14.0 *	23/ 6.644	114/ 0.853	222/ 1.099
15.0 *	22/ 0.061	106/ 0.197	208/ 3.060
16.0 *	20/ 0.043	100/ 0.048	195/ 0.465
17.0 *	19/ 0.234	94/ 3.295	183/ 1.434
18.0 *	18/ 0.197	89/ 3.278	173/ 1.423
19.0 *	17/ 0.033	84/ 0.042	164/ 0.471
20.0 *	16/ 0.034	80/ 0.157	156/ 3.058

DATA OF MAX SYNTHESIS STRESS
 STRESS VALUE : 89.6 N/MM2
 SPEED : 46.3 RPM
 SHAFT NO. : 10

TABLE OF MAX TORSIONAL STRESS SYNTHESIS (N/MM2)

SHAFT NO.												
RPM	*	1	2	3	4	5	6	7	8	9	10	11

20	*	0.0	5.0	5.1	4.9	5.1	5.1	5.0	1.9	1.9	5.7	3.1
22	*	0.0	5.0	5.2	5.0	5.2	5.4	5.3	2.3	2.2	6.9	3.8
24	*	0.0	5.2	5.5	5.4	5.7	6.0	5.9	2.6	2.6	8.0	4.4
26	*	0.0	5.2	5.4	5.3	5.4	5.7	5.5	2.3	2.3	7.0	3.8
28	*	0.0	5.2	5.5	5.4	5.5	5.8	5.6	2.5	2.5	7.6	4.2
30	*	0.0	5.3	5.7	5.6	5.5	5.8	5.7	2.8	2.8	8.8	4.8
32	*	0.0	5.4	5.8	5.7	5.7	6.1	5.9	3.2	3.2	10.2	5.6
34	*	0.0	5.5	6.0	5.8	6.0	6.5	6.3	3.5	3.5	11.0	6.0
36	*	0.0	5.6	6.2	6.1	6.3	6.9	6.8	4.0	4.0	12.8	7.0
38	*	0.0	5.7	6.5	6.5	6.7	7.5	7.4	4.8	4.9	15.7	8.6
40	*	0.0	6.0	7.0	7.1	7.5	8.5	8.5	6.2	6.3	20.8	11.4
42	*	0.0	6.4	7.8	8.2	8.9	10.3	10.7	8.6	8.8	29.2	16.1
44	*	0.1	7.1	9.4	10.6	12.1	14.4	15.5	14.3	14.7	49.5	27.2
46	*	0.2	7.6	11.6	14.4	17.9	21.9	24.8	25.2	26.1	89.4	49.2
46	*	0.2	7.6	11.6	14.3	18.3	22.3	24.8	25.2	26.1	89.6	49.3
48	*	0.1	6.1	9.0	10.6	13.6	16.5	17.9	15.1	15.7	54.7	30.1
50	*	0.1	5.2	7.2	7.9	9.9	12.1	12.6	8.4	8.9	31.3	17.2
52	*	0.1	5.1	6.5	6.7	8.4	10.3	10.3	5.7	6.1	21.8	12.0
54	*	0.1	5.2	6.2	6.2	7.7	9.4	9.1	4.3	4.6	17.1	9.4
56	*	0.1	5.4	6.3	5.9	7.3	8.8	8.4	3.3	3.6	13.6	7.5
58	*	0.1	5.6	6.4	5.7	7.1	8.6	8.0	2.7	3.0	11.4	6.3
60	*	0.0	5.7	6.5	5.6	7.0	8.5	7.8	2.3	2.6	10.0	5.5
62	*	0.0	5.8	6.7	5.6	6.9	8.4	7.7	2.1	2.3	9.1	5.0
64	*	0.0	6.0	7.0	5.6	6.9	8.5	7.6	1.9	2.1	8.5	4.7
66	*	0.0	6.1	7.2	5.7	7.0	8.6	7.5	1.7	1.9	8.0	4.4
68	*	0.0	6.3	7.5	5.9	7.1	8.7	7.6	1.6	1.8	7.6	4.2
70	*	0.0	6.4	7.8	6.1	7.2	8.9	7.7	1.5	1.8	7.5	4.1
72	*	0.0	6.6	8.1	6.3	7.3	9.1	7.9	1.5	1.8	7.5	4.1
74	*	0.1	6.8	8.4	6.6	7.5	9.4	8.1	1.6	1.9	7.7	4.3
76	*	0.1	7.0	8.8	6.8	7.8	9.7	8.4	1.8	2.1	8.5	4.7
78	*	0.1	7.2	9.1	7.0	8.0	9.8	8.5	2.1	2.3	9.6	5.3
80	*	0.1	7.3	9.4	6.9	7.9	9.6	8.3	2.3	2.6	10.9	6.0

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DATA OF MAX SYNTHESIS STRESS
STRESS VALUE : 89.6 N/MM2
SPEED : 46.3 RPM
SHAFT NO. : 10

TABLE OF MAX TORSIONAL STRESS SYNTHESIS (N/MM2)

SHAFT NO.												
RPM	*	1	2	3	4	5	6	7	8	9	10	11

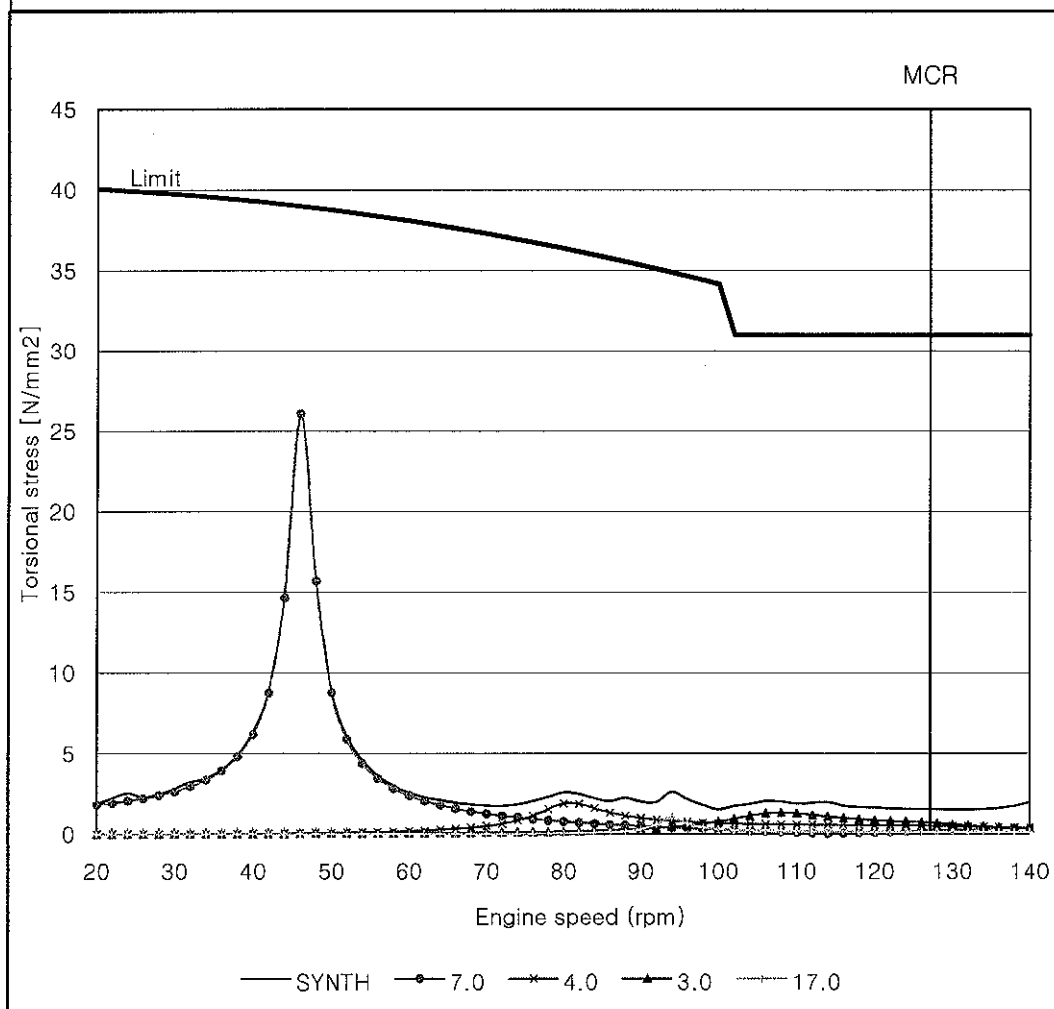
82	*	0.1	7.5	9.5	6.9	7.7	9.4	7.8	2.3	2.5	10.9	6.0
84	*	0.1	7.8	9.9	7.4	7.9	9.9	7.9	2.1	2.2	9.8	5.4
86	*	0.1	8.4	10.9	8.3	8.7	11.0	8.7	2.1	2.1	8.9	5.0
88	*	0.1	9.3	12.6	10.2	10.5	12.8	10.3	2.6	2.3	8.7	4.8
90	*	0.1	9.6	12.7	9.9	10.5	13.1	10.5	2.3	2.0	8.1	4.5
92	*	0.1	9.9	12.8	10.8	11.0	13.5	10.0	2.0	2.0	7.3	4.0
94	*	0.2	10.4	14.0	12.9	12.7	14.9	11.4	2.9	2.6	7.0	3.9
96	*	0.1	10.1	13.5	11.3	11.2	13.8	11.0	2.3	2.2	6.9	3.8
98	*	0.1	9.9	13.2	11.2	10.7	13.3	10.1	2.0	1.8	7.4	4.1
100	*	0.1	10.2	12.9	10.2	10.1	12.5	10.0	1.6	1.5	7.3	4.1
102	*	0.1	10.6	13.3	10.4	10.7	13.5	10.6	1.7	1.7	7.3	4.0
104	*	0.1	10.9	13.7	10.6	10.9	13.9	10.9	1.9	1.9	7.5	4.1
106	*	0.1	11.3	13.9	11.1	11.3	14.4	11.3	2.0	2.1	7.7	4.3
108	*	0.1	11.8	14.5	11.7	11.6	14.6	11.6	2.0	2.0	8.0	4.4
110	*	0.1	12.0	15.1	12.4	11.7	14.9	11.8	2.1	1.9	7.8	4.3
112	*	0.1	12.6	16.2	13.3	12.0	15.3	12.1	2.3	2.0	7.8	4.3
114	*	0.1	13.6	17.3	14.4	13.2	16.1	12.4	2.4	2.0	7.7	4.3
116	*	0.1	13.6	17.5	14.3	12.9	16.2	13.1	2.1	1.8	7.2	4.0
118	*	0.1	13.5	17.4	13.6	13.4	17.3	13.8	1.8	1.7	6.8	3.8
120	*	0.1	14.1	18.2	14.4	13.7	17.8	13.9	1.8	1.7	6.7	3.7
122	*	0.1	14.5	18.9	15.2	14.1	18.5	14.2	1.7	1.6	6.6	3.7
124	*	0.1	14.9	19.5	15.8	14.7	19.0	14.6	1.7	1.6	6.6	3.6
126	*	0.1	15.3	20.3	16.3	15.1	19.6	15.0	1.7	1.6	6.5	3.6
128	*	0.1	15.5	20.8	16.6	15.3	20.0	15.2	1.7	1.5	6.4	3.5
130	*	0.1	15.4	20.9	16.6	15.2	20.1	15.2	1.7	1.5	6.1	3.4
132	*	0.1	15.4	21.0	16.7	15.1	20.1	15.3	1.7	1.5	5.9	3.3
134	*	0.1	15.5	21.1	16.7	15.1	20.2	15.4	1.8	1.6	5.7	3.2
136	*	0.1	15.6	21.3	16.9	15.1	20.3	15.7	2.0	1.6	5.5	3.1
138	*	0.1	15.8	21.5	17.1	15.3	20.9	16.2	2.2	1.8	5.3	3.0
140	*	0.2	16.2	21.7	17.7	16.0	22.0	17.0	2.6	2.0	5.3	3.0

VALUES WITH NEGATIVE SIGN ARE VIBRATORY TORQUES IN (KNM).

Engine type	: 7S50MC-C	Tuning wheel	: NONE
Rating at MCR	: 11060 kW x 127 rpm	Turning wheel	: 3442.3 kgm ²
Firing condition	: Normal firing	Gas harmonics	: 242503

BARRED SPEED RANGE : 41 ~ 51 RPM

Torsional vibratory stress of shaft no. 9 CRANK
DIA. = 599.9 mm UTS = 610.0 N/mm²

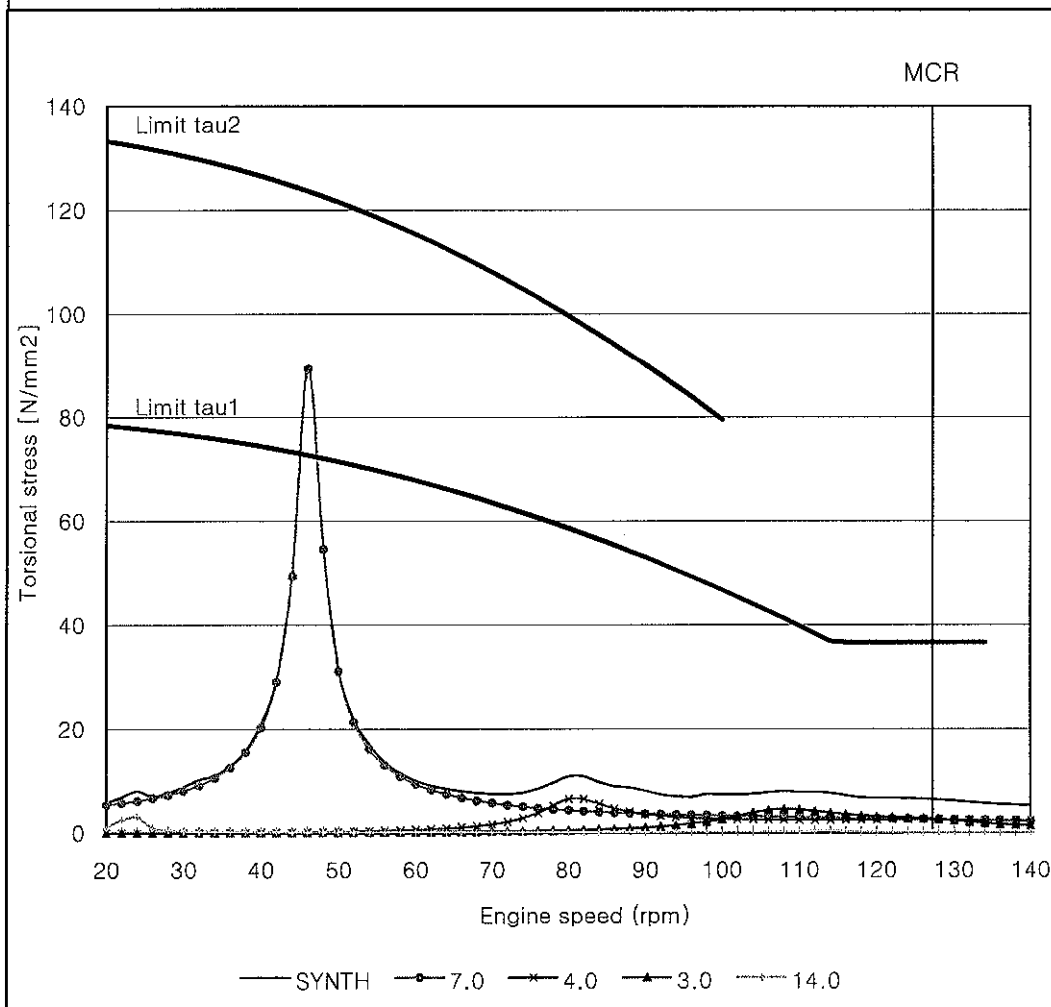


Limit : MAN Diesel Crankshaft Limits
Crankshaft : FSB 242457 / SOLID CRANK PIN

Engine type	: 7S50MC-C	Tuning wheel	: NONE
Rating at MCR	: 11060 kW x 127 rpm	Turning wheel	: 3442.3 kgm ²
Firing condition	: Normal firing	Gas harmonics	: 242503

BARRED SPEED RANGE : 41 ~ 51 RPM

Torsional vibratory stress of shaft no. 10 INTER
DIA. = 410.0 mm UTS = 600.0 N/mm²



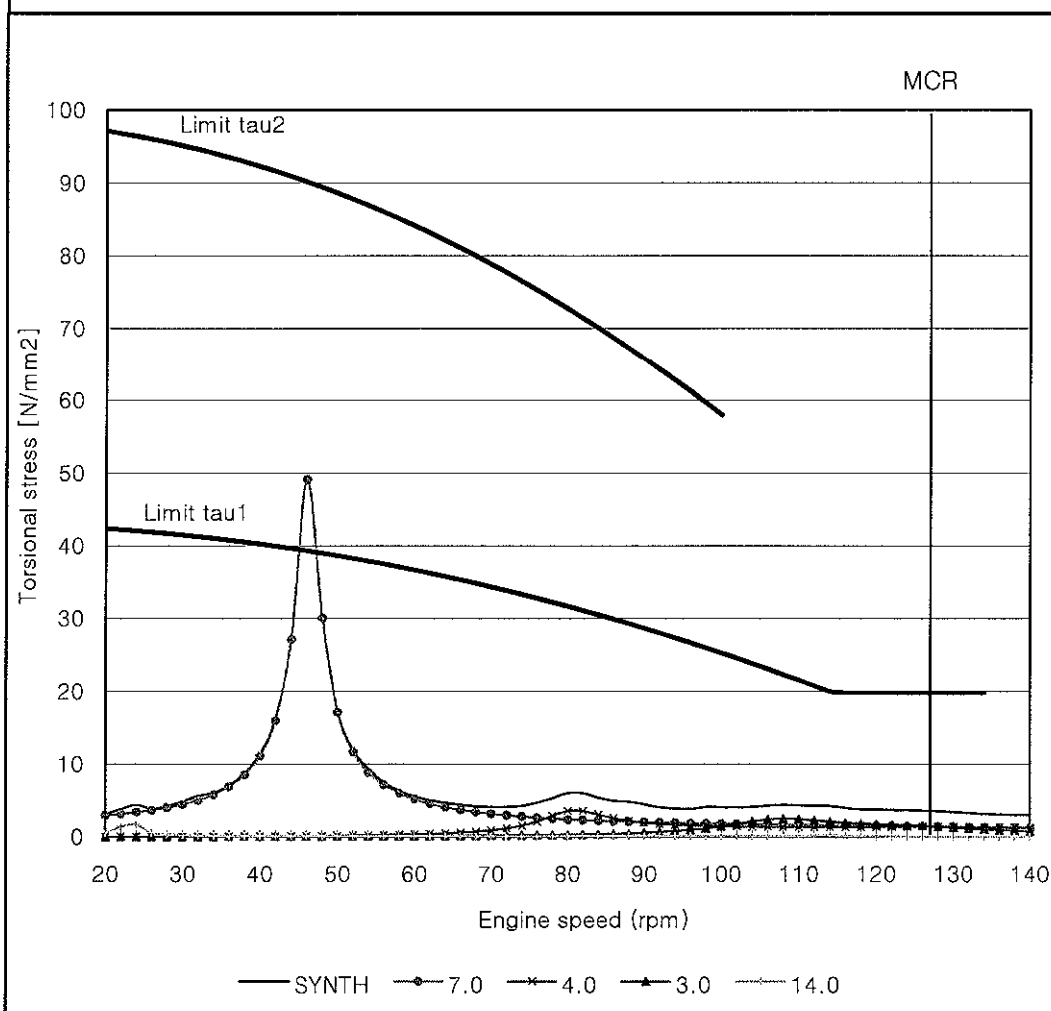
Limit tau1 : Limit for continuous operation with 1.0 Ck factor
Limit tau2 : Limit for transient operation

Engine type : 7S50MC-C
Rating at MCR : 11060 kW x 127 rpm
Firing condition : Normal firing

Tuning wheel : NONE
Turning wheel : 3442.3 kgm²
Gas harmonics : 242503

BARRED SPEED RANGE : 41 ~ 51 RPM

Torsional vibratory stress of shaft no. 11 PROP
DIA. = 500.0 mm UTS = 600.0 N/mm²



Limit tau1 : Limit for continuous operation with 0.55 Ck factor
Limit tau2 : Limit for transient operation

Engine type : 7S50MC-C

Tuning wheel : NONE

Rating at MCR : 11060 kW x 127 rpm

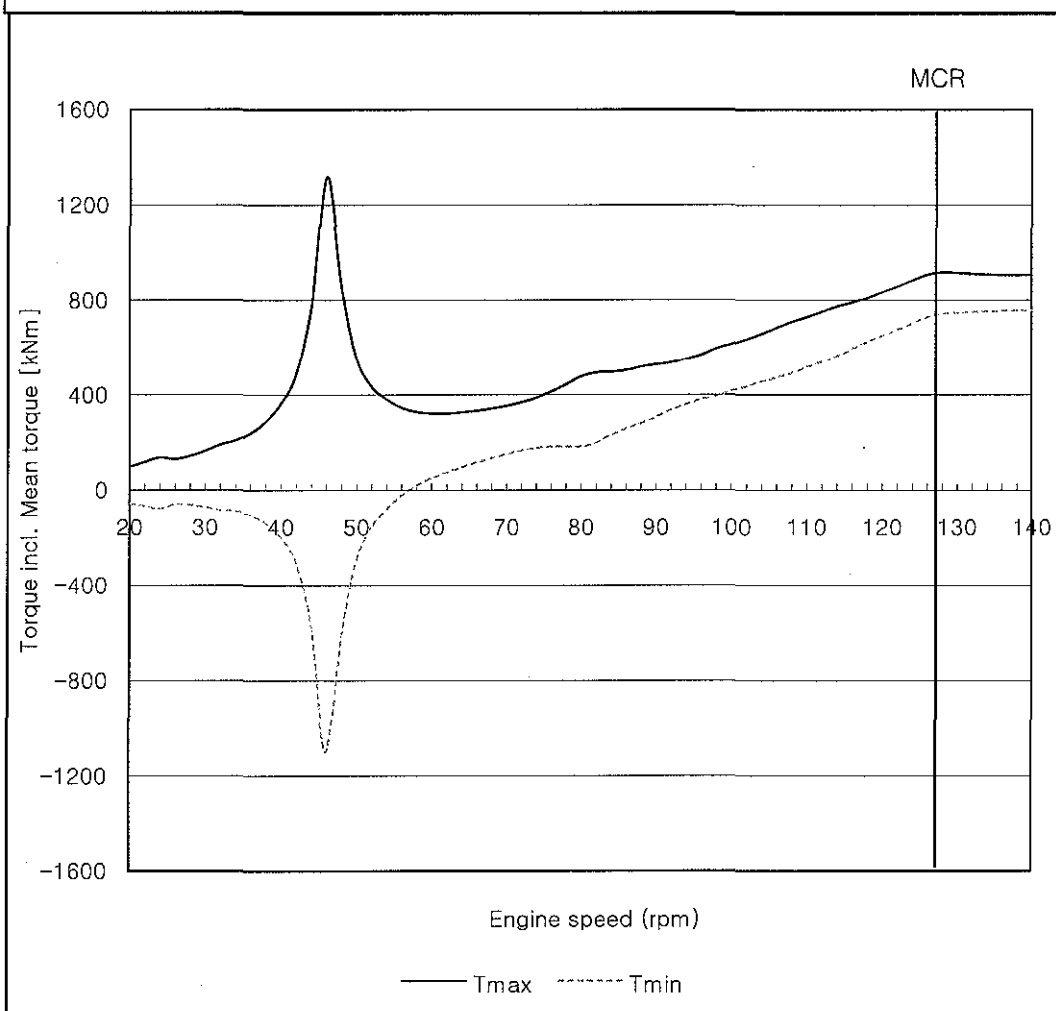
Turning wheel : 3442.3 kgm²

Firing condition : Normal firing

Gas harmonics : 242503

BARRED SPEED RANGE : 41 ~ 51 RPM

Max and Min torque incl. Mean troque of shaft no. 11 PROP



41 RPM	:	424 kNm
46 RPM	:	1319 kNm
51 RPM	:	493 kNm
127 RPM	:	913 kNm

RESONANCE ENGINE SPEED & VECTOR SUMMATION

		NATURAL FREQUENCY (CPM)					
ORDER	*	323.9		1595.7		3113.9	

1.0	*	324/	0.917	1596/	0.684	3114/	2.725
2.0	*	162/	0.891	798/	0.811	1557/	0.153
3.0	*	108/	1.085	532/	2.461	1038/	1.791
4.0	*	81/	1.078	399/	2.456	778/	1.789
5.0	*	65/	0.888	319/	0.813	623/	0.150
6.0	*	54/	0.912	266/	0.690	519/	2.725
7.0	*	46/	5.774	228/	1.709	445/	1.472
8.0	*	40/	0.921	199/	0.678	389/	2.725
9.0	*	36/	0.894	177/	0.810	346/	0.156
10.0	*	32/	1.092	160/	2.465	311/	1.793
11.0	*	29/	1.071	145/	2.452	283/	1.788
12.0	*	27/	0.885	133/	0.814	259/	0.147
13.0	*	25/	0.908	123/	0.695	240/	2.726
14.0	*	23/	5.774	114/	1.709	222/	1.472
15.0	*	22/	0.925	106/	0.672	208/	2.725
16.0	*	20/	0.897	100/	0.808	195/	0.160
17.0	*	19/	1.099	94/	2.470	183/	1.794
18.0	*	18/	1.065	89/	2.447	173/	1.786
19.0	*	17/	0.882	84/	0.816	164/	0.145
20.0	*	16/	0.904	80/	0.701	156/	2.726

DATA OF MAX SYNTHESIS STRESS
 STRESS VALUE : 92.7 N/MM2
 SPEED : 46.0 RPM
 SHAFT NO. : 10

TABLE OF MAX TORSIONAL STRESS SYNTHESIS (N/MM2)

SHAFT NO.												
RPM	*	1	2	3	4	5	6	7	8	9	10	11

20	*	0.0	5.1	5.2	5.1	5.2	5.1	4.9	2.7	2.7	8.1	4.4
22	*	0.0	5.2	5.3	5.2	5.3	5.3	5.2	3.1	3.0	9.3	5.1
24	*	0.0	5.4	5.7	5.6	5.9	5.9	5.6	3.2	3.2	9.7	5.3
26	*	0.0	5.4	5.7	5.6	5.7	5.4	5.2	2.9	2.8	8.5	4.7
28	*	0.0	5.4	5.8	5.7	5.7	5.6	5.3	3.1	3.1	9.6	5.3
30	*	0.0	5.5	6.0	5.9	5.6	5.7	5.5	3.6	3.6	11.3	6.2
32	*	0.0	5.6	6.0	5.9	6.3	6.2	6.0	4.1	4.2	13.3	7.3
34	*	0.0	5.7	6.2	6.0	6.4	6.5	6.2	4.3	4.3	13.6	7.5
36	*	0.0	5.8	6.2	6.6	6.6	7.1	7.2	5.5	5.5	17.4	9.6
38	*	0.0	6.0	6.8	6.8	7.1	7.9	7.4	6.1	6.2	19.8	10.9
40	*	0.0	6.4	7.4	7.3	8.1	9.0	9.0	7.7	7.8	25.4	14.0
42	*	0.1	6.7	8.2	8.6	9.2	10.6	11.3	10.3	10.5	34.5	19.0
44	*	0.1	7.4	9.8	11.1	12.3	14.5	15.8	15.6	16.0	53.5	29.4
46	*	0.2	8.0	12.0	14.9	18.3	22.3	25.0	26.3	27.2	92.7	51.0
46	*	0.2	8.0	12.0	14.8	18.6	22.6	25.3	26.2	27.1	92.5	50.9
48	*	0.1	6.5	9.4	11.2	14.0	16.4	17.8	16.4	17.0	58.7	32.3
50	*	0.1	5.7	7.6	8.5	10.3	11.8	12.3	9.9	10.4	35.8	19.7
52	*	0.1	5.5	7.0	7.3	8.9	10.4	9.9	7.4	7.7	26.5	14.6
54	*	0.1	5.7	7.3	6.9	8.4	9.6	7.4	6.2	6.6	23.9	13.2
56	*	0.1	6.0	7.1	7.0	8.0	7.9	6.9	5.1	5.5	20.0	11.0
58	*	0.1	6.2	7.1	6.8	7.6	7.3	7.1	4.6	4.8	17.6	9.7
60	*	0.1	6.2	7.3	6.9	7.5	7.2	7.2	4.7	4.9	17.0	9.4
62	*	0.1	6.3	7.6	7.3	7.7	7.0	7.2	5.1	5.3	18.7	10.3
64	*	0.1	6.3	8.0	7.6	8.6	7.6	6.3	5.6	6.0	21.6	11.9
66	*	0.1	6.7	8.3	7.3	9.3	8.9	5.4	5.4	5.5	19.8	10.9
68	*	0.1	7.1	8.5	7.0	9.2	9.0	5.3	4.8	4.9	16.9	9.3
70	*	0.1	7.4	8.9	7.1	9.2	8.8	6.0	4.7	4.8	16.2	8.9
72	*	0.1	7.7	9.3	7.3	9.3	8.8	6.6	4.8	4.9	16.1	8.9
74	*	0.1	7.9	9.8	7.6	9.5	8.7	7.2	5.3	5.4	17.5	9.6
76	*	0.1	8.2	10.0	7.7	9.8	9.1	8.3	6.2	6.3	21.0	11.6
78	*	0.1	8.4	10.4	8.0	9.9	9.4	9.2	8.0	8.2	27.2	14.9
80	*	0.1	8.4	10.4	7.6	9.4	9.6	9.5	9.7	10.0	33.5	18.4

DATA OF MAX SYNTHESIS STRESS
 STRESS VALUE : 92.7 N/MM2
 SPEED : 46.0 RPM
 SHAFT NO. : 10

TABLE OF MAX TORSIONAL STRESS SYNTHESIS (N/MM2)

SHAFT NO.												
RPM	*	1	2	3	4	5	6	7	8	9	10	11

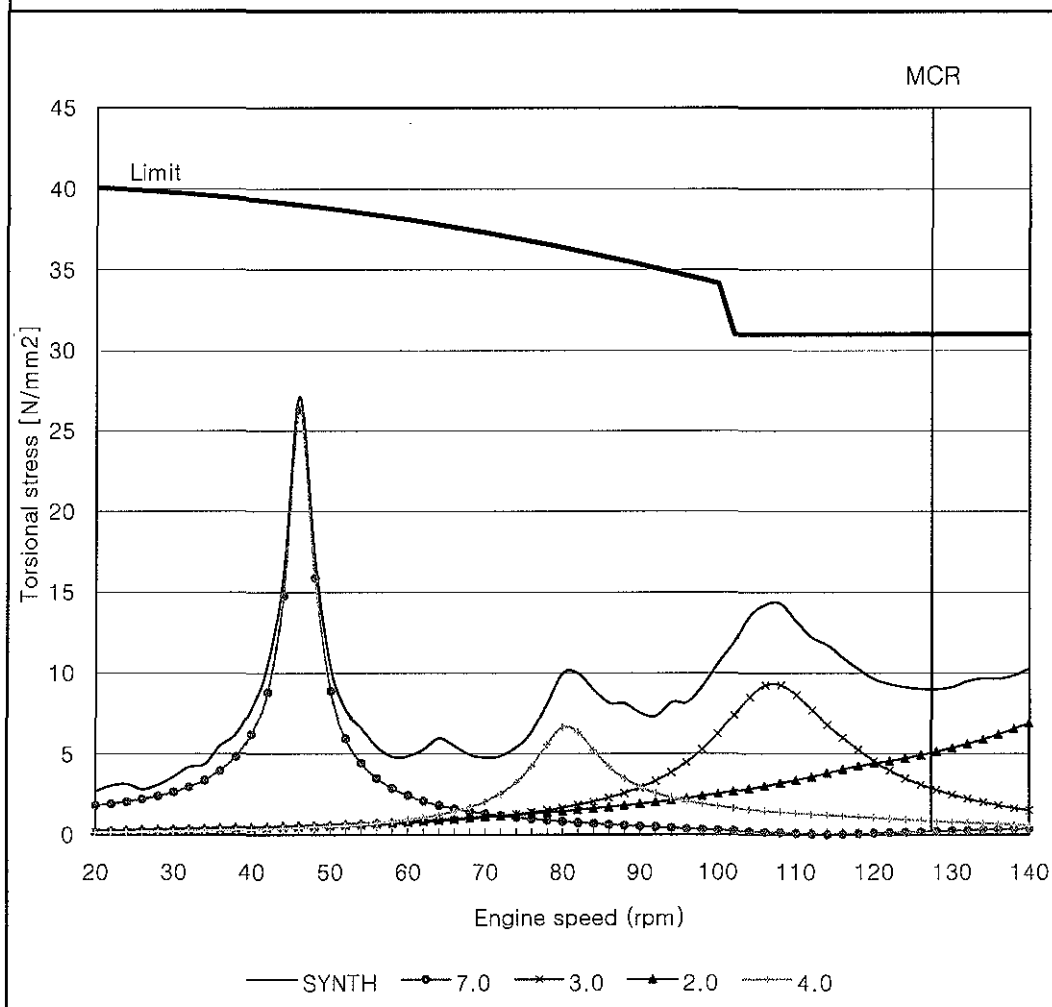
82	*	0.1	8.3	10.0	7.3	8.2	9.0	9.3	9.6	10.0	34.6	19.0
84	*	0.1	8.7	10.4	8.5	8.6	8.3	7.9	8.6	8.9	31.4	17.3
86	*	0.1	9.7	11.2	9.9	9.1	8.7	7.6	8.0	8.1	28.6	15.8
88	*	0.1	10.7	12.8	11.5	10.1	10.6	8.7	8.2	8.1	26.9	14.9
90	*	0.1	10.7	12.9	12.1	11.1	12.1	8.0	7.5	7.5	25.9	14.3
92	*	0.1	11.3	13.7	12.1	11.2	12.6	8.6	7.3	7.3	25.5	14.0
94	*	0.2	12.3	14.9	14.0	12.3	13.6	10.1	8.4	8.2	26.5	14.6
96	*	0.1	11.8	14.1	13.1	11.9	13.5	9.9	8.4	8.2	28.3	15.6
98	*	0.1	12.0	14.7	13.0	13.0	14.7	10.2	9.3	9.2	30.7	16.9
100	*	0.1	12.4	15.5	14.2	13.9	15.8	11.6	10.7	10.6	34.1	18.8
102	*	0.1	12.8	16.1	13.8	14.5	16.6	12.1	11.8	11.9	38.6	21.2
104	*	0.1	13.4	16.2	14.8	13.9	16.2	12.7	13.4	13.5	43.5	23.9
106	*	0.2	13.8	16.8	15.8	14.9	17.1	13.1	14.1	14.2	48.4	26.7
108	*	0.2	13.5	16.9	15.2	15.7	17.7	14.5	13.8	14.3	49.1	27.0
110	*	0.2	14.4	17.4	15.9	14.3	17.2	15.9	12.8	13.2	48.1	26.5
112	*	0.2	15.4	19.0	17.4	15.7	18.0	17.1	12.0	12.2	45.3	25.0
114	*	0.2	16.3	20.6	19.3	17.3	17.9	17.1	11.7	11.7	42.6	23.5
116	*	0.2	15.7	19.6	18.1	17.3	16.8	16.0	10.7	10.9	39.8	21.9
118	*	0.2	15.6	19.2	17.1	16.6	15.7	15.2	9.9	10.2	37.2	20.5
120	*	0.2	15.6	19.0	16.3	16.3	15.8	15.0	9.3	9.6	35.1	19.4
122	*	0.1	15.7	18.7	16.3	16.4	16.7	15.1	9.1	9.3	33.6	18.5
124	*	0.2	15.6	18.6	15.6	15.9	17.5	14.6	9.0	9.1	32.4	17.8
126	*	0.1	15.6	18.6	15.6	15.0	17.5	13.9	8.7	9.0	31.7	17.4
128	*	0.1	15.8	18.6	16.0	14.5	17.8	13.6	8.8	9.0	31.3	17.2
130	*	0.2	16.2	18.3	16.8	14.8	18.6	13.7	9.0	9.1	31.1	17.1
132	*	0.2	17.0	18.4	18.9	15.3	19.9	14.1	9.5	9.5	31.0	17.1
134	*	0.2	16.9	19.7	19.4	16.3	19.0	14.2	9.6	9.7	31.6	17.4
136	*	0.2	16.1	20.0	17.8	15.3	17.9	14.3	9.6	9.6	32.6	18.0
138	*	0.2	15.8	20.3	17.0	14.8	18.5	14.3	9.8	9.9	33.6	18.5
140	*	0.2	16.1	20.6	16.7	15.1	19.4	14.3	10.4	10.3	34.8	19.2

VALUES WITH NEGATIVE SIGN ARE VIBRATORY TORQUES IN (KNM).

Engine type	: 7S50MC-C	Tuning wheel	: NONE
Rating at MCR	: 11060 kW x 127 rpm	Turning wheel	: 3442.3 kgm ²
Firing condition	: No.7 Cylinder Misfiring	Gas harmonics	: 242503

BARRED SPEED RANGE : 41 ~ 51 RPM
MAX. SPEED 102 RPM (Misfiring only)

Torsional vibratory stress of shaft no. 9 CRANK
DIA. = 599.9 mm UTS = 610.0 N/mm²

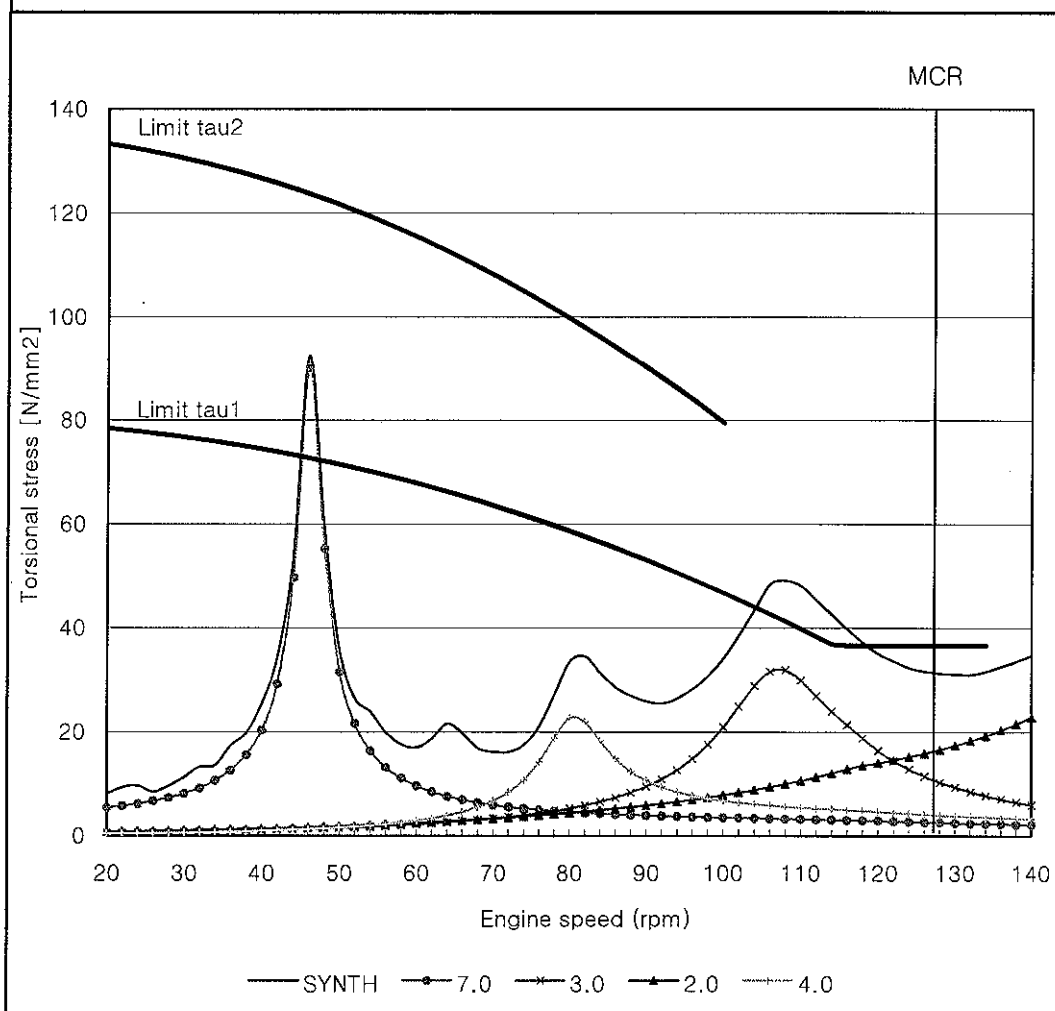


Limit : MAN Diesel Crankshaft Limits
Crankshaft : FSB 242457 / SOLID CRANK PIN

Engine type	: 7S50MC-C	Tuning wheel	: NONE
Rating at MCR	: 11060 kW x 127 rpm	Turning wheel	: 3442.3 kgm ²
Firing condition	: No.7 Cylinder Misfiring	Gas harmonics	: 242503

BARRED SPEED RANGE : 41 ~ 51 RPM
MAX. SPEED 102 RPM (Misfiring only)

Torsional vibratory stress of shaft no. 10 INTER
DIA. = 410.0 mm UTS = 600.0 N/mm²

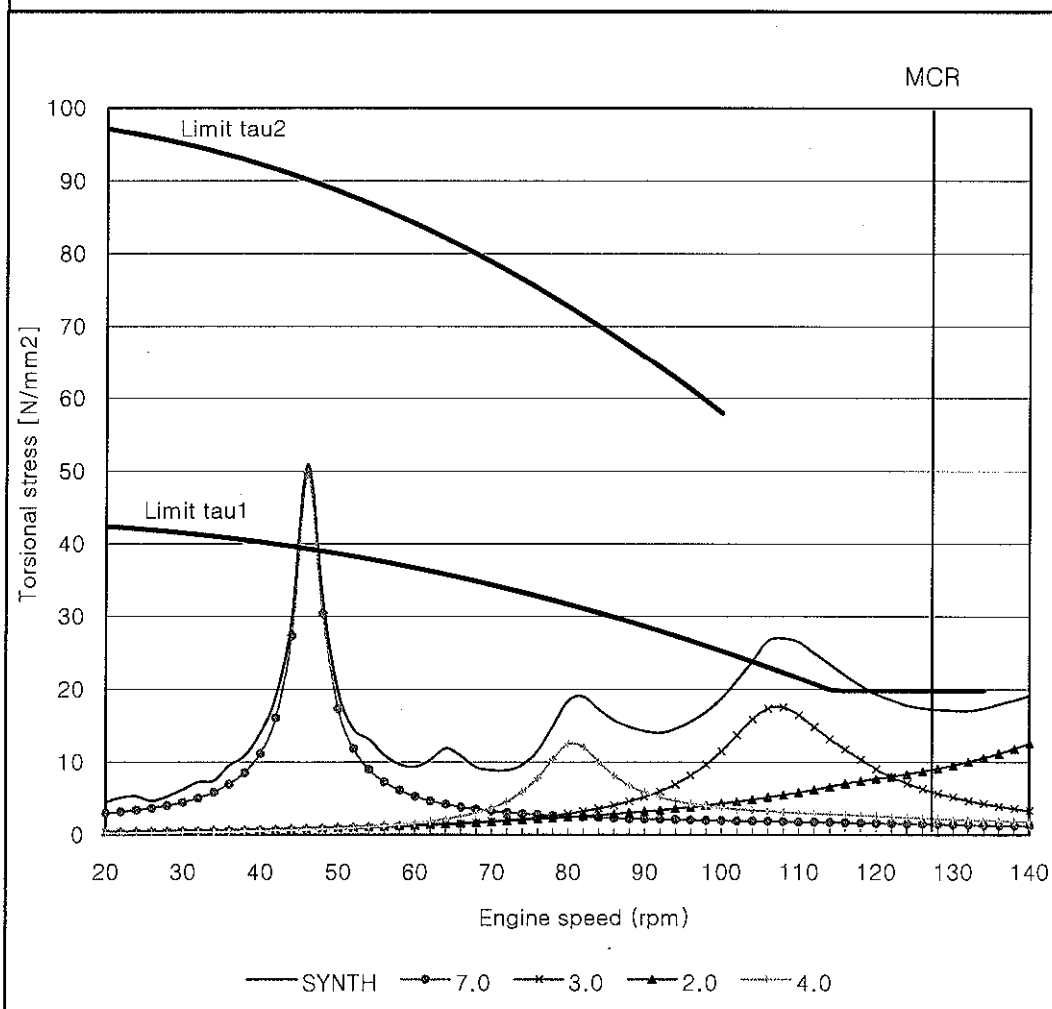


Limit tau1 : Limit for continuous operation with 1.0 Ck factor
Limit tau2 : Limit for transient operation

Engine type	: 7S50MC-C	Tuning wheel	: NONE
Rating at MCR	: 11060 kW x 127 rpm	Turning wheel	: 3442.3 kgm ²
Firing condition	: No.7 Cylinder Misfiring	Gas harmonics	: 242503

BARRED SPEED RANGE : 41 ~ 51 RPM
MAX. SPEED 102 RPM (Misfiring only)

Torsional vibratory stress of shaft no. 11 PROP
DIA. = 500.0 mm UTS = 600.0 N/mm²

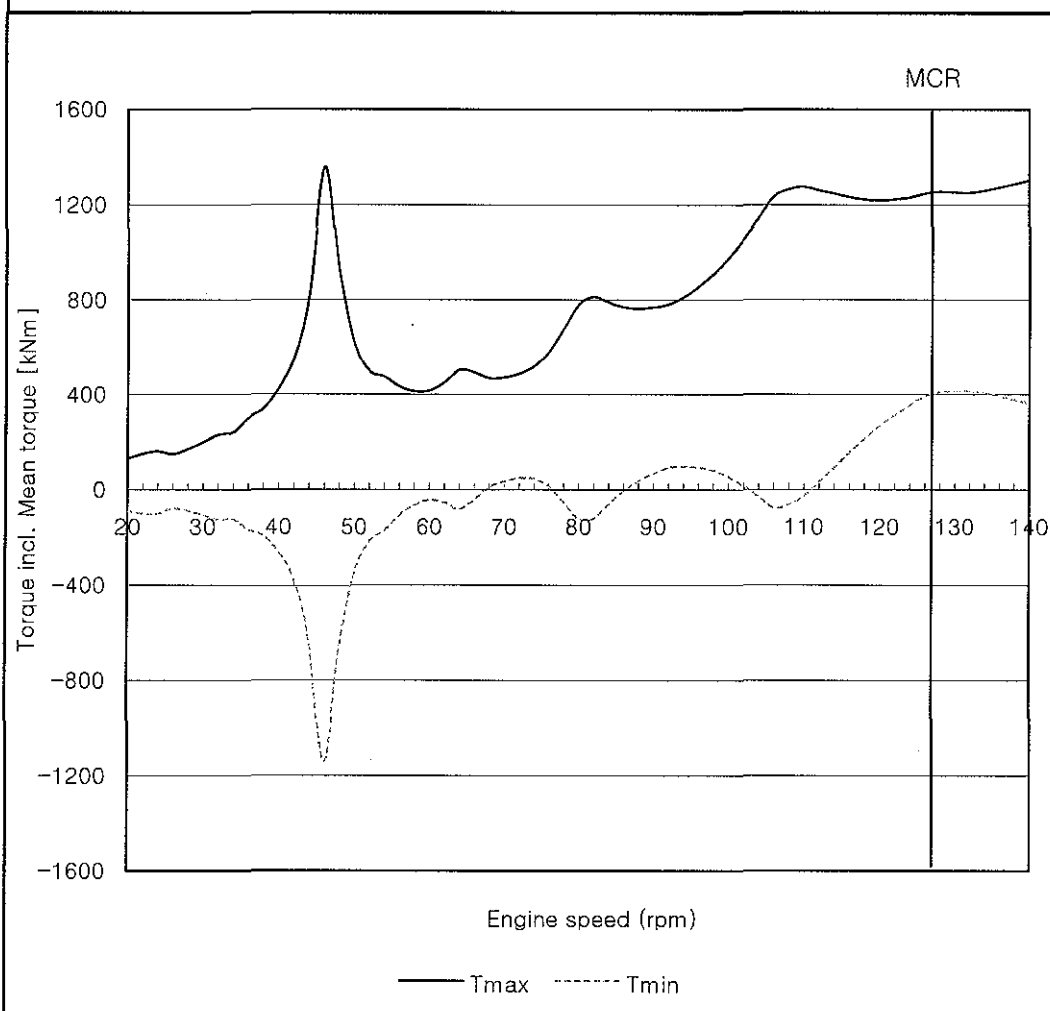


Limit tau1 : Limit for continuous operation with 0.55 Ck factor
Limit tau2 : Limit for transient operation

Engine type	: 7S50MC-C	Tuning wheel	: NONE
Rating at MCR	: 11060 kW x 127 rpm	Turning wheel	: 3442.3 kgm ²
Firing condition	: No.7 Cylinder Misfiring	Gas harmonics	: 242503

BARRED SPEED RANGE : 41 ~ 51 RPM
MAX. SPEED 102 RPM (Misfiring only)

Max and Min torque incl. Mean troque of shaft no. 11 PROP



41 RPM	:	491 kNm
46 RPM	:	1358 kNm
51 RPM	:	555 kNm
127 RPM	:	1251 kNm

PLAN HISTORY					
REV.	DATE	DESCRIPTION	DRAWN	CHECKED	APPROVED
0	2009. 10.01.	ISSUED & SUBMITTED TO CLASS	M.S. KIM	H.J. KIM	G.J. KOO
NO. OF PAGES (INCL. COVER) : 9					
YARD	STX OFFSHORE & SHIPBUILDING		HULL NO.	S-4006/12/14/29	
VESSEL TYPE	81K DWT B/C		CLASS	LR	
PLANT	7S50MC-C MAIN PROPULSION				
APPRD : Geum-jae Koo		TITLE : CALCULATION OF AXIAL VIBRATION			
CHKED : Hyung-jin Kim					
DRAWN : Min-seok Kim					
DATE : 2009. 10. 01.					
STX Heavy Industries Co., Ltd. Startower 10F, 73-37, Yongho-dong, Changwon Gyeongsangnam-do, Korea Post. 641-842 Tel: +82-55-278-9676 Fax: +82-55-284-2190			DOCUMENT NO. AV-0933-NB733		

PLANT INFORMATION

1. ENGINE

MODEL : 7S50MC-C
POWER AT MCR (KW) : 11060.00
SPEED AT MCR (RPM) : 127.00

1.1 CRANK SHAFT : FSB 242457 / SOLID CRANKPIN
1.2 AXIAL VIBRATION DAMPER : FITTED, STANDARD AXIAL DAMPER
1.3 TUNING WHEEL : NONE
1.4 TURNING WHEEL : M = 2822.0 KG

2. SHAFTING & PROPELLER

MANUFACTURER :
GENERAL ARRANGEMENT DRAWING :

2.1 INTER. SHAFT : D 410.0 / 0.0, UTS = MIN. 600 N/MM2
2.2 PROPE. SHAFT : D 500.0 / 0.0, UTS = MIN. 600 N/MM2
2.3 PROPELLER : 4-BLADE F.P. PROPELLER
: M (IN AIR) = 17490.0 KG

CONCLUSION

THE ENGINE IS EQUIPPED WITH A STANDARD AXIAL VIBRATION DAMPER.

PERMISSIBLE DISPLACEMENTS AT CRANKSHAFT FORE END (PEAK TO PEAK) ARE :

- OVERHAUL : 2.170 MM
- ALARM : 2.900 MM
- SLOW-DOWN : 3.620 MM

THERE HAS BEEN FOUND SATISFACTORY AXIAL VIBRATION BEHAVIOUR IN NORMAL OPERATING CONDITIONS.

IN CASE OF AXIAL DAMPER FAILURE, THE ENGINE SPEED SHOULD NOT EXCEED 70 RPM (55% OF MCR) IN ORDER TO AVOID TORSIONAL AND AXIAL RESONANCES.

PARTICULARS OF SYSTEM

DESCRIPTION ON SYSTEM

ENGINE : 7S50MC-C : TURNWHL(2822.0KG)
PROPELLER : 4BLADE-FPP : 17490.0KG(IN AIR)

ENGINE

MODEL : 7S50MC-C
OUTPUT AT MCR (KW) : 11060.00
SPEED AT MCR (RPM) : 127.00
BORE (CM) : 50.00
STROKE (CM) : 200.00
NO. OF CYLINDER : 7
RECIPROCATING WEIGHT (KG) : 2721.00
CON-ROD LENGTH (CM) : 205.00
THICKNESS OF CRANK ARM (CM) : 23.30
BREDTH OF CRANK ARM (CM) : 99.50
DAMPING COEFF. OF DAMPER (KGS/CM) : 2651.17
FIRING ORDER : 1 7 2 5 4 3 6

PROPELLER

NO. OF BLADE : 4
PROPELLER DIA. (CM) : 620.000
PITCH DIA. (CM) : 393.700
PITCH RATIO : 0.635
EXPANDED AREA RATIO : 0.515

STX HEAVY INDUSTRIES CO., LTD.
AXIAL VIBRATION CALCULATION

DOCU. NO. : AV-0933-NB733 | REV. : 0
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MASS NO	DESCRIPTION	MASS (KGS2/CM)	AXIAL STIFFNESS (*E6KG/CM)	THRUST STIFFNESS (*E6KG/CM)	SHAFT DIA (CM)	SHAFT LENGTH (CM)
1	FL+CW	2.1158	101.9680	0.0000	59.99	38.00
2	CYL1	3.2314	0.3295	0.0000	59.99	85.00
3	CYL1+CYL2	6.4617	0.3499	0.0000	59.99	85.00
4	CYL2+CYL3	6.4617	0.4184	0.0000	59.99	85.00
5	CYL3+CYL4	6.4617	1.2345	0.0000	59.99	85.00
6	CYL4+CYL5	6.4617	0.4359	0.0000	59.99	85.00
7	CYL5+CYL6	6.4617	0.3565	0.0000	59.99	85.00
8	CYL6+CYL7	6.4617	0.6124	0.0000	59.99	85.00
9	CYL7	3.2314	169.9466	0.0000	59.99	28.90
10	THRUST CAM	4.8598	84.9733	0.8929	59.99	86.00
11	TURNWHL	6.8786	4.0702	0.0000	41.00	692.30
12	FLANGE	9.0612	6.2684	0.0000	50.00	742.30
13	PROPELLER	35.0345	0.0000	0.0000	0.00	0.00

NATURAL FREQUENCY

0 NODE NATURAL FREQUENCY 53.7 (RAD/SEC) 8.6 (HZ) 513.2 (CPM)

1 NODE NATURAL FREQUENCY 115.1 (RAD/SEC) 18.3 (HZ) 1098.8 (CPM)

STX HEAVY INDUSTRIES CO., LTD.
AXIAL VIBRATION CALCULATION

Docu. NO. : AV-0933-NB733 | REV. : 0
DATE : 2009. 10. 01 | PAGE : 4 / 8

HOLZER TABLE FOR AXIAL VIBRATION

0 NODE AXIAL VIBRATION

P= 53.7 (RAD/SEC)

MASS NO	MASS (KGS2/CM)	AMP (CM)	AXIAL STIFFNESS (*E6KG/CM)	THRUST STIFFNESS (*E6KG/CM)	TOTAL FORCE (KG)	DELTA AMP (CM)
1	2.1158	1.0000	101.9680	0.0000	0.611E+04	0.0001
2	3.2314	0.9999	0.3295	0.0000	0.154E+05	0.0469
3	6.4617	0.9531	0.3499	0.0000	0.332E+05	0.0950
4	6.4617	0.8581	0.4184	0.0000	0.492E+05	0.1177
5	6.4617	0.7404	1.2345	0.0000	0.631E+05	0.0511
6	6.4617	0.6894	0.4359	0.0000	0.759E+05	0.1742
7	6.4617	0.5152	0.3565	0.0000	0.855E+05	0.2399
8	6.4617	0.2753	0.6124	0.0000	0.907E+05	0.1481
9	3.2314	0.1272	169.9466	0.0000	0.919E+05	0.0005
10	4.8598	0.1267	84.9733	0.8929	-0.195E+05	-0.0002
11	6.8786	0.1269	4.0702	0.0000	-0.170E+05	-0.0042
12	9.0612	0.1311	6.2684	0.0000	-0.135E+05	-0.0022
13	35.0345	0.1332	0.0000	0.0000	-0.496E+02	0.0000

STX HEAVY INDUSTRIES CO., LTD.
AXIAL VIBRATION CALCULATION

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0 NODE EXCITING ENERGY

ORDER	CRITIC SPEED (RPM)	INDICATED PRESSURE (KG/CM2)	HARMONIC COEFF (KG/CM2)	VECTOR SUM (CM)	ENG EXT ENERGY (KG-CM)	PROP EXT ENERGY (KG-CM)	TOT EXT ENERGY (KG-CM)
1	513.2	19.38	60.12	0.0165	0.611E+04	0.000E+00	0.611E+04
2	256.6	19.38	44.65	0.0375	0.103E+05	0.000E+00	0.103E+05
3	171.1	19.38	22.75	0.0677	0.949E+04	0.000E+00	0.949E+04
4	128.3	19.38	14.24	0.0677	0.594E+04	0.343E+04	0.937E+04
5	102.6	12.66	7.02	0.0375	0.162E+04	0.000E+00	0.162E+04
6	85.5	8.79	3.70	0.0165	0.377E+03	0.000E+00	0.377E+03
7	73.3	6.46	2.08	0.1522	0.196E+04	0.000E+00	0.196E+04
8	64.1	4.94	1.26	0.0165	0.128E+03	0.234E+03	0.362E+03
9	57.0	3.91	0.74	0.0375	0.172E+03	0.000E+00	0.172E+03
10	51.3	3.16	0.44	0.0677	0.184E+03	0.000E+00	0.184E+03
11	46.7	2.62	0.31	0.0677	0.129E+03	0.000E+00	0.129E+03
12	42.8	2.20	0.24	0.0375	0.549E+02	0.000E+00	0.549E+02
13	39.5	1.87	0.19	0.0165	0.192E+02	0.000E+00	0.192E+02
14	36.7	1.61	0.17	0.1522	0.161E+03	0.000E+00	0.161E+03
15	34.2	1.41	0.16	0.0165	0.160E+02	0.000E+00	0.160E+02
16	32.1	1.24	0.14	0.0375	0.320E+02	0.000E+00	0.320E+02
17	30.2	1.10	0.13	0.0677	0.532E+02	0.000E+00	0.532E+02
18	28.5	0.98	0.11	0.0677	0.466E+02	0.000E+00	0.466E+02
19	27.0	0.88	0.10	0.0375	0.233E+02	0.000E+00	0.233E+02
20	25.7	0.79	0.09	0.0165	0.906E+01	0.000E+00	0.906E+01

0 NODE DAMPING AND AMPLITUDE

ORDER	BRG DAMPER DAMP ENERGY (KG-CM)	HYSTERESIS DAMP ENERGY (KG-CM)	PROPELLER DAMP ENERGY (KG-CM)	TOTAL DAMP ENERGY (KG-CM)	ACT AMP MASS 1 (MM)
1	0.465E+06	0.220E+04	0.247E+04	0.470E+06	0.1301
2	0.465E+06	0.220E+04	0.124E+04	0.469E+06	0.2205
3	0.465E+06	0.220E+04	0.824E+03	0.468E+06	0.2027
4	0.465E+06	0.220E+04	0.618E+03	0.468E+06	0.2003
5	0.465E+06	0.220E+04	0.495E+03	0.468E+06	0.0347
6	0.465E+06	0.220E+04	0.412E+03	0.468E+06	0.0081
7	0.465E+06	0.220E+04	0.353E+03	0.468E+06	0.0418
8	0.465E+06	0.220E+04	0.309E+03	0.468E+06	0.0077
9	0.465E+06	0.220E+04	0.275E+03	0.468E+06	0.0037
10	0.465E+06	0.220E+04	0.247E+03	0.468E+06	0.0039
11	0.465E+06	0.220E+04	0.225E+03	0.468E+06	0.0028
12	0.465E+06	0.220E+04	0.206E+03	0.468E+06	0.0012
13	0.465E+06	0.220E+04	0.190E+03	0.468E+06	0.0004
14	0.465E+06	0.220E+04	0.177E+03	0.468E+06	0.0035
15	0.465E+06	0.220E+04	0.165E+03	0.468E+06	0.0003
16	0.465E+06	0.220E+04	0.155E+03	0.468E+06	0.0007
17	0.465E+06	0.220E+04	0.145E+03	0.468E+06	0.0011
18	0.465E+06	0.220E+04	0.137E+03	0.468E+06	0.0010
19	0.465E+06	0.220E+04	0.130E+03	0.468E+06	0.0005
20	0.465E+06	0.220E+04	0.124E+03	0.468E+06	0.0002

STX HEAVY INDUSTRIES CO., LTD.
AXIAL VIBRATION CALCULATION

Docu. No. : AV-0933-NB733 | REV. : 0
DATE : 2009. 10. 01 | PAGE : 6 / 8

HOLZER TABLE FOR AXIAL VIBRATION

1 NODE AXIAL VIBRATION

P= 115.1 (RAD/SEC)

MASS NO	MASS (KGS2/CM)	AMP (CM)	AXIAL STIFFNESS (*E6KG/CM)	THRUST STIFFNESS (*E6KG/CM)	TOTAL FORCE (KG)	DELTA AMP (CM)
1	2.1158	1.0000	101.9680	0.0000	0.280E+05	0.0003
2	3.2314	0.9997	0.3295	0.0000	0.708E+05	0.2148
3	6.4617	0.7849	0.3499	0.0000	0.138E+06	0.3942
4	6.4617	0.3906	0.4184	0.0000	0.171E+06	0.4096
5	6.4617	-0.0189	1.2345	0.0000	0.170E+06	0.1375
6	6.4617	-0.1564	0.4359	0.0000	0.156E+06	0.3587
7	6.4617	-0.5152	0.3565	0.0000	0.112E+06	0.3150
8	6.4617	-0.8301	0.6124	0.0000	0.413E+05	0.0674
9	3.2314	-0.8975	169.9466	0.0000	0.286E+04	0.0000
10	4.8598	-0.8975	84.9733	0.8929	0.746E+06	0.0088
11	6.8786	-0.9063	4.0702	0.0000	0.664E+06	0.1631
12	9.0612	-1.0694	6.2684	0.0000	0.536E+06	0.0855
13	35.0345	-1.1549	0.0000	0.0000	-0.953E+02	0.0000

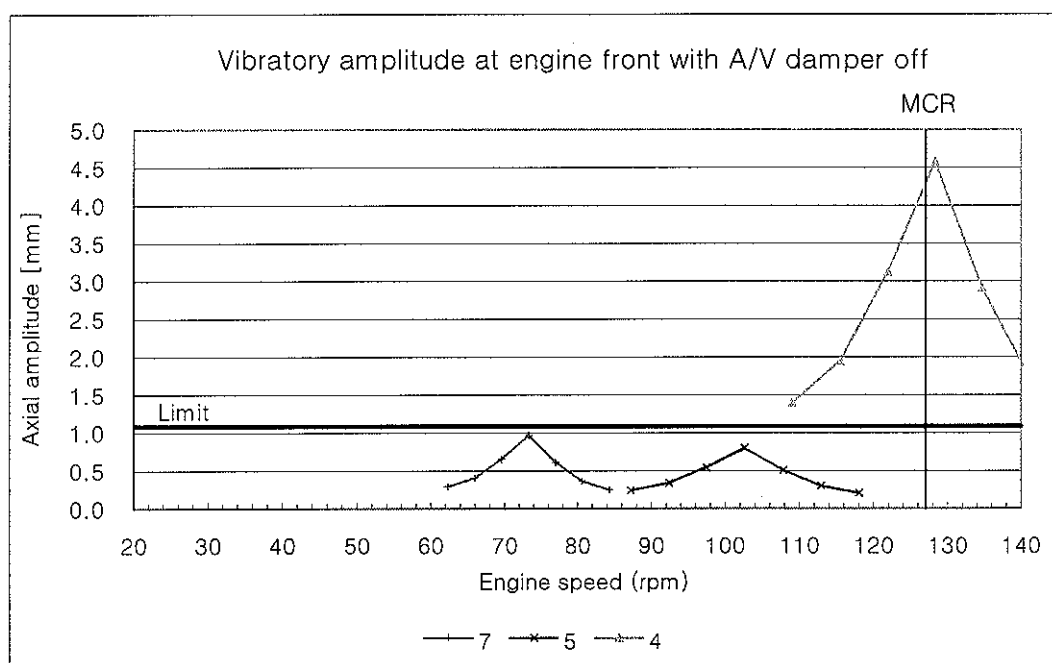
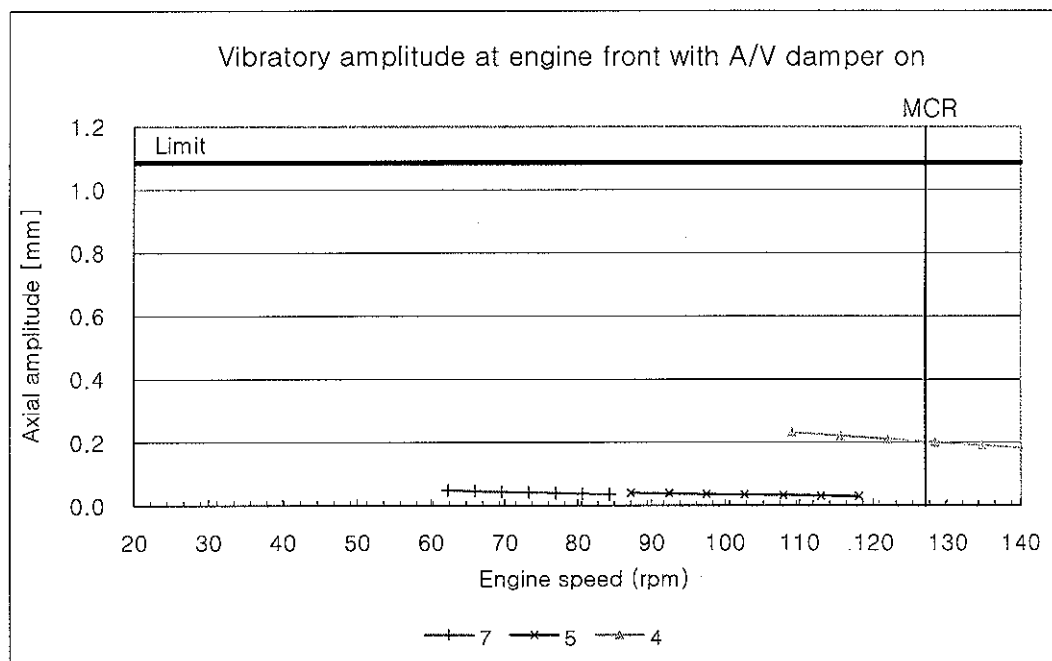
1 NODE EXCITING ENERGY

ORDER	CRITIC SPEED (RPM)	INDICATED PRESSURE (KG/CM2)	HARMONIC COEFF (KG/CM2)	VECTOR SUM (CM)	ENG EXT ENERGY (KG-CM)	PROP EXT ENERGY (KG-CM)	TOT EXT ENERGY (KG-CM)
1	1098.8	19.38	228.78	0.0814	0.115E+06	0.000E+00	0.115E+06
2	549.4	19.38	173.57	0.0913	0.978E+05	0.000E+00	0.978E+05
3	366.3	19.38	70.44	0.0792	0.344E+05	0.000E+00	0.344E+05
4	274.7	19.38	13.27	0.0792	0.648E+04	0.136E+06	0.143E+06
5	219.8	19.38	9.65	0.0913	0.544E+04	0.000E+00	0.544E+04
6	183.1	19.38	5.41	0.0814	0.272E+04	0.000E+00	0.272E+04
7	157.0	19.38	2.70	0.3843	0.641E+04	0.000E+00	0.641E+04
8	137.4	19.38	1.28	0.0814	0.644E+03	0.930E+04	0.994E+04
9	122.1	17.91	0.67	0.0913	0.378E+03	0.000E+00	0.378E+03
10	109.9	14.51	0.98	0.0792	0.479E+03	0.000E+00	0.479E+03
11	99.9	11.99	1.05	0.0792	0.512E+03	0.000E+00	0.512E+03
12	91.6	10.08	0.98	0.0913	0.553E+03	0.000E+00	0.553E+03
13	84.5	8.59	0.83	0.0814	0.418E+03	0.000E+00	0.418E+03
14	78.5	7.40	0.72	0.3843	0.170E+04	0.000E+00	0.170E+04
15	73.3	6.45	0.62	0.0814	0.310E+03	0.000E+00	0.310E+03
16	68.7	5.67	0.49	0.0913	0.276E+03	0.000E+00	0.276E+03
17	64.6	5.02	0.41	0.0792	0.199E+03	0.000E+00	0.199E+03
18	61.0	4.48	0.34	0.0792	0.165E+03	0.000E+00	0.165E+03
19	57.8	4.02	0.28	0.0913	0.157E+03	0.000E+00	0.157E+03
20	54.9	3.63	0.24	0.0814	0.118E+03	0.000E+00	0.118E+03

1 NODE DAMPING AND AMPLITUDE

ORDER	BRG DAMPER DAMP ENERGY (KG-CM)	HYSTERESIS DAMP ENERGY (KG-CM)	PROPELLER DAMP ENERGY (KG-CM)	TOTAL DAMP ENERGY (KG-CM)	ACT AMP MASS 1 (MM)
1	0.999E+06	0.151E+05	0.852E+06	0.187E+07	0.6159
2	0.999E+06	0.151E+05	0.426E+06	0.144E+07	0.6790
3	0.999E+06	0.151E+05	0.284E+06	0.130E+07	0.2652
4	0.999E+06	0.151E+05	0.213E+06	0.123E+07	1.1642
5	0.999E+06	0.151E+05	0.170E+06	0.118E+07	0.0459
6	0.999E+06	0.151E+05	0.142E+06	0.116E+07	0.0235
7	0.999E+06	0.151E+05	0.122E+06	0.114E+07	0.0564
8	0.999E+06	0.151E+05	0.106E+06	0.112E+07	0.0887
9	0.999E+06	0.151E+05	0.946E+05	0.111E+07	0.0034
10	0.999E+06	0.151E+05	0.852E+05	0.110E+07	0.0044
11	0.999E+06	0.151E+05	0.774E+05	0.109E+07	0.0047
12	0.999E+06	0.151E+05	0.710E+05	0.109E+07	0.0051
13	0.999E+06	0.151E+05	0.655E+05	0.108E+07	0.0039
14	0.999E+06	0.151E+05	0.608E+05	0.107E+07	0.0158
15	0.999E+06	0.151E+05	0.568E+05	0.107E+07	0.0029
16	0.999E+06	0.151E+05	0.532E+05	0.107E+07	0.0026
17	0.999E+06	0.151E+05	0.501E+05	0.106E+07	0.0019
18	0.999E+06	0.151E+05	0.473E+05	0.106E+07	0.0016
19	0.999E+06	0.151E+05	0.448E+05	0.106E+07	0.0015
20	0.999E+06	0.151E+05	0.426E+05	0.106E+07	0.0011

Engine type	: 7S50MC-C MK7	Tuning wheel	: NONE
Rating at MCR	: 11060 kW x 127 rpm	Turning wheel	: 2822.0 kg
Damper condition	: ON & OFF	Gas harmonics	: 242504



Limit : 1.085 mm (0-peak) Overhaul limit according to MAN Diesel



SHOP TRIAL RECORD OF DIESEL ENGINE

ORDERER : STX Offshore & Shipbuilding Co., Ltd.

OWNER : National Navigation Company

SHIPYARD : STX Offshore & Shipbuilding Co., Ltd.

SHIP NO. : S-4012

TYPE : 7S50MC-C7

ENGINE NO. : SB7S50-09511

WITNESSED BY :

11/08/10 NNC

APPROVED BY : K. W. Lee

[Signature]

WITNESSED BY :

S.M. Kim 11. Aug. 2010. (Shipyard)

REVIEWED BY : M. K. Park

[Signature]

WITNESSED BY :

CHECKED BY : K. S. Yang

[Signature]

WITNESSED BY :

Lloyd's Register Asia
S. J. Han
Changwon 11.08.10

PREPARED BY : K. J. Lim

[Signature]

BUS1053548

STX Heavy Industries Co., Ltd.



1. Shop Test Protocol

Engine No. : SB7S50-09511

Date : 2010. 08. 10.

Contents

Sheet No.

Main specification of equipment	MA-H-804C-17W-02
Function test	MA-H-804C-17W-03
-. Starting test	
-. Governor test	
-. Safety device test	
-. Minimum revolution test	
Summary data of shop trial	MA-H-804C-17W-04
I.S.O correction Data	MA-H-804C-17W-04.1
Performance data sheet	
- 25 % Load	MA-H-804C-17W-05.1
- 50 % Load	MA-H-804C-17W-05.2
- 75 % Load	MA-H-804C-17W-05.3
- 90 % Load	MA-H-804C-17W-05.4
- 100% Load 1st	MA-H-804C-17W-05.5
- 100% Load 2nd	MA-H-804C-17W-05.5.1
- 110% Load	MA-H-804C-17W-05.6
Fuel oil consumption table	MA-H-804C-17W-06
Adjustment data sheet	MA-H-804C-17W-07
Performance graph	MA-H-804C-17W-08
Presumed horse power curve	MA-H-804C-17W-09
Graph For EL-Governor Limit	MA-H-804C-17W-10
Appendices (Record of inspection)	
-. Bed plate alignment	
-. Clearance of bearing	
-. Clearance of piston ring	
-. Piston centering	

stx		2. Main Specification of Equipment		Prepared by: K. S. YANG	
Engine No. : SB7S50-09511					Date : 2010. 08. 10.
Engine	Engine Type	7S50MC-C7			
	No. of cylinder	7	ea		
	Diameter of cylinder	500	mm		
	Stroke of piston	2,000	mm		
	Rated output at (MCR)	11,060	kW		
	Rated speed at (MCR)	127	rpm		
	Max.combustion pressure	150	bar		
	Mean effective pressure	19.0	bar		
	Piston speed at (MCR)	8.47	m/s		
	Firing order (Ahead)	1 - 7 - 2 - 5 - 4 - 3 - 6			
	Rotational direction	Clock wise (view from flywheel)			
Dynamometer	Maker	ZOELLNER			
	Type	19N234F3			
	Maximum capacity	30,000 kW	200 rpm		
	Date of calibration	2010. 06. 03.			
Turbocharger	Maker	STX Metal MAN			
	Type	TCA66-20063			
	Serial No.	SQN 006			
	Max.design value	16,000 rpm	Tmax. 500 °C		
Governor/Actuator	Maker	Kongsberg			
	Type	AC-C20	SGMGH-13Q5A-OM11		
	Serial No.	928	B0102F508210009		
Aux.blower	Maker	TMMCO			
	Type	HAA-334/120N			
	Serial No.	SSA 01430101	SSA 01430102		
	Capacity	2.24 / 4.55 m³/s			
	Speed	3,580 rpm			
	Pressure	571/327 mmAq			
Blower motor	Maker	ABB			
	Type	M2QA225M2B			
	Serial No.	3GC10500262333001001	3GC105002623330021001		
	Voltage / Frequency	440 V	60 Hz		
	Power / Speed	55.0 kW	3,580 rpm		
Air cooler	Maker	DONGHWA ENTEC			
	Type	LKMY-JT			
	Serial No.	DHL31280-A			
	Cooling surface	747.8 m²			
Lubricator	Maker	SHIN HEUNG			
	Type	ATLAS			
	Pump unit	21 * 2			
	Serial No.	10070801	10070802		
Turning gear	Maker	YU RIM PRECISION			
	Type	YRTG 20-27			
	Serial No.	2010-125			

stx		3. Function Test				Prepared by: K. S. YANG		
Engine No. : SB7S50-09511		Owner : NNC		Output : 11,060 kW				
Hull No. : S-4012		Shipyard : STX		Test date : 2010. 08. 10				
Starting Test						Unit : kg/cm ²		
Initial air pressure	26.0	6	Astern	22.2	12	Astern		
1 Ahead	25.4	7	Ahead	21.5	13	Ahead		
2 Astern	24.8	8	Astern	Omit	14	Astern		
3 Ahead	24.2	9	Ahead		15	Ahead		
4 Astern	23.6	10	Astern		16	Astern		
5 Ahead	22.9	11	Ahead		18	Ahead		
Test condition :		Air reservoir capacity : 4,500 ℓ		Room temperature : 28 °C				
		Cooling water temp. : 30 °C		Lub oil temperature : 32 °C				
Governor Test								
(Full Load ----> 15~20 %)								
Setting(MCR) 127.0 rpm		Instant 132.1 rpm		Instant Variation 4.02%		21 s		
				Stability 127.0 rpm				
Safety Device Test								
Test Item (Shut Down)		Design value		Actual				
Over speed		138.4 rpm		137.0 rpm				
Main lub oil low pressure of engine inlet		1.2 kg/cm ²		1.2 kg/cm ²				
Thrust bearing temperature high (Only Simulation Test)		(90±2) °C		43°C / 43°C simulation				
Oil mist detector level high		Check		O.K All cylinders				
Turning gear interlock		Check		O.K				
Emergency stop		Check		O.K				
Minimum Revolution Test								
Engine speed	Output (kW)	Load (ton)	Load (kNm)	Governor Index	Scav' Air Air	Fuel pump Index	T/C-1 speed	T/C-2 speed
30.6 rpm	339.7	-	106.0	23.0	0.05 kg/cm ²	22.0 mm	1785 rpm	-
Auxiliary Blower Control								
Test item		Design value		Actual				
Auxiliary blower control (#1 Auto Start)		0.45 kg/cm ²		0.45 kg/cm ²				
Auxiliary blower control (#2 Auto Stop)		0.70 kg/cm ²		0.70 kg/cm ²				
Thrust Bearing Axial Clearance		0.5 ~ 1.0 mm		0.60 mm				

MA-H-804C-17W-03

STX Heavy Industries Co.,Ltd.

A4 (210x297)

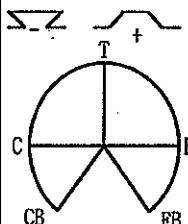


4. Summary Data of Shop Trial

Prepared by:
K. S. YANG

Engine No. : SB7S50-09511	Owner	NNC			Output	11,060 kW		
Hull No. : S-4012	Shipyard	STX			Test Date	2010. 08. 10		
	Unit	1	2	3	4	5	6	7
Load	%	25	50	75	90	100-1	100-2	110
Engine speed	rpm	80.0	100.8	115.4	122.6	127.0	127.0	131.1
Engine output	kW	2,765	5,530	8,293	9,954	11,060	11,060	12,166
Room temperature	℃	31.3	31.8	32.2	32.8	33.3	33.4	33.6
Room humidity	%	64.0	63.0	61.0	57.0	56.0	54.0	53.0
Barometer pressure	mbar	1016	1016	1015	1015	1014	1014	1014
Fuel oil temperature	℃	38.0	38.0	38.0	40.0	40.0	40.0	40.0
Fuel oil consumption	kg/h	508.8	982.8	1437.0	1730.4	1934.4	1939.2	2151.6
Fuel oil consumption	g/kW/h	184.01	177.72	173.28	173.84	174.90	175.33	176.85
Fuel oil consumption (ISO.)	g/kW/h	182.63	176.26	171.73	172.16	173.29	173.72	175.17
Max.Combustion pressure	bar	71.7	98.7	128.7	143.1	148.6	148.4	153.9
Compression pressure	bar	42.7	65.0	99.6	115.4	125.3	125.4	134.4
Mean effective pressure	bar	7.5	12.0	15.7	17.7	19.0	19.0	20.3
Fuel injection pump index	mm	32.0	45.0	58.0	66.0	72.0	72.0	77.0
Exh.temp cylinder outlet	℃	293.6	306.4	301.4	317.1	338.6	338.6	353.6
Exh.temp T/C inlet	℃	330.0	370.0	375.0	405.0	435.0	435.0	460.0
Exh.temp T/C outlet	℃	280.0	280.0	235.0	250.0	260.0	260.0	265.0
Turbocharger speed (X 1000)	rpm	5.2	9.5	12.6	13.7	14.4	14.4	14.9
Scav.air temperature	℃	36.0	31.0	36.0	39.0	40.0	40.0	41.0
Scav.air pressure	kg/cm ²	0.25	0.90	1.80	2.30	2.55	2.55	2.80
Turbocharger gas inlet press	kg/cm ²	0.19	0.80	1.60	2.10	2.35	2.35	2.55

Crank Shaft Deflection

 CRANK PIN POSITION VIEW FROM FLY WHEEL (unit : 1/100 mm)	Cold Condition (before engine running with dynamometer)								Room Temp.	26 ℃
	Cylinder No.	1	2	3	4	5	6	7	-	
	fore/aft most	CB	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
	max:±47.0	C	1.0	-8.0	-6.0	0.0	-1.0	-3.0	7.0	
	others	T	6.0	-10.0	-5.0	5.0	0.0	3.0	15.0	
	max:±18.0	E	1.0	-7.0	-3.0	2.0	-2.0	3.0	5.0	
		EB	-1.0	0.0	1.0	0.0	0.0	1.0	0.0	
	Hot Condition (after engine running with dynamometer)								Lub oil Temp.	44 ℃
	fore/aft most	CB	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
	max:±47.0	C	1.0	1.0	-4.0	-5.0	-5.0	2.0	6.0	
	others	T	8.0	8.0	1.0	-7.0	-4.0	6.0	25.0	
	max:±18.0	E	0.0	4.0	-2.0	-5.0	-2.0	5.0	6.0	
		EB	-3.0	-2.0	-1.0	-3.0	1.0	4.0	1.0	

Bearing Temperature After Running

Unit : ℃

Bearing No.	1	2	3	4	5	6	7	8	9
Main bearing	49	49	49	49	50	49	49	49	49
Crank bearing	49	49	50	49	48	49	49		
Cross head bearing	49	48	50	49	50	49	49		
Lub oil temperature	44								

stx		ISO correction Data					Prepared by: K. S. YANG	
Engine No. : SB7S50-09511		Owner		NNC		Output		11,060 kW
Hull No. : S-4012		Shipyard		STX		Test Date		2010. 08. 10
	Unit	1	2	3	4	5	6	7
Load	%	25	50	75	90	100-1	100-2	110
Engine speed	rpm	80.0	100.8	115.4	122.6	127.0	127.0	131.1
Engine output	kW	2,765	5,530	8,293	9,954	11,060	11,060	12,166
Barometer pressure	mbar	1016	1016	1015	1015	1014	1014	1014
T/C inlet temp.	°C	33.8	33.3	34.8	35.8	35.8	36.2	36.2
Scav.air temperature	°C	36.0	31.0	36.0	39.0	40.0	40.0	41.0
T/C outlet back-press	mmAq	10	60	170	200	240	230	270
Max.Combustion pressure	bar	71.7	98.7	128.7	143.1	148.6	148.4	153.9
Max.Combustion pressure (ISO)	bar	72.7	99.8	130.6	145.3	150.9	150.8	156.5
Compression pressure	bar	42.7	65.0	99.6	115.4	125.3	125.4	134.4
Compression pressure (ISO)	bar	43.5	65.9	101.3	117.5	127.8	128.0	137.5
Scav.air pressure	kg/cm ²	0.25	0.90	1.80	2.30	2.55	2.55	2.80
Scav.air pressure (ISO)	kg/cm ²	0.27	0.92	1.84	2.35	2.61	2.61	2.88
Exh.temp T/C inlet	°C	330.0	370.0	375.0	405.0	435.0	435.0	460.0
Exh.temp T/C inlet (ISO)	°C	318.2	358.2	359.5	388.8	419.2	418.9	444.9
Exh.temp T/C outlet	°C	280.0	280.0	235.0	250.0	260.0	260.0	265.0
Exh.temp T/C outlet (ISO)	°C	265.4	266.8	219.5	234.4	244.8	244.7	250.5
T/C speed (X 1000)	rpm	5.2	9.5	12.6	13.7	14.4	14.4	14.9
T/C speed (X 1000) (ISO)	rpm	5.2	9.5	12.6	13.7	14.3	14.3	14.9
ISO reference conditions								
Load	%	25	50	75	90	100	100	110
Barometer pressure	mbar	1000	1000	1000	1000	1000	1000	1000
T/C inlet temp.	°C	25	25	25	25	25	25	25
Scav.air temperature	°C	33.5	27.4	31.2	34.4	37.0	37.0	39.9
T/C outlet back-press	mmAq	25	86	179	248	300	300	355
Correction equations (based on MAN B&W documentation)								
$CorrP_{max} = P_{max,m} \cdot (100 - 0.2198 \cdot \Delta T_{inl} + 0.081 \cdot \Delta T_{sc} + 0.022 \cdot \Delta P_{amb} - 0.005278 \cdot \Delta P_{back}) / 100$ $CorrP_{comp} = P_{comp,m} \cdot (100 - 0.2954 \cdot \Delta T_{inl} + 0.153 \cdot \Delta T_{sc} + 0.0301 \cdot \Delta P_{amb} - 0.007021 \cdot \Delta P_{back}) / 100$ $CorrP_{sc} = P_{sc,m} \cdot (100 - 0.2856 \cdot \Delta T_{inl} + 0.222 \cdot \Delta T_{sc} + 0.0293 \cdot \Delta P_{amb} - 0.006788 \cdot \Delta P_{back}) / 100$ $CorrT_{exh,inl} = T_{exh,inl,m} \cdot (100 + 0.2466 \cdot \Delta T_{inl} + 0.059 \cdot \Delta T_{sc} - 0.0225 \cdot \Delta P_{amb} + 0.006127 \cdot \Delta P_{back}) / 100$ $CorrT_{exh,out} = T_{exh,out,m} \cdot (100 + 0.316 \cdot \Delta T_{inl} + 0.018 \cdot \Delta T_{sc} - 0.0037 \cdot \Delta P_{amb} + 0.009741 \cdot \Delta P_{back}) / 100$ $CorrTCrpm = TCrpm,m \cdot \sqrt{(298.15 \cdot P_{amb}) / (T_{inl} + 273.15)} \cdot (CorrP_{sc} / P_{sc,m})$								
where : Δ refers to (reference - measured) subscript m measured T _{inl} T/C inlet temperature (°C) T _{sc} Scav.air temperature (°C) T _{exh,inl} Exh.temp T/C inlet (K) T _{exh,out} Exh.temp T/C outlet (K) P _{amb} Barometer pressure (mmHg) P _{max} Max.Combustion pressure (barabs) P _{comp} Compression pressure (barabs) P _{sc} Scav.air pressure (barabs) P _{back} T/C outlet back-press (mmAq) TCrpm T/C speed (rpm)								



5. Performance Data of 25 % Load

Prepared by:
K. S. YANG

Engine No.	SB7S50-09511	Test Duration	30 min	Weather	cloudy	Test Date	2010. 08. 10
Owner	NNC	Ship yard	STX	Hull No.	S-4012	Measuring time	11:10
Ambient temp.	31.3 °C	T/C inlet temp.	33.8 °C	Humidity	64.0 %	Barom.press	1016 mbar
Engine Speed	Dynamometer type		Cylinder oil		Circulation oil		
80.0 rpm	ZOELLNER/19N234F3		S-oil, Dn.seamaster A-50		S-oil, Daphne marine SX30		
Engine output	Dynamometer weight		Gov' index	VIT Control	Turbine oil		
2,765 kW	330.0 kNm		35.0	- kg/cm ²	S-oil, Daphne marine SX30		

Fuel Oil Consumption

Kind of oil	Bunker - A	viscosity	14.1cst at 40°C	Gravity	0.8684 at 15°C	L.C.V	10,151 Kcal/kg
Measuring	Consumption (kg/h)		Consumption (g/kW/h)		ISO.conversion (g/kW/h)		
84.8 kg	10'00"	508.8	184.01	182.63			

Inlet	Main Lub oil	Cooling Oil	Cam Shaft Lub Oil	Cooling L/T water	Cooling H/T water	Fuel Oil	Exh'valve Actuator Oil
Pressure	2.20 kg/cm ²	2.20 kg/cm ²	- kg/cm ²	1.00 kg/cm ²	3.50 kg/cm ²	6.80 kg/cm ²	- kg/cm ²
Temperature	42 °C	42 °C	- °C	28 °C	70 °C	38 °C	- °C
Thrust bearing temp/press	43 °C	2.20 kg/cm ²	Crank shaft Axial	- mm	v/v Spring air	7.80 kg/cm ²	

Cylinder Number	Unit	1	2	3	4	5	6	7	-	-	Ave.
Maximum combustion pressure	bar	73	73	71	72	71	71	71			71.7
Compression pressure	bar	43	43	43	43	43	42	42			42.7
ΔP (Pmax - Pcom)	bar	30	30	28	29	28	29	29			29.0
V.I.T adjustment index	mm	-	-	-	-	-	-	-			-
Fuel pump index	mm	32.0	32.0	32.0	32.0	32.0	32.0	32.0			32.0
Actuator index	Pos	36.0									36.0
Exhaust gas temp.cyl' outlet	°C	290	300	290	300	295	290	290			293.6
Cooling water temp.cyl' outlet	°C	76	76	76	76	76	76	76			76.0
Cooling oil temp.piston outlet	°C	44	44	44	44	44	44	44			44.0

Air Cooler / Scavenging Air

Temp.	Air	Cooler inlet	50 °C	Press.	Scavenge air pressure	Engine inlet	0.25 kg/cm ²
		Cooler outlet	28 °C			Engine inlet	- kg/cm ²
		Engine inlet	36 °C				
	Cooling water	Cooler inlet	28 °C		Diff. pressure (Air cooler)		60 mmAq
		Cooler outlet	29 °C		Diff. Pressure(Exh.gas-air reservoir)		0.06 kg/cm ²

Turbocharger

Speed (rpm)		Exhaust Inlet temp	Exhaust Outlet temp	Exhaust Inlet press	Outlet Back-press	Lub oil Inlet press	Lub oil Out let temp	Filter Droop
#.1	5,169	330 °C	280 °C	0.19 kg/cm ²	10 mmAq	2.20 kg/cm ²	46 °C	64 mmAq
#.2		°C	°C	kg/cm ²	mmAq	kg/cm ²	°C	mmAq

Remark :



5. Performance Data of 50 % Load

Prepared by:
K. S. YANG

Engine No.	SB7S50-09511	Test Duration	30 min	Weather	cloudy	Test Date	2010. 08. 10
Owner	NNC	Ship yard	STX	Hull No.	S-4012	Measuring time	11:40
Ambient temp.	31.8 °C	T/C inlet temp.	33.3 °C	Humidity	63.0 %	Barom.press	1016 mbar
Engine Speed	Dynamometer type		Cylinder oil		Circulation oil		
100.8 rpm	ZOELLNER/19N234F3		S-oil, Dn.seamaster A-50		S-oil, Daphne marine SX30		
Engine output	Dynamometer weight		Gov' index	VIT Control	Turbine oil		
5,530 kW	523.9 kNm		50.0	- kg/cm ²	S-oil, Daphne marine SX30		

Fuel Oil Consumption

Kind of oil	Bunker - A	viscosity	14.1cst at 40°C	Gravity	0.8684 at 15°C	L.C.V	10,151 Kcal/kg
Measuring	Consumption (kg/h)		Consumption (g/kW/h)		ISO.conversion (g/kW/h)		
163.8 kg	10'00"	982.8	177.72	176.26			

Inlet	Main Lub oil	Cooling Oil	Cam Shaft Lub Oil	Cooling L/T water	Cooling H/T water	Fuel Oil	Exh'valve Actuator Oil
Pressure	2.20 kg/cm ²	2.20 kg/cm ²	— kg/cm ²	1.00 kg/cm ²	3.50 kg/cm ²	6.80 kg/cm ²	— kg/cm ²
Temperature	42 °C	42 °C	— °C	29 °C	70 °C	38 °C	— °C
Thrust bearing temp/press	44 °C	2.20 kg/cm ²	Crank shaft Axial	— mm	v/v Spring air	7.80 kg/cm ²	

Cylinder Number	Unit	1	2	3	4	5	6	7	-	-	Ave.
Maximum combustion pressure	bar	99	98	98	99	98	99	100			98.7
Compression pressure	bar	65	65	65	65	65	65	65			65.0
ΔP (Pmax - Pcom)	bar	34	33	33	34	33	34	35			33.7
V.I.T adjustment index	mm	-	-	-	-	-	-	-			-
Fuel pump index	mm	45.0	45.0	45.0	45.0	45.0	45.0	45.0			45.0
Actuator index	Pos	51.0									51.0
Exhaust gas temp.cyl' outlet	°C	310	310	300	310	310	300	305			306.4
Cooling water temp.cyl' outlet	°C	77	77	77	77	77	77	76			76.9
Cooling oil temp.piston outlet	°C	46	46	46	46	46	46	46			46.0

Air Cooler / Scavenging Air

Temp.	Air	Cooler inlet	100 °C	Press.	Scavenge air pressure	Engine inlet	0.90 kg/cm ²
		Cooler outlet	30 °C			Engine inlet	— kg/cm ²
		Engine inlet	31 °C				
	Cooling water	Cooler inlet	29 °C		Diff. pressure (Air cooler)		120 mmAq
		Cooler outlet	32 °C		Diff. Pressure(Exh.gas-air reservoir)		0.10 kg/cm ²

Turbocharger

Speed (rpm)		Exhaust Inlet temp	Exhaust Outlet temp	Exhaust Inlet press	Outlet Back-press	Lub oil Inlet press	Lub oil Out let temp	Filter Droop
#.1	9,490	370 °C	280 °C	0.80 kg/cm ²	60 mmAq	2.20 kg/cm ²	51 °C	40 mmAq
#.2		°C	°C	kg/cm ²	mmAq	kg/cm ²	°C	mmAq

Remark :

stx		5. Performance Data of 75 % Load				Prepared by: K. S. YANG	
Engine No.	SB7S50-09511	Test Duration	30 min	Weather	cloudy	Test Date	2010. 08. 10
Owner	NNC	Ship yard	STX	Hull No.	S-4012	Measuring time	14:00
Ambient temp.	32.2 °C	T/C inlet temp.	34.8 °C	Humidity	61.0 %	Barom.press	1015 mbar
Engine Speed	Dynamometer type		Cylinder oil		Circulation oil		
115.4 rpm	ZOELLNER/19N234F3		S-oil, Dn.seamaster A-50		S-oil, Daphne marine SX30		
Engine output	Dynamometer weight		Gov' index	VIT Control	Turbine oil		
8,293 kW	686.5 kNm		63.0	- kg/cm ²	S-oil, Daphne marine SX30		

Fuel Oil Consumption							
Kind of oil	Bunker - A	viscosity	14.1cst at40°C	Gravity	0.8684 at15°C	L.C.V	10,151 Kcal/kg
Measuring	Consumption (kg/h)		Consumption (g/kW/h)		ISO.conversion (g/kW/h)		
239.5 kg	10'00"	1437.0	173.28		171.73		

Inlet	Main Lub oil	Cooling Oil	Cam Shaft Lub Oil	Cooling L/T water	Cooling H/T water	Fuel Oil	Exh'valve Actuator Oil
Pressure	2.40 kg/cm ²	2.40 kg/cm ²	— kg/cm ²	1.00 kg/cm ²	3.50 kg/cm ²	6.80 kg/cm ²	— kg/cm ²
Temperature	42 °C	42 °C	— °C	30 °C	73 °C	38 °C	— °C
Thrust bearing temp/press	45 °C	2.40 kg/cm ²	Crank shaft Axial	— mm	v/v Spring air	7.50 kg/cm ²	

Cylinder Number	Unit	1	2	3	4	5	6	7	-	-	Ave.
Maximum combustion pressure	bar	129	129	129	129	129	128	128			128.7
Compression pressure	bar	100	100	100	99	100	99	99			99.6
ΔP (Pmax - Pcom)	bar	29	29	29	30	29	29	29			29.1
V.I.T adjustment index	mm	-	-	-	-	-	-	-			-
Fuel pump index	mm	58.0	58.0	58.0	58.0	58.0	58.0	58.0			58.0
Actuator index	Pos	64.0									64.0
Exhaust gas temp.cyl' outlet	°C	300	305	300	305	300	300	300			301.4
Cooling water temp.cyl' outlet	°C	80	80	80	80	80	80	80			80.0
Cooling oil temp.piston outlet	°C	47	47	47	47	47	47	47			47.0

Air Cooler / Scavenging Air							
Temp.	Air	Cooler inlet	155 °C	Press.	Scavenge air pressure	Engine inlet	1.80 kg/cm ²
		Cooler outlet	34 °C			Engine inlet	— kg/cm ²
		Engine inlet	36 °C				
	Cooling water	Cooler inlet	30 °C		Diff. pressure (Air cooler)		180 mmAq
		Cooler outlet	39 °C		Diff. Pressure(Exh.gas-air reservoir)		0.20 kg/cm ²

Turbocharger								
Speed (rpm)		Exhaust Inlet temp	Exhaust Outlet temp	Exhaust Inlet press	Outlet Back-press	Lub oil Inlet press	Lub oil Out let temp	Filter Droop
#.1	12,611	375 °C	235 °C	1.60 kg/cm ²	170 mmAq	2.20 kg/cm ²	56 °C	90 mmAq
#.2		°C	°C	kg/cm ²	mmAq	kg/cm ²	°C	mmAq

Remark :							



5. Performance Data of 90 % Load

Prepared by:
K. S. YANG

Engine No.	SB7S50-09511	Test Duration	60 min	Weather	cloudy	Test Date	2010. 08. 10
Owner	NNC	Ship yard	STX	Hull No.	S-4012	Measuring time	14:30
Ambient temp.	32.8 °C	T/C inlet temp.	35.8 °C	Humidity	57.0 %	Barom.press	1015 mbar
Engine Speed	Dynamometer type		Cylinder oil		Circulation oil		
122.6 rpm	ZOELLNER/19N234F3		S-oil, Dn.seamaster A-50		S-oil, Daphne marine SX30		
Engine output	Dynamometer weight		Gov' index	VIT Control	Turbine oil		
9,954 kW	775.2 kNm		72.0	- kg/cm ²	S-oil, Daphne marine SX30		

Fuel Oil Consumption							
Kind of oil	Bunker - A	viscosity	14.1cst at 40°C	Gravity	0.8684 at 15°C	L.C.V	10,151 Kcal/kg
Measuring	Consumption (kg/h)		Consumption (g/kW/h)		ISO.conversion (g/kW/h)		
288.4 kg	10'00"		1730.4		173.84		
					172.16		

Inlet	Main Lub oil	Cooling Oil	Cam Shaft Lub Oil	Cooling L/T water	Cooling H/T water	Fuel Oil	Exh'valve Actuator Oil
Pressure	2.40 kg/cm ²	2.40 kg/cm ²	— kg/cm ²	1.00 kg/cm ²	3.50 kg/cm ²	6.80 kg/cm ²	— kg/cm ²
Temperature	42 °C	42 °C	— °C	31 °C	71 °C	40 °C	— °C
Thrust bearing temp/press	46 °C	2.40 kg/cm ²	Crank shaft Axial	— mm	v/v Spring air	7.80 kg/cm ²	

Cylinder Number	Unit	1	2	3	4	5	6	7	-	-	Ave.
Maximum combustion pressure	bar	143	142	143	144	143	144	143			143.1
Compression pressure	bar	115	115	116	115	116	115	116			115.4
ΔP (Pmax - Pcom)	bar	28	27	27	29	27	29	27			27.7
V.I.T adjustment index	mm	-	-	-	-	-	-	-			-
Fuel pump index	mm	66.0	66.0	66.0	66.0	66.0	66.0	66.0			66.0
Actuator index	Pos	73.0									73.0
Exhaust gas temp.cyl' outlet	°C	320	325	310	320	320	310	315			317.1
Cooling water temp.cyl' outlet	°C	79	79	79	79	79	79	79			79.0
Cooling oil temp.piston outlet	°C	48	48	48	48	48	48	48			48.0

Air Cooler / Scavenging Air							
Temp.	Air	Cooler inlet	180 °C	Press.	Scavenge air pressure	Engine inlet	2.30 kg/cm ²
		Cooler outlet	37 °C			Engine inlet	— kg/cm ²
		Engine inlet	39 °C				
	Cooling water	Cooler inlet	31 °C		Diff. pressure (Air cooler)		210 mmAq
		Cooler outlet	43 °C		Diff. Pressure(Exh.gas-air reservoir)		0.20 kg/cm ²

Turbocharger							
Speed (rpm)	Exhaust Inlet temp	Exhaust Outlet temp	Exhaust Inlet press	Outlet Back-press	Lub oil Inlet press	Lub oil Out let temp	Filter Droop
#.1	13,747	405 °C	250 °C	2.10 kg/cm ²	200 mmAq	2.20 kg/cm ²	58 °C
#.2		°C	°C	kg/cm ²	mmAq	kg/cm ²	°C
							mmAq

Remark :										
Fuel oil leakage check => 880 g/ 10 min.				* Stuffing Box Drain						
=> 880 * 6 = 5,280 g/hr. (0.53 g/kW/hr)				Cyl.	No.1	No.2	No.3	No.4	No.5	No.6
=> 5,280 * 24 = 126720 g/24hr				g/10min	6	6	4	18	2	6
				Litter/day	1.0	1.0	0.7	2.9	0.3	1.0



5. Performance Data of 100% Load 1st

Prepared by:
K. S. YANG

Engine No.	SB7S50-09511	Test Duration	30 min	Weather	cloudy	Test Date	2010. 08. 10
Owner	NNC	Ship yard	STX	Hull No.	S-4012	Measuring time	15:00
Ambient temp.	33.3 °C	T/C inlet temp.	35.8 °C	Humidity	56.0 %	Barom.press	1014 mbar
Engine Speed	Dynamometer type		Cylinder oil		Circulation oil		
127.0 rpm	ZOELLNER/19N234F3		S-oil, Dn.seamaster A-50		S-oil, Daphne marine SX30		
Engine output	Dynamometer weight		Gov' index	VIT Control	Turbine oil		
11,060 kW	831.6 kNm		78.0	- kg/cm ²	S-oil, Daphne marine SX30		

Fuel Oil Consumption

Kind of oil	Bunker - A	viscosity	14.1cst at40°C	Gravity	0.8684 at15°C	L.C.V	10,151 Kcal/kg
Measuring	Consumption (kg/h)		Consumption (g/kW/h)		ISO.conversion (g/kW/h)		
322.4 kg	10'00"	1934.4	174.90	173.29			

Inlet	Main Lub oil	Cooling Oil	Cam Shaft Lub Oil	Cooling L/T water	Cooling H/T water	Fuel Oil	Exh'valve Actuator Oil
Pressure	2.40 kg/cm ²	2.40 kg/cm ²	— kg/cm ²	1.00 kg/cm ²	3.50 kg/cm ²	6.80 kg/cm ²	— kg/cm ²
Temperature	43 °C	43 °C	— °C	30 °C	70 °C	40 °C	— °C
Thrust bearing temp/press	47 °C	2.40 kg/cm ²	Crank shaft Axial	— mm	v/v Spring air	7.80 kg/cm ²	

Cylinder Number	Unit	1	2	3	4	5	6	7	-	-	Ave.
Maximum combustion pressure	bar	149	148	148	149	149	148	149			148.6
Compression pressure	bar	125	125	125	125	126	125	126			125.3
ΔP (Pmax - Pcom)	bar	24	23	23	24	23	23	23			23.3
V.I.T adjustment index	mm	-	-	-	-	-	-	-			-
Fuel pump index	mm	72.0	72.0	72.0	72.0	72.0	72.0	72.0			72.0
Actuator index	Pos	79.0									79.0
Exhaust gas temp.cyl' outlet	°C	340	345	340	345	340	330	330			338.6
Cooling water temp.cyl' outlet	°C	80	80	80	80	80	80	80			80.0
Cooling oil temp.piston outlet	°C	50	50	50	50	50	50	50			50.0

Air Cooler / Scavenging Air

Temp.	Air	Cooler inlet	192 °C	Press.	Scavenge air pressure	Engine inlet	2.55 kg/cm ²
		Cooler outlet	38 °C			Engine inlet	— kg/cm ²
		Engine inlet	40 °C				
	Cooling water	Cooler inlet	30 °C		Diff. pressure (Air cooler)	220 mmAq	
		Cooler outlet	45 °C		Diff. Pressure(Exh.gas-air reservoir)	0.20 kg/cm ²	

Turbocharger

Speed (rpm)		Exhaust Inlet temp	Exhaust Outlet temp	Exhaust Inlet press	Outlet Back-press	Lub oil Inlet press	Lub oil Out let temp	Filter Droop
#.1	14,370	435 °C	260 °C	2.35 kg/cm ²	240 mmAq	2.20 kg/cm ²	60 °C	130 mmAq
#.2		°C	°C	kg/cm ²	mmAq	kg/cm ²	°C	mmAq

Remark :

Fuel oil leakage check => 912 g/ 10 min.

=> 912 * 6 = 5,472 g/hr. (0.49 g/kW/hr)

=> 5,472 * 24 = 131,328 g/24hr



5. Performance Data of 100% Load 2nd

Prepared by:
K. S. YANG

Engine No.	SB7S50-09511	Test Duration	30 min	Weather	cloudy	Test Date	2010. 08. 10
Owner	NNC	Ship yard	STX	Hull No.	S-4012	Measuring time	15:30
Ambient temp.	33.4 °C	T/C inlet temp.	36.2 °C	Humidity	54.0 %	Barom.press	1014 mbar
Engine Speed	127.0 rpm	Dynamometer type	ZOELLNER/19N234F3	Cylinder oil	S-oil, Dn.seamaster A-50	Circulation oil	S-oil, Daphne marine SX30
Engine output	11,060 kW	Dynamometer weight	831.6 kNm	Gov' index	VIT Control	Turbine oil	S-oil, Daphne marine SX30
					78.0 - kg/cm ²		

Fuel Oil Consumption							
Kind of oil	Bunker - A	viscosity	14.1cst at 40°C	Gravity	0.8684 at 15°C	L.C.V	10,151 Kcal/kg
Measuring	Consumption (kg/h)	Consumption (g/kW/h)	ISO.conversion (g/kW/h)				
323.2 kg	10'00"	1939.2	175.33	173.72			

Inlet	Main Lub oil	Cooling Oil	Cam Shaft Lub Oil	Cooling L/T water	Cooling H/T water	Fuel Oil	Exh'valve Actuator Oil
Pressure	2.40 kg/cm ²	2.40 kg/cm ²	— kg/cm ²	1.00 kg/cm ²	3.50 kg/cm ²	6.80 kg/cm ²	— kg/cm ²
Temperature	43 °C	43 °C	— °C	30 °C	70 °C	40 °C	— °C
Thrust bearing temp/press	48 °C	2.40 kg/cm ²	Crank shaft Axial	— mm	v/v Spring air	7.80 kg/cm ²	

Cylinder Number	Unit	1	2	3	4	5	6	7	—	—	Ave.
Maximum combustion pressure	bar	149	149	147	149	148	149	148			148.4
Compression pressure	bar	126	125	125	125	126	126	125			125.4
ΔP (Pmax - Pcom)	bar	23	24	22	24	22	23	23			23.0
V.I.T adjustment index	mm	—	—	—	—	—	—	—			—
Fuel pump index	mm	72.0	72.0	72.0	72.0	72.0	72.0	72.0			72.0
Actuator index	Pos	79.0									79.0
Exhaust gas temp.cyl' outlet	°C	340	345	340	345	340	330	330			338.6
Cooling water temp.cyl' outlet	°C	79	79	79	79	79	79	80			79.1
Cooling oil temp.piston outlet	°C	50	50	50	50	50	50	50			50.0

Air Cooler / Scavenging Air							
Temp.	Air	Cooler inlet	192 °C	Press.	Scavenge air pressure	Engine inlet	2.55 kg/cm ²
		Cooler outlet	38 °C			Engine inlet	— kg/cm ²
		Engine inlet	40 °C				
	Cooling water	Cooler inlet	30 °C		Diff. pressure (Air cooler)		220 mmAq
		Cooler outlet	45 °C		Diff. Pressure(Exh.gas-air reservoir)		0.20 kg/cm ²

Turbocharger								
Speed (rpm)		Exhaust Inlet temp	Exhaust Outlet temp	Exhaust Inlet press	Outlet Back-press	Lub oil Inlet press	Lub oil Out let temp	Filter Droop
#.1	14,370	435 °C	260 °C	2.35 kg/cm ²	230 mmAq	2.20 kg/cm ²	60 °C	130 mmAq
#.2		°C	°C	kg/cm ²	mmAq	kg/cm ²	°C	mmAq

Remark :							
* Stuffing Box Drain							
Cyl.	No.1	No.2	No.3	No.4	No.5	No.6	No.7
g/30min	14	20	18	62	14	34	24
Litter/day	0.8	1.1	1.0	3.4	0.8	1.9	1.3



5. Performance Data of 110% Load

Prepared by:
K. S. YANG

Engine No.	SB7S50-09511	Test Duration	30 min	Weather	cloudy	Test Date	2010. 08. 10
Owner	NNC	Ship yard	STX	Hull No.	S-4012	Measuring time	16:00
Ambient temp.	33.6 °C	T/C inlet temp.	36.2 °C	Humidity	53.0 %	Barom.press	1014 mbar
Engine Speed		Dynamometer type		Cylinder oil		Circulation oil	
131.1 rpm		ZOELLNER/19N234F3		S-oil, Dn.seamaster A-50		S-oil, Daphne marine SX30	
Engine output		Dynamometer weight		Gov' index	VIT Control	Turbine oil	
12,166 kW		886.1 kNm		83.0	- kg/cm ²	S-oil, Daphne marine SX30	

Fuel Oil Consumption

Kind of oil	Bunker - A	viscosity	14.1cst at 40°C	Gravity	0.8684 at 15°C	L.C.V	10,151 Kcal/kg
Measuring		Consumption (kg/h)		Consumption (g/kW/h)		ISO.conversion (g/kW/h)	
358.6 kg	10'00"	2151.6		176.85		175.17	

Inlet	Main Lub oil	Cooling Oil	Cam Shaft Lub Oil	Cooling L/T water	Cooling H/T water	Fuel Oil	Exh'valve Actuator Oil
Pressure	2.40 kg/cm ²	2.40 kg/cm ²	— kg/cm ²	1.00 kg/cm ²	3.50 kg/cm ²	6.80 kg/cm ²	— kg/cm ²
Temperature	44 °C	44 °C	— °C	30 °C	69 °C	40 °C	— °C
Thrust bearing temp/press	48 °C	2.40 kg/cm ²	Crank shaft Axial	— mm	v/v Spring air	7.80 kg/cm ²	

Cylinder Number	Unit	1	2	3	4	5	6	7	-	-	Ave.
Maximum combustion pressure	bar	155	154	153	153	154	154	154			153.9
Compression pressure	bar	135	134	134	135	135	133	135			134.4
ΔP (Pmax - Pcom)	bar	20	20	19	18	19	21	19			19.4
V.I.T adjustment index	mm	-	-	-	-	-	-	-			-
Fuel pump index	mm	77.0	77.0	77.0	77.0	77.0	77.0	77.0			77.0
Actuator index	Pos	86.0									86.0
Exhaust gas temp.cyl' outlet	°C	360	365	350	350	350	350	350			353.6
Cooling water temp.cyl' outlet	°C	80	80	80	80	80	80	81			80.1
Cooling oil temp.piston outlet	°C	50	50	50	50	50	50	50			50.0

Air Cooler / Scavenging Air

Temp.	Air	Cooler inlet	205 °C	Press.	Scavenge air pressure	Engine inlet	2.80 kg/cm ²
		Cooler outlet	39 °C			Engine inlet	— kg/cm ²
		Engine inlet	41 °C				
	Cooling water	Cooler inlet	30 °C		Diff. pressure (Air cooler)		230 mmAq
		Cooler outlet	46 °C		Diff. Pressure(Exh.gas-air reservoir)		0.25 kg/cm ²

Turbocharger

Speed (rpm)		Exhaust Inlet temp	Exhaust Outlet temp	Exhaust Inlet press	Outlet Back-press	Lub oil Inlet press	Lub oil Out let temp	Filter Droop
#.1	14,930	460 °C	265 °C	2.55 kg/cm ²	270 mmAq	2.20 kg/cm ²	60 °C	144 mmAq
#.2		°C	°C	kg/cm ²	mmAq	kg/cm ²	°C	mmAq

Remark :



6. Calculation sheet of fuel oil consumption

Prepared by:
K. S. YANG

Date : 20010-08-10
 Ship owner : National Navigation Company
 Ship yard : STX Offshore & Shipbuilding Co., Ltd.
 Ship No. : S-4012
 Engine model : 7S50MC-C7
 Engine No. : SB7S50-09511
 Engine output : 11,060 kW
 Engine speed : 127 rpm
 Fuel oil specific gravity at 15 °C : 0.8684
 Fuel oil lower calorific value : 10,151 Kcal/kg

Engine load (%)

25%	50%	75%	90%	100%-1	100%-2	110%
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Engine output

kW	2,765 kW	5,530 kW	8,293 kW	9,954 kW	11,060 kW	11,060 kW	12,166 kW
bhp	3,759 bhp	7,519 bhp	11,275 bhp	13,534 bhp	15,037 bhp	15,037 bhp	16,541 bhp

Measuring quantity (Kg), AL

84.8 kg	163.8 kg	239.5 kg	288.4 kg	322.4 kg	323.2 kg	358.6 kg
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Measuring time (s), B

600 s	600 s	600 s	600 s	600 s	600 s	600 s
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Specific fuel oil consumption (Measured, g/kW-h), B3=(ALX3600X1000)/(BXkW)

184.01	177.72	173.28	173.84	174.90	175.33	176.85
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Ambient temperature (°C), T1

31.3 °C	31.8 °C	32.2 °C	32.8 °C	33.3 °C	33.4 °C	33.6 °C
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Charge air coolant temperature (°C), T2

28.0 °C	29.0 °C	30.0 °C	31.0 °C	30.0 °C	30.0 °C	30.0 °C
---------	---------	---------	---------	---------	---------	---------

Ambient pressure (mbar), P

1016 mbar	1016 mbar	1015 mbar	1015 mbar	1014 mbar	1014 mbar	1014 mbar
-----------	-----------	-----------	-----------	-----------	-----------	-----------

Conversion factor by ambient temperature, T1F=0.002X(25-T1)/10

-0.00126	-0.00136	-0.00144	-0.00156	-0.00166	-0.00168	-0.00172
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Conversion factor by boost Air coolant temperature, T2F=0.006X(25-T2)/10

-0.00180	-0.00240	-0.00300	-0.00360	-0.00300	-0.00300	-0.00300
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Conversion factor by ambient pressure, PF=0.0002X(P-1000)/10

0.00032	0.00032	0.00030	0.00030	0.00028	0.00028	
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Conversion factor by lower calorific value, HF=(H-10200)/10200

-0.00480392	-0.00480392	-0.00480392	-0.00480392	-0.00480392	-0.00480392	-0.00480392
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Total conversion factor, TCF=T1F+T2F+PF+HF

-0.00754	-0.00824	-0.00894	-0.00966	-0.00918	-0.00920	-0.00952
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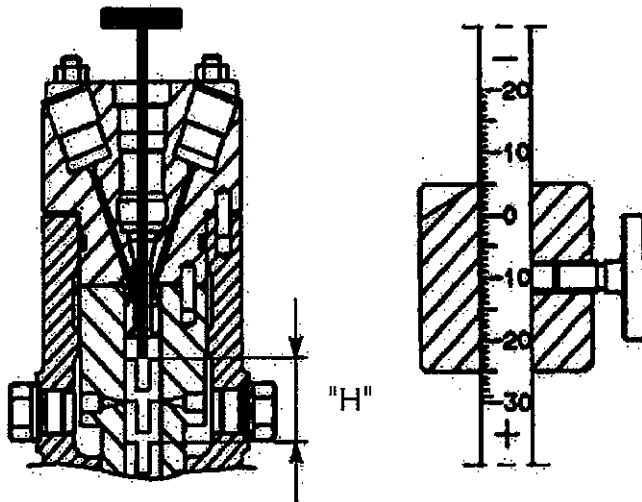
Converted value of F.O consumption based on ISO (g/kW-h), B1=B3+(B3*TCF)

g/kW-h	182.63	176.26	171.73	172.16	173.29	173.72	175.17
g/bhp-h	134.32	129.64	126.31	126.62	127.46	127.77	128.84
ton/day	129.12	123.39	119.18	119.13	120.00	120.11	121.15

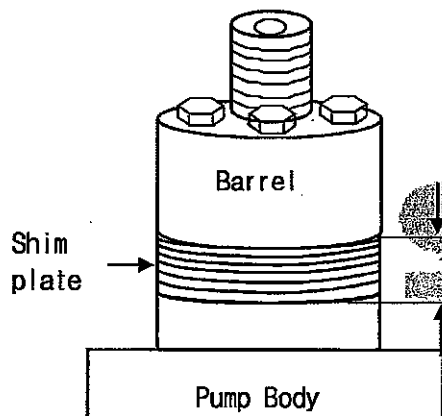
Guarantee of fuel oil consumption at SMCR (100% Load)	171.0 g/kW-h + 5.0% =(179.6g)
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Engine No. : SB7S50-09511

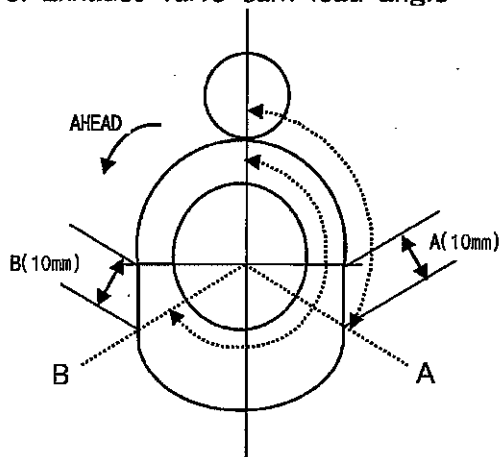
Date : 2010. 08. 10.

1. Fuel cam lift (Lead at before "TDC")


Cyl' No.	(H) mm	CA °
1	16.08	12.5
2	15.50	12.2
3	15.10	12.0
4	14.87	11.9
5	15.36	12.1
6	15.88	12.4
7	15.16	12.0
Mean	15.42	12.2

2. Shim thickness of fuel injection pump


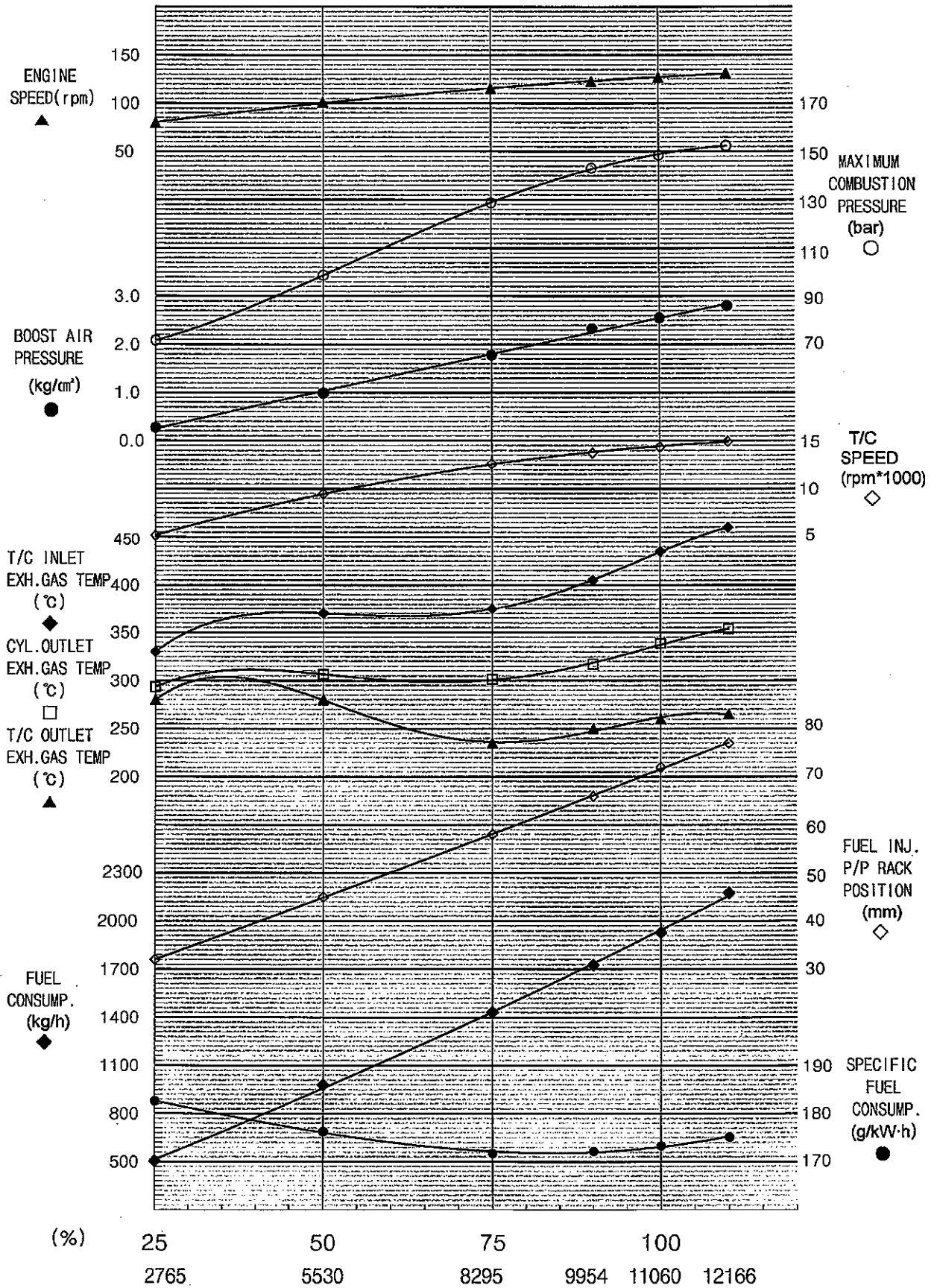
Cyl' No.	(T) mm
1	6.0
2	6.0
3	6.0
4	6.0
5	6.0
6	6.0
7	6.0
Mean	6.0
Standard : 6.0 ± 0.5 mm	

3. Exhaust valve cam lead angle


Cyl'No.	A.	B.
1	112.9	254.1
2	112.9	254.1
3	112.9	254.1
4	112.9	254.1
5	112.9	254.1
6	112.9	254.1
7	112.9	254.1
Mean	112.9	254.1
Standard : A-112.9, B-254.1 ± 0.5 °		

Checked by : _____

Verified by : _____

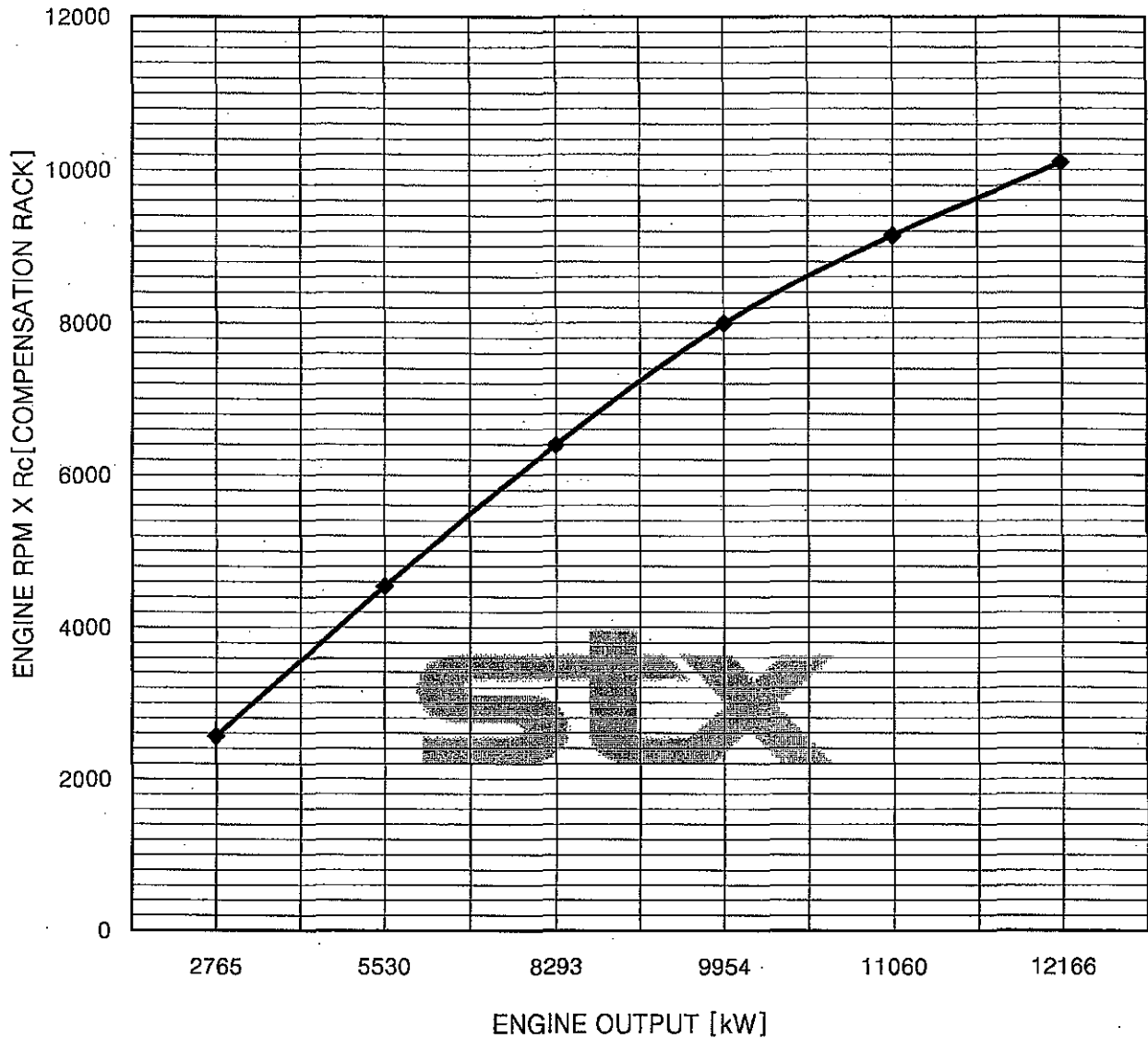




ENGINE NO. SB7S50-09511

PRESUMED HORSE POWER CURVE

DATE. 2010.08.10



$$* R_c = R' \times S_2 / S_1 \times Q_2 / Q_1$$

R_c = COMPENSATION FUEL RACK

R' = SEA TRIAL & MEASURED FUEL RACK IN SERVICE

S₁ = FUEL OIL GRAVITY USED IN SHOP TEST

S₂ = FUEL GRAVITY USED IN SEA TRIAL AND SERVICE

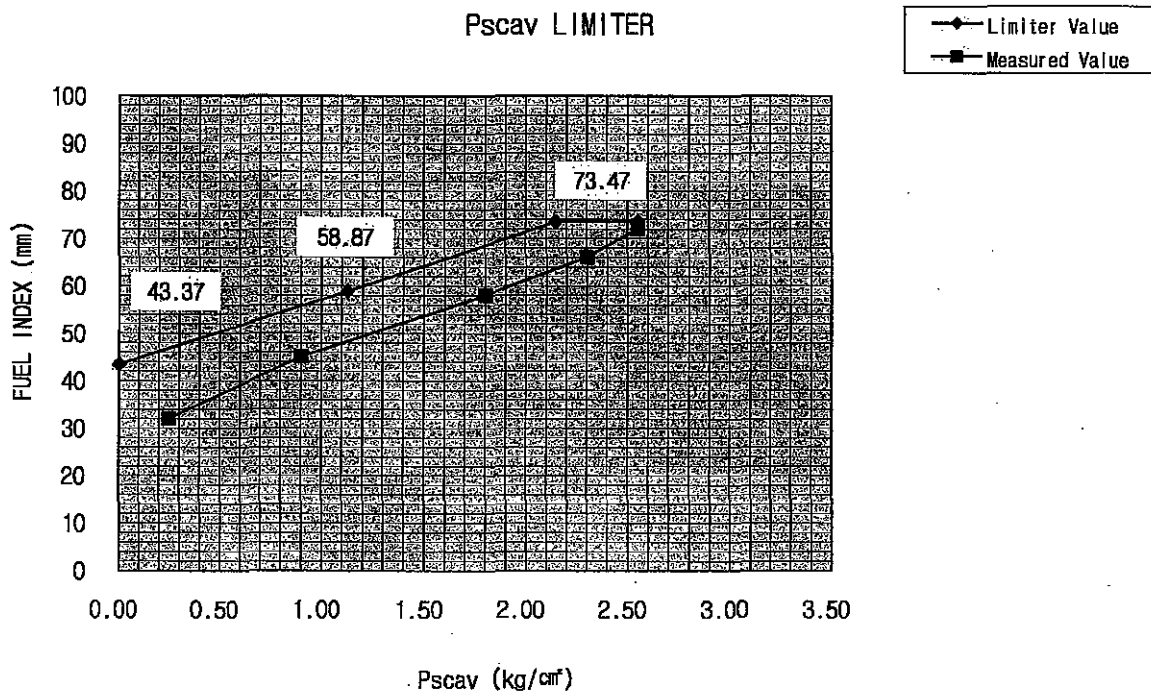
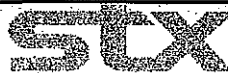
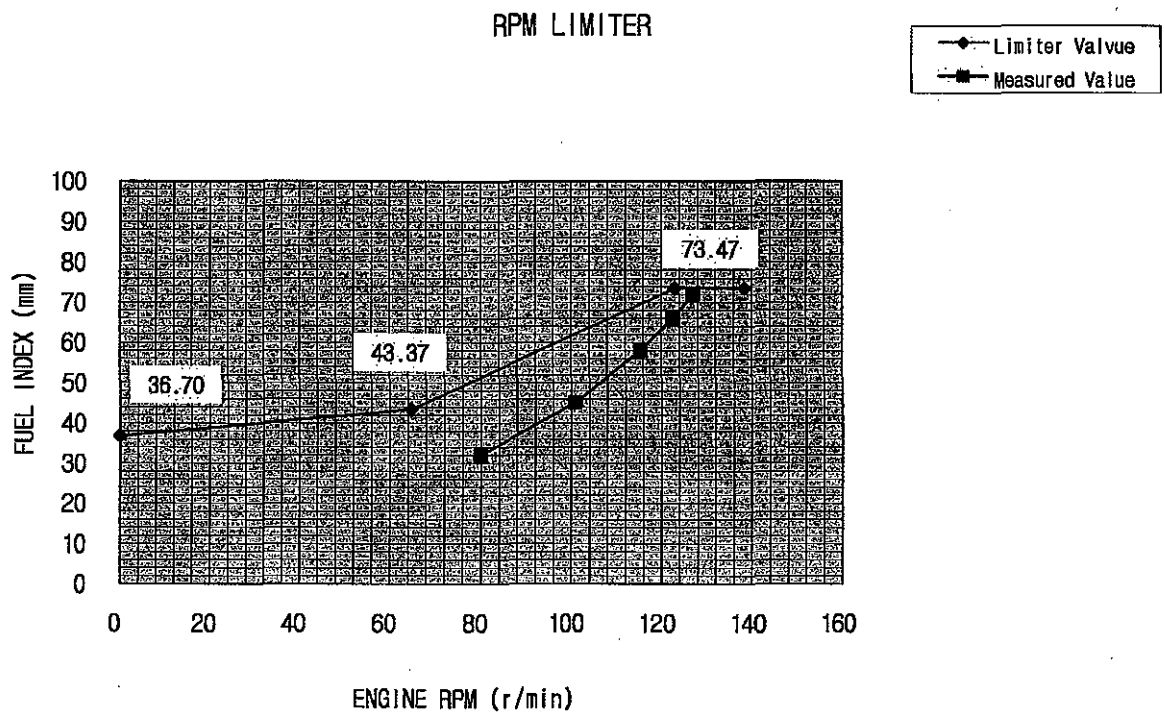
Q₁ = LOWER CALORIFIC VALUE USED IN SHOP TEST (Kcal/kg)

Q₂ = LOWER CALORIFIC VALUE USED IN SEA TRIAL AND SERVICE (Kcal/kg)



Graph For EL-Governor Limit

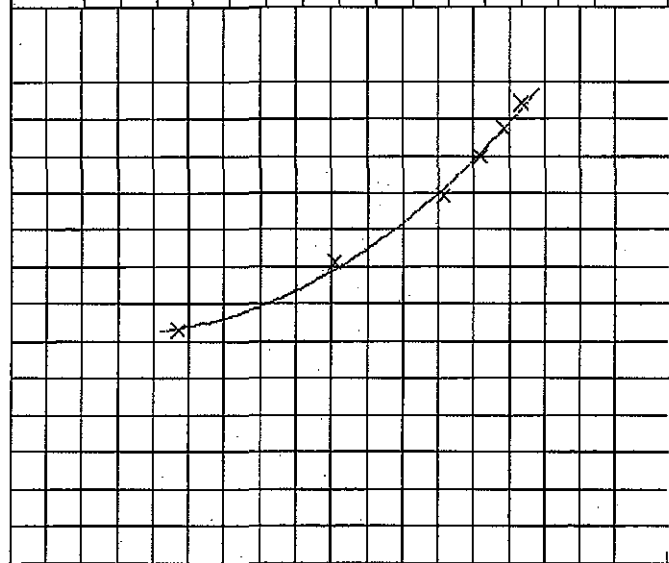
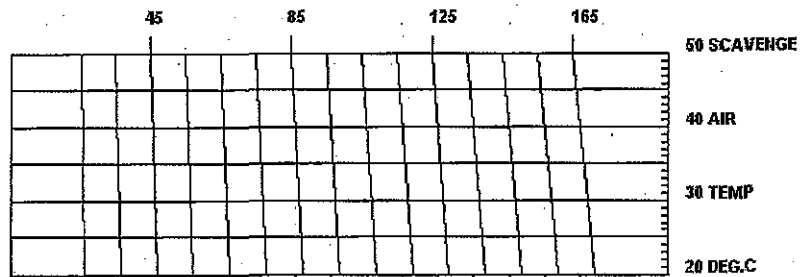
Owner	NNC
Ship No.	S-4012
Engine No.	SB7S50-09511



Power Estimation for

SB7S50-09511(S-4012)

T/C SPEED / 100



FUEL PUMP

INDEX

90

80

70

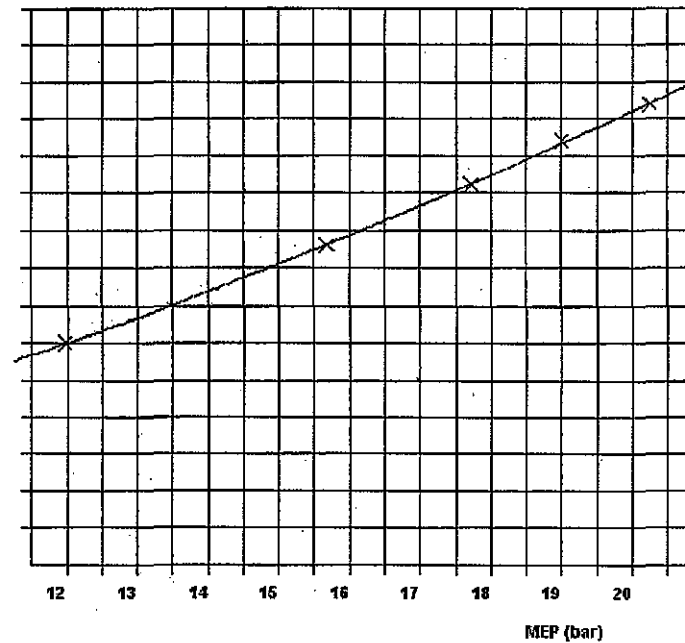
60

50

40

30

20



BHP

17000

13000

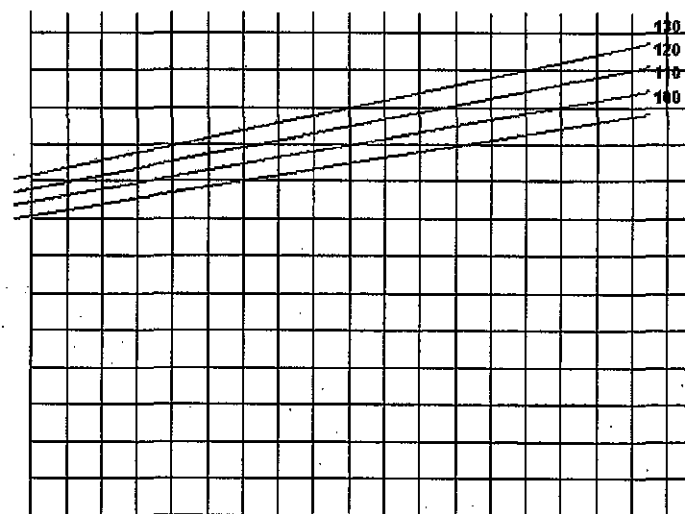
9000

5000

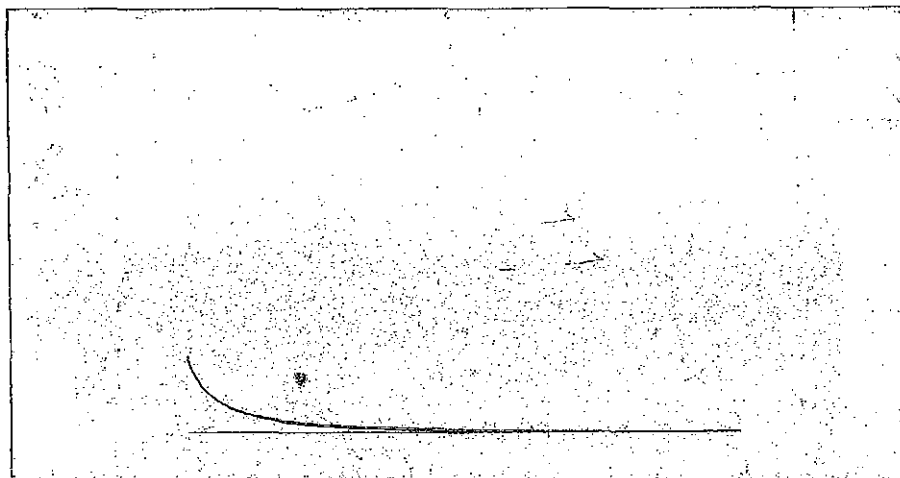
1000

-3000

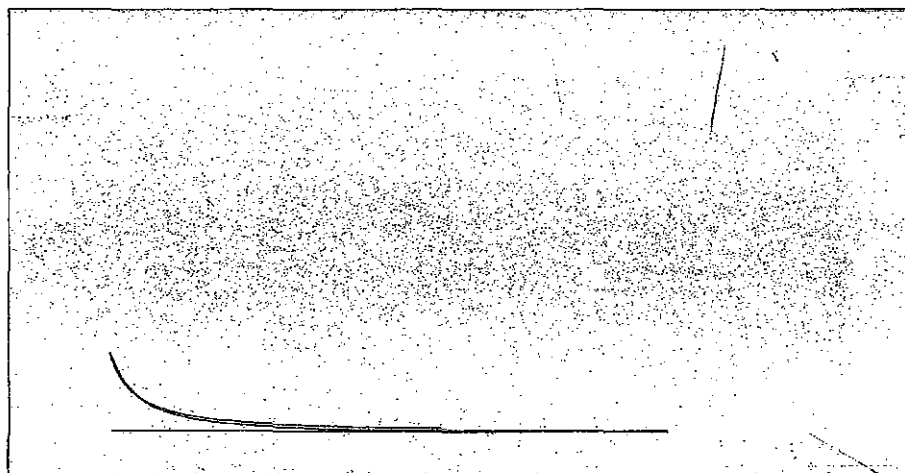
-7000



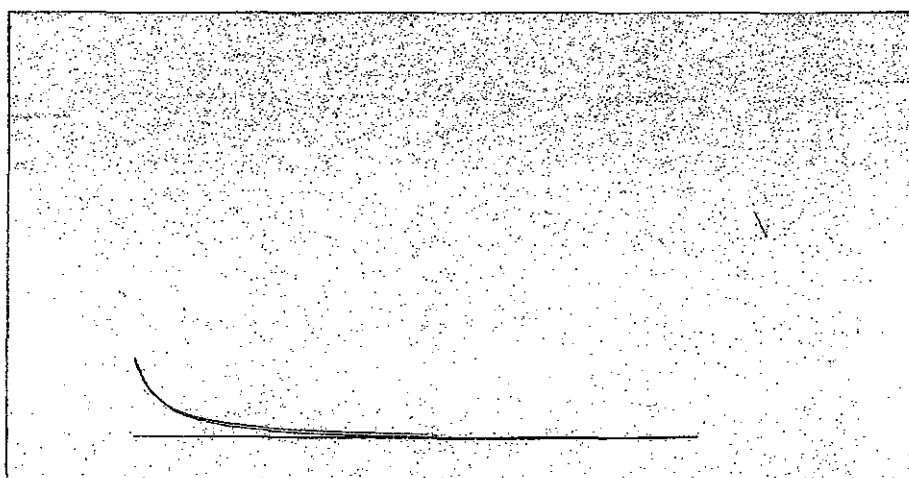
Zero set of No.1 Cylinder



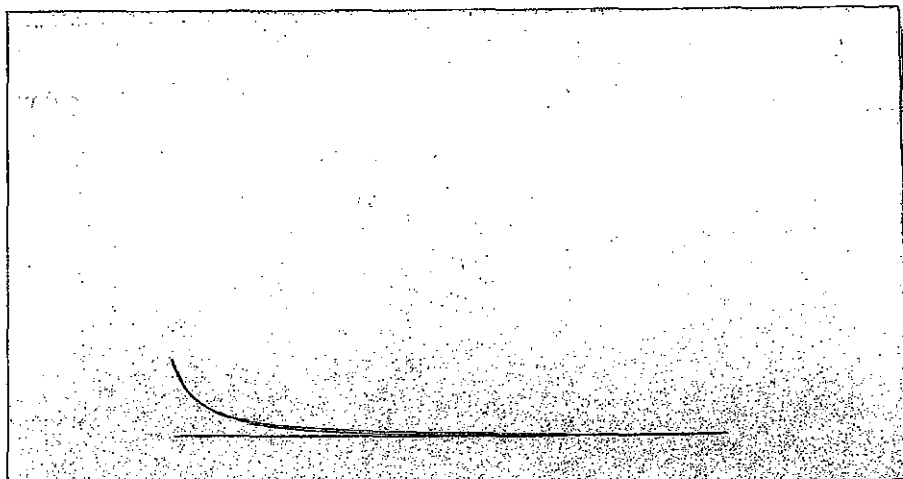
Zero set of No.2 Cylinder



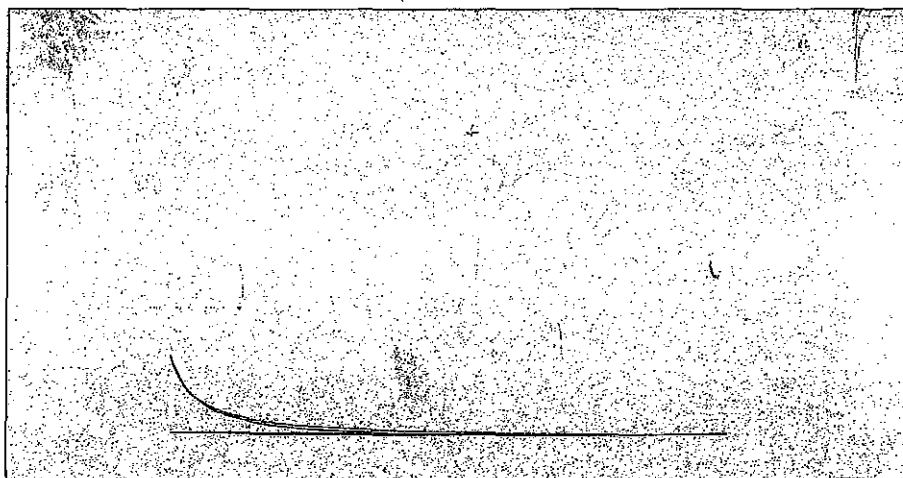
Zero set of No.3 Cylinder



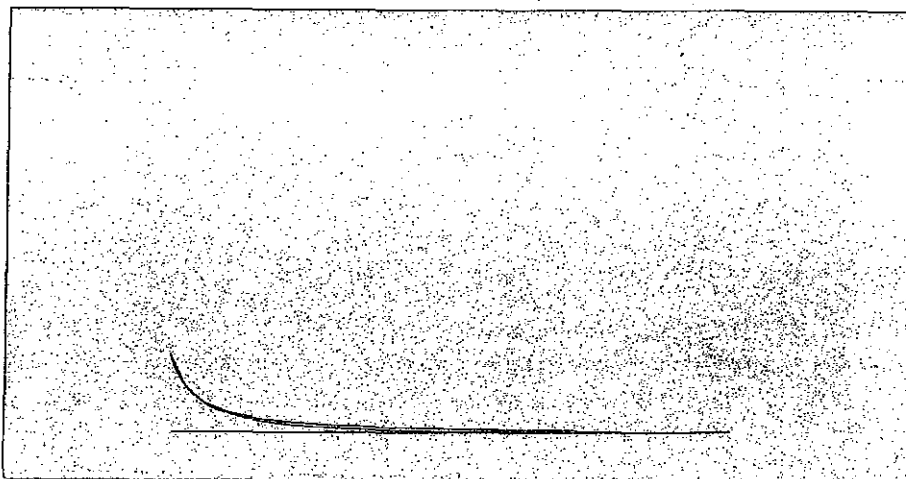
Zero set of No.4 Cylinder



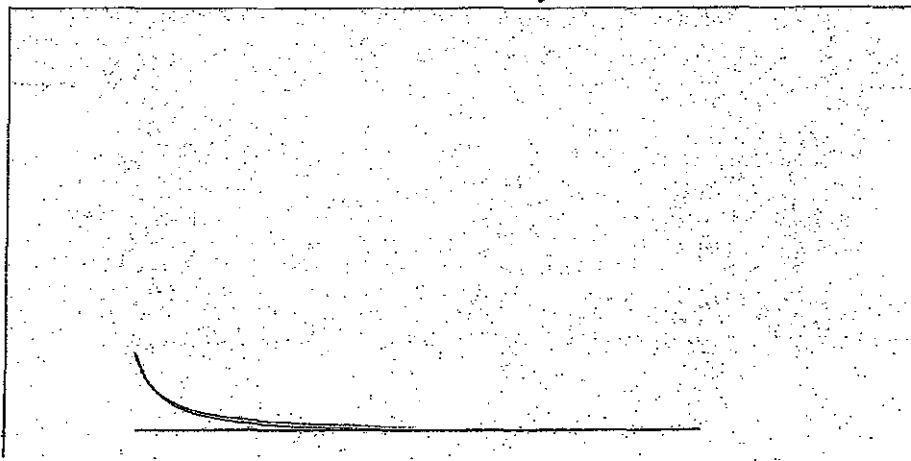
Zero set of No.5 Cylinder



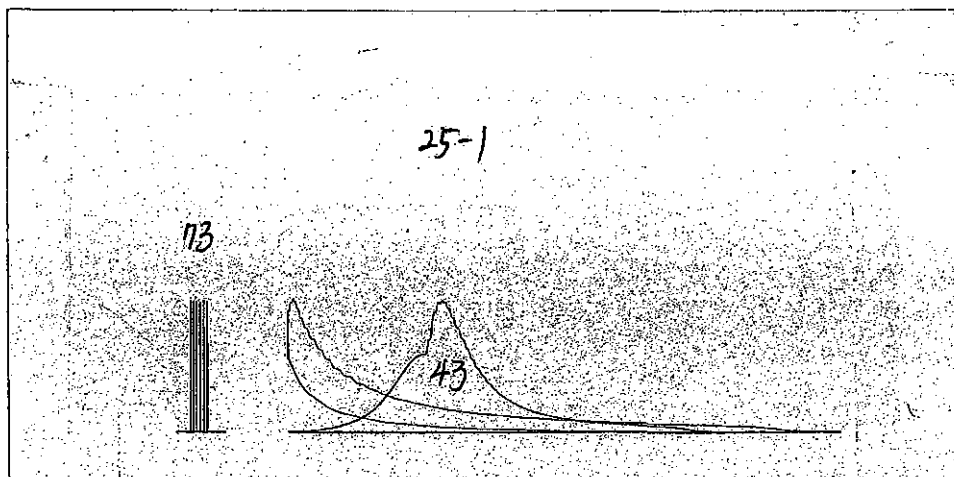
Zero set of No.6 Cylinder



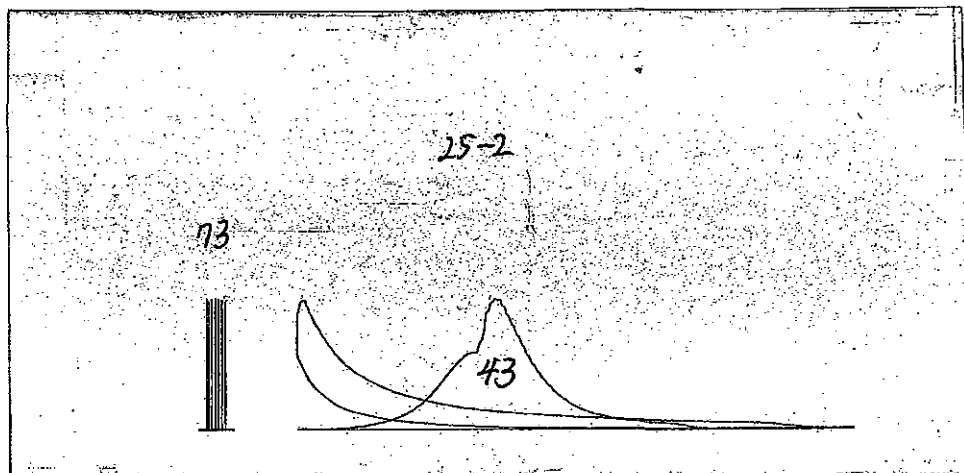
Zero set of No.7 Cylinder



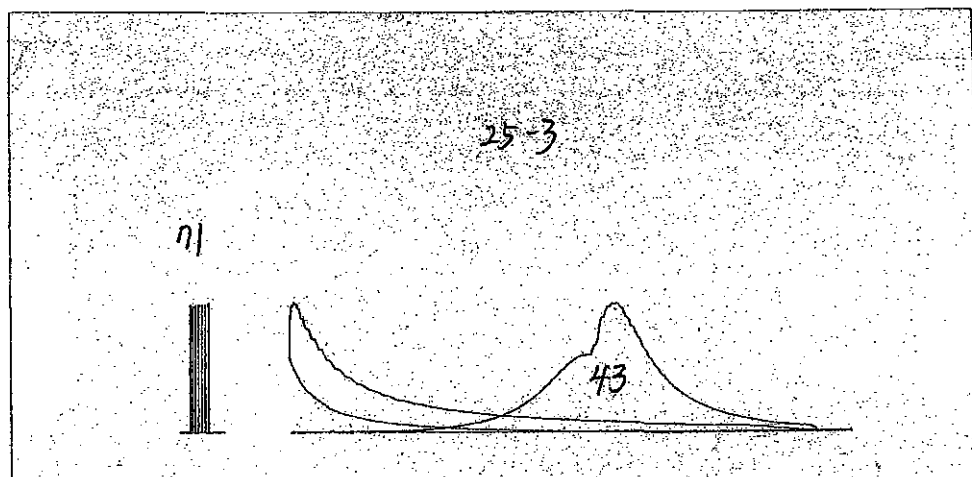
25%t of No.1 Cylinder



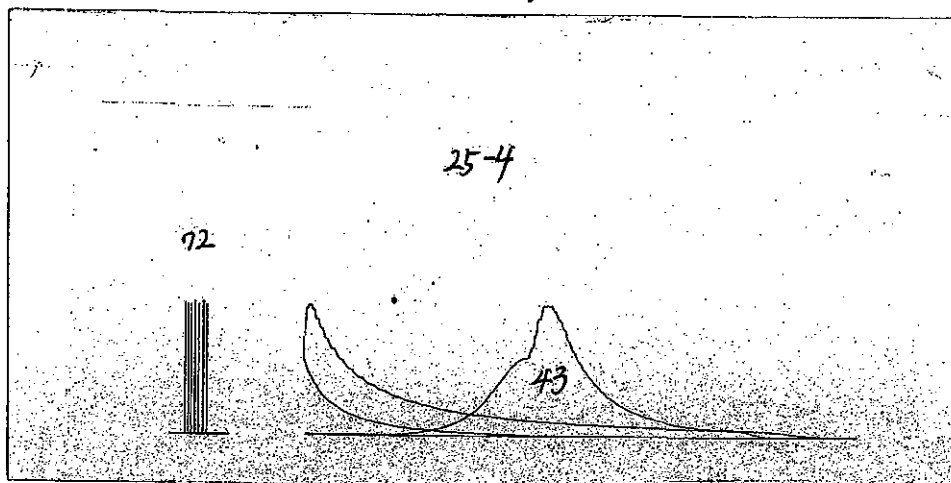
25% of No.2 Cylinder



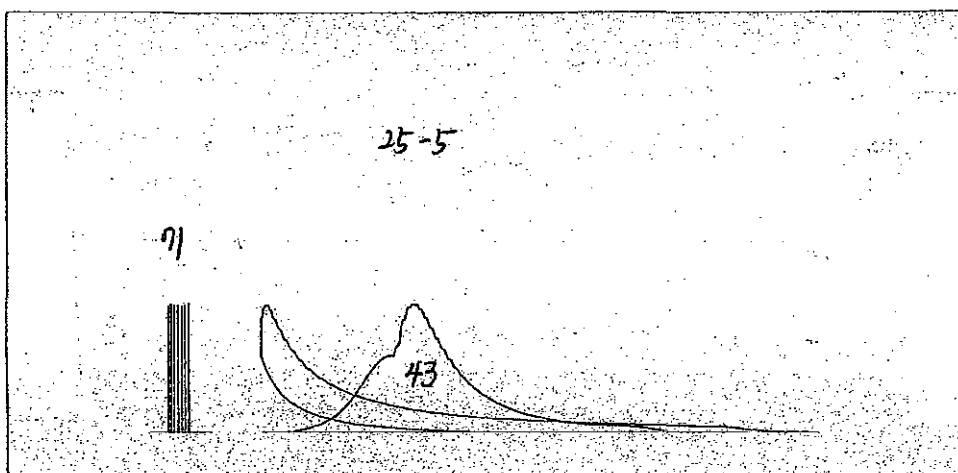
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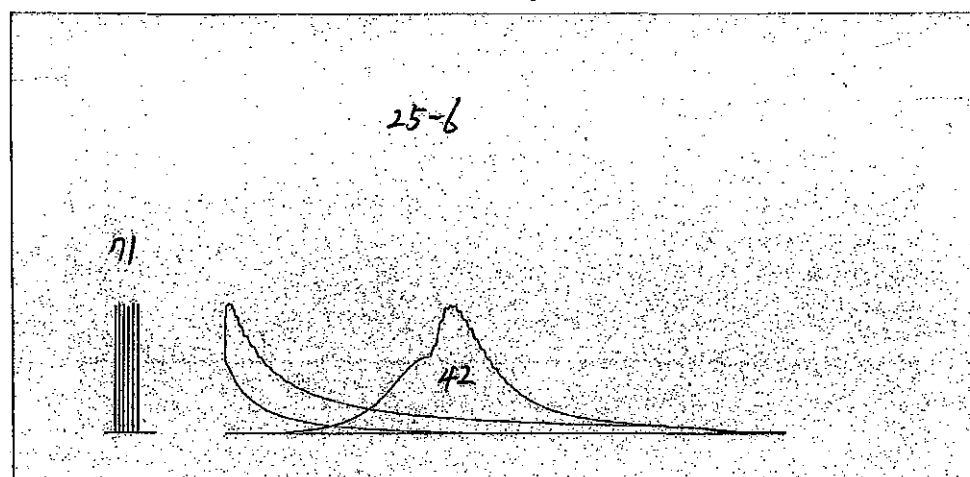
25% of No.4 Cylinder



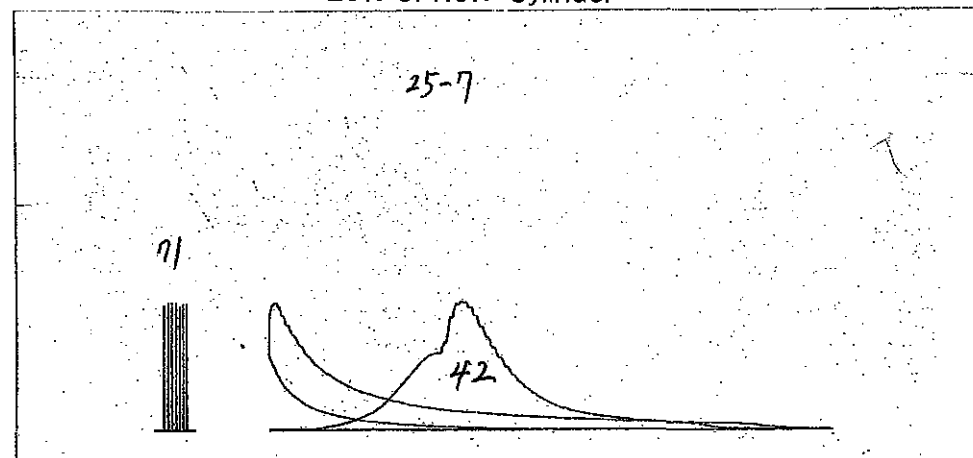
25% of No.5 Cylinder



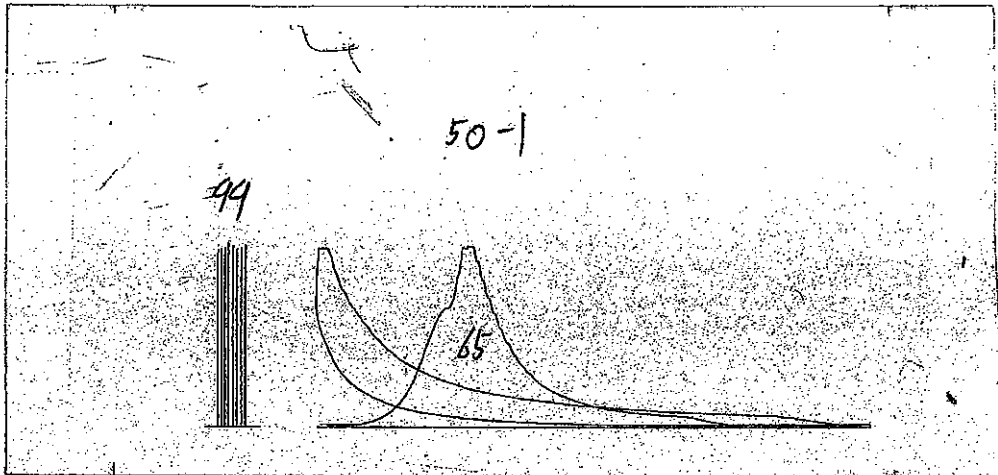
25% of No.6 Cylinder



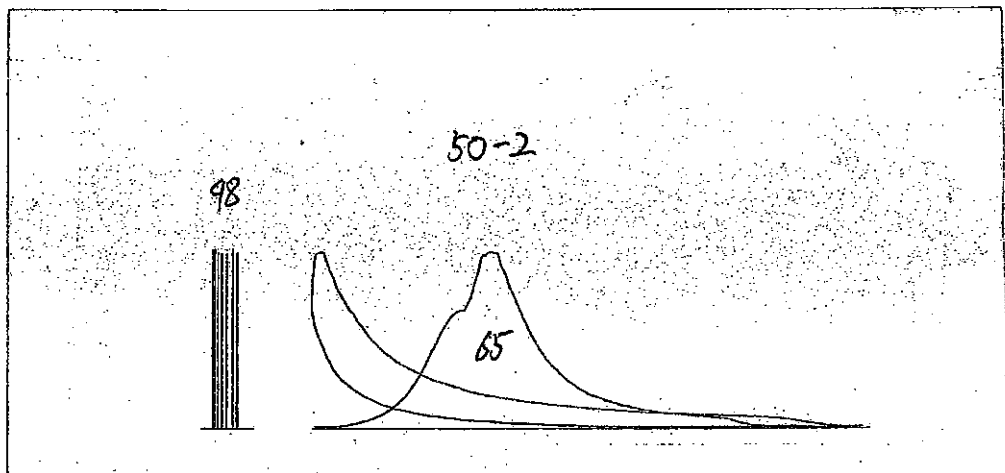
25% of No.7 Cylinder



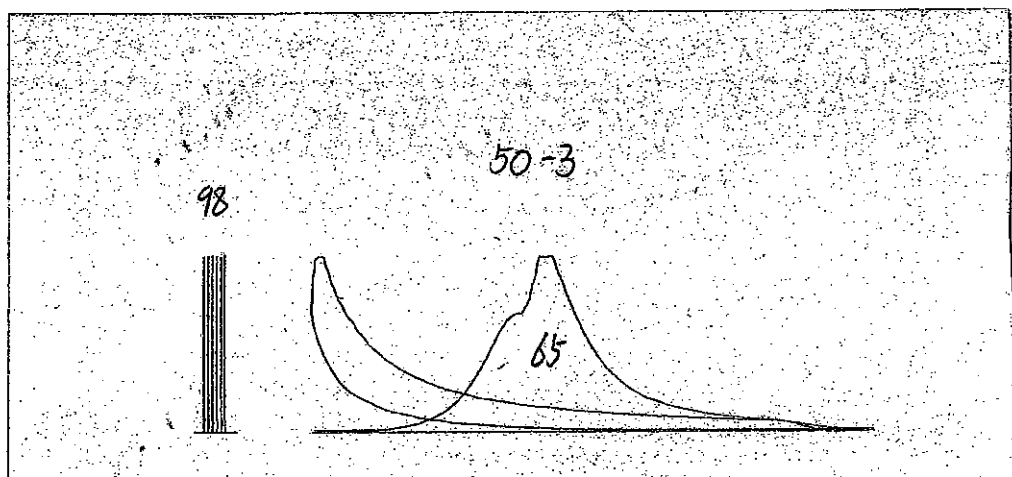
50% of No.1 Cylinder



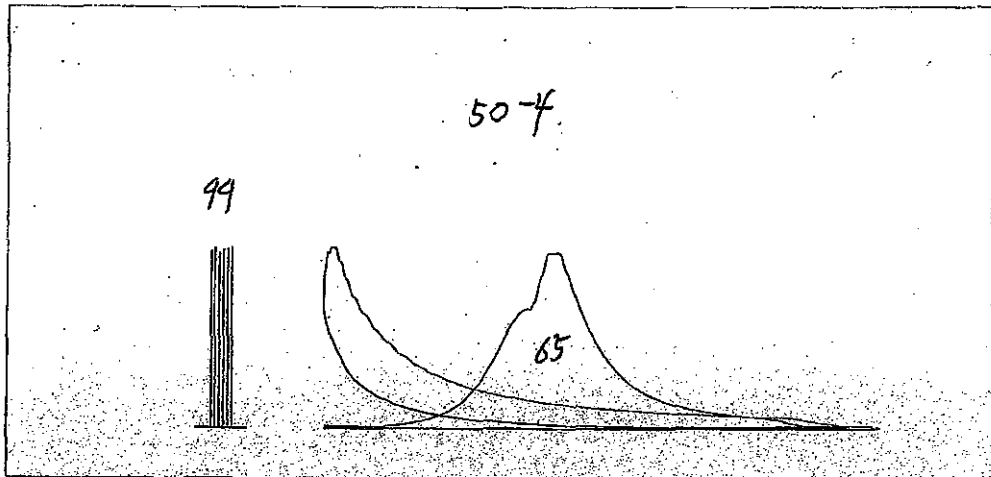
50% of No.2 Cylinder



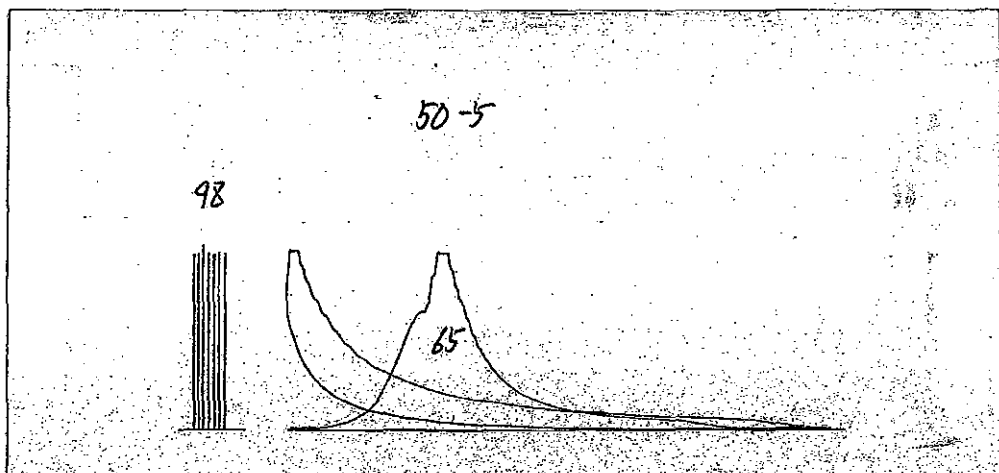
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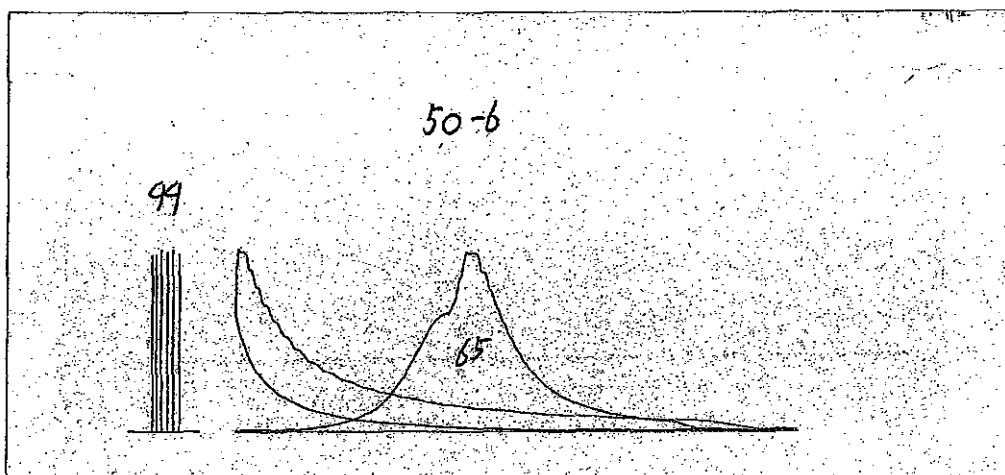
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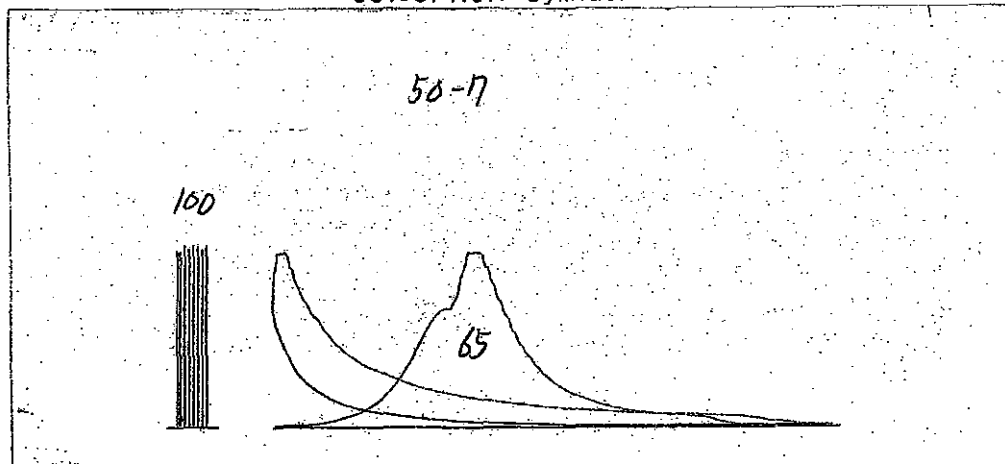
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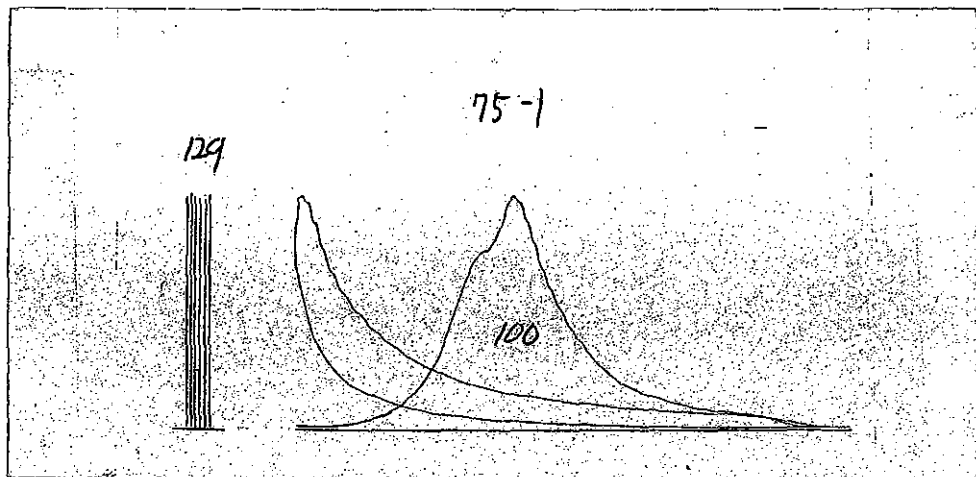
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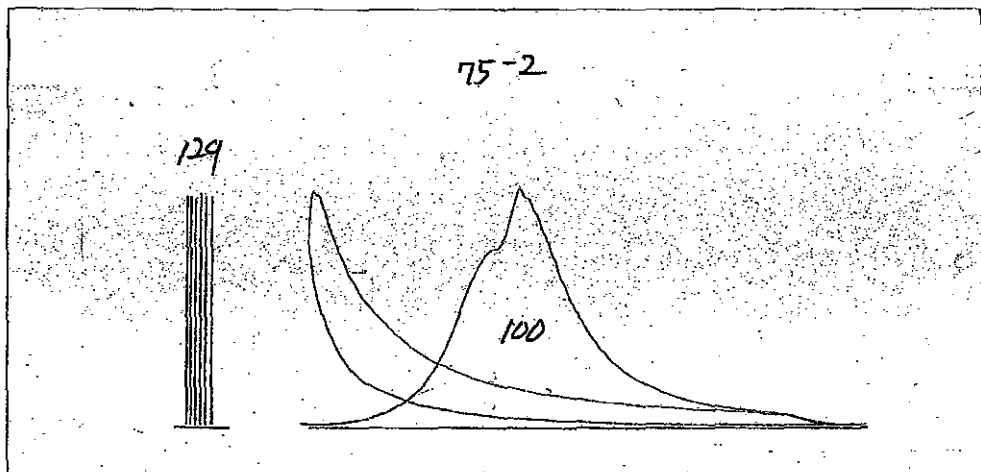
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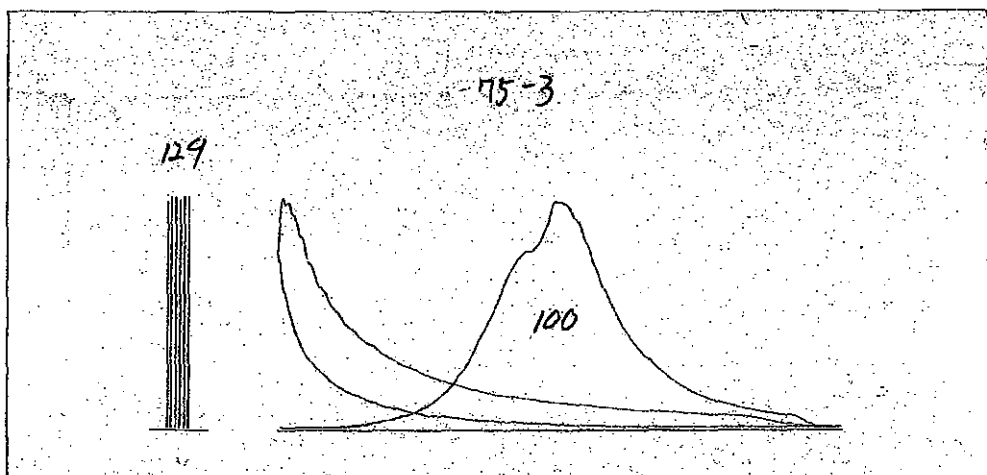
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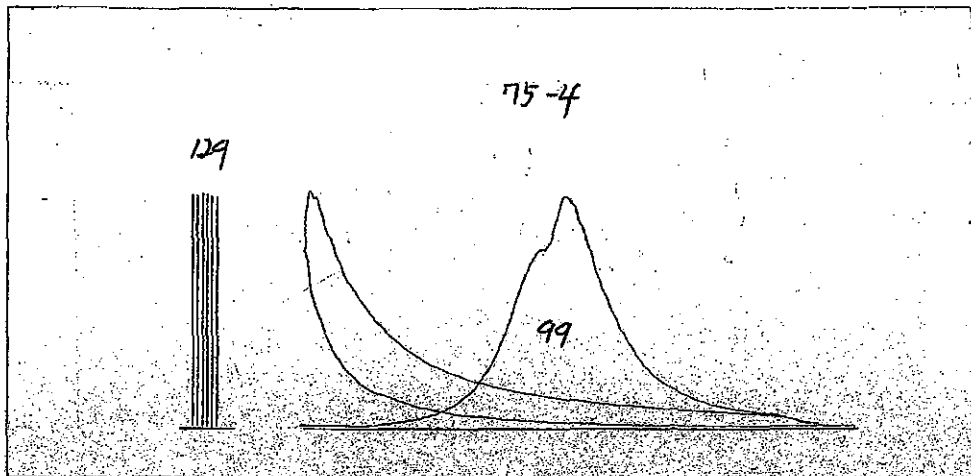
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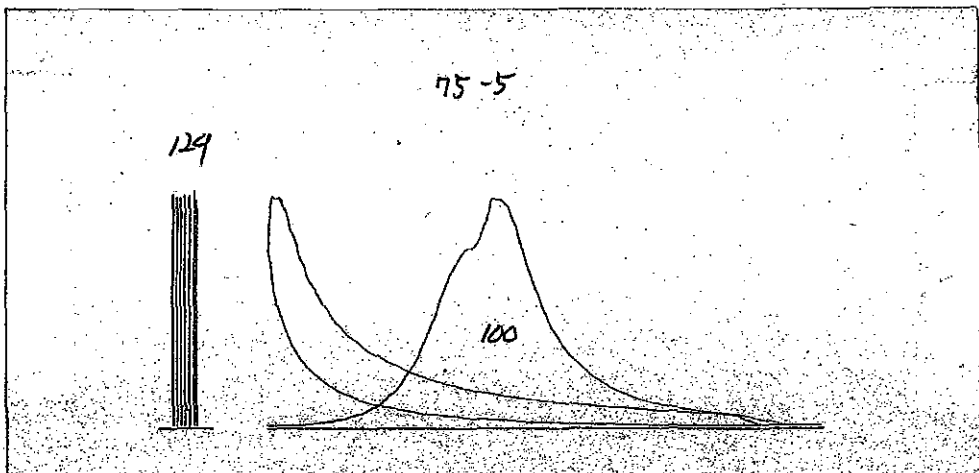
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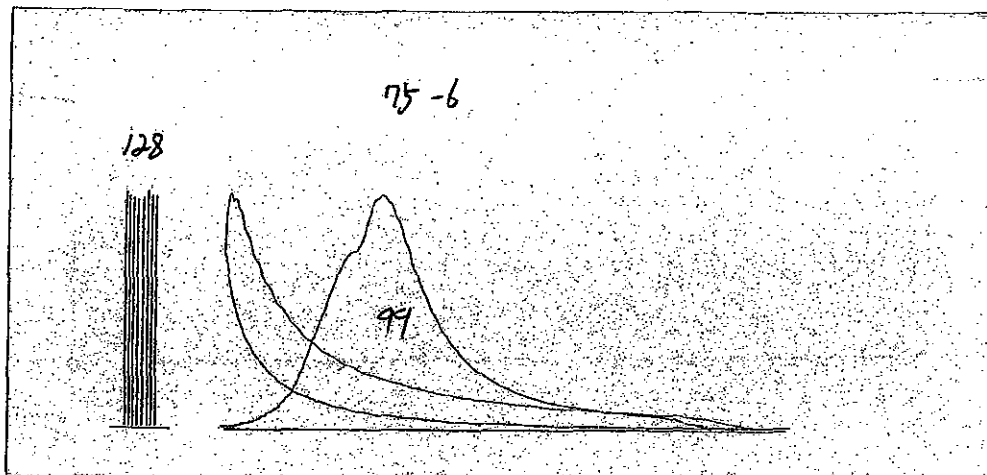
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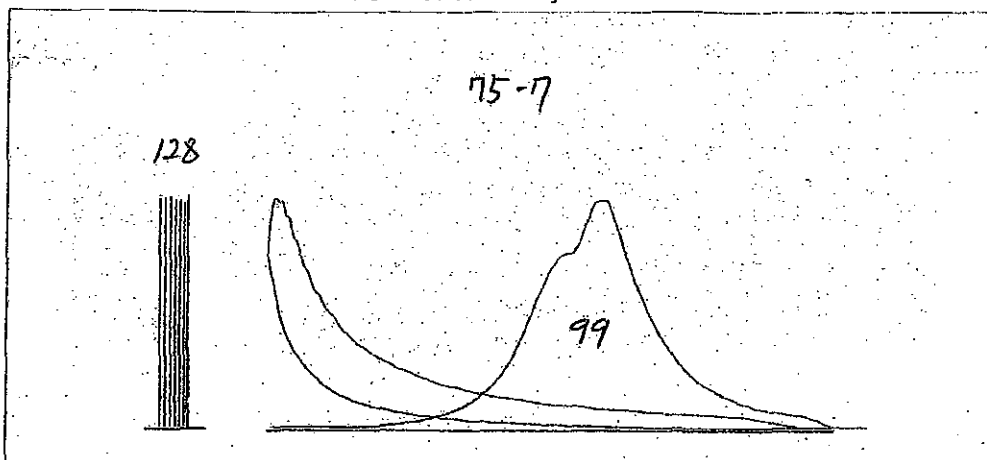
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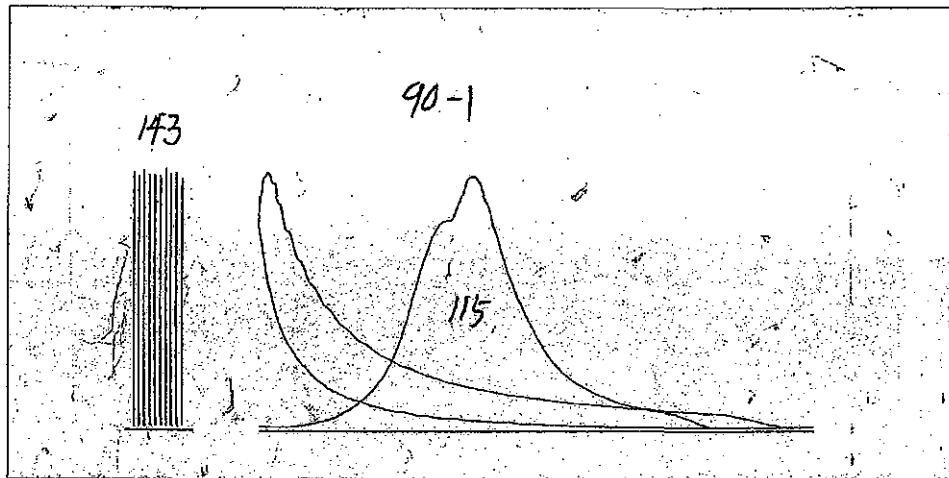
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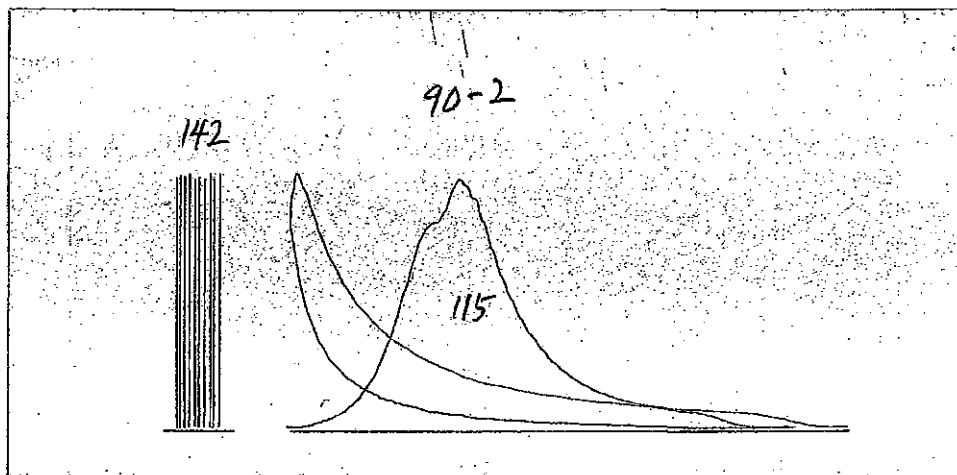
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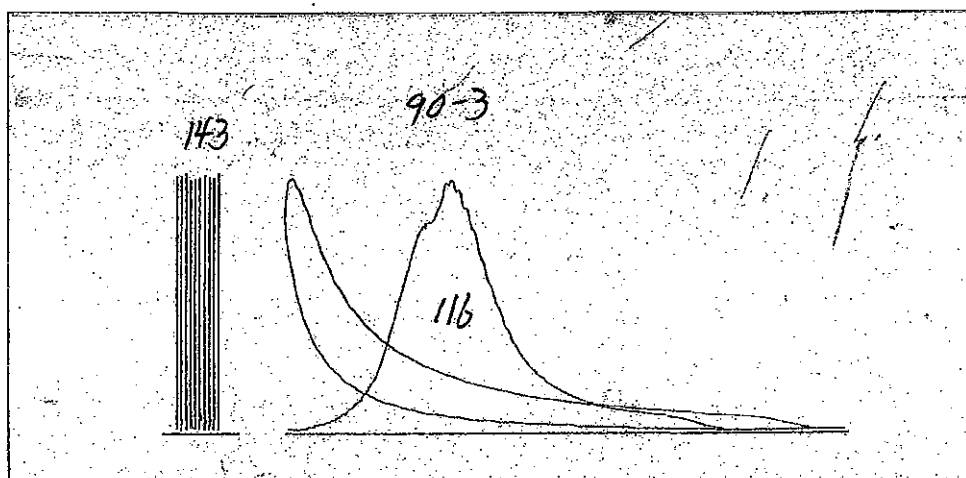
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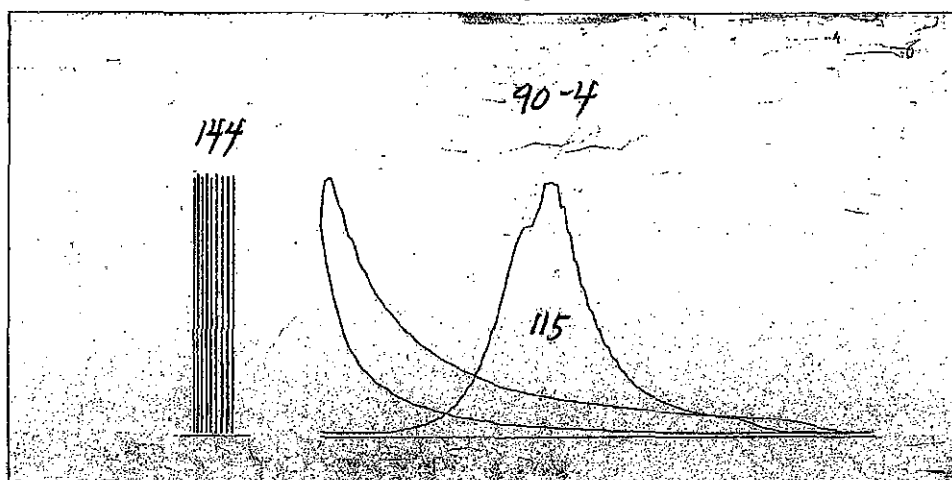
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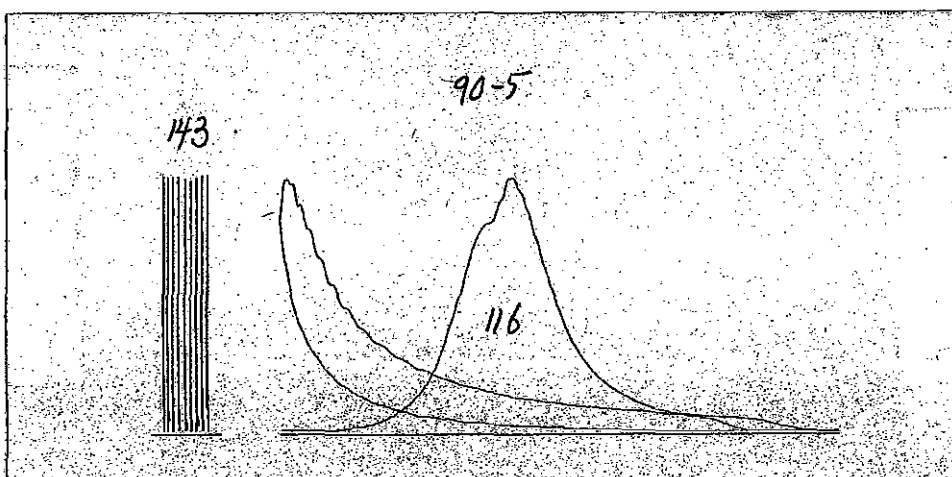
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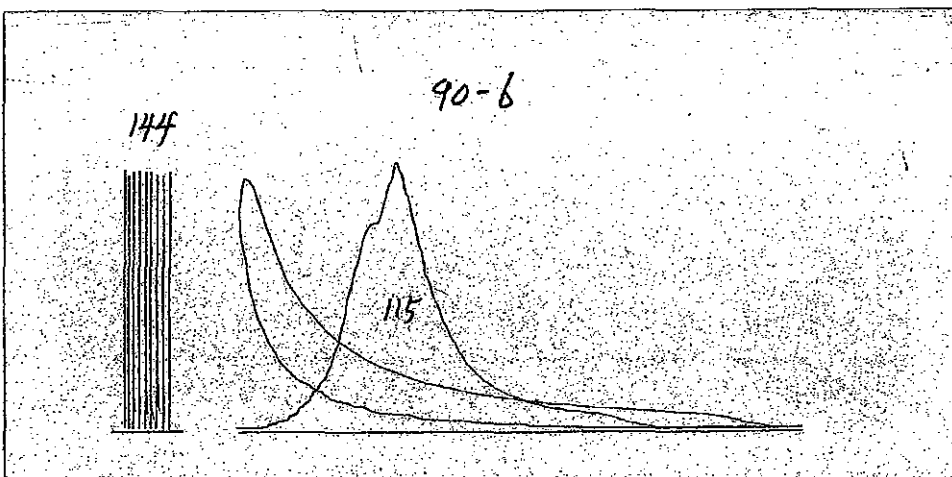
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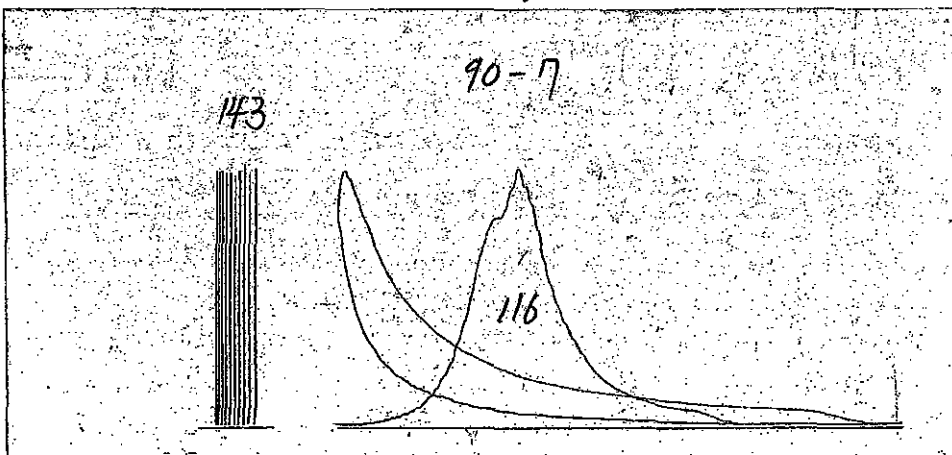
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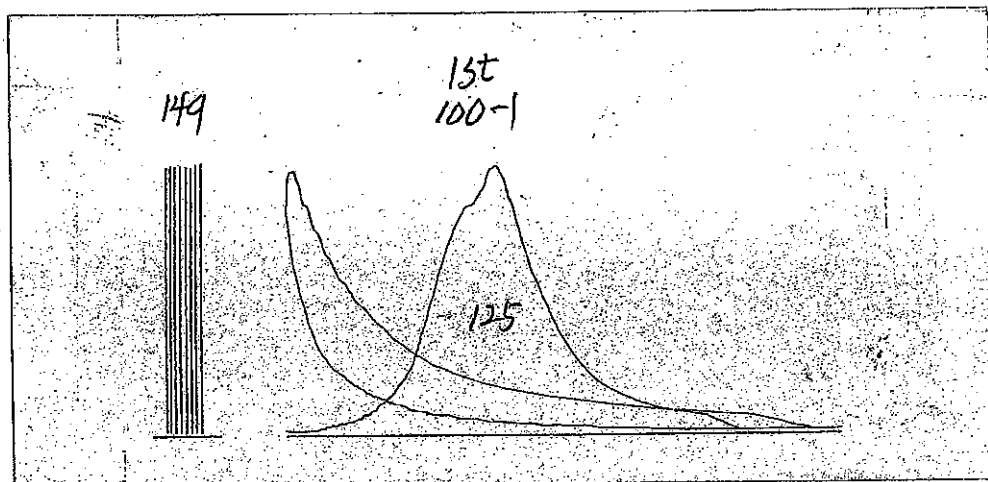
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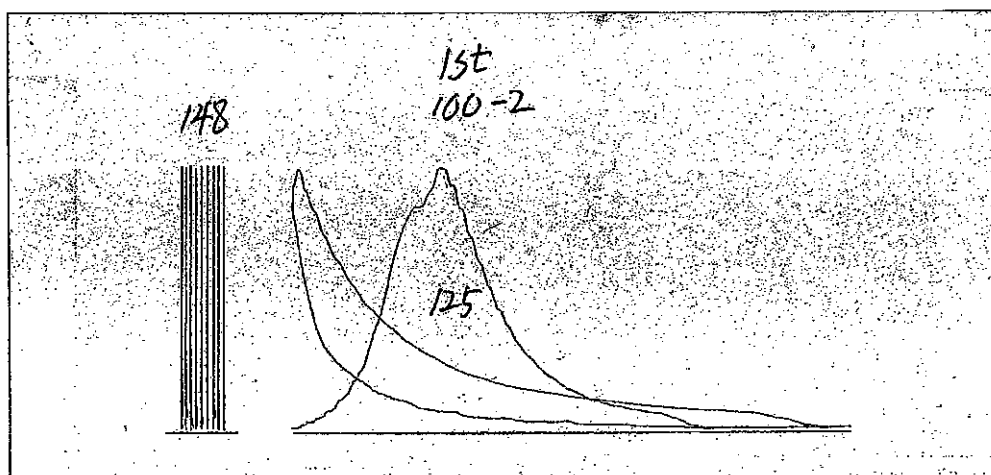
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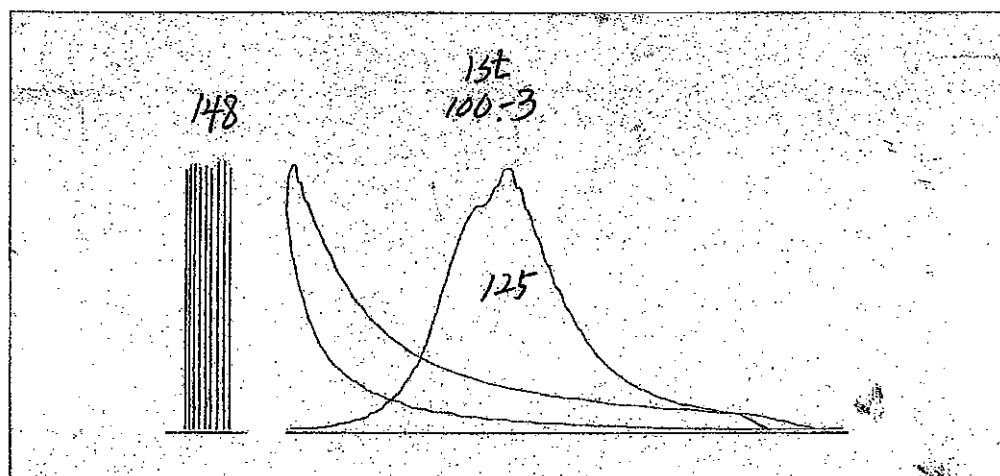
100% of No.1 Cylinder-1st



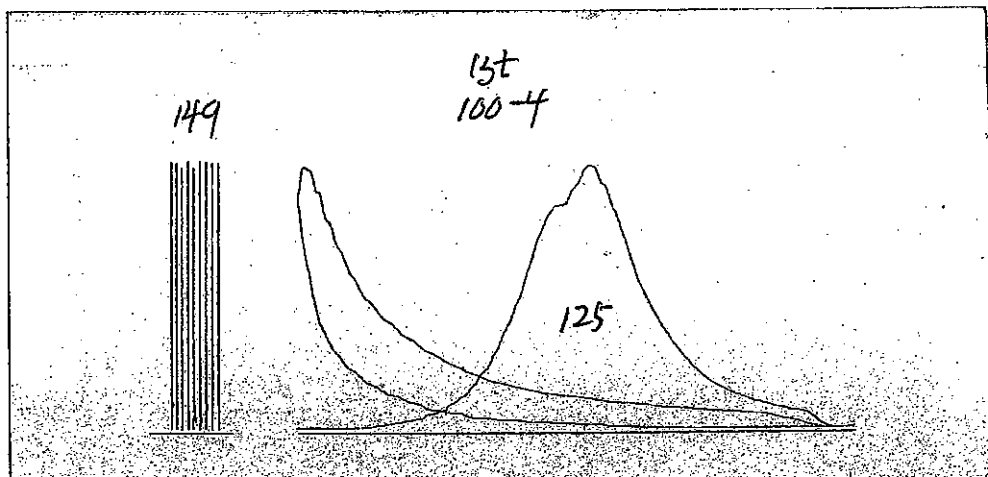
100% of No.2 Cylinder-1st



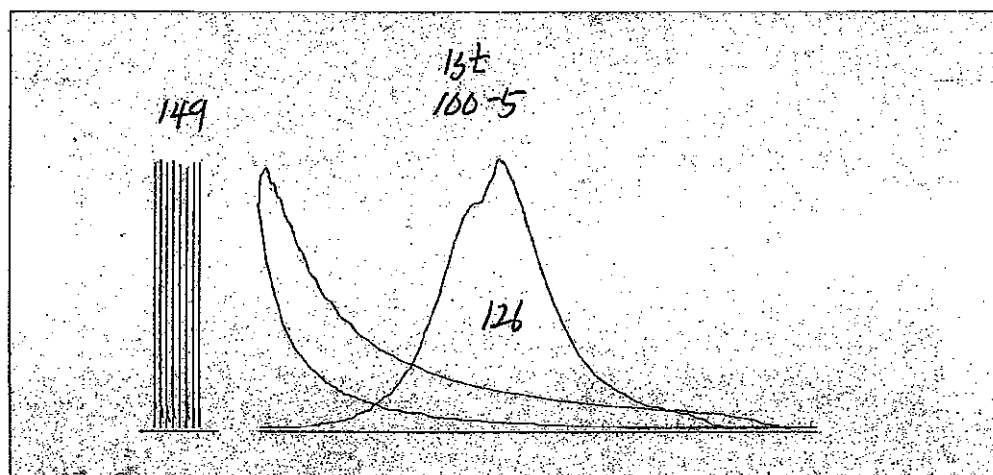
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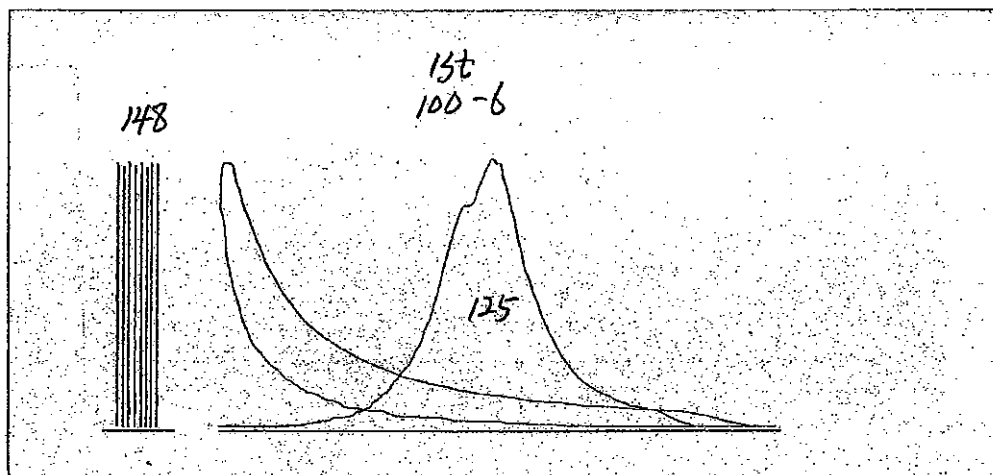
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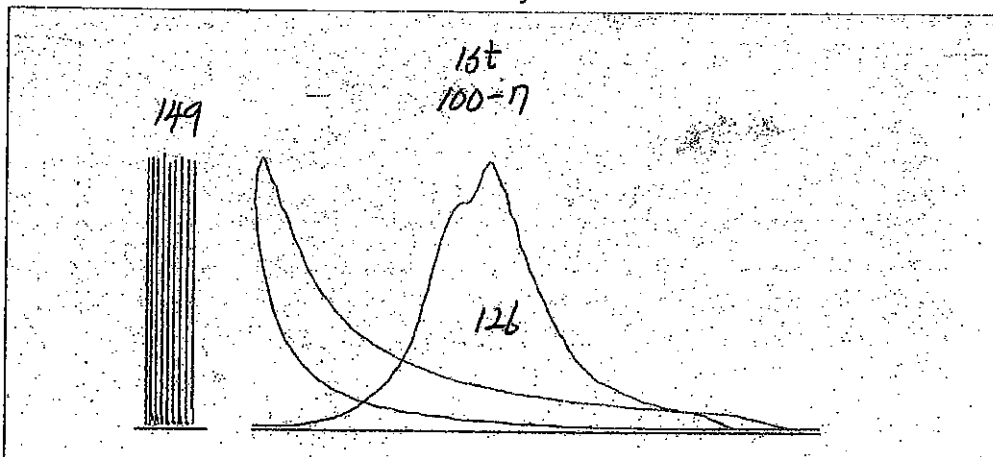
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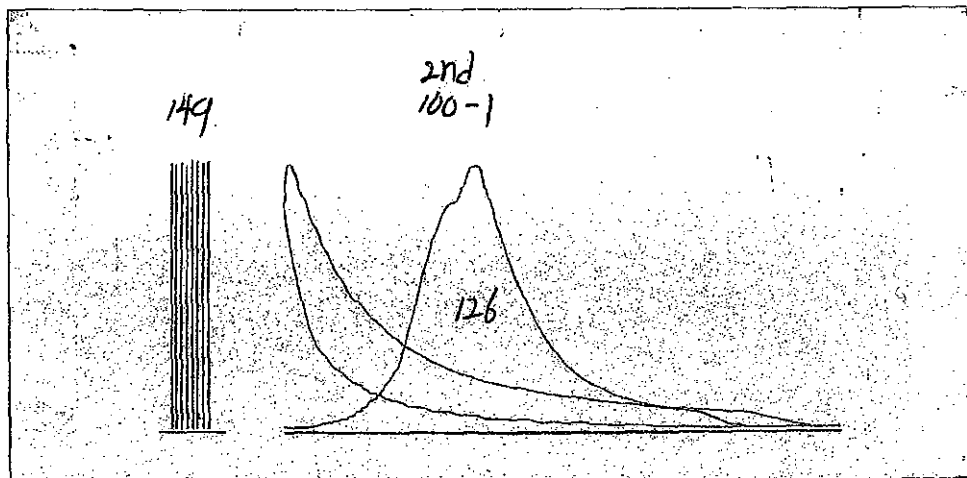
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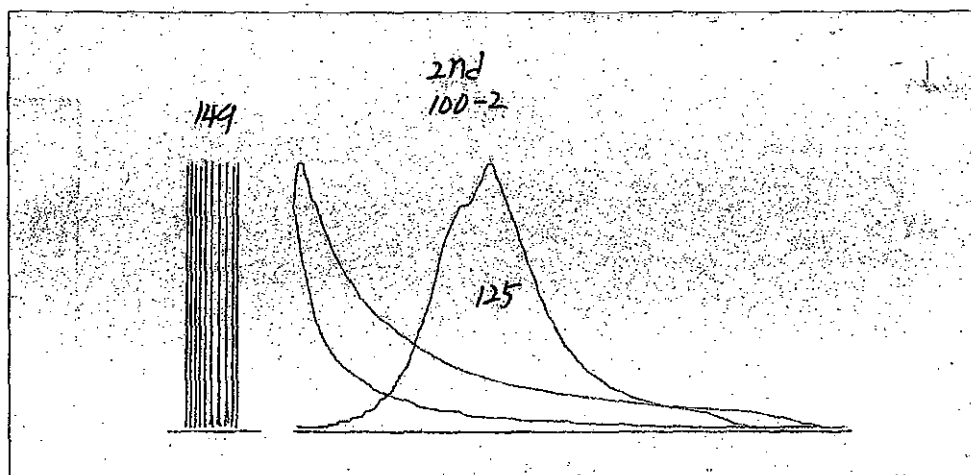
100% of No.7 Cylinder-1st



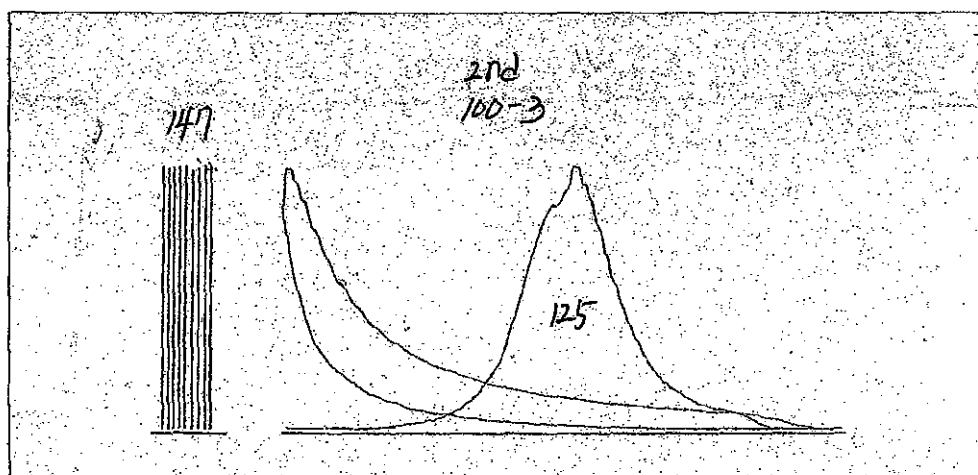
100% of No.1 Cylinder-2nd



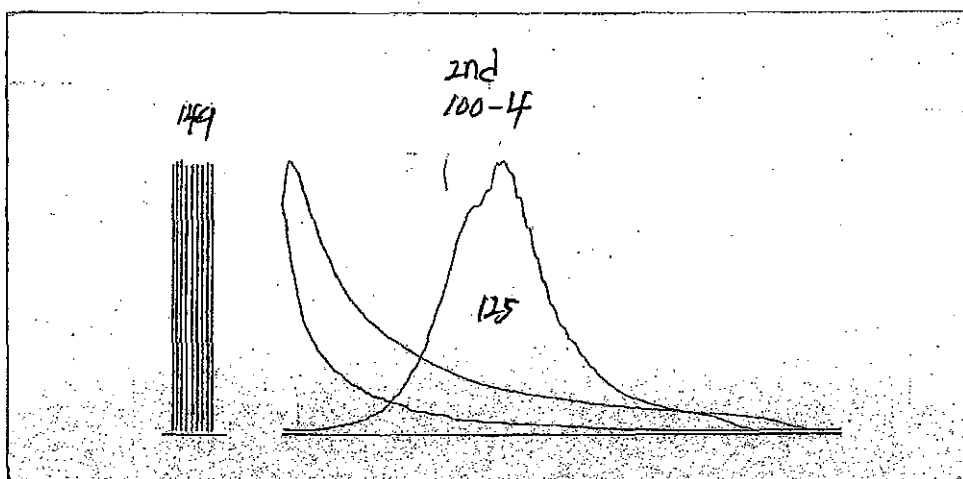
100% of No.2 Cylinder-2nd



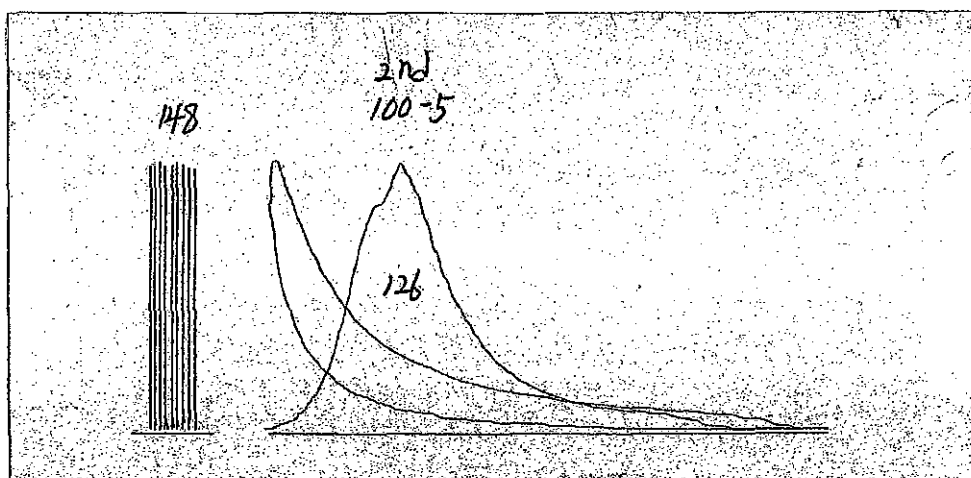
100% of No.3 Cylinder-2nd



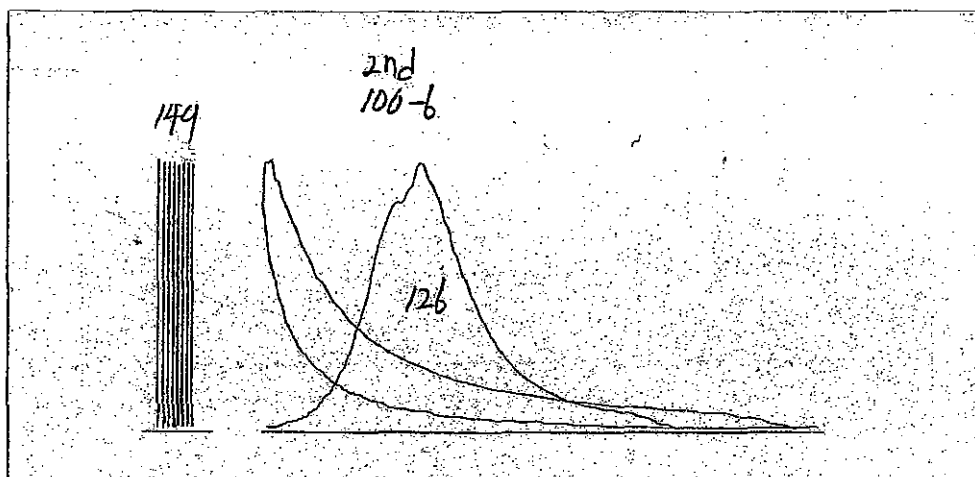
100% of No.4 Cylinder-2nd



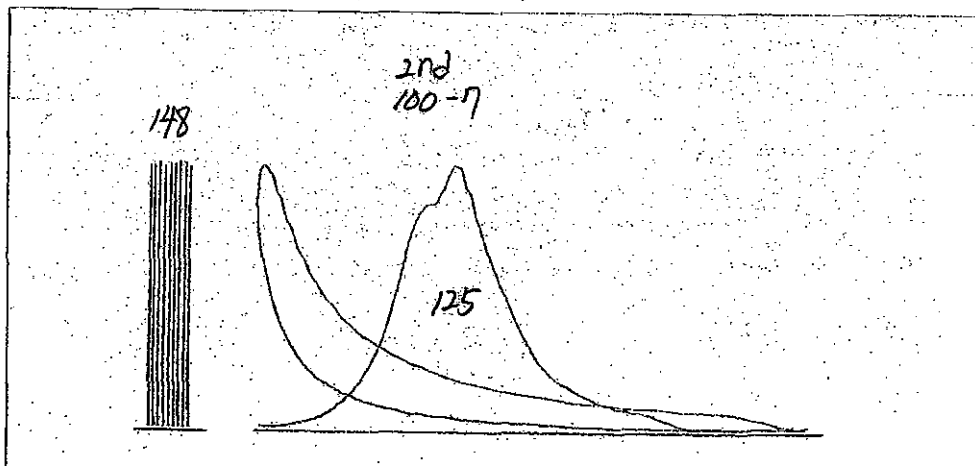
100% of No.5 Cylinder-2nd



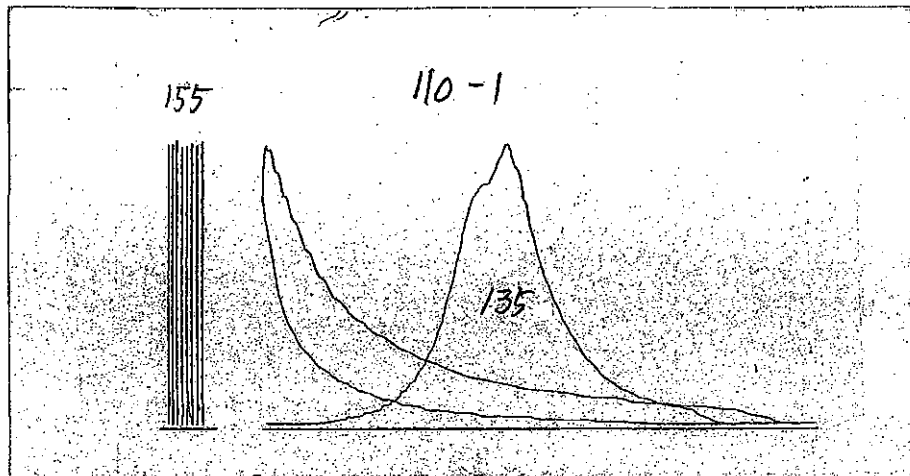
100% of No.6 Cylinder-2nd



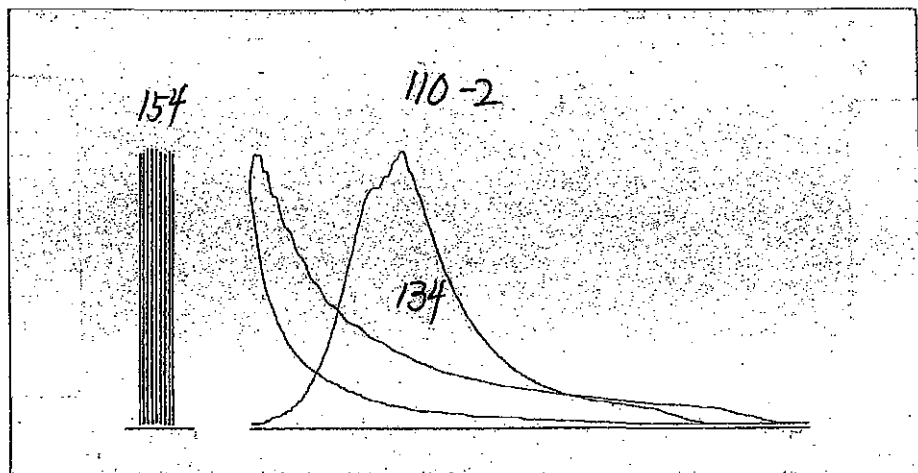
100% of No.7 Cylinder-2nd



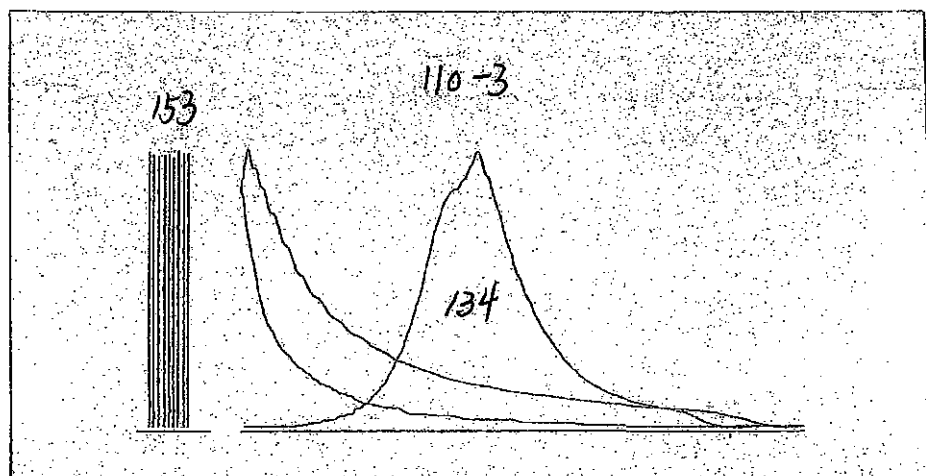
110% of No.1 Cylinder



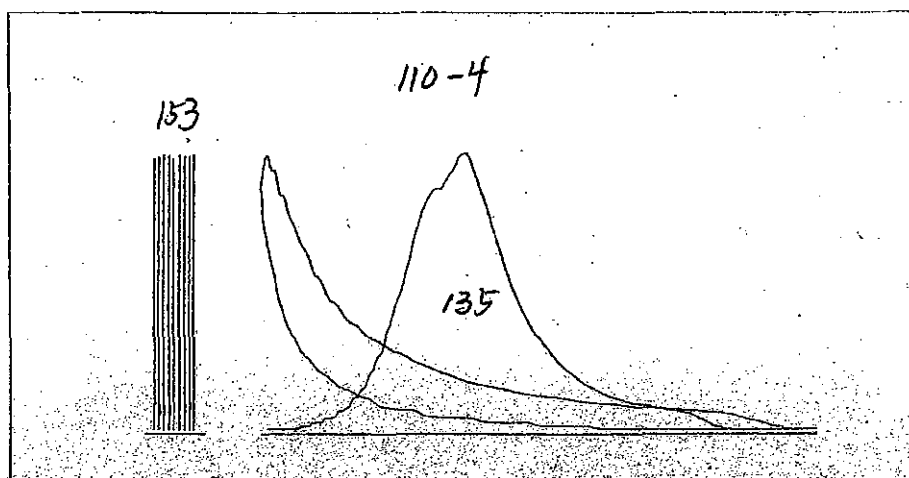
110% of No.2 Cylinder



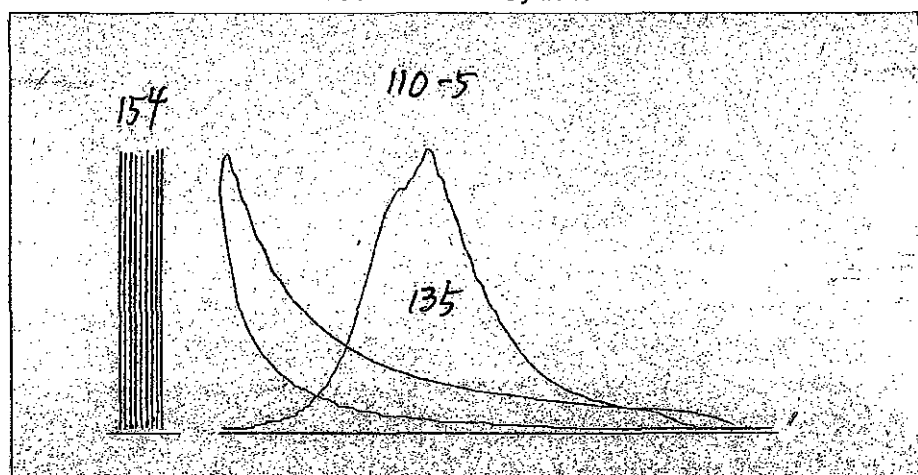
110% of No.3 Cylinder



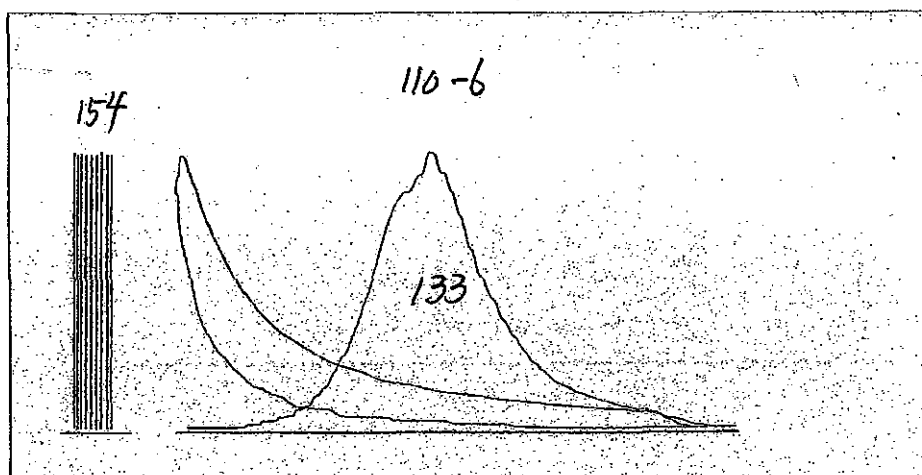
110% of No.4 Cylinder



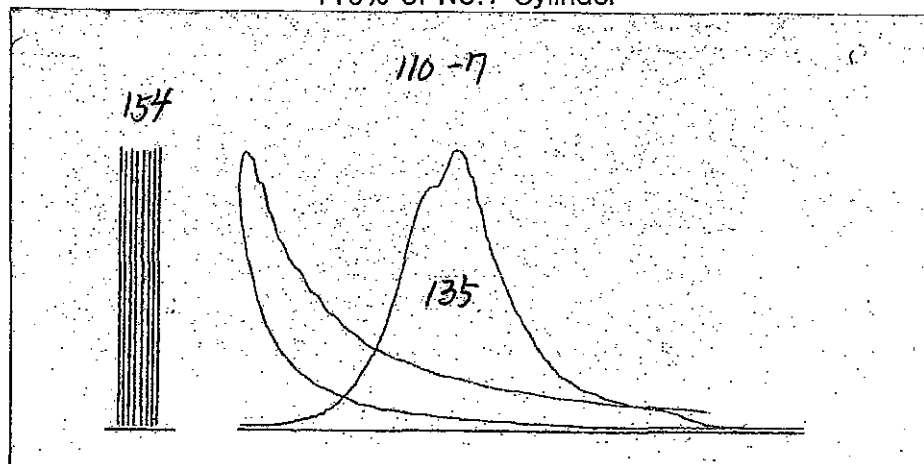
110% of No.5 Cylinder



110% of No.6 Cylinder



110% of No.7 Cylinder





INSPECTION REPORT

Report No.

STXQA-10-451

Page

1 OF 1

Hull No.

S-4012

Engine Type

7S50MC-C7

Unit

1/100 mm

Owner

NNC

Engine No.

SB7S50-09511

Date

2010. 07. 20

Shipyard

STX

Class

LR

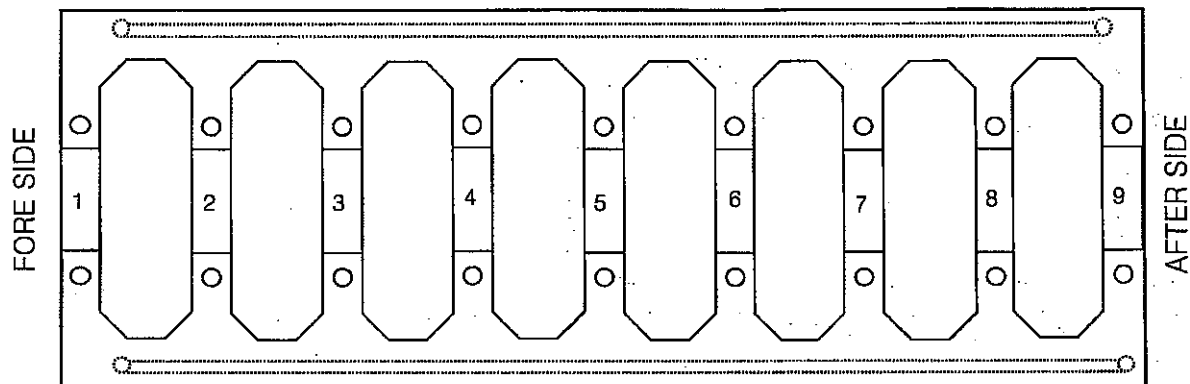
Result

Acceptable

Kind of Inspection : BED PLATE LEVELING & CRANK SHAFT DEFLECTION

EXHAUST SIDE

(Room Temp : 28 °C)



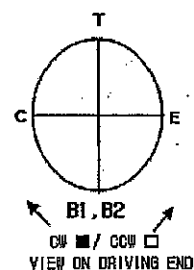
CAM SIDE

1. Alignment of bedplate (after tightened) Spec : support to support ; 0.05mm. Total ; 0.1mm/10m.

Support No.	1	2	3	4	5	6	7	8	9
Position									
CAM SIDE	0	-1	-1	-4	-5	-3	-2	0	
EXHAUST SIDE	-1	-2	-2	-2	-2	-2	-3	-2	-2
REMARK	DEVIATION VALUE (Reference Height : 6.85 mm = 0.0)								

2. Crankshaft deflection Spec : after most Max 0.47mm, others Max 0.18mm

Position	Cyl No.	1	2	3	4	5	6	7
	B1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	C	3.0	3.0	2.0	1.0	-2.0	2.0	0.0
	T	6.0	6.0	5.0	3.0	-1.0	5.0	-1.0
	E	3.0	3.0	4.0	3.0	2.0	4.0	0.0
	B2	1.0	0.0	0.0	1.0	0.0	0.0	1.0
REMARK								



3. Thrust Bearing High Temperature Trip. Set, Value : 90 ± 2.0 °C

Actual : 90 °C

4. Fwd side axial Damper clearance(mm)

Spec : (1.7±0.1mm)

Top	1.70	Bottom	1.75	Cam'	1.75	Exh'	1.70
-----	------	--------	------	------	------	------	------

Witnessed by :

Approved by : H. J. Shin

Witnessed by :

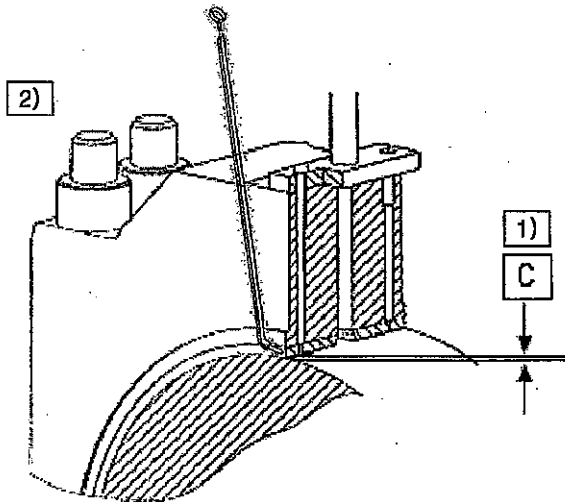
Reviewed by : K. S. Yang

Witnessed by :

Prepared by : S. H. Lee

stx	INSPECTION REPORT			Report No.	STXQA-10-452
				Page	1 OF 1
Hull No.	S-4012	Engine Type	7S50MC-C7	Unit	1/100mm
Owner	NNC	Engine No.	SB7S50-09511	Date	2010. 07. 28.
Shipyard	STX	Class	LR	Result	Acceptable

Kind of Inspection : MAIN BEARING CLEARANCE & TIGHTENING TORQUE



Spec Condition	Vertical (C)	
	min	max
Before Tightening Stay Bolt	35	60
After Tightening Stay Bolt	35	60
Measurement for upper shell		

1. Main Bearing clearance

Unit : 1/100 mm

Division	Cyl/No	1	2	3	4	5	6	7	8	9
Before Tightening Stay Bolt	Fore	47	47	47	46	44	43	42	45	47
	After	48	47	47	44	44	44	43	46	47
After Tightening Stay Bolt	Fore	43	43	43	42	42	42	41	42	42
	After	44	43	44	41	42	42	41	41	43

2. Tightening torque of main bearing cap

Unit : bar

Bearing No.		1	2	3	4	5	6	7	8	9
Spec. : 1500 ~ 1530 bar	Left	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500
	Right	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500

3. Tightening torque of Tie-rod bolt

Unit : bar

Bearing No.		1	2	3	4	5	6	7	8	9
Spec. : 1500 ~ 1530 bar	Left	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500
	Right	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500	1,500

Witnessed by :	Approved by : H. J. Shin
Witnessed by :	Reviewed by : K. S. Yang
Witnessed by :	Prepared by : S. H. Lee



INSPECTION REPORT

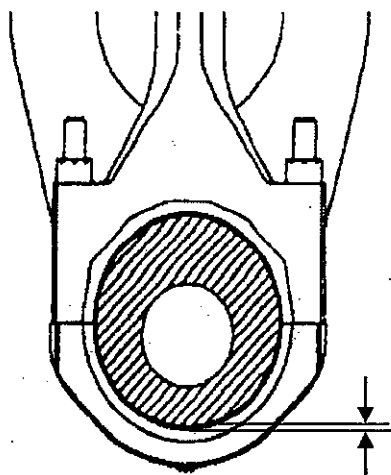
Report No. STXQA-10-453

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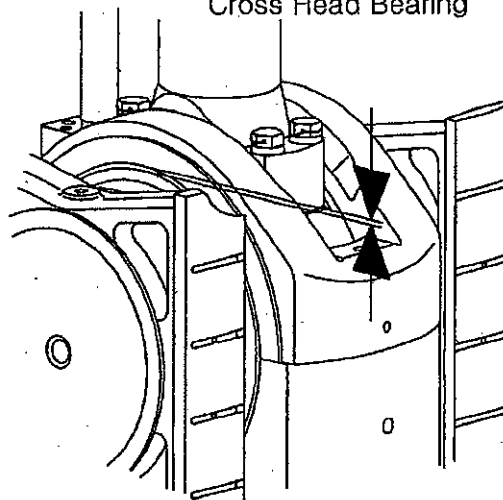
Hull No.	S-4012	Engine Type	7S50MC-C7	Unit	1/100mm
Owner	NNC	Engine No.	SB7S50-09511	Date	2010. 07. 28.
Shipyard	STX	Class	LR	Result	Acceptable

Kind of Inspection : CRANK & CROSS BEARING CLEARANCE

Crank Bearing



Cross Head Bearing



1. Cross Head Bearing Clearance

(Spec : 0.15 ~ 0.50mm)

Position \ Cyl'No.	1	2	3	4	5	6	7	-	-
Fore	24	26	28	26	28	30	25		
After	23	26	29	26	27	29	25		

2. Crank Bearing Clearance

(Spec : 0.25 ~ 0.50mm)

Position \ Cyl'No.	1	2	3	4	5	6	7	-	-
Fore	38	32	36	35	37	36	30		
After	37	33	35	35	37	36	31		

3. Tightening torque of Bearing Cap

(Spec. : 1500 ~ 1530 bar)

Division	BRG No.	1	2	3	4	5	6	7	-	-
Crank Bearing	Left	1,500	1,500	1,500	1,500	1,500	1,500	1,500		
	Right	1,500	1,500	1,500	1,500	1,500	1,500	1,500		
Crosshead Bearing	Left	1,500	1,500	1,500	1,500	1,500	1,500	1,500		
	Right	1,500	1,500	1,500	1,500	1,500	1,500	1,500		

Witnessed by :

Approved by : H. J. Shin

Witnessed by :

Reviewed by : K. S. Yang

Witnessed by :

Prepared by : S. H. Lee



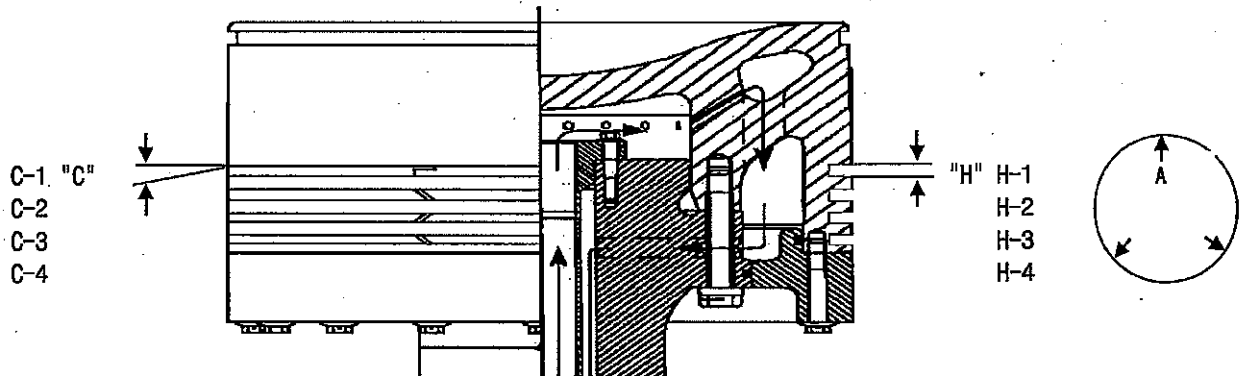
INSPECTION REPORT

Report No. STXQA-10-454

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Hull No.	S-4012	Engine Type	7S50MC-C7	Unit	mm
Owner	NNC	Engine No.	SB7S50-09511	Date	2010. 07. 20
Shipyard	STX	Class	LR	Result	Acceptable

Kind of Inspection : PISTON RING CLEARANCE



1). Ring groove (Spec : H-1 : 12.5,+0.240~0.265mm / H-2,3,4 : 9.5,+0.240~0.265mm)

RING NO.	Point	CYLINDER NO.								
		1	2	3	4	5	6	7	SPARE	
H-1	A	+0.255	+0.265	+0.260	+0.260	+0.260	+0.250	+0.250	+0.265	
	B	+0.255	+0.265	+0.260	+0.260	+0.260	+0.250	+0.250	+0.265	
	C	+0.255	+0.265	+0.260	+0.260	+0.260	+0.250	+0.250	+0.265	
H-2	A	+0.255	+0.260	+0.250	+0.255	+0.250	+0.255	+0.265	+0.260	
	B	+0.255	+0.260	+0.250	+0.255	+0.250	+0.255	+0.265	+0.260	
	C	+0.255	+0.260	+0.250	+0.255	+0.250	+0.255	+0.265	+0.260	
H-3	A	+0.250	+0.250	+0.255	+0.265	+0.245	+0.260	+0.260	+0.260	
	B	+0.250	+0.250	+0.255	+0.265	+0.245	+0.260	+0.260	+0.260	
	C	+0.250	+0.250	+0.255	+0.265	+0.245	+0.260	+0.260	+0.260	
H-4	A	+0.255	+0.260	+0.245	+0.260	+0.255	+0.260	+0.255	+0.260	
	B	+0.255	+0.260	+0.245	+0.260	+0.255	+0.260	+0.255	+0.260	
	C	+0.255	+0.260	+0.245	+0.260	+0.255	+0.260	+0.255	+0.260	

2). Ring clearance (Spec : 0.35 ~ 0.40 mm)

RING NO.	Point	CYLINDER No.								
		1	2	3	4	5	6	7	8	9
C-1	A	0.37	0.37	0.35	0.37	0.36	0.37	0.38		
	B	0.37	0.37	0.35	0.37	0.36	0.37	0.38		
	C	0.37	0.37	0.35	0.37	0.36	0.37	0.38		
C-2	A	0.38	0.35	0.36	0.38	0.38	0.37	0.39		
	B	0.38	0.35	0.36	0.38	0.38	0.37	0.39		
	C	0.38	0.35	0.36	0.38	0.38	0.37	0.39		
C-3	A	0.36	0.35	0.37	0.36	0.37	0.36	0.38		
	B	0.36	0.35	0.37	0.36	0.37	0.36	0.38		
	C	0.36	0.35	0.37	0.36	0.37	0.36	0.38		
C-4	A	0.38	0.36	0.38	0.38	0.35	0.37	0.36		
	B	0.38	0.36	0.38	0.38	0.35	0.37	0.36		
	C	0.38	0.36	0.38	0.38	0.35	0.37	0.36		

Witnessed by :

Approved by : H. J. Shin

Witnessed by :

Reviewed by : K. S. Yang

Witnessed by :

Prepared by : S. H. Lee



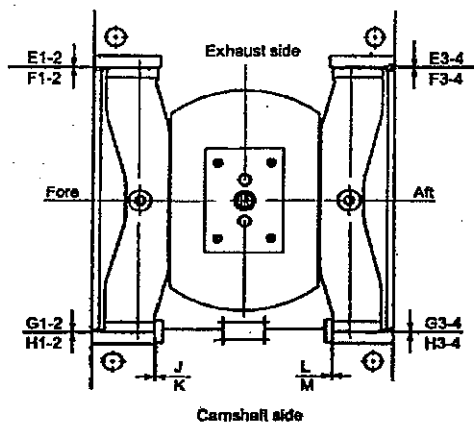
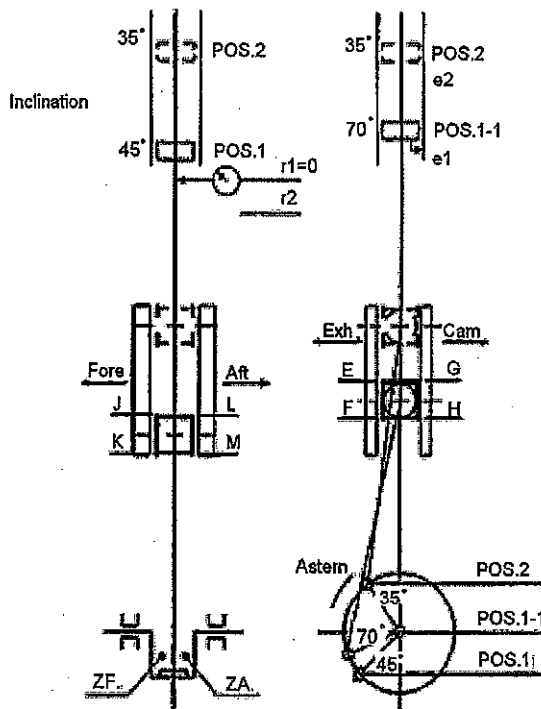
INSPECTION REPORT

Report No. STXQA-10-455

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Hull No.	S-4012	Engine Type	7S50MC-C7	Unit	1/100 mm
Owner	NNC	Engine No.	SB7S50-09511	Date	2010. 07. 28.
Shipyard	STX	Class	LR	Result	Acceptable

Kind of Inspection : PISTON ROD ALIGNMENT



Measure point	Value		mm
PF + PA	N max		0.50
	O max		0.70
E + G H + F	N	max	0.50
		min	0.20
	O max		0.80
J K L M	N	max	1.05
		min	0.50
	O max		1.10
ZF, ZA	SEMI	N min	5.0
	BUILT	O min	4.0
R1 ± R2	N/O max		0.35

N : New engine (less than 100 running hours)

O : Others (engine in service)

R1±R2 : Piston rod Inclination

Total : "F" means piston pointing forward

: "A" means piston pointing backward

Witnessed by :

Approved by : H. J. Shin

Witnessed by :

Reviewed by : K. S. Yang

Witnessed by :

Prepared by : S. H. Lee

*** Attach 3 Page.(Check sheet)



INSPECTION REPORT

Report No. STXQA-10-455

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Hull No.	S-4012	Engine Type	7S50MC-C7	Unit	1/100 mm
Owner	NNC	Engine No.	SB7S50-09511	Date	2010. 07. 28.
Shipyard	STX	Class	LR	Result	Acceptable

Kind of Inspection : PISTON ROD ALIGNMENT (ACTUAL MEASUREMENT)

ITEM	Crank position	Check points	Cylinder No.							
			1	2	3	4	5	6	7	-
Clearance of piston/liner	1.	P F	30	30	30	20	30	30	30	
		P A	10	10	10	20	10	10	10	
		PF+PA	40	40	40	40	40	40	40	
Clearance between con rod and crank	45 °	ZF	1100	1050	990	1100	1000	1050	1090	
		ZA	900	950	1010	900	1000	950	920	
Clearance Between guide Strip and Guide		J	70	70	70	80	60	70	80	
		K	70	70	70	80	60	70	80	
		L	80	80	90	60	80	70	80	
		M	80	80	90	60	80	70	80	
Clearance Between Guide shoe and Cross head Guide	1-1.	E1	25	35	30	30	35	35	30	
		E2	25	35	30	30	35	35	30	
		E3	30	30	30	25	35	35	30	
		E4	30	30	30	25	35	35	30	
		G1	0	0	0	0	0	0	0	
		G2	0	0	0	0	0	0	0	
		G3	0	0	0	0	0	0	0	
		G4	0	0	0	0	0	0	0	
		E1+G1	25	35	30	30	35	35	30	
		E2+G2	25	35	30	30	35	35	30	
		E3+G3	30	30	30	25	35	35	30	
		E4+G4	30	30	30	25	35	35	30	
	"ABDC"	H1	0	0	0	0	0	0	0	
		H2	0	0	0	0	0	0	0	
		H3	0	0	0	0	0	0	0	
		H4	0	0	0	0	0	0	0	
		F1	30	35	30	35	40	35	35	
		F2	25	35	35	35	40	35	35	
		F3	40	35	35	35	40	35	35	
		F4	40	35	35	35	40	35	35	
		H1+F1	30	35	30	35	40	35	35	
		H2+F2	25	35	35	35	40	35	35	
		H3+F3	40	35	35	35	40	35	35	
		H4+F4	40	35	35	35	40	35	35	
Inclination (r1=0, e1=0)	piston rod	R1 ± R2	-13	-12	-10	-14	-4	-5	-10	
	piston	e2	12	10	7	8	13	12	13	
	Total		F25	F22	F17	F22	F17	F17	F23	



INSPECTION REPORT

Report No. STXQA-10-455

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Hull No.	S-4012	Engine Type	7S50MC-C7	Unit	1/100 mm
Owner	NNC	Engine No.	SB7S50-09511	Date	2010. 07. 28.
Shipyard	STX	Class	LR	Result	Acceptable

Kind of Inspection : PISTON ROD ALIGNMENT (ACTUAL MEASUREMENT)

ITEM	Crank position	Check points	Cylinder No.							
			1	2	3	4	5	6	7	-
Clearance of piston/liner		PF	18	20	23	12	17	18	17	
		PA	22	20	17	28	23	22	23	
		PF+PA	40	40	40	40	40	40	40	
Clearance Between guide Strip and Guide		J	70	90	90	70	80	60	80	
		K	70	90	90	70	80	60	80	
		L	70	70	70	70	70	80	90	
		M	70	70	70	70	70	80	90	
		E1	30	35	35	35	40	35	35	
		E2	30	35	35	35	40	35	35	
		E3	30	35	35	35	40	35	35	
		E4	30	35	35	35	40	35	35	
		G1	0	0	0	0	0	0	0	
		G2	0	0	0	0	0	0	0	
		G3	0	0	0	0	0	0	0	
		G4	0	0	0	0	0	0	0	
	2.	E1+G1	30	35	35	35	40	35	35	
		E2+G2	30	35	30	35	40	35	35	
		E3+G3	30	35	35	35	40	35	35	
	"BTDC"	E4+G4	30	35	35	35	40	35	35	
		H1	0	0	0	0	0	0	0	
		H2	0	0	0	0	0	0	0	
	35 °	H3	0	0	0	0	0	0	0	
		H4	0	0	0	0	0	0	0	
		F1	25	30	30	30	35	35	30	
		F2	25	35	30	30	35	35	30	
		F3	30	30	30	25	30	30	30	
		F4	30	30	30	25	30	30	30	
		H1+F1	25	30	30	30	35	35	30	
		H2+F2	25	35	30	30	35	35	30	
		H3+F3	30	30	30	25	30	30	30	
		H4+F4	30	30	30	25	30	30	30	
Thickness of Shims between Guide strips/Crosshead Guide		BF	120	210	260	160	180	180	130	
		BA	290	240	200	260	250	240	330	



INSPECTION REPORT

Report No. STXQA-10-455

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Hull No.	S-4012	Engine Type	7S50MC-C7	Unit	1/100 mm
Owner	NNC	Engine No.	SB7S50-09511	Date	2010. 07. 28.
Shipyard	STX	Class	LR	Result	Acceptable

Kind of Inspection : CALCULATED MEASUREMENT

POS.	Cylinder No. 1			Cylinder No. 2			Cylinder No. 3			Cylinder No. 4		
2. "BTDC" 35 °	18	2	22	20	0	20	23	3	17	12	8	28
	(20)	->	(20)	(20)	<->	(20)	(20)	<-	(20)	(20)	->	(20)
	70		70	90		70	90		70	70		70
	(72)		(68)	(90)		(70)	(87)		(73)	(78)		(62)
1. "ABDC" 45 °	70		70	90		70	90		70	70		70
	(72)		(68)	(90)		(70)	(87)		(73)	(78)		(62)
	30	10	10	30	10	10	30	10	10	20	0	20
	(20)	<-	(20)	(20)	<-	(20)	(20)	<-	(20)	(20)	<->	(20)
1-1. "ABDC" 70 °	70		80	70		80	70		90	80		60
	(60)		(90)	(60)		(90)	(60)		(100)	(80)		(60)
	70		80	70		80	70		90	80		60
	(60)		(90)	(60)		(90)	(60)		(100)	(80)		(60)
Piston(e2)		12	Piston(e2)		10	Piston(e2)		7	Piston(e2)		8	
Piston rod(R1±R2)		-13	Piston rod(R1±R2)		-12	Piston rod(R1±R2)		-10	Piston rod(R1±R2)		-14	
Total		F25	Total		F22	Total		F17	Total		F22	
POS.	Cylinder No. 5			Cylinder No. 6			Cylinder No. 7					
2. "BTDC" 35 °	17	3	23	18	2	22	17	3	23			
	(20)	->	(20)	(20)	->	(20)	(20)	->	(20)	()		()
	80		70	60		80	80		90			
	(83)		(67)	(62)		(78)	(83)		(87)	()		()
1. "ABDC" 45 °	80		70	60		80	80		90			
	(83)		(67)	(62)		(78)	(83)		(87)	()		()
	30	10	10	30	10	10	30	10	10			
	(20)	<-	(20)	(20)	<-	(20)	(20)	<-	(20)	()		()
1-1. "ABDC" 70 °	60		80	70		70	80		80			
	(50)		(90)	(60)		(80)	(70)		(90)	()		()
	60		80	70		70	80		80			
	(50)		(90)	(60)		(80)	(70)		(90)	()		()
Piston(e2)		13	Piston(e2)		12	Piston(e2)		13				
Piston rod(R1±R2)		-4	Piston rod(R1±R2)		-5	Piston rod(R1±R2)		-10				
Total		F17	Total		F17	Total		F23				



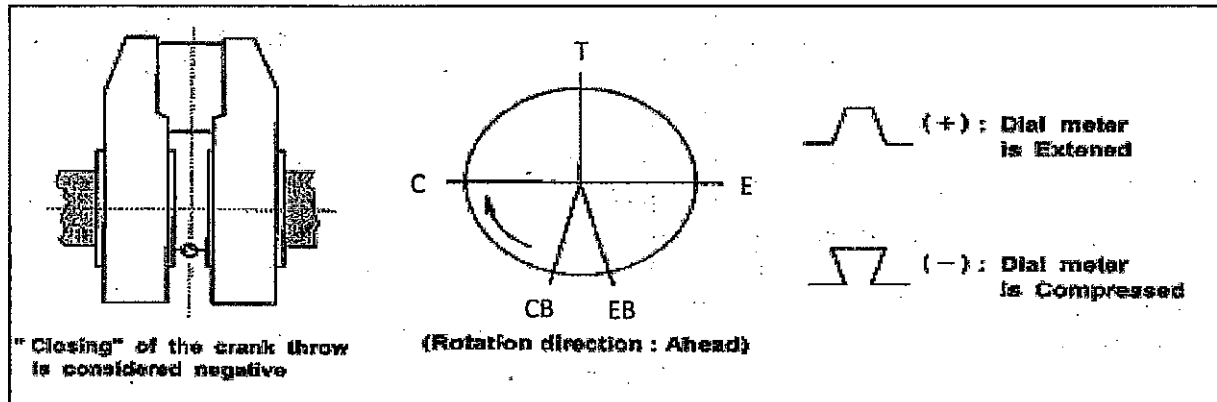
INSPECTION REPORT

Report No. STXQA-10-455-1
Page 1 OF 1

Hull No.	S-4012	Engine Type	7S50MC-C7	Unit	1/100mm
Owner	NNC	Engine No.	SB7S50-09511	Date	2010. 07. 28.
Shipyard	STX	Class	KR	Result	Acceptable

Kind of Inspection : CRANKSHAFT DEFLECTION AFTER TIGHTENING STAY BOLT

(Room Temp : 31 °C)



Spec : after most : Max 0.47, others : Max 0.18

CYLINDER NUMBER		1	2	3	4	5	6	7	-	-
POSITION	CB	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
After Tightening Stay bolt	C	3.0	0.0	1.0	-3.0	-5.0	-5.0	-3.0		
	T	7.0	6.0	6.0	0.0	-2.0	0.0	3.0		
	E	4.0	2.0	3.0	-1.0	1.0	6.0	7.0		
	EB	0.0	0.0	2.0	0.0	1.0	3.0	6.0		

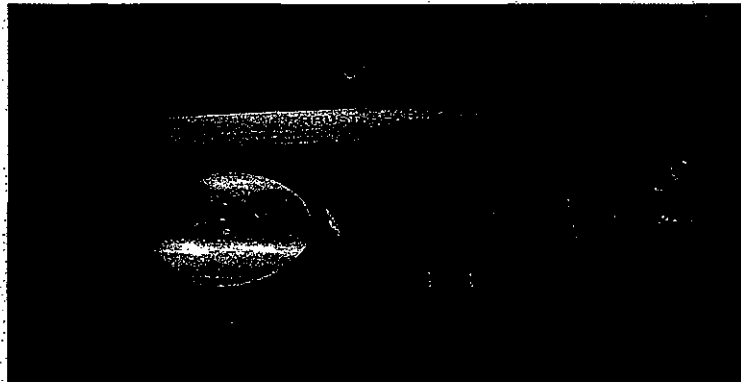
CYLINDER NUMBER		1	2	3	4	5	6	7	-	-
POSITION	CB									
After Tightening Stay bolt	C									
	T									
	E									
	EB									

Witnessed by :	Approved by : H. J. Shin
Witnessed by :	Reviewed by : K. S. Yang
Witnessed by :	Prepared by : S. H. Lee

Eng' Type : 7S50MC-C7
Hull No : S-4012
PJT No : H10B591
Date : 2010. 07. 20

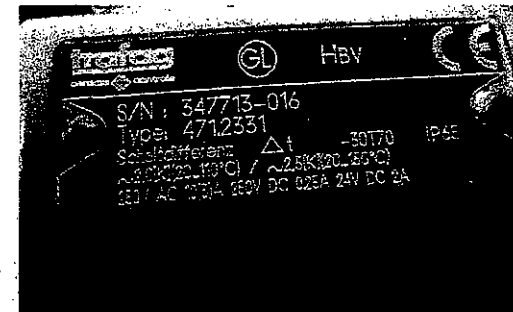
THRUST PAD TRIP TEST

stx



STEP 1

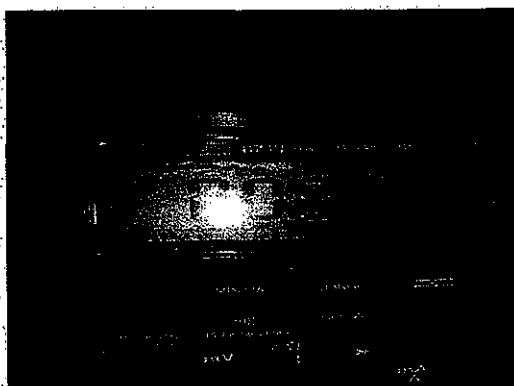
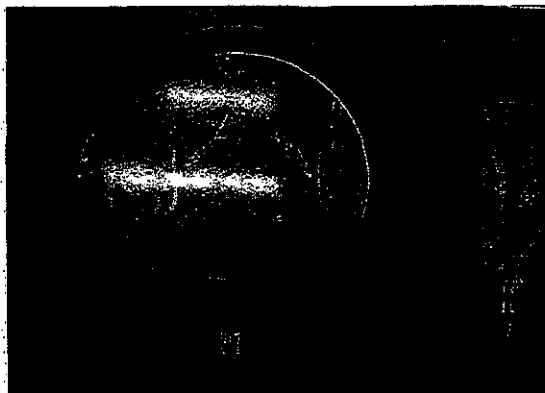
INSTALL THE THERMOMETER



STEP 2

SERIAL NO CHECK

SERIAL NO. : 347713-016
Set, Value : 90 ± 2.0 °C
Actual : 90 °C



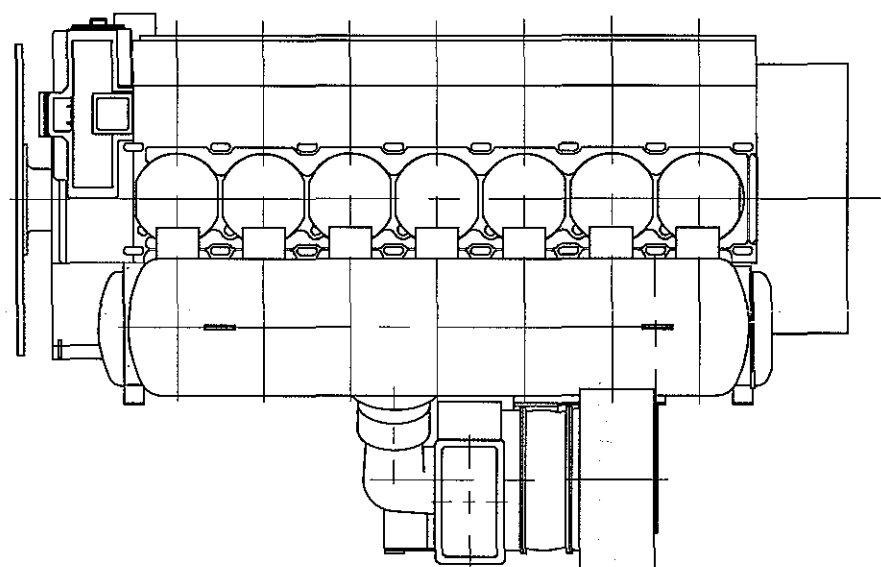
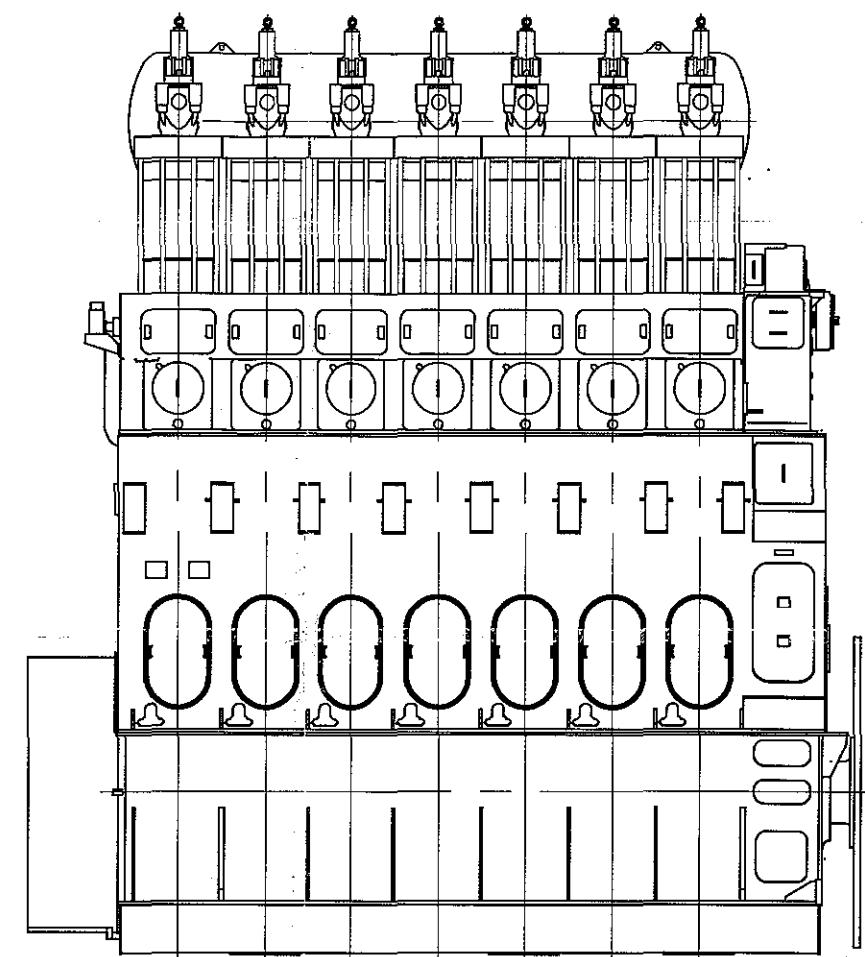
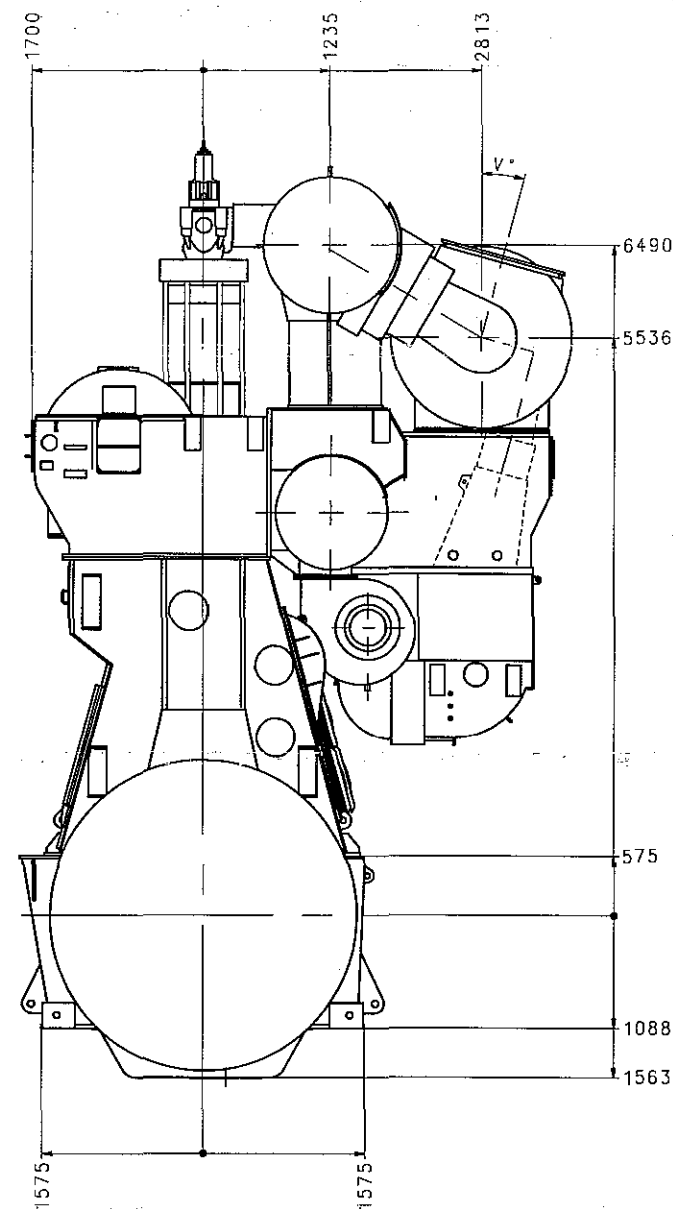
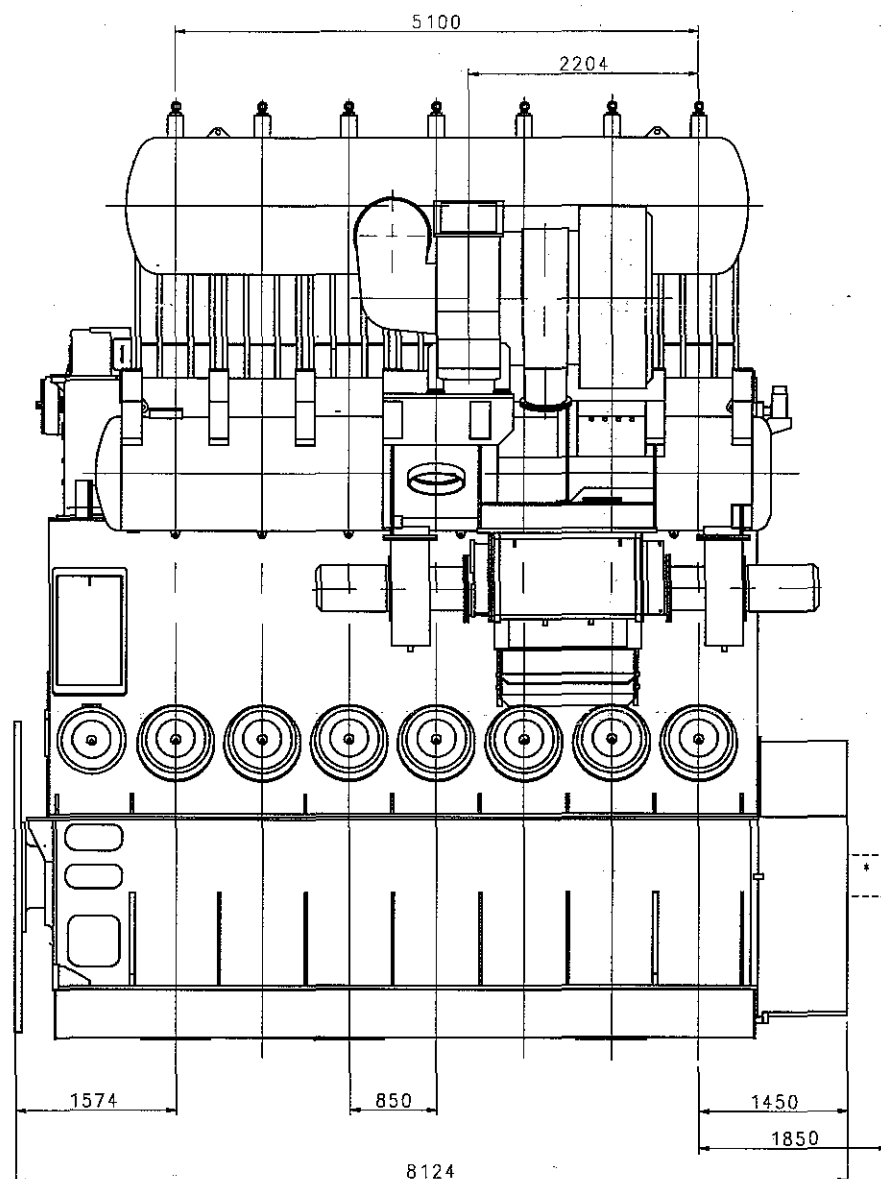
STEP 4

TRIP

Checked by: K. S. YANG

Approved by:

Signature



* Free space required for installation of equipment for measuring torsional and axial vibrations on engines without PTO
If the free space shown is not available, the shipyard should contact the engine supplier
On engines equipped with PTO, special measuring equipment is required

Please note.
The dimensions given are subject to revision without notice.
For platforms dimensions see "Gallery outline"

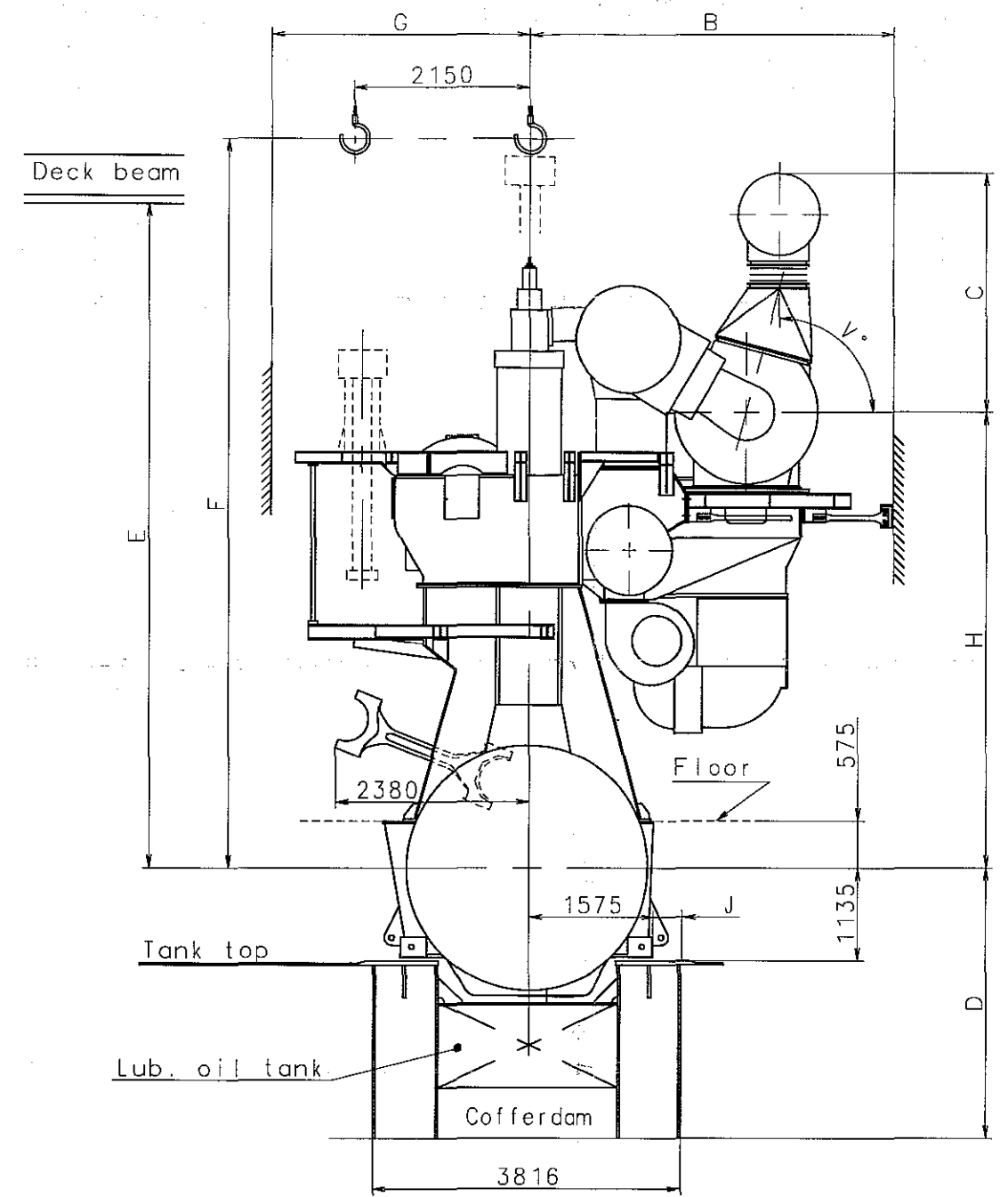
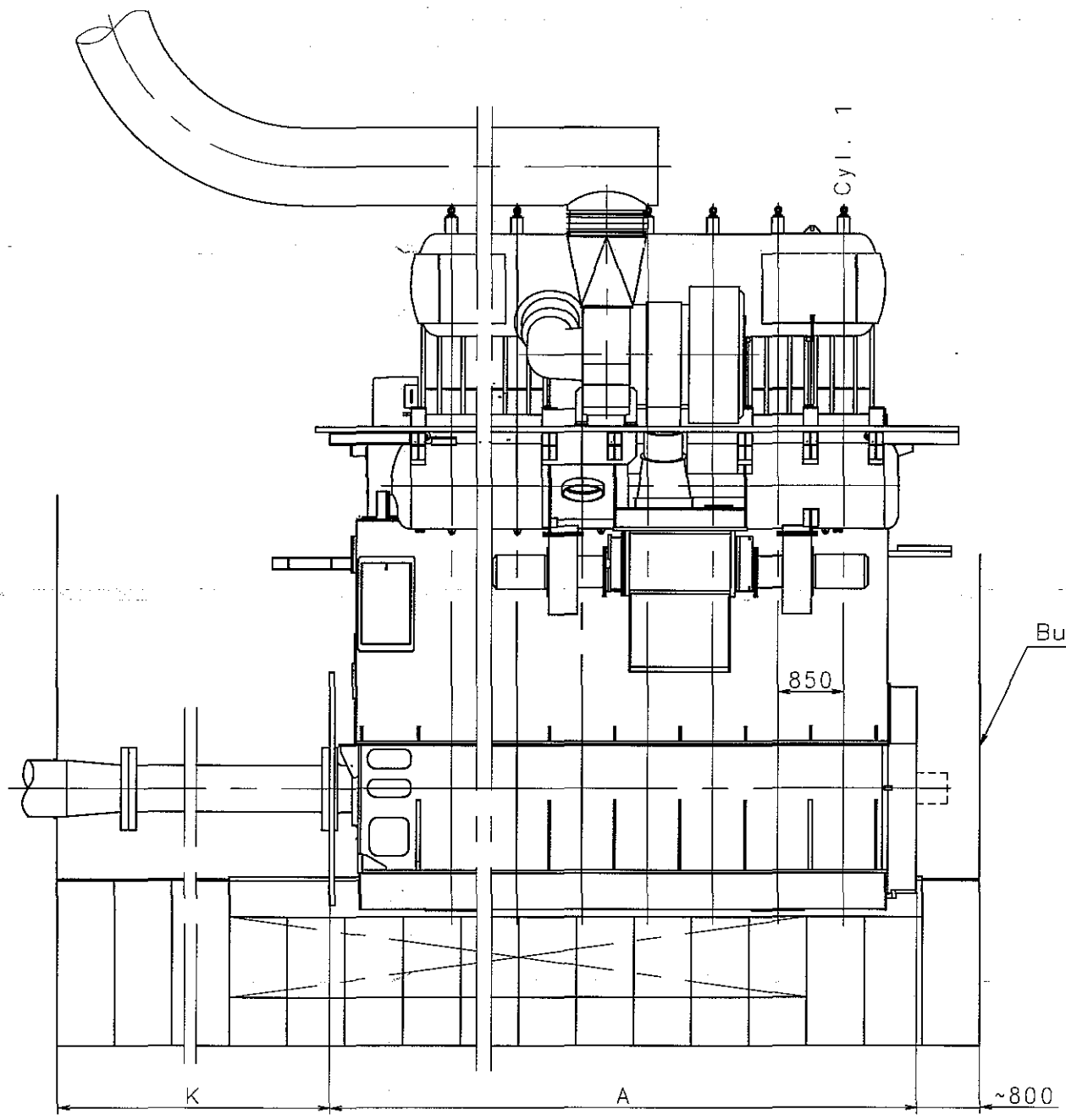
T/C TYPE : TCA66

Basic Standards (MBO SB) & Suppl. Drawing No.:		EN21C Surf. roughness		Projection: 		Material / Blank:		
1176199-7.7		EN21F-m Tolerances		Mass (kg):		Final User Material:		
Date	Des.	Chg.	Appd.	A.C.	Change / Replacement			C
				†	Replaced by Ident. No.:			
				†				
				†				
				†				
				†				
				†				
				†				
				†				
				†				
20041125JTS CSP UGJ								†
Similar Drawing No.:					Replacement for Ident. No.:			
Scale:	1:25	Size:	A0	Type:	7S50MC-C	Page No.:	01 (01)	 SOX Engine Co., Ltd.
Ident. No.:		Description:			Engine outline		Ident. No.:	
372033							2805007560	
Final User Info. No.:		Final User Description:					Final User Ident. No.:	

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V° = 15° 30° 45° 60° 75° 90



Normal centreline distance for twin engine installation: 5800 mm.

The dimensions are given in mm, and are for guidance only. If the dimensions cannot be fulfilled, please contact STX Corporation or our local representative.

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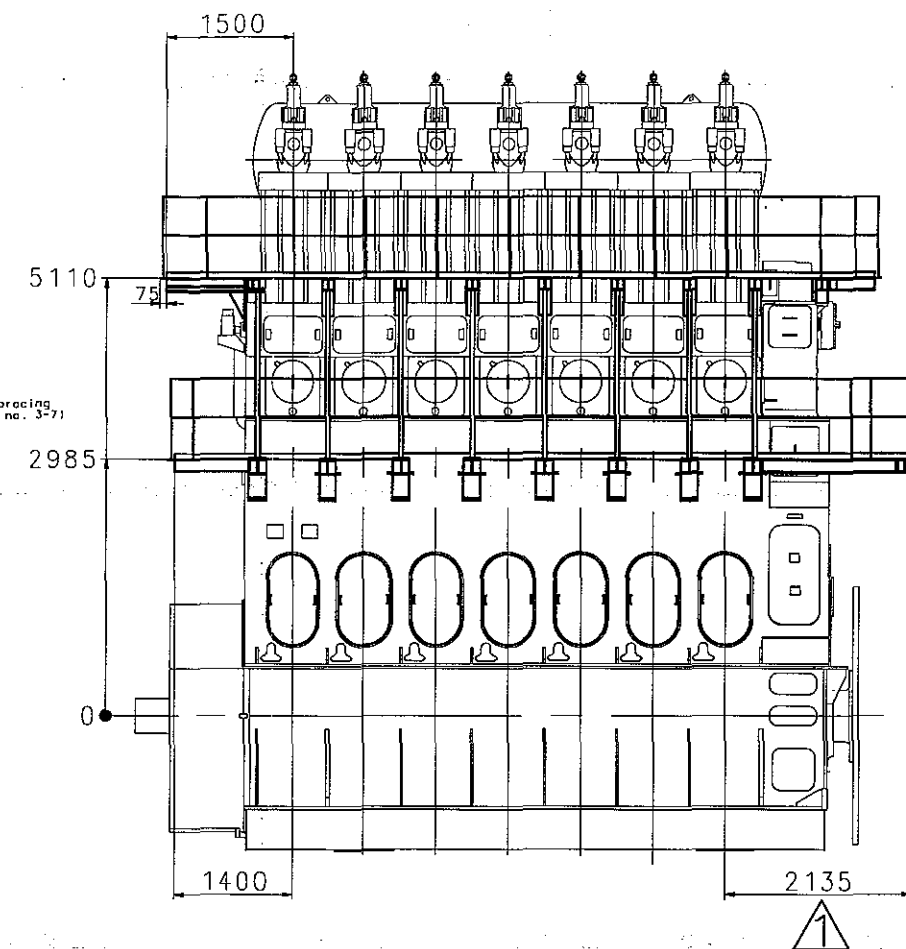
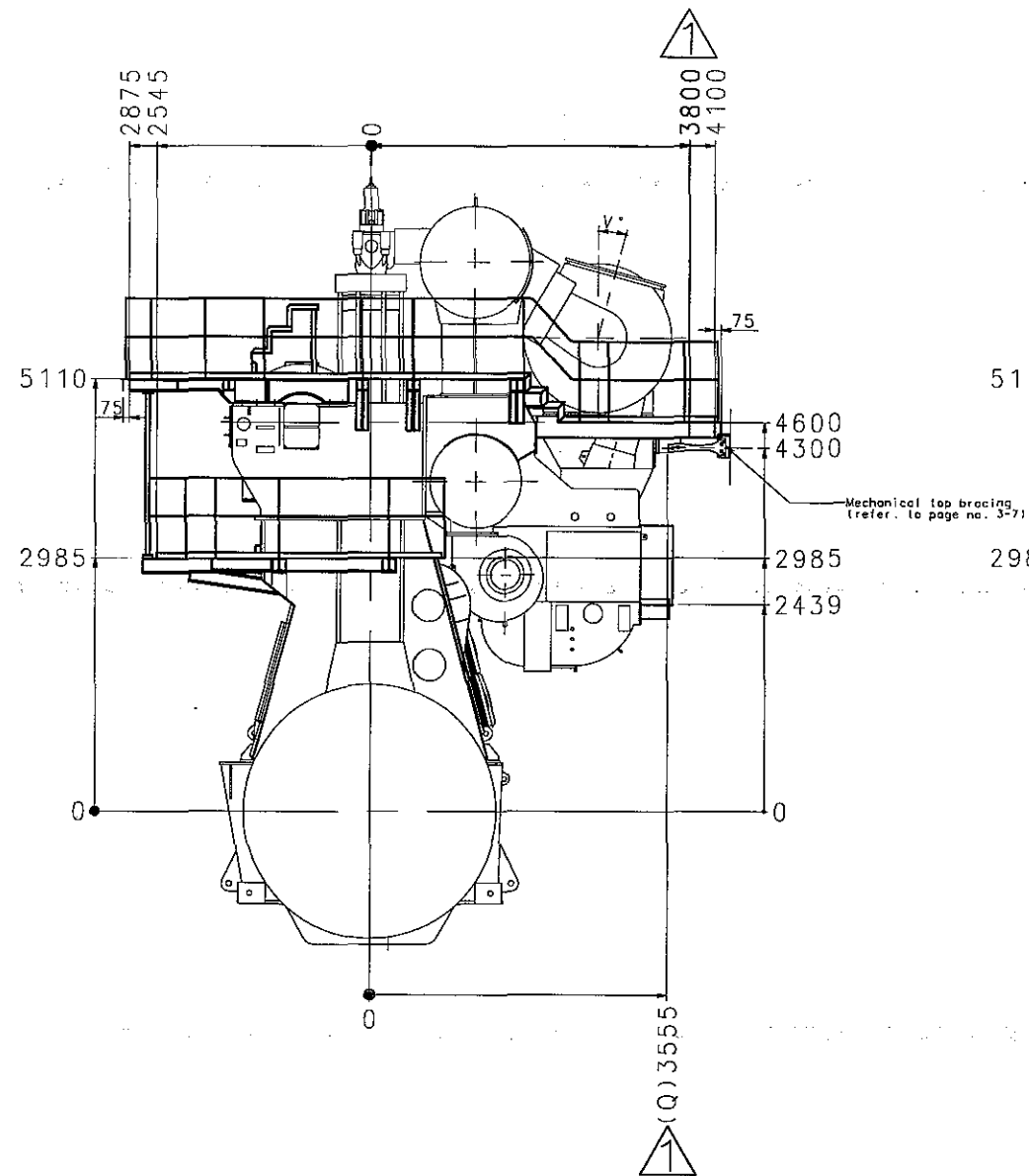
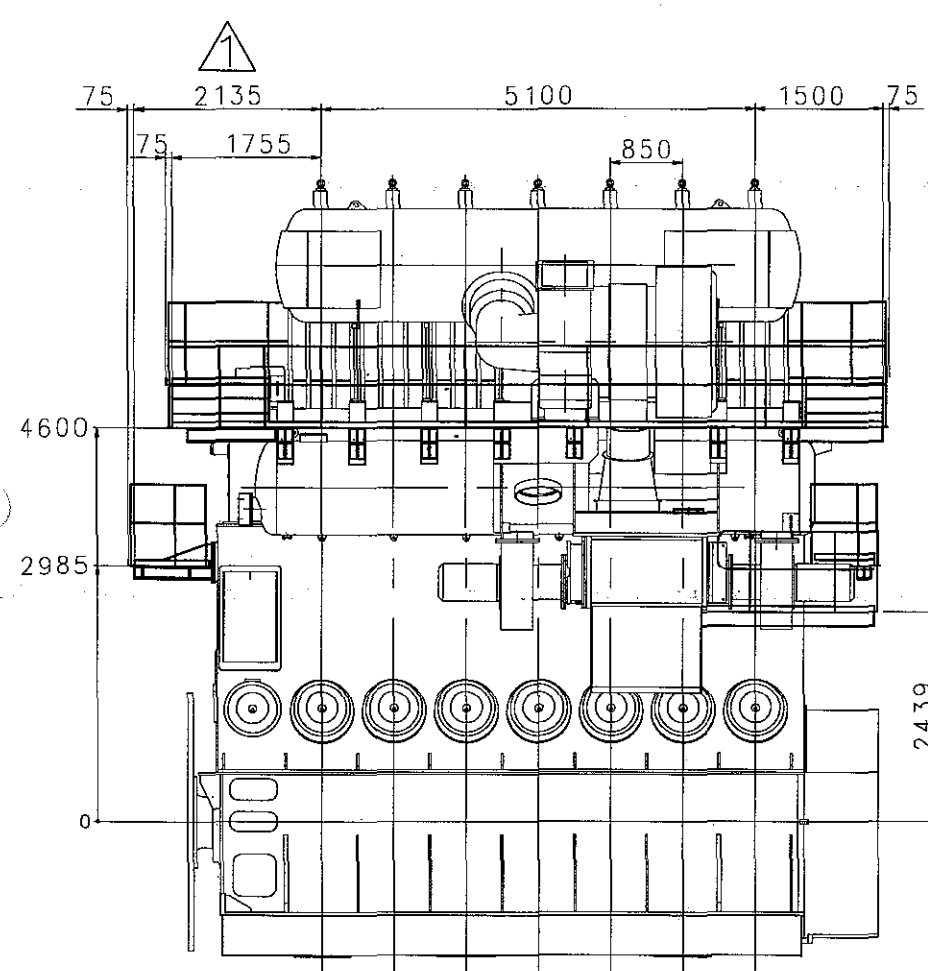
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Basic Standards (MBD SB) & Suppl. Drawing No.: 0781074-5.0		EN21C Surf. roughness		Projection:		Material / Blank:	
		EN21F-m Tolerances		Mass (kg):		Final User Material:	
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement		C.No
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							7
							6
							5
							4
							3
							2
							1
							0
20030220							
Similar Drawing No.:				Replacement for Ident. No.:			
Scale:	Size: A1	Type: S50MC-C (TC EXH.)		Page No.: 01 (02)		STX Engine Co., Ltd.	
Info. No.: 386600		Description: Engine space requirement		Ident. No.: 2805007570			
Final User Info. No.:		Final User Description:		Final User Ident. No.:			

Cyl. No.	4	5	6	7	8		
A	min.	5074	5924	6774	7624	8474	Fore end: A min. shows basic engine.
	max.	5574	6424	7274	8124	8974	A max. shows engine with built-on tuning wheel. For PTO: See corresponding space requirement.
B	4270	4270	4270	4050	-	MAN B&W TCA turbocharger	The required space to the engine room casing includes mechanical top bracing.
	4270	-	-	-	-	ABB TPL turbocharger	
	4270	4495	4495	-	-	ABB VTR turbocharger	
	4270	4270	4495	-	-	Mitsubishi turbocharger	
C	2945	3247	3347	3647	3825	MAN B&W TCA turbocharger	Dimensions according to turbocharger choice at nominal MCR.
	2816	3120	3220	3626	3626	ABB TPL turbocharger	
	2747	3074	3174	3632	3632	ABB VTR turbocharger	
	2810	3010	3245	3671	3671	Mitsubishi turbocharger	
D	3158	3198	3278	3308	3358	The dimension includes a cofferdam of 600 mm and must fulfil min. height to tank top according to classification rules.	
E	8250					The minimum distance from crankshaft centreline to lower edge of deck beam, when using MAN B&W Double Jib Crane.	
F	9000					Minimum overhaul height, normal lifting procedure.	
	8475					Minimum overhaul height, reduced height lifting procedure.	
G	3175					See "Top bracing arrangement", if top bracing fitted on camshaft side.	
H	5495	5595	5536	5595	6010	MAN B&W TCA turbocharger	Dimensions according to turbocharger choice at nominal MCR.
	5561	5561	5561	5557	5557	ABB TPL turbocharger	
	5591	5527	5527	5894	5894	ABB VTR turbocharger	
	5650	5650	5571	-	-	Mitsubishi turbocharger	
J	350					Space for tightening control of holding down bolts.	
K	See text					K must be equal to or larger than the propeller shaft, if the propeller shaft is to be drawn into the engine room.	
V	0°, 15°, 30°, 45°, 60°, 75°, 90°					Max. 30° when engine room has min. headroom above turbocharger.	

Basic Standards (M&B SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:
0781074-5.0		EN21F-m Tolerances	Mass (kg):	Final User Material:
Date	Des.	Chk.	Appd.	A.C.
Change / Replacement				C.No.
Replaced by Ident. No.:				9
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				2
				1
				0
20041125				Replacement for Ident. No.:
Scale:	A1	Type:	S50MC-C (TC EXH.)	Page No.:
Info. No.:	386600	Description:	Engine space requirement	Ident. No.:
Final User Info. No.:		Final User Description:		Final User Ident. No.:
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Note

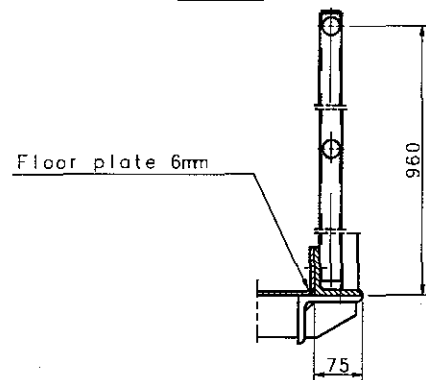
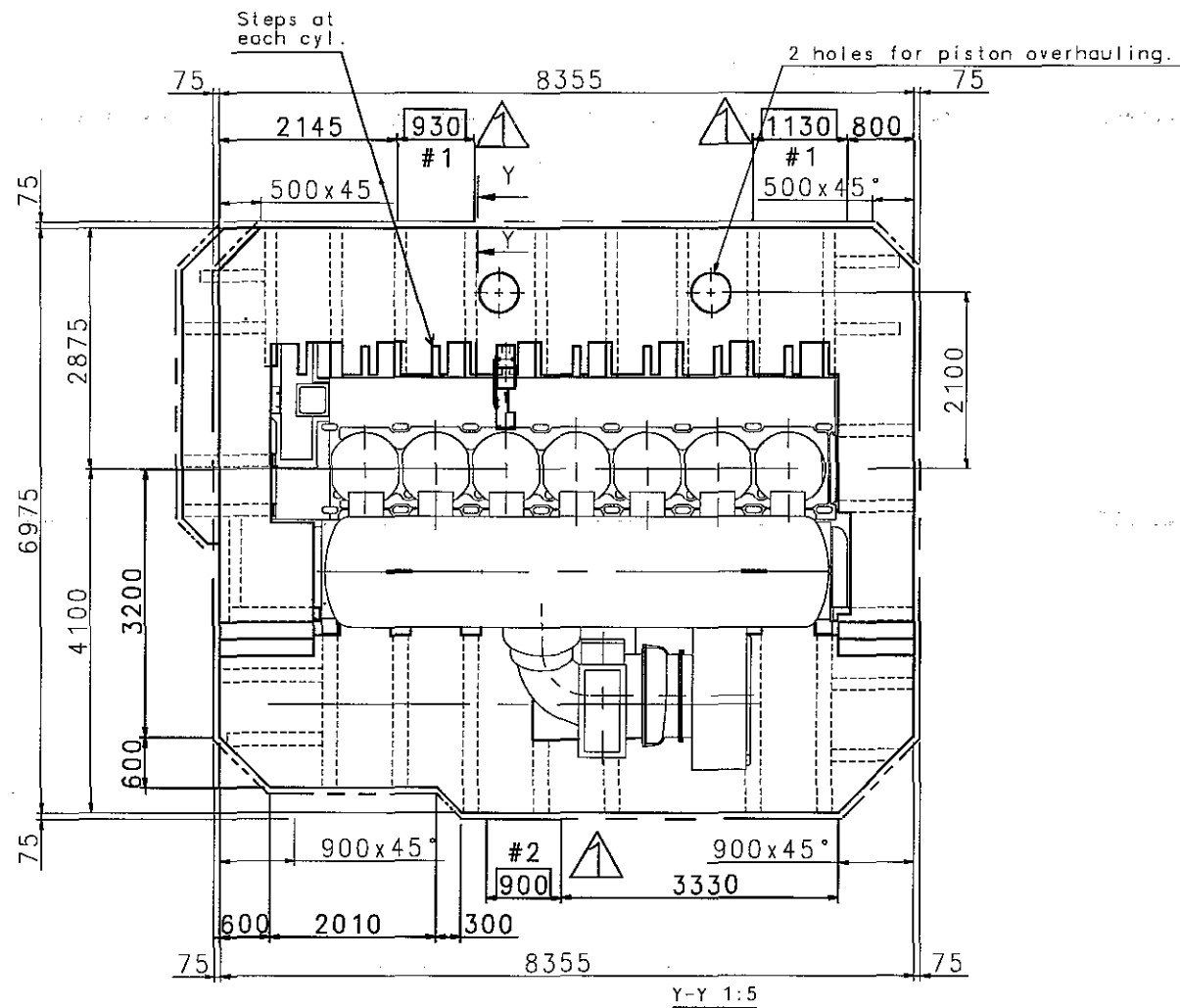
1. The dimensions are in mm and drawing to revision without notice.
2. For engine dimensions see "Engine outline"
3. The type of T/C : TCA66, prepared 2nd order moment compasator on fore. end.
4. The angle "V" = 15°

V° = 0° 15° 30° 45° 60° 75° 90°

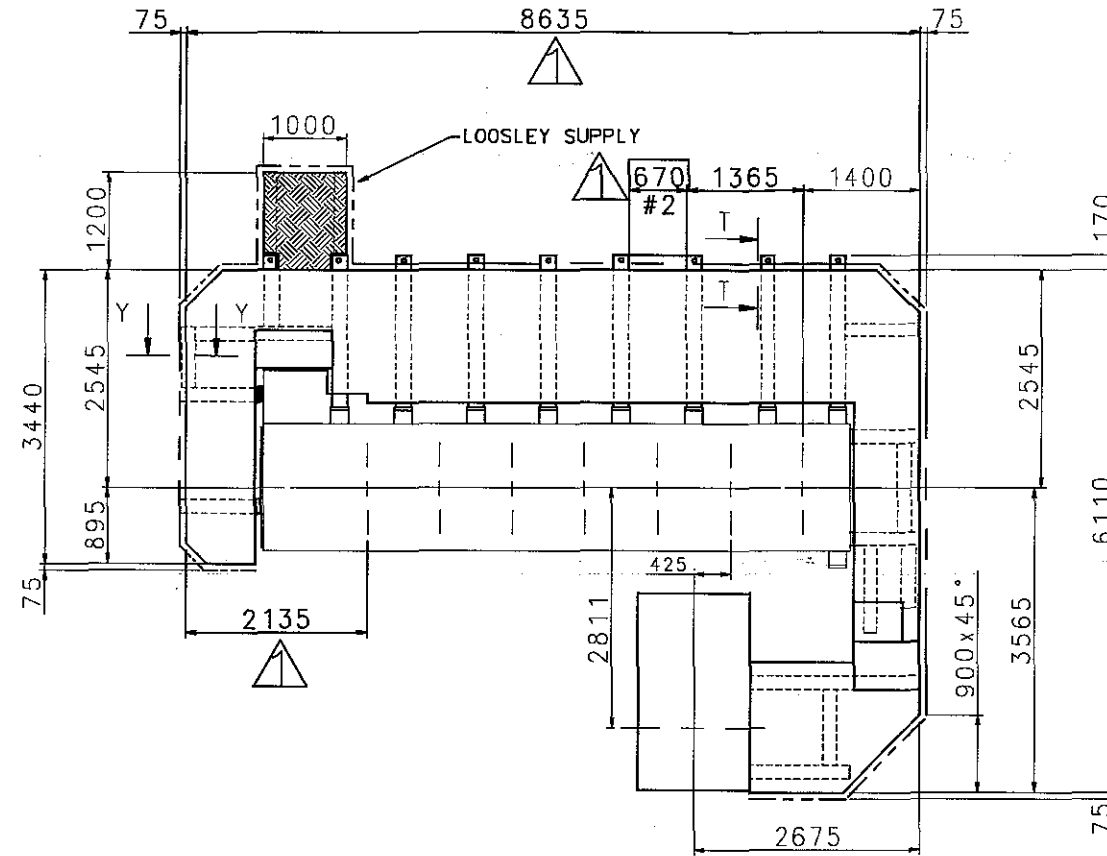
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Basic Standards (MED SB) & Suppl. Drawing No.: 1176241-6.8		EN21C Surf. roughness	Projection: 1st angle	Material / Blank:
		EN21F-m Tolerances	Most (kg):	Final User Material:
Date	Dep.	Chk.	Appd.	A.C.
Change / Replacement				C. No.
Replaced by Ident. No.:				9
				8
				7
				6
				5
				4
				3
				2
				1
				0
20090910 SMLJTSTBK		Yard comment applied		
20090205 SMLJTSTBK		First issue		
Similar Drawing No.:		Replacement for Ident. No.:		
Scale:	1:25	Size:	A0	Type:
7S50MC-C + TCA66		01 (02)		Page No.:
369871		GALLERY OUTLINE		Ident. No.:
Final User Info. No.:		Final User Description:		Final User Ident. No.:
				2-3-1

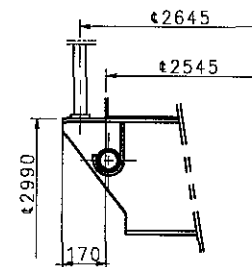
Upper platform



Lower platform



T-T 1:10



#1 : Coaming up
#2 : Coaming down
[] : Handrail cutting

Note

1. The dimensions are in mm and drawing to revision without notice.
2. For engine dimensions see 'Engine outline'

Basic Standards (MMD SD) & Suppl. Drawing No.:		EN21C Surf. roughness		Projection:		Material / Blank:	
1176241-6.8		EN21F-m Tolerances		Mass (kg):		Final User Material:	
Date	Des.	CHK.	Appd.	A.C.	Change / Replacement		
				†	Replaced by Ident. No.:		
					9		
					8		
					7		
					6		
					5		
					4		
					3		
					2		
20090910	SML JTSTBK			†	Yard comment applied		
20090205	SML JTSTBK			†	First issue		
Star/for Drawing No.:				Replacement for Ident. No.:			
Scale:	Size:	Type:			Page No.:		
1:25	A0	7550MC-C + TCA66			02 (02)	SOX Engine Co.Ltd.	
Ident. No.:		Description:				Ident. No.:	
369871		GALLERY OUTLINE				2005029420.1A	
Final Desr Info. No.:		Final User Description:				Final User Ident. No.:	
						2-3-2	

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AFT

DEAD CENTER POINTER

FORE

TURNING WHEEL

4xM12-25
TURNING WHEEL COVER
MOUNTING PLATE

VIEW "A"

VIEW "B"

* Pick-ups Arrangement

	A	B	C
LYNGSO	Control System	Safety System	Governor System
NABTESCO	Control System	Governor System	Safety System
KONGSBERG	Control System	Governor System	Safety System

GENERAL TOLERANCE FOR MACHINED MEASURES				REV	LOC	DATE	MADE	APPD	NOTE	
NOMINAL MEASURE		TOLERANCES		MATERIAL		SCALE		SIZE		MODEL
OVER	MAX.			N/S		N/S		A1		SSOMC-C
1	4	±0.1		DWN	20060824	H.C. YUN				TURNING GEAR ARRANGEMENT
4	16	±0.2		CHKD	20060824	C.S. PARK				
16	63	±0.3		APPD	20060824	J.S. PARK				
63	250	±0.5								
250	1000	±0.8								

Class No.	N4	N5	N6	N7	N8	N9
Ra Max. Value(μm)	0.2	0.4	0.6	0.8	1.6	3.2

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STX Engine Co., Ltd. 2805014880



유림정밀공업

PURCHASE ORDER SPECIFICATION

POS NO	YRTG 20-27
REV NO	0
PAGE	1 OF 11

MAIN ENGINE TURNING GEAR

YURIM TYPE : YRTG 20-27 (50MC-C)

FOR APPROVAL

NO	DIVISION	DRAWING NO.	ETC.
1	PARTS LIST	YU50MC-C-PLST	
2	ASSEMBLY	YU50MC-C-001	
3	ASSEMBLY	YU50MC-C-002	
4	ASSEMBLY	YU50MC-C-003	
5	SPECIFICATION TURNING GEAR REDUCER		
6	GEAR WHEEL	YU50MC-C-43	
7	SPARE PARTS LIST FOR REDUCTION GEAR	YU50MC-C-SPLST1	
8	INDUCTION MOTOR (2.2kW,440V,60Hz)	YU50MC-C-57	SPACE HEATER
9	DATA SHEET FOR MOTOR		
	SPARE PARTS LIST FOR MOTOR	YU50MC-C-SPLST2	

1	2009.01.14	CHANGED SLOT HOLE			
0	2008.07.17	FIRST ISSUE			
REV NO	DATE	DESCRIPTION	PREP'D BY	CHECKED BY	APPROVED BY



YURIM PRECISION INDUSTRIAL CO.

36-5, SANKYOK-DONG, BUK-GU, DAEGU, KOREA

TEL : (053)384-4894

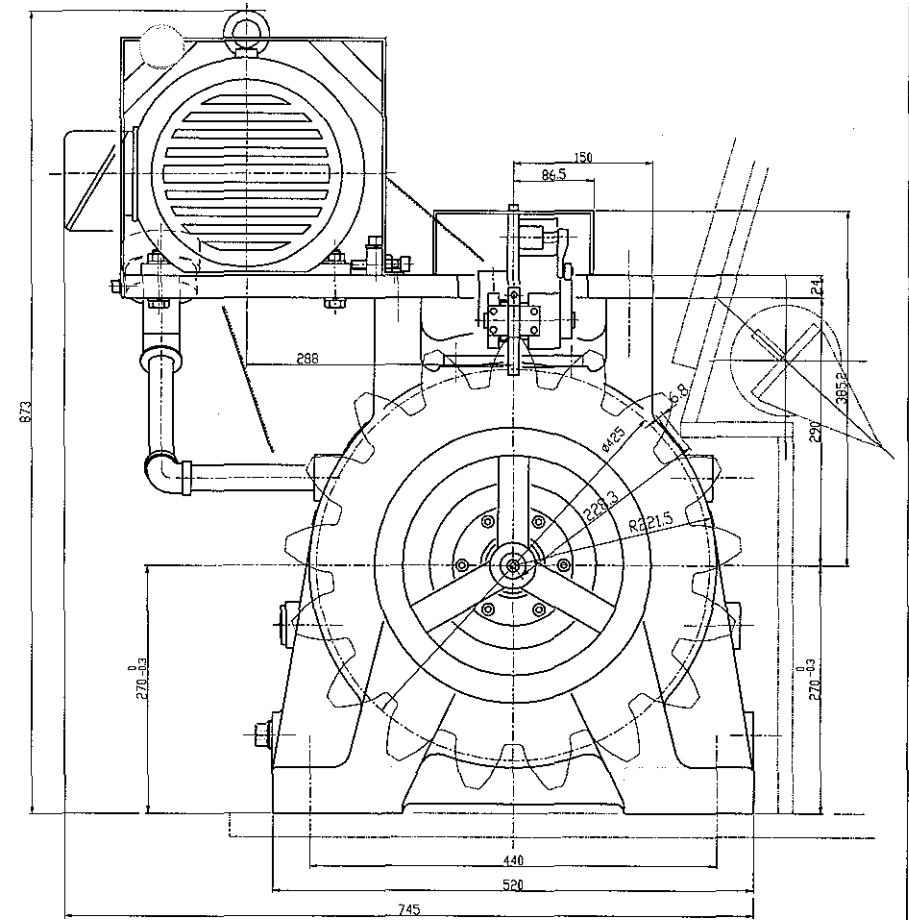
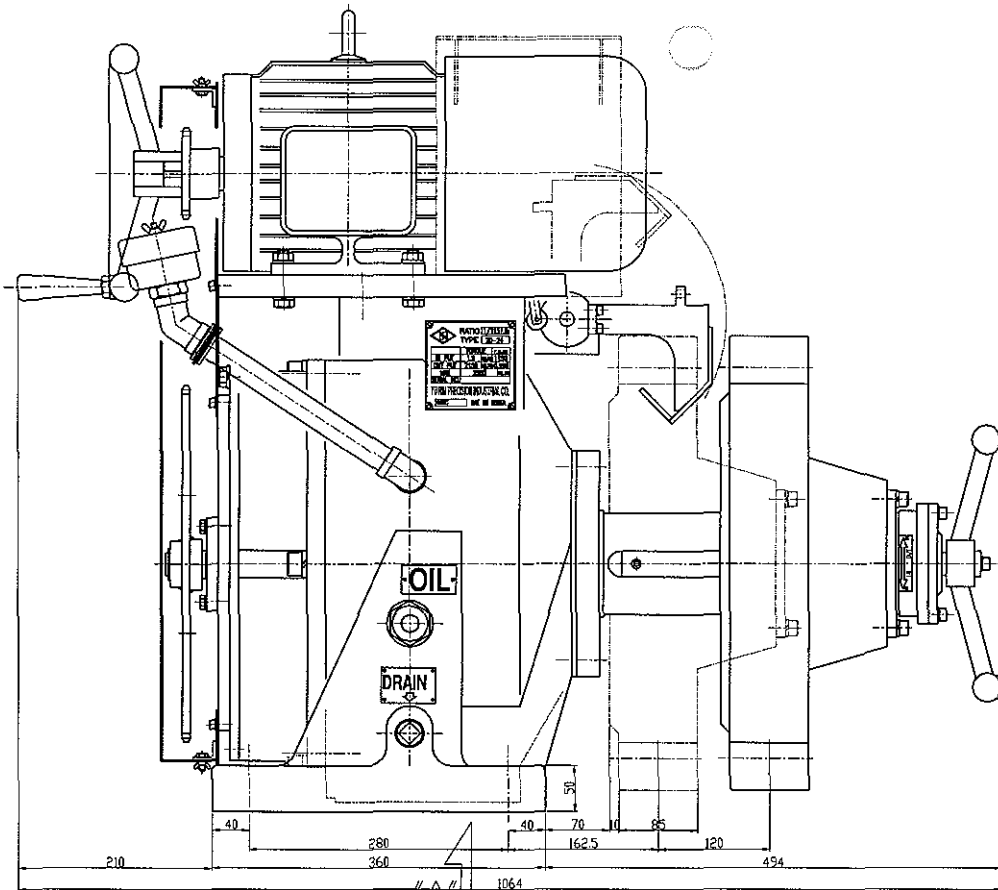
FAX : (053)384-0030

271	HEX'SOCKET BOLT,\$W	8.8T	6		M8X20L		YU- 50 S-27
263	STOP RING	PUR	1		R #47		YU- 50 S-26
262	HEX'SOCKET BOLT,\$W	8.8T	6		M8X35L		YU- 50 S-26
261	BALL BEARING	PUR	1		#6005 ZZ		YU- 50 S-26
251	HEX'SOCKET BOLT,\$W	8.8T	4		M8X20L		YU- 50 S-25
241	HEX'SOCKET BOLT,\$W	8.8T	2		M6X15L		YU- 50 S-24
231	HEX'SOCKET BOLT	8.8T	6		M8X15L		YU- 50 S-24
222	PIN	S45C-H	4		ø12X40L(M6 TAP)		YU- 50 S-22
221	SP'L R/BEARING	PUR	1		#23024		YU- 50 S-22
211	HEX'SOCKET BOLT,\$W	8.8T	8		M12X80L		YU- 50 S-21
202	PIN	S45C-H	4		ø16X60L		YU- 50 S-20
201	HEX'HEAD BOLT	8.8T	10		M12X85L(S=50)		YU- 50 S-20
171	NEEDLE BEARING	PUR	8		NA #5909		YU- 50 S-16
162	PIN	S45C-H	4		ø10X30L		YU- 50 S-16
161	HEX'SOCKET BOLT,\$W	8.8T	8		M10X40L		YU- 50 S-16
151	NEEDLE BEARING	PUR	4		TAFI #304530		YU- 50 S-15
131	STOP RING	SK-5	4		S #30		YU- 50 S-13
124	DU BUSH	PUR	2		MB 3030 DU		YU- 50 S-12
123	PIN	S45C-H	6		ø10X45L		YU- 50 S-12
122	SET SCREW	PUR	6		M8X15L		YU- 50 S-12
121	SET SCREW	PUR	4		M10X15L		YU- 50 S-12
112	PIN	S45C-H	4		ø8X30L		YU- 50 S-11
111	HEX'SOCKET BOLT,\$W	8.8T	8		M8X40L		YU- 50 S-11
101	NEEDLE BEARING	PUR	3		NA #4905		YU- 50 S-10
081	STOP RING	SK-5	3		S #25		YU- 50 S-08
074	DU BUSH	PUR	2		MB 3030 DU		YU- 50 S-07
073	PIN	S45C-H	4		ø6X25L		YU- 50 S-07
072	SET SCREW	PUR	4		M6X10L		YU- 50 S-07
071	SET SCREW	PUR	3		M8X15L		YU- 50 S-07
064	STOP RING	SK5-H	1		S #25		YU- 50 S-06
063	KEY (원반)	S45C	1		8X7X35L		YU- 50 S-06
062	BALL BEARING	PUR	1		#6305		YU- 50 S-06
061	BALL BEARING	PUR	1		#6206		YU- 50 S-06
052	OIL SEAL	PUR	1		D 30458		YU- 50 S-05
051	HEX'HEAD BOLT,\$W	8.8T	6		M8X25L		YU- 50 S-05
042	OIL SEAL	PUR	1		D 11514514		YU- 50 S-04
041	HEX'SOCKET BOLT	8.8T	6		M12X30L		YU- 50 S-04
031	HEX'HEAD BOLT,\$W	8.8T	8		M12X30L		YU- 50 S-03
023	SPRING PIN	SK-5	4		ø6X30L		YU- 50 S-02
022	BALL BEARING	PUR	1		#6224		YU- 50 S-02
021	HEX'SOCKET BOLT,\$W	8.8T	8		M16X40L		YU- 50 S-02
013	DRAIN PLUG	SS400	1		3/4"		YU- 50 S-01
012	OIL GAUGE	PUR	1		PF 1" ROUND		YU- 50 S-01
011	AIR BREATHER	PUR	1		3/4" STEEL		YU- 50 S-01

40	CHAIN COVER(A)	SPHC	1				YU- 50 S-40
39	CHAIN SPROCKET	S45C	1				YU- 50 S-39
38	SPACE	BC3	3				YU- 50 S-38
37	CHAIN SPROCKET	S45C	1				YU- 50 S-37
36	TENSION STOPPER	SS400	1				YU- 50 S-36
35	HAND WHEEL	PUR	1				YU- 50 S-35
34	MOTOR BASE	SS400	1				YU- 50 S-34
33	MAGNET	PUR	1				YU- 50 S-33
32	MAGNET SEAT	MC NYLON	2				YU- 50 S-32
31	HANDLE COVER	SS41	1				YU- 50 S-31
30	SCREW	S45C-H	1				YU- 50 S-30
29	NUT HOLDER	SS400	1				YU- 50 S-29
28	HAND WHEEL	PUR	1				YU- 50 S-28
27	BEARING COVER	SS400	1				YU- 50 S-27
26	SCREW CASE	SS400	1				YU- 50 S-26
25	SCREW NUT	S45C-H	1				YU- 50 S-25
24	L/S COVER	SS400	1				YU- 50 S-24
23	SLIDING KEY	S45C	2				YU- 50 S-23
22	OUT PUT SHAFT	S45C-H	1				YU- 50 S-22
21	CARRIER	FCD450	1			3 RD	YU- 50 S-21
20	INTERNAL GEAR	S45C-H	1			3 RD.	YU- 50 S-20
19	COLLAR	S45C	8			3 RD	YU- 50 S-19
18	IDLE GEAR PIN	SCM440	4			3 RD.	YU- 50 S-18
17	IDLE GEAR	SCM440	4			3 RD.	YU- 50 S-17
16	INTERNAL GEAR	S45C-H	1			2 ND.	YU- 50 S-16
15	IDLE GEAR	SCM440	4			2 ND.	YU- 50 S-15
14	COLLAR	S45C	8			2 ND.	YU- 50 S-14
13	IDLE GEAR PIN	SCM440	4			2 ND.	YU- 50 S-13
12	CARRIER	S45C-H/SCM440	1			2 ND.	YU- 50 S-12
11	INTERNAL GEAR	S45C-H	1			1 ST.	YU- 50 S-11
10	IDLE GEAR	SCM440	3			1 ST.	YU- 50 S-10
09	COLLAR	S45C	6			1 ST.	YU- 50 S-09
08	IDLE GEAR PIN	SCM440	3			1 ST.	YU- 50 S-08
07	CARRIER	S45C-H/SCM440	1			1 ST.	YU- 50 S-07
06	SUN GEAR	SCM440	1			1 ST.	YU- 50 S-06
05	BEARING COVER	SS400	1				YU- 50 S-05
04	BEARING COVER	SS400	1				YU- 50 S-04
03	BEARING HOUSING	FCD450	1				YU- 50 S-03
02	GEAR HOUSING	FCD450	1				YU- 50 S-02
01	MAIN CASE	FCD450	1				YU- 50 S-01
003	ASSEMBLY						YU- 50 S-003
002	ASSEMBLY						YU- 50 S-002
001	ASSEMBLY						YU- 50 S-001

	GEAR OIL	PUR					
621	ROUNDHEAD SCREW	PUR	12		M4X8L		YU- 50 S-58
581	HEX'SOCKET BOLT,\$/W,P/W		4		M8X15L		YU- 50 S-57
531	HEX'SOCKET BOLT,\$W	8.8T	2		M6X40L		YU- 50 S-53
521	CHAIN	PUR	150	mm			YU- 50 S-52
461	HEX'SOCKET BOLT	8.8T	4		M6X20L		YU- 50 S-46
451	HEX'SOCKET BOLT,\$W	8.8T	4		M6X20L		YU- 50 S-45
442	SPRING PIN	SK-5	2		ø6X25L		YU- 50 S-44
441	HEX'HEAD BOLT,\$W	8.8T	4		M12X30L		YU- 50 S-44
422	FLY NUT	PUR	6		M6		YU- 50 S-42
421	HEX'HEAD BOLT	8.8T	6		M6X15L		YU- 50 S-42
411	HEX'SOCKET BOLT,\$/W,P/W		8		M6X15L		YU- 50 S-41
392	SET SCREW	PUR	2		M8X12L		YU- 50 S-39
391	CHAIN	SK-5	2.1	M	RS 50		YU- 50 S-39
371	SET SCREW	PUR	2		M6X15L		YU- 50 S-37
362	HEX'SOCKET BOLT	8.8T	2		M10X35L		YU- 50 S-36
361	HEX'SOCKET BOLT,NUT	8.8T	2		M8X60L		YU- 50 S-36
343	EYE BOLT	PUR	2		M16		YU- 50 S-34
342	HEX'SOCKET BOLT	8.8T	6		M16X30L		YU- 50 S-34
321	HEX'SOCKET BOLT	8.8T	1		M6X20L		YU- 50 S-32
311	HEX'HEAD BOLT,\$W	8.8T	1		M8X15L		YU- 50 S-31
301	KEY	S45C	1		6X6X30L		YU- 50 S-30
291	HEX'SOCKET BOLT,\$W	8.8T	6		M10X25L		YU- 50 S-29
273	STOP RING	PUR	1		R #47		YU- 50 S-27
272	BALL BEARING	PUR	1		#6005 ZZ		YU- 50 S-27

NO.	NAME	MATERIAL	Q'TY	W'HT	SIZE	REMARKS	DWG.NO	NO.	NAME	MATERIAL	Q'TY	W'HT	SIZE	REMARKS	DWG.NO	NO.	NAME	MATERIAL	Q'TY	W'HT	SIZE	REMARKS	DWG.NO
PROJECT TITLE CUSTOMER DATE  YURIM PRECISION INDUSTRIAL CO. TURNING GEAR WITH MOTOR YRTG 20-27 STX 05.07.06 DWG.NO APPROVED BY CHECKED BY DESIGNED BY DRAWN BY LIST NO YU-50MC-C-PLST																							

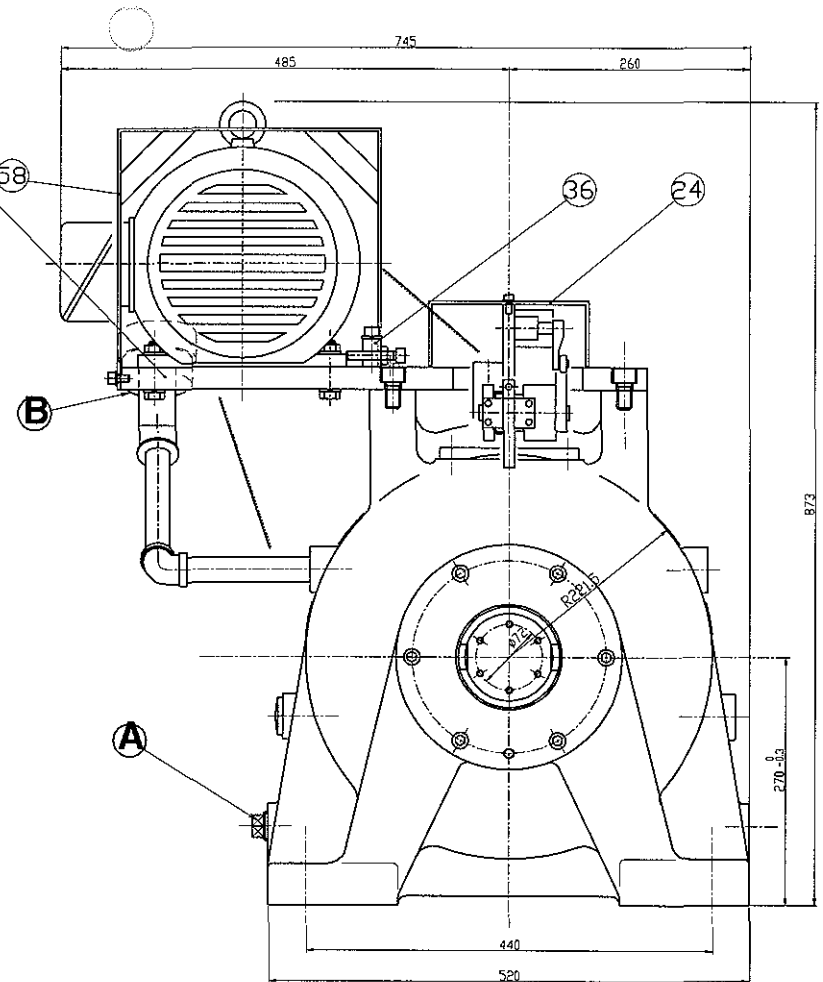
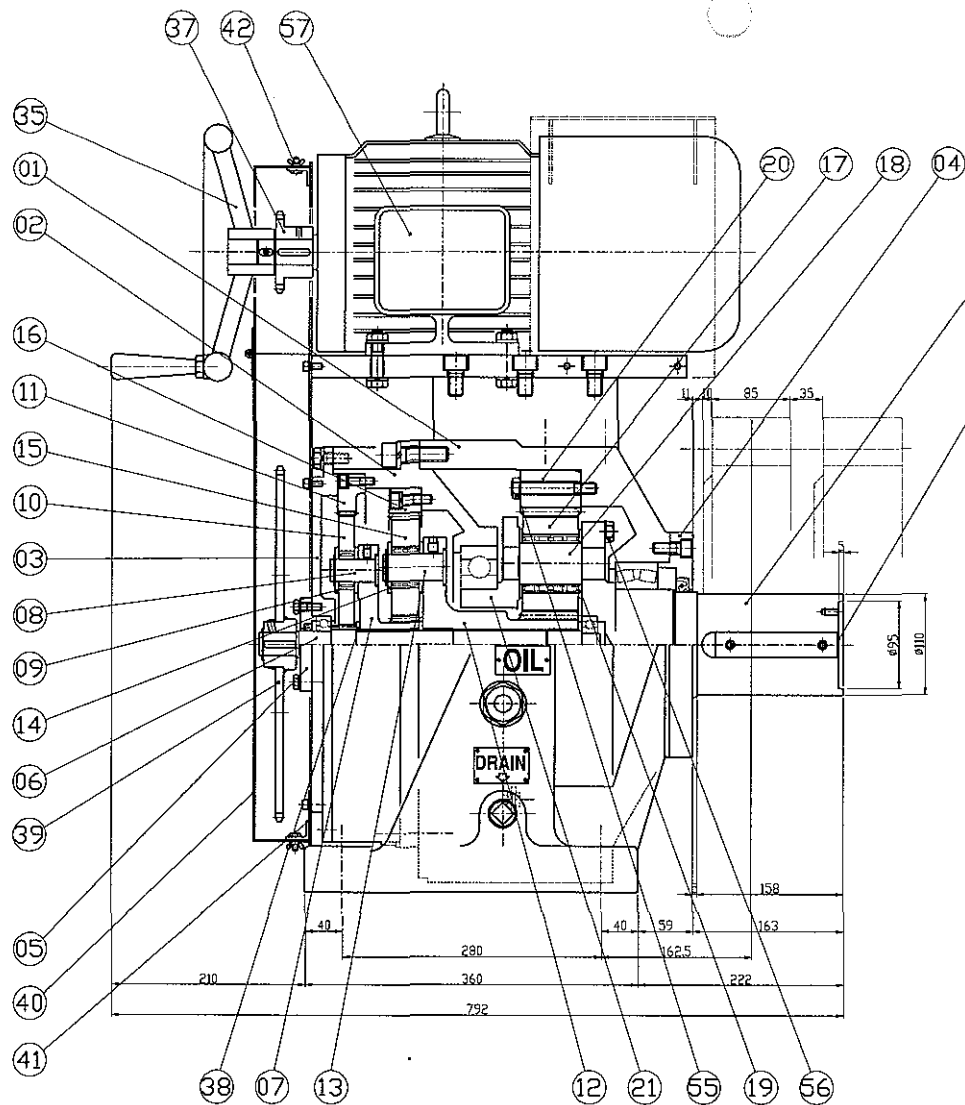


	RATIO 1/1151.8
	TYPE 20-27
IN PUT	1.9 kg.m 950
OUT PUT	2150 kg.m 0.825
MAX	2580 kg.m
SERIAL NO.	
DATE	DAE GU KOREA

LUBRICATION OIL SPECIFICATION

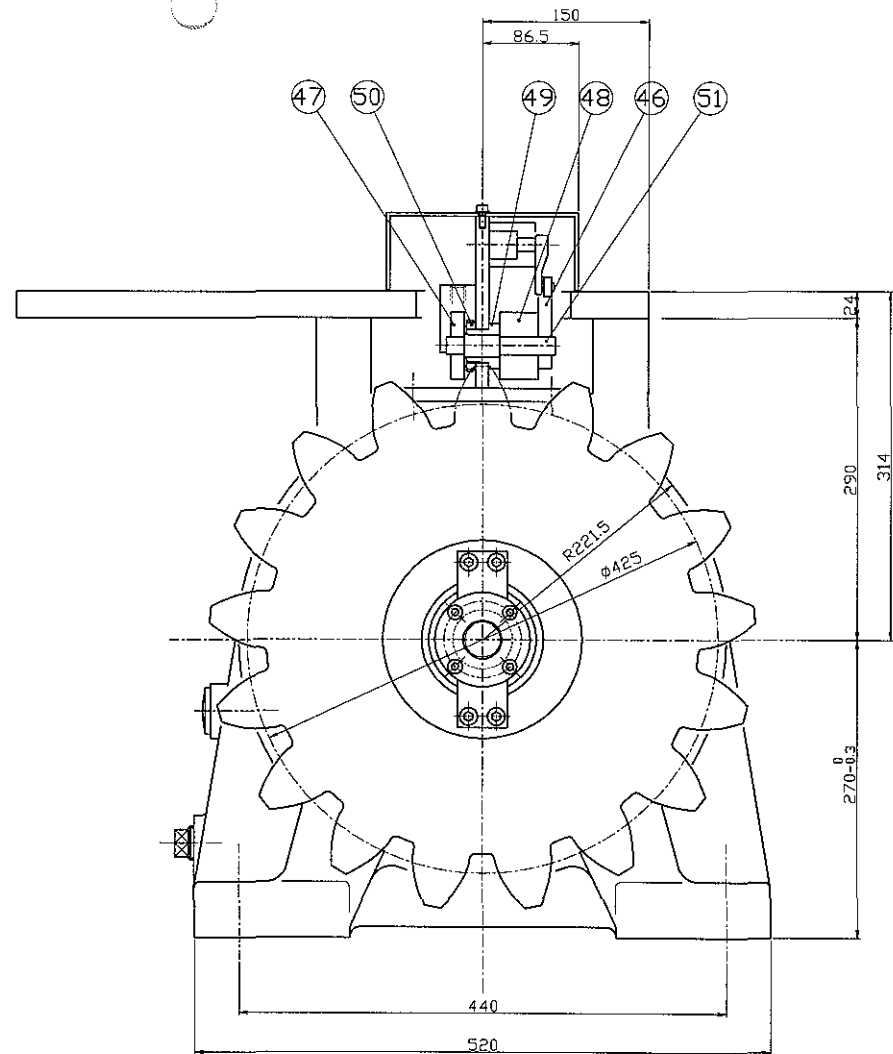
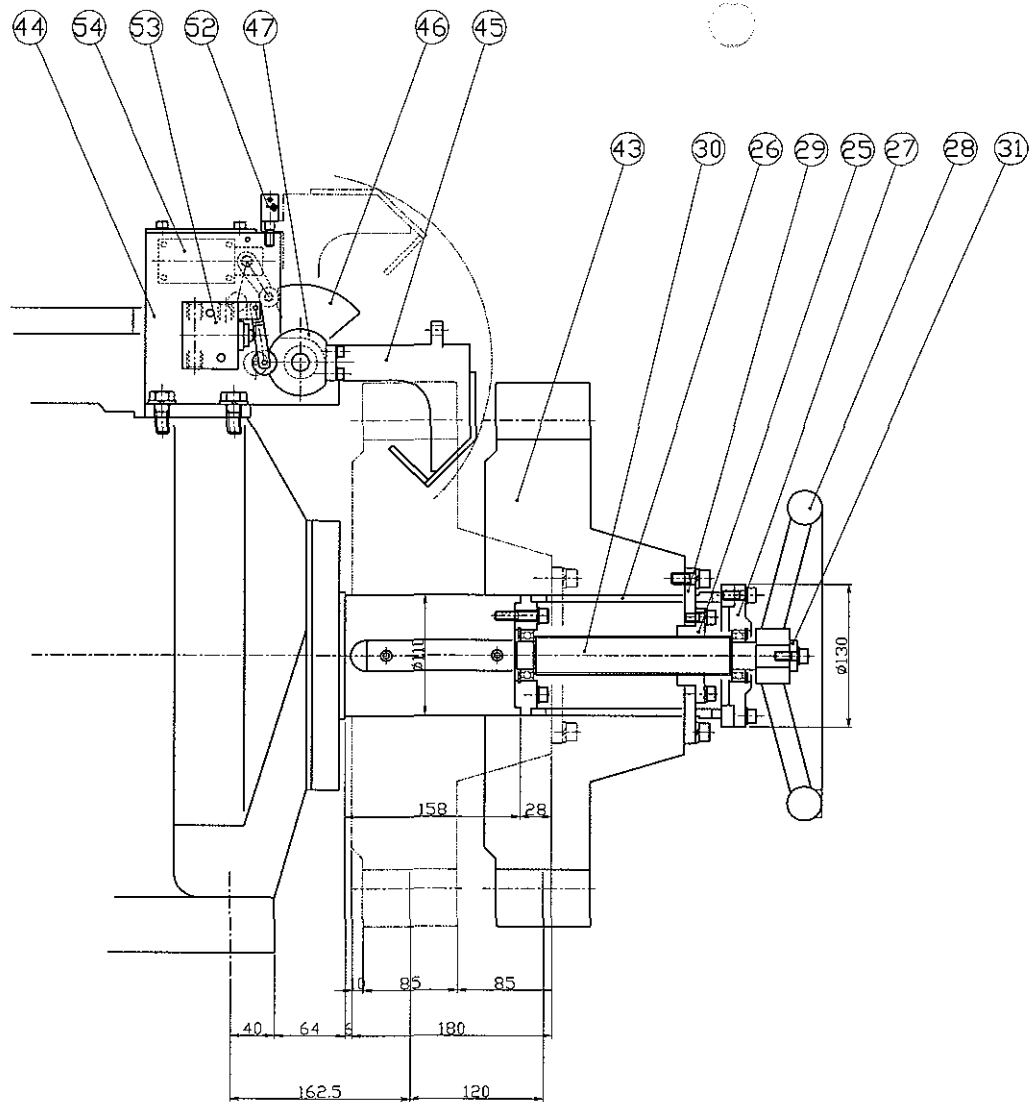
MAKER	LUBRICATION OIL	REMARK
CALTEX	MEROPA 460	1. US STEEL 222,224
YUKONG (SK)	SUPER GEAR EP 460	meet standard requirements
ESSO	SPARTAN EP 460	2. AGMA 250.04
GULF	EP LUBE H.D	meet standard requirements
SHELL OIL	OMALA 460	3. DIN 51517 PART-3
		meet standard requirements
		4. CINCINNATI- MILACRON
		P-59 meet standard requirements

001	ASSEMBLY						
NO.	NAME		MATERIAL	Q'TY	W'HT	SIZE	REMARKS
PROJECT						APPROVED BY	
TITLE		TURNING GEAR WITH MOTOR YRTG 20-27			SCALE	CHECKED BY	
CUSTOMER		STX			1/6	DESIGNED BY	
DATE		05.07.06			LIST NO		
		YURIM PRECISION INDUSTRIAL CO.			DWG.NO	YU-50MC-C-001	



SPECIFICATION									
MOTOR	2.2kW, 6P, 440V, 60Hz								
BRAKE	DC 90V								
REDUCTION RATIO	$(18/75) \times (19/19+143) \times (19/19+93) \times (16/16+72) = 1/1151.8$								
OUTPUT SHAFT SPEED	$1150/1151.8 = 0.998$ r.p.m								
OUTPUT TORQUE	NORMAL 2000 kg.m MAX 2700 kg.m								
STEP	1 ST.		2 ND.		3 RD.				
NAME	PART	SUN	IDLE	RING	SUN	IDLE	RING	SUN	IDLE
MODULE		M2			M3			M4	
NO. OF TEETH	19	62	143	19	37	93	16	28	72
PITCH DIAMETER	Ø38	Ø144	Ø286	Ø57	Ø111	Ø279	Ø64	Ø112	Ø288
WIDTH OF TEETH	27	17	20	30	30	35	60	60	65

002 ASSEMBLY										
NO.	NAME		MATERIAL	Q'TY	W'HT	SIZE	REMARKS			
PROJECT							APPROVED BY			
TITLE		TURNING GEAR WITH MOTOR			SCALE		CHECKED BY			
		YRTG 20-27			1/6		DESIGNED BY			
CUSTOMER		STX					DRAWN BY			
DATE		08.07.17			LIST NO					
		YURIM PRECISION INDUSTRIAL CO.			DWG.NO		YU-50MC-C-002			



003 ASSEMBLY							
NO.	NAME		MATERIAL	Q'TY	W'HT	SIZE	REMARKS
PROJECT						APPROVED BY	
TITLE		TURNING GEAR WITH MOTOR YRTG 20-27			SCALE	CHECKED BY	
CUSTOMER		STX			1/5	DESIGNED BY	
DATE		05.07.06			LIST NO	DRAWN BY	
		YURIM PRECISION INDUSTRIAL CO.			DWG.NO	YU-50MC-C-003	

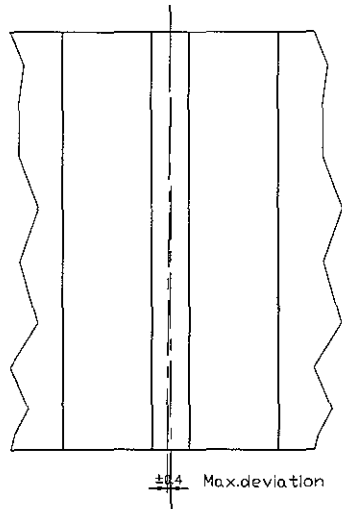


YURIM PRECISION INDUSTRIAL CO.

FOR APPROVAL		SPECIFICATION OF TURNING GEAR REDUCER	
CUSTOMER	STX	PROJECT NAME	
REDUCER TYPE	PLANETARY GEAR	QUANTITY	SET

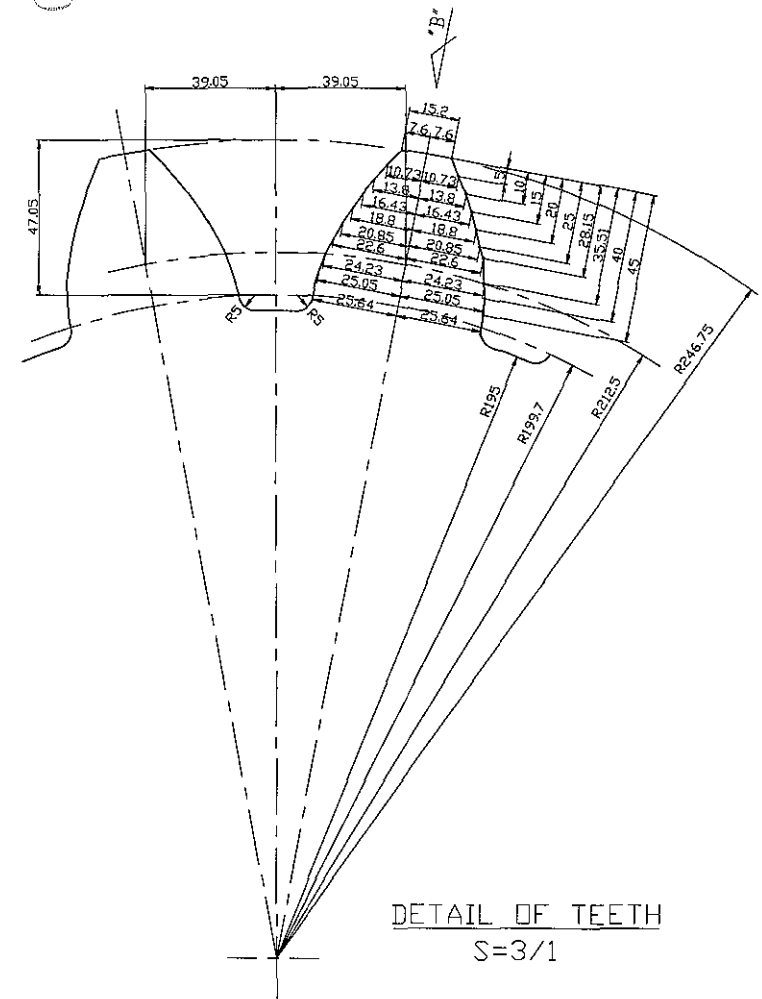
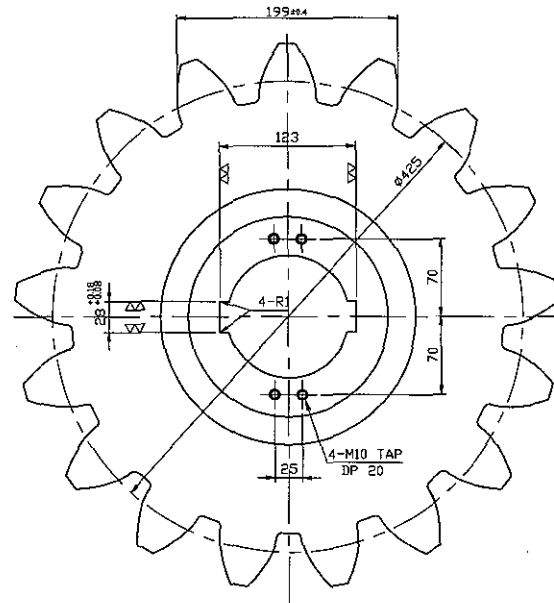
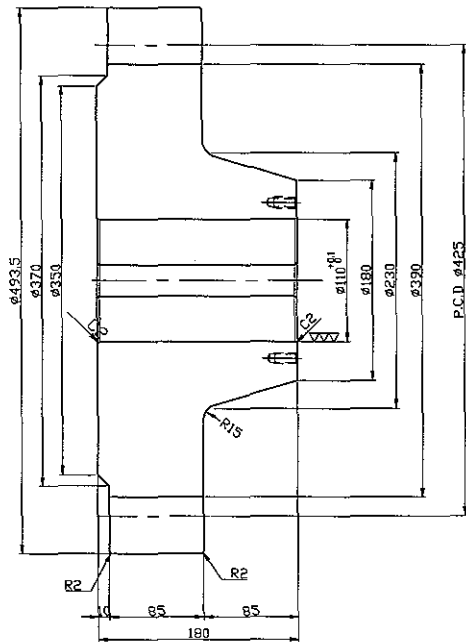
DRIVEN MACHINE		DRIVER		
KINDS	ENGINE TURNING GEAR	KINDS	ELECTRIC A.C MOTOR	
NO. OF REV.	0.998 r.p.m	MAKER	OTIS-LG	
CONDITION OF SERVICE		POWER	2.2 kW(3HP)	
NORMAL RATIO	1/1151.8	TYPE	TEFC	
DURATION	1 hour	PHASE	3 PHASE	
AMBIENT TEMP	50 °C	POLES	6 POLES	
MAX. TORQUE	120 %	INSULATION CLASS	F	
FREQUENCY OF START	OVER 10 / h	TEMP RISED	B	
INSTALLATION	IN DOOR	VOLTAGE	A.C 440 V	
IMPACT LOAD	UNIFORM	SPEED	1150 r.p.m	
SERVICE FACTOR	1.0	ACCESSORIES		
FINAL MACHINE	GEAR	COUPLING	CHAIN	
COOLING METHOD	OIL BATH	BRAKE	WITH	
LUBRICANT	MAX.OIL TEMP.: 80°C	DATE	2009.01.14	
	QUANTITY :ABOUT 12L	DESIGNED	CHECKED	APPROVED
	COOL.METHOD :OIL BATH			
DIRECTION OF REVOLUTION		REMARKS		
VIEW F SHAFT	CLOCK OR COUNTER CLOCK WISE			
METERIAL (HEAT TREATMENT)				
SUN GEAR	NITRIDING			
PLANETARY GEAR	NITRIDING			
INTERNAL GEAR	S45C-H (Q.T)			
CASING	FCD 450			
OUT PUT SHAFT	S45C-H(Q.T)			

43 (Wavy line symbol)

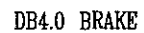
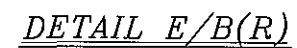
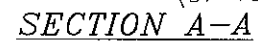
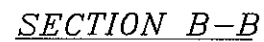


VIEW "B"

Number of teeth	z	17
Module	m	25
Pressure angle		20°
Displacement of profile x * m		12.5

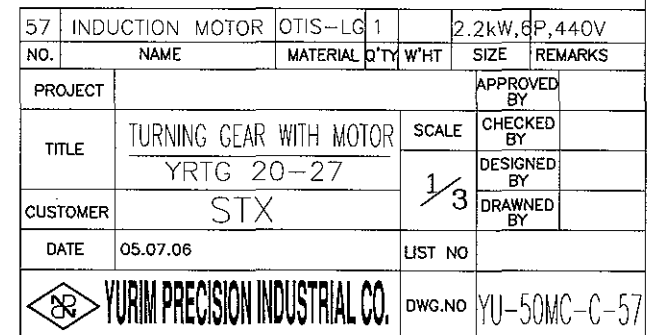


43	GEAR WHEEL	FCD500	1			
NO.	NAME	MATERIAL	Q'TY	W'HT	SIZE	REMARKS
PROJECT					APPROVED BY	
TITLE		TURNING GEAR WITH MOTOR YRTG 20-27			SCALE	CHECKED BY
CUSTOMER		STX			1/5	DESIGNED BY
DATE		05.07.06			LIST NO	DRAWN BY
YURIM PRECISION INDUSTRIAL CO.					DWG.NO	YU-50MC-C-43



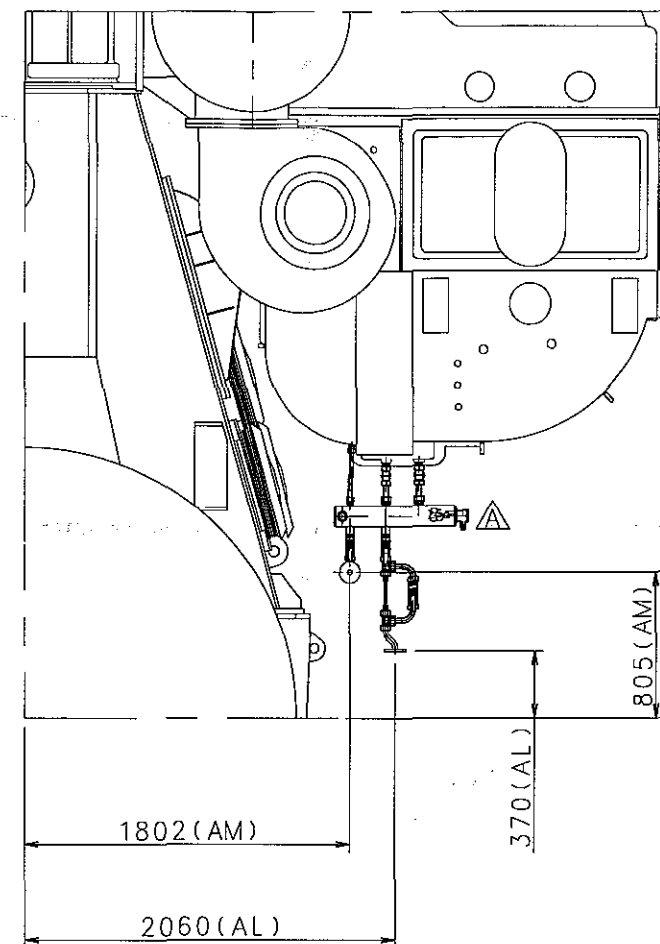
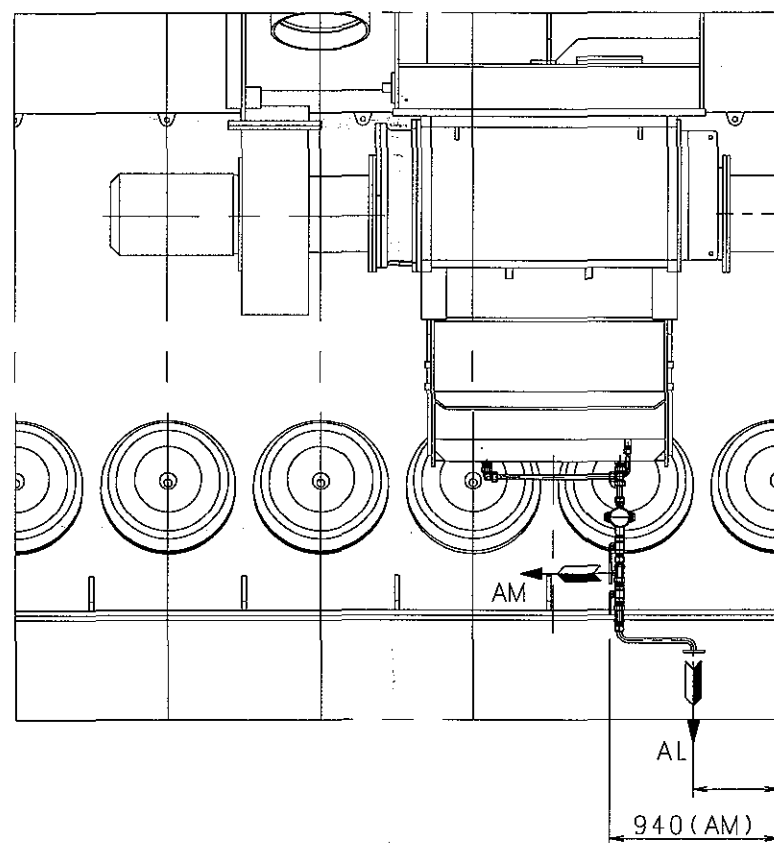
GENERAL SPECIFICATIONS

MOTOR	FRAME NO.	kw	POLE	VOLTS	HZ	INS. CLASS	SERVICE FACTOR	RATING	BEARINGS		APPROX WEIGHT	PROTECTION	MODEL NAME
									DRIVE	OPP.DRIVE			
	112M	2.2	6	440	60	F	1.0	CONT.	6206ZZ	6206ZZ	50KG	IP55	I03HV8YUMH



3 Phase Induction Motor Data Sheet

Customer : STX			Output	2.2 KW (3 HP)
Item No. : I03HV8YUMH			Frame No. : 112M	Pole : 6
Service : FOR TURNING GEAR			Service Factor : 1.0	
Quantity : 1 SET			Insulation Class : F	
Motor Type ☒ Squirrel Cage ☐ Wound Rotor			Temp Rise : 100 deg	
Enclose ☐ Drip-Proof ☒ TEFC			Voltage : 440 V	
☐ Weather Protected			Frequency : 60 HZ	
☐ Increased Safety Exp.-proof			Efficiency (rated) : 81.0 %	
Installation ☐ Indoor ☒ Outdoor			Power factor (rated) : 72.0 %	
Starting ☒ Direct on Line ☐ Star-Delta			Current (rated) : 4.9 A	
Method ☐ Reactor ☐ Resister			Current (starting) : 25.5 A	
☐			Slip : 5.4 %	
Drive System ☒ Direct Coupled ☐ V-Belt			Torque (rated) : 1.88 Kg.m ²	
☐ Geared			torque (starting) : 200 %	
Mounting ☐ Horizontal ☒ Vertical			torque (Maximum) : 220 %	
☐ Flange				
Duty Cycle ☒ Continuous ☐ 60 Min			GD ² of Motor : 0.058Kg.m ²	
Rotation (Facing Drive end)			ALLOWable Load GD ² Referred Torque	
☒ CW ☒ CCW			Motor Shaft : 29.3 Kg.m ²	
Ambient Temperature : -10°C ~ 50°C			Outline Drawing No. :	
Site Atitude (under) : 1000m			Painting :	
Noise & Vibration Level: 63 db / 30u				
Standards : IEC ,KS ,NEMA				
Accessory Equipment			Remark : IP55	
(☒ SPACE HEATER SPEC. : 1PH,240V,24W)				
			Tolerance : According to KSC-4002	
PREPARED	CHECKED	APPROVED BY		
			LG·OTIS	



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Basic Standards (MBD SB) & Suppl. Drawing No.: 3174889-3.0		EN21C Surf. roughness		Projection:		Material / Blank:		
		EN21F-m Tolerances		Mass (kg):		Final User Material:		
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement			C.No
				*	Replaced by Ident. No.:			H
				*				G
				*				F
				*				E
				*				D
				*				C
				*				B
				*				B
				*				A
20050926	JTSCSP	JSP		*	LEVEL SWITCH POSITION MODIFIED..			0
20010321				*				
Similar Drawing No.:				Replacement for Ident. No.:				
Scale:	Size:	Type:	S46-50MC-C TC:EXH		Page No.:		01 (01)	STX Engine Co., Ltd.
1:20	A1							
Info. No.:	Description:		375405		Drain pipe - air cooler		280500146A	
Final User Info. No.:	Final User Description:						Final User Ident. No.:	2-6

CAD Drawing - do not correct by hand

A

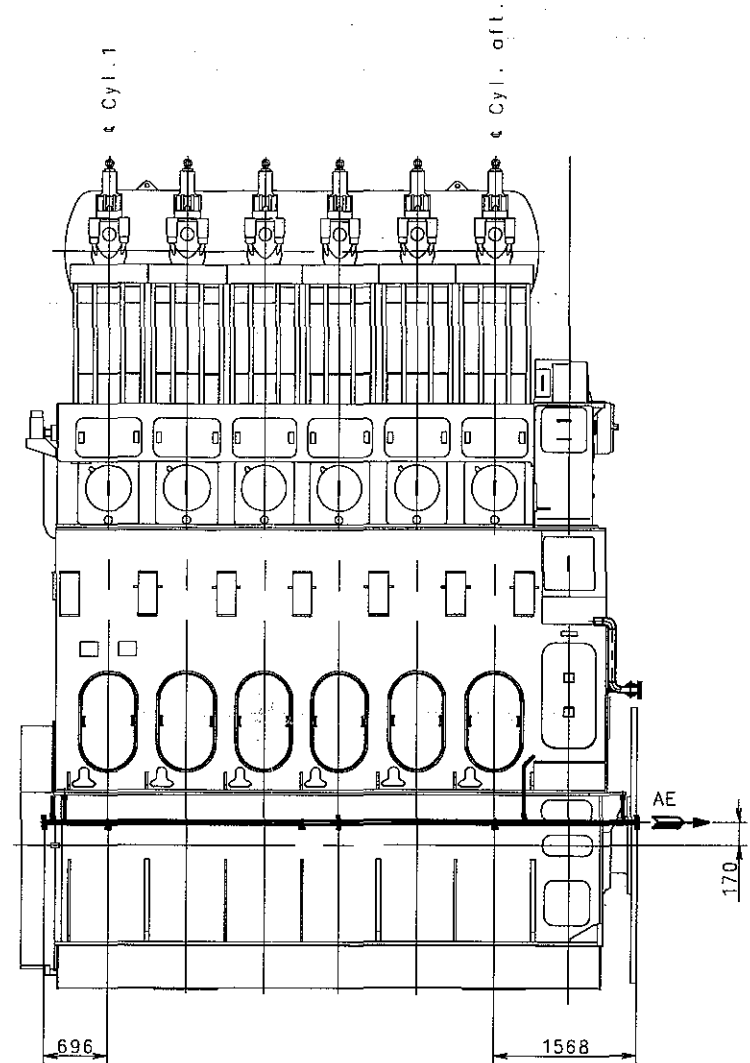
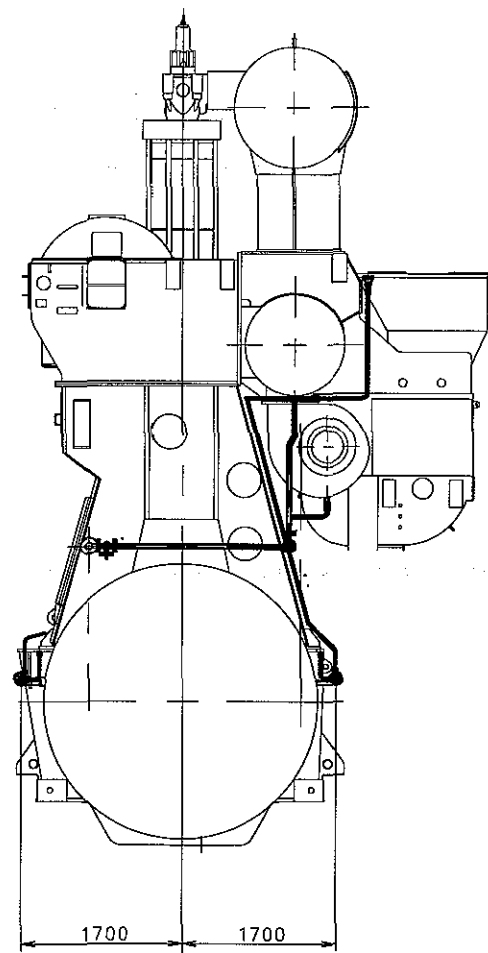
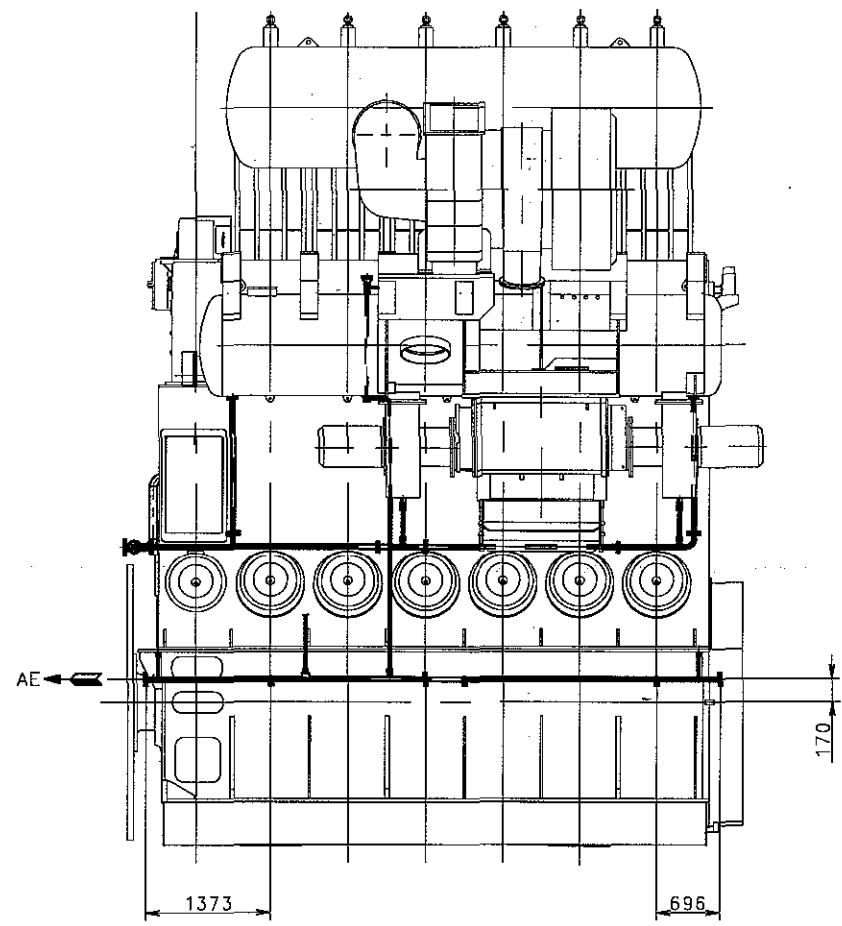
B

C

F

D

E



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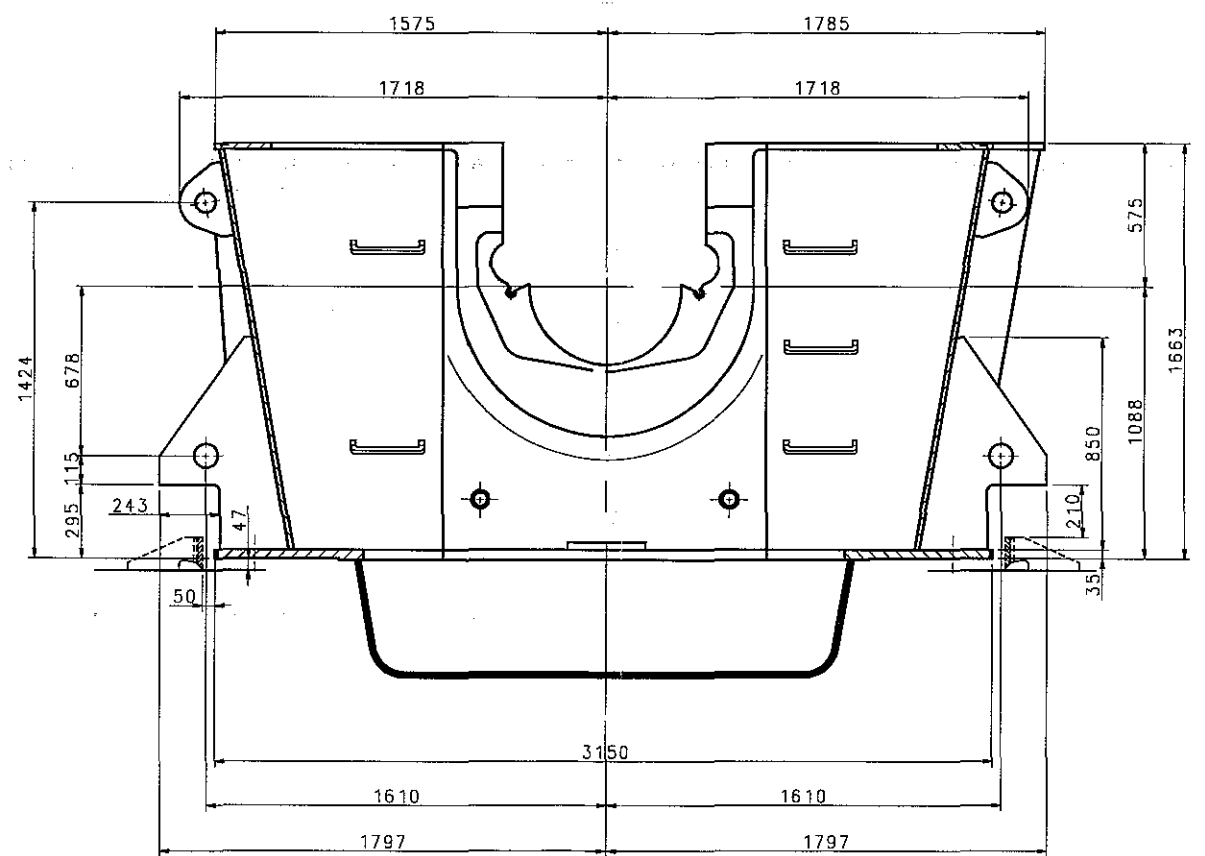
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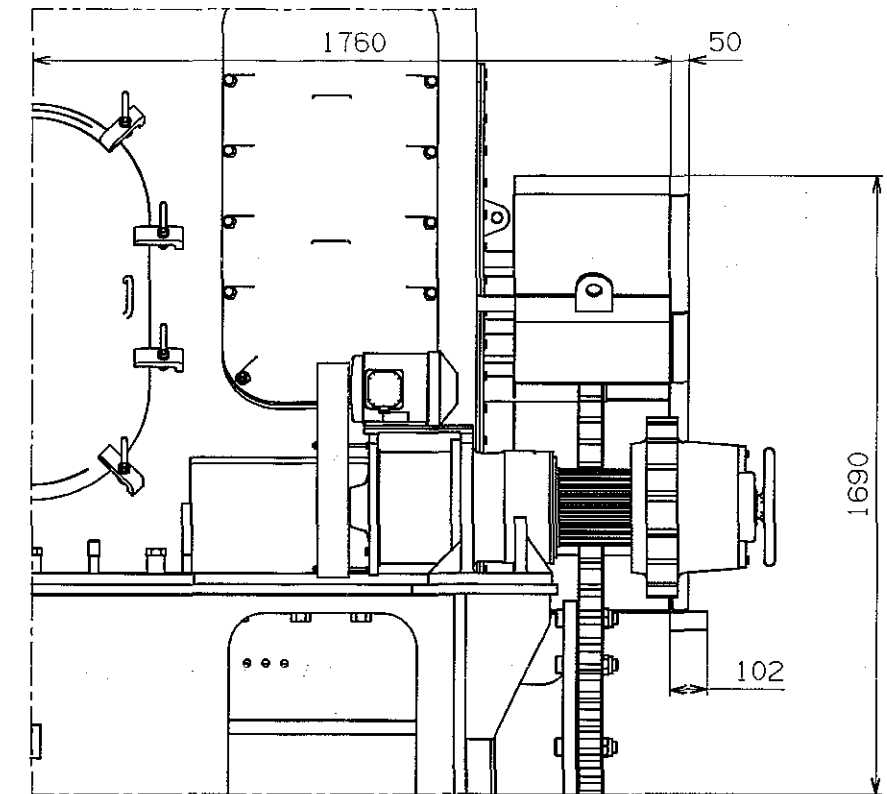
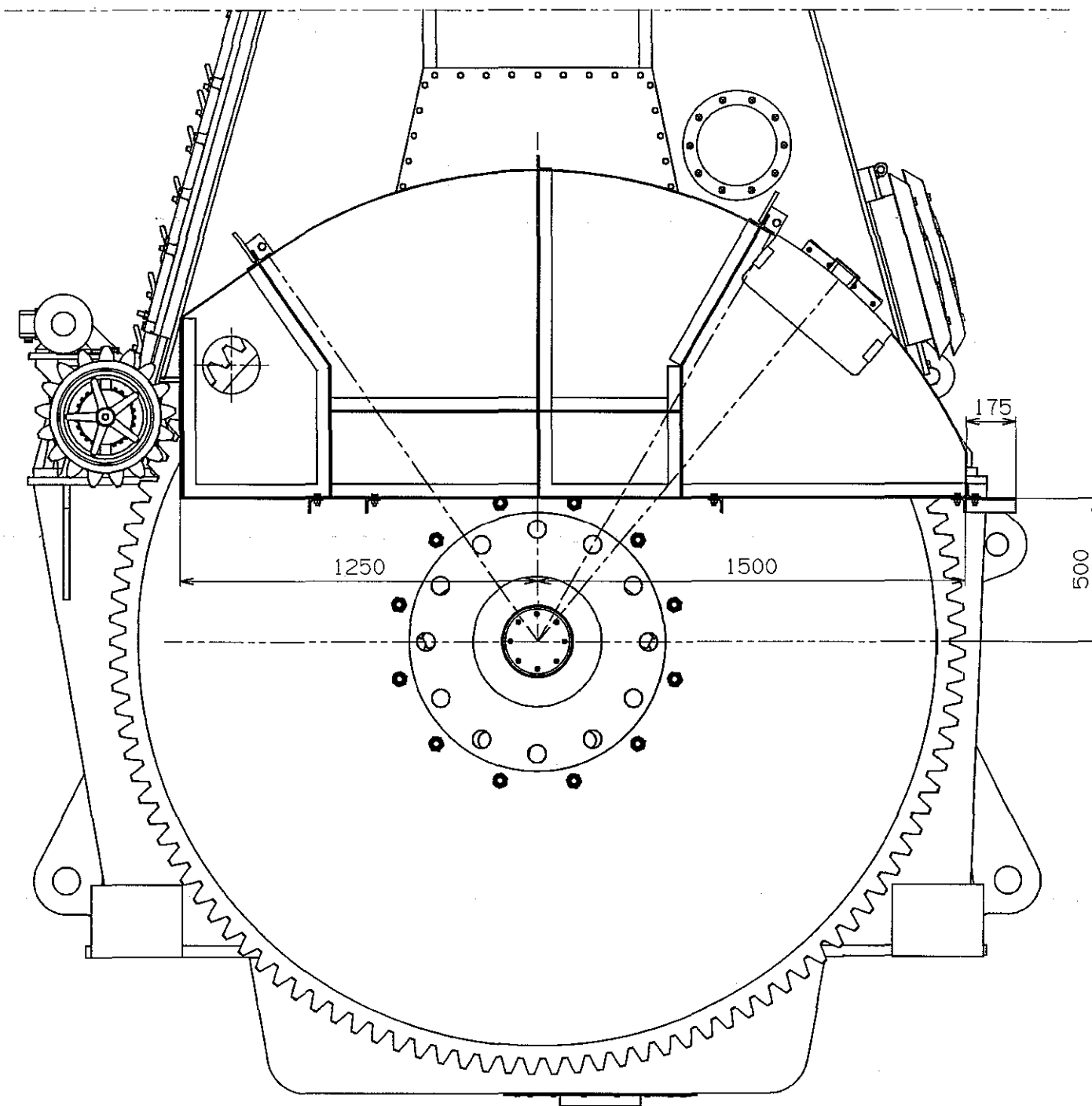


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				*	Replaced by Ident. No.:					9	
				*						8	
				*						7	
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				*						5	
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				*						3	
				*						2	
				*						1	
20050222				JTSCSPJSP	*						0
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Info. No.:		Description:					Ident. No.:				
375014		Various pipes-exh.side					2805009260				
Final User Info. No.:		Final User Description:					Final User Ident. No.:				

CAD Drawing - do not correct by hand



Basic Standards (USD SB) & Suppl. Drawing No.:		EN21C Surf. roughness		Preparation:	Material: Blank:																					
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Similar Drawing No.:		1176174-5		Replacement for Ident. No.:																						
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Info. No.:	Description:			Ident. No.:																						
301071	Jack-up bracket arrangement			2805002400																						
Final User Info. No.:	Final User Description:			Final User Ident. No.:																						



Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:
		EN21F-m Tolerances	Mass (kg):	Final User Material:
Date	Des.	Chk.	Appd.	A.C.
Change / Replacement				C.No.
Replaced by Ident. No.:				I
				H
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				D
				C
				B
				A
				O
20090730WGKHSKSKN - 신규 도면 제작.				
Similar Drawing No.:		Replacement for Ident. No.:		
Scale:	Size:	Type:	Page No.:	
1:10	A1	TS50MC-C	01 (01)	stx Engine Co., Ltd.
Info. No.:	Description:			Ident. No.:
	ARR. OF FLY WHEEL COVER			2801024530
Final User Info. No.:	Final User Description:			Final User Ident. No.:

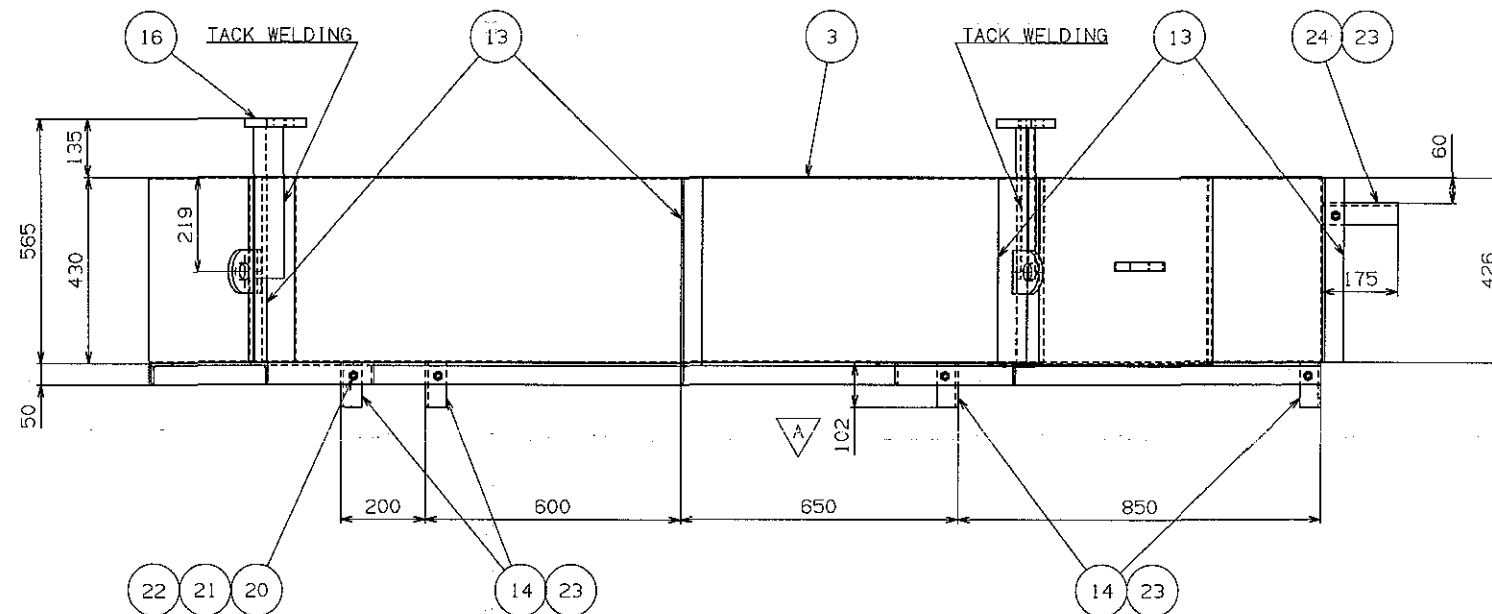
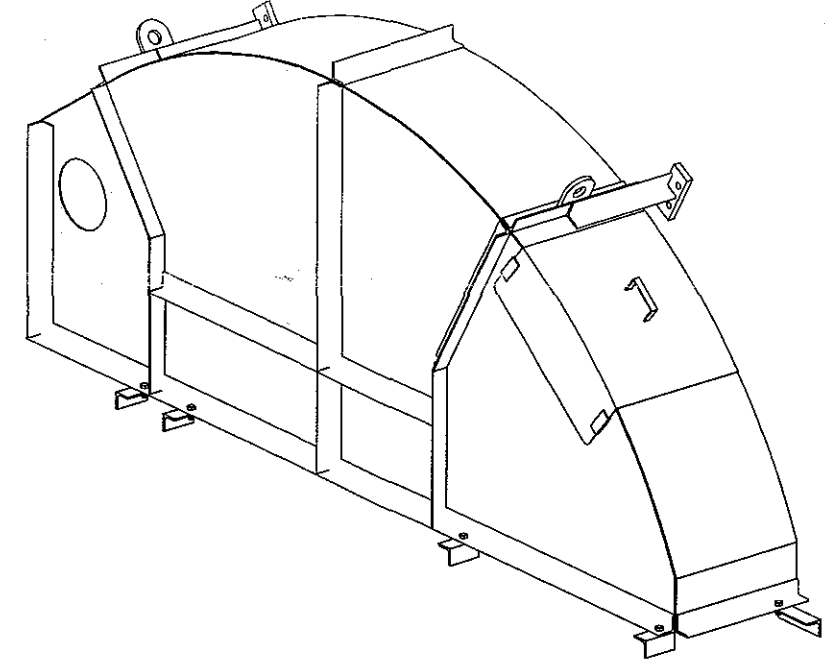
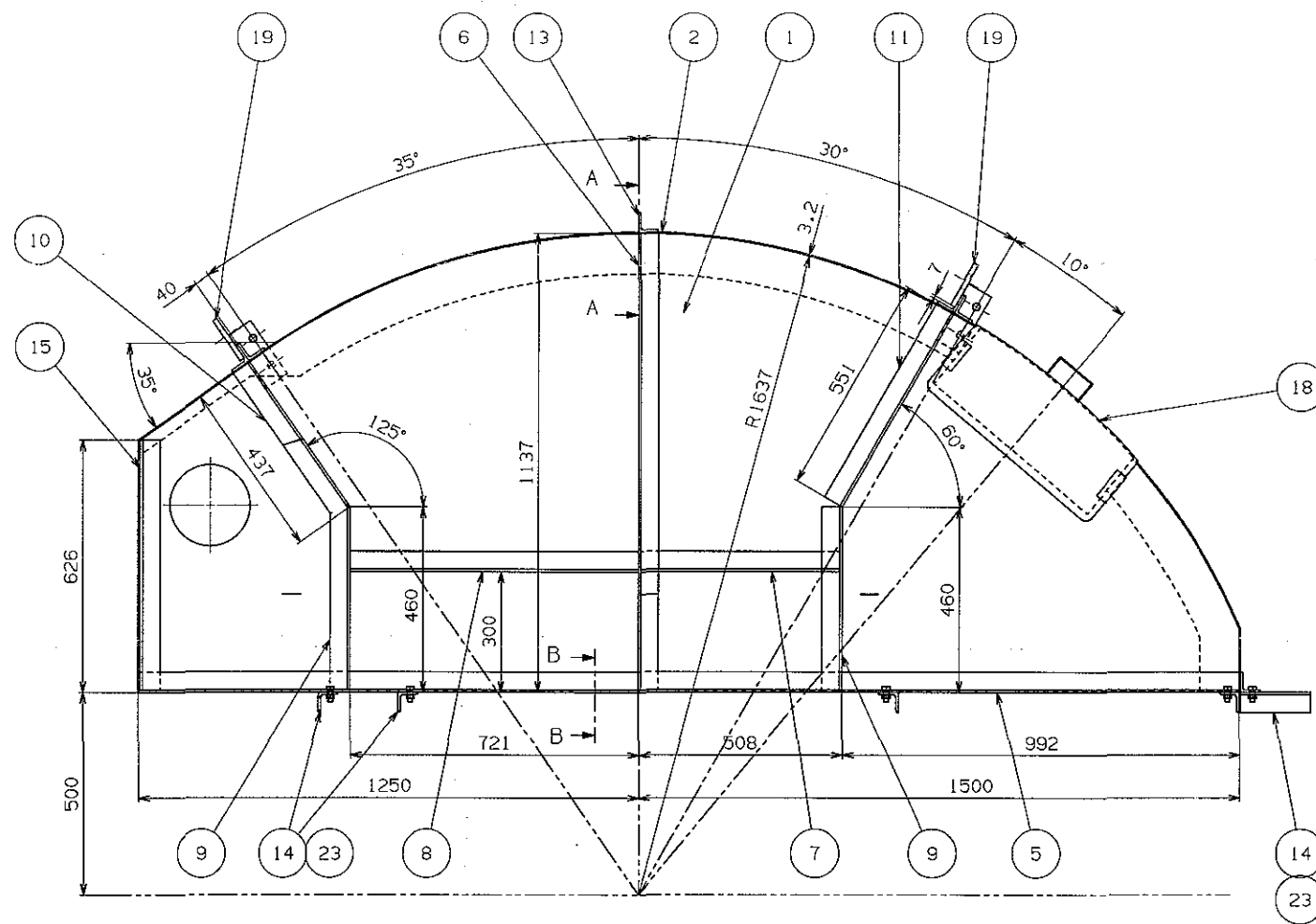
본 도면은 UG 도면이므로 수정함에 의해 수정될 수 있음. This is a UG drawing and should not be corrected by hand.

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MAN B&W

not correct by hand - UG Drawing



NOTE

1. 모든 접합부는 ∇ , ∇ 으로 견고히 용접 할 것.
2. 날카로운 모서리는 제거 할 것.
3. P05.23은 제작 후 밀도 검증 할 것.

-	-	BOLT	-	2	-	M16-50L
24	-	ANGLE	-	1	-	50x50x6-175
23	-	PLATE	-	30	-	10Tx50x50
22	-	WASHER	-	5	-	M12
21	-	NUT	-	5	-	M12
20	-	BOLT	-	5	-	M12-70L
19	280200122A-03	LUG	-	2	-	12Tx100x120
18	-	DOOR SUPPORT	-	1	-	3.2Tx221x530
17	-	-	-	-	-	-
16	-	SUPPORT	-	2	-	-
15	-	ANGLE	-	1	-	50x50x6-626
14	-	ANGLE	-	4	-	50x50x6-102
13	-	ANGLE	-	4	-	50x50x6-426
12	-	-	-	-	-	-
11	-	ANGLE	-	1	-	50x50x6-551
10	-	ANGLE	-	1	-	50x50x6-437
9	-	ANGLE	-	2	-	50x50x6-460
8	-	ANGLE	-	1	-	50x50x6-721
7	-	ANGLE	-	1	-	50x50x6-508
6	-	ANGLE	-	1	-	50x50x6-1137
5	-	ANGLE	-	1	-	50x50x6-2750
-	-	-	-	-	-	-
3	-	PLATE	-	1	-	3.2Tx2750xR1637
2	-	PLATE	-	1	-	3.2Tx2750xR1637
1	-	PLATE	-	1	-	3.2Tx2750xR1637

NO.	PART NO.	NAME	MATERIAL	Q'TY	WEIGHT	REMARKS
Basic Standards (MEO SB) & Suppl. Drawing No.1 EN210 Surf. roughness EN211-m Tolerances						
Date Des. Q.A. Appt. A.C. Change / Replacement						
Replaced by Ident. No.1						
20090730WGHKSKBN - 신규 도면 제작.						
Scale: 1:6 A0 Type: 7S50MC-C Page No.: 01 (02) stx Engine Co., Ltd.						
Description: FLY WHEEL COVER Ident. No.: 2801024540						
Final User Info. No.: Final User Description: Final User Ident. No.:						

If measuring pins are required, we recommend that 4 pieces are installed at the positions marked by #.

All hot work on the tanktop must be finished before the epoxy is cast.

Note:
The lower flanges
of the bedplate
are not machined
to 1:100 taper.

2x1 off Ø66 holes

The width of
machining on
the underside
of bedplate

Epoxy wedges to be
chiselled after
curing to enable
mounting of side
chock liners

- 1) the engine builder drills the holes for holding down bolts in the bedplate while observing the toleranced locations indicated on STX Corporation drawings for machining the bedplate and
- 2) the shipyard drills the holes for holding down bolts in the top plates while observing the toleranced locations given on the present drawing, and
- 3) the holding down bolts are made in accordance with STX Corporation drawings of these bolts.



All welding workmanship
equal to boiler shop
standard e.g. EN287-1.

- 4028 This space to be kept free from pipes etc. along both sides of the engine in order to facilitate the overhaul work on holding down bolts, supporting chocks and side chocks.

4032 Middle of main bearing.

Centreline cylinder

4317
Corners of floorplates to
be cut to enable proper
welding of girders.
e.g. as shown.

4046
Thickness of floorplates
between main engine girders
22mm.

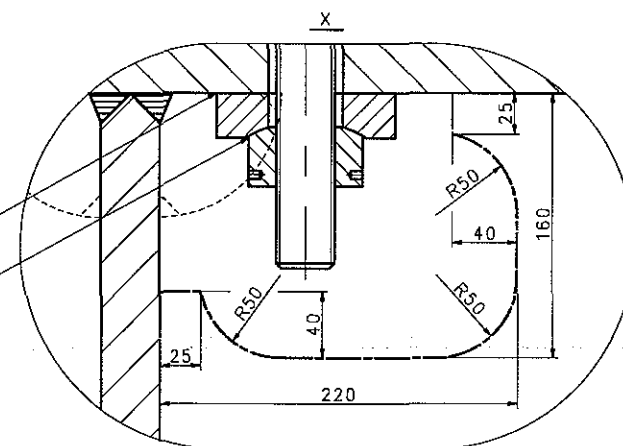
4292
Continuous girder to
extend with full
dimensions at least
to ship's frame
forward of engine
and at least to
ship's frame aft of
the aft end of end
chock.

4059
Slots to be cut in vertical floorplates to clear nuts when necessary.

4282
Contact face is to be plane
and to be carefully cleaned
and lightly greased. Paint
need not to be removed.

4047
Spherical washers
greased before mounting.

1141
If required by classification
society, apply this bracket.
Thickness of bracket is the
same as thickness of floorplates

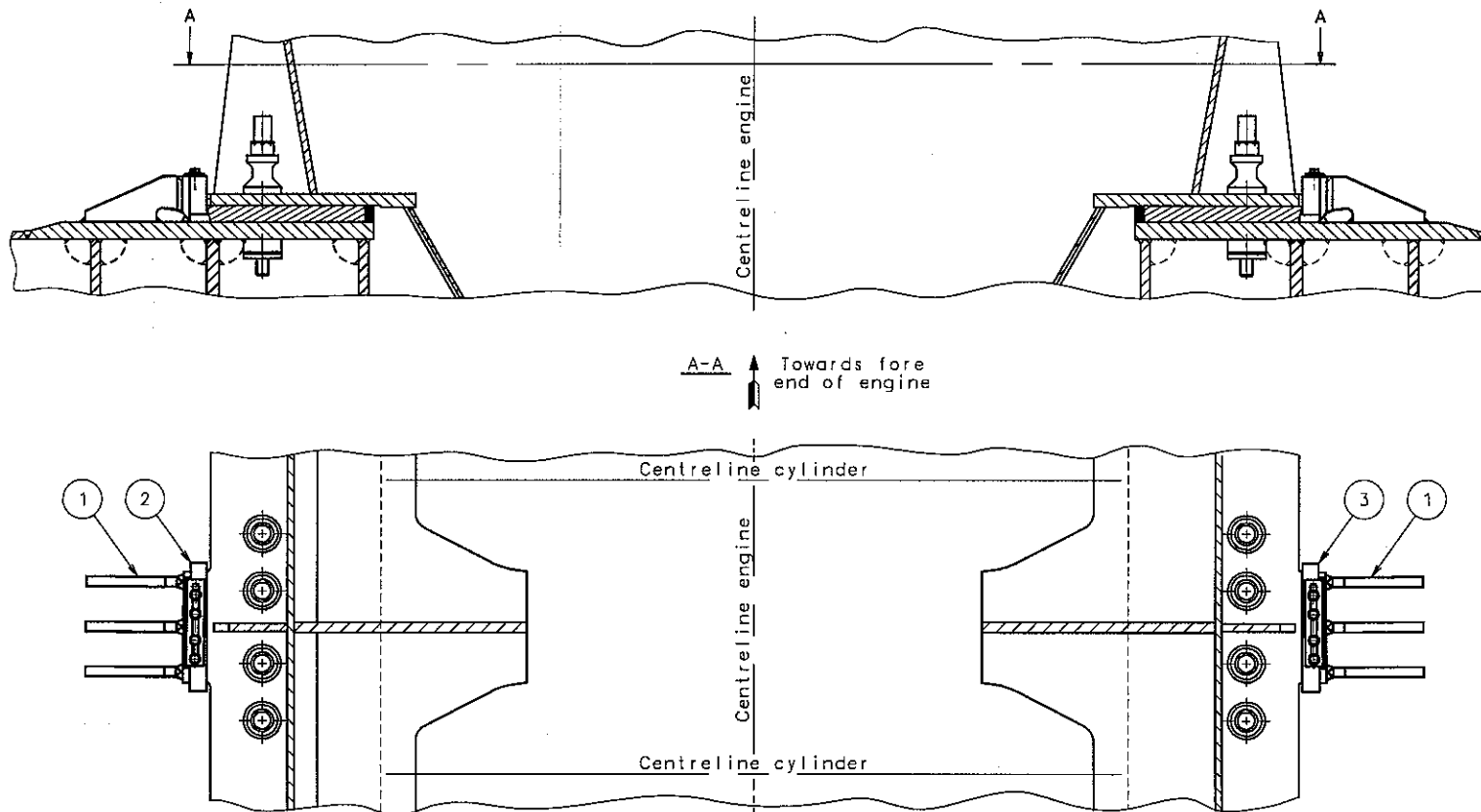


이 문서 GDSN 10-272 0425 001 01 00 00
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Basic Standards (M05 SB) & Suppl. Drawing No.:				EN21C Surf. roughness		Projection: 		Material / Blank:	
0787280-2.5				EN21F-m Tolerances		Mass (kg):		Final User Material:	
Date		Des.	Chk.	Appd.	A.C.	Change / Replacement			C.No.
					✓	Replaced by Ident. No.:			9
					✓				8
					✓				7
					✓				6
				Z4					5
				Z4					4
				Z4					3
				Z4					2
				Z4					1
20040406					✓				0
Similar Drawing No.:						Replacement for Ident. No.:			
Scale:	Size:	Type:				Page No.:			
1:4	A0		S50MC-C			01 (01)	STX Engine Co., Ltd.		
Ident. No.:		Description:					Ident. No.:		
387600		Engine seating w.epoxy chock					2805004390		
Final User Info. No.:		Final User Description:					Final User Ident. No.:		



A set of side chocks consist of two side chocks, one starboard liner and one portside liner.
For number of side chocks and liners, see relevant drawing: "Arr. of epoxy chocks" or "Arr. of holding down bolts".

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1	Liner, starboard	3		2805000420	2805000420	SS400	16.3
1	Liner, port	2		2805000410	2805000410	SS400	16.3
2	Side chock	1		2805000400	2805000400	SS400	20.6
QTY	NAME OF PART	POS	INFO NO	PART NO	DRG.	NO MATERIAL	KG
Basic Standards (M&B) & Suppl. Drawing No.:				EN21C Surf. roughness	Projection	Material / Blank:	
0781948-2.1				EN21F-m Tolerances	Mass (kg):	Final User Material:	
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement		C.No
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							8
							7
							6
							5
							4
							3
							2
							1
19970714				Z4			0
Similar Drawing No.:				Replacement for Ident. No.:			
Scale:	Size:	Type:	S50MC-C		Page No.:	01 (01)	
1:10	A2				SBX Engine Co., Ltd.		
287661		Description:				Ident. No.:	
		Assembly of side chock				2805000390	
Final User Info. No.:		Final User Description:				Final User Ident. No.:	

1 2 3 4 5 6

A

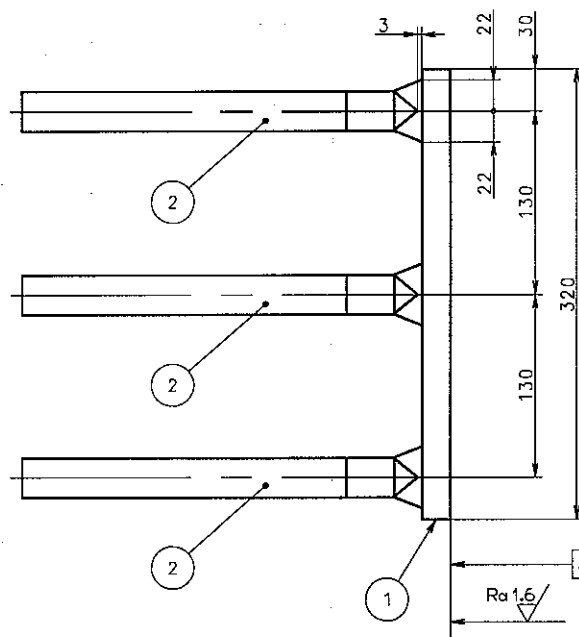
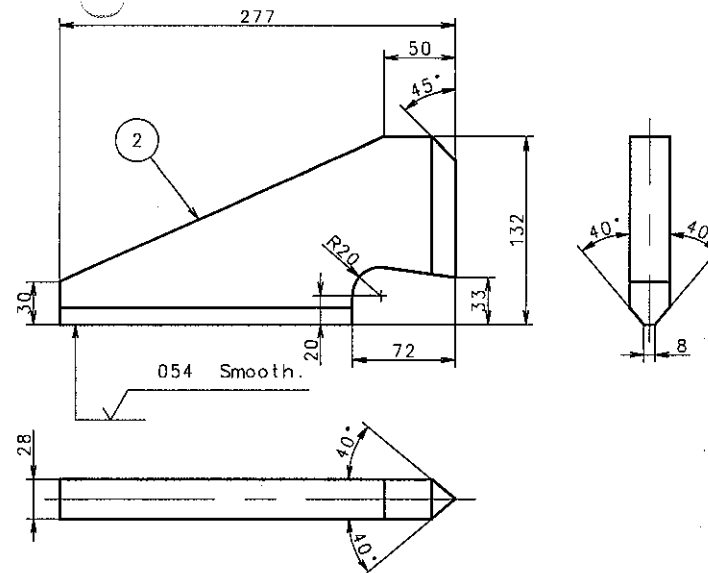
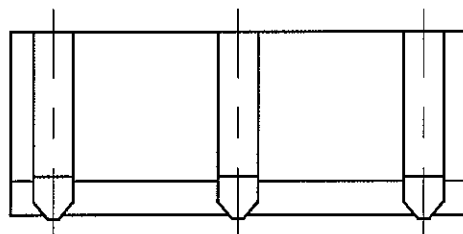
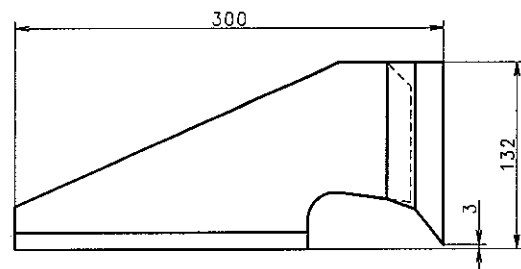
F

B

C

D

1-2-2



0.05

Ra 1.6

145 To be finish-machined after welding.

689 Sharp edges to be removed.

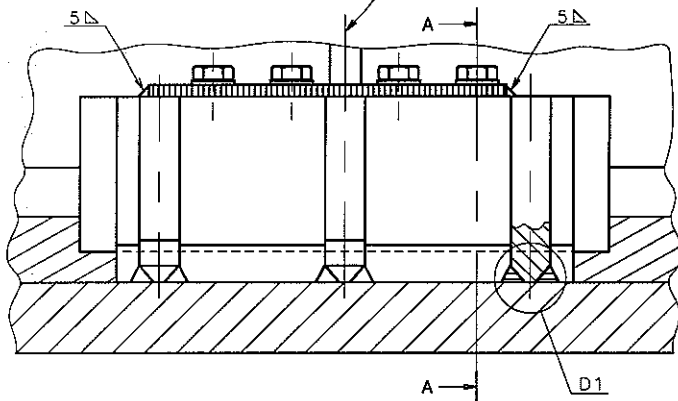
4223 The side chock will correspond to a height of the supporting chocks of 50-25 mm

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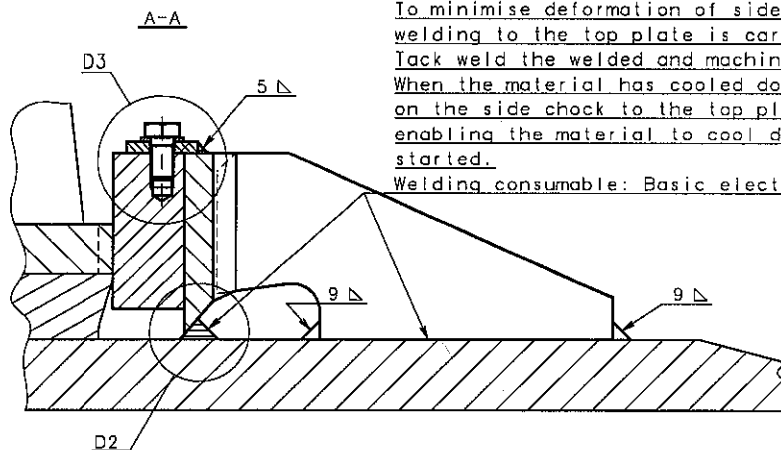
3	Plate, shaped	2							SS400	4.9
1	Plate, shaped	1							SS400	5.9
QTY	NAME OF PART	POS	INFO NO	PART NO	DRG. NO	MATERIAL	KG			
0781938-6.2	Basic Standards (MED SB) & Suppl. Drawing No.: EN31C Surf. roughness	Projection:	EN31F-m Tolerances	Material / Blom:	Mass (kg):	Final User Material:				
19970711										
Similar Drawing No.:										
Scale:	1:2.5	A2	Type:	S50MC-C	Page No.:	01 (01)				
Info. No.:	287700		Description:	Side chock	Ident. No.:	2805000400				
Final User Info. No.:			Final User Description:		Final User Ident. No.:					

4032 Middle of main bearing.



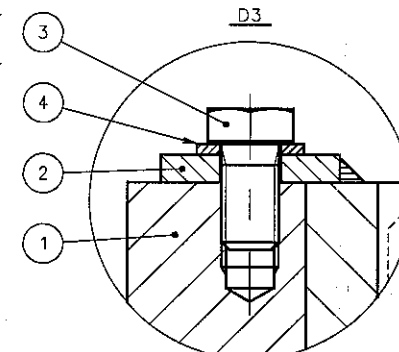
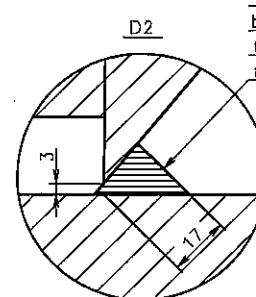
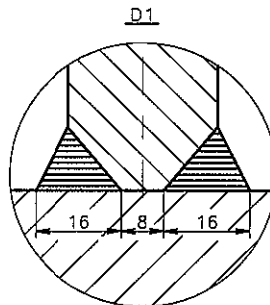
4036

To minimise deformation of side chocks owing to welding stresses, welding to the top plate is carried out as follows:
 Jack weld the welded and machined side chock to the top plate.
 When the material has cooled down, carry out welding of all seams on the side chock to the top plate in steps at suitable intervals enabling the material to cool down before the next welding step is started.
 Welding consumable: Basic electrode.

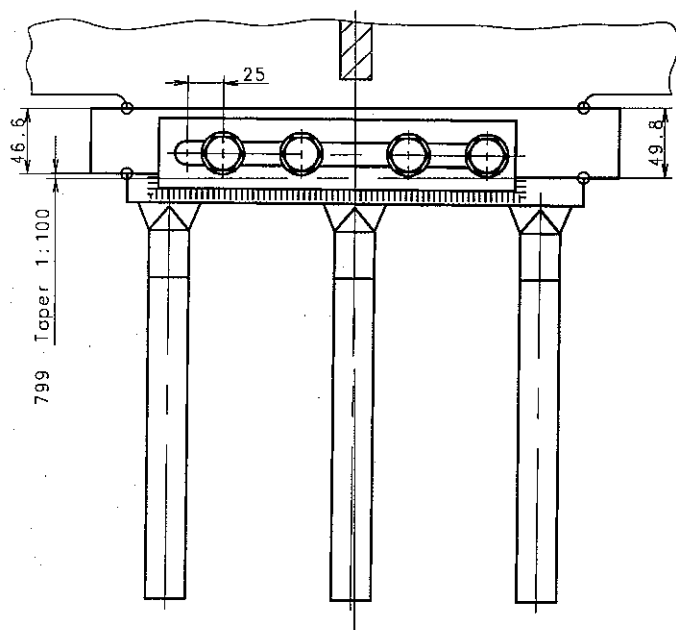


4178

The root pass to be welded with full penetration from this side.



4294 Towards fore end of engine.



4227

Mount the liners from aft on both sides of the engine. After adaptation, hammer the liners in line with each engine frame into place simultaneously. Liner surfaces must have at least 80% contact.

4295

Weld the side chock to the top plate to permit the liner, pos 1, to be fitted-in from aft (towards fore of engine), making it possible to knock the liner further forward, and thus to compensate for possible wear of the contact surfaces.

4067

All welding workmanship equal to boiler shop standard e.g. EN287-1.

4	Washer	4				SS400	0.011
4	Hexagon head screw	3				SCM435	0.077
1	Lock plate	2	2805000450	2805000450		SS400	0.55
1	Liner f. side chock, port	1	2805000430	2805000430		SS400	15.4
QTY	NAME OF PART	POS	INFO NO	PART NO	DRG. NO	MATERIAL	KG

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0781967-3.3	EN210-m Tolerances	Final User Material:

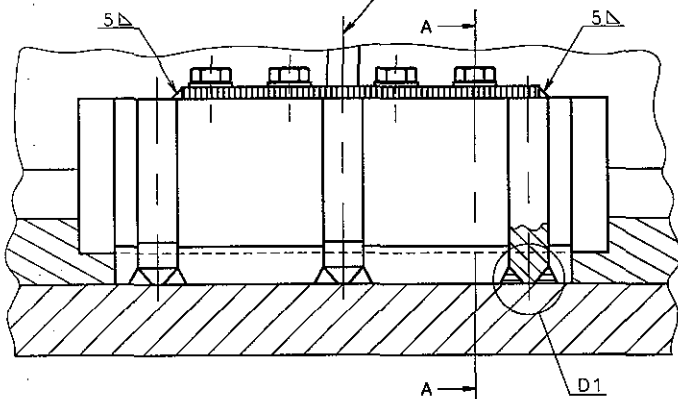
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						2
						1
						0

Similar Drawing No.:	Replacement for Ident. No.:
Scale: 1:2.5	Site: A2
Info. No.:	Type: S50MC-C
287730	Description: Liner f. side chock, port
Final User Info. No.:	Final User Description:
Ident. No.:	2805000410
Final User Ident. No.:	

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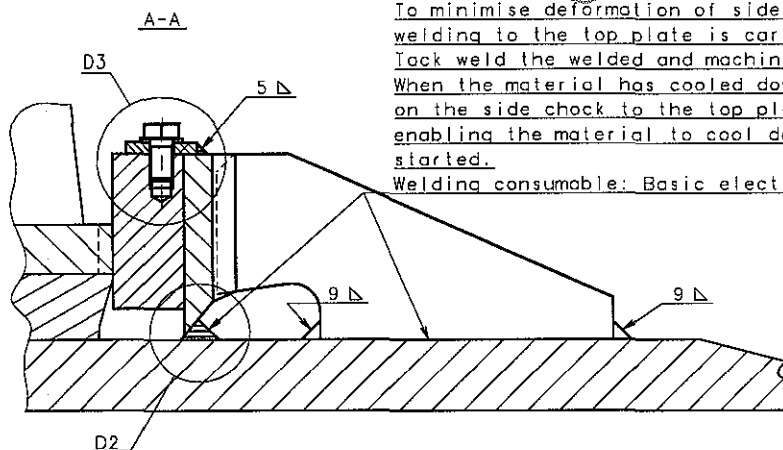


4032 Middle of main bearing.

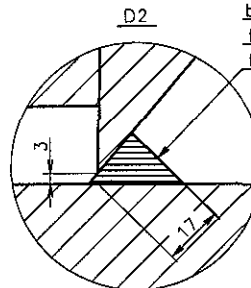
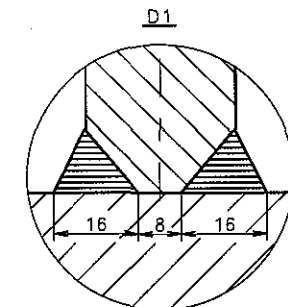
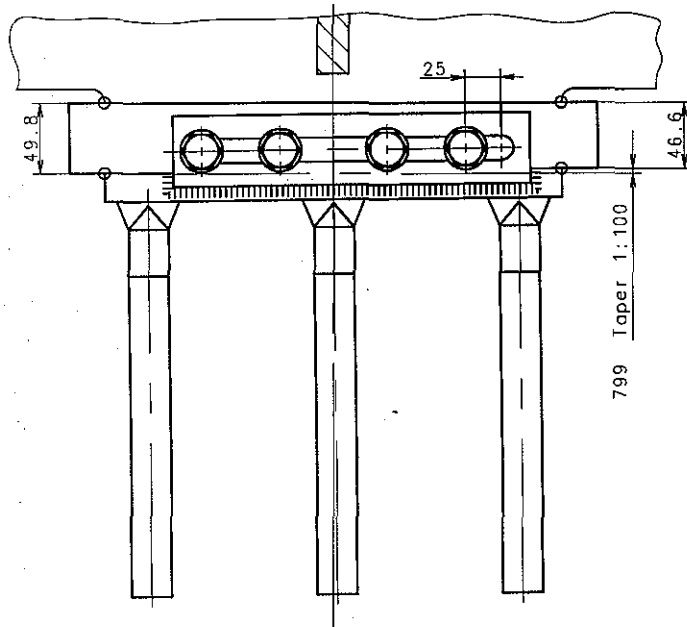


4036

To minimise deformation of side chocks owing to welding stresses, welding to the top plate is carried out as follows:
 Tack weld the welded and machined side chock to the top plate.
 When the material has cooled down, carry out welding of all seams on the side chock to the top plate in steps at suitable intervals enabling the material to cool down before the next welding step is started.
 Welding consumable: Basic electrode.

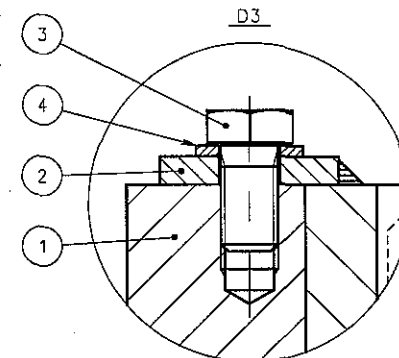


4294 Towards fore end of engine.



4178

The root pass to be welded with full penetration from this side.



4227

Mount the liners from aft on both sides of the engine. After adaptation, hammer the liners in line with each engine frame into place simultaneously. Liner surfaces must have at least 80% contact.

4295

Weld the side chock to the topplate to permit the liner, pos 1, to be fitted-in from aft (towards fore of engine), making it possible to knock the liner further forward, and thus to compensate for possible wear of the contact surfaces.

4067

All welding workmanship equal to boiler shop standard e.g. EN287-1.

4	Washer	4				SS400	0.011
4	Hexagon head screw	3				SCM435	0.077
1	Lock plate	2	2805000450	2805000450		SS400	0.55
1	Liner f.side chock, starboard	1	2805000440	2805000440		SS400	15.4
QTY	NAME OF PART	POS	INFO NO	PART NO	DRG. NO	MATERIAL	KG

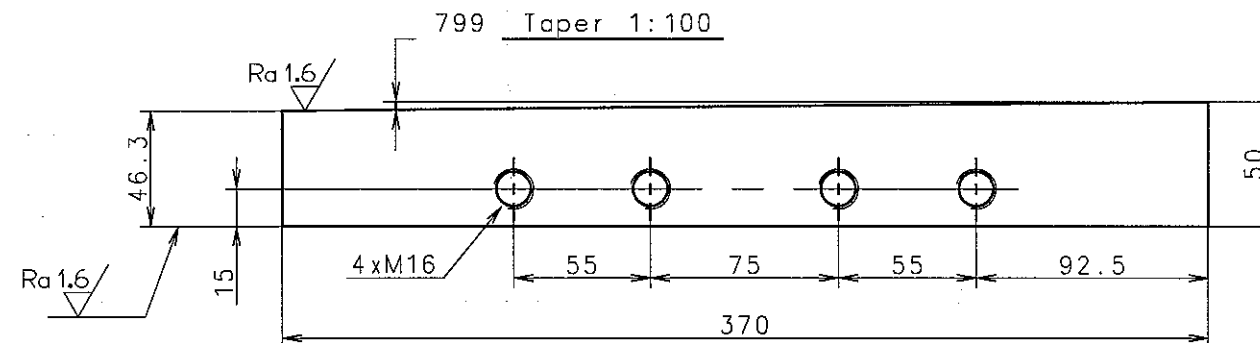
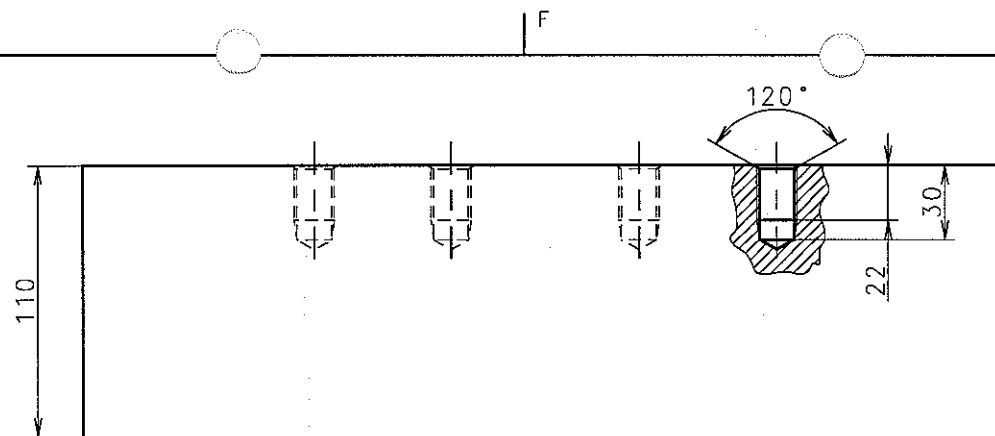
Basic Standards (MBS) & Suppl. Drawing No.: EN210 Surf. roughness	Projection:	Material / Blank:
0781966-1.3	EN21F-m Tolerances	Mass (kg):
		Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
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						7
						6
						5
						4
						3
						2
						1
						0

19970714							
Scale:	Size:	Type:	Page No.:	SBX Engine Co., Ltd.			
1:2.5	A2	S50MC-C	01 (01)	Ident. No.:			
287720	Liner f.side chock, starboard			2805000420			
Final User Info. No.:		Final User Description:		Final User Ident. No.:			

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689 Sharp edges to be removed.

Ra 12.5/ 164 All over, except where otherwise stated.

Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:
0789726-1.1		EN21F-m Tolerances	SS400	Final User Material:
		Mass (kg):	15.4	
Date	Des.	Chk.	Appd.	A.C.
Change / Replacement				C.No
Replaced by Ident. No.:				9
				8
				7
				6
				5
				4
				3
				2
				1
19970710				0
Similar Drawing No.:		Replacement for Ident. No.: 0782131-4.		
Scale:	Size:	Type:	Page No.:	STX Engine Co., Ltd.
1:2	A3	L-S50MC/S50MC-C	01 (01)	
Info. No.:	Description:			Ident. No.:
187730	Liner for side chock, port			2805000430
Final User Info. No.:	Final User Description:			Final User Ident. No.:

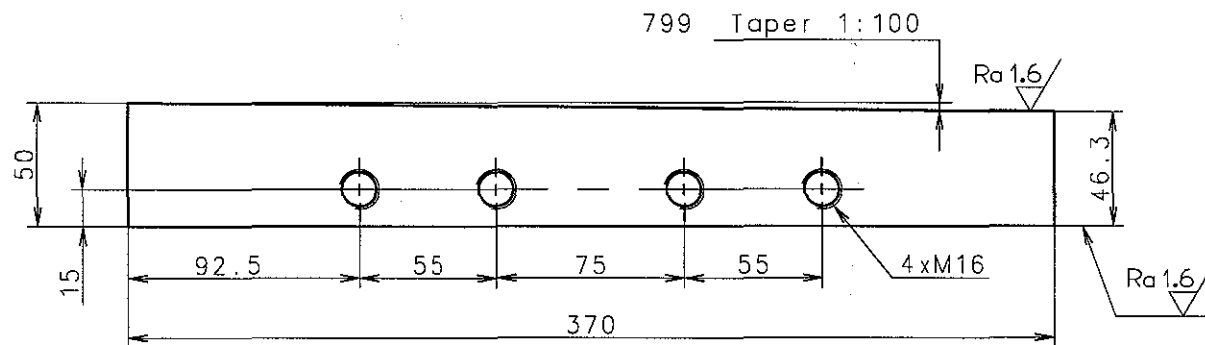
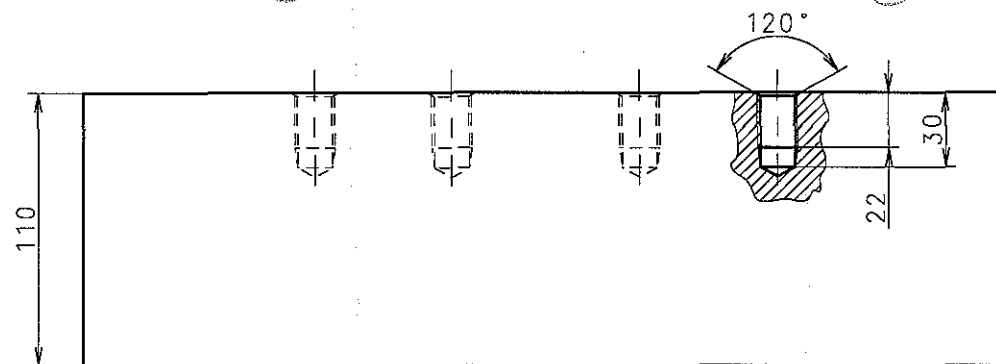
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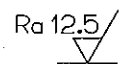


CAD Drawing - do not correct by hand

3-3-4



689 Sharp edges to be removed.



164 All over, except where otherwise stated.

Basic Standards (MBD SB) & Suppl. Drawing No.:				EN21C Surf. roughness		Projection: 		Material / Blank:	
0789724-8.1				EN21F-m Tolerances		Mass (kg): 15.4		SS400	
								Final User Material:	
Date		Des.	Chk.	Appd.	A.C.	Change / Replacement			
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					*				
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					*				
					*				
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					*				
					Z4				
19970710					*				
Similar Drawing No.:						Replacement for Ident. No.: 0782130-2.			

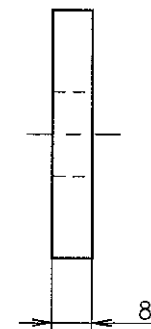
Scale:	Size:	Type:	Page No.:	STX Engine Co., Ltd.
1:2	A3	L-S50MC/S50MC-C	01 (01)	
Info. No.:	Description:			Ident. No.:
187720	Liner for side chock, starboard			2805000440
Final User Info. No.:	Final User Description:			Final User Ident. No.:

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CAD Drawing - do not correct by hand



Basic Standards (MBD SB) & Suppl. Drawing No.:					EN21C Surf. roughness	Projection: 	Material / Blank:
0789727-3.1					EN21F-m Tolerances	Mass (kg): 0.55	SS400
						Final User Material:	
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement		C.R.
				*	Replaced by Ident. No.:		9
				*			8
				*			7
				*			6
				*			5
				*			4
				*			3
				*			2
				Z4			1
19970710				*			0

Similar Drawing No.:		Replacement for Ident. No.:	
Scale: 1:1	Size: A3	Type: L-S50MC/S50MC-C	Page No.: 01 (01)
Info. No.: 187725	Description: Lock plate, side chock liner		Ident. No.: 2805000450
Final User Info. No.:	Final User Description:		Final User Ident. No.:



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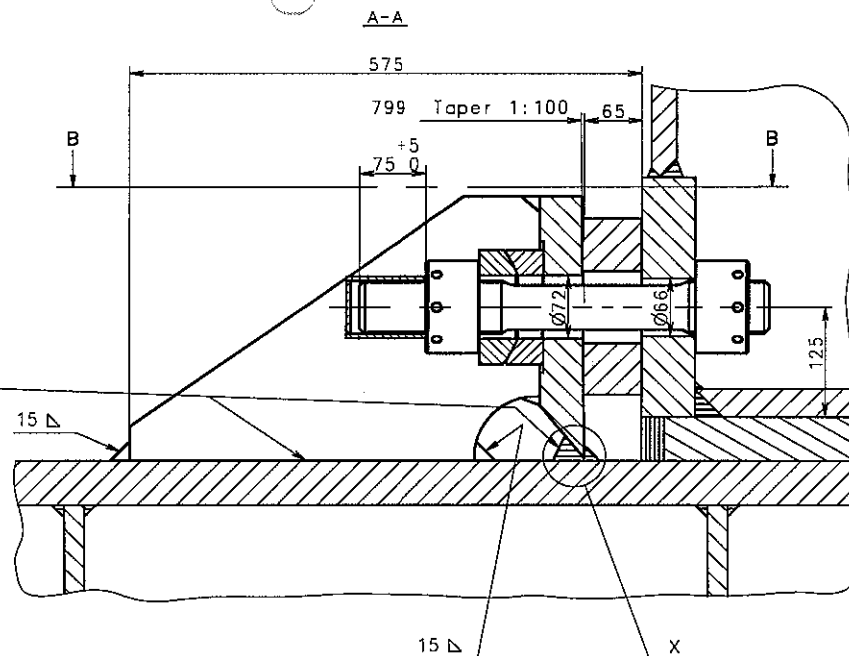
3-3-6

4316

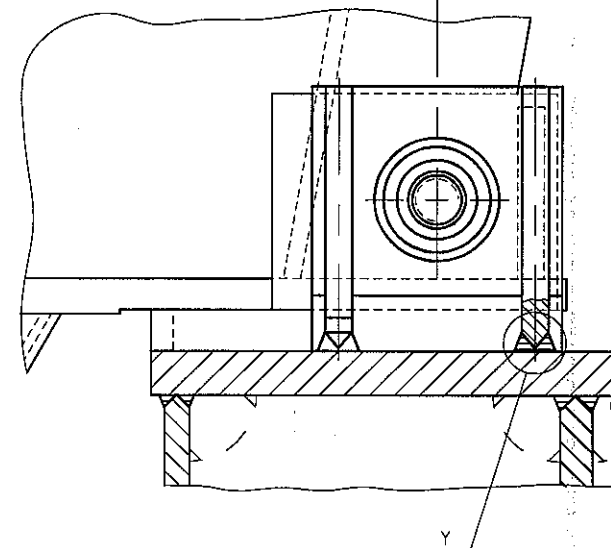
To minimise deformation of end chocks owing to welding stresses, welding to the top plate is carried out as follows: Tack weld the welded and machined end chock to the top plate. When the material has cooled down, carry out welding of all seams on the end chock to the top plate in steps at suitable intervals enabling the material to cool down before the next welding step is started. Welding consumable: Basic electrode.

4067

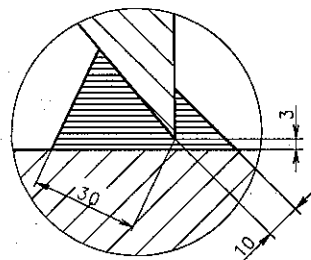
All welding workmanship equal to boiler shop standard e.g. EN287-1.



1430 to centreline engine



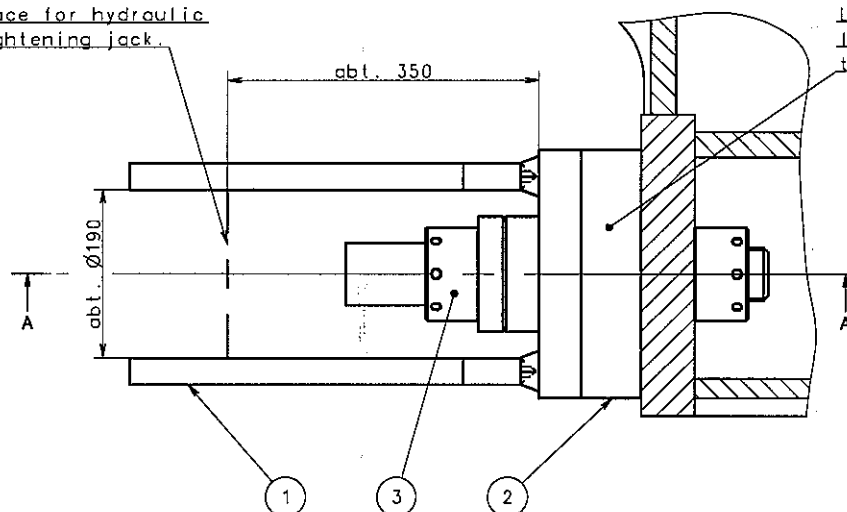
X



4030

Space for hydraulic tightening jack.

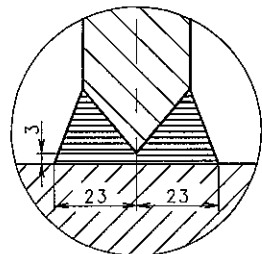
B-B



4318

Mount the liner from top. Liner surfaces must have at least 80% contact. Hammer the liner into place.

Y



1	End chock bolt	3		2805000490	2805000490		15.5
1	Liner for end chock	2		2805000480	2805000480	FC200	21.4
1	End chock	1		2805000470	2805000470	SS400	68.0
Qty	Name of part	Pos	Info no	Part No	Drq. No	Material	kg

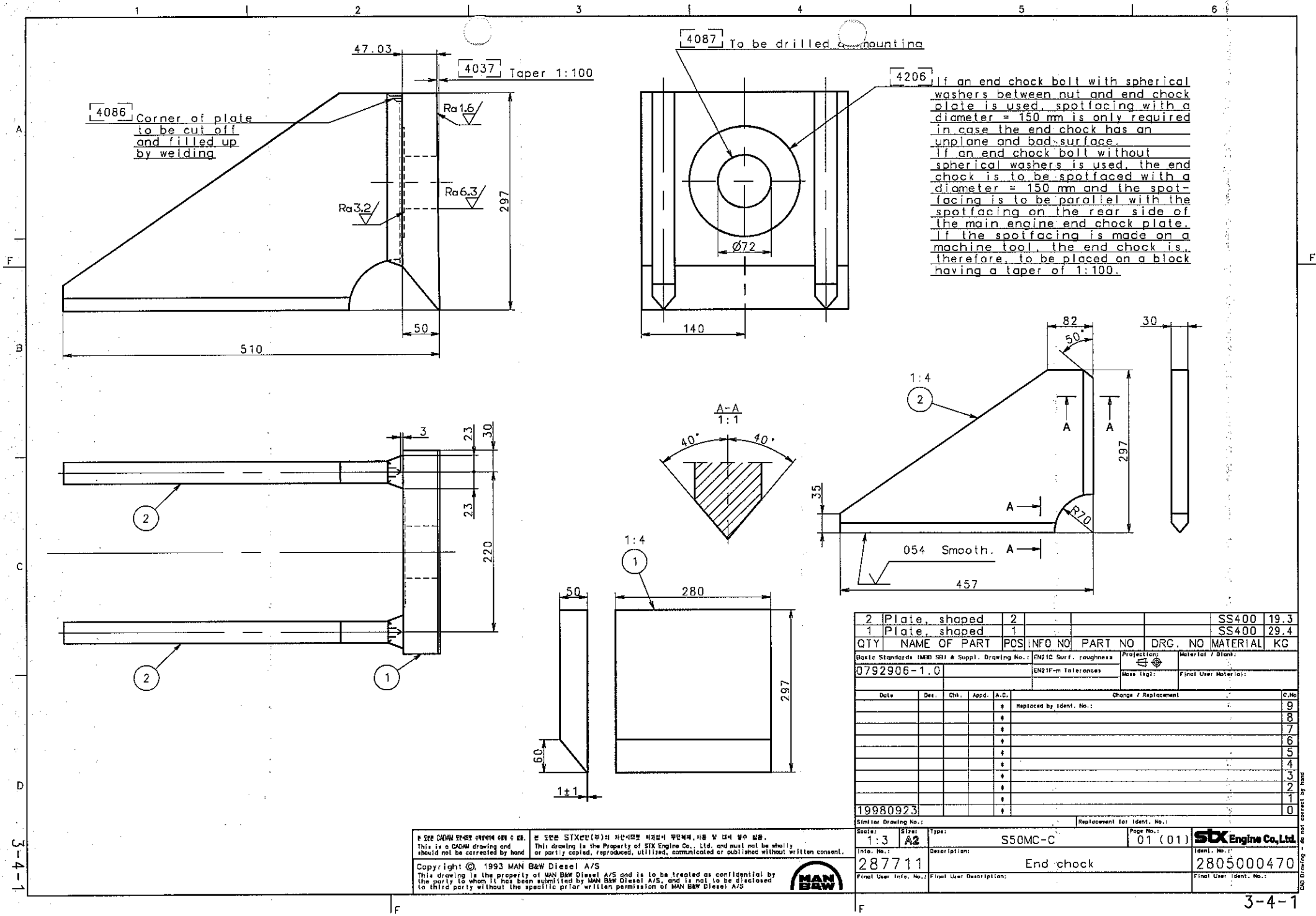
Basic Standards (MBO SB) & Suppl. Drawing No.			EN210 Surf. roughness		Projection		Material / Blank	
0792922-7.1			EN21F-m Tolerances		Moss (tip)		Final User Material	

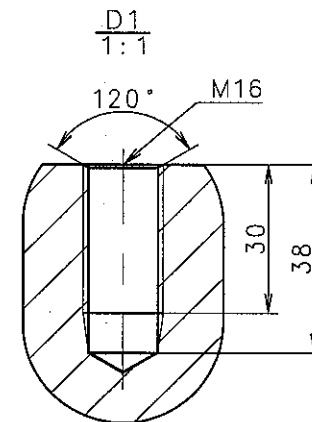
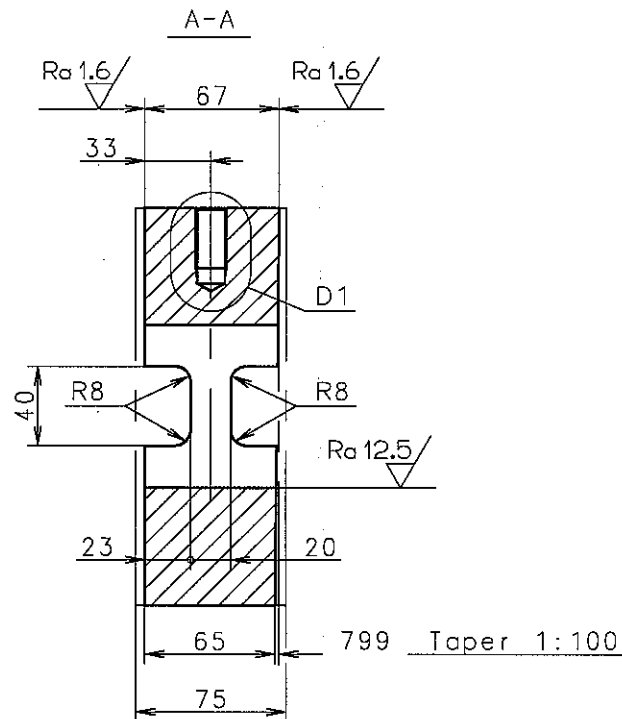
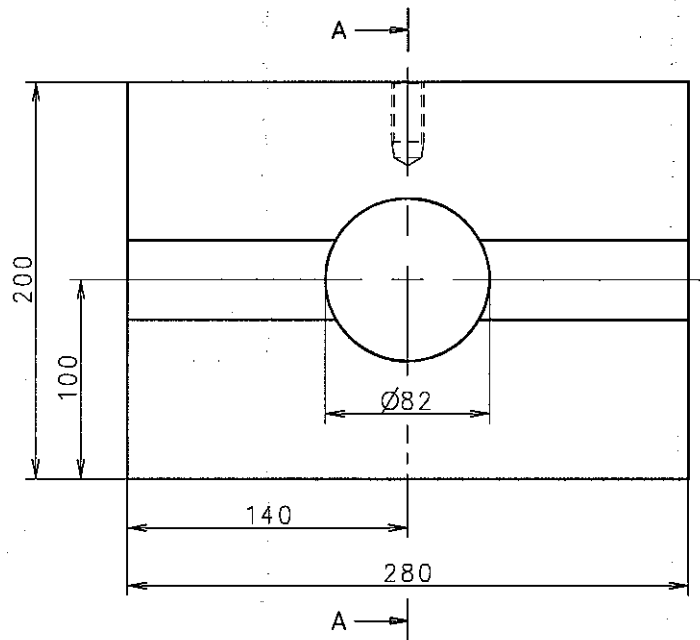
Date	Des.	Dis.	Appd.	A.C.	Change / Replacement	C.No
					Replaced by Ident. No.:	9
						8
						7
						6
						5
						4
						3
						2
						1
19990128						0

Similar Drawing No.:		Replacement for Ident. No.: 0789778-7	
Scale:	Size:	Type:	Page No.:
1:4	A2	S50MC-C	01 (01)
Info. No.:		Description:	
287682		Assembly of end chock	
Final User Info. No.:		Final User Description:	
Ident. No.:		Ident. No.:	
2805000460		2805000460	

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689 Sharp edges to be removed.

054 Smooth. 164 All over, except where otherwise stated.

Basic Standards (MBD SB) & Suppl. Drawing No.:					EN21C Surf. roughness	Projection: 	Material / Blank:	
0789779-9.2					EN21F-m Tolerances	Mass (kg): 21.4	FC200	
							Final User Material:	
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement			C.N
				*	Replaced by Ident. No.:			9
				*				8
				*				7
				*				6
				*				5
				*				4
				*				3
				*				2
				Z4				1
19980907				*				0

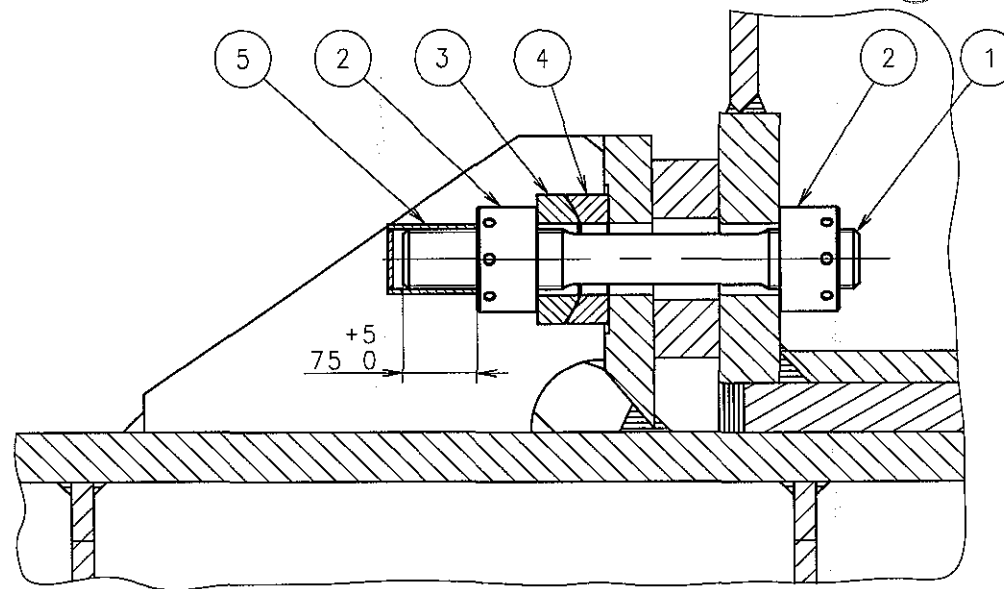
Similar Drawing No.:		Replacement for Ident. No.:	
Scale:	Size:	Type:	Page No.:
1:2.5	A3	S50MC-C	01 (01)
Info. No.:	Description:		Ident. No.:
187740	Liner for end chock		2805000480
Final User Info. No.:	Final User Description:		Final User Ident. No.:

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CAD Drawing - do not correct by hand



4129 Oil pressure for tightening up 1500 bar (with piston area of hydraulic jack = 5640 mm²).

4130 For initial tightening with new bolts the oil pressure is increased to 1650 bar.

1	Protecting cap	5			2805000540	RUBBER	0.129
1	Spherical washer	4			2805000530	S45C	2.57
1	Spherical washer	3			2805000520	S45C	2.55
2	Round nut	2			2805000510	S45C	2.97
1	Stud f. end chock bolt	1			2805000500	SCM435	6.73
QTY	NAME OF PART	POS	INFO NO	PART NO	DRG. NO	MATERIAL	KG
Basic Standards (MBD SB) & Suppl. Drawing No.:			EN21C Surf. roughness		Projection:	Material / Blank:	
0789776-3.0			EN21F-m Tolerances		Mass (kg):	Final User Material:	
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement		C.No
				*	Replaced by Ident. No.:		9
				*			8
				*			7
				*			6
				*			5
				*			4
				*			3
				*			2
				*			1
19980922				*			0

Similar Drawing No.:		Replacement for Ident. No.:	
Scale:	Size:	Type:	Page No.:
1:5	A3	S50MC-C	01 (01)
Info. No.:		Description:	
287680		Assembly of end chock bolt	
Final User Info. No.:		Ident. No.:	
		2805000490	
Final User Description:		Final User Ident. No.:	

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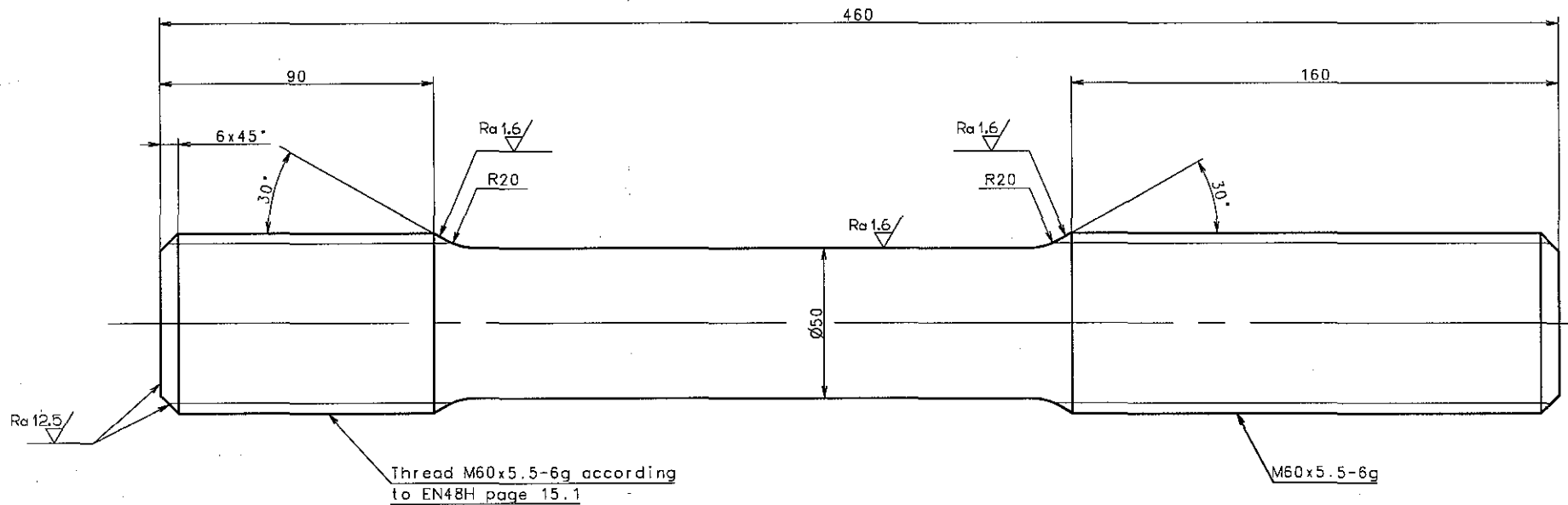
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CAD Drawing - do not correct by hand

1 2 3 4 5 6

A
F
B
C
D
3-4-4



Material specification f. SCM435:

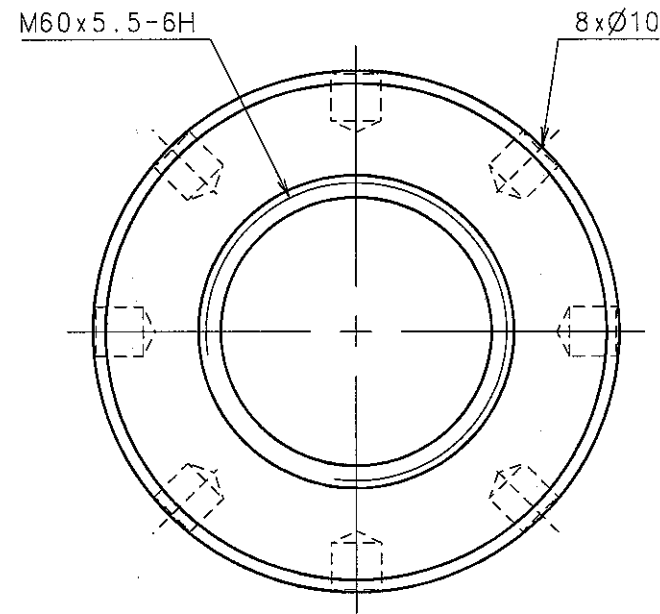
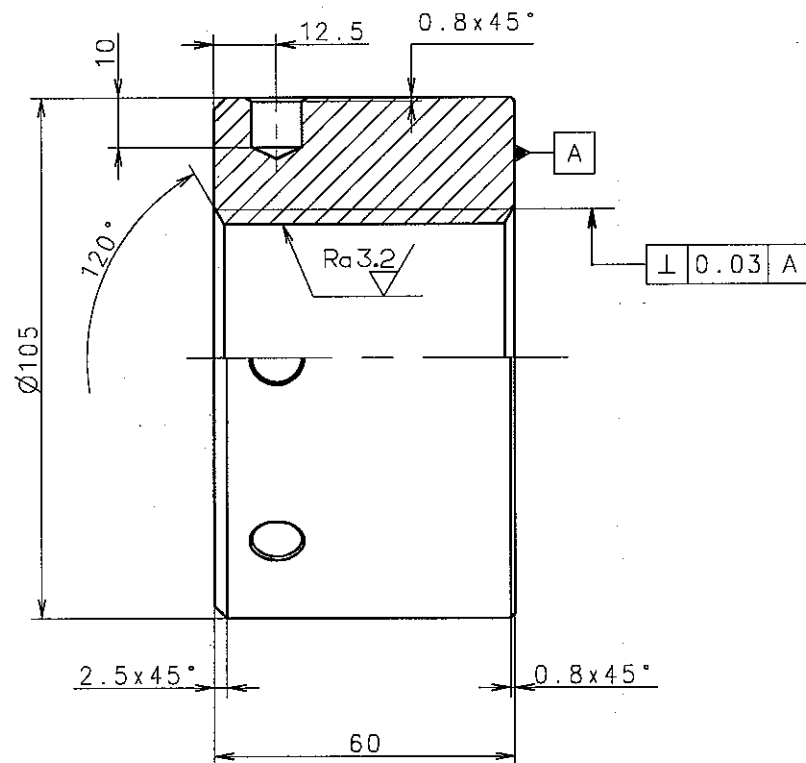
Tensile strength : 780-930 N/mm²
Yield stress : min. 560 N/mm²
Elongation : min. 14%
Reduction of area : min. 50%
Brinell hardness : 230-275 HB
Impact strength : min. 30J

Basic Standards (ISO) & Suppl. Drawing No.: 0792907-3.1		EN21C Surf. roughness		Projection:		Material / Blank: SCM435	
		EN21F-m Tolerances		Mass (kg): 6.73		Final User Material:	
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement		C.No.
					Replaced by Ident. No.:		9
							8
							7
							6
							5
							4
							3
							2
							1
							0
19980922					Z4		
Similar Drawing No.:					Replacement for Ident. No.:		
Scale: 1:1	Size: A2	Type: S50MC-C	Page No.: 01 (01)		SDX Engine Co., Ltd.		
Info. No.: 187681		Description: Stud for end check bolt		Ident. No.: 2805000500		Final User Ident. No.:	
Final User Info. No.:		Final User Description:					

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126 Make or similar: 1421, MAN B&W Diesel A/S

455 Type: EN73R60

Ra 6.3

164 All over, except where otherwise stated.

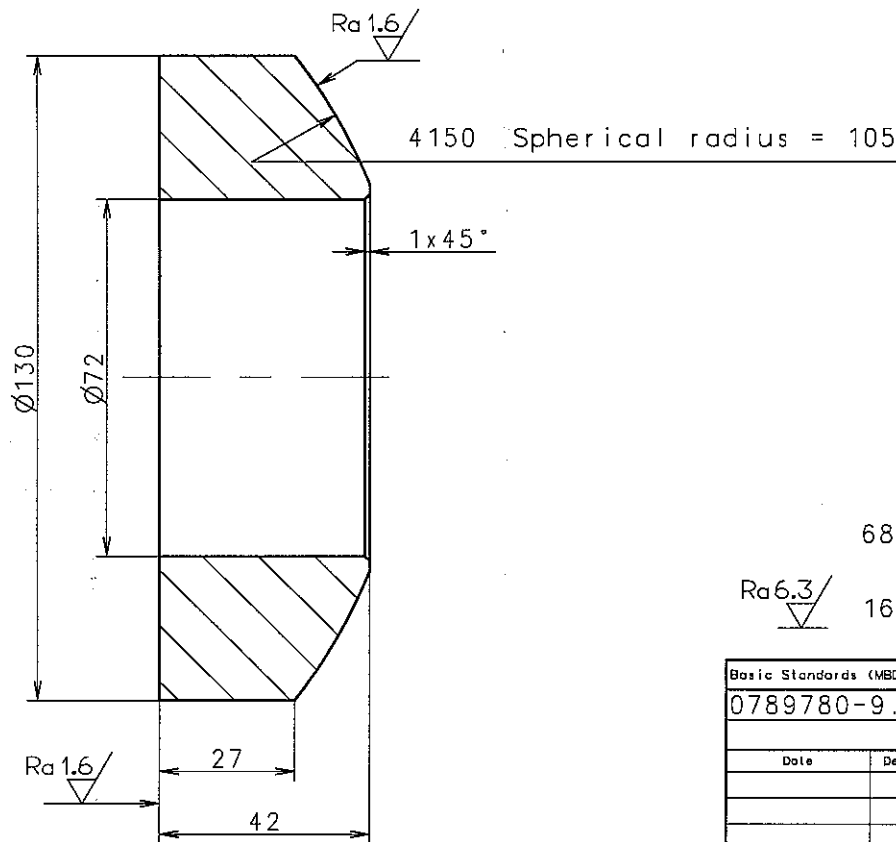
Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:
0789777-5.2		EN21F-m Tolerances		S45C
			Mass (kg):	Final User Material:
			2.97	
Date	Des.	Chk.	Appd.	A.C.
				Change / Replacement
				Replaced by Ident. No.:
				9
				8
				7
				6
				5
				4
				3
				2
				1
				0
19980922				
Similar Drawing No.:		Replacement for Ident. No.: 0788036-5		
Scale:	Size:	Type:	Page No.:	Stx Engine Co., Ltd.
1:1	A3	EN73R60	01 (01)	
Info. No.:		Description:		Ident. No.:
187652		ROUND NUT		2805000510
Final User Info. No.:		Final User Description:		Final User Ident. No.:

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689 Sharp edges to be removed.

Ra6.3/ 164 All over, except where otherwise stated

Basic Standards (MBD SB) & Suppl. Drawing No.:					EN21C Surf. roughness	Projection: 	Material / Blank: S45C	
0789780-9.1					EN21F-m Tolerances	Mass (kg): 2.55	Final User Material:	
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement			C.No
				*	Replaced by Ident. No.:			9
				*				8
				*				7
				*				6
				*				5
				*				4
				*				3
				*				2
				Z4				1
19980922				*				0

Similar Drawing No.:			Replacement for Ident. No.:		
Scale:	Size:	Type:	Page No.:		STX Engine Co., Ltd.
1:1	A3	S50MC-C	01 (01)		
Info. No.:	Description:			Ident. No.:	
187654	Spherical washer			2805000520	
Final User Info. No.:	Final User Description:			Final User Ident. No.:	

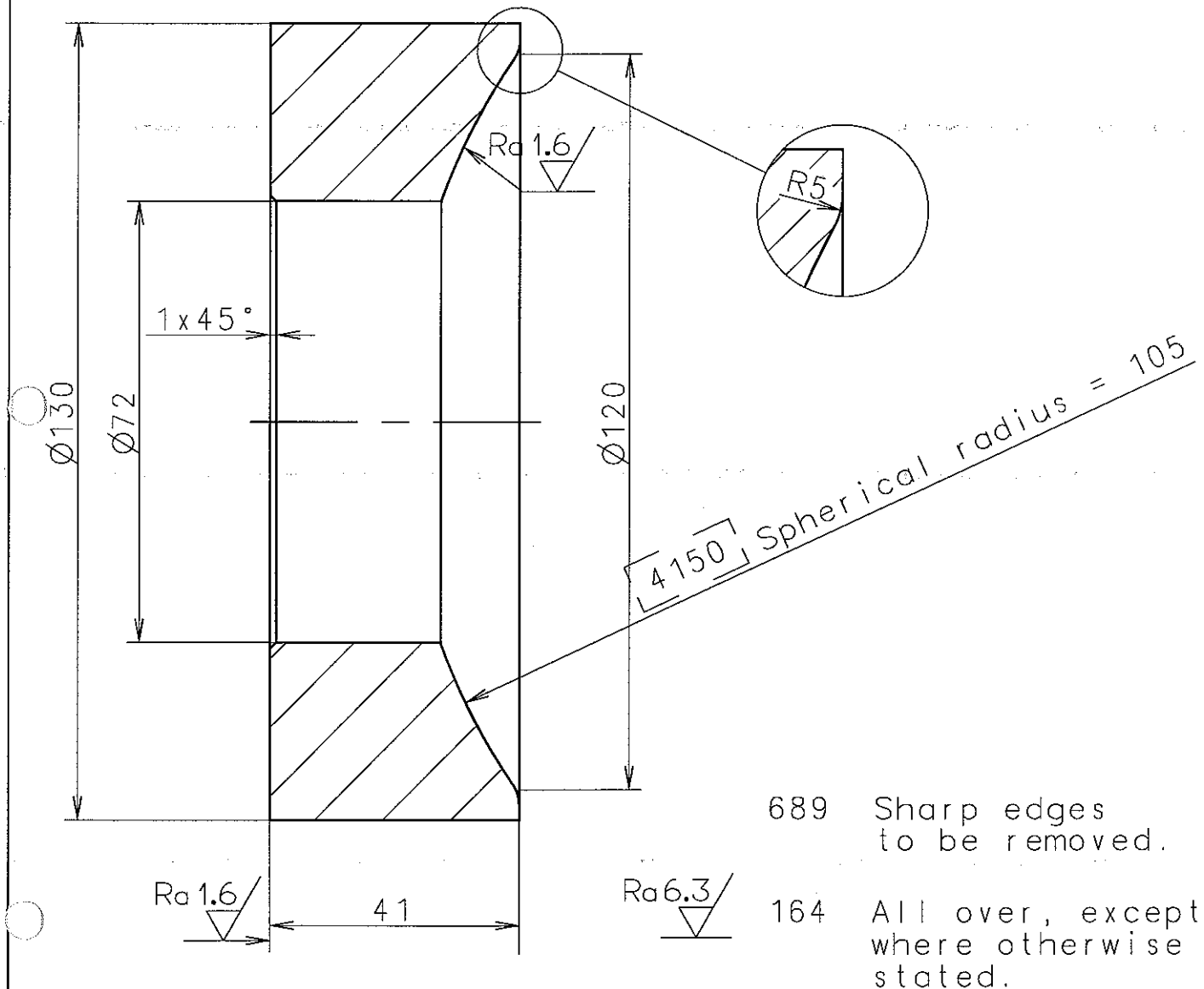
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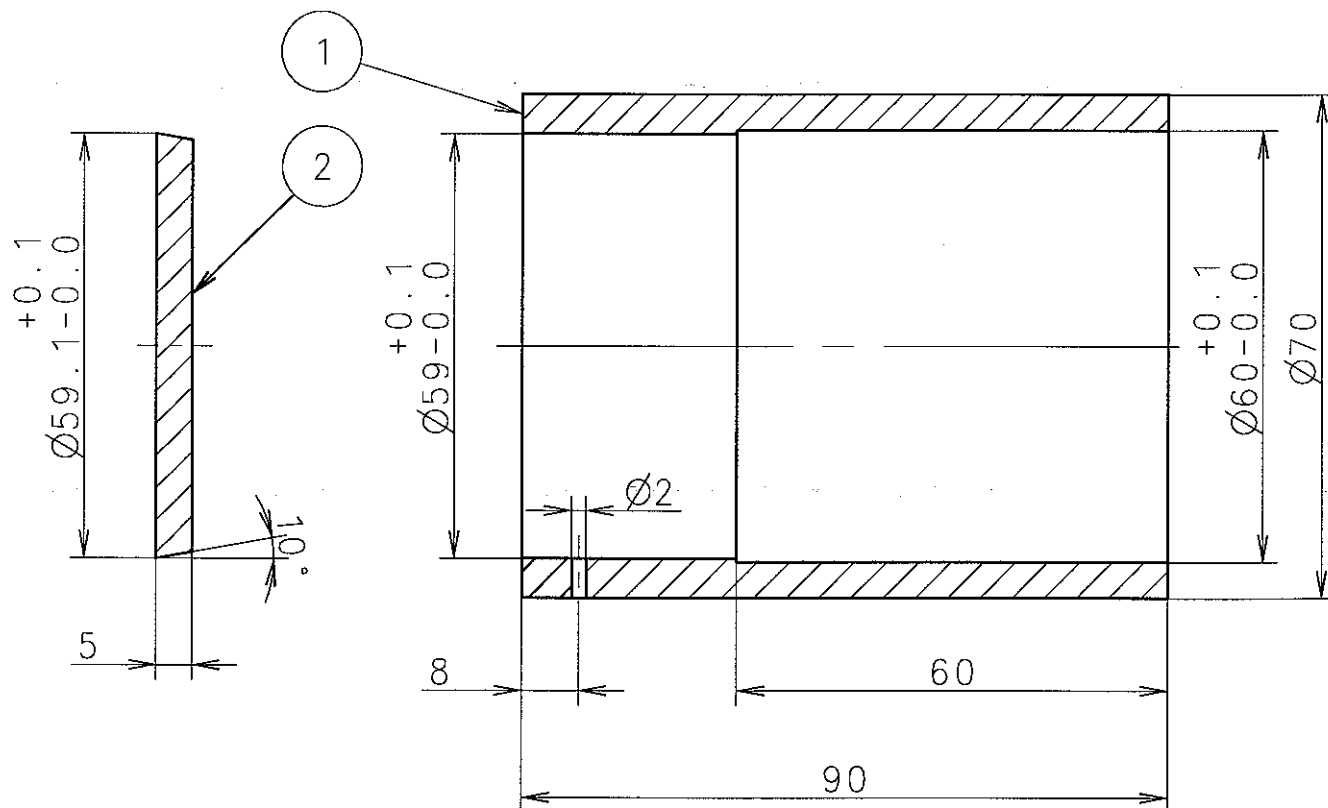
Basic Standards (MBD SB) & Suppl. Drawing No.:				EN21C Surf. roughness		Projection: 		Material / Blank: S45C		
0792064-7.0				EN21F-m Tolerances		Mass (kg): 2.57		Final User Material:		
Date		Des.	Chk.	Appd.	A.C.	Change / Replacement				C.No
					*	Replaced by Ident. No.:				5
					*					4
					*					3
					*					2
					*					1
19980922					*					0
Similar Drawing No.:						Replacement for Ident. No.:				
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 01 (01)		STX Engine Co., Ltd.	
Info. No.: 187654		Description: Spherical washer						Ident. No.: 2805000530		
Final User Info. No.:		Final User Description:						Final User Ident. No.:		



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Cover to be glued into cap.
(With glue for "PA6" plastic).

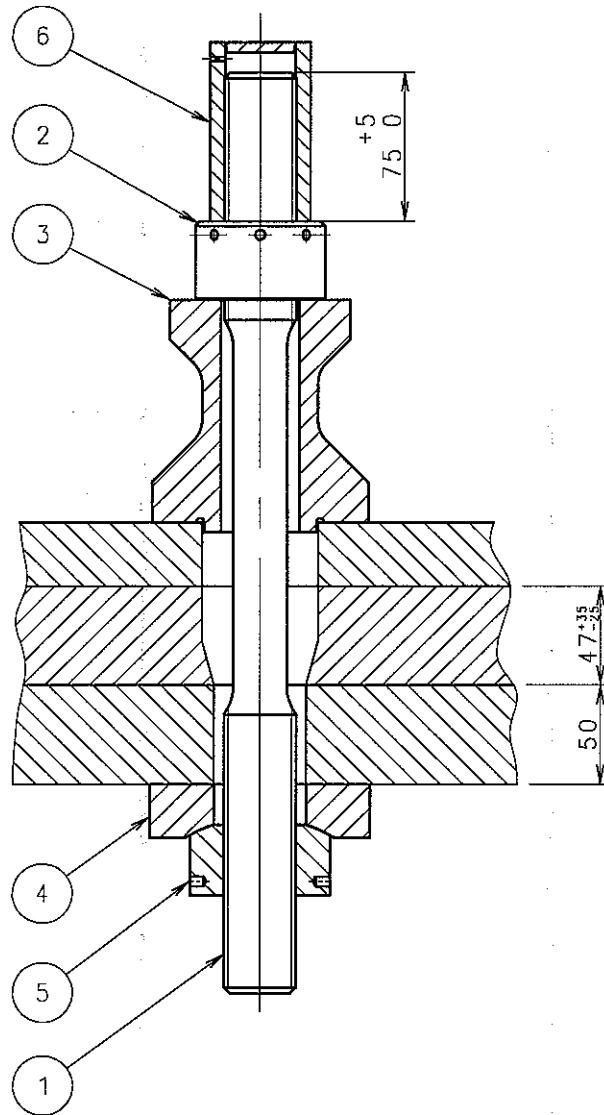


Material 6250 equals "PA6" plastic.

Ra3.2/

164 All over, except where otherwise stated.

01	Cover	2				RUBBER	0.015
1	Cap	1				RUBBER	0.104
QTY	NAME OF PART	POS	INFO NO	PART NO	DRG. NO	MATERIAL	KG
Basic Standards (MBD SB) & Suppl. Drawing No.:			EN21C Surf. roughness		Projection:		Material / Blank:
0787945-4.1			EN21F-m Tolerances				Final User Material:
					Mass (kg):		
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement		C.No
				*	Replaced by Ident. No.:		5
				*			4
				*			3
				*			2
980915				Z4			1
				*			0
Similar Drawing No.:				Replacement for Ident. No.:			
Scale:	Size:	Type:		Page No.:		STX Engine Co., Ltd.	
1:1	A4			01 (01)			
Info. No.:	Description:					Ident. No.:	
287603	Protecting cap					2805000540	
Final User Info. No.:		Final User Description:				Final User Ident. No.:	



4308

Oil pressure for initial and for normal
tightening to be 1500 bar
with piston area of hydraulic jack:
 $A_H = 1885 \text{ mm}^2$
when epoxy supporting chocks are used.

1	Protecting cap	6			2805000610	RUBBER	0.12
1	Spherical nut	5			2805000600	S45C	0.71
1	Spherical washer	4			2805000590	S45C	1.60
1	Distance pipe	3			2805000580	FC300	3.88
1	Round nut	2			2805000570	S45C	0.74
1	Holding down bolt	1			2805000560	SCM435	3.02
QTY	NAME OF PART	POS	INFO NO	PART NO	DRG. NO	MATERIAL	kg

Basic Standards (MBO SB) & Suppl. Drawing No.:	EN21C Surf. roughness	Projection:	Material / Blank:
0789751-1.1	EN21F-m Tolerances	Mass (kg):	Final User Material:

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	9
				*		8
				*		7
				*		6
				*		5
				*		4
				*		3
				*		2
				Z4		1
19971104				*		0

Similar Drawing No.: Replacement for Ident. No.: 0789711-6.

Scale:	Size:	Type:	Page No.:	STX Engine Co., Ltd.
1:2.5	A3	S50MC-C	01 (01)	
Info. No.:	Description:	Ident. No.:		
287602	Assy of holding down bolt	2805000550		
Final User Info. No.:	Final User Description:	Final User Ident. No.:		

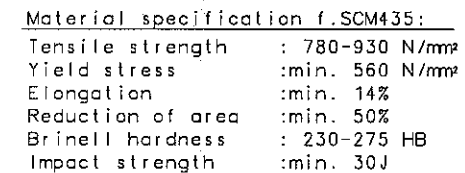
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CAD Drawing - do not correct by hand



Basic Standards (MBC SB) & Suppl. Drawing No.:				EN21C Surf. roughness		Projection: 		Material / Blank:	
0781983-9.1				EN21F-m Tolerances		Mass (kg): 3.02		SCM435 Final User Material:	
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement				C.No.
				†	Replaced by Ident. No.:				9
				†					8
				†					7
				†					6
				†					5
				†					4
				†					3
				†					2
				Z4					1
19971104				†					0
Similar Drawing No.:					Replacement for Ident. No.:				
Serial:	Size:	Type:				Page No.:			
1:1	A2		S50MC-C			01 (01)	 Engine Co., Ltd.		
Info. No.:		Description:				Ident. No.:			
187610		Holding down bolt				2805000560			
Final User Info. No.:		Final User Description:				Final User Ident. No.:			

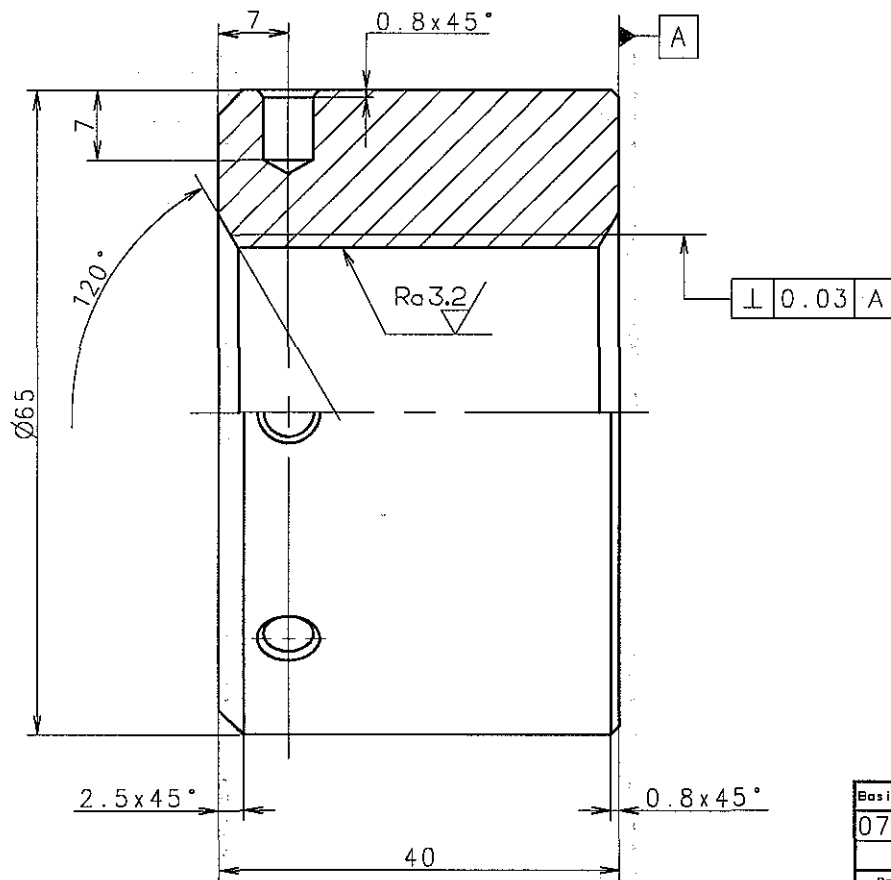
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126 Make or similar: 1421, MAN B&W Diesel A/S

455 Type: EN73R36.

Ra6.3

164 All over, except where otherwise stated.

Basic Standards (MBD SB) & Suppl. Drawing No.:					EN21C Surf. roughness	Projection:	Material / Blank:
0788011-3.4					EN21F-m Tolerances	0.74	S45C
					Mass (kg):	Final User Material:	
					0.74		
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement		
					Replaced by Ident. No.:		
							9
							8
							7
							6
							5
							4
							3
							2
							1
							0
Similar Drawing No.:					Replacement for Ident. No.:		
Scale:	Size:	Type:	EN73R36		Page No.:	01 (01)	
2:1	A3				STX Engine Co., Ltd.		
Info. No.:		Description:			Ident. No.:		
187652		Round nut			2805000570		
Final User Info. No.:		Final User Description:			Final User Ident. No.:		

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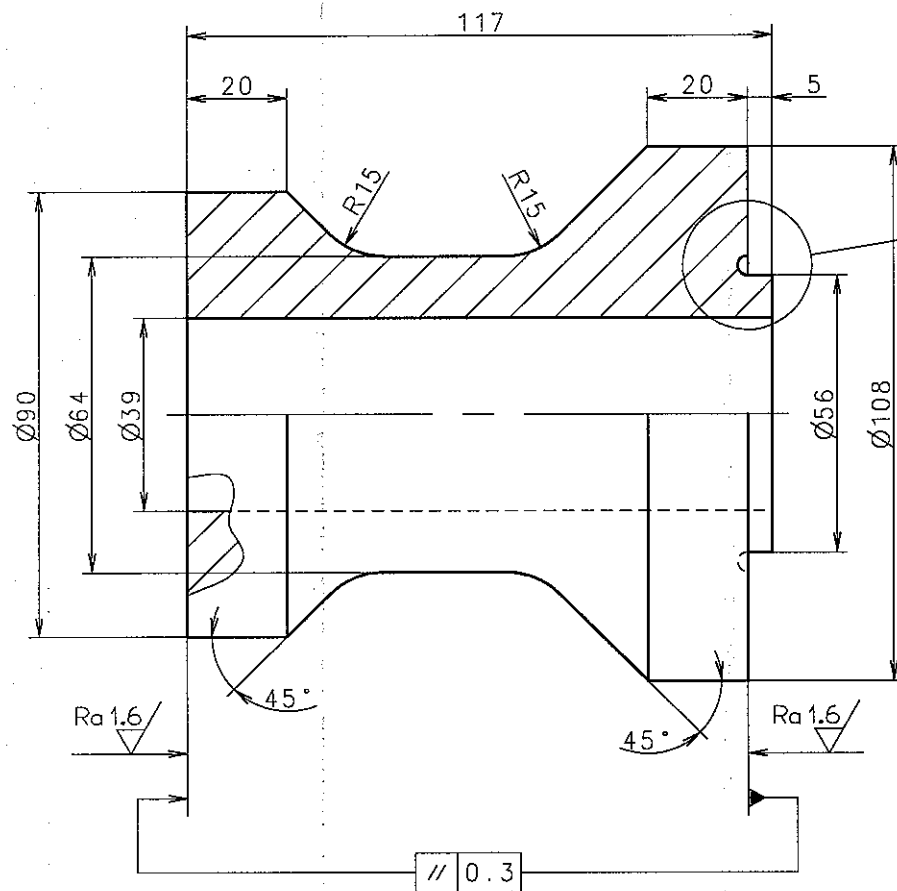
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CAD Drawing - do not correct by hand



689 Sharp edges to be removed.

Ra 25/

164 All over, except where otherwise stated.

Basic Standards (MBO SB) & Suppl. Drawing No.:				EN21C Surf. roughness	Projection: 	Material / Blank:	
0788012-5.4				EN21F-m Tolerances	Mass (kg): 3.88	FC300 Final User Material:	
Date	Des.	Chk.	Appd.	A.C. Change / Replacement			C.No
			*	Replaced by Ident. No.:			9
			*				8
			*				7
			*				6
			*				5
			Z4				4
			Z4				3
			Z4				2
			Z4				1
			*				0

Similar Drawing No.:		Replacement for Ident. No.:	
Scale:	Size:	Type:	Page No.:
1:1	A3	L-S50MC, S50MC-C	01 (01)
Info. No.:		Description:	Ident. No.:
187612		Distance pipe	2805000580
Final User Info. No.:		Final User Description:	Final User Ident. No.:

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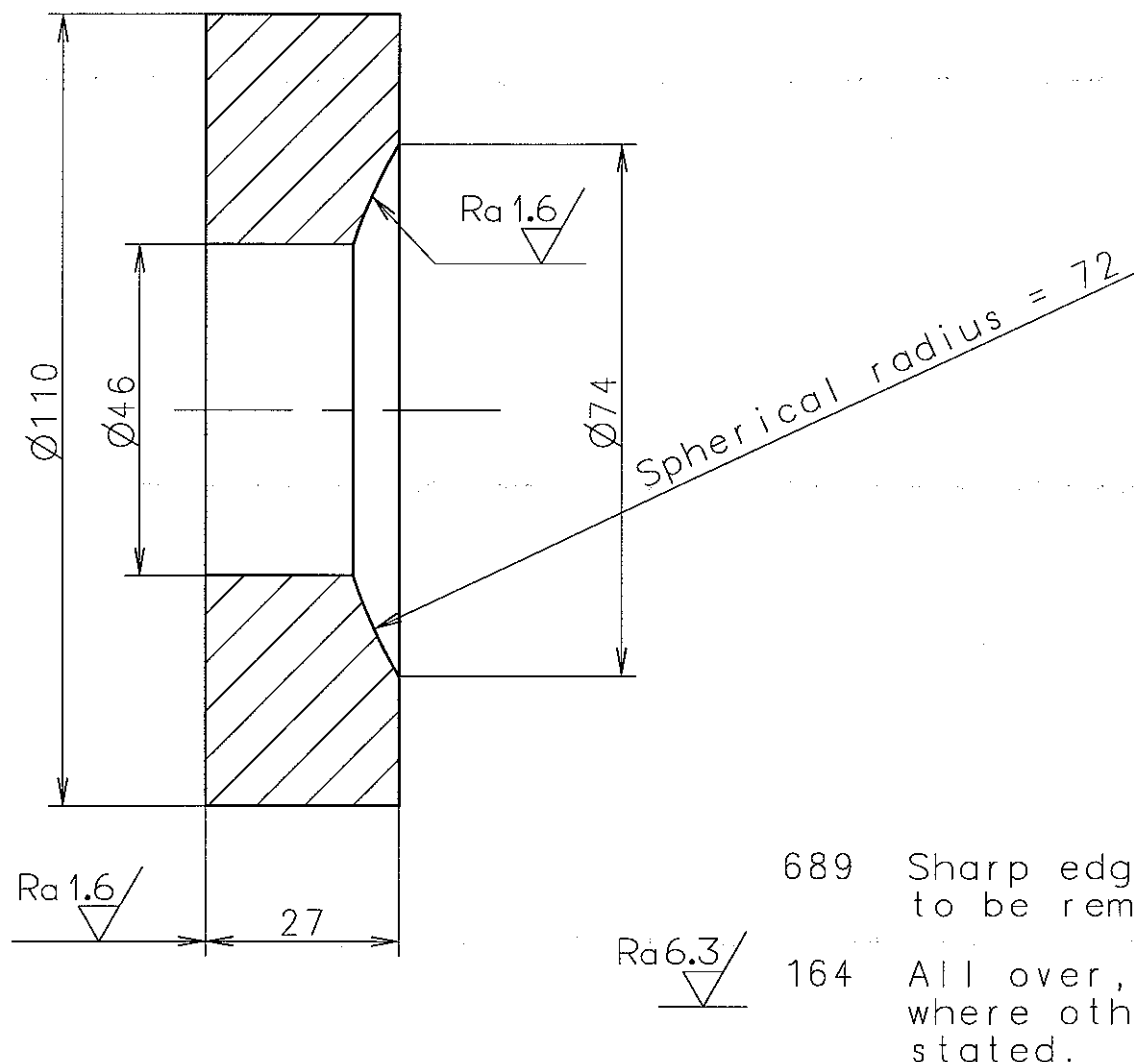
CAD Drawing - do not correct by hand

3-5-3

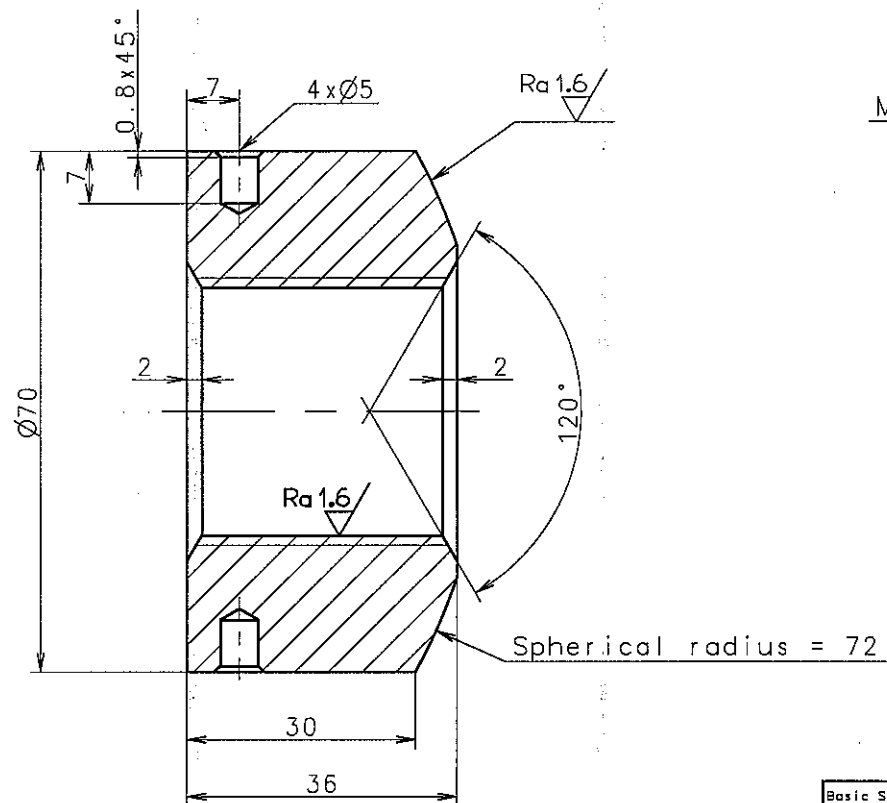


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Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness		Projection:		Material / Blank:		
0787910-6.2		EN21F-m Tolerances				S45C		
				Mass (kg):		Final User Material:		
				1.60				
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement			C.No
				*	Replaced by Ident. No.:			5
				*				4
				*				3
				Z4				2
				Z4				1
941122				*				0
Similar Drawing No.:				Replacement for Ident. No.:				
Scale:	Size:	Type:			Page No.:	STX Engine Co., Ltd.		
1:1	A4	L-S50MC, S50MC-C			01 (01)			
Info. No.:		Description:				Ident. No.:		
187654		Spherical washer				2805000590		
Final User Info. No.:		Final User Description:				Final User Ident. No.:		



689 Sharp edges to be removed.

Ra 6.3/ 164 All over, except where otherwise stated.

Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:
0788013-7.5		EN21F-m Tolerances	First Angle	S45C
			Mass (kg):	Final User Material:
			0.71	
Date	Des.	Chk.	Appd.	A.C.
Change / Replacement				C.No
Replaced by Ident. No.:				9
				8
				7
				6
				5
				4
				3
				2
				1
				0
Similar Drawing No.:		Replacement for Ident. No.:		
Scale:	Size:	Type:	Page No.:	stxEngine Co., Ltd.
~	A3	L-S50MC, S50MC-C	01 (01)	
Info. No.:	Description:		Ident. No.:	
187655	Spherical nut		2805000600	
Final User Info. No.:	Final User Description:		Final User Ident. No.:	

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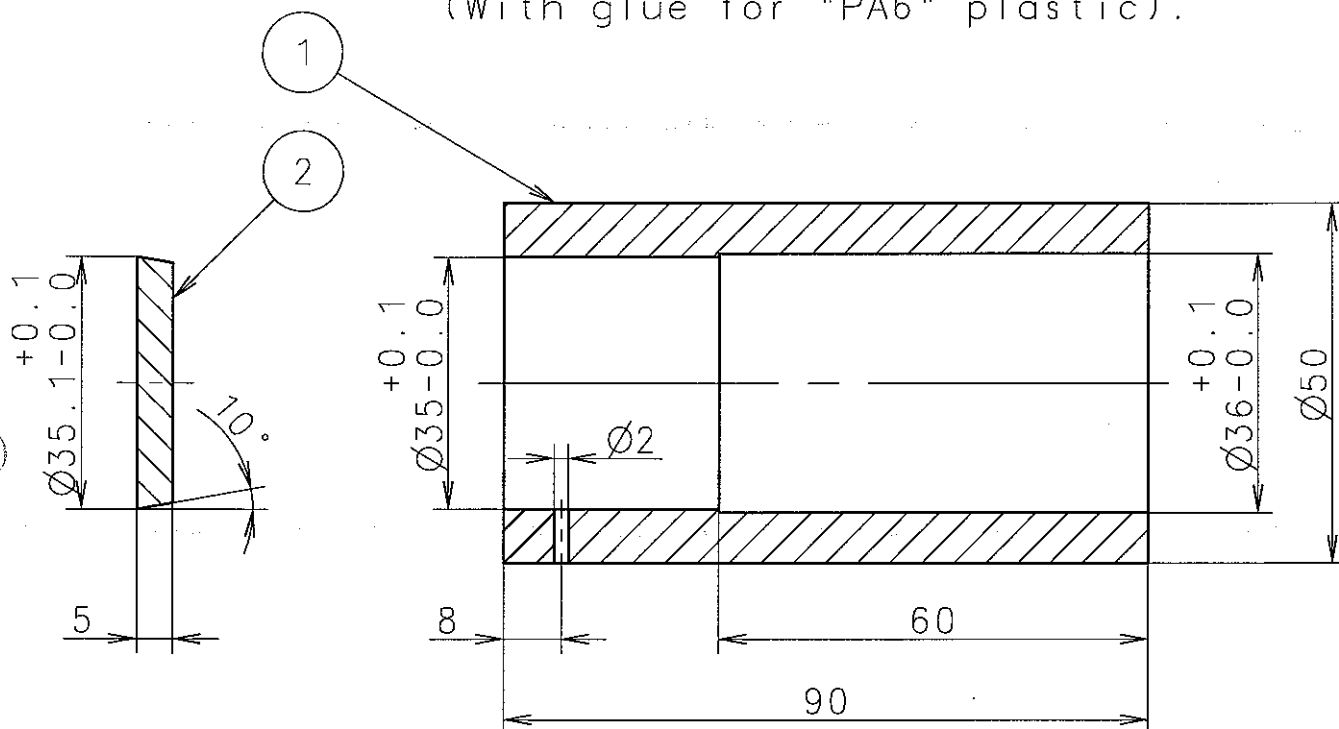
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Cover to be glued into cap.
(With glue for "PA6" plastic).



Material 6250 equals "PA6" plastic. $Ra3.2/\sqrt{163}$ All over.

1	Cover	2				RUBBER	0.010
1	Cap	1				RUBBER	0.114

QTY	NAME OF PART	POS	INFO NO	PART NO	DRG. NO	MATERIAL	KG
-----	--------------	-----	---------	---------	---------	----------	----

Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness		Material / Blank:
0792049-3.0		EN21F-m Tolerances		Final User Material:
			Mass (kg):	

Date	Des.	Chk.	Appd.	A.C.	Change / Replacement	C.No
				*	Replaced by Ident. No.:	5
				*		4
				*		3
				*		2
				*		1
19971103				*		0

Similar Drawing No.:	Replacement for Ident. No.:
----------------------	-----------------------------

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 01 (01)	STX Engine Co., Ltd.
------------	----------	---------------	-------------------	----------------------

Info. No.: 287690	Description: Protecting cap (M36x4)	Ident. No.: 2805000610
-------------------	-------------------------------------	------------------------

Final User Info. No.:	Final User Description:	Final User Ident. No.:
-----------------------	-------------------------	------------------------

595 Thread and contact face
for nuts/bolts have to be
greased with Molycote
before tightening-up.

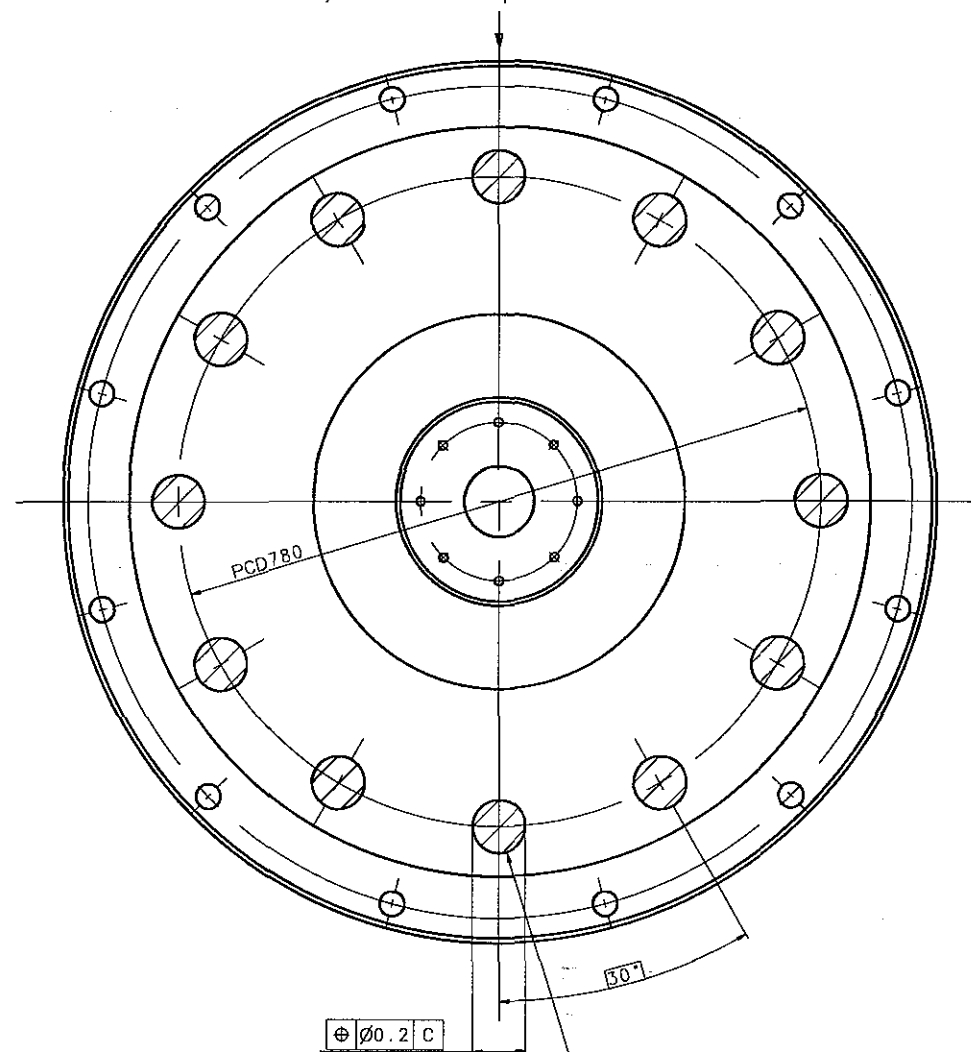
453 Tightening-up torque 4000 Nm

or

488 Proceed in indicated sequence:

1. 453 Tightening-up torque 100 Nm
2. 460 Tightening-up angle 25°.

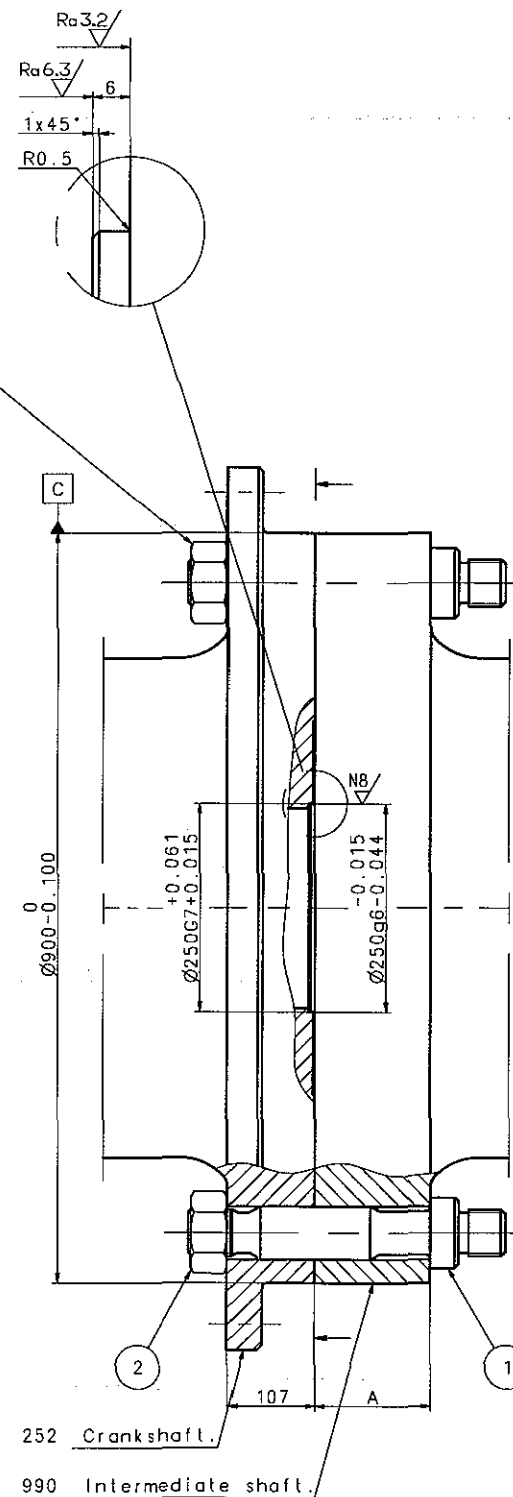
Cyl. 1 in top dead centre



1095 12 holes for 64mm
cylindrical fitted bolts.

147 FINAL-Drill Ø64H7

1053 To be reamed Ø64 together with crankshaft.



252 Crankshaft.

990 Intermediate shaft.

FITTEDBOLT & NUT : SHAFT MAKER OR SHIPYARD SUPPLY

12	Nut	2					17.0
12	Fitted bolt	1					107.5
Qty	Name of part	Pos	Info.no.	Part no.	Drg.no.	Material	Mass
Basic Standards (ISO 52) & Suppl. Drawing No.:			EN21C Surf. roughness		Projection: 	Material / Blank:	
1170655-4.2			EN21F-m Tolerances		Mass (kg):	Final User Material:	
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement		C.No.
				0	Replaced by Ident. No.:		9
				0			8
				0			7
				0			6
				0			5
				0			4
				0			3
				0			2
				0			1
				0			0
20090205 SML JTS TBK			* First issued				
Similar Drawing No.:					Replacement for Ident. No.:		
Scale:	Size:	Type:	S50MC-C		Page No.:	01 (01) 주식회사 STX	
1:1.25	A0						
Info. No.:		Description:			Ident. No.:		
200116		Shaft line			2805029430		
Final User Info. No.:		Final User Description:			Final User Ident. No.:		

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Exhaust side



1

5



1

К

1

Calculation of Top Bracing Length

In order to determinate the individual length L of each top bracing bar, the procedure below can be used...

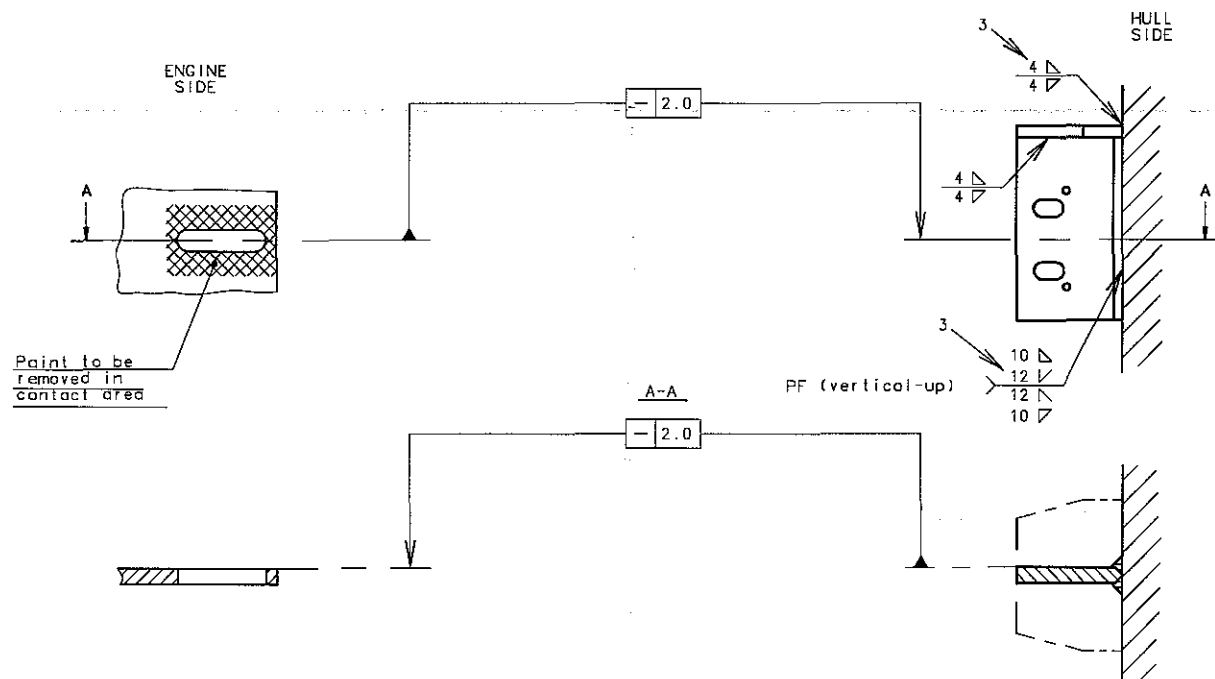
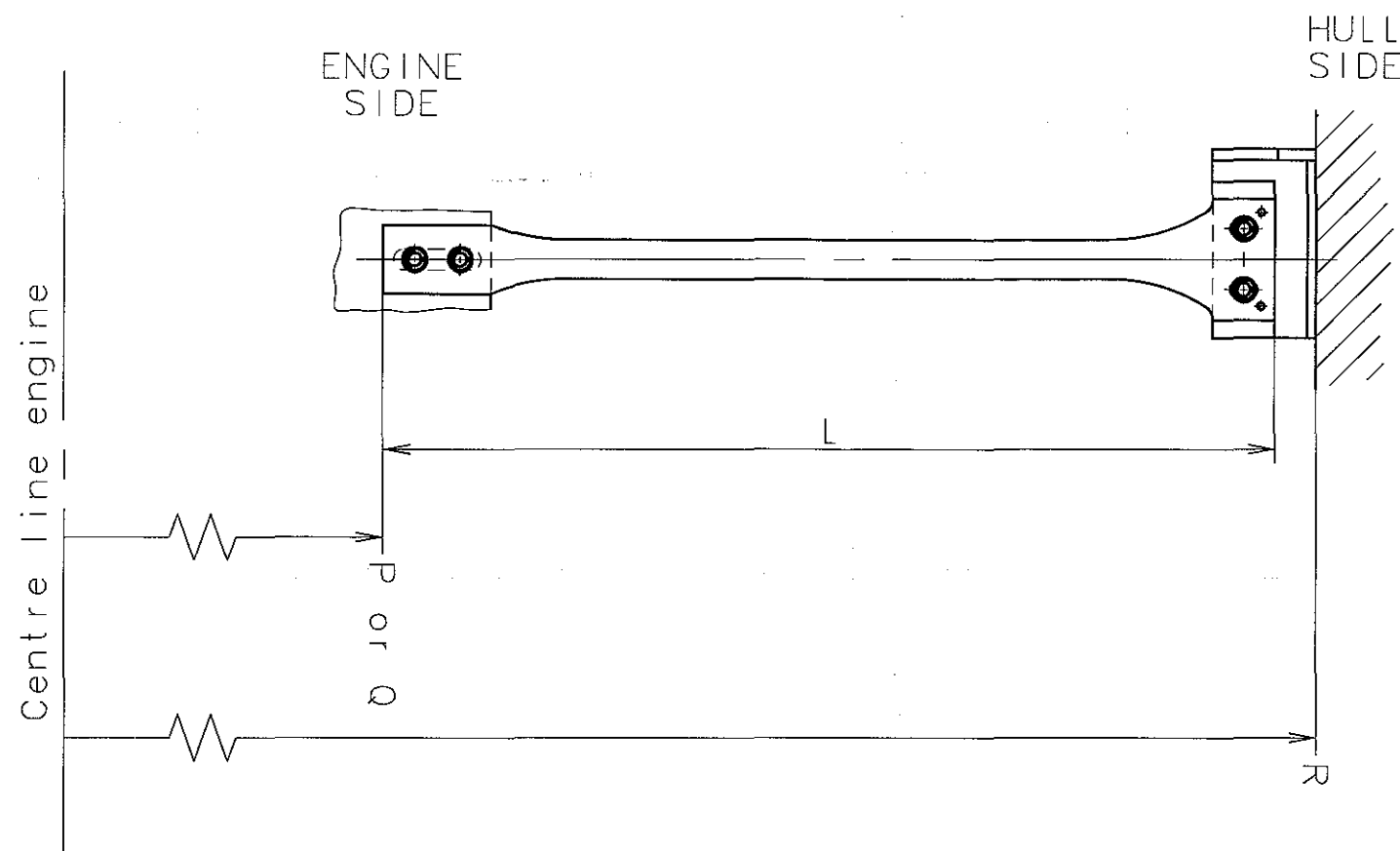
When the length L is determined the relevant outline drawing, short, medium or long can be chosen.

$L = R - P - 90$ (LONG) For top bracing attached at point P on the engine

$L = R - Q - 90$ (SHORT) For top bracing attached at point Q on the engine

The figures P and Q on engine side are the distances from engine centre line to point of attachment on engine side. Please see top bracing arrangement drawing.

R is the distance from the engine centre line to point of attachment at hull side.



Basic Standards (MBD SB) & Suppl. Drawing No.:				EN21C Surf. roughness	Projection: 	Material / Blank:	
0787696-1.3				EN21F-m Tolerances	Mass (kg):	Final User Material:	
0793622-5.0							
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement		C.N.
				*	Replaced by Ident. No.:		9
				*			8
				*			7
				*			6
				*			5
				*			4
				Z4			3
				Z4			2
				Z4			1
19980415				*			0

Similar Drawing No.:			Replacement for Ident. No.:	
Scale: 1:10	Size: A3	Type: L50MC/S50MC-C	Page No.: 01 (01)	STX Engine Co.,Ltd
Info. No.: 384606		Description: Preparations f. mech. topbracing		
Final User Info. No.:		Final User Description:		Final User Ident. No.: 3-7-1

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Transverse top bracing

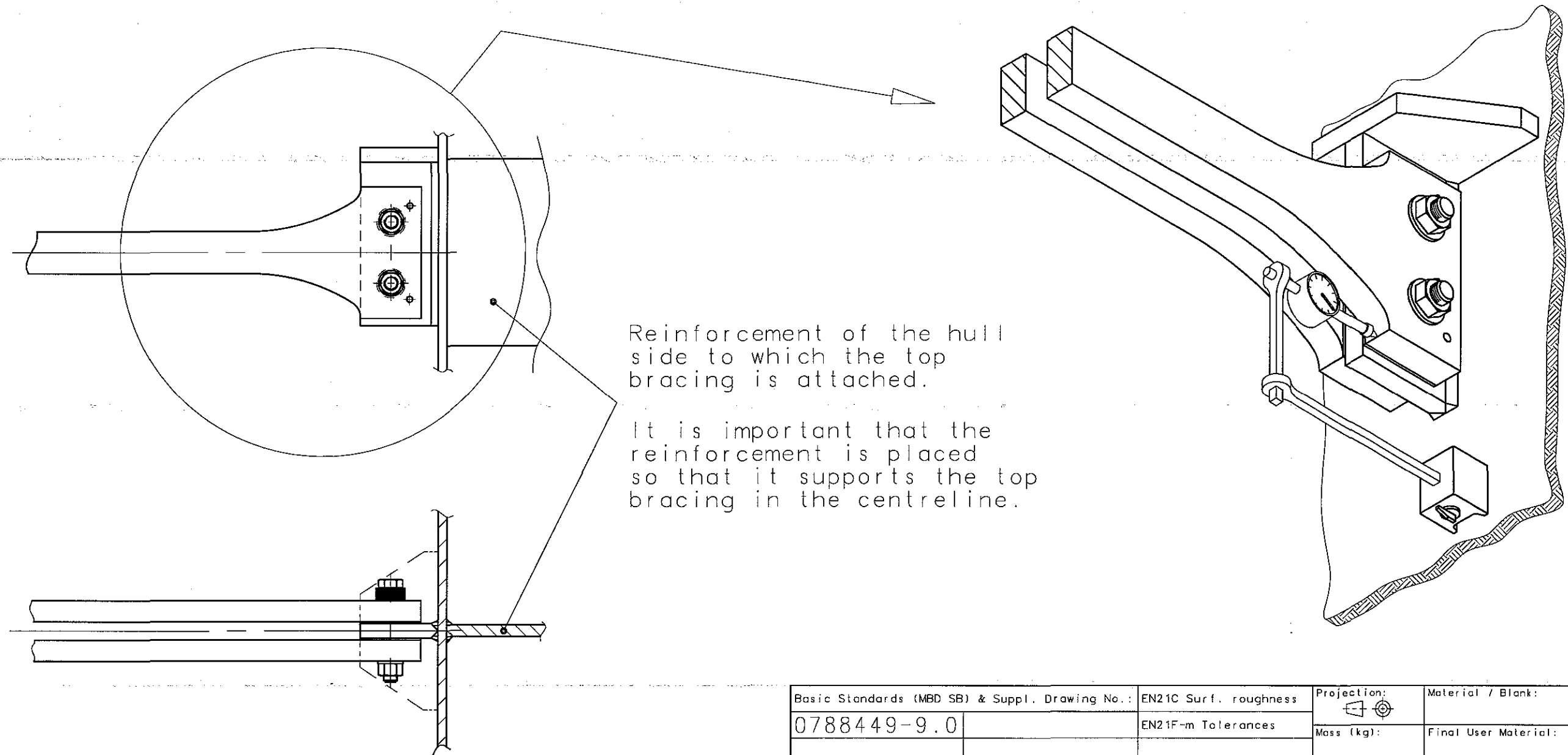
Engine bore	a)	b)
	± kN	MN/m
98	248	230
90	209	210
80	165	190
70	126	170
60	93	140
50	64	120
46	55	110
42	45	100
35	32	85

Forces from engine and minimum required rigidity of hull side.

a): Force per bracing at specified number.

b): Minimum horizontal rigidity.
Horizontal installation of the Top bracing is required.

Basic Standards (MBD SB) & Suppl. Drawing No.:					EN21C Surf. roughness	Projection:	Material / Blank:	
0792027-7.0					EN21F-m Tolerances			
						Mass (kg):	Final User Material:	
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement			C.No
				*	Replaced by Ident. No.:			5
				*				4
				*				3
				*				2
				*				1
19970814				*				0
Similar Drawing No.:					Replacement for Ident. No.:			
Scale:	Size:	Type:			Page No.:		STX Engine Co., Ltd.	
	A4	K-L-S/MC/MC-C/ME-C			01 (01)			
Info. No.:		Description:					Ident. No.:	
384616		Top bracing-forces					2805000650	
Final User Info. No.:		Final User Description:					Final User Ident. No.:	



Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness		Projection:		Material / Blank:	
0788449-9.0		EN21F-m Tolerances		Mass (kg):		Final User Material:	
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement		C.No
				*	Replaced by Ident. No.:		9
				*			8
				*			7
				*			6
				*			5
				*			4
				*			3
				*			2
				*			1
19971112				*			0
Similar Drawing No.:					Replacement for Ident. No.:		
Scale:	Size:	Type:		Page No.:		stx Engine Co., Ltd.	
~	A3			01 (01)			
Info. No.:		Description:				Ident. No.:	
384608		Data for top bracing				2805000660	
Final User Info. No.:		Final User Description:				Final User Ident. No.:	
						3-7-3	

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Bolts for top bracing are to be tightened by means of a torque wrench.

The bolts at the engine side are to be tightened to 470 Nm. The bolts at the hull side are to be tightened to 120 Nm.

When during the trial trip the engine has reached its working temperature, the two bolts for the frictional assembly of the top bracing at the hull side are to be loosened. After approximately one minute the bolts are tightened again. This procedure is to be carried out for each individual top bracing.

Make sure that the beams are supported in the vertical direction during this operation.

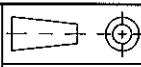
The tightening should be checked as follows:
Check if relative movements occur between top bracing and fastening plate (casing side or girder). This can be carried out by placing a dial gauge (or similar) as shown on drwg. No. 2805000660. Carry out checking of the top bracing at the forward end, the centre, and the aft end of the engine.

If relative movements larger than ± 0.02 mm are ascertained, increase the tightening torque at the bolts at the hull side to 215 Nm (at all top bracings). At the same time, observe at which tightening torque the nuts are loosened at all top bracings. If the relative movement (after having increased the tightening torque) has still not disappeared, increase the tightening torque to 310 Nm, and again observe the loosening torque at each bolt.

After some time in service top bracing might become ineffective (due to setting or wear of the friction material). The tightening should therefore be checked (as described above) if:

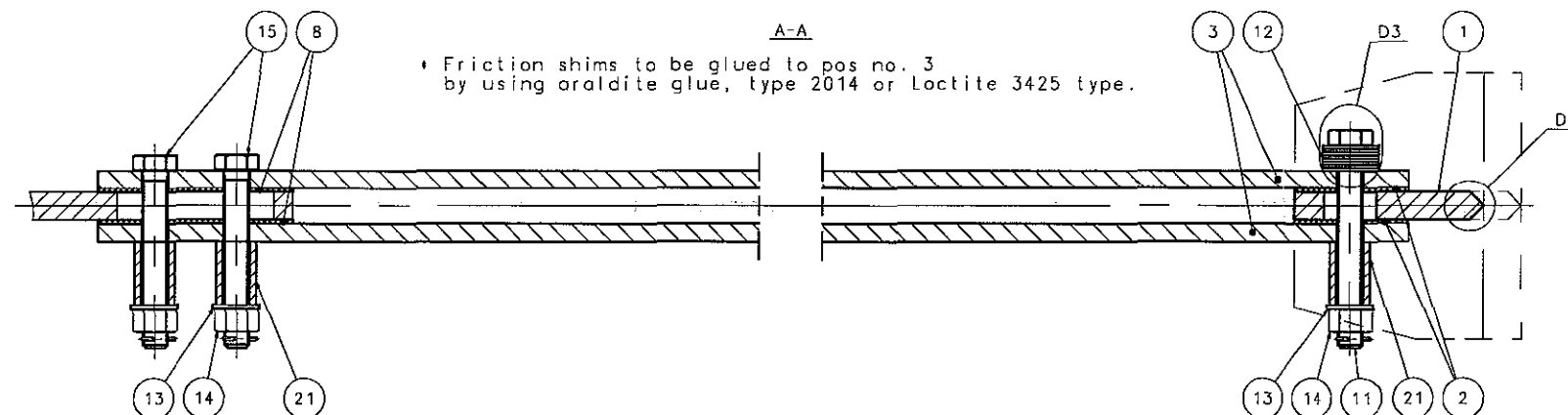
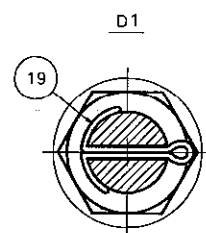
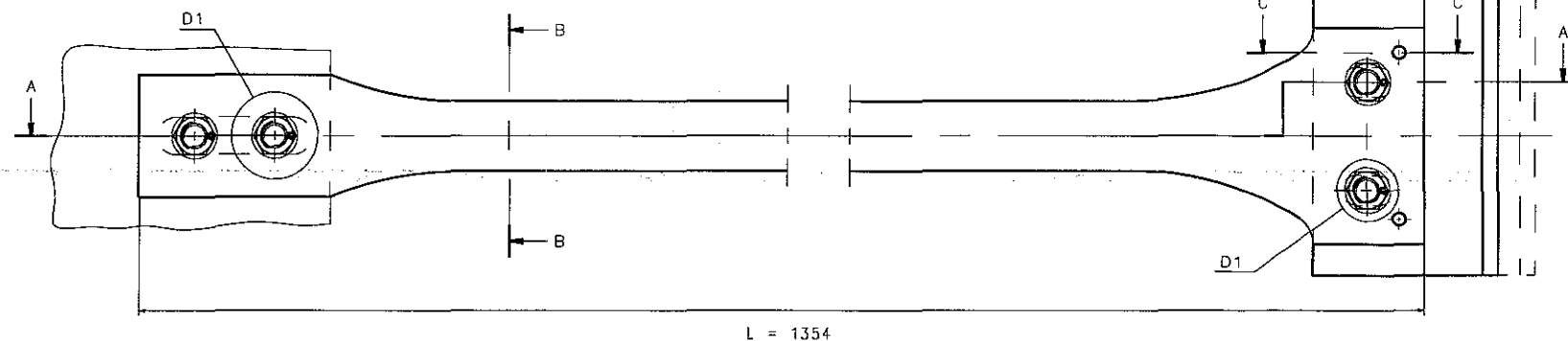
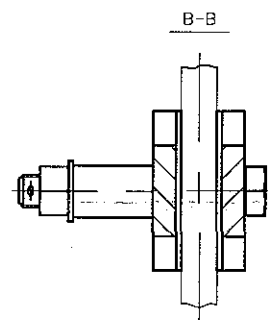
- 1) Unexpected change of hull level - vibrations are observed,
or
- 2) The turbocharger(s) start to vibrate intensely.

If the above-mentioned is not observed, the checking should be carried out once or twice a year.

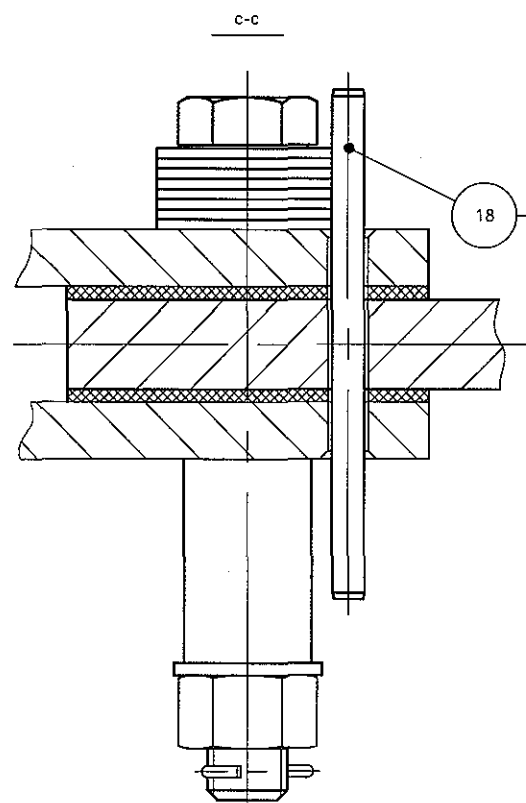
GENERAL TOLERANCE FOR MACHINED MEASURES			REV	LOC	DATE	MADE	APPD	NOTE	
			MATERIAL			SCALE		 CAD	
NOMINAL MEASURE		TOLERANCES				WEIGHT		kg	
OVER	MAX.							S50MC-C	
1	4	± 0.1	DWN	20020329	K.W.SEOL	SIZE	TOP BRACING INSTRUCTIONS		
4	16	± 0.2	CHKD	20020329	H.G.KIM	A4			
16	63	± 0.3	APPD	20020329	J.S.PARK				
63	250	± 0.5					DRAWING NO.		
250	1000	± 0.8					2805000670		

NOTE

1. 각 SET 별로 ITE LIST 첨부하여 견고히 PACKING 할 것.
2. ASS'Y 도면 참조하여 용접부위 선가공 할 것.
3. 용접부를 제외한 모든 PLATE에 한색으로 도색하여 납품 할 것.



* Friction shims to be glued to pos no. 3
by using araldite glue, type 2014 or Loctite 3425 type.

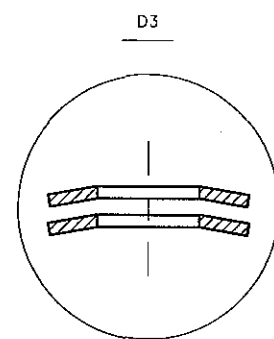


pins to be installed in
the friction connection
during mounting.
THE PINS MUST BE REMOVED
AFTER MOUNTING IS
COMPLETED.

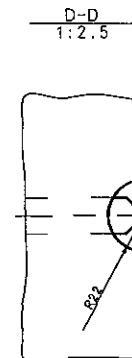
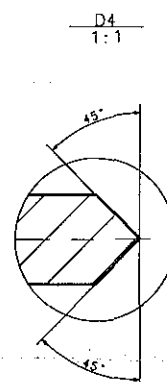
Item No. 1 and 9: Delivered
length = 180mm.
To be adapted by mounting,
though minimum 120mm.

Regarding item No. 1:
Carry out the chamfered
detail D4 after fitting.

Regarding item No. 9:
Carry out the circular cut
section D-D after fitting.



The disc
springs
are to be
positioned
front to
back.



For tightening-up torque of all nuts, see instruction.

NO.	DWG NO.	NAME	MATERIAL	Q'TY	WEIGHT	REMARKS
21	2805000830	SPACER TUBULAR	STS480	4		SHORT
19	2805000820	SPLIT PIN	SM400B	4		
18	2805000810	GUIDE PIN	SS400	2		
15	2805000800	HEX. HEAD SCREW	SCM440	2		
14	2805000790	HEX. NUT	8.8T	4		
13	2805000780	WASHER	SM400B	4		
12	2805000770	DISC SPRING	SUP10	16		
11	2805000760	HEX. HEAD SCREW	SCM440	2		
9	2805000750	PLATE	SM520B	1		
8	2805000740	FRICT. SHIM	NON-ASBESTOS	2		
3	2802042420	PLATE	SM520B	2		MEDIUM
2	2805000700	FRICT. SHIM	NON-ASBESTOS	2		
1	2805000690	PLATE	SM400B	1		

Class No.	N4	N5	N6	N7	N8	N9
Re Wt. Value(m)	0.2	0.4	0.8	1.6	3.2	6.3
	250	1000	2500	10000	25000	100000

GENERAL TOLERANCE FOR MACHINED MEASURES	REY	LOC	DATE	MADE	APPD	NOTE
NOMINAL MEASURE						
TOLERANCES						
OVER	MAX.					
1	4	±0.1				
4	16	±0.2				
15	63	±0.3				
63	250	±0.5				
250	1000	±0.8				

SCALE	DATE	MADE	APPD	NOTE
N / S				
DWN	20090610	S.M.LEE		
CHKD	20090610	J.T.SEO		
APPD	20090610	T.B.KIM		

NAME	SIZE	MODEL
TOP BRACING OUTLINE	A2	SS0MC-C

NAME	SIZE	MODEL
STX Engine Co., Ltd.		

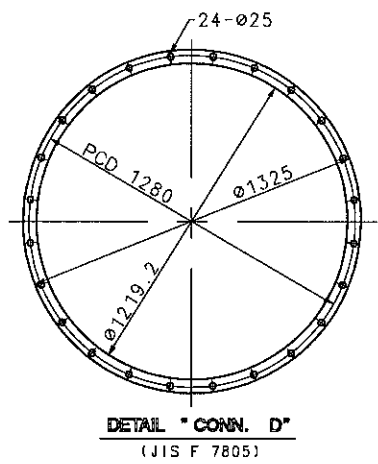
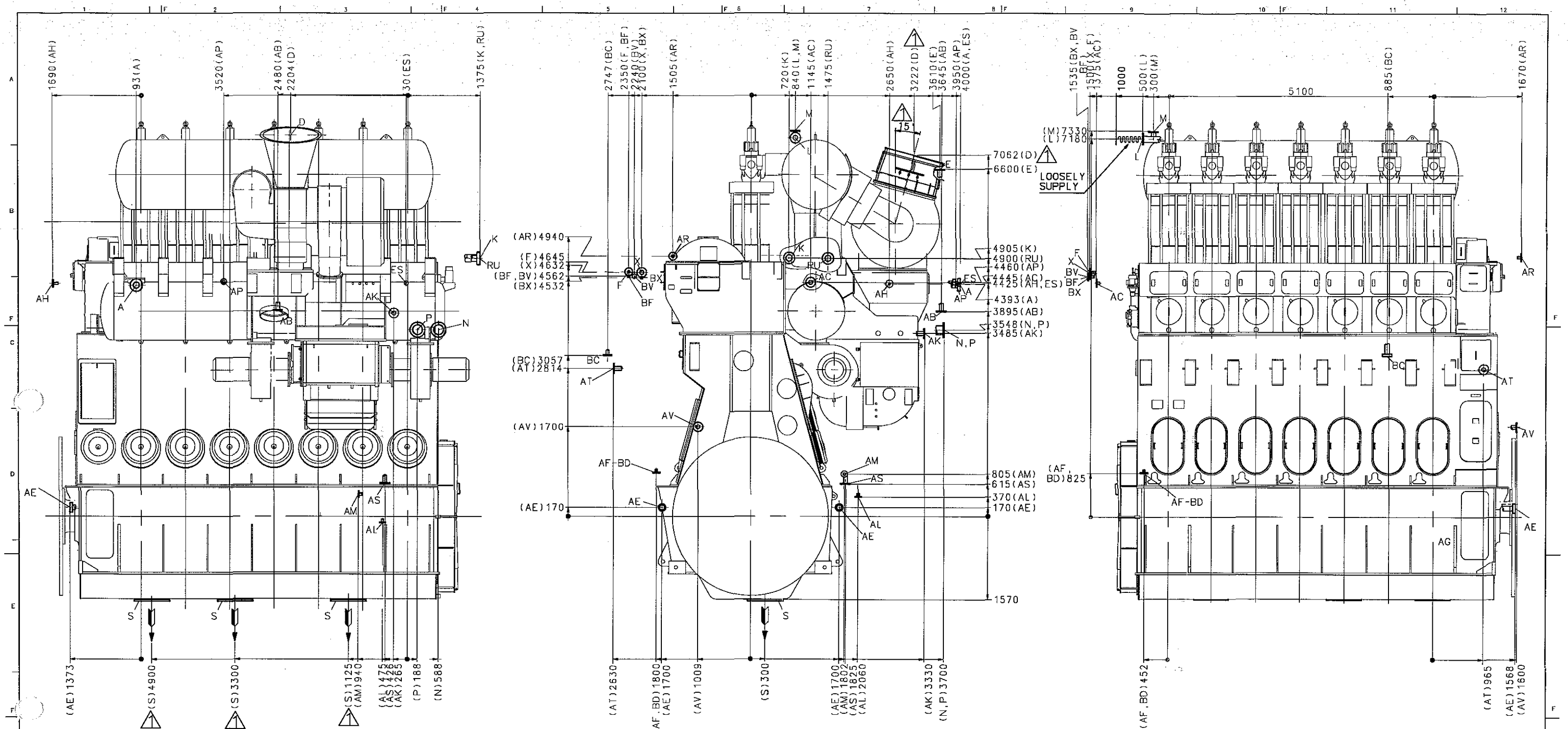
NAME	SIZE	MODEL
2801023690		

For tightening-up torque of all nuts, see 2805000670

reference dwg. no. : 0792234-9.5

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Please note.
The letters refer to "List of flanges"
The dimensions given are subject
to revision without notice.
For engine dimensions see "Engine outline"
and "Gallery outline"

T/C TYPE : TCA66-20
AIR COOLER TYPE : LKMY-JT

SEE CADAM SYSTEM DRAWING FOR A/E.
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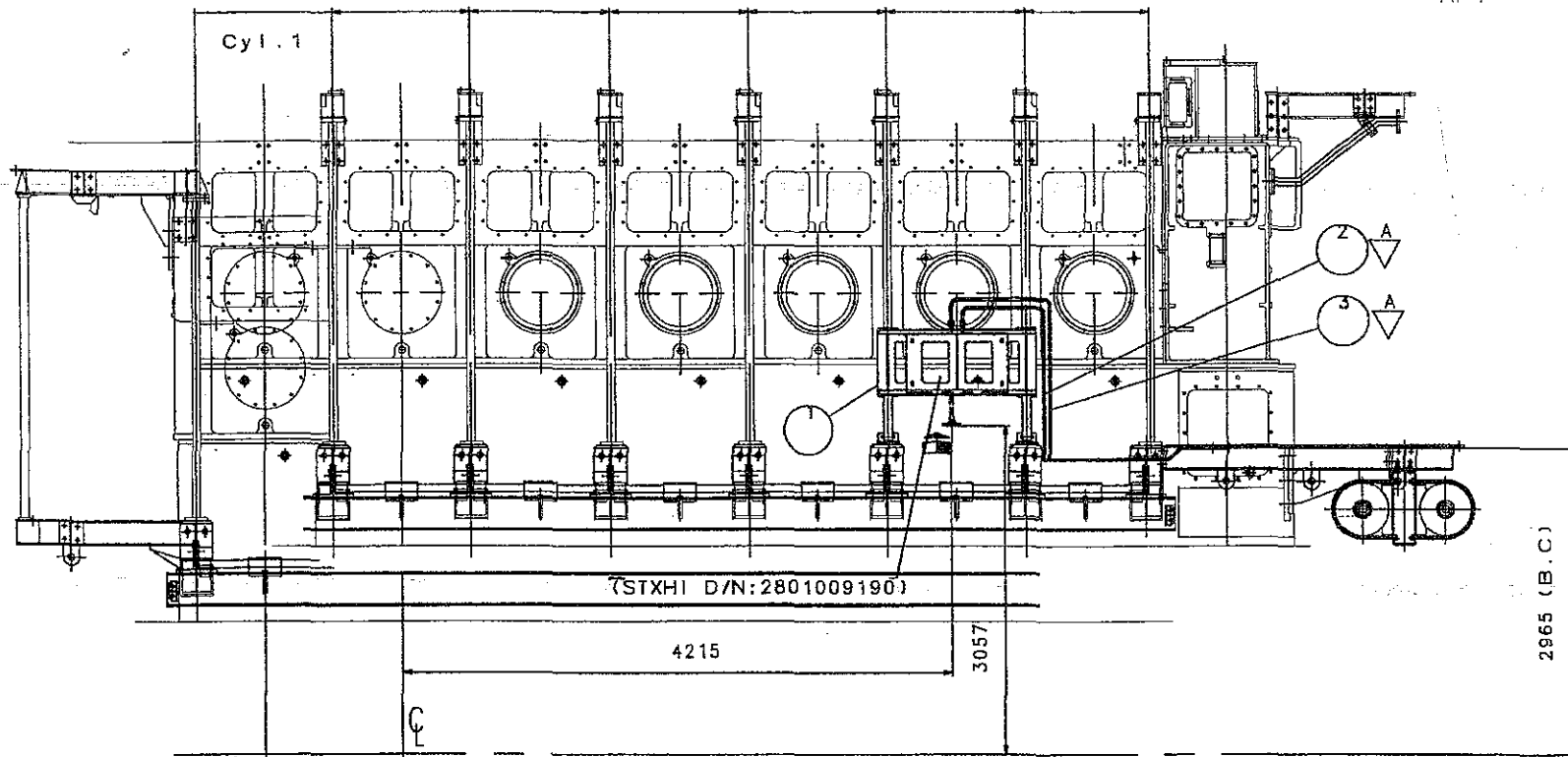
Basic Standards (MBS) & Suppl. Drawing No.:					EN21C Surf. roughness		Projection: 		Material / Blank:																										
3170570-6.0					EN21F-m Tolerances		Mass (kg):		Final User Model:																										
Date					Des.					Chg.					Appd.					A.C.					Change / Replacement										C.No.
																									Replaced by Ident. No.:										8
																																			7
																																			6
																																			5
																																			4
																																			3
																																			2
																																			1
																																			0
26090727					SML					JTS					TBK					Yard comments applied															
20090213					SML					JTS					TBK					First issued															
Similar Drawing No.:																									Replacement for Ident. No.:										
Scale:					Size:					Type:					Page No.:																				
1:25					A0					7S50MC-C					01 (01)																				
Info. No.:					Description:																				Ident. No.:										
372034					Engine pipe connections																				2805030080.1A										
Final User Info. No.:										Final User Description:															Final User Ident. No.:										

FORE

850 850 850 850 850 850 770

AFT

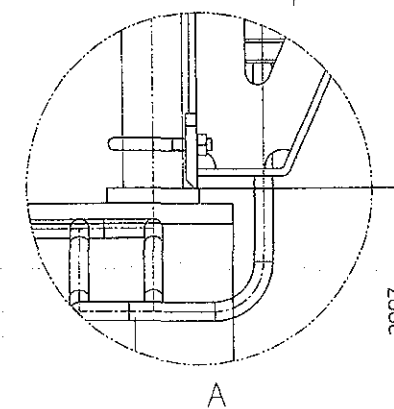
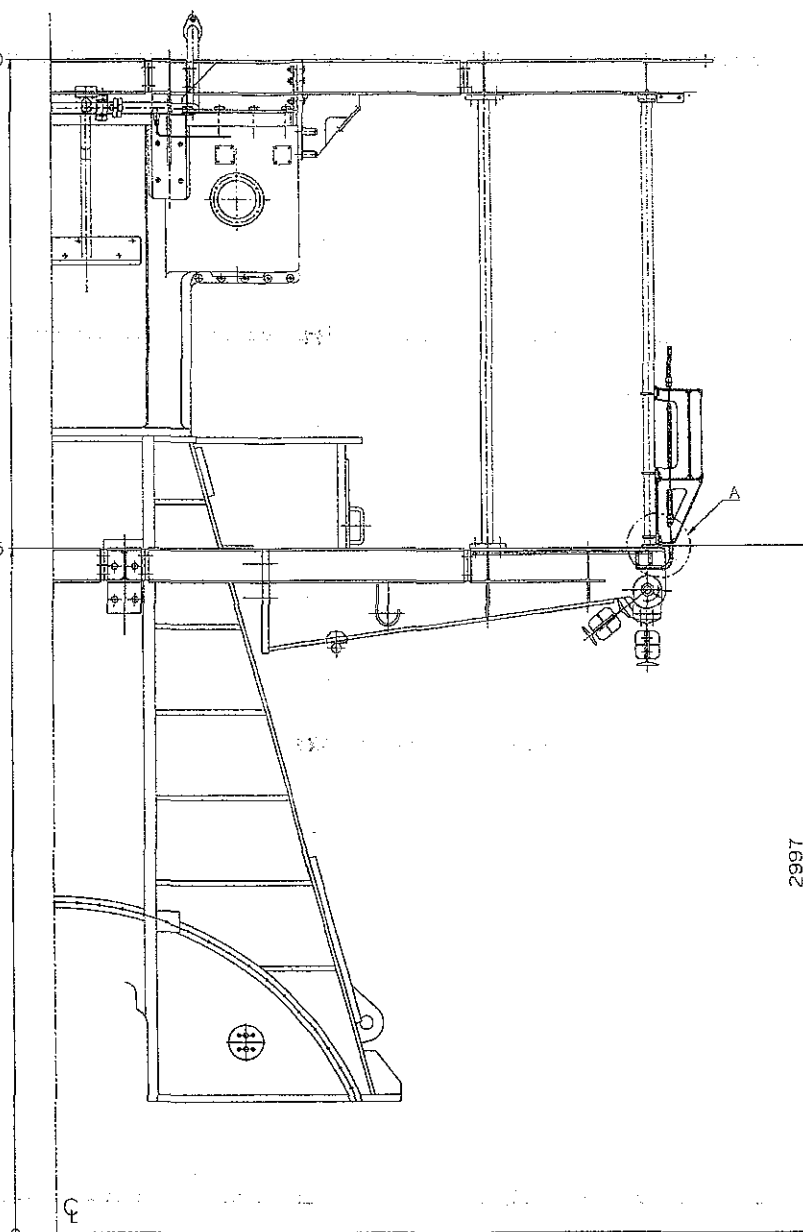
Cyl. 1



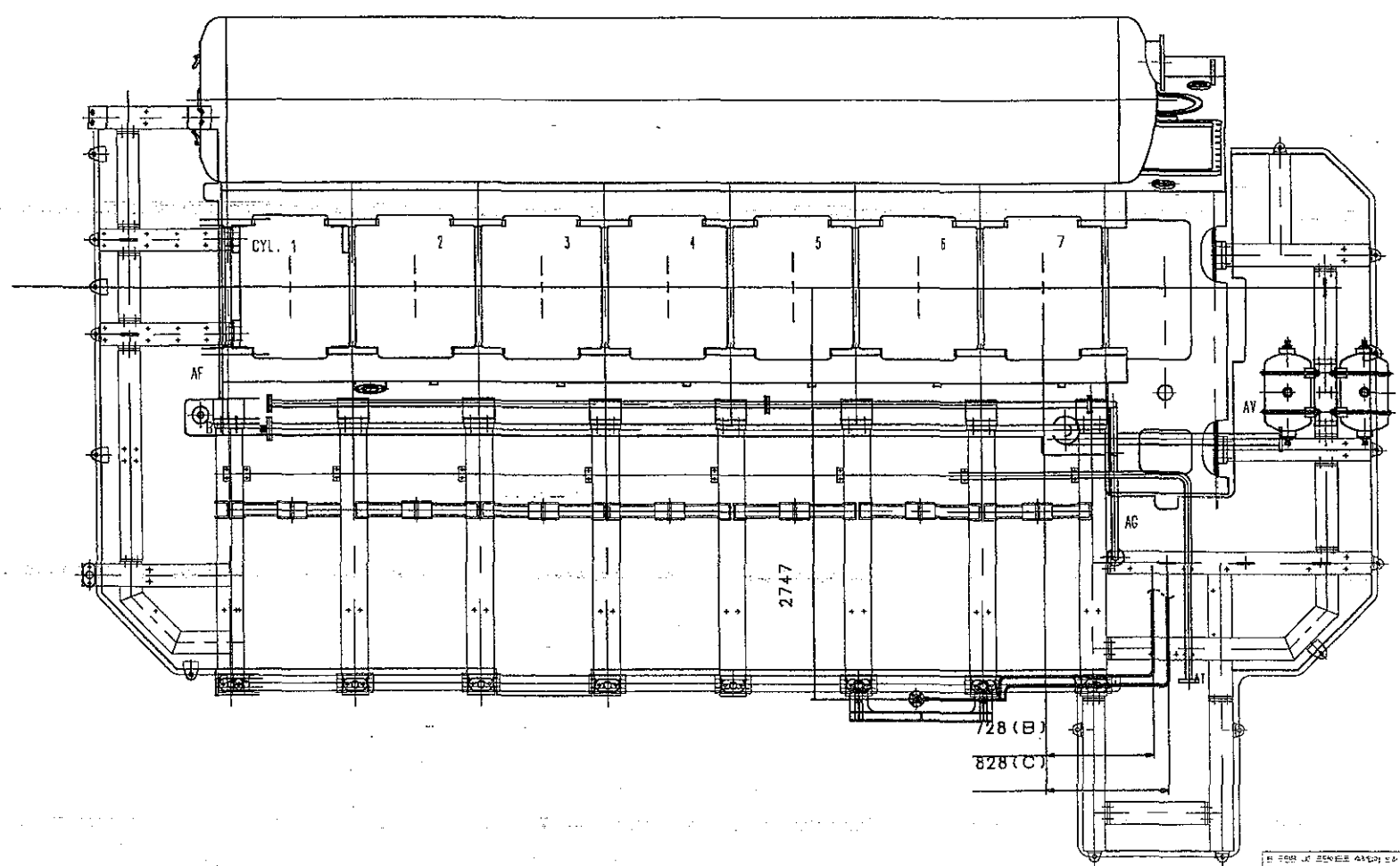
5:10

2985

2997



2997



MAN B&W
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Besta Slenderos (MSB) & Suppl. Drawing No.:		EN21C Surf. roughness		Projection:		Material / Blank:	
		EN21F-m Tolerances		Mass (kg):		Final User Material:	
Date	Des.	Chr.	Appr.	A.C.	Change / Replacement		
					Replaced by (date, Rev.):		
20090602 SML JTS TEK - ALGERIA							
Drawing No.:		Page No.:		Rev.:			
1:15		A0		S50MC-C		01 (01)	
Disposition:				STX Engine Co., Ltd.			
AIR REDUCING UNIT SUPPORT ASS'Y				2801023640			
Final User: (date, Rev.):				Final User: (date, Rev.):			

060990908Z

DRAWING NO.

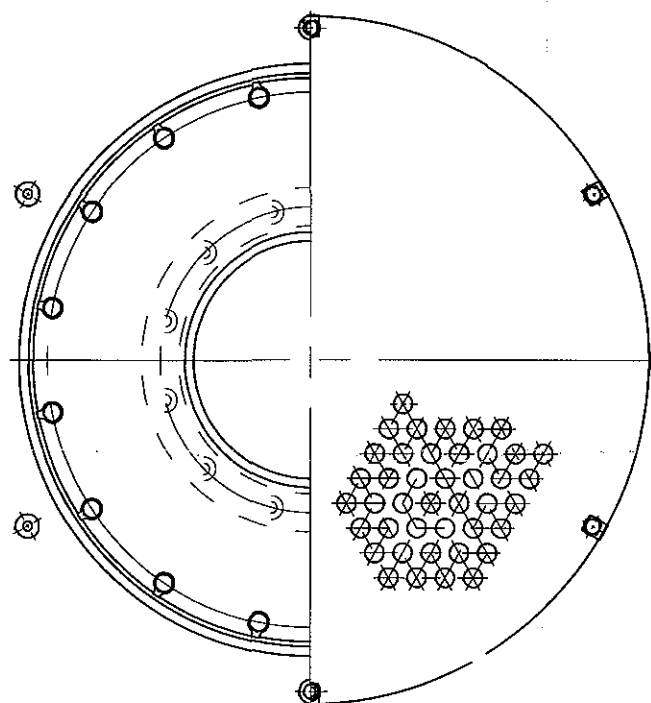
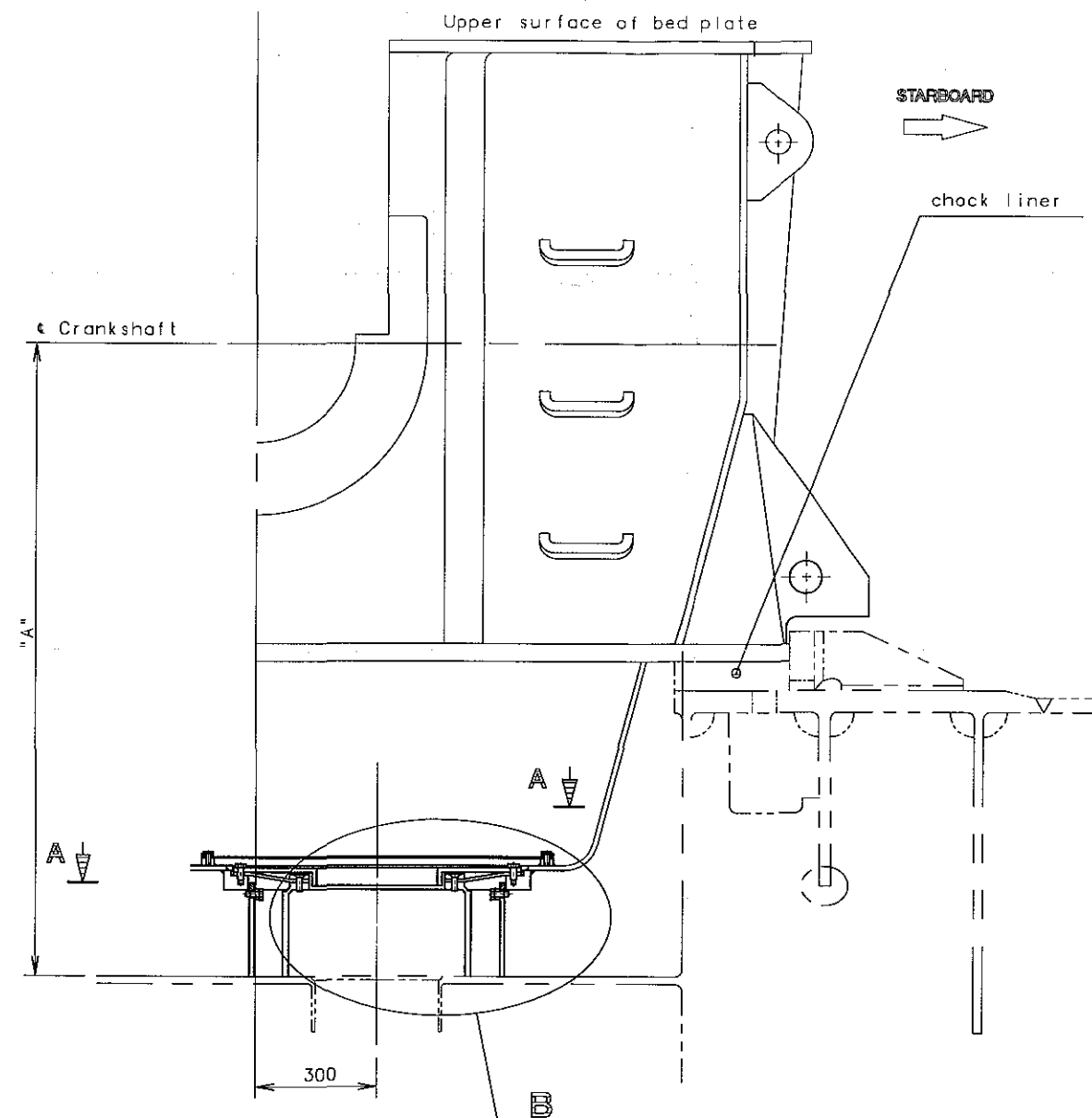
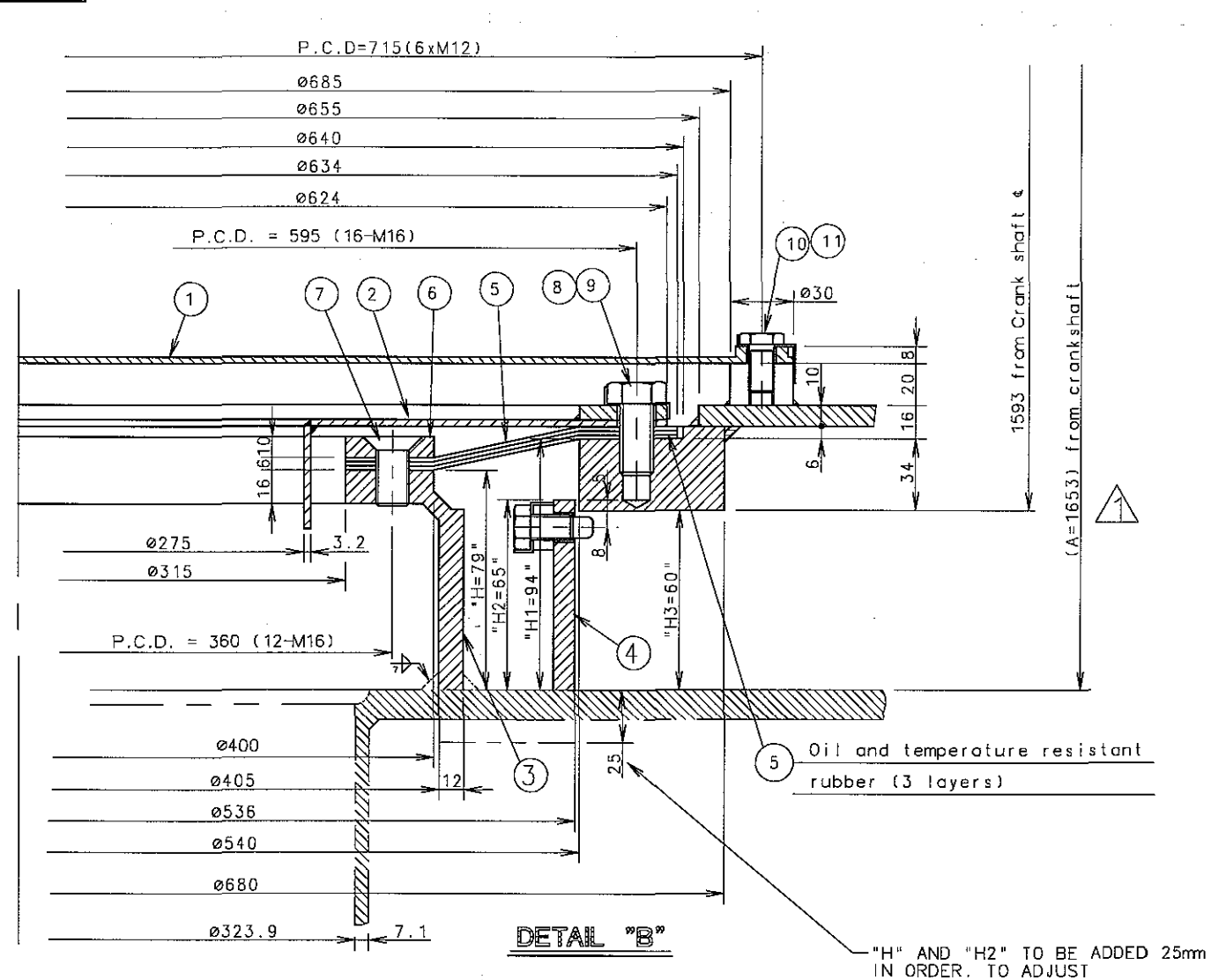
BV	STEAM INLET FOR CLEANING DRAIN SCAVENGING AIR CHAMBERS	ø15	COPPER PIPE				BITE TYPE UNION	
BF	STEAM OUTLET FOR HEATING FUEL OIL PIPES	ø15	COPPER PIPE				BITE TYPE UNION	
BX	STEAM INLET FOR HEATING FUEL OIL PIPES	ø15	COPPER PIPE				BITE TYPE UNION	
BD	F.O DRAIN HOT WATER TRACING	ø15	COPPER PIPE				BITE TYPE UNION	
ES	TO ECONOMIZER FOR SOOT BLOWER SEALING AIR	25A	JIS 5K	34.5	95	75	M10	4
AV	DRAIN FROM SCAVENG. AIR CHAMBERS TO CLOSED DRAINTANK	65A	JIS 5K	77.1	155	130	M12	4
AT	FIRE EXTINGUISHING IN SCAVENGE AIR BOX	32A	JIS 10K	43.2	135	90	M12	4
AS	COOLING WATER DRAIN AIR COOLER	25A	JIS 5K	34.5	95	75	M10	4
AR	OIL VAPOUR DISCHARGE	40A	JIS 5K	49.1	120	95	M12	4
AP	AIR INLET FOR DRY CLEANING OF T/C	20A	JIS 10K	27.7	100	65	M12	4
AM	OUTLET AIR COOLER TO CHEMICAL CLEANING TANK	40A	JIS 5K	49.1	120	95	M12	4
AL	DRAIN FROM CLEANING AC./WATER MIST CATCHER	40A	JIS 5K	49.1	120	95	M12	4
AK	INLET CLEANING AIR COOLER	25A	JIS 5K	34.5	95	75	M10	4
AH	COOLING WATER DRAIN	32A	JIS 5K	43.2	115	90	M12	4
AG	CONNECT WITH "AE" CONN.							
AF	FUEL OIL DRAIN OUTLET	32A	JIS 5K	43.2	115	90	M12	4
AE-1	DRAIN FROM BEDPLATE	32A	JIS 5K	43.2	115	90	M12	4
AE-2	DRAIN FROM BEDPLATE	32A	JIS 5K	43.2	115	90	M12	4
AC	LUBRICATING OIL INLET TO CYLINDER LUBRICATORS	25A	JIS 5K	34.5	95	75	M10	4

AB	LUBRICATING OIL OUTLET FROM TURBOCHARGER	80A	JIS 5K	90.0	180	145	M16	4
X	FUEL OIL INLET (NECK FLANGE FOR WELDING SUPPLIED)	65A	JIS 16K	77.1	175	140	M16	8
RU	P.C.O & M.L.O INLET	200A	JIS 5K	218.0	320	280	M20	8
S	SYSTEM OIL OUTLET TO BOTTOM TANK	REFET TO ARRANGE OF L.O OUTLET-VERTICAL (PAGE 4-3)						
P	COOLING WATER OUTLET FROM AIR COOLER	150A	JIS 5K	166.6	265	230	M16	8
N	COOLING WATER INLET TO AIR COOLER	150A	JIS 5K	166.6	265	230	M16	8
M	COOLING WATER DE-AERATION	25A	JIS 5K	34.5	95	75	M10	4
L	COOLING WATER OUTLET	100A	JIS 5K	115.4	200	165	M16	8
K	COOLING WATER INLET	100A	JIS 5K	115.4	200	165	M16	8
F	FUEL OIL OUTLET	40A	JIS 16K	49.1	140	105	M16	4
E	VENTING OF LUB. OIL PIPE TURBOCHARGER	80A	JIS 5K	90.0	180	145	M16	4
D	EXHAUST OUTLET	REFER TO ENGINE PIPE CONNECTIONS (PAGE 4-1)						
BC	CONTROL & SAFETY AIR INLET	25A	JIS 30K	34.0	130	95	M16	4
A	STARTING AIR INLET (NECK FLANGE FOR WELDING SUPPLIED)	100A	JIS 30K	115.4	240	195	M22	8
NO.	ITEMS	PIPE	Kg/Cm ²	DIA. IN	DIA. OUT	PCD	DIA x L	Q'TY
				FLANGE			BOLT	

GENERAL TOLERANCE FOR MACHINED MEASURES			REV	LOC	DATE	MADE	APPD	NOTE	
NOMINAL MEASURE	TOLERANCES	OVER	MATERIAL		SCALE		WEIGHT	CAD	MODEL
			N / S		kg				
1	4	±0.1	DWN	20091221	S.M. LEE	SIZE	NAME		
4	16	±0.2	CHKD	20091221	J.T. SEO	A3	LIST OF FLANGES		
16	63	±0.3	APPD	20091221	T.B. KIM				
63	250	±0.5	DRAWING NO.						2805033090
250	1000	±0.8	STX Engine Co., Ltd.						

Class No.	N4	N5	N6	N7	N8	N9
Ro Max. Value(μm)	0.2	0.4	0.8	1.6	3.2	6.3
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11	P065D0012000	LOCKING WASHER	SPCC	6		M12
10	P000F0012020	HEX. BOLT	8.8T	6		M12x20
9	P065D0016000	LOCKING WASHER	SPCC	16		M16
8	P000F0016035	HEX. BOLT	8.8T	16		M16x35
7	EN59S1632	SCREW	8.8T	12		M16x32
6	2602000100	FLANGE	SS400	1		
5	2602000090	RUBBER PLATE	NBR	3		
4		PROTECTING RING	SS400	1		M12
3		FLANGE	SS400	1		
2	2602000060	COVER	SS400	1		
1	2602000050	GRATE		1		
NO.	DRAWING No.	NAME	MATERIAL	Q'TY	WEIGHT	REMARKS

REMARK

- (1) Standard height "A" is 1643mm which depends on standard 50mm thickness chock liner, thus it is subjected to change with the thickness of chock liner
- (2) All parts in this drawing are supplied by engine maker
- (3) "H1" is measured after placing and alignment of bed plate
"H" is determined as "H1"-15mm

GENERAL TOLERANCE FOR MACHINED MEASURES				1		060210	JTS	JSP	YARD COMMENT : "A" - 1653mm		
REV		LOC		DATE		MADE		APPD		NOTE	
NOMINAL MEASURE		TOLERANCES		MATERIAL		SCALE N / S		 CAD		MODEL	
								WEIGHT		kg	
OVER		MAX.		DWN		20050516		J.T.SEO		SIZE	
1		4		±0.1		CHKD		20050516		C.S.PARK	
4		16		±0.2		APPD		20050516		J.S.PARK	
16		63		±0.3		A2		ARRANGE OF L.O OUTLET-VERTICAL			
63		250		±0.5							
250		1000		±0.8							
STX Engine Co., Ltd.								DRAWING NO. 2805010091			

Class No.	N4	N5	N6	N7	N8	N9	16	63	±0.3
Ra Max. Value(um)	0.2	0.4	0.8	1.6	3.2	6.3	63	250	±0.5
							250	1000	±0.8

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No.	Symbol	Symbol designation	No.	Symbol	Symbol designation
1. GENERAL CONVENTIONAL SYMBOLS			3. VALVES, GATE VALVES, COCKS AND FLAPS		
1.1		Pipe	3.1		Valve, stright through
1.2		Pipe with indication of direction of flow	3.2		Valve, angle
1.3		Valves, gate valves, cocks and flaps	3.3		Valve, three-way
1.4		Appliances	3.4		Non-return valve(flaf), straight
1.5		Indicating and measuring instruments	3.5		Non-return valve(flaf), angle
2. PIPES AND PIPE JOINT			3.6		Non-return valve(flaf), straight, screw down
2.1		Cossing pipes, not connected	3.7		Non-return valve(flaf), angle, screw down
2.2		Cossing pipes, connected	3.8		flap, straight throught
2.3		Tee pipe	3.9		flap, angle
2.4		Flexible pipe	3.10		Reducing valve
2.5		Expansion pipe (corrugated) general	3.11		Safety valve
2.6		Joint, screwed	3.12		Angle safety valve
2.7		Joint, flanged	3.13		Self-closing valve
2.8		Joint, sleeve	3.14		Quick-openning valve
2.9		Joint, quick-releasing	3.15		Quick-closing valve
2.10		Expansion joint with gland	3.16		Regulating valve
2.11		Expansion pipe	3.17		Kingston valve
2.12		Cap nut	3.18		Ball valve (-cock)
2.13		Blank flange	3.19		Butterfly valve
2.14		Spectacle flange	3.21		Double-seated changeover valve
2.15		Bulkhead fitting watertight, flanged	3.22		Suction valve chest
2-16		Bulkhead crossing, non-watertight	3.23		Suction valve chest with non return valves
2-17		Pipe going upwards	3.24		Double-seated changeover valve, straight
2-18		Pipe going downwards	3.25		Double-seated changeover valve, angle
2-19		Orifice	3.26		Cock, straight through

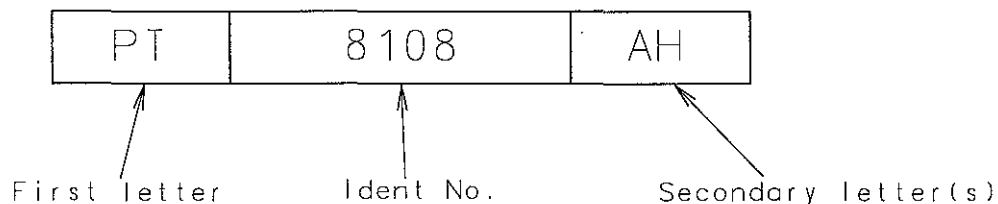
No.	Symbol	Symbol designation	No.	Symbol	Symbol designation
3.27		Cock, angle	5.10		Various accessories (text to be added)
3.28		Cock, three-way, L-port in plug	5.11		Piston pump
3.29		Cock, three-way, T-port in plug	6. FITTINGS		
3.30		Cock, four-way, straight through in plug	6.1		Funnel
3.31		Cock with bottom connection	6.2		Bell-mounthed pipe end
3.32		Cock, stright through, with bottom connection	6.3		Air pipe
3.33		Cock, angle, with bottom connection	6.4		Air pipe with net
3.34		Joint, quick-releasing	6.5		Air pipe with cover
4. CONTROL AND REGULATION PARTS			6.6		Air pipe with cover and net
4.1		Hand-operated	6.7		Air pipe with pressure-vacuumvalve
4.2		Remote control	6.8		Air pipe with pressure-vacuumvalve and net
4.3		Spring	6.9		Deck fitting for sounding or fitting pipe
4.4		Mass	6.10		Short sounding pipe with selfclosing cock
4.5		Float	6.11		Stop for sounding rod
4.6		Piston	7. READING INSTRUMENTS WITH ORDINARY SYMBOL DESIGNATIONS		
4.7		Membrane	7.1		Sight flow indicator
4.8		Electric motor	7.2		Observasion glass
4.9		Electro-magnetic	7.3		Level indicator
5. APPLIANCES			7.4		Distance level indicator
5.1		Mudbox	7.5		Counter (indicate function)
5.2		Filter or strainer	7.6		Recorder
5.3		Magnetic filter	7.7		Flow meter
5.4		Seperator	8. VALVE CONDITION		
5.5		Steam trap	8.1		Normal open
5.6		Centrifugal pump	8.2		Nomal close
5.7		Gear or screw pump	9. SCOPE OF SUPPLY		
5.8		Hand pump (bucket)	9.1		By engine maker
5.9		Ejector			



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Codes for identification of instruments and signal-related functions



Measured or Indicating variable (First letter(s))		Ident. number The first number digits indicate the point of measurement, the next two are serial numbers.		Function(secondary letter(s))	
DS	Density switch	11	Manoeuvring system	A	Alarm
DT	Density transmeter	12	hydraulic power supply	C	Control
GT	Gauging transmeter (load/index transmeter)	14	Combustion pressure supervision	H	High
FT	Flow transmeter	20	ECS to/from safety system	I	indication (remote)
FS	Flow switch	21	ECS to/from remote control system	L	Low
LS	Level switch			R	Recording
LI	Level indication(local)	22	ECS to/from alarm system	S	Switching
LT	Level transmeter	30	ECS Miscellaneous input/output values	Y	Slow-down/ Load reduction
PDI	Pressure difference indication(local)	40	Tacho/crankshaft pos. system	X	Unclassified function
PDS	Pressure difference switch	41	Engine cylinder components		
		50	VOC:supply system	Z	Shut-down
PDT	Pressure difference transmeter	51	VOC:sealing oil system		
		52	VOC:control oil system		
PI	Pressure indication (local)	53	Other VOC related systems		
		54	VOC engine related components		
PS	Pressure switch	80	Fuel oil system		
PT	Pressure transmeter	81	Lubrication oil system		
ST	Speed transmeter	82	Cylinder lub. oil system		
TC	Thermo couple(NiCr-Ni)	83	Stuffing box drain system		
TE	Temperature element (pt-100)				
TI	Temperature indication (local)				

Basic standard (MBD) & supply dwg. no. : 0788933-9.3

Replacement for Ident. No.:

Scale:	Size: A4	Type: ALL ENGINES	Page No.: 01 (03)	STX Engine Co., Ltd.
Info. No.: 730071	Description: IDENTIFICATION CODES			Ident. No.: 900500024A
Final User Info. No.:	Final User Description:			Final User Ident. No.:



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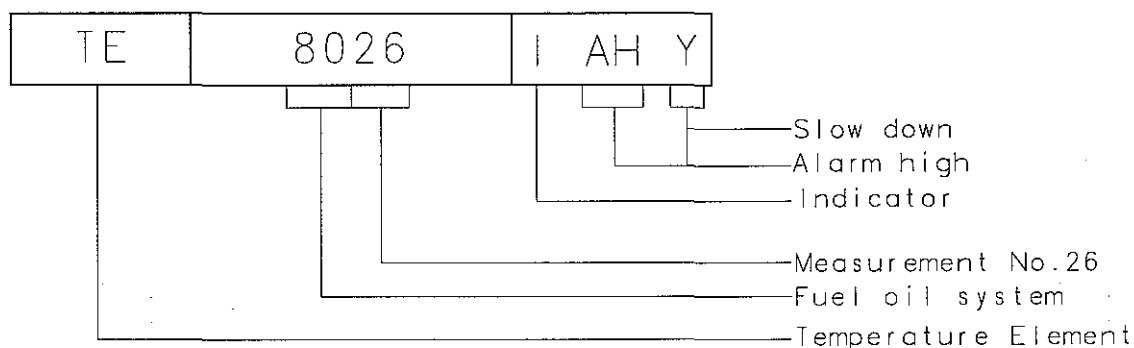
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Measured or Indicating variable (First letter(s))		Ident. number The first number digits indicate the point of measurement, the next two are serial numbers.		Funtion(secondary letter(s))
TS	Temperature switch	84	Cooling water system e.g. central cooling water e.g. sea cooling water e.g. jacket cooling water	
WS	Vibration switch			
WT	Vibration transmeter			
VS	Viscosity switch			
VT	Viscosity transmeter	85	Compressed air supply system e.g. control air e.g.starting air	
ZV	Position valve (solenoid valve)			
ZS	Position switch (limit switch)	86	Scavenge air	
		87	Exhaust gas system	
ZT	Position transmeter (e.g.proximity sensor)	88	Miscellaneous functions e.g. axial vibration	
XC	Unclassified control	90	Project specific Manoeuvring system Manoeuvring system	
XS	Unclassified switch			
XT	Unclassified transmeter			
		Note: ECS:Engine control system VOC:Volatile Organic Compound		

The first link(first letter) indicates what is measured or the indicating variable. The second link is the ident. No., in which the first two digits indicate the point of measurement of the indicating variable, followed by a serial number. The third link(secondary letter(s)) indicates the function of the measured value

Example



Basic standard (MBD) & supply dwg no. : 0788933-9

Replacement for Ident. No.:

Scale:	Size: A4	Type: ALL ENGINES	Page No.: 02 (03)	STX Engine Co., Ltd.
Info. No.: 730071	Description: IDENTIFICATION CODES			Ident. No.: 900500024A
Final User Info. No.:	Final User Description:			Final User Ident. No.:



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Repeated signals:

Signal which are repeated, such as per cylinder measurement of per turbocharger measurement, etc. are provided with a suffix number. The suffix number is identical with the place of measurement, such as 1 cylinder 1, etc. Where signals are redundant, suffix A or B may be used.

Example

Cylinder or Turbocharger-Related Signals

ZV	1120-1	C
----	--------	---

e.g. Cyl. No. / TC No.
Aux. Blower No.

Redundant Signals

PT	8603-A	IAHC
----	--------	------

System A

PT	8603-B	IAHC
----	--------	------

System B

Cylinder-Related Redundant Signal

ZV	8203-A-1	C
----	----------	---

e.g. Cyl. No. / TC No.

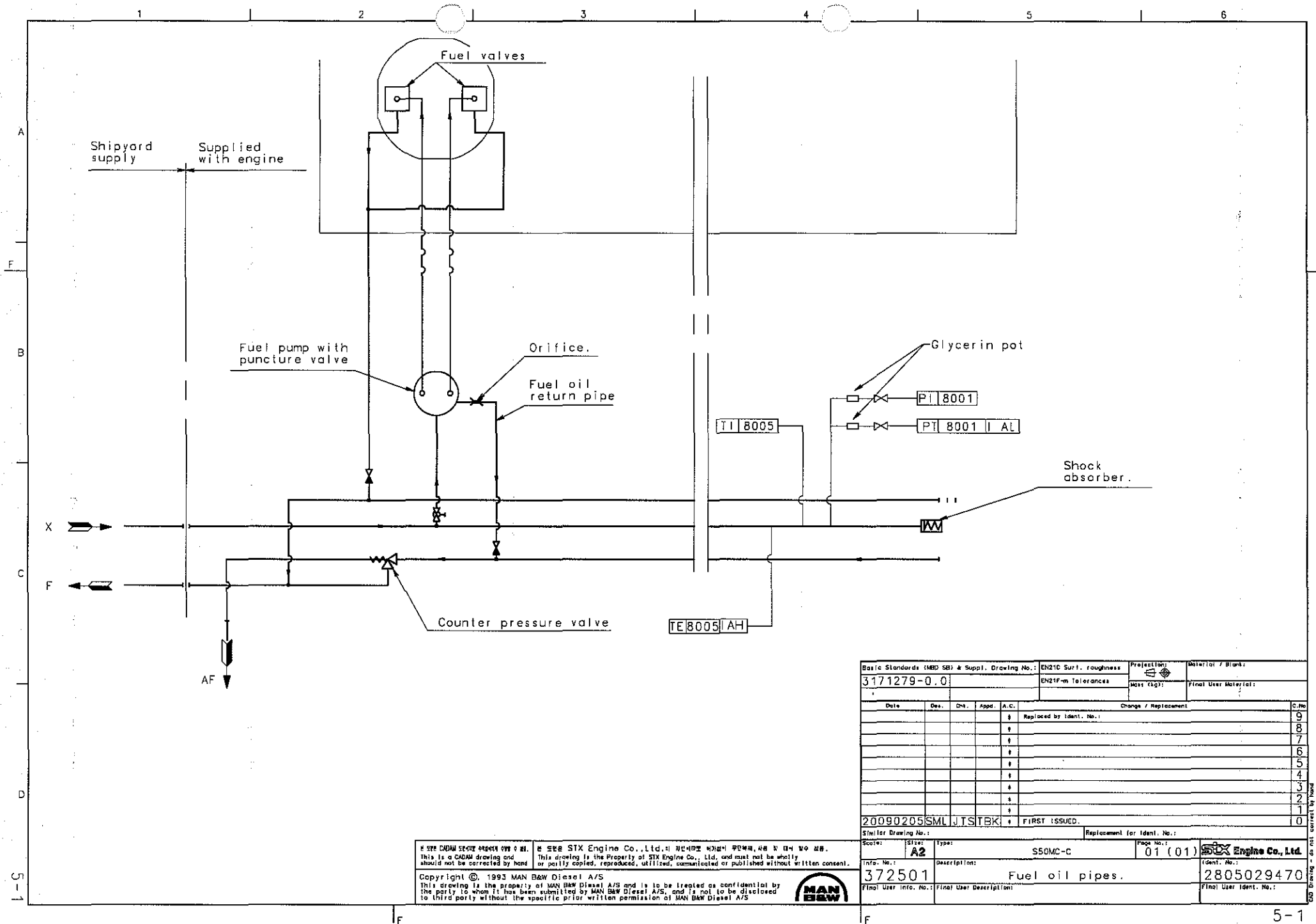
System A

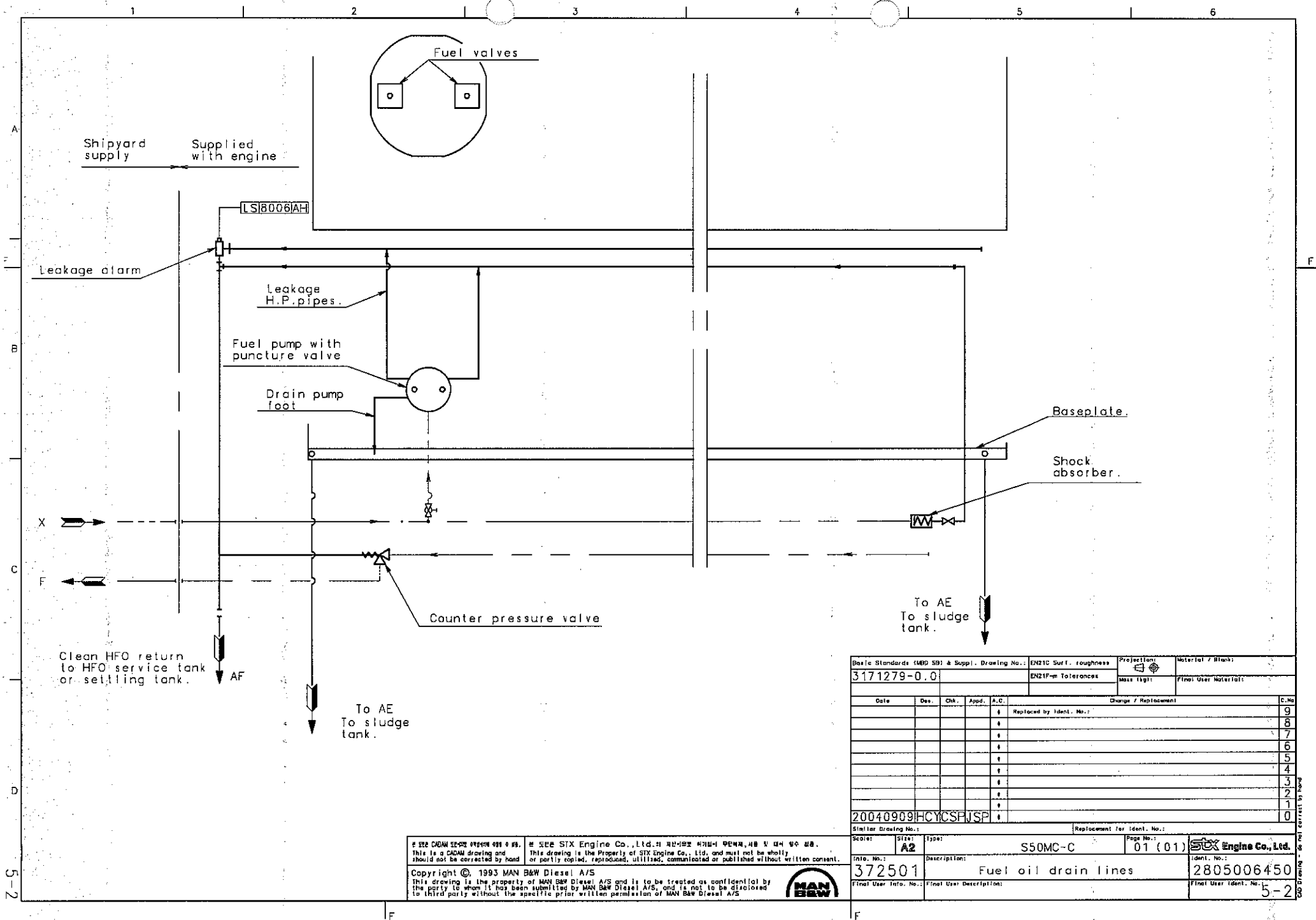
Graphical presentation in PI-diagrams according to ISO 1219 1-II

Basic standard (MBD) & supply dwg no.: 0788933-9

Replacement for Ident. No.:

Scale:	Size: A4	Type: ALL ENGINES	Page No.: 03 (03)	STX Engine Co., Ltd.
Info. No.: 730071	Description: IDENTIFICATION CODES			Ident. No.: 900500024A
Final User Info. No.:	Final User Description:			Final User Ident. No.:





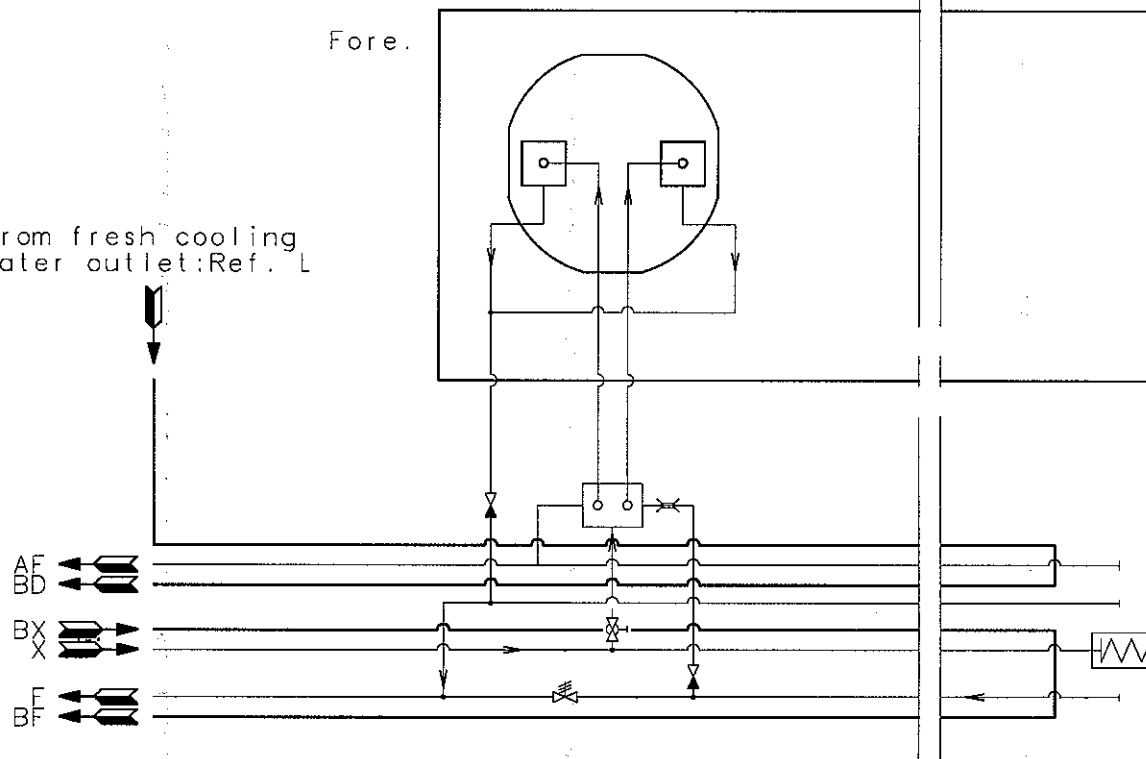
Basic Standards (MBD 59) & Suppl. Drawing No.:		EN21C Surf. roughness		Projection:	Material / Blank:
3171279-0.0		EN21F-m Tolerances		First User Material:	
Date		Des.	Chk.	Appd.	A.C.
Change / Replacement					C.No.
Replaced by Ident. No.:					9
					8
					7
					6
					5
					4
					3
					2
					1
					0
Similar Drawing No.:					Replacement for Ident. No.:
Scale:	Size:	Type:	Page No.:	S50MC-C 01 (01)	
Info. No.:	Description:			Ident. No.:	
372501	Fuel oil drain lines			2805006450	
Final User Info. No.:			Final User Description:		Final User Ident. No.:
					5-2

* SEE CADAN SE-002 0000 000 000.
 This is a CADAN drawing and should not be corrected by hand.
 * SEE STX Engine Co., Ltd.의 제1차 설계도서 제1차 설계도서 제1차 설계도서.
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Drawing No. 2805006450
 Date 1993.05.20
 Drawn by 5-2

From fresh cooling
water outlet:Ref. L



The letters refer to "List of flanges"

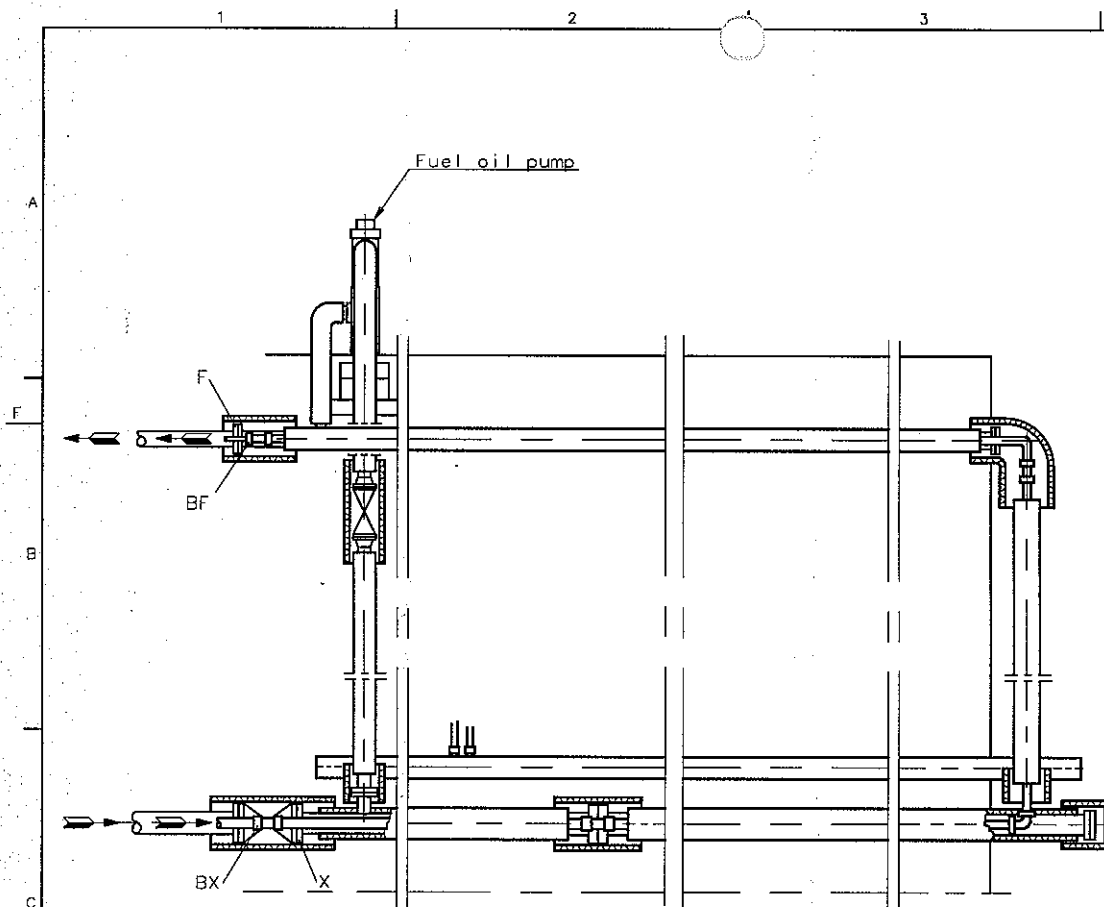
Suppl. Drawing No.:		1831533-1.1		Similar No.:		Material:	
				Scale:		Mass. (kg):	
				Blank:			
Date	Des.	Chk.	Photo.	Prod. P.	A.C.	Replaced by Ident. No.:	C
					*		9
					*		8
					*		7
					*		6
					*		5
					*		4
					*		3
					*		2
					*		1
					*	Replacement for Ident. No.:	0
MBD Info. No.:		Size		Type:		STX Engine Co., Ltd.	
399176		A 3		Description:		Ident. no.:	
				Fuel oil pipes, heating		2805000890	
Final User Info. No.:		Final User Description:		Final User Ident. no.:		5-3	

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Surface roughness and tolerances see MAN B&W Diesel A/S SB Standards Book; EN21C and EN21F medium.

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INSULATION OF PIPES

General information:

Insulation of fuel oil pipes and drain of fuel oil pipes should not be carried out until the piping systems have been subjected to the pressure tests specified and approved by the respective Classification Societies and/or Authorities.

The directions mentioned below include insulations of hot pipes, flanges and valves, with a surface temperature of the complete insulations of +55°C at a room temperature of +38°C. As for the choice of material, and if required, approval for the specific purpose, reference is made to the respective Classification Society.

Fuel oil pipes:

The pipes are to be insulated with 20 mm mineral wool of min. 150 kg/m³ and covered with glass cloth min. 400 g/m².

Fuel oil pipes & heating pipes together:

Two or more pipes can be insulated with 30 mm wired mats of mineral wool of min. 150 kg/m³ covered with glass cloth min. 400 g/m².

Valves, flanges:

The valves and flanges are to be insulated by means of removable pads.

Flange and valve pads are made of glass cloth, min. 400 g/m², containing mineral wool stuffed to min. 150 kg/m³.

Thickness of the mats to be:

Fuel oil pipes: 20 mm

Fuel oil pipes & heating pipes together: 30 mm

The pads are to be fitted so that they overlap the pipe insulating material by the pad thickness.

At flanged joints, insulating material on pipes should not be fitted closer than corresponding to the min. bolt length.

Mounting:

Mounting of the insulation is to be made in accordance with the supplier's instructions.

028 The letters refer to "List of flanges".

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DIN 15-Standard for surface roughness												DIN 15-Tolerances for dimensions without tolerance indication											
The basic symbols and class numbers are in accordance with ISO 1302(15-07) and VDA 1932(15-08)												This standard, which is in accordance with ISO 1789, states the permissible deviations for dimensions without direct tolerance indications on metal components manufactured by machining											
Class No.	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12	Max. measurement error in mm	1.5 up to 3	over 3 up to 30	over 30 up to 100	over 100 up to 300	over 300 up to 1000	over 1000 up to 3000	over 3000 up to 10000	over 10000 up to 30000	over 30000 up to 100000	over 100000 up to 300000
R _a max. value in µm	0,025	0,05	0,1	0,2	0,4	0,6	1,0	1,6	2,5	4,0	6,3	10,0	Medium series	±0,1	±0,2	±0,3	±0,5	±0,8	±1,2	±1,6	±2,5	±4,0	±6,3

Suppl. Drawing No.: 1701891-9		Similar No.:		Material:	
		Scale:		Mass (kg):	
				Blanks:	
Date	Des.	Dis.	App.	Prod. P.	A.C.
Replaced by Ident. No.:					C
					9
					8
					7
					6
					5
					4
					3
					2
					1
					0
Info No.: 399173		Size: A 2		Type:	
Description: Fuel oil pipes-insulation		Ident. no.: 2805001410		STX Engine Co., Ltd.	
Final User Info. No.:		Final User Description:		Final User Ident. no.:	
				5-4	

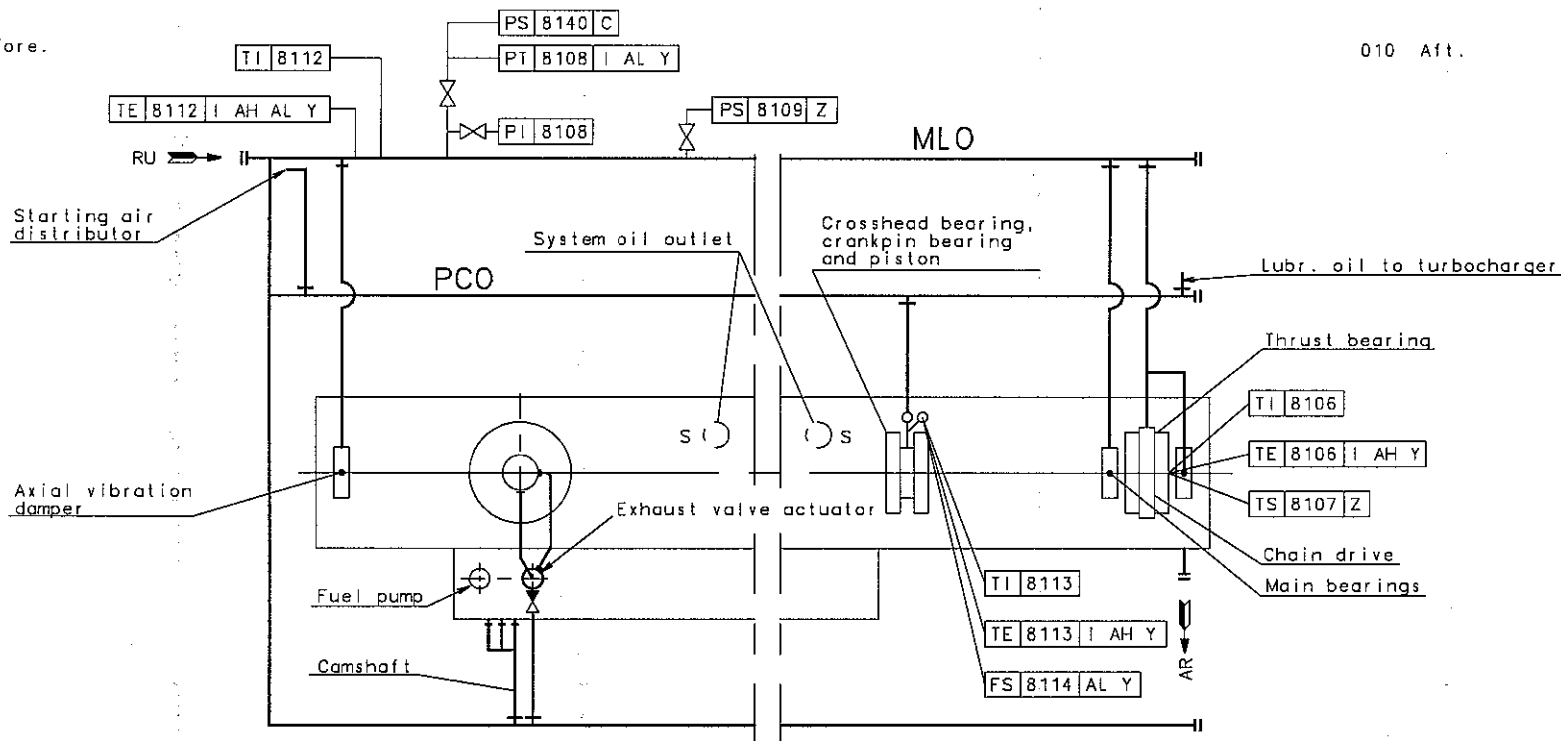
Property	Units	Value
Density at 15°C	kg/m ³	<u>≤991*</u>
Kinematic viscosity at 100°C	cSt	≤55
at 50°C	cSt	≤700
Flash point	°C	≥60
Pour point	°C	≤30
Carbon residue	%(m/m)	≤22
Ash	%(m/m)	≤0.15
Total sediment after aging	%(m/m)	≤0.10
Water	%(V/V)	≤1.0
Sulphur	%(m/m)	≤5.0
Vanadium	mg/kg	≤600
Aluminum + Silicon	mg/kg	≤80

m/m = mass V/V = volume

* 1010 kg/m³ : Provided automatic modern clarifies are installed.

130 Fore.

010 Aft.



ELECTRIC GOVERNOR

T/C TYPE : TCA-66

Basic Standards (ISO 9001) & Suppl. Drawing No.:					ENDIC Surf. roughness	Projection	Material / Stamp
3171276-5.0					DN21F-m Tolerances	None (sp)	Final User Material:
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement		C.No.
					Replaced by Ident. No.:		9
							8
							7
							6
							5
							4
							3
							2
							1
20090205SMLJTSTBK					FIRST ISSUED.		0

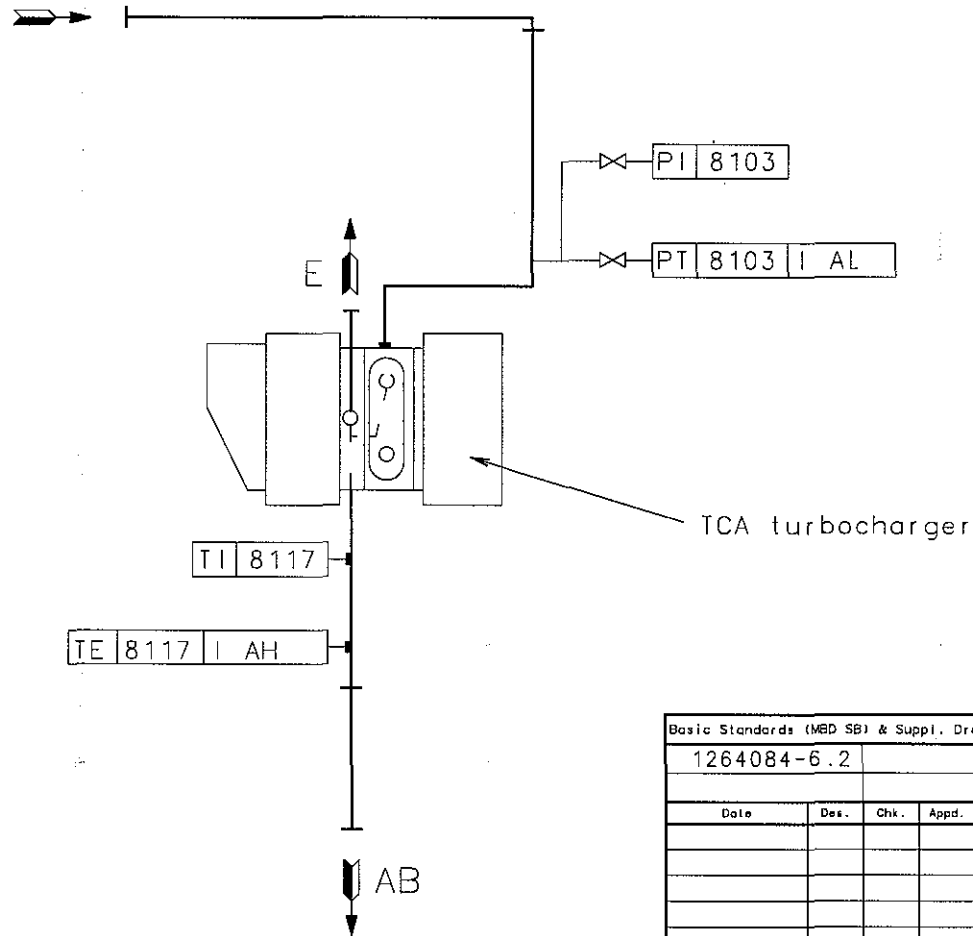
Similar Drawing No.:		Replacement for Ident. No.:	
3171276-5.0		01 (01)	
Scale:	Size:	Type:	Page No.:
	A2		
Ident. No.:		Description:	
373080		Lub. & cool. oil pipes	
Final User Info. No.:		Final User Description:	
Ident. No.:		Final User Ident. No.:	
2805029480			

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From system oil



Basic Standards (MBD SB) & Suppl. Drawing No.:				EN21C Surf. roughness		Projection: 		Material / Blank:	
1264084-6.2				EN21F-m Tolerances		Mass (kg):		Final User Material:	

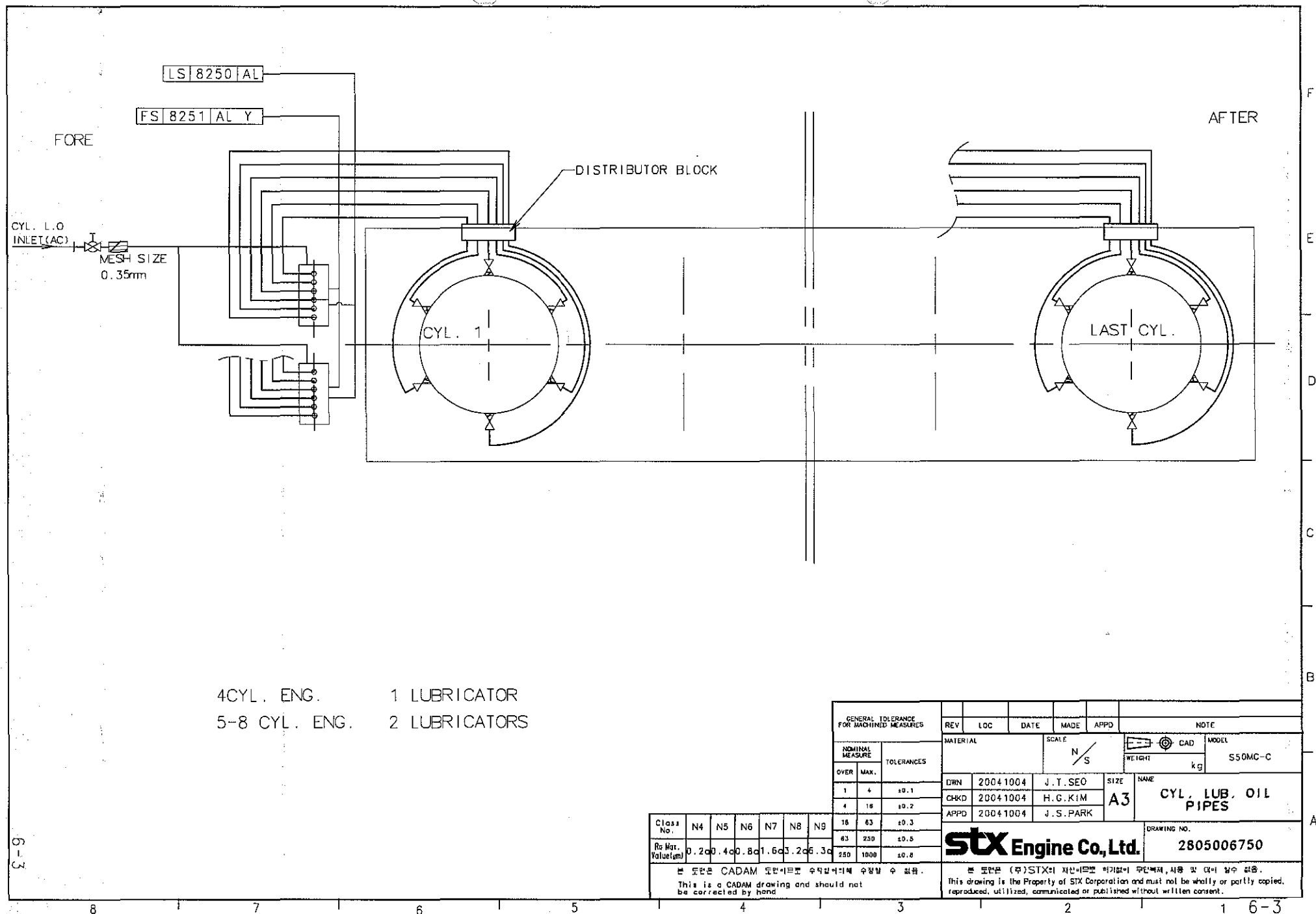
Similar Drawing No.:			Replacement for Ident. No.:		
Scale:	Size: A3	Type:	Page No.:	01 (01)	
Info. No.:		Description:		Ident. No.:	
373570		Turboch. lub.oil pipes		2805020250	
Final User Info. No.:		Final User Description:		Final User Ident. No.:	

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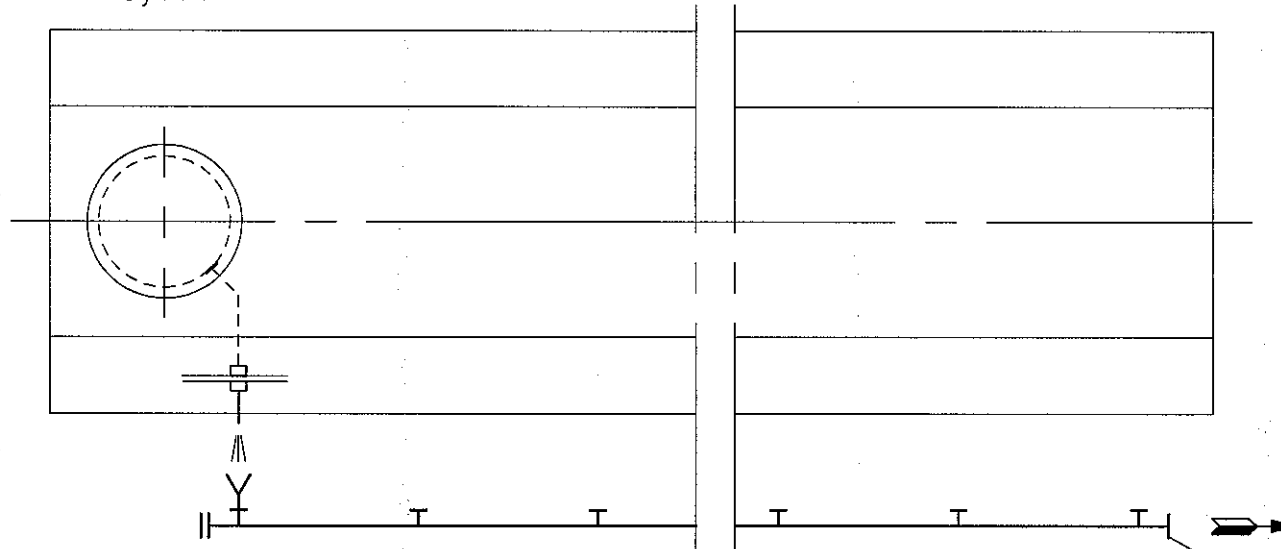
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CAD Drawing - do not correct by hand



Cyl. 1



AG (TO "AE" CONNECTION)
"AG + "AE" COMMON

028 The letters refer to "List of flanges"

Suppl. Drawing No.:		1430880-9.2		Similar No.:		Material:			
				Scale:		Mass. (kg):			
				~		Blank:			
Date	Des.	Chk.	Appd.	AC	Replaced by Ident. No.:			C	
				*				9	
				*				8	
				*				7	
				*				6	
				*				5	
				*				4	
				*				3	
				*				2	
				*				1	
20090206				SML	JTS	TBK	*	Replacement for Ident. No.:	0
MBD Info. No.:		Size		Type:		주식회사 STX			
373310		A 3		Description:		Ident. no.:			
				Stuffing box drain pipes		2805029700			
Final User Info. No.:		Final User Description:		Final User Ident. no.:					

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1. Circulating oil (Lubricating and cooling oil)

The circulating oil (Lubricating and cooling oil) must be a rust and oxidation inhibited engine oil of SAE 30 viscosity grade.

In order to keep the crankcase and piston cooling space clean of deposits, the oils should have adequate dispersion and detergent properties.

Alkaline circulating oils are generally superior in this respect.

Company	Circulating oil SAE30/TBN5-10
BP	Energol OE-HT-30
Total	Atlanta Marine D-3005
Castrol	Marine CDX-30
Chevron	Veritas 800 Marine 30
Exxon	Exxmar XA
SK	Supermar AS
Mobil	Mobilgard 300
Shell	Melina 30/30S
Texaco	Doro AR 30

The list must not be considered complete, and other brands may be chosen with satisfactory results.

The list is to be used as a guide. We undertake no responsibility for difficulties that might be caused by the oils.

The firms are stated in alphabetical order.

2. Cylinder lub. oil

Cylinder oils should, preferably, be of the SAE50 viscosity grade.

During the shop trial on Marine Diesel Oil, as well as during sea trial, a cylinder oil with a Total Base Number(TBN) of 10-20, and a viscosity of SAE40 or SAE50, will normally be used until the changeover to heavy fuel oil is effected. When burning residual fuel(1-5% sulphur), normally a cylinder oil of about 70TBN will give good results. higher TBN oils (up to 100) may be used when extremely severe running conditions are prevailing.

It should be noted that some high alkaline cylinder oils are not compatible with certain low sulphur fuels(having poor combustion properties), or with some diesel oils.

If any such incompatibility, expressed by poor cylinder condition, is revealed when inspecting through the scavenge air ports, it may be advisable to change to a cylinder oil with a TBN40~50

Changing over to a low-alkaline oil is recommended when more than 10 hours continuous running on gas oil or low-sulphur diesel oil is anticipated.

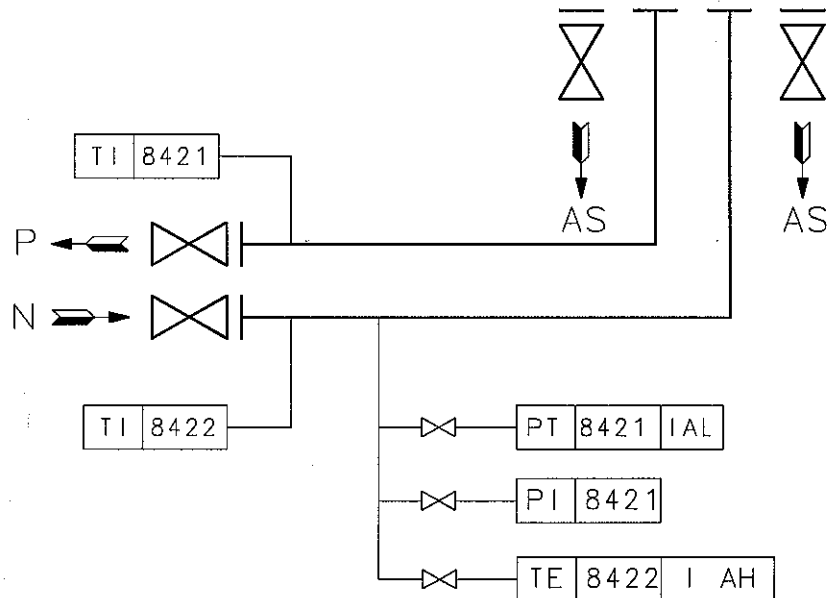
Company	Cylinder oil	
	SAE50/TBN70	SAE50/TBN40~50
BP	CLO-50 M	CL/CL-DX 405
Castrol	Cyltech 70	CL/CL-DX 405
Chevron	Delo Cyloil Special	Taro special HT 50
SK	Supermar Cyl. 70 Plus	-
Exxon	Exxmar X 70	Mobilgard L 540
Mobil	Mobilgard 570	Mobilgard L 540
Shell	Alexia 50	Alexia LS
Texaco	Taro Special HT70	Taro Special HT50
Total	Talusia HR70	Talusia LS 40

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The firms are stated in alphabetical order.

Scavenge
air cooler



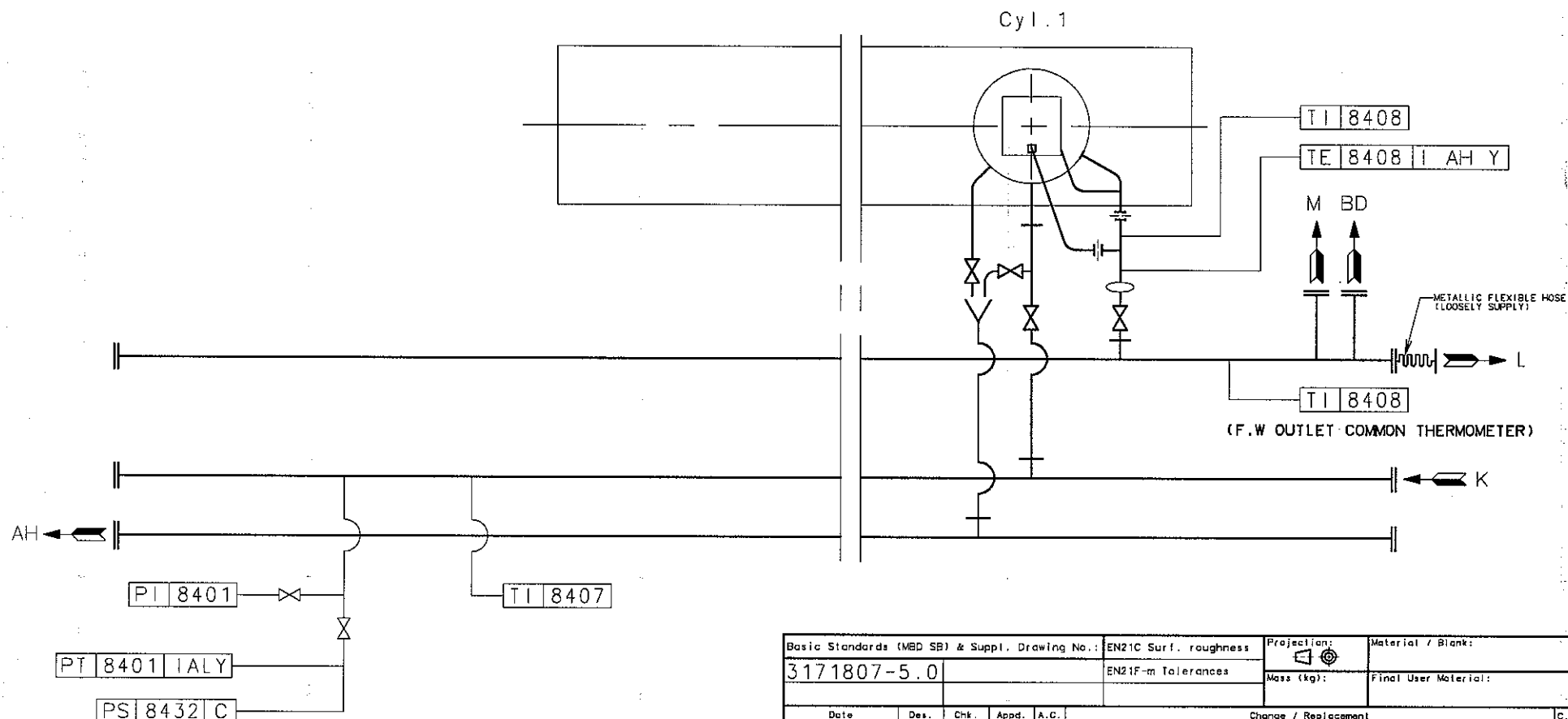
Basic Standards (MBD SB) & Suppl. Drawing No.:				EN21C Surf. roughness		Projection: 		Material / Blank:		
3174731-1.0				EN21F-m Tolerances		Mass (kg):		Final User Material:		
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement					C.No
				*	Replaced by Ident. No.:					9
				*						8
				*						7
				*						6
				*						5
				*						4
				*						3
				*						2
				*						1
20090205SMLJTSBK				*	FIRST ISSUED.					0

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Similar Drawing No.:				Replacement for Ident. No.:		
Scale:	Size:	Type:		Page No.:		
	A3			01 (01)	STX Engine Co., Ltd.	
Info. No.:		Description:				Ident. No.:
375491		LT cooling pipes, 1 to				2805029490
Final User Info. No.:		Final User Description:				Final User Ident. No.:



Basic Standards (MBD SB) & Suppl. Drawing No.:				EN21C Surf. roughness		Projection: 		Material / Blank:		
3171807-5.0						EN21F-m Tolerances		Mass (kg):		
								Final User Material:		
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement					C.No
				*	Replaced by Ident. No.:					9
				*						8
				*						7
				*						6
				*						5
				*						4
				*						3
				*						2
				*						1
20070703JTSCSPSKN				*	FIRST ISSUED.					0

Similar Drawing No.:			Replacement for Ident. No.:		
Scale:	Size: A3	Type:	Page No.: 01 (01)	 Stx Engine Co., Ltd.	
Info. No.: 375070		Description: Fresh cool. water pipes			Ident. No.: 2805019210
Final User Info. No.:		Final User Description:			Final User Ident. No.:

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CAD Drawing - do not correct by hand

Recommended Nitrite-borate corrosion inhibitors for cooling water treatment

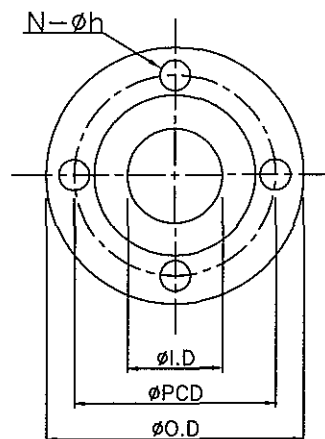
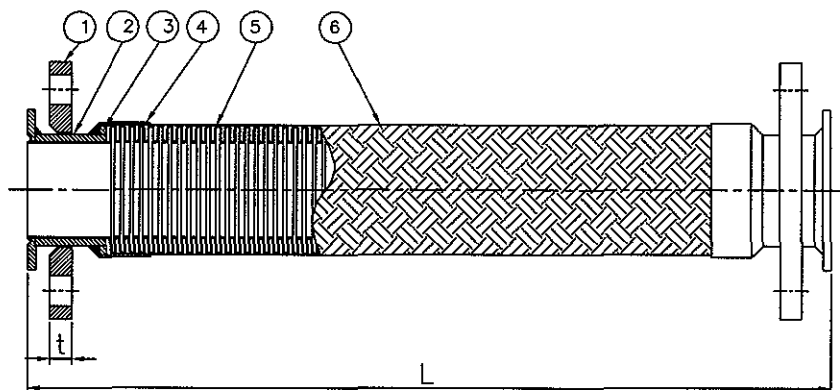
Firm	Name of Inhibitor	Delivery Form	Maker's minimum Recommended Dosage, kg/ton Cooling water
Atlas Products & Services Ltd. Erith, Kent, England	Atlas C 11 Atlas C 12 Atlas C 14	Liquid Liquid Powder	12 23 2.5
DIA Prosim Vitry-sur-Seine, France	RD 11 m	Powder	2.4
DREW Chemical Corp. New York, U.S.A	Ameroid DEWT-NC	Powder	3.5
Gamlen Chemical Company, Uxbridge, England	Gamcor NB	Powder	3
Houseman Hegro Burnham, Slough, England	Cooltreat 101	Powder	2.5
Magnes Maritec International Inc. Edgewater, New Jersey U.S.A	Boronite	Powder	2.3
Nalfloc Limited Northwich, Cheshire, England	Nalfleet 9-12	Powder	2.5
The Perolin Company, Inc Wilton, Connecticut, U.S.A	Formet 326	Powder	2.4
Rochem Ships Eq. A/S Oslo, Norway	Rocor NB	Powder	3
"VECOM" B.V., Maassluis, Holland	Vecom D99	Powder	2.5

The list is to be used as a guide and must not be considered complete. We undertake no responsibility for difficulties that might be caused by the these or other water treatments

the firms are stated in alphabetical order.

Suitable cleaners can normally be supplied by these firms.

2705001000	RECOMM. INHIBITOR F. C.W.	STX Engine Co., Ltd. 7-3
------------	---------------------------	------------------------------------



SPECIFICATION.

1. Max. working pressure : 5 kg/cm²
2. Hose assembly test pressure : 7.5 kg/cm²
3. Max. Working temperature : 300°C
4. Flexible hose to be pressure tested according to the demands of the classification society.
5. Marking - [DN, PN, TP, Maker, Ship-No., Class., Cert-No., Date].
6. Steel part to be painted with YEOL COAT QT606(silver) after sanding & picking except machined part.
7. Flange connection are transform by request of customer (JIS,ANSI,DIN,etc).
8. Bellows material can be changed by request of customer (SUS304,316L,321).

STX ENGINE
DWG. NO.

2802014240

Flange(JIS STANDARD 5K, See No. 7 of SPECIFICATION)

NOMINAL SIZE	L (mm)	Flange(JIS 5K)					Medium
		øO.D	øPCD	t	N-øh	øI.D	
5K-100A	500	200	165	16	8-19	115.4	Control Air Fresh Water Sea Water Cooling Water

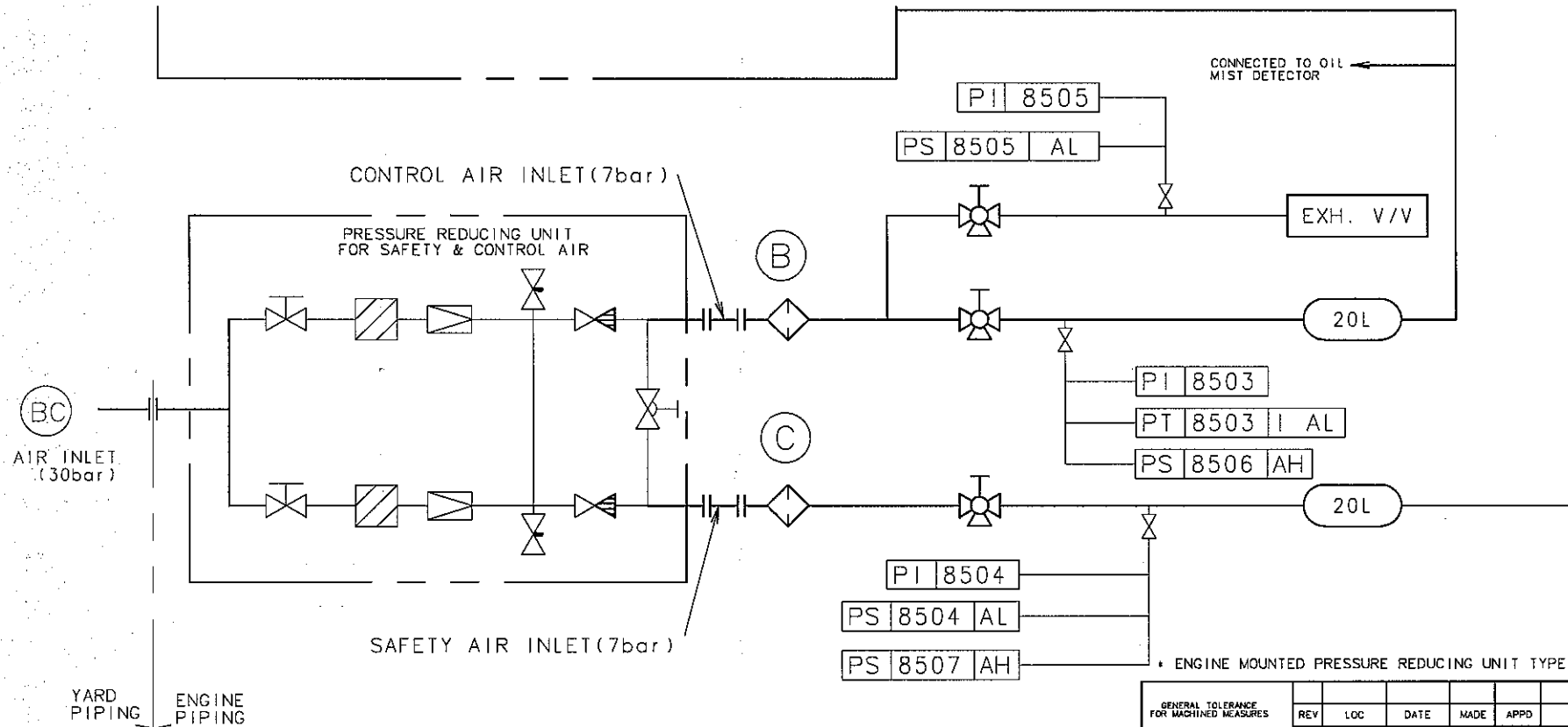
6	WIRE BRAID	1	SUS304	
5	BELLOWS	1	SUS316L	
4	BRAID BAND	2	SUS304	
3	LAP	RING PLATE	2	SS400
2	JOINT	END PIPE	2	STPG370
1	FLANGE	2	SS400	JIS 5K
NO.	DESCRIPTION	QUANTITY	MATERIAL	REMARK
3				
2				
1				
0	2005. 04. 06.	FIRST ISSUE		Y H LEE J H KOH J H KOH
REV.	DATE	REASON FOR REVISION	DRAWN	CHECKED

FLUID NAME	-
MOVEMENT	
AXIAL EXTENSION	10 mm
AXIAL COMPRESSION	10 mm
LATERAL DEFLECTION	±20 mm
ANGULAR DEGREE	mm
TEMPERATURE	DESIGN max.300°C OPERATION °C
PRESSURE	DESIGN kg/cm ² G OPERATION kg/cm ² G
TEST PRESS	PNEUM. kg/cm ² G HYDRO. kg/cm ² G
SURFACE	SILVER
TEST METH.	HYDRO. TEST
WEIGHT	kg/ea
QUANTITY	SETS

PROJECT			
METALLIC FLEXIBLE HOSE			
TYPE : HMF-501			
SUBJECT	METALLIC FLEXIBLE HOSE	DWG NO.	HKF1-100500-5040
SCALE	NONE	한 국 후 력 시 블	HAN KOOK FLEXIBLE CO.

SEE DETAIL DRAWING

"PNEUMATIC MANOEUVRING SYSTEM"



Reducing unit be supplied
by engine maker

* ENGINE MOUNTED PRESSURE REDUCING UNIT TYPE

GENERAL TOLERANCE FOR MACHINED MEASURES				REV	LOC	DATE	MADE	APPD	NOTE	
NOMINAL MEASURE		TOLERANCES		MATERIAL		SCALE N S		 CAD		MODEL S50MC-C
OVER	MAX.							WEIGHT kg		
1	4	±0.1		DWN	20070703	J. T. SEO		SIZE A3	NAME CONTROL & SAFETY AIR PIPES	
4	16	±0.2		CHKD	20070703	C. S. PARK				
16	63	±0.3		APPD	20070703	J. S. PARK				
63	250	±0.5		stx Engine Co., Ltd.						DRAWING NO. 2805019470
250	1000	±0.8								

Class No.	N4	N5	N6	N7	N8	N9
Ra Max. Value(μm)	0.2	0.4	0.8	1.6	3.2	6.3
	63	250	1000	2500	6300	16000

본 도면은 CADAM 소프트웨어로 작성되었으며 수정할 수 없음.
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8-2

8

7

6

5

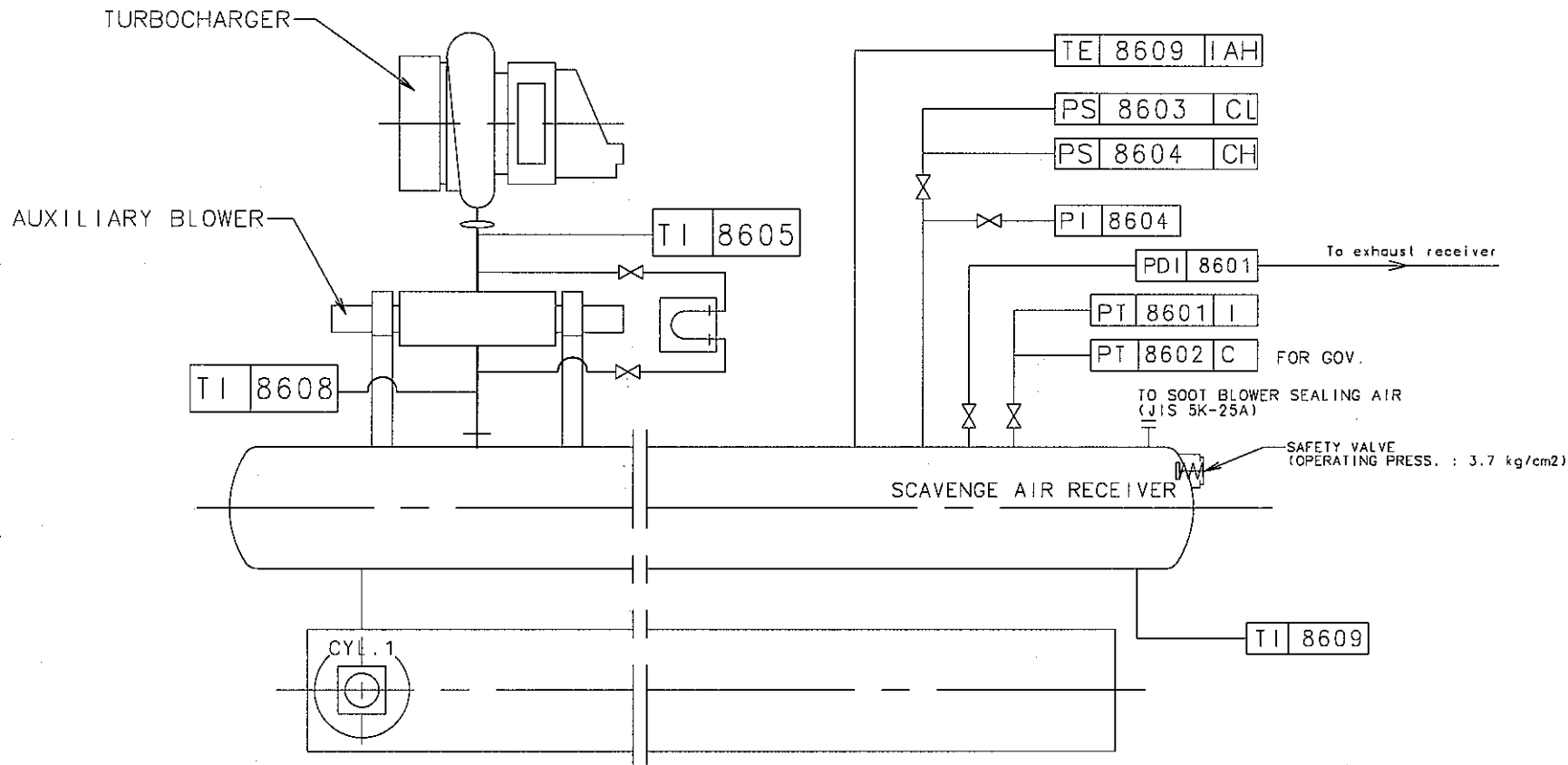
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3

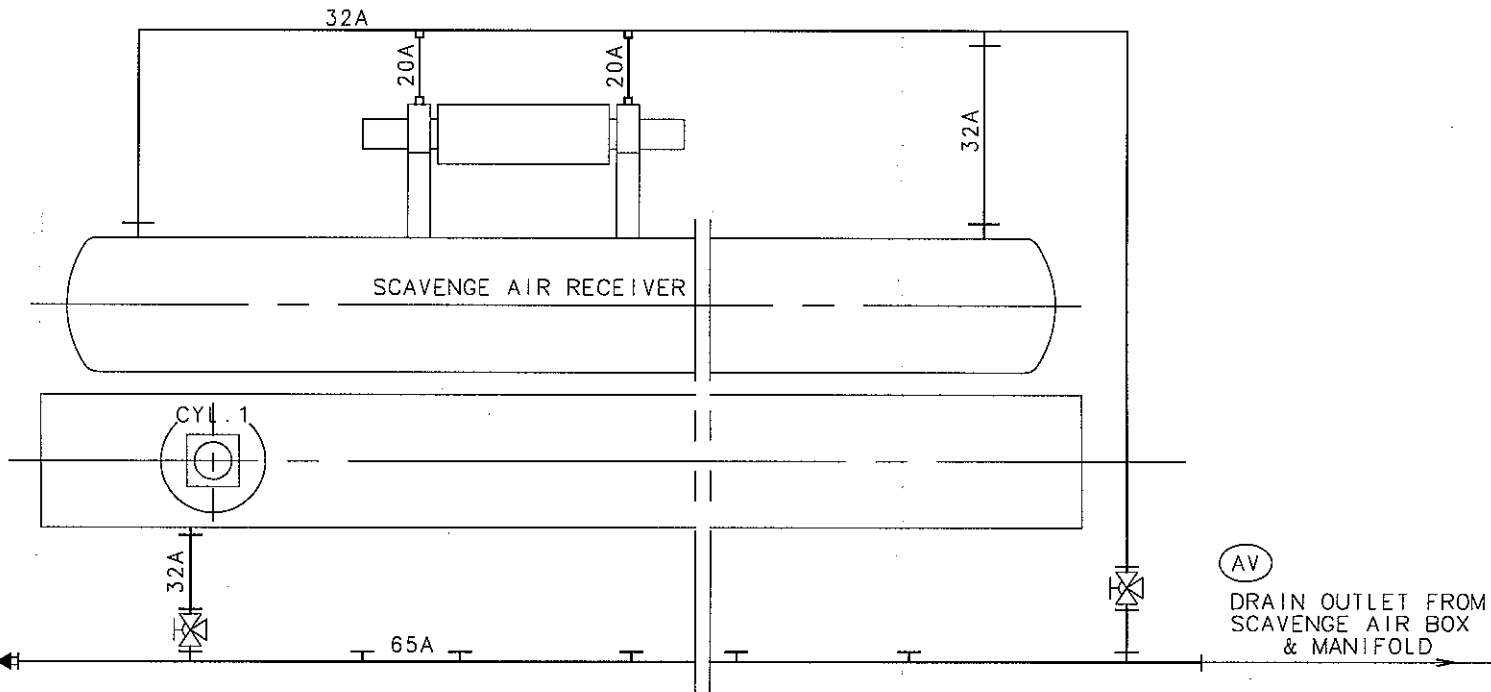
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1

8-2



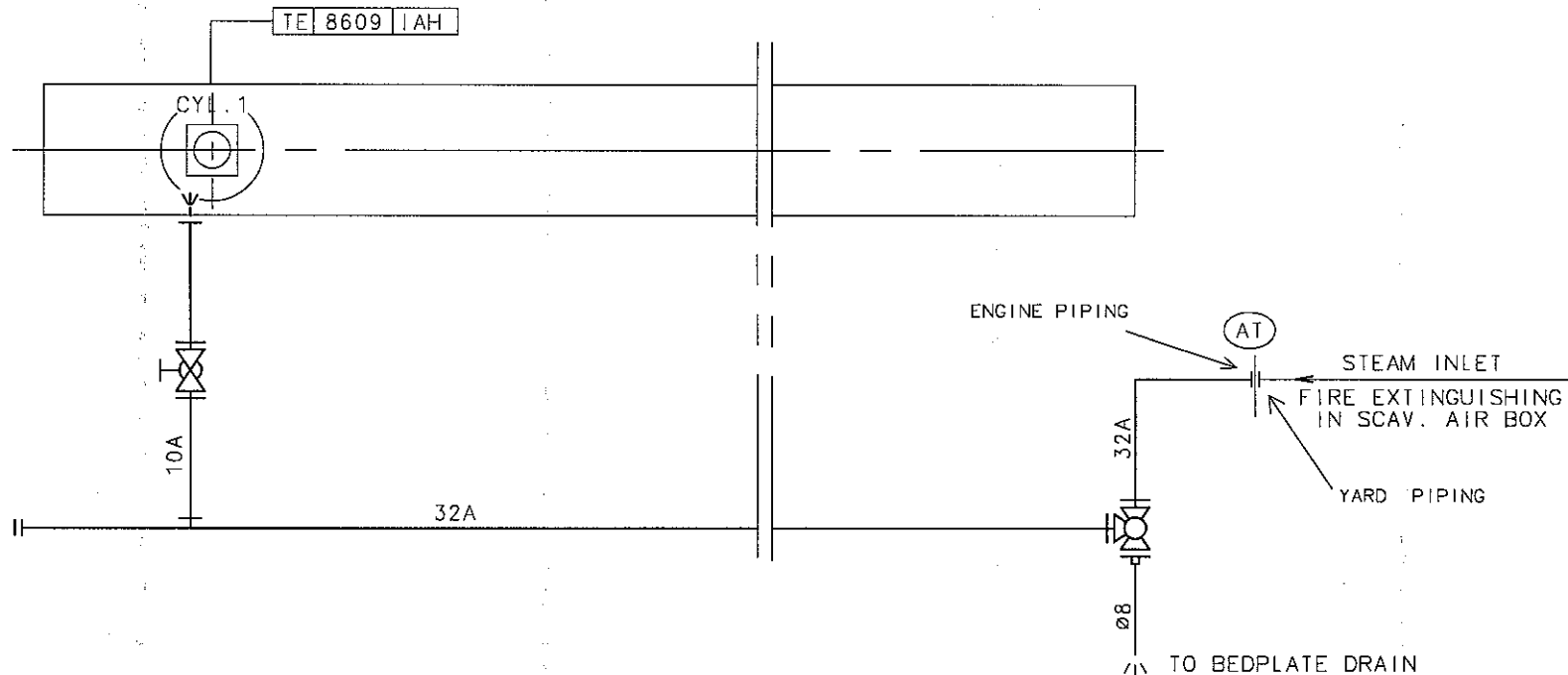
GENERAL TOLERANCE FOR MACHINED MEASURES				REV	LDC	DATE	MADE	APPD	NOTE	
NOMINAL MEASURE		TOLERANCES		MATERIAL		SCALE		 CAD  WEIGHT		MODEL
OVER	MAX.			N		S		kg		S50MC-C
1	4	±0.1		DWN	20070703	J.T.SEO	SIZE	NAME		
4	10	±0.2		CHKD	20070703	C.S.PARK	A3	SCAVENGE AIR PIPES		
16	83	±0.3		APPD	20070703	S.K.NAM	DRAWING NO.			
83	250	±0.5		STX Engine Co., Ltd.				2805019220		
250	1000	±0.8								
Class No. N4 N5 N6 N7 N8 N9 Ra Max. Value(μm) 0.2 0.4 0.8 1.6 3.2 6.3 12.5				본 도면은 (주)STX에 저작권이 있으며 허가없이 무단복제, 사용 및 대외 발수 없음. This drawing is the Property of STX Corporation and must not be wholly or partly copied, reproduced, utilized, communicated or published without written consent.						



GENERAL TOLERANCE FOR MACHINED MEASURES		1	061228	JIS	JSP	YARD COMMENT: NORMAL CLOSE VALVE ADDED.	
REV	LOC	DATE	MADE	APPD	NOTE		
NOMINAL MEASURE		SCALE		N / S		MODEL	
OVER	MAX.	WEIGHT		kg		S50MC(E)-C	
1	4	±0.1		DWN		20090206	S.M.LEE
4	16	±0.2		CHKD		20090206	J.T.SEO
63	250	±0.5		APPD		20090206	T.B.KIM
250	1000	±0.8		SIZE		A3	NAME
DRAWING NO.				2805029710			

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Basic Standards (MBO SB) & Suppl. Drawing No.:					EN21C Surf. roughness		Projection: 		Material / Blank:		
3133531-1.1						EN21F-m Tolerances		Mass (kg):		Final User Material:	
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement						C.No
			*		Replaced by Ident. No.:						9
			*								8
			*								7
			*								6
			*								5
			*								4
			*								3
			*								2
			*								1
20070918JTS		CSPSKN			FIRST ISSUE.						0

Similar Drawing No.:		Replacement for Ident. No.:	
Scale:	Size: A3	Type:	Page No.: 01 (01)
Info. No.: 399001	Description: Steam exting.in scav.box		Ident. No.: 2805020260
Final User Info. No.:		Final User Description:	
		Final User Ident. No.:	

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CUSTOMER

STX ENGINE

ARTICLE

AUX. BLOWER WITH MOTOR

SHIP NO.

S-4006/12/29/14

PROJECT NO.

H10B733/591/686/687

ENGINE TYPE

7S50MC-CCLASS : **LR**

SHIP OWNER

STX/NNC

PAINT COLOR

IN SIDE :

MUNSELL NO. **7.5 BG 7/2**

OUT SIDE :

MUNSELL NO. **7.5 BG 7/2**

FAN SPECIFICATION

MOTOR SPECIFICATION

MODEL	HAA-334/120N			FRAME NO.	225M2A	
DRAWING NO.	H-0143L-A	REV. NO.	0	MAKER	ABB	
AIR CAPACITY	2.24/4.55	m ³ /sec	1.225 kg/m ³	OUT PUT & NO. OF POLES	55 kW	2 Poles
AIR PRESSURE	571/327	mmAq		VOLTAGE & CYCLES	440 V	60 Hz
AIR DENSITY	1.225 ~ 2.1		kg/m ³	RATING CURRENT	86.4	A
REVOLUTION	3578		rpm	REVOLUTION	3578	rpm
AIR TEMP.	15		℃	AMB. TEMP & CLASS / GRADE	50 ℃, F Class,	IP 55

NO. OF PER SHIP

2

Set / Engine

TOTAL WEIGHT

ab.

850

Kg/Set

HISTORY OF DRAWING

APPROVED

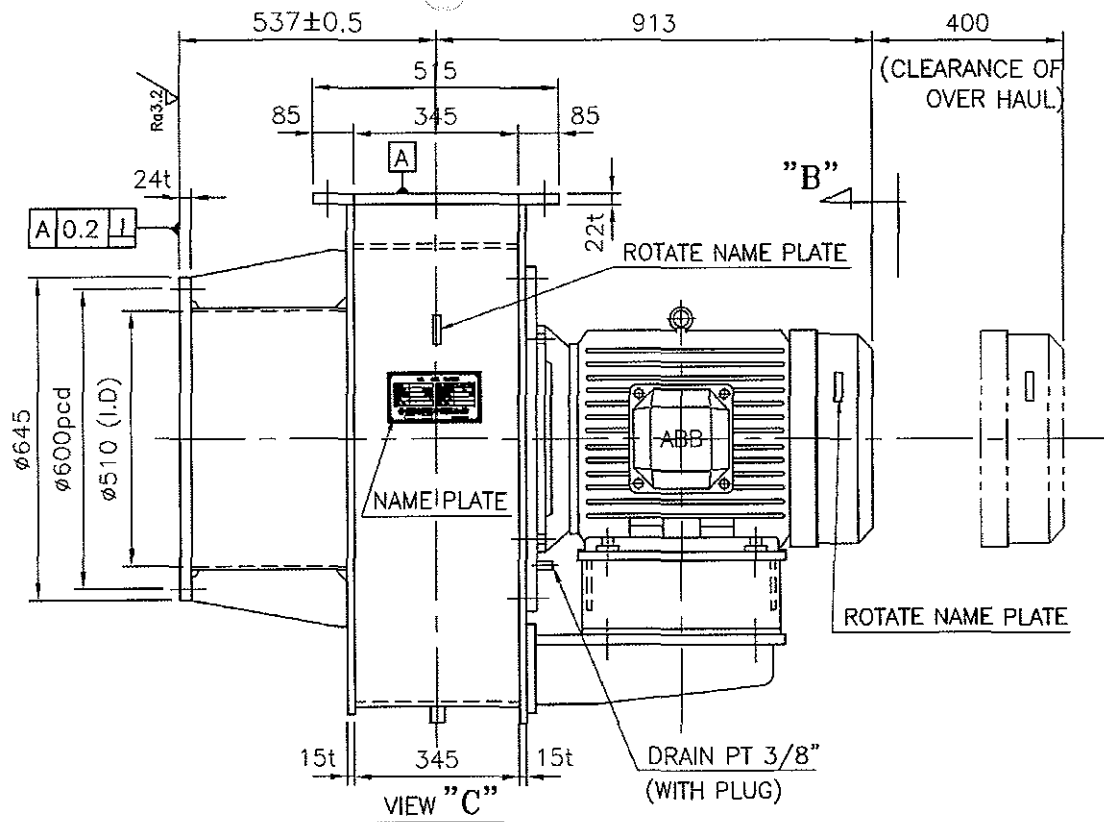
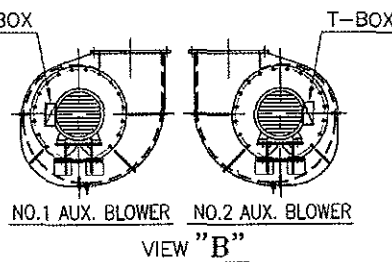
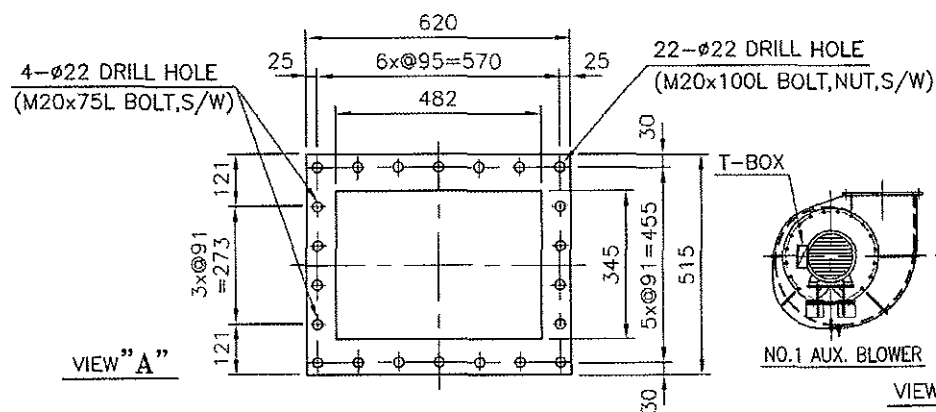
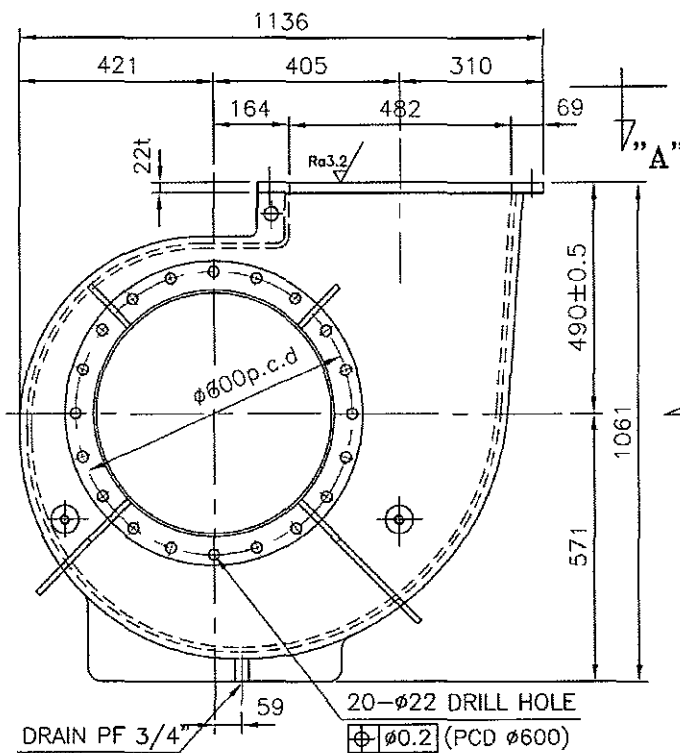
CHECKED

DRAWN

REV.	DATE	HISTORY	J.H.KIM	J.S.KIM	S.M.LEE	Y.S.YOU
0	MAY.27.2008	FOR WORKING DWG.	FILE NO.	HMR- 5635-R0	A43A	
1	SEP.01.2008	FOR REWORKING DWG.				
2			 HMMCO Hyundai Marine Machinery Co.,Ltd.			
3						
4						

DRAWING LIST

[illegible]



SPECIFICATION

TYPE	HAA-334/120N	ENGINE TYPE	7S50MC-C
AIR VOLUME	2.24/4.55 m ³ /sec	TEMP.	15 °C
PRESSURE	571/327 mmAq	AIR GRAVITY	1.225 kg/m ³
REVOLUTION	3578 RPM	SUC. AIR DENSITY	1.225~2.1 kg/m ³
MOTOR	55.0 kW 2P 440 V 60 Hz	IP 55	50 °C

Q'ty	SCALE	TITLE	LAYOUT DWG. OF.
2 Sets	N.S		AUX. BLOWER WITH MOTOR

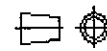
DWG. NO.	REV.	SHIP NO.	3RD ANGLE
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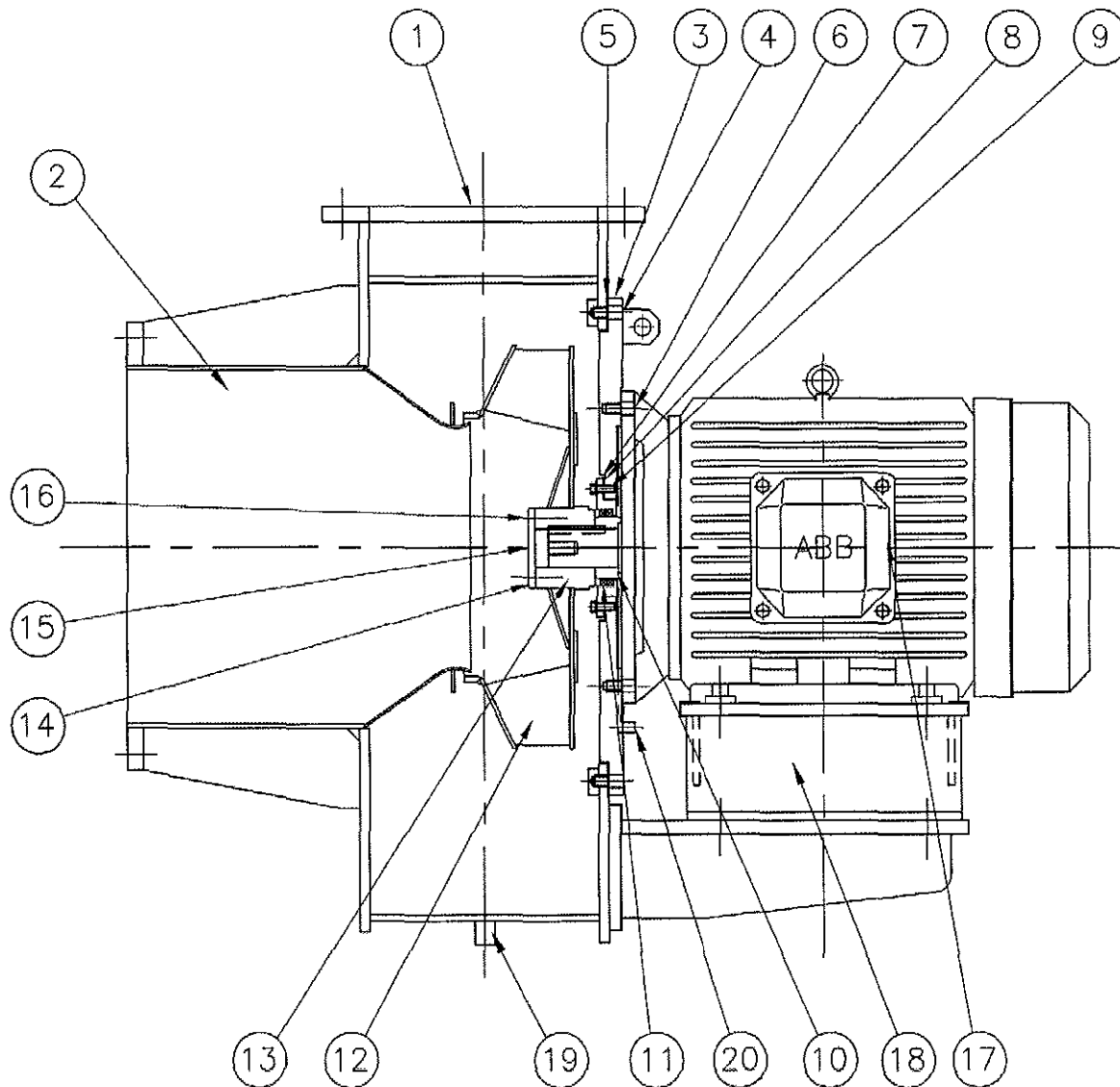
H-0143L-A	0	S-4006/12/29/14	
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A43A

DATE	CODE	MEMO	SIGN

AMEND





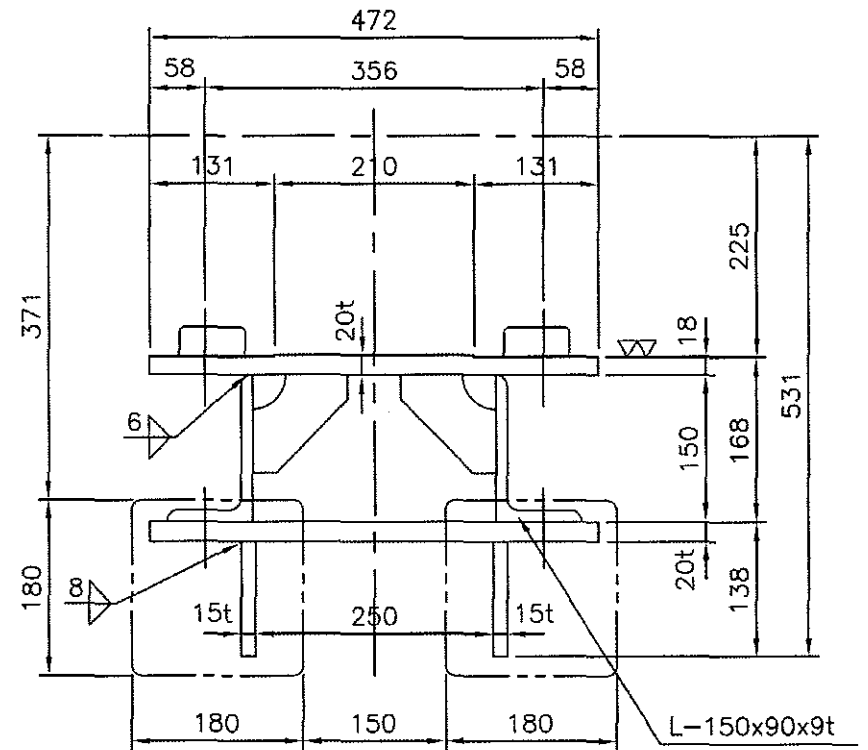
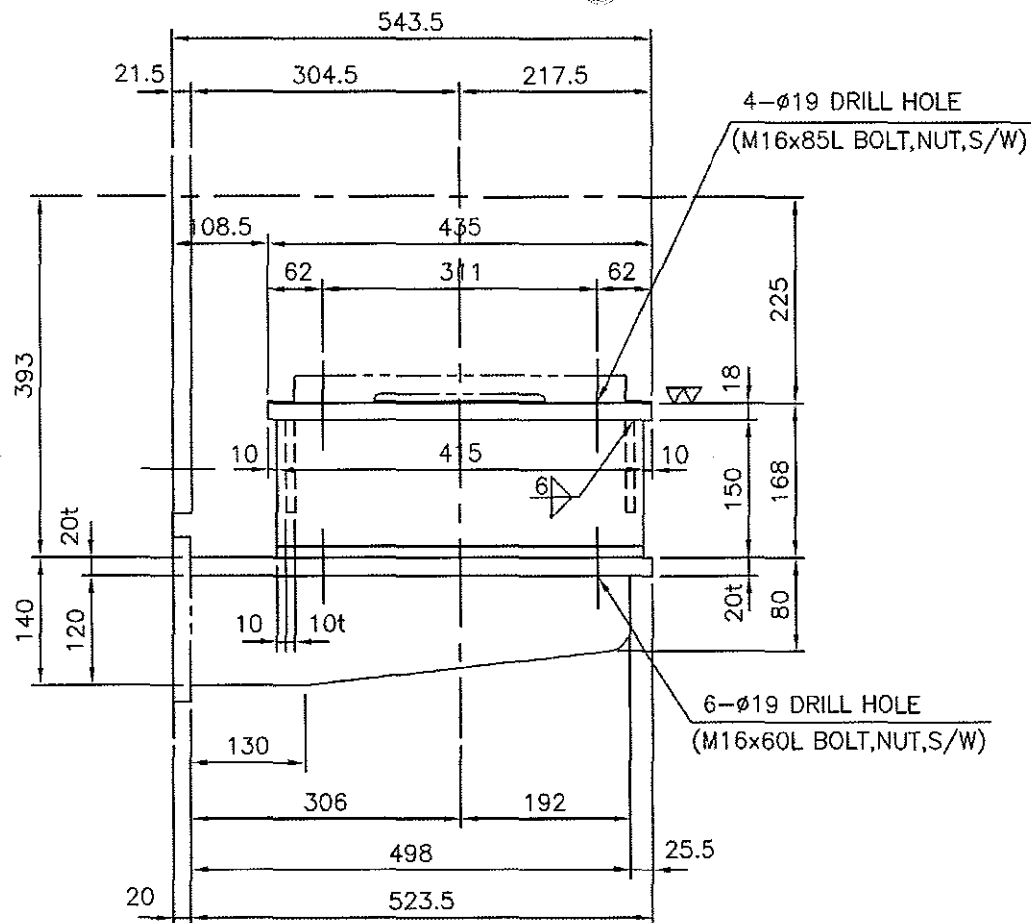
* MAKER SUPPLY

1. DISCHARGE GASKET : 620x515x1.5t - 2EA + 1SPARE
2. DISCHARGE BOLT, NUT, S/W : M20x100L - 18EA/Set
3. DISCHARGE BOLT, S/W : M20X75L - 4EA/Set
4. DISTANCE PIPE : $\phi 32 \times \phi 22 \times 30L$ -18ea/Set

19	DRAIN & PLUG	SS400	1		
18	DRAIN	SS400	1		
17	MOTOR		1		
16	BOLT, S/W	8.8T	2		
15	BOLT, S/W	8.8T	1		
14	SHAFT CAP	SS400	1		
13	BOSS	A6061	1		
12	IMPELLER	A5083P	1		
11	GASLOCK SEAL		2		
10	SEALING RING	S45C	1		
9	BOLT, S/W	8.8T	6		
8	GASKET		1		
7	STUFFING BOX	SS400	1		
6	MOTOR FIXING BOLT, S/W	8.8T	8		
5	GASKET		1		
4	BOLT, S/W	8.8T	20		
3	MOTOR SEAT	SS400	1		
2	SUCTION CONE	SS400	1		
1	FAN CASING	SS400	1		
NO.	DESCRIPTION	MAT'L	Q'ty	PART NO.	REMARK
Q'ty		SCALE		TITLE	
2 Sets					
DWG. NO.		REV.		SHIP NO.	
H-0143S-A		0		S-4006/12/29/14	
				3RD ANGLE	
					

DATE	CODE	MEMO	SIGN

AMEND



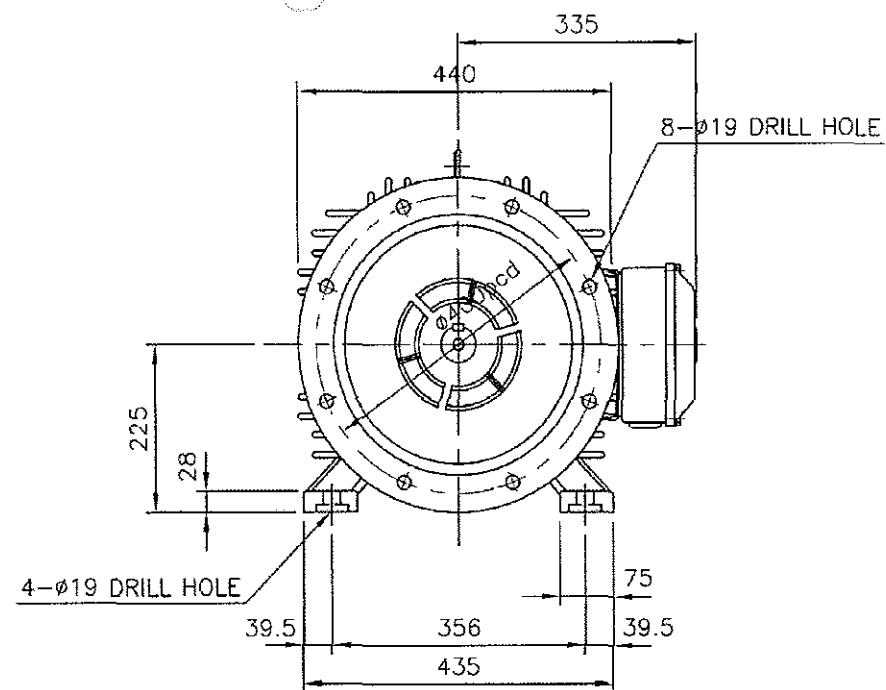
MAKER	FRAME	POWER
HHI	225S	55
ABB	225M	51.75, 55

NO.	DESCRIPTION	MAT'L	Q'ty	PART NO.	REMARK
Q'ty	SCALE	TITLE			
2 Sets	N.S	DETAIL DWG OF MOTOR BED			
DWG. NO.	REV.	SHIP NO.			3RD ANGLE
H-0011MB	0	S-2006/12/29/14			

A43A

DATE	CODE	MEMO	SIGN

AMEND



* SPECIFICATION

Technical drawing of a circular component with the following dimensions and specifications:

- Top flange width: 16Wx10H
- Outer diameter: $\phi 350js6 = \phi 350 \pm 0.016$
- Inner diameter: $\phi 55m6 = \phi 55 \begin{matrix} +0.030 \\ +0.011 \end{matrix}$
- Height: 49
- Thread: M20 TAP (DI¹: 39mm)
- Weight: **WEIGHT : 328Kg**

REMARK : 1. ELEMENT SPACE HEATER CAPACITY : 220V, 1 ϕ , 60W

Q'ty	SCALE	TITLE	LAYOUT DWG OF HHI MOTOR (55kW-2P Fr.225M2A-B35)
2 Sets	N.S		
DWG. NO.	REV.	SHIP NO.	3RD ANGLE
H-0575MT	0	S-4006/12/29/14	

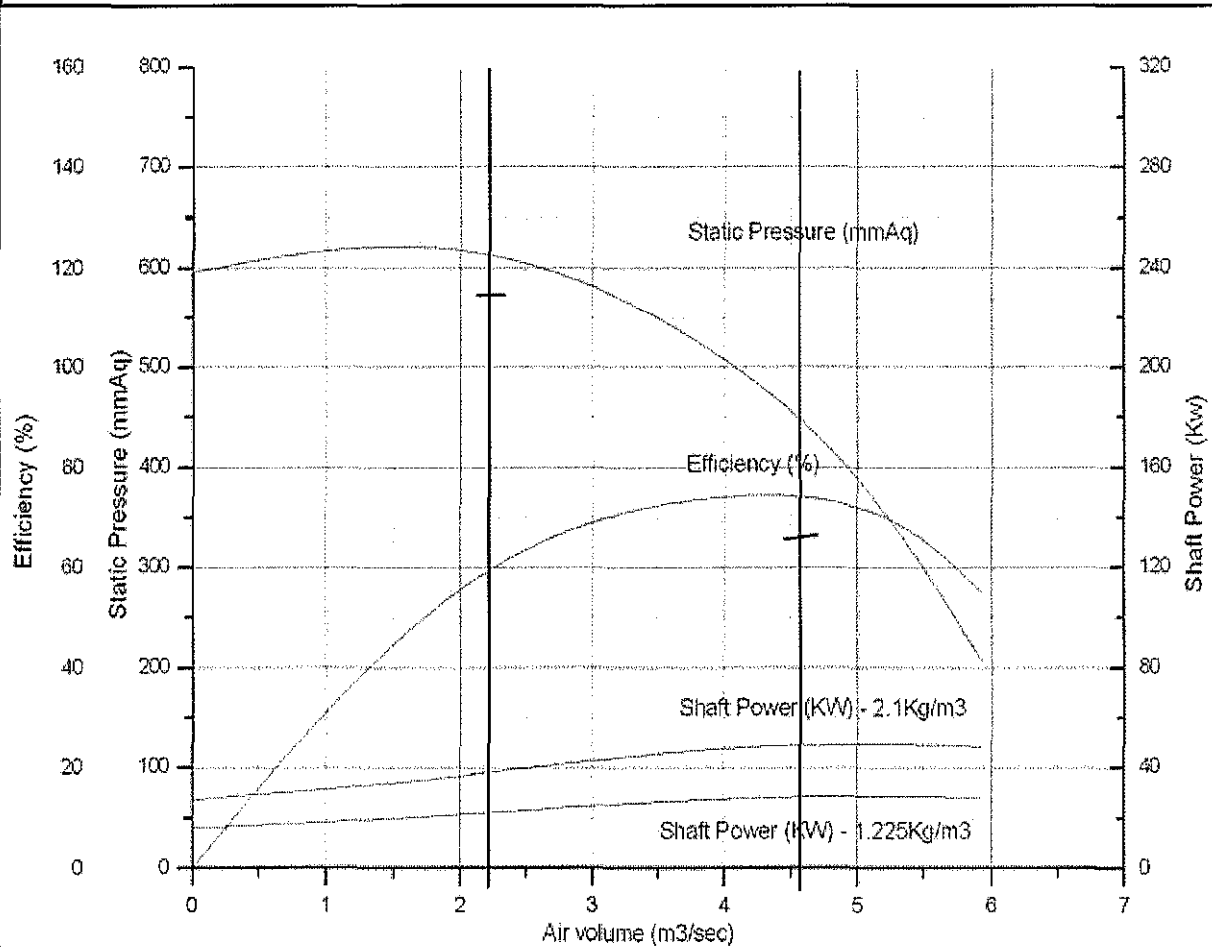
ESTIMATED PERFORMANCE CURVED

CLIENT	STX ENGINE		
SHIP YARD	STX/NNC		
SERVICE	AUX. BLOWER WITH MOTOR		
NUMBER OF UNIT	2 Sets/Ship	ENGINE TYPE	7S50MC-C
SHIP NO.	S-4006/12/29/14	TYPE	N 27

OPERATING CONDITIONS

FAN & BLOWER MODEL	HAA-334/120N	FRAME NO.	225M2A
CAPACITY	2.24/4.55 m³/sec	MOTOR CAPACITY	55 kW
FAN STATIC PRESSURE	571/327 mmAq	POLE / CYCLE	2 P / 60 Hz
AIR DENSITY	1.225 kg/m³	VOLTAGE	440 V
ACTUAL SPEED	3578 RPM	Power Consumption	49.0 kW

ESTIMATED PERFORMANCE CURVED



CUSTOMER

S T X

USED FOR STX MAN B&W 7S50MC-C MAIN DIESEL ENGINE

SUBJECT

747.8 m² AIR COOLER

DONGHWA DRAWING NO. : DHCA-0748T4Y3A

DongHwa Entec

HEAD OFFICE & FACTORY

1575-6 SONGJEONG-DONG, GANGSEO-GU,
BUSAN, KOREA (ZIP. NO. 618-270)
MAIN PHONE : (051) 974-4718
F A X : (051) 974-4750
WEB SITE : www.dh.co.kr

SEOUL OFFICE

#1316 CHAMSIL I-SPACE, 11-10 SINCHEON-DONG,
SONGPA-GU, SEOUL, KOREA (ZIP. NO. 138-240)
PHONE : (02) 421-4707
F A X : (02) 421-4708

CLASS : refer to pos

DATE : 2009. 10. 29.

REV. MARK : 1

MANAGER

CHECKED

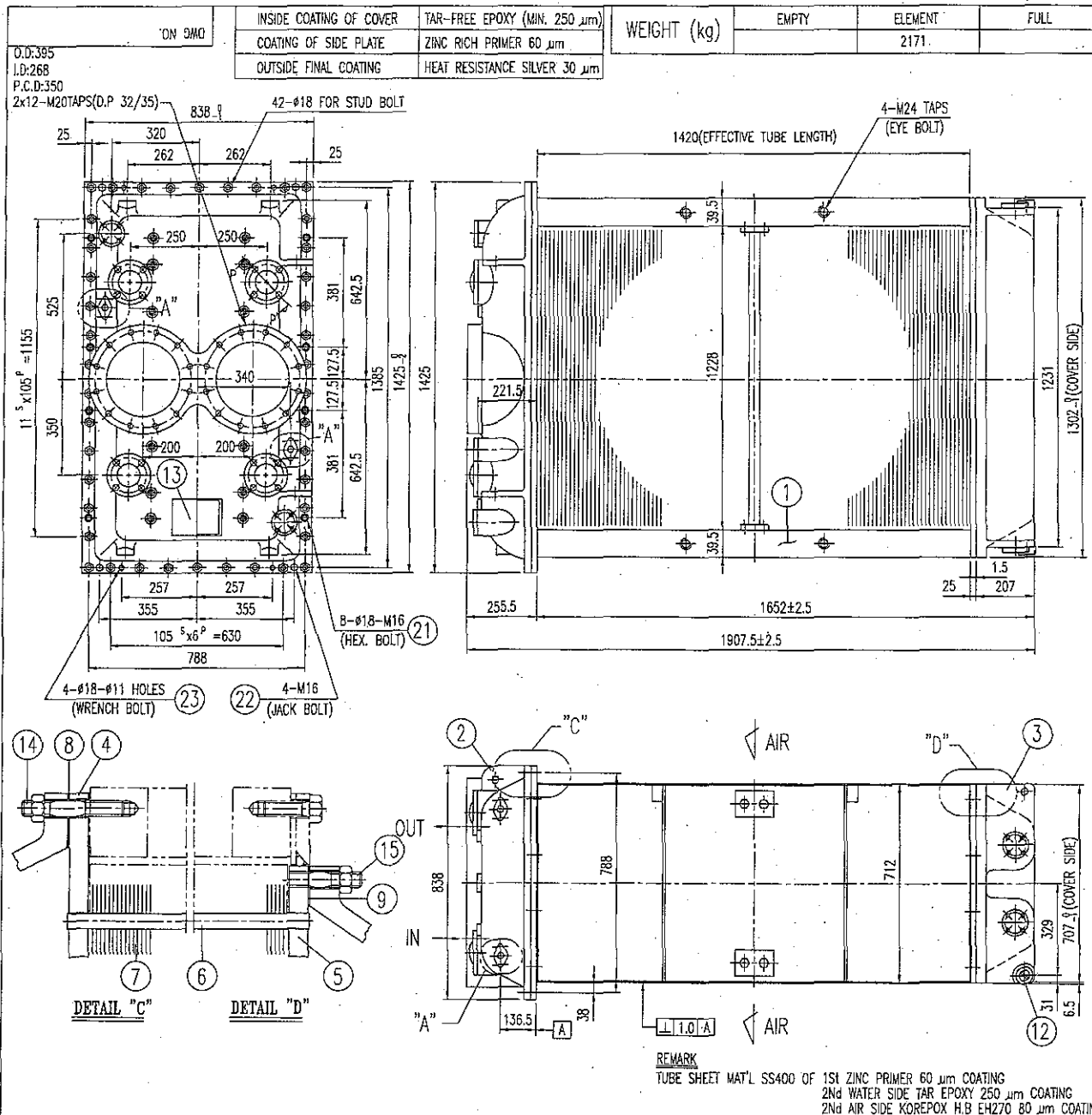
DRAWN

B. J. Kim

TECHNICAL SPECIFICATION OF AIR COOLER

- | | |
|--------------------------|-------------------------------------|
| 1. SHIP NO. refer to pos | 2. ENG. TYPE: 7S50MC-C |
| 3. CLASS : refer to pos | 4. COOLER TYPE LKM-(Y)JT |
| 5. Q'TY OF AIR COOLER : | 1 SET/ENG stx part no. : 2801006200 |

6. SPECIFICATION			
DESCRIPTION	UNIT	ISO CONDITION	TROPICAL CONDITION
HEAT DISSIPATION	kW	4256	4354
AIR AMOUNT	kg/s	27.42	25.76
TEMP. - AIR INLET	°C	189.0	214.7
TEMP. - AIR OUTLET	°C	35	48
MAX. ALLOWABLE WORKING PRESSURE - AIR	bar.G	2.65	2.53
TEST PRESSURE - AIR	bar.G	3.31	3.16
NO. OF PASS - AIR		1	
ALLOWBLE PRESS. DROP- AIR	mbar	24	
REQUIRED WATER QUANTITY	m³/hr	170	
TEMP. -WATER INLET	°C	25	36
TEMP. -WATER OUTLET	°C	46.5	58.0
MAX. ALLOWABLE WORKING PRESSURE - WATER	bar.G	4.5	
TEST PRESSURE - WATER	bar.G	6.75	
MAX. PRESS. DROP OF WATER	bar.G	0.8	
NO. OF PASS - WATER		4	
COOLANT		F.W	
AIR SIDE FOULING FACTOR	m². °C/W	0	0
WATER SIDE FOULING FACTOR	m². °C/W	0	0
7. CALCULATION			
LMTD :	°C	48.4	54.6
AIR SIDE MAX. MASS VELOCITY	m/s	8.8	9.0
AIR SIDE PRESSURE DROP	mbar	20.5	19.8
WATER SIDE VELOCITY	m/s	2.0	2.0
WATER SIDE PRESSURE DROP.	bar.G	0.32	0.32
U-VALUE	kW/m². °C	0.1365046	0.1336718
REQ. SURFACE AREA:	m²	644.6	596.3
ACT. SURFACE AREA	m²	747.8	747.8
SURFACE AREA MARGIN	%	16	25
8. TYPE OF ALTERNATIV:			
	138/38/20/4		
TUBE EFFECTIVE LENGTH	mm	1420	
TUBE LONGI. ROW NOS.		38	
TUBE VERTICAL ROW NOS.		20	
TUBE O.D	mm	14.3	
TUBE I.D	mm	12.7	
TUBE WALL TH'K	mm	0.8	
TUBE LONGI. PITCH.	mm	32	
TUBE VERTICAL PITCH.	mm	27.7	
TUBE MAT'L	mm	C7060T(CU-NI)	
FIN LENGTH	mm	1226	
FIN WITH	mm	138	
FIN TH'K	mm	0.15	
FIN PITCH	mm	2.2	
FIN MAT'L		COPPER	



M.X	DATE	REASON FOR REVISION	OWN	DES	CHK	APP
△						
△						

708 $\times 135$ (TUBE SHEET)

R25 $\times 135$

27 PF3/4 WITH PLUG

DETAIL "A"

AIR VENT & DRAIN

17 19 18

1303 $\times 135$ (TUBE SHEET)

MACHINING OF FLOATING TUBE SHEET

SECTION P-P'

27	PLUG	BS	6	PF3/4
26	BLANK			
25	BLANK			
24	BLANK			
23	WRENCH BOLT	8.8	4	M10 $\times$ 45L
22	JACK BOLT	8.8	8	M16
21	HEX. BOLT	8.8	8	M16 $\times$ 50L
20	BLANK			
19	PROTECTING COVER	FC250/SS400	4/4	
18	HEX. BOLT	8.8	16/16	M16/M12
17	GASKET	ASBESTOS-F	4/4	
16	BLANK			
15	STUT BOLT & NUT	SUS304	40	M16 $\times$ 75L
14	STUD BOLT & NUT	8.8	42	M16 $\times$ 95L
13	NAME PLATE	SUS304	1	
12	ROLLER	SUS304	1SET	
11	BLANK			
10	BLANK			
9	GASKET	ASBESTOS-F	1	1.5L
8	GASKET	ASBESTOS-F	1	1.5L
7	FIN	COPPER	1SET	2.2p.
6	TUBE	C7060T	760	$\phi 14 \times 0.8t$
5	FLOATING TUBE SHEET	SS400	1	1303 $\times$ 708 $\times$ 25t
4	FIXED TUBE SHEET	SS400	1	1425 $\times$ 838 $\times$ 25t
3	REVERSING COVER	FC250	1	WEIGHT: 292 kg
2	CHANNEL COVER	FC250	1	WEIGHT: 403 kg
1	SIDE FRAME	SS400	1sets	6t

PART NO.	PART NAME	MATERIAL	WORK Q'TY/SET	SPARE Q'TY/SET	REMARK
TYPE	LKMY-JT	CUSTOMER	STX	NO. REQUIRED	1 SET/ENG
APPROVED		SCALE	STX PART NO.	2801006200	
CHECKED		NONE	TITLE	747.8 m ² AIR COOLER	
DRAWN	B.J.Kim2009.12.11	PROJECTION	3RD	DWG NO.	DHCA-0748T4Y3A

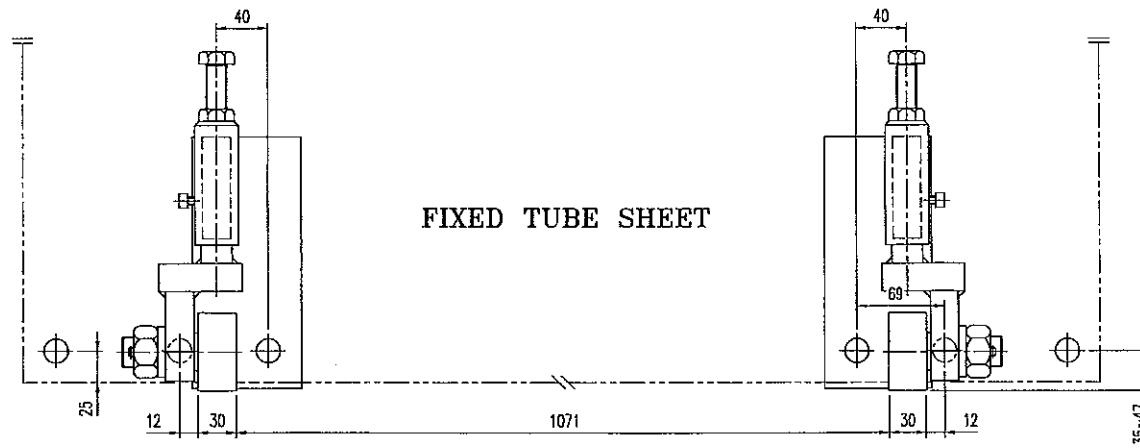
0 REV

DongHwa Entec

SAVE NAME: 0748T4Y3A

ON DWG

M.K	DATE	REASON FOR REVISION	DWN	DES	CHK	APP
△						
△						
△						



APPROVED		SCALE	TYPE	(A)/B/C/N/Q
CHECKED		NONE	TITLE	ROLLER PITCH of AIR COOLER INSERT
DRAWN	H.C.Oh 2004.09.21	PROJECTION	3RD	DWG NO. A00-AA-01A
DongHwa Entec				0
				REV

SAVE NAME:A00AA01A

ON 5MD

125

117

R3

4-ø4

DongHwa Entec

	A
SHIP NO.	B
PART NO.	C
AIR COOLER TYPE	D
CLASSIFICATION SOCIETIES	E
WATER SIDE MAX. ALLOWABLE WORKING PRESS. (bar)	F
WATER SIDE HYDRO TEST PRESS (bar)	G
COOLING SURFACE AREA (m ²)	H
MFG. NO.	I
WEIGHT EMPTY (kg)	J
DATE	K

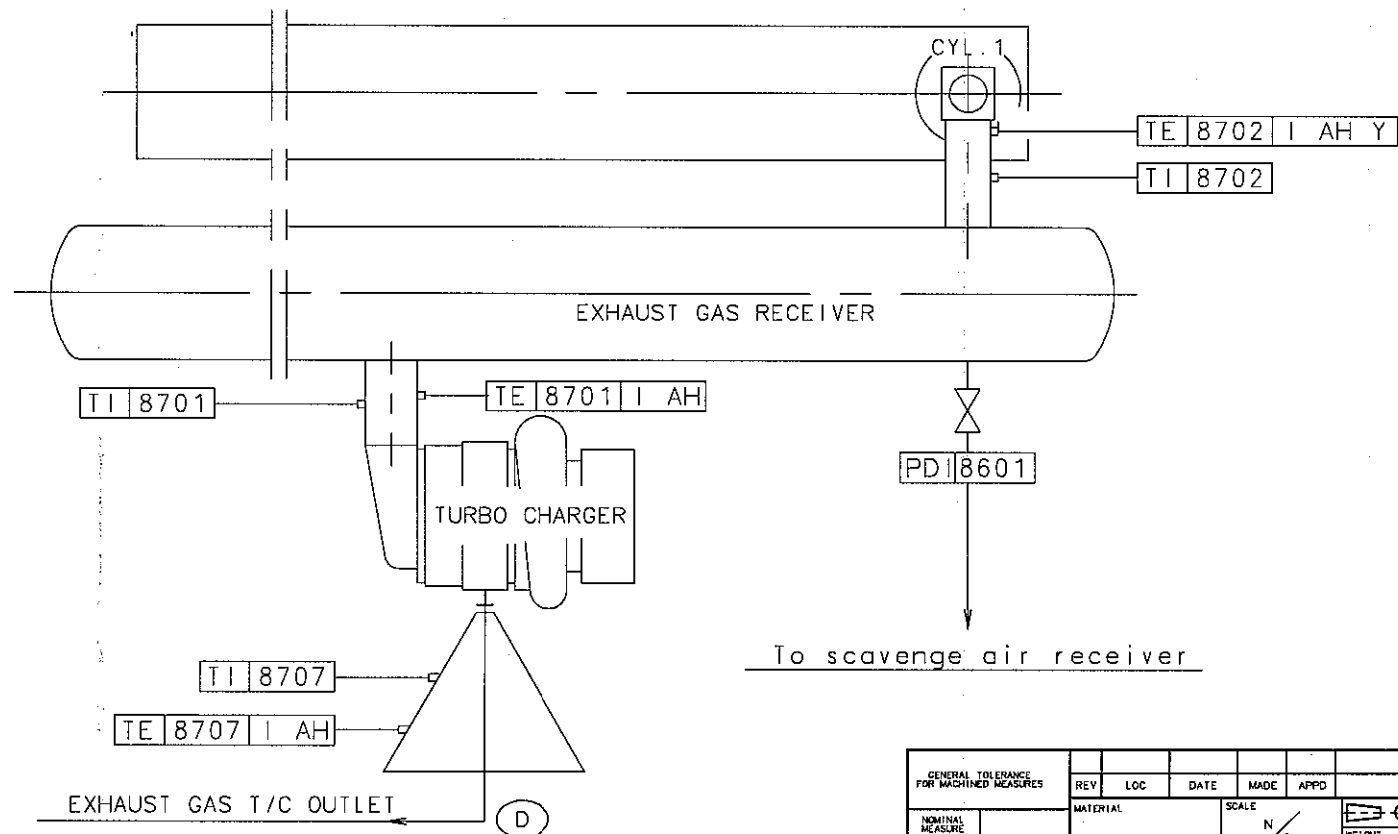
72 80

Web Site : www.dh.co.kr

A	AIR COOLER FOR 7S50MC-C ENGINE
B	refer to pos
C	2801006200
D	LKMY-JT
E	refer to pos
F	4.5
G	6.75
H	747.8
I	DH*****A
J	2171
K	refer to pos

CUSTOMER		STX		NO. REQUIRED 1SET/ENG	
APPROVED		SCALE	STX PART NO. 2801006200		
CHECKED		NON	TITLE NAME PLATE		
DRAWN	B.J.Kim 2009.10.29	PROJECTION	3RD	DWG NO. DHCA-0748T4Y3A	
DongHwa Entec					0 REV

SAVE NAME : S-4030NP

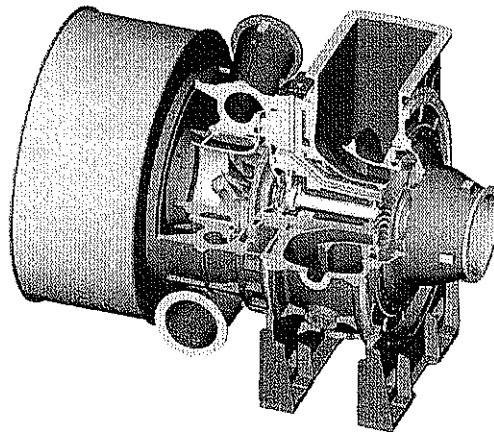


GENERAL TOLERANCES FOR MACHINED MEASURES		REV	LOC	DATE	MADE	APPD	NOTE	
NOMINAL MEASURE		MATERIAL		SCALE		WEIGHT		MODEL
TOLERANCES		N / S		N / S		kg		S50MC-C
OVER	MAX.	DWN	20070906	J. T. SEO	SIZE	NAME		
1	±0.1	CHKD	20070906	C. S. PARK	A3	EXHAUST GAS PIPES		
6	±0.2	APPD	20070906	S. K. NAM	DRAWING NO.			
10	±0.3	STX Engine Co., Ltd. 2805019900						
83	±0.5	This drawing is the Property of STX Corporation and must not be wholly or partly copied, reproduced, utilized, communicated or published without written consent.						
250	±0.8							
1000	±1.0							

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Technical Delivery Specification for Exhaust Gas Turbochargers



TCA66-20063
for engine 7S50MC-C7

stx Metal Co., Ltd.

Technical Delivery Specification for Exhaust Gas Turbochargers

Index of Contents

- Figures of Turbochargers
- Scope of Supply
- Sectional View
- Casing Position
- Weights
- Speed Measuring System
- Cleaning Systems
- List of Standard Spare Parts
- List of Standard Tools
- Connection Dimension Drawing
- U-Tube Manometer

FIGURES OF TURBOCHARGER TCA66-20063

ENGINE DATA Data for OPT=MCR

Engine	7S50MC-C	Project	WJZ / KIRAN	Manufacturer	STX
Bore (mm)	500	Stroke (mm)	2000.02	Engine-Application	Propeller
Power OPT (kW)	11060	Speed OPT (rpm)	127.0	Optimisation Power(%)	100
Power MCR (kW)	11060	Speed MCR (rpm)	127.0	MEP (bar)	
Spec. air cons. (kg/kWh)	8.92	Scav. Air. P. (bara)	3.65	Scav. Air. Temp. (°C)	37
Turbocharging system	Constant pressure			Number of TC	1
Amb. Press. (bar)	1.000	Amb. Temp. (°C)	25	Exhaust backpressure (mbar)	30

TURBOCHARGING DATA

Comp.-Press. ratio	3.72	Comp. mass-flow rate (kg/s)	27.40	Comp.-vol. flow rate (m³/s)	23.55
Turb.-Press. ratio	3.22	red.Turb.vol.-fl. rate ((m³/s)/sqr(K))	0.630	Comp.-vol. flowrate20°C (m³/s)	23.36
Turb.mass-flow rate(kg/s)	27.92	T-Bypass (%)	0.0	Temp. b. Turbine. (°C)	406
Speed (rpm)	14550	Overload Speed (rpm)	15250	Temp. Typeplate (°C)	500
				Speed Typeplate (rpm)	16000

TURBINE WHEEL

DR (mm)	406	AR (cm²)	792.3	HR (mm)	100.9
---------	------------	----------	--------------	---------	--------------

NOZZLE RING

AD (cm²)	572.0	Installed		Variation 1		Variation 2	
----------	--------------	-----------	--	-------------	--	-------------	--

TURBINE CASINGS

INLET(S)	1	90°	A (cm²)	2005	C (m/s)	
OUTLET		Diffusor				

COMPRESSOR WHEEL

D2 (mm)	633.0	D1A (mm)	474.8	B2 (mm)	48.1
Number of Blades	11/22				

DIFFUSOR

A4K (cm²)	444.0	Installed		Variation 1		Variation 2	
4 (deg)	22.7						
Number of vanes	17						

COMPRESSOR CASINGS INLET Silencer

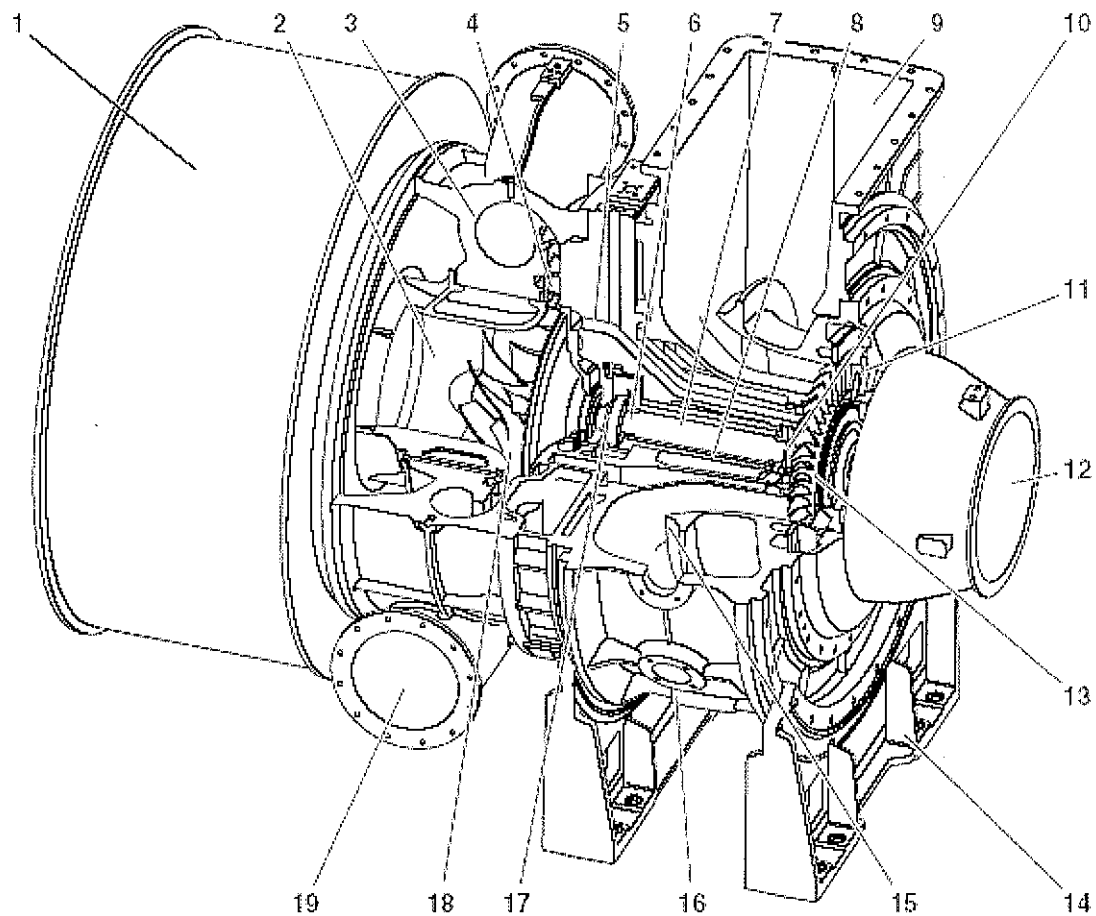
OUTLET(S)	1	A (cm²)	1064
-----------	----------	---------	-------------

Scope of Supply

TCA66-20063

- plate type silencer with soft air filter(working 1 + spare 1)
- u-tube manometer
- dry cleaning device for turbine
- pick-up for speed measurement and digital indicating instrument
- mounted lube oil tank
- sound insulation of compressor casing
- IMO-NO_x-certificate
- acceptance by classification society _____
- 1 set of standard tools in box including locking device for rotor
- 1 set of standard spare parts in box
- 5 instruction books in English language
- colour: silver/blue
- oil supply to be connected on the left side (view from turbine)

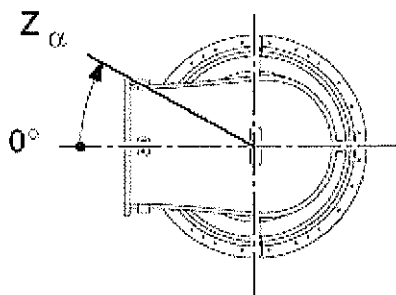
Sectional View



- | | | |
|---------------------------------|-------------------------------|------------------------------------|
| 1 Silencer | 8 Turbine rotor | 15 Gas-outlet diffuser |
| 2 Insert | 9 Gas outlet casing | 16 Outlet, washing water |
| 3 Compressor casing | 10 Bearing bush, turbine side | 17 Thrust bearing |
| 4 Diffuser | 11 Nozzle ring | 18 Compressor wheel |
| 5 Bearing casing | 12 Gas-admission casing | 19 Discharge, compressed fresh air |
| 6 Bearing bush, compressor side | 13 Turbine blades | |
| 7 Bearing body | 14 Casing foot | |

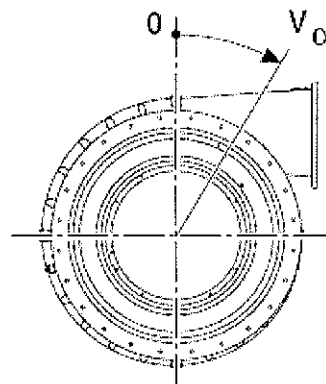
Casing positions seen from turbine side

Turbine Inlet Casing



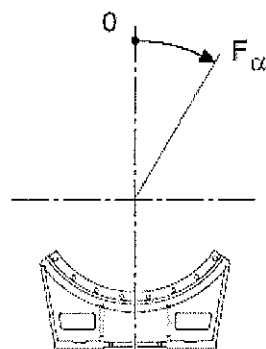
$$Z_\alpha = 30^\circ$$

Compressor Casing



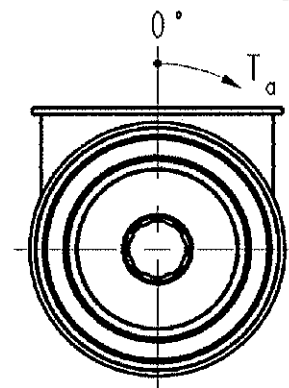
$$V_\alpha = 105^\circ$$

Casing Foot



$$F_\alpha = 0^\circ$$

Turbine Outlet Casing



$$T_\alpha = 15^\circ$$

You are kindly requested to inform us immediately in case of any changes.

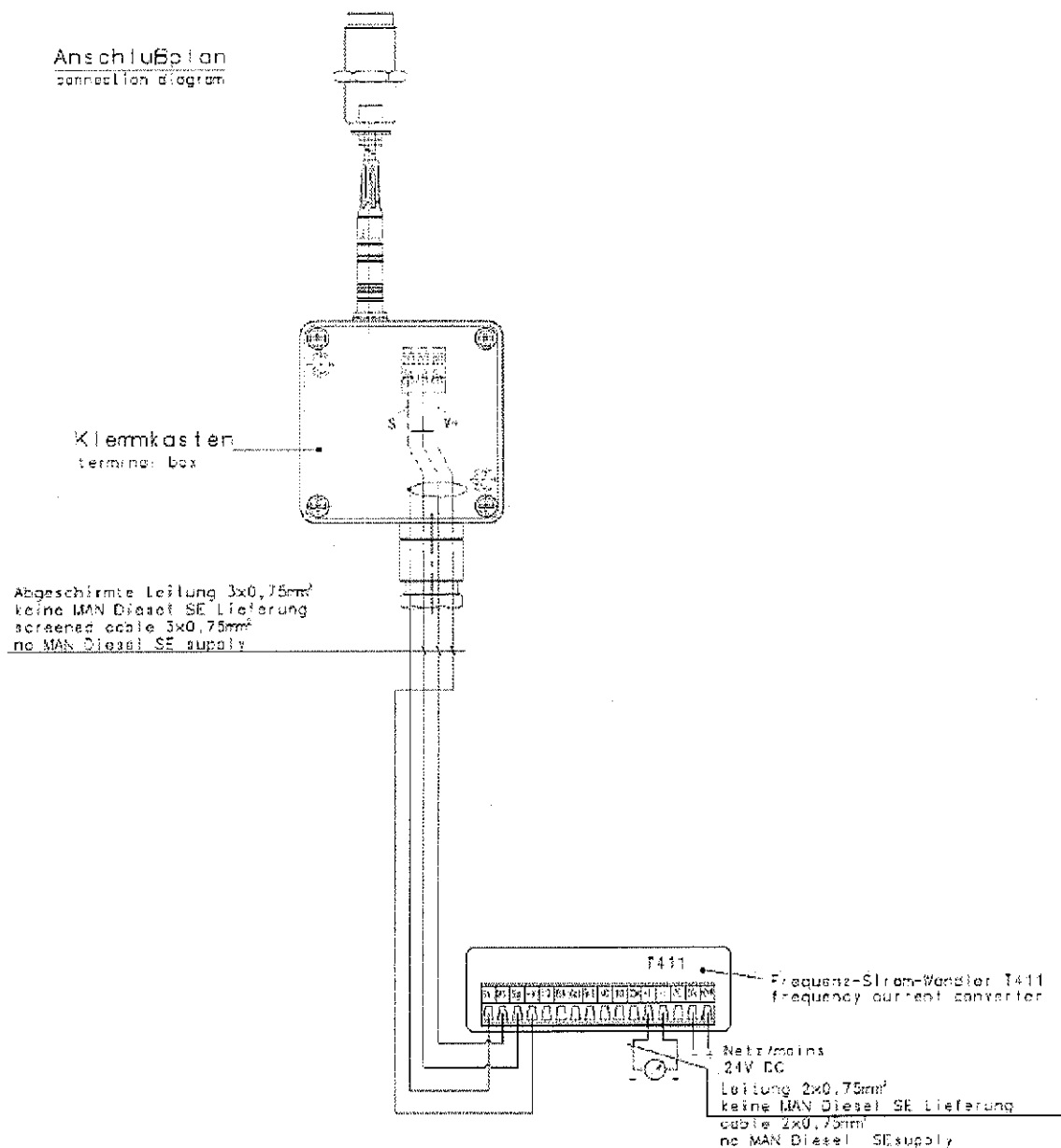
Weights of Assemblies TCA66

500	Exhaust gas turbocharger, incl. silencer	approx. 5 500 kg
501	Gas-admission casing, complete	
	0°	270 kg
	90°	380 kg
508	Gas outlet casing, complete	1 200 kg
509	Gas outlet diffuser	232 kg
	Shroud ring	20 kg
513	Nozzle ring	37 kg
517	Bearing casing, complete	2 000 kg
518	Casing feet, complete	340 kg
520	Rotor gear, complete	200 kg
	Turbine rotor	130 kg
	Compressor wheel	66 kg
540	Insert	198 kg
542	Diffuser	64 kg
544	Silencer, complete (if existing)	900 kg
545	Intake casing, complete (if existing)	xxxx kg
548	Compressor casing	single socket, complete 800 kg
554	Post lubrication tank, complete	140 kg
595	Spare parts with box	approx. 13.5 kg
598	Tools with box	approx. 46 kg

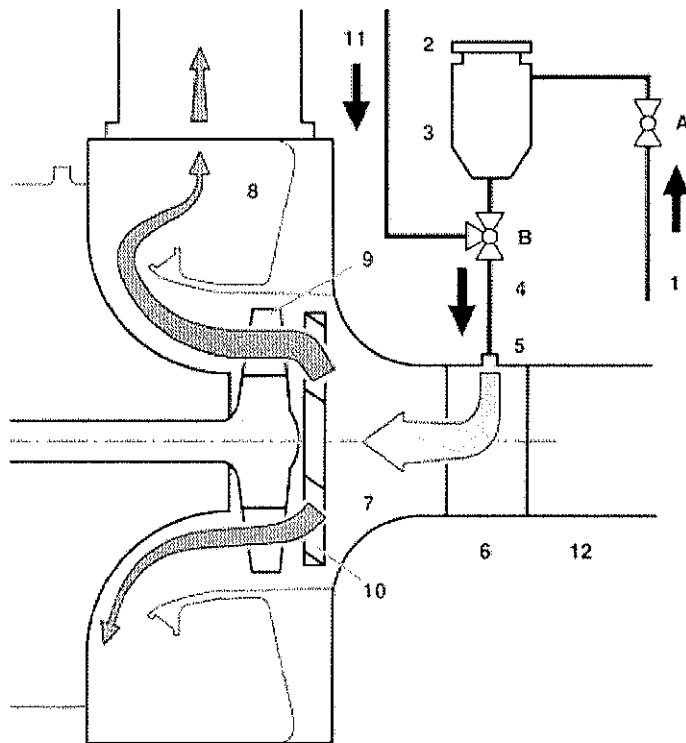
Speed Measuring System with Digital Tachometer

- Speed Transmitter
- Terminal Box
- Digital Tachometer
- Additional Analog Indicator (option)

Anschlußplan
connection diagram



Turbine Dry Cleaning

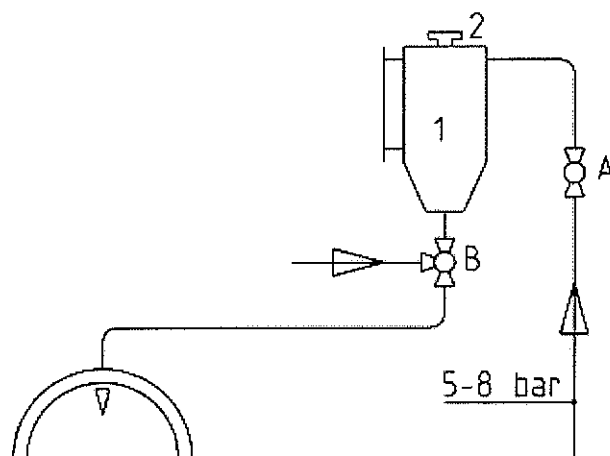


1. Compressed air pipe (5 – 8 bar)
Φ 15 x 2 mm
 2. Screw plug
 3. Granulate container
 4. Piping
 5. Connection flange
 6. Adapter
 7. Gas-admission casing
 8. Gas-outlet casing
 9. Turbine wheel
 10. Nozzle ring
 11. Sealing air (withdrawal after
charge air cooler
 12. Exhaust gas pipe
- A. Stop cock (compressed-air
connection G1/2)
- B. Three-way cock with sealing air
connection

The granulate container (2,3), the air supply cock (A) and the 3-way valve (B) will be delivered with the turbocharger. The compressed air supply pipe (G1/2", air pressure 5-8 bar) as well as the granulate pipe (25x2) have to be applied by the engine maker. The granulate container must not be positioned more than 1m below the connection flange.

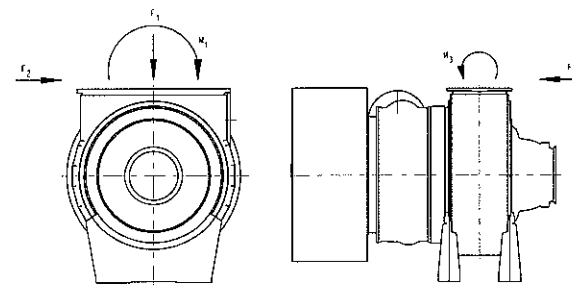
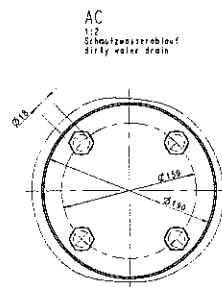
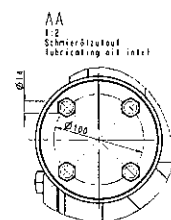
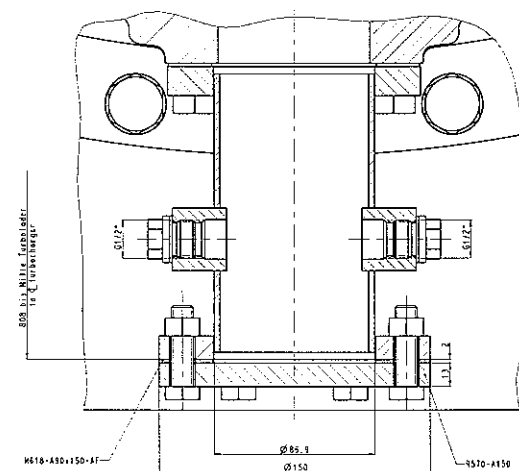
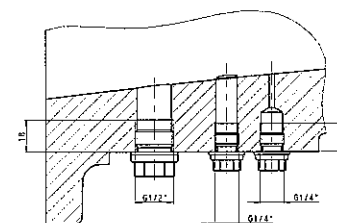
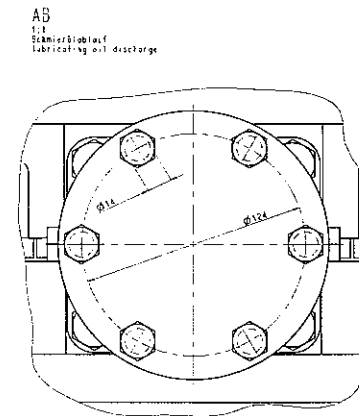
Instruction Plate for Turbine Dry Cleaning

DRY CLEANING FOR TURBINE AT NORMAL ENGINE LOAD



1. COCK (A) MUST BE CLOSED.
COCK (B) MUST BE ON POS "0" (BLOWING-OUT).
2. UNSCREW SCREW PLUG (2).
FILL GRANULATED MATERIAL INTO THE CONTAINER (1)
AND CLOSE WITH SCREW PLUG (2).
FILL-IN QUANTITY:

TCA33...0.5 LITRES	TCA77...2.0 LITRES
TCA44...0.5 LITRES	TCA88...2.5 LITRES
TCA55...1.0 LITRES	TCA99...3.0 LITRES
TCA66...1.5 LITRES	
3. OPEN COCK (A). OPEN COCK (B) SLOWLY TO POS. "1"
UNTIL A WHISTLING SOUND INDICATES THAT
BLOWING-IN OF GRANULATED MATERIAL TAKES
PLACE. INJECTION TIME APPROX. 30 SECONDS.
4. CLOSE COCK (A).
PUT COCK (B) TO POS "0" (BLOWING-OUT).

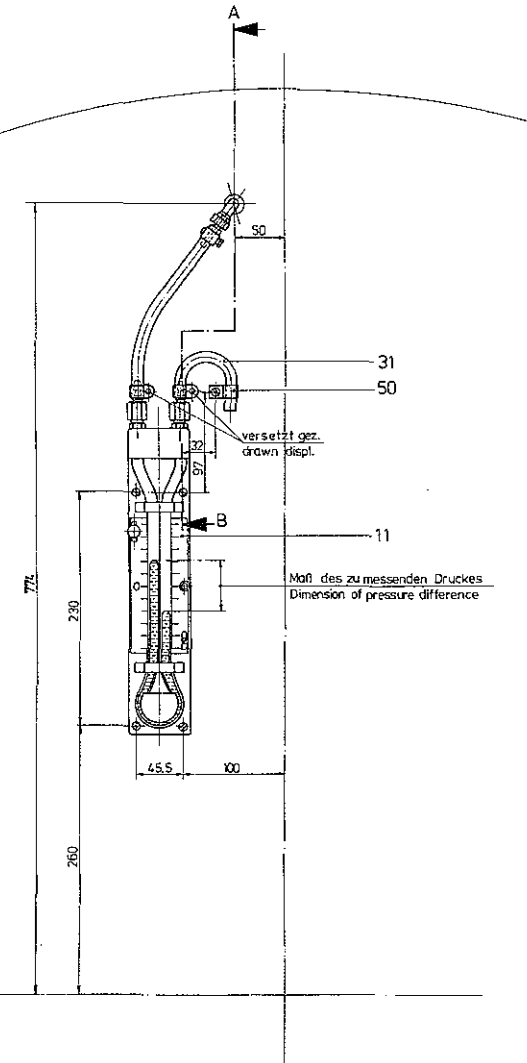


F_1 H	M_1 mm	F_2 H	F_3 H	M_3 mm
9900	7500	9900	4300	3700

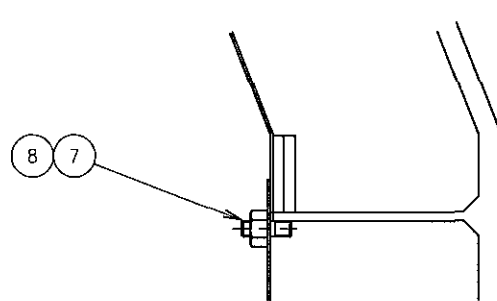
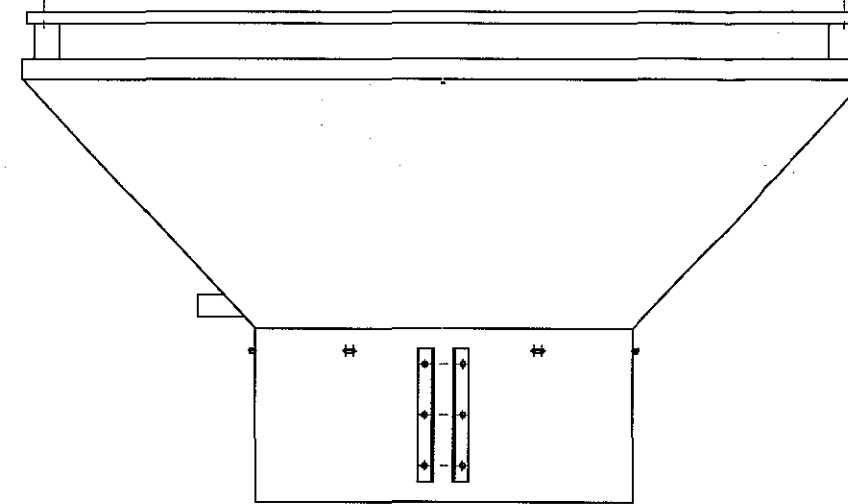
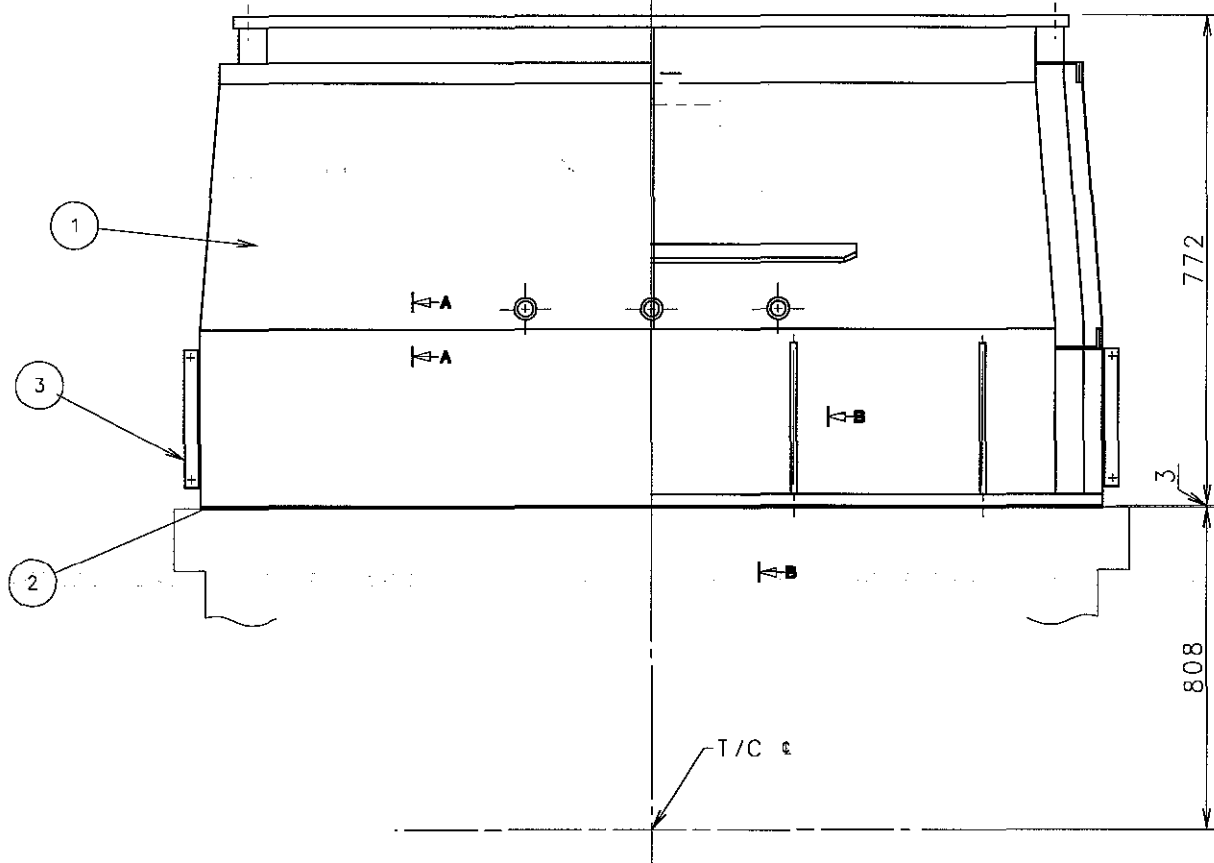
Alle anderen Gehäuseanschlüsse müssen frei von äußeren Kräften und Momenten sein.
all screwing casing connections must not be loaded by external forces and moments.

[illegible]

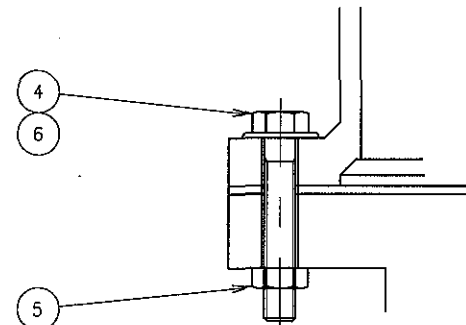
X11.56000-0028



1. Name/Firma/Fabrikat NA 57	2. Bezeichnung Druck- Meßeinrichtung für Schalldämpfer	3. Zeichn. N. & Unterschrift C11.56000 - 0027
4. Bauart NA 57	5. Auftragsnr. Ent. Abw. B malat GIN 7168	6. Zeichn. N. & Auftragsnr. 125111
7. Bauart NA 57	8. Bauart Ent. Abw. B malat GIN 7168	9. Bauart 125111
10. Bauart NA 57	11. Bauart Ent. Abw. B malat GIN 7168	12. Bauart 125111
13. Bauart NA 57	14. Bauart Ent. Abw. B malat GIN 7168	15. Bauart 125111
16. Bauart NA 57	17. Bauart Ent. Abw. B malat GIN 7168	18. Bauart 125111
19. Bauart NA 57	20. Bauart Ent. Abw. B malat GIN 7168	21. Bauart 125111
22. Bauart NA 57	23. Bauart Ent. Abw. B malat GIN 7168	24. Bauart 125111
25. Bauart NA 57	26. Bauart Ent. Abw. B malat GIN 7168	27. Bauart 125111
28. Bauart NA 57	29. Bauart Ent. Abw. B malat GIN 7168	30. Bauart 125111
31. Bauart NA 57	32. Bauart Ent. Abw. B malat GIN 7168	33. Bauart 125111
34. Bauart NA 57	35. Bauart Ent. Abw. B malat GIN 7168	36. Bauart 125111
37. Bauart NA 57	38. Bauart Ent. Abw. B malat GIN 7168	39. Bauart 125111
40. Bauart NA 57	41. Bauart Ent. Abw. B malat GIN 7168	42. Bauart 125111
43. Bauart NA 57	44. Bauart Ent. Abw. B malat GIN 7168	45. Bauart 125111
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58. Bauart NA 57	59. Bauart Ent. Abw. B malat GIN 7168	60. Bauart 125111
61. Bauart NA 57	62. Bauart Ent. Abw. B malat GIN 7168	63. Bauart 125111
64. Bauart NA 57	65. Bauart Ent. Abw. B malat GIN 7168	66. Bauart 125111
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79. Bauart NA 57	80. Bauart Ent. Abw. B malat GIN 7168	81. Bauart 125111
82. Bauart NA 57	83. Bauart Ent. Abw. B malat GIN 7168	84. Bauart 125111
85. Bauart NA 57	86. Bauart Ent. Abw. B malat GIN 7168	87. Bauart 125111
88. Bauart NA 57	89. Bauart Ent. Abw. B malat GIN 7168	90. Bauart 125111
91. Bauart NA 57	92. Bauart Ent. Abw. B malat GIN 7168	93. Bauart 125111
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97. Bauart NA 57	98. Bauart Ent. Abw. B malat GIN 7168	99. Bauart 125111
100. Bauart NA 57	101. Bauart Ent. Abw. B malat GIN 7168	102. Bauart 125111
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106. Bauart NA 57	107. Bauart Ent. Abw. B malat GIN 7168	108. Bauart 125111
109. Bauart NA 57	110. Bauart Ent. Abw. B malat GIN 7168	111. Bauart 125111
112. Bauart NA 57	113. Bauart Ent. Abw. B malat GIN 7168	114. Bauart 125111
115. Bauart NA 57	116. Bauart Ent. Abw. B malat GIN 7168	117. Bauart 125111
118. Bauart NA 57	119. Bauart Ent. Abw. B malat GIN 7168	120. Bauart 125111



SECTION A - A



SECTION B - B

8	P060F0000600	PLAT WASHER		12		
7	P040H0006000	HEX. NUT		12		
6	P060F0001600	PLAT WASHER		26		
5	P040H0016000	HEX. NUT		26		
4	EN61Y1665	SCREW		26		
3	2801006660	PROTECTION COVER		1		
2	2802009730	GASKET		1		
1	2802019280	EXH. PIPE		1		
NO	PART NO	NAME	MATERIAL	QTY	WEIGHT	REMARK

GENERAL TOLERANCE FOR MACHINED MEASURES		REV	LOC	DATE	MADE	APPD	NOTE	
NOMINAL MEASURE		TOLERANCES		MATERIAL		SCALE		MODEL
OVER	MAX.							S50MC-C
1	4	±0.1		DWN	20090210	S.M.LEE	SIZE	NAME
4	16	±0.2		CHKD	20090210	J.T.SEO	A2	EXH. PIPE ASS'Y T/C OUTLET
				APPD	20090210	T.B.KIM		
16	63	±0.3		STX Engine Co., Ltd.				
63	250	±0.5		DRAWING NO. 2801022480				
250	1000	±0.8						

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F

AIR SUPPLY
(from the manoeuvring system.)

Safety relief valve
air spring

Setting point : 23bar

Safety relief valve

Controlled Oil Level
(for exh. v/v sealing)

Basic Standards (MBD SB) & Suppl. Drawing No.:					EN21C Surf. roughness	Projection:	Material / Blank:
					EN21F-m Tolerances	Mass (kg):	Final User Material:
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement		C.No
				*	Replaced by Ident. No.:		9
				*			8
				*			7
				*			6
				*			5
				*			4
				*			3
				*			2
				*			1
20090205	SML	JTS	TBK	*	First issue		0
Similar Drawing No.:					Replacement for Ident. No.:		

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Scale:	Size: A3	Type: COL	Page No.: 01 (01)	STX Engine Co., Ltd.
Info. No.: 374202	Description: Spring.air pipes-exh.valve			Ident. No.: 2805029520
Final User Info. No.:	Final User Description:			Final User Ident. No.: 11-1

Drawing is not correct by hand

F

Water mist catcher

Sight glass

Scavenge
air cooler

LS 8611 AH

The letters refer to "List of flanges"
The item No. refer to "List of instruments"

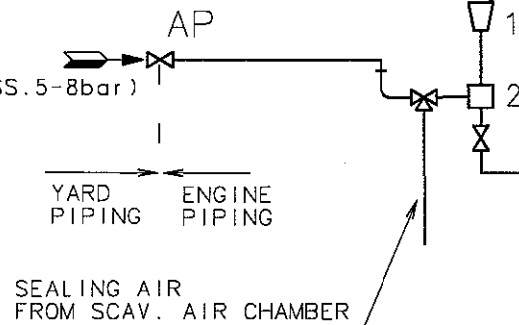
AM AL

Basic Standards (MBD SB) & Suppl. Drawing No.:				EN21C Surf. roughness		Projection:		Material / Blank:	
3171686-3.0				EN21F-m Tolerances		Mass (kg):		Final User Material:	
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement				C.No
				*	Replaced by Ident. No.:				9
				*					8
				*					7
				*					6
				*					5
				*					4
				*					3
				*					2
				*					1
20061020JTSCSPJSP				*	FIRST ISSUE				0
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Scale:	Size:	Type:			Page No.:		STX Engine Co., Ltd.		
	A3	UMS			01 (01)				
Info. No.:		Description:				Ident. No.:			
375405		Air-cool.clean.pipes				2805015640			
Final User Info. No.:		Final User Description:				Final User Ident. No.:			

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AIR INLET FOR SOFT BLAST
CLEANING OF T/C(WORKING PRESS.5-8bar)



Dry cleaning turbine side

To bedplate drain

AE

MAN B&W TCA turbocharger

1. TRAY FOR SOLID GRANULES
2. CONTAINER FOR GRANULES

The letters refer to "List of flanges"

Basic Standards (MBD SB) & Suppl. Drawing No.:				EN21C Surf. roughness	Projection: 	Material / Blank:	
1264080-9.3				EN21F-m Tolerances	Mass (kg):	Final User Material:	
1264094-2.3							
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement		C.N.
				*	Replaced by Ident. No.:		9
				*			8
				*			7
				*			6
				*			5
				*			4
				*			3
				*			2
				*			1
20090210	SML	JTS	TBK	*	FIRST ISSUE		0

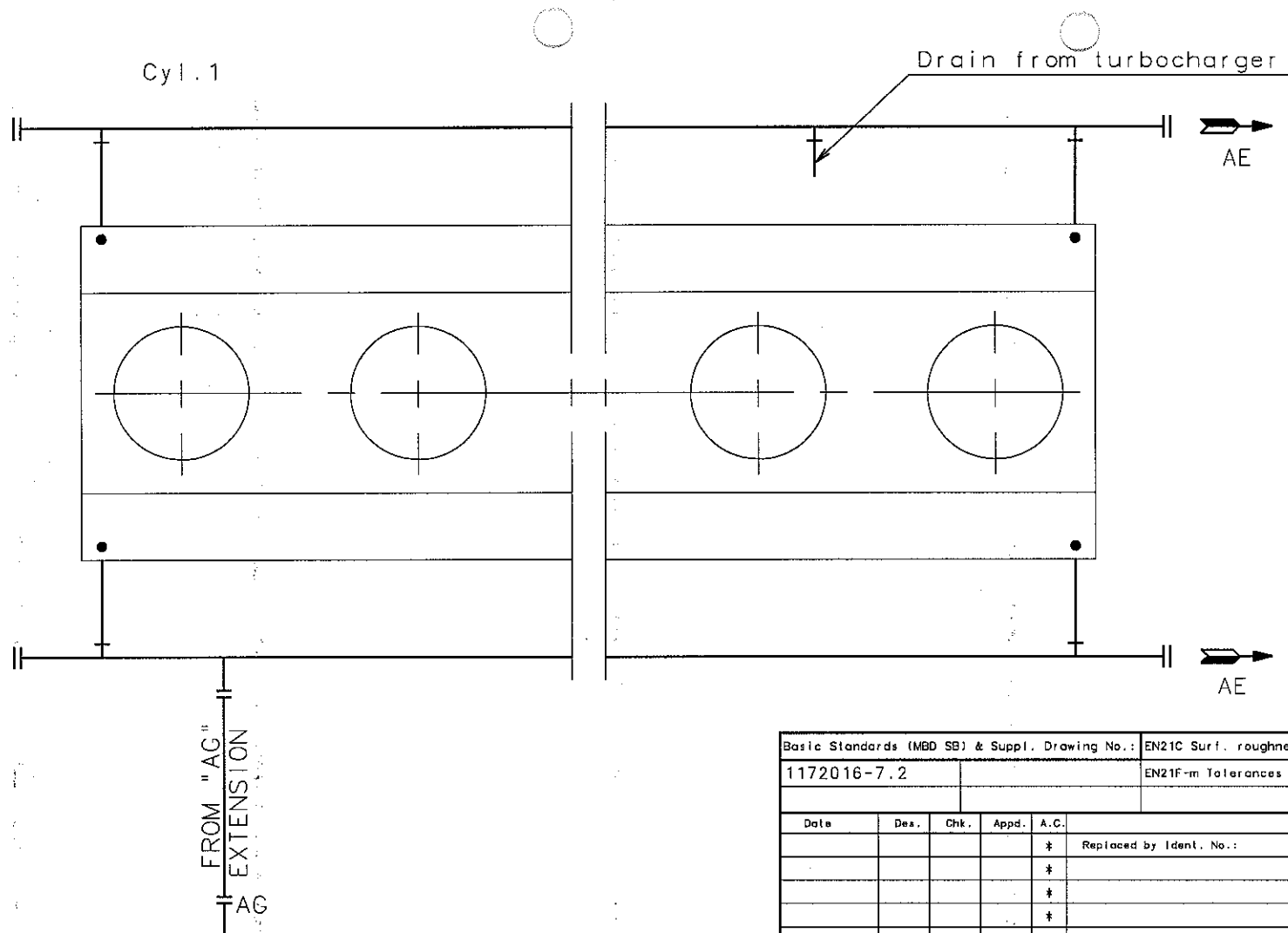
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	A3				01 (01)	
Info. No.:		Description:			Ident. No.:	
375323		T/C CLEANING PIPES			2805030000	
Final User Info. No.:		Final User Description:			Final User Ident. No.:	

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CAD Drawing - do not correct by hand



Basic Standards (MBD SB) & Suppl. Drawing No.:		EN21C Surf. roughness	Projection:	Material / Blank:
1172016-7.2		EN21F-m Tolerances		
			Mass (kg):	Final User Material:
Date	Des.	Chk.	Appd.	A.C.
Change / Replacement				C.No
			*	Replaced by Ident. No.:
			*	
			*	
			*	
			*	
			*	
			*	
			*	
			*	
			*	
			*	
20090202	SML	JTS	TBK	* First issued
Similar Drawing No.:				Replacement for Ident. No.:
Scale:	Size:	Type:	Page No.:	
	A3		01 (01)	
Info. No.:	Description:			Ident. No.:
373280	Bedplate drain pipes			2805029720
Final User Info. No.:	Final User Description:			Final User Ident. No.:

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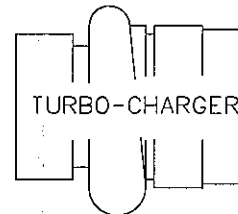


CAD Drawing - do not correct by hand

PRESSURE DROP, EXHAUST RECEIVER
& SCAVENGING MANIFOLD

press gauge

PD 8601



TURBO-CHARGER

PRESSURE DROP
AIR COOLER (W.C)

ø10X1.5

AIR COOLER

EXHAUST RECEIVER

ø10X1.5

SCAVENGING MANIFOLD

CYL. 1

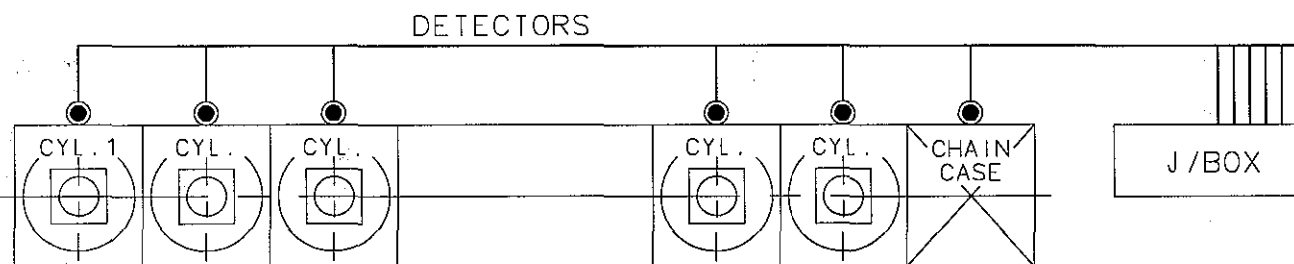
TC : TCA

GENERAL TOLERANCE FOR MACHINED MEASURES			REV	LOG	DATE	MADE	APPD	NOTE		
NOMINAL MEASURE		TOLERANCES	MATERIAL		SCALE N S		 CAD		MODEL	
OVER	MAX.				WEIGHT		kg	S50MC-C		
1	4	±0.1	DWN	20041125	J. T. SEO	SIZE	NAME			
4	16	±0.2	CHKD	20041125	C. S. PARK	A3	U-TUBE MANOMETER PIPES			
			APPD	20041125	J. S. PARK					
STX Engine Co., Ltd.							DRAWING NO.			
							2805007760			

Class No.	N4	N5	N6	N7	N8	N9	16	63	±0.3
Ra Max. Value(um)	0.2	0.4	0.8	1.6	3.2	6.3	250	1000	±0.8

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GENERAL TOLERANCES FOR MACHINED MEASURES				REV	LOC	DATE	MADE	APPD	NOTE	
NOMINAL MEASURE				MATERIAL		SCALE		WEIGHT		MODEL
TOLERANCES						N / S		kg		
OVER	MAX.			DWN	20090210	S. M. LEE		SIZE	NAME	
1	4	±0.1		CHKD	20090210	J. T. SEO		A3	OIL MIST DETECTOR PIPES	
4	16	±0.2		APPD	20090210	I. B. KIM				
16	63	±0.3								
63	250	±0.5								
250	1000	±0.8								
Class No. N4 N5 N6 N7 N8 N9 Re Max. Value (mm) 0.2 0.4 0.8 1.6 3.2 6.3 12.5				STX Engine Co., Ltd.				DRAWING NO. 2805030010		

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MAIN ENGINE CONTROL SYSTEM

stx Heavy Industries Co., Ltd.

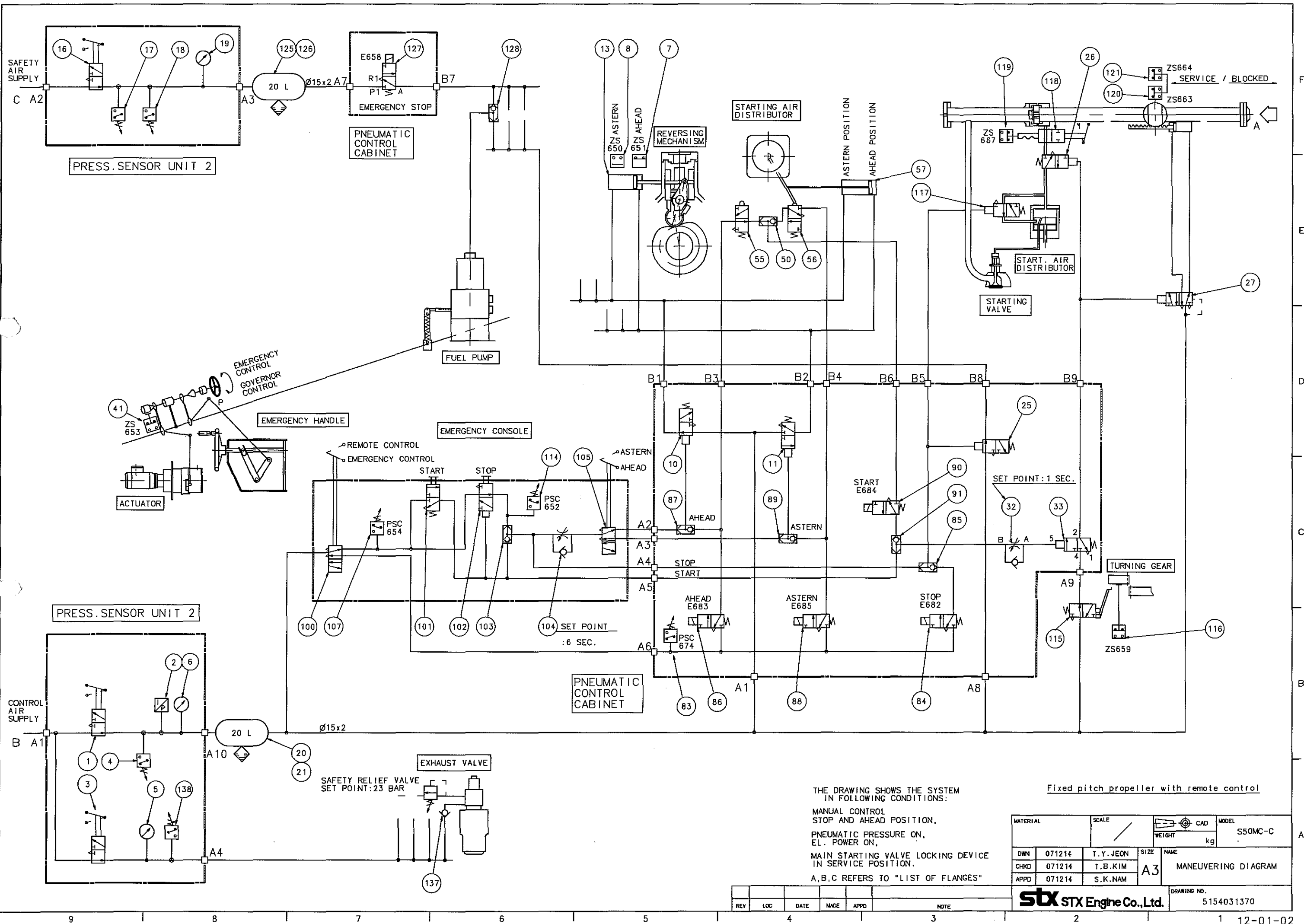
SHIP YARD		STX		APPROVED BY		T.B.KIM	
OWNER		NNC		CHECKED BY		M.H.CHOI	
HULL NO.		S-4006/12/14/29		DRAWN BY		T.Y.JEON	
MODEL		7S50MC-C		DATE		2009.01.20	
CLASSIFICATION		LR		LIST NO.		SEA09-7023	
PJT. NO.		H10B733/591/687/686				SEW09-7351	
REV.	DESCRIPTION	DATE	DWG	CHECK	APPD		
①	FOR WORKING	2009. 11. 26	T.Y.JEON	M.H.CHOI	J.B.Kim		
②	FOR FINAL	2010. 06. 24	T.Y. JEON	✓	M.H.CHOI		

CONTENTS

NO	DESCRIPTION
12-01	PNEUMATIC MANEUVERING SYSTEM
12-02	SENSOR LIST
12-03	INSTRUMENTATION & LIGHTING ARRANGEMENT
12-04	WIRING DIAGRAM
12-05	T/C TACHO SYSTEM
12-06	E.C.R PRESSURE INDICATORS
12-07	OMD REMOTE CONTROL PANEL
12-08	BEARING CONDITION MONITORING SYSTEM
12-09	AUX. BLOWER STARTER & T/G MOTOR STARTER
12-10	ECR MANEUVERING HANDLE
12-11	SPARE PARTS

12. 01 PNEUMATIC MANEUVERING SYSTEM

STX Heavy Industries Co., Ltd



THE DRAWING SHOWS THE SYSTEM
IN FOLLOWING CONDITIONS:
MANUAL CONTROL
STOP AND AHEAD POSITION,
PNEUMATIC PRESSURE ON,
EL. POWER ON,
MAIN STARTING VALVE LOCKING DEVICE
IN SERVICE POSITION.
A,B,C REFERS TO "LIST OF FLANGES"

Fixed pitch propeller with remote control

MATERIAL		SCALE		CAD		MODEL	
DWN 071214		T.Y. JEON		SIZE		S50MC-C	
CHKD 071214		T.B. KIM		NAME		MANEUVERING DIAGRAM	
APPD 071214		S.K. NAM		A3		DRAWING NO.	
						5154031370	

REV	LOC	DATE	MADE	APPD	NOTE

STX STX Engine Co., Ltd.

1. SYMBOL DESCRIPTION

The symbols consist of one or more square fields.

The number of fields corresponds to the number of valve positions.

The connecting lines are connected to the field which represents the valve position at a given moment of the process.

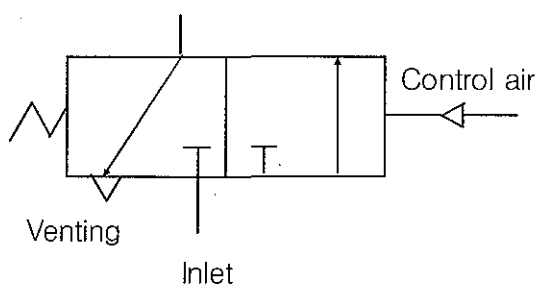
The change of position is conceived to take place by the fields being displaced at right angles to the connecting lines, which are assumed to have a stationary position on the paper.

A short cross line on a broken line indicates a closed path.

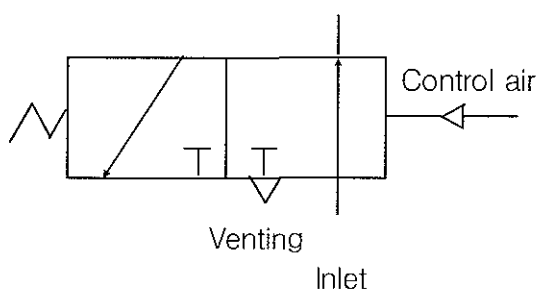
Example of pressure controlled 2-position valve with spring return :

1 = Initial position, 2 = Changed position

1



2



2. COMPONENTS

Pos. 1 : Ball valve

For manual cutting-off of control air supply.

Pos. 2 : Pressure transmitter

For alarm and indication of the control air supply pressure.

Alarm point 5.5 kg/cm².

Pos. 3 : Ball valve

For manual cutting-off of air to exhaust valve spring.

Pos. 4 : Pressure switch

For alarm if the control air pressure is not vented

during FINISHED WITH ENGINE. Alarm point 0.5 kg/cm².

Pos. 5 : Pressure gauge

Indicates exhaust valve spring air supply pressure.

Pos. 6 : Pressure gauge

Indicates control air supply pressure.

Pos. 7 : Magnet switch

Activated when reversing cylinder (13) is in AHEAD position.

Pos. 8 : Magnet switch

Activated when reversing cylinder (13) is in ASTERN position.

Pos. 10 : Two-position, Three-way valve

Leads air to reversing cylinders (13) and (57) for reversing to AHEAD position.

Pos. 11 : Two-position, Three-way valve

Leads air to reversing cylinders (13) and (57) for reversing to ASTERN position.

Pos. 13 : Air cylinder

Reverses the roller for the fuel pump to AHEAD and ASTERN, respectively.

Pos. 16 : Ball valve

For manual cutting-off of safety air supply.

Pos. 17 : Pressure switch

For alarm of the safety air supply pressure.

Alarm point 5.5 kg/cm².

Pos. 18 : Pressure switch

For alarm if the safety air pressure is not vented during FINISHED WITH ENGINE. Alarm point 0.5 kg/cm².

Pos. 19 : Pressure gauge

Indicates safety air supply pressure.

Pos. 20 : Air receiver 20ℓ

Reduces time lags in the manoeuvring system.

Pos. 21 : Drain valve

For draining water from the manoeuvring system.

Pos. 25 : Two-position, three-way valve

Leads air to puncture valves in fuel pumps during STOP and START.

Pos. 26 : Two-position, three-way valve

Prevents air inlet to starting air distributor in case of leaking main starting valve.

It is actuated by START signal and allows air to valve (117) and the second part of the distributing groove in the starting air distributor.

Pos. 27 : Two-position, five-way valve

Controls the main starting valve and, if installed, slow-turning valve (open or closed).

Pos. 32 : Throttle non-return valve

Delays venting of the pilot signal to valve (33). The delay is adjustable.

Delay about 1 second.

Pos. 33 : Two-position, three-way valve

Leads pilot signal to valves (26) and (27) when turning gear is disengaged and pilot signal is given from valve (101) through valve (32).

Pos. 41 : Switch

Gives signal to manoeuvring system when the change-over Mechanism is in governor control mode.

Pos. 50 : Double non-return valvePos. 55 : Two-position, three-way valve

Blocks the remote START AHEAD signal until the starting air distributor is in AHEAD position.

Pos. 56 : Two-position, three-way valve

Blocks the remote START ASTERN signal until the starting air distributor is in ASTERN position.

Pos. 57 : Air cylinder

Reverses the starting air distributor from AHEAD to ASTERN and vice versa.

Pos. 83 : Pressure switch

Actuated when engine is on remote control.

Pos. 84 : Two-position, three-way solenoid valve

Gives pilot signal to valves (25) and (117) when STOP is ordered from remote control system.

Pos. 85 : Double non-return valvePos. 86 : Two-position, three-way solenoid valve

Gives pilot signal to valve (10) and supply air to valve (55) when AHEAD is ordered from remote control system.

Pos. 87 : Double non-return valve

Pos. 88 : Two-position, three-way solenoid valve

Gives pilot signal to valve (11) and supply air to valve (56) when ASTERN is ordered from remote control system.

Pos. 89 : Double non-return valve

Pos. 90 : Two-position, three-way solenoid valve

Leads pilot signal to valve (33) when START is ordered from remote Control system and the starting air distributor is in the wanted position.

Pos. 91 : Double non-return valve

Pos. 100 : Two-position, five-way valve, hand operated

Leads air to the emergency control system or to the remote (bridge And control room) control systems, respectively.

Pos. 101 : Two-position, three-way valve, hand operated

Leads pilot signal to valves (25), (33) and (117) and supply air to valve (105) when activated.

Pos. 102 : Two-position, three-way valve, hand operated

Leads pilot signal to valves (25) and (117) and supply air to valve (105), when activated.

Pos. 103 : Double non-return valve

Pos. 104 : Throttle non-return valve

Keeps the AHEAD or ASTERN signal actuated for 6 seconds after START during local control.

Pos. 105 : Two-position, four-way valve, hand operated

Leads pilot signal to valves (10) and (11), corresponding to order (AHEAD or ASTERN), when reversing during emergency control.

Pos. 107 : Press switch

Gives signal to manoeuvring system when engine is on emergency control.

Pos. 114 : Pressure switch

Resets shut-down function in safety system when STOP valve (102) is activated.

Pos. 115 : Two-position, three-way valve

Blocks the start possibility when the turning gear is engaged.

Pos. 116 : Switch

Gives signal to lamp in manoeuvring console, if turning gear is engaged.

Pos. 117 : Two-position, three-way valve

Activated by STOP signal and leads air to the first part of the distributing groove in the starting air distributor.

Pos. 118 : Shut-off valve

For manual cutting-off of control air to the starting air distributor.

Pos. 119 : Switch

Gives signal to lamp on manoeuvring console when starting air distributor is blocked.

Pos. 120 : Switch

Gives signal to lamps on manoeuvring console to indicate whether main starting valve is in SERVICE position or BLOCKED position.

Pos. 121 : Switch

Gives signal to telegraph system/communication system when main starting valve is blocked.

Pos. 125 : Air receiver 20ℓ

Reduces time lags in the safety system.

Pos. 126 : Drain valve

For draining water from the safety system

Pos. 127 : Two-position, three-way solenoid valve

Leads air to puncture valve in fuel pumps when shut-down signal is Given (from safety system)

Pos. 128 : Double non-return valve

Pos. 137 : Non-return valve

Prevents back-flow of air from exhaust valve spring.

Pos. 138 : Pressure switch

For alarm if exhaust valve spring air pressure

Alarm point 5.5 kg/cm².

3. FUNCTIONING OF THE ENGINE CONTROL SYSTEM

For plants equipped with fixed-pitch propeller, the following modes of control are available :

- A. Remote control
- B. Emergency control

A. Remote control

The change-over valve (100) must be in its REMOTE CONTROL position.

Stop

In this condition the STOP solenoid valve (84) is to be actuated.

Valve (84) leads a pilot signal to valve (25) and valve (117).

Valve (25) leads air to the fuel pumps to activate the puncture valves in the high pressure fuel system. Valve (117) supplies starting air to the first part of the groove in the starting air distributor disc.

Start in ahead direction

In this condition the STOP solenoid valve (84), the AHEAD solenoid valve (86) and the START solenoid valve (90) are to be actuated.

Valve (86) leads a pilot signal to valve (10) and air to valve (55).

Valve (10) supplies air to the reversing cylinder (13) and (57) for reversing the roller guides and the starting air distributor.

When the starting air distributor is in ahead position, valve (55) leads air to valve (90). Valve (90) leads via valve (32) a pilot signal to valve (33).

This causes a pilot signal, if the turning is disengaged, to be led to the valves (26) and (27). Valve (27) will open the main starting air valve and valve (26) supplies air to the starting air distributor.

The engine will now rotate on starting air.



When the engine has reached the start level rpm, valve (84) and (90) are to be released. This will at once release the valves (25) and (117).

Valve (25) will deaerate the puncture valves in the fuel pumps, thereby allowing the fuel to be injected.

Valve (117) will change the timing of the starting air distributor.
One second later, delayed by valve (32), valve (90) will deaerate valve (33), causing the pilot signals to valves (26) and (27) to be deaerated.
Valve (26) will close the air supply to the starting air distributor.
Valve (27) will close the main starting valve.
Six seconds later the solenoid valve (86) is to be released.
This will deaerate the pilot air to valve (10), whereby the reversing cylinders (13) and (57) will be deaerated.

Speed setting

During START the governor is to be supplied with a pre-setting signal corresponding to 45 ~ 50% MCR rpm. This value is to be kept as long as start solenoid valve (86) is actuated. After this the speed settings signal is set by the regulating handle.

Repeated start

In case of start failure, i.e. if the engine stops after the starting sequence has finished, it is possible to increase the governor output at a new start attempt.
This is done by activating "CANCEL OF LIMITERS" function in governor during the starting sequence at a repeated start.

Reversing and start in astern direction

The procedure for reversing and start ASTERN is in principle the same as described during AHEAD direction, except after stop solenoid valve (88) is to be actuated instead of ahead solenoid valve (86), whereby an air signal is lead to start solenoid valve (90) via interlock valve (56), when the starting air distributor is in correct position.

Emergency stop

The automatic emergency stop via the safety system operates by activating valve (127) in the safety air system, independent of the control air system.
When activated, safety air is led via valve (128) to the puncture valve, stopping the high pressure fuel supply. Besides this, the engine can in an emergency situation be stopped manually by pulling the emergency handle to STOP position.

B. Emergency Control

In case of breakdown of the normal pneumatic manoeuvring system, the Governor or its electronics, the engine can be operated from the emergency console on the engine.

Changing-over from NORMAL to EMERGENCY control is carried out by turning the impact hand wheel, P, anticlockwise.

Now, the governor is disconnected from the fuel pumps, and the regulating handle on emergency console is connected to them.

Furthermore, valve (100) has to be changed from NORMAL to EMERGENCY control.

Hereby the supply to the valves in the emergency control is pressurized and the supply to the solenoid valves (84), (86) and (88) are deaerated.

Before changing-over valve (100), it has to be checked that valve (102) is depressed and valve (105) is in the wanted position.

In stop condition with valve (102) depressed the same components are activated

as during stop from remote control.

In order to start the engine the emergency handle is to be moved to a suitable position and after this valve (101) is to be depressed.

The start sequence is the same as described under remote start, except :

- 1) Activating valve (33) is by-passing interlock valves (55) and (56).
- 2) Delayed deaeration of the valves (10) and (11) is caused by check valve (104).

When the engine has reached start level rpm valve (101) is released and the emergency handle is moved to a position corresponding to the wanted rpm.

4. ENGINE SAFETY SYSTEM

The system is separately supplied with air via valve (16), and is controlled by the safety system (with separate power supply) in the manoeuvring console (not shown in diagram).

In case of shut-down, the safety system actuates valve (127).

Then an air supply is led to the fuel pumps for actuating the puncture valve in the high pressure fuel system, whereby the fuel injection is stopped.

The system is connected during all modes of engine control.

12.02 SENSOR LIST

STX Heavy Industries Co., Ltd

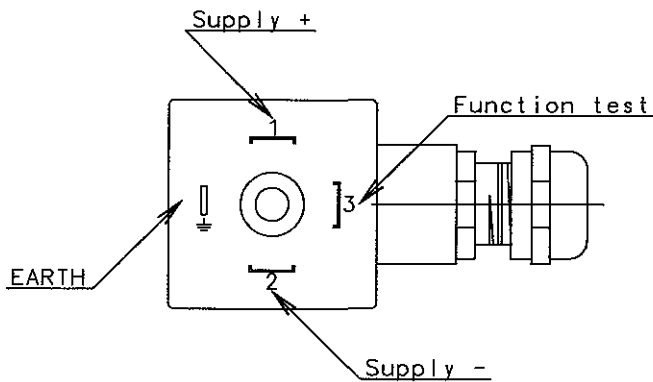
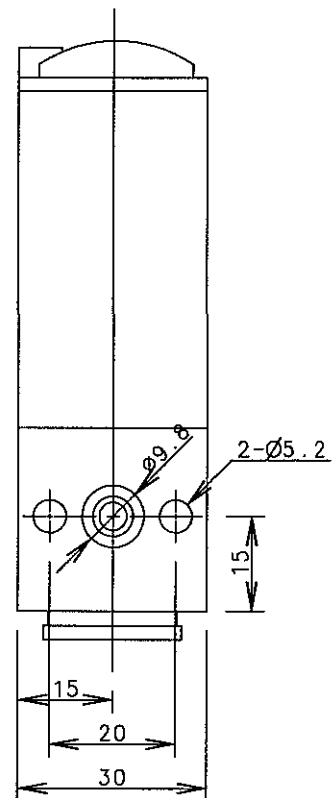
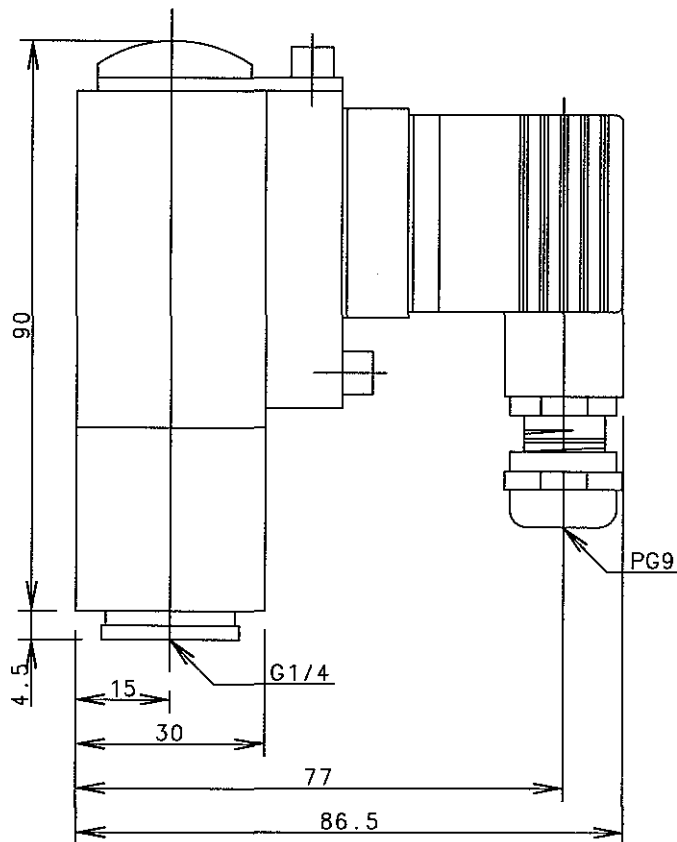
SENSOR LIST FOR STX-MAN B&W 7S50MC-C MAIN ENGINE

SYMBOL	Q'TY	DESCRIPTION	UNIT	ALARM		SLOWDOWN		SHUTDOWN		TIME DELAY (SEC)	PUMP AUTO START	DWG. NO.	REMARK
		PURPOSE	RANGE	LOW	HIGH	LOW	HIGH	LOW	HIGH				
FUEL OIL SYSTEM													
PT 8001	1	F.O INLET PRESS.	kg/cm ²	1) 6.5						6-8		5154037720	
		INDICATION/ALARM	0~16										
TE 8005	1	F.O INLET TEMP.	℃	2) 135	2) 155	△						5154007620	
		INDICATION/ALARM	0~200										
LS 8006	1	F.O LEAKAGE HIGH LEVEL	-		0					6-8		0997371-2.1	
		ALARM	-										
LUBRICATING OIL SYSTEM													
PT 8103	1	T/C L.O INLET PRESS.	kg/cm ²	1.2						6-8		5154037720	
		INDICATION/ALARM	0~6										
TE 8106	1	THRUST BEARING TEMP.	℃		75		80			6-8		5154037050	
		INDICATION/ALARM/SLOWDOWN	0~100										
TS 8107	1	THRUST BEARING HIGH TEMP.	℃					90		6-8		5154037050	
		SHUTDOWN	20~150										
PT 8108	1	MAIN L.O & P.C.O INLET PRESS.	kg/cm ²	3) 1.6		3) 1.4				6-8		5154037720	
		INDICATION/ALARM/SLOWDOWN	0~6										
PS 8109	1	MAIN L.O & P.C.O INLET LOW PRESS.	kg/cm ²					3) 1.2		6-8		5154037730	
		SHUTDOWN	-0.2~4										
TE 8112	1	MAIN L.O & P.C.O INLET TEMP.	℃	35	55		60			6-8		5154007620	
		INDICATION/ALARM/SLOWDOWN	0~200										
TE 8113	7	P.C.O OUTLET TEMP.	℃		70		75			6-8		5154007620	
		INDICATION/ALARM/SLOWDOWN	0~200										
FS 8114	7	P.C.O OUTLET NON-FLOW	-	0		0				6-8		5154017890	
		ALARM/SLOWDOWN	-										
TE 8117	1	T/C L.O OUTLET TEMP.	℃		95					6-8		5154007620	
		INDICATION/ALARM	0~200										
PS 8140	1	MAIN L.O & P.C.O INLET PRESS.	kg/cm ²							3) 1.8		5154037730	
		STAND-BY PUMP AUTO START	1~10										
△ BEARING WEAR MONITORING SYSTEM													
XC 8126	14	BEARING HIGH WEAR	mm		0.5		0.7					-	
		INDICATION/ALARM/SLOWDOWN	-										
-	-	BEARING HIGH WEAR(SENSOR DEVIATION)	mm		0.4							-	
		INDICATION/ALARM	-										
-	-	BEARING HIGH WEAR(CYLINDER DEVIATION)	mm		0.3		0.5					-	
		INDICATION/ALARM/SLOWDOWN	-										
-	-	SYSTEM FAILURE	-									-	
		ALARM	-										
-	2	RTD SENSOR	℃									-	
		INDIATION	-										
CYLINDER LUBRICATING OIL SYSTEM													
LS 8250	2	CYLINDER L.O LOW LEVEL	-	0						6-8		-	
		ALARM	-										
FS 8251	2	CYLINDER L.O NON-FLOW	-	0		0				6-8		-	
		ALARM/SLOWDOWN	-										
COOLING WATER SYSTEM													
PT 8401	1	JACKET C.W INLET PRESS.	kg/cm ²	2.0		1.5				6-8		5154037720	
		INDICATION/ALARM/SLOWDOWN	0~6										
TE 8408	7	JACKET C.W OUTLET TEMP.	℃		85		90			6-8		5154007620	
		INDICATION/ALARM/SLOWDOWN	0~200										
PT 8421	1	AIR COOLER C.W INLET PRESS.	kg/cm ²	1.0						6-8		5154037720	
		INDICATION/ALARM	0~6										
TE 8422	1	AIR COOLER C.W INLET TEMP.	℃		40					6-8		5154007620	
		INDICATION/ALARM	0~200										
PS 8432	1	JACKET C.W INLET PRESS.	kg/cm ²							6-8	2.5	5154037730	
		STAND-BY PUMP AUTO START	1~10										

SYMBOL	Q'TY	DESCRIPTION	UNIT	ALARM		SLOWDOWN		SHUTDOWN		TIME DELAY (SEC)	PUMP AUTO START	DWG. NO.	REMARK
		PURPOSE	RANGE	LOW	HIGH	LOW	HIGH	LOW	HIGH				
COMPRESSED AIR SUPPLY SYSTEM													
PT 8501	1	STARTING AIR INLET PRESS.	kg/cm ²	15						6~8		5154037720	
		INDICATION/ALARM	0~40										
PT 8502	1	STARTING AIR INLET PRESS.	kg/cm ²									5154037720	
		INDICATION ON BRIDGE	0~40										
PT 8503	1	CONTROL AIR INLET PRESS.	kg/cm ²	5.5						6~8		5154037720	
		INDICATION/ALARM	0~10										
PS 8504	1	SAFETY AIR INLET LOW PRESS.	kg/cm ²	5.5						6~8		5154037730	
		ALARM	1~10										
PS 8505	1	EXH. V/V SPRING AIR LOW PRESS.	kg/cm ²	5.5						6~8		5154037730	
		ALARM	1~10										
PS 8506	1	CONTROL AIR NOT VENTED	kg/cm ²		0.5					6~8		5154037730	
		ALARM AT F.W.E	-0.2~4										
PS 8507	1	SAFETY AIR NOT VENTED	kg/cm ²		0.5					6~8		5154037730	
		ALARM AT F.W.E	-0.2~4										
SCAVENGE AIR SYSTEM													
PT 8601	1	SCAVENGE AIR INLET PRESS.	kg/cm ²									5154037720	
		INDICATION	0~4										
PT 8602	1	SCAVENGE AIR INLET PRESS.	kg/cm ²									5154037720	ELECTRIC GOV. ONLY
		GOVERNOR CONTROL	0~4										
PS 8603	1	SCAVENGE AIR INLET PRESS.	kg/cm ²							0.45		5154037730	
		AUX.BLOWER AUTO START	-0.2~4										
PS 8604	1	SCAVENGE AIR INLET PRESS.	kg/cm ²							0.7		5154037730	
		AUX.BLOWER AUTO STOP	-0.2~4										
TE 8609	1	SCAVENGE AIR RECEIVER TEMP.	℃		6) 55					6~8		5154007620	
		INDICATION/ALARM	0~200										
TE 8610	7	SCAVENGE AIR BOX FIRE DETECTION	℃		80		120			6~8		5154007620	
		INDICATION/ALARM/SLOWDOWN	0~200										
LS 8611	1	WATER MIST CHATCHER HIGH LEVEL	-		0					6~8		1472156-6.1	
		ALARM	-										
EXHAUST GAS SYSTEM													
TE 8701	1	T/C INLET EXH.GAS TEMP.	℃		530					6~8		900200550	
		INDICATION/ALARM	-50~600										
TE 8702	7	CYL.OUTLET EXH. GAS TEMP.	℃		430		450			6~8		900200550	
		INDICATION/ALARM/SLOWDOWN	-50~600										
△ -	-	CYL.OUTLET EXH. GAS DEVIATION TEMP.	℃	-50	50	-60	60			6~8		-	
		INDICATION/ALARM/SLOWDOWN											
TE 8707	1	T/C OUTLET EXH.GAS TEMP.	℃		350					6~8		900200550	
		INDICATION/ALARM	-50~600										
MANOEUVRING SYSTEM													
PSC 654	1	ENGINE SIDE CONTROL POSITION	kg/cm ²		2							1101000550-05	
		CONTROL	1~16										
PSC 674	1	REMOTE CONTROL POSITION	kg/cm ²		2							1101000550-05	
		CONTROL	1~16										
PSC 652	1	RESET SHUTDOWN DURING EM'CY STOP	kg/cm ²		2							1101000550-05	
		CONTROL	1~16										
ZS 663	1	MAIN STARTING V/V SERVICE	-										
		CONTROL	-										
ZS 664	1	MAIN STARTING V/V BLOCKED	-										
		CONTROL	-										
ZS 659	1	TURNING GEAR ENGAGED/DISENGAGED	-										
		CONTROL	-										
ZS 653	1	EM'CY CONTROL POSITION	-										
		CONTROL	-										
ZS 667	1	STARTING AIR DISTRIBUTOR BLOCKED	-										
		CONTROL	-										
ZS 651	7	FUEL CAM AHEAD POSITION	-										F.P.P ONLY
		CONTROL	-										
ZS 650	7	FUEL CAM ASTERN POSITION	-										F.P.P ONLY
		CONTROL	-										

SYMBOL	Q'TY	DESCRIPTION	UNIT	ALARM		SLOWDOWN		SHUTDOWN		TIME DELAY (SEC)	PUMP AUTO START	DWG. NO.	REMARK
		PURPOSE	RANGE	LOW	HIGH	LOW	HIGH	LOW	HIGH				
MISCELLANEOUS													
XC 8813-1	1	CRANK CASE OIL MIST HIGH DENSITY	mg/ ℓ		0.5					6-8			
		ALARM	-										
XC 8813-2	1	CRANK CASE OIL MIST HIGH DENSITY	mg/ ℓ				0.5			6-8			
		SLOWDOWN	-										
XC 8814	1	OIL MIST DETECTOR FAILURE	-										
		ALARM	-										
RPM DETECTOR													
ZT 4020	1	MAIN ENGINE RPM OVER SPEED	rpm						MCR*1.09				
		SHUTDOWN	-										
ZT 8801	1	TURBOCHARGER RPM OVER SPEED	rpm		MAX*0.97								
		ALARM	-										
PS : PRESSURE SWITCH PT : PRESSURE TRANSMITTER TS : TEMPERATURE SWITCH TE : TEMPERATURE TRANSMITTER(PT-100) TC : THERMOCOUPLE WITH AMPLIFIER FS : FLOW(FLOAT) SWITCH LS : LEVEL SWITCH ZS : LIMIT SWITCH XC : ELECTRIC INTERNAL CONTACT WT : APPROXIMITY SWITCH													

- 1) : AT FUEL OIL VISCOSITY MAX. 700cst/50°C MEASURED AT CAMSHAFT LEVEL
- 2) : ALARM SETPOINT HAS TO BE DECIDED ACCORDING TO FUEL SPECIFICATION
- 3) : MEASURED ON PRESSURE GAUGE PLACED 1800mm ABOVE CRANKSHAFT CENTER LINE
- 4) : WITH STOPPED COOLING WATER PUMP, THE SETPOINT IS THE STATIC PRESSURE PLUS THE STATED VALUE
- 5) : LOWER 0.2bar / 0.2kg/cm² / 0.02Mpa THAN NORMAL SERVICE VALUE. THE SET POINT FOR NORMAL SERVICE VALUE TO STATED AT THE SEA TRIAL.
- 6) : TO BE CUT DURING ENGINE STOP AND TO REMAIN CUT-OUT UNTIL 3-5 MIN. AFTER ENGINE START



MAKER : DANFOSS A/S
(P/N : 10H1403-0)

PRESSURE RANGE (bar)	TYPE NO.	CODE NO.
0 TO 4	MBS5100-1611	060N1259
0 TO 6	MBS5100-1811	060N1260
0 TO 10	MBS5100-2011	060N1261
0 TO 16	MBS5100-2211	060N1262
0 TO 25	MBS5100-2411	060N1263
0 TO 40	MBS5100-2611	060N1270

Performance

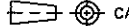
Hysteresis and repeability	< 0.1%FS	
Max. overload pressure	Press. range FS < 300bar	min. 2 x FS
	Press. range FS > 300bar	min. 1.5 x FS

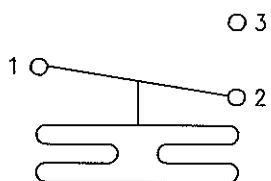
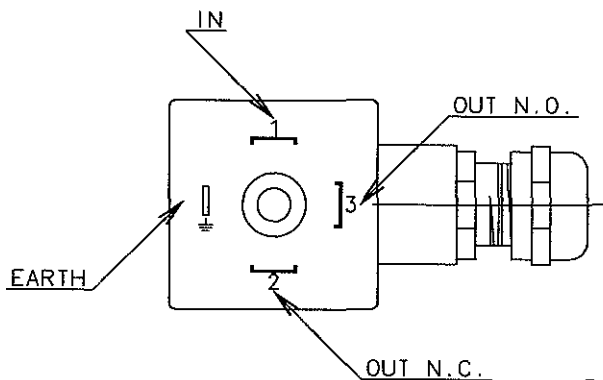
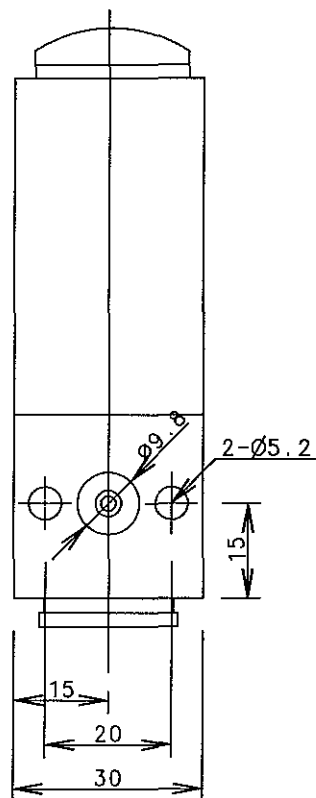
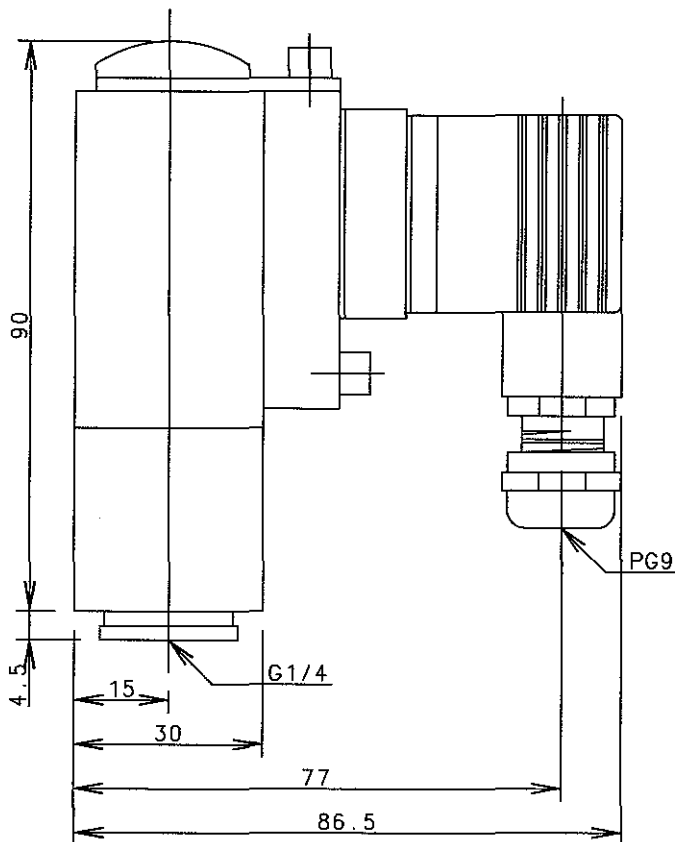
Electrical specifications

Output signal	4 - 20 mA	
Supply voltage	10 to 30v d.c.	
Current limitation (linear output signal up to 1.5 x nom range)	28 mA	

Environmental conditions

EMC	Emmission	EN 50081-1
Temperature	Operating	-40 to 85°C
Enclosure	IP 65	
Vibration stability	Sinusoidal 20g/ 20Hz-2kHz	IEC 68-2-6
Shock resistance	Shock 500g/1 ms, 100g/6ms	IEC 68-2-27
	Free fall	IEC 68-2-32

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	20080405	Y.S.LEE	SIZE	NAME PRESSURE TRANSMITTER
			CHKD	20080405	W.K.PARK	A4	
			APPD	20080405	T.B.KIM		
SCALE 1/1.2		MODEL ALL	 STX Engine Co.,Ltd.			DRAWING NO. 5154037720	



1 : INPUT
2 : OUTPUT N.C.
3 : OUTPUT N.O.

AC15 : 0.1A, 250V
CD13 : 12W, 125V

MAKER : DANFOSS A/S
(P/N : 10H1308-1)

PRESSURE RANGE (bar)	TYPE NO.	CODE NO.
-0.2 TO 4	MBC5100-1211-3DB04	061B0006
-0.2 TO 10	MBC5100-1411-3DB04	061B0052
1 TO 10	MBC5100-2431-3DB04	061B1006
5 TO 20	MBC5100-3231-3DB04	061B1035
5 TO 40	MBC5100-3641-1DB04	061B1005
		ONLY PG 11

Performance

Repeatability	< 1% FS
Response time	< 4 ms
Max. switch freq.	10/min (0.16Hz)

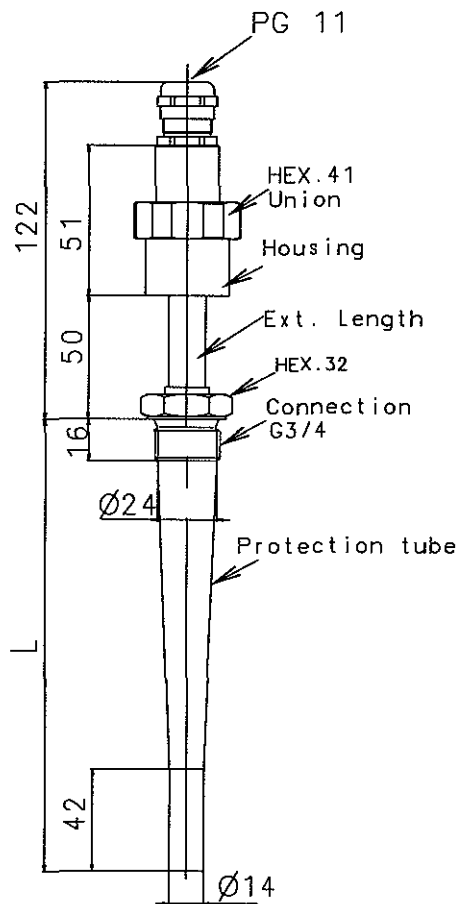
Electrical specifications

Switch	Microswitch	SPDT
Contact load	AC 11	0.1A 250V
	DC 11	12W 125V
Electrical conn.	Pole plug	

Environmental conditions

Temperature	Operating	-40 to 85°C
	Emission	-50 to 85°C
Enclosure	IP 65	
Vibration stability	Transport	4g/25Hz ~ 100Hz
Shock resistance	Shock 50g/6ms	IEC 68-2-27
	Free fall	IEC 68-2-32

REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL		CAD	DWN	20080405	Y.S.LEE	SIZE NAME A4 PRESSURE SWITCH (MBC5100)
WEIGHT		kg	CHKD	20080405	W.K.PARK	
SCALE		MODEL	APPD	20080405	T.B.KIM	
1/1.2		ALL	STX STX Engine Co., Ltd.			DRAWING NO. 5154037730



Technical data

- 1.Measuring range : -50°C to 600°C
- 2.Pt 100
- 3.Tolerance : EN60751 CLASS B
- 4.Extension : 50mm
- 5.Electrical connection : PG13.5 (IP65)

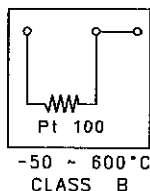
Material

- 1.Housing & union : Nickel plated brass
- 2.Spring : W.no.1.4568
- 3.Extension length : SUS316
- 4.Protection tube : SUS316 (Solid Drilled)

STANDARD LENGTH

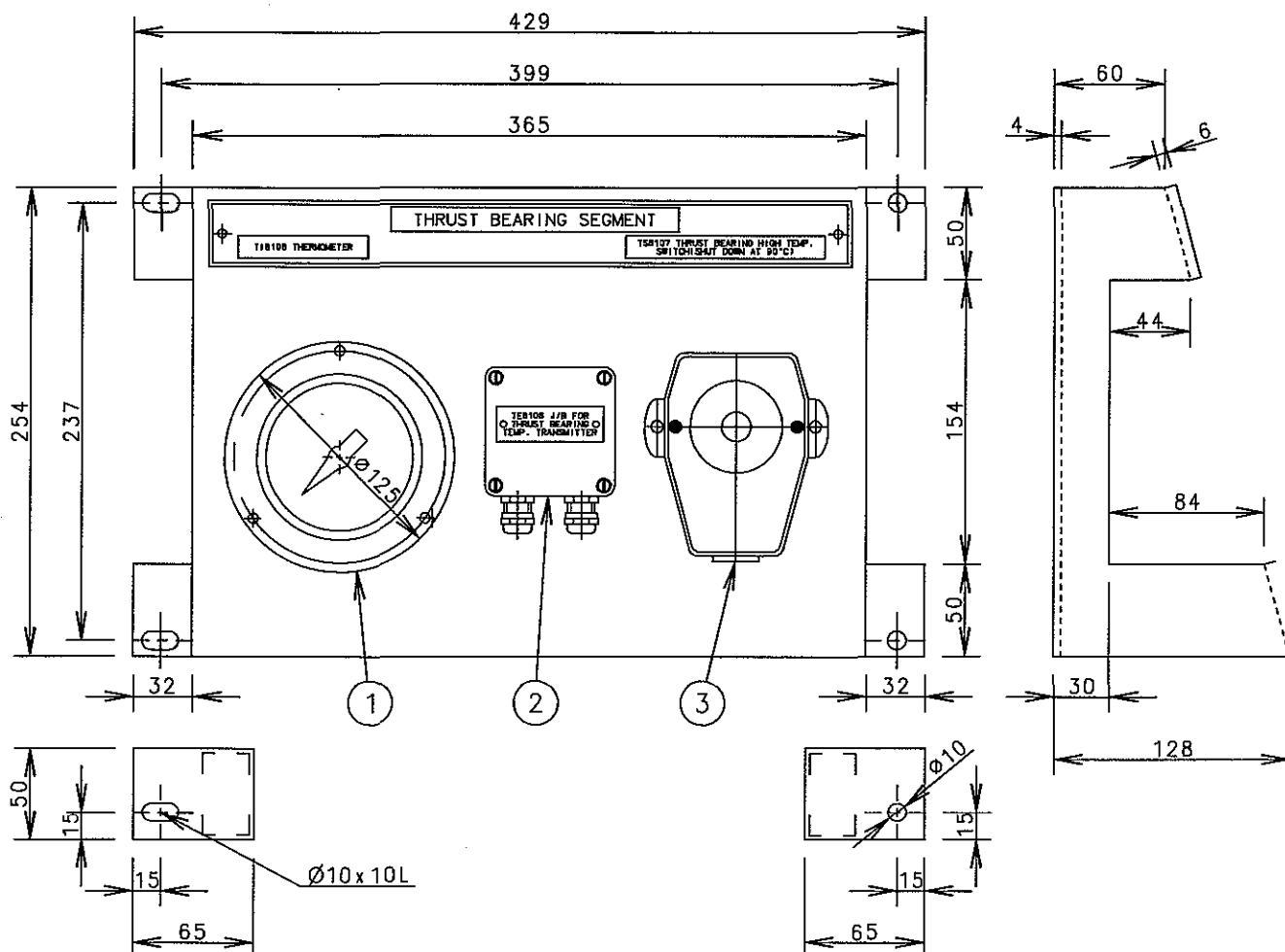
NO.	PART NO.	"L"	CODE NO.
1	MBT5116-080	80	084Z6110
2	MBT5116-100	100	084Z8150
3	MBT5116-120	120	084Z8151
4	MBT5116-150	150	084Z8152
5	MBT5116-200	200	084Z8154
6	MBT5116-250	250	084Z8155
7	MBT5116-300	300	084Z8156

TERMINAL



REV	LOC	DATE	MADE	APPD	NOTE
MATERIAL	WEIGHT	MODEL	DWN	080407	T.Y. JEON
			CHKD	080407	D.H. LEE
			APPD	080407	T.B. KIM
SCALE	N/S	ALL	DRAWING NO.		9002000550

STX STX Engine Co.,Ltd.

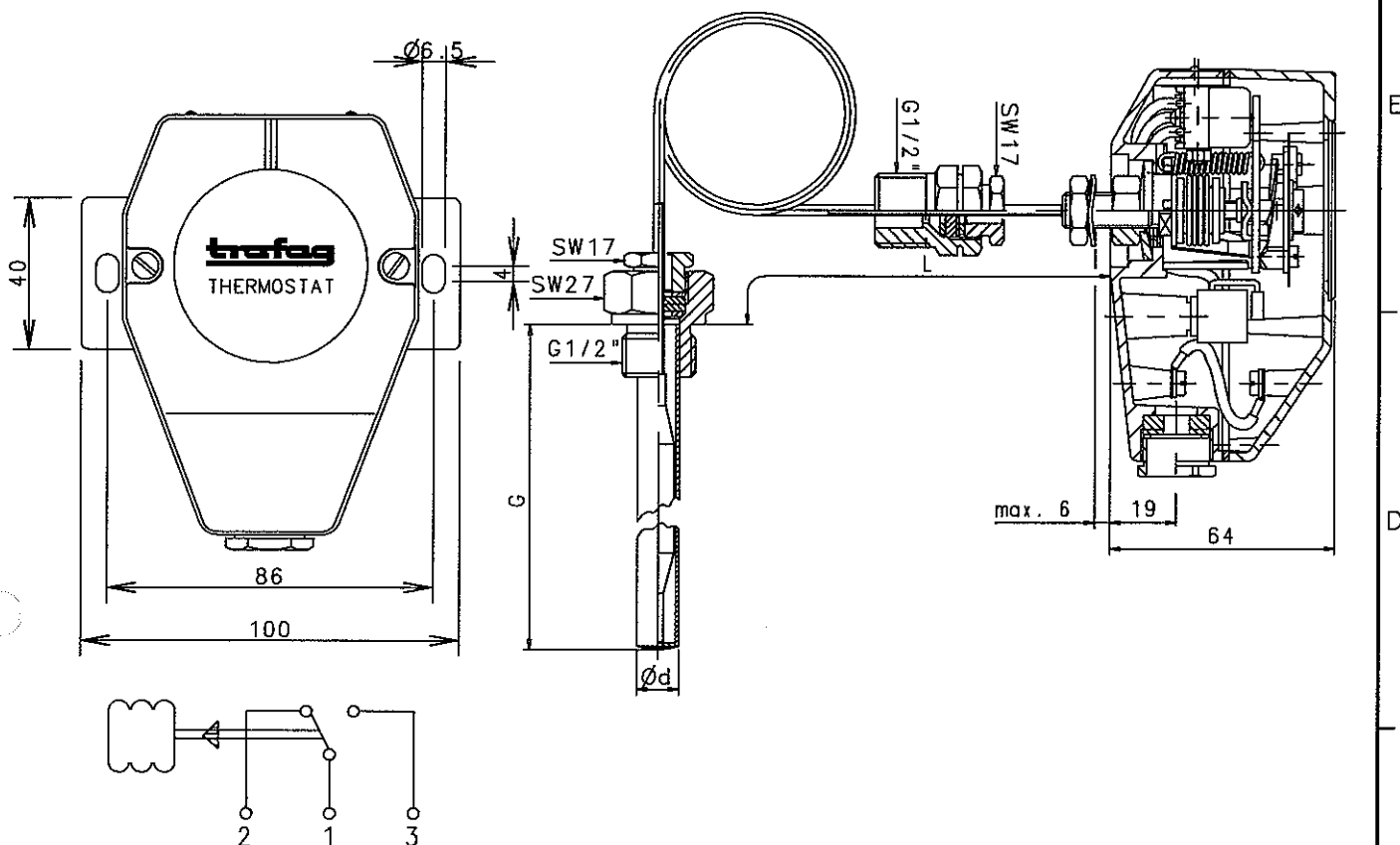


NOTES

- CONTACT MAX. 250V AC or DC
AMBIENT TEMP. 50°C
SHUT DOWN
CONNECT 1-3 (NO) CLOSED AT HIGH TEMP
- MECHANICAL REMOTE THERMOMETER
- SHUT DOWN THERMOSTAT TRAFAG NAVISTAT TYPE G471 2331
4m CAP. TUBE WITH PVC PROTECTION
CAPILLARY TUBE LEAD - IN BUSH NO.80 TEMPERATURE
TEMPERATURE RANGE 20°C - 150°C, SET POINT 90°C
- PAINTS
-1st : EPOXY PRIMER
-2nd : HAMMER TONE
- NAME PLATE SIZE
-MAKER STANDARD

3	TEMP. SWITCH	TS8107	5151003530-01	CALBE L=5000
2	TEMP. TRANSMITTER	TE8106	MP39-3/8-125-01	CALBE L=5000
1	THERMOMETER	T18106	5154037070-01	CALBE L=5000
NO.	DESCRIPTION	SYMBOL	PART NO.	REMARK

REV	LOC	DATE	MADE	APPD	NOTE				
MATERIAL		 CAD	DWN	080401	Y.S. LEE	SIZE A4	NAME THRUST BEARING SEGMENT		
			CHKD	080401	W.K. PARK				
		WEIGHT kg	APPD	080401	T.B. KIM				
SCALE 1/4		MODEL S50~70MC/MC-C		STX STX Engine Co., Ltd.		DRAWING NO. 5154037050-01			



Connections

(Specifications)

Range in °C	Sensor max. °C	Switch type No.	Switching diff. K (°C)	Sensor length (G)	Cable length (L)
20 ... 110	115	23 26	2.0 3.5	G=65 G=110 G=130 G=150	L=1000 L=3000 L=5000 L=8000
20 ... 150	165	23 26	2.0 5.0		
40 ... 300	330	23	7.0		
		23	7.0		
Max. housing temperature : -30 ... +70°C					
Pressure proofness of protection tube : max. 25bar					

(Information)

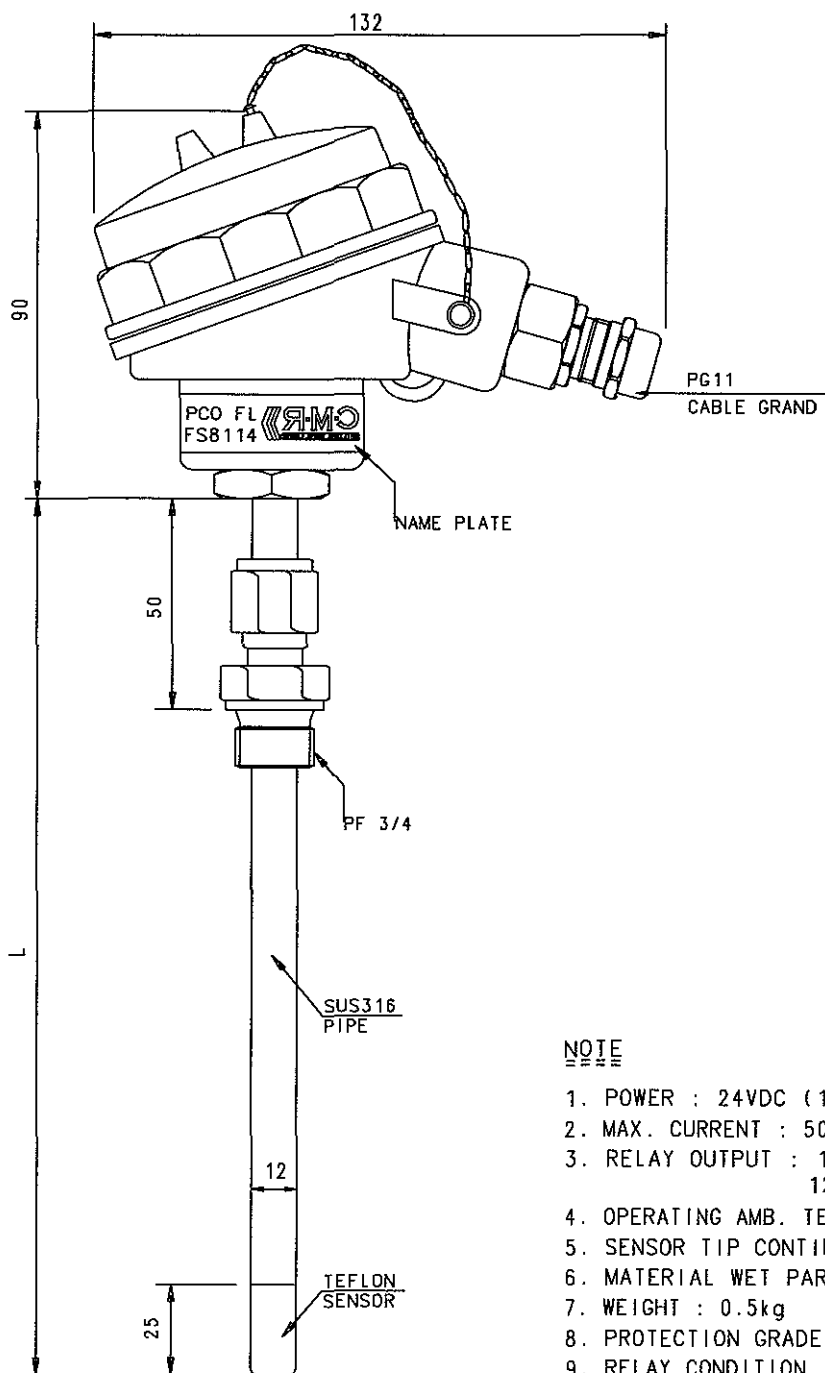
- Salt mist tested
- Oil proof sealings
- Accuracy $\pm 2\%$ of full scale
- Vibration resistance to 4g (GL)
- Set point drift due to ambient temperature drift from 20°C to 55°C, smaller than 2.5°C.

(Microswitch ratings)

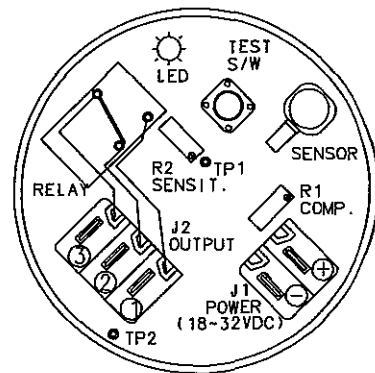
- Switch type No. 23, 26 and 12
- AC 250V - 10(3)A
- DC 12V - 2A, 24V - 2A, 110V - 0.5A, 250V - 0.25A

3			
2	5151003530-02	L : 6000	
1	5151003530-01	L : 5000	
NO.	PART NO.	LENGTH	REMARK

REV	LOC	DATE	MADE	APPD	NOTE			
MATERIAL		 CAD WEIGHT kg	DWN	080401	Y.S.LEE	SIZE A4	NAME TEMP. SWITCH FOR THRUST BEARING (TYPE : 471.23)	
			CHKD	080401	W.K.PARK			
			APPD	080401	T.B.KIM			
SCALE 1/2		MODEL ALL	STX STX Engine Co.,Ltd.			DRAWING NO. 5154037050-02		



CONNECTION =====



NO.	L	
1	150	
2	200	←
3	250	

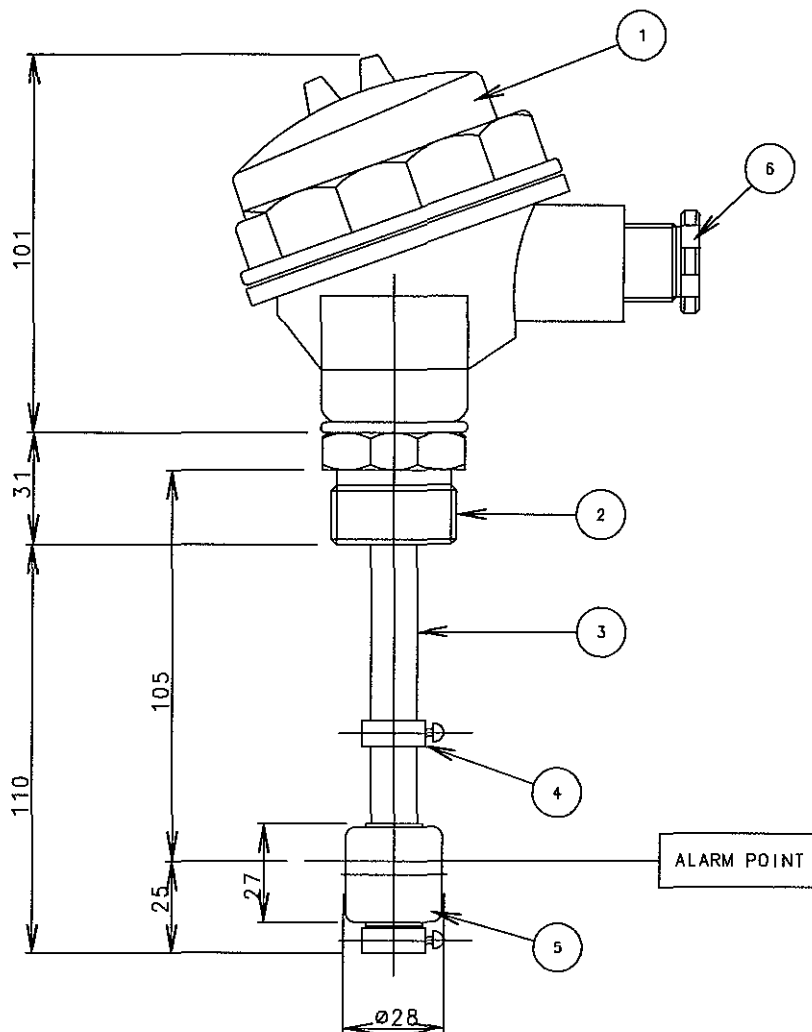
NOTE

1. POWER : 24VDC (18 to 32VDC)
2. MAX. CURRENT : 50mA
3. RELAY OUTPUT : 1 POTENTIAL FREE CONTACT
125VAC/0.5A or 30VDC/1.0A
4. OPERATING AMB. TEMP. : -10 to 70°C
5. SENSOR TIP CONTINUOUS TEMP. : -10 to 150°C
6. MATERIAL WET PART : SUS 316 and TEFLON
7. WEIGHT : 0.5kg
8. PROTECTION GRADE : IP56
9. RELAY CONDITION
 - DRY : 3-2 CLOSE
 - WET : 3-1 CLOSE

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	080407	T.Y. JEON	SIZE A4	NAME P.C.O NON-FLOW LEVEL SWITCH
			CHKD	080407	D.H. LEE		
			APPD	080407	T.B.KIM		
SCALE N/S		MODEL ALL	 STX Engine Co.,Ltd.			DRAWING NO. 5154017890	

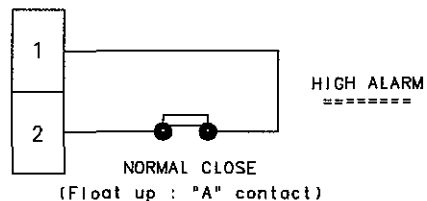
STX STX Engine Co., Ltd.

P.C.O NON-FLOW
LEVEL SWITCH



CONNECTION

=====



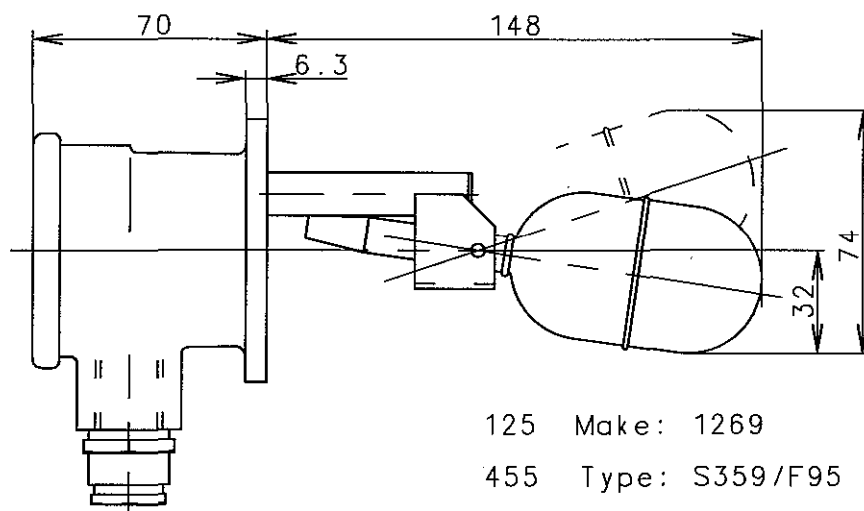
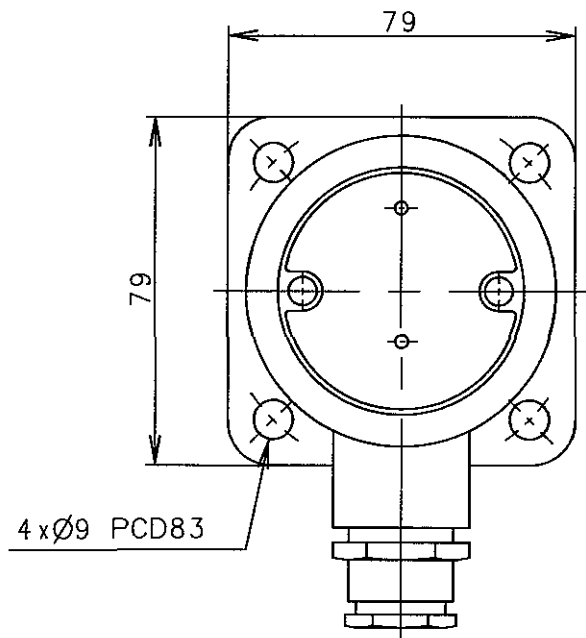
SPECIFICATION

=====

1. VOLUME : 34.3
2. WEIGHT : 18..8g
3. BUOYANCY : 15.5
4. SPECIFIC GRAVITY : 0.55
5. MATERIAL : SUS304

6	CABLE GLAND	BS	1	PG9
5	FLOAT	SUS304	1	Ø28
4	STOPPER	SUS304	2	
3	SWITCH PIPE	SUS304	1	
2	SWITCH SOCKET	SUS304	1	
1	HEAD	AL, DC	1	
NO.	N A M E	MAT'L	Q'TY	REMARKS

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	080407	T.Y. JEON	SIZE	NAME LEVEL SWITCH
			CHKD	080407	D.H. LEE	A4	
		WEIGHT	kg	APPD	080407		
		SCALE N/S		MODEL ALL	DRAWING NO. 0997371-2.1		
STX STX Engine Co., Ltd.							



<Terminal Connections>

Red- Supply

Blue- Normally Open

Black- Normally Closed

A.C. 250V 0,25A
10VA Maximum

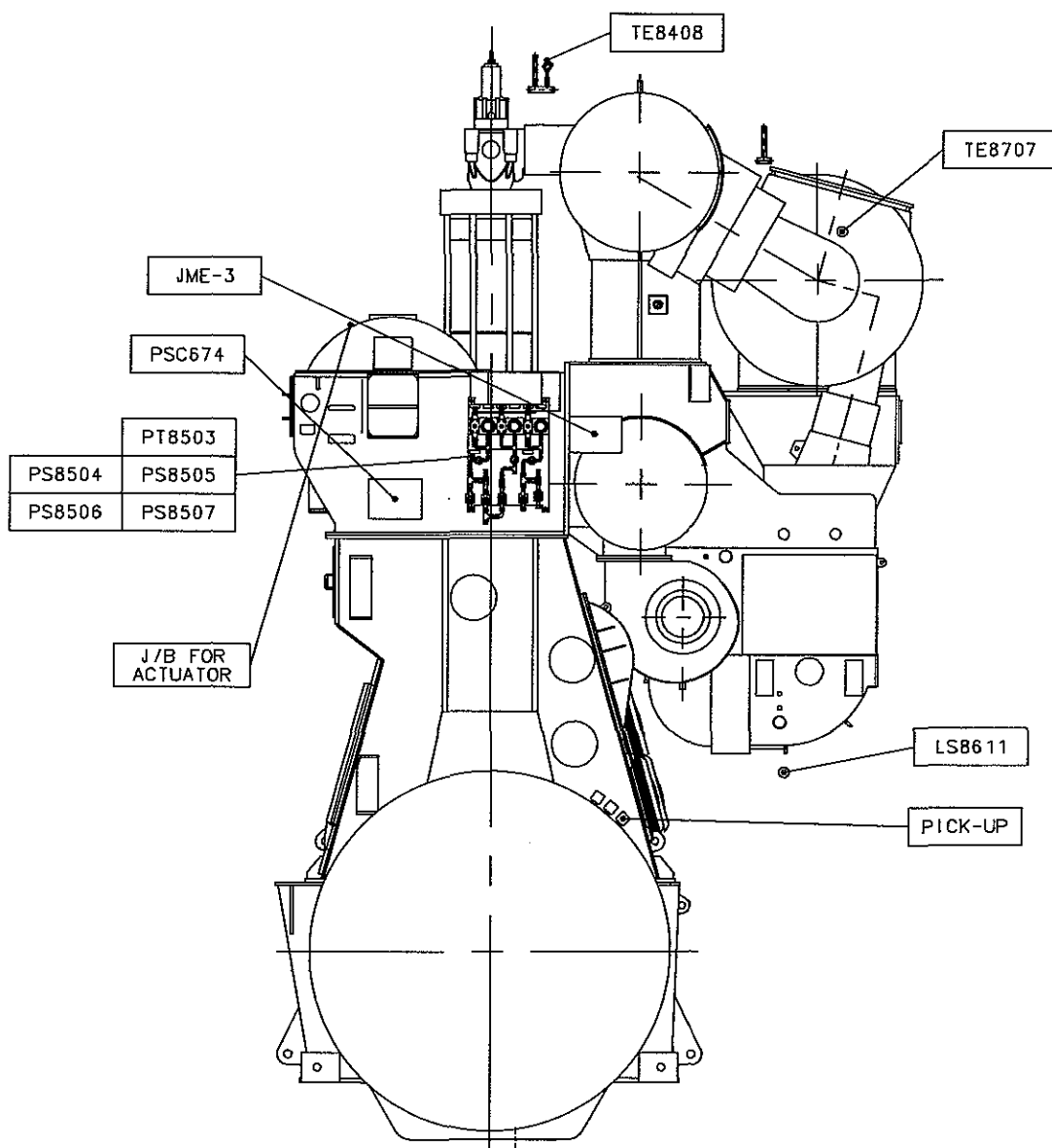
REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL		CAD	DWN	080407	T.Y. JEON	SIZE
		WEIGHT	CHKD	080407	D.H. LEE	A4
		kg	APPD	080407	T.B. KIM	NAME
SCALE		MODEL	DRAWING NO.			
N/S		ALL	1472156-6.1			

stx STX Engine Co., Ltd.

12. 03 INSTRUMENTATION & LIGHTING ARRANGEMENT

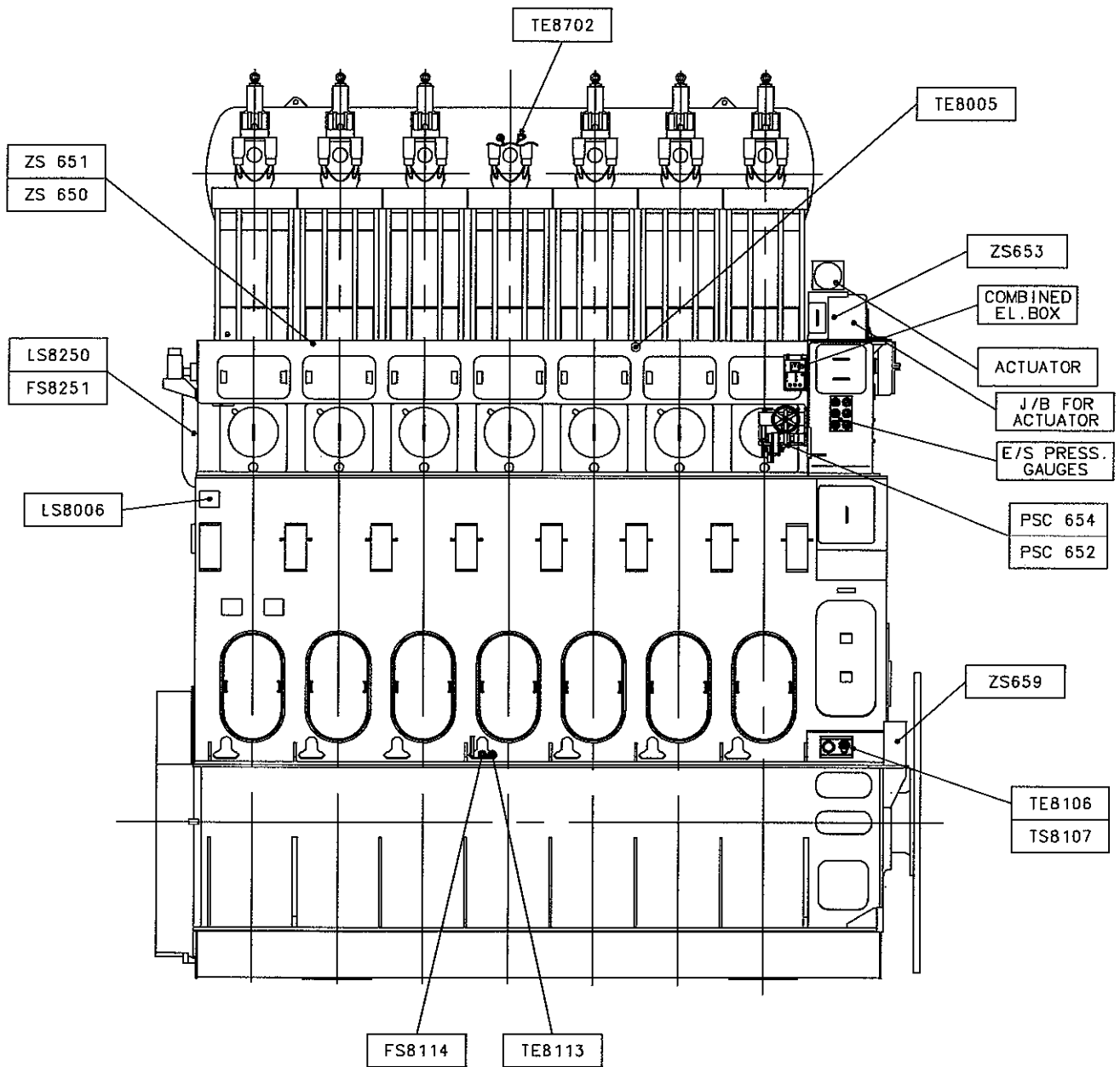
STX Heavy Industries Co., Ltd

<AFT END SIDE>



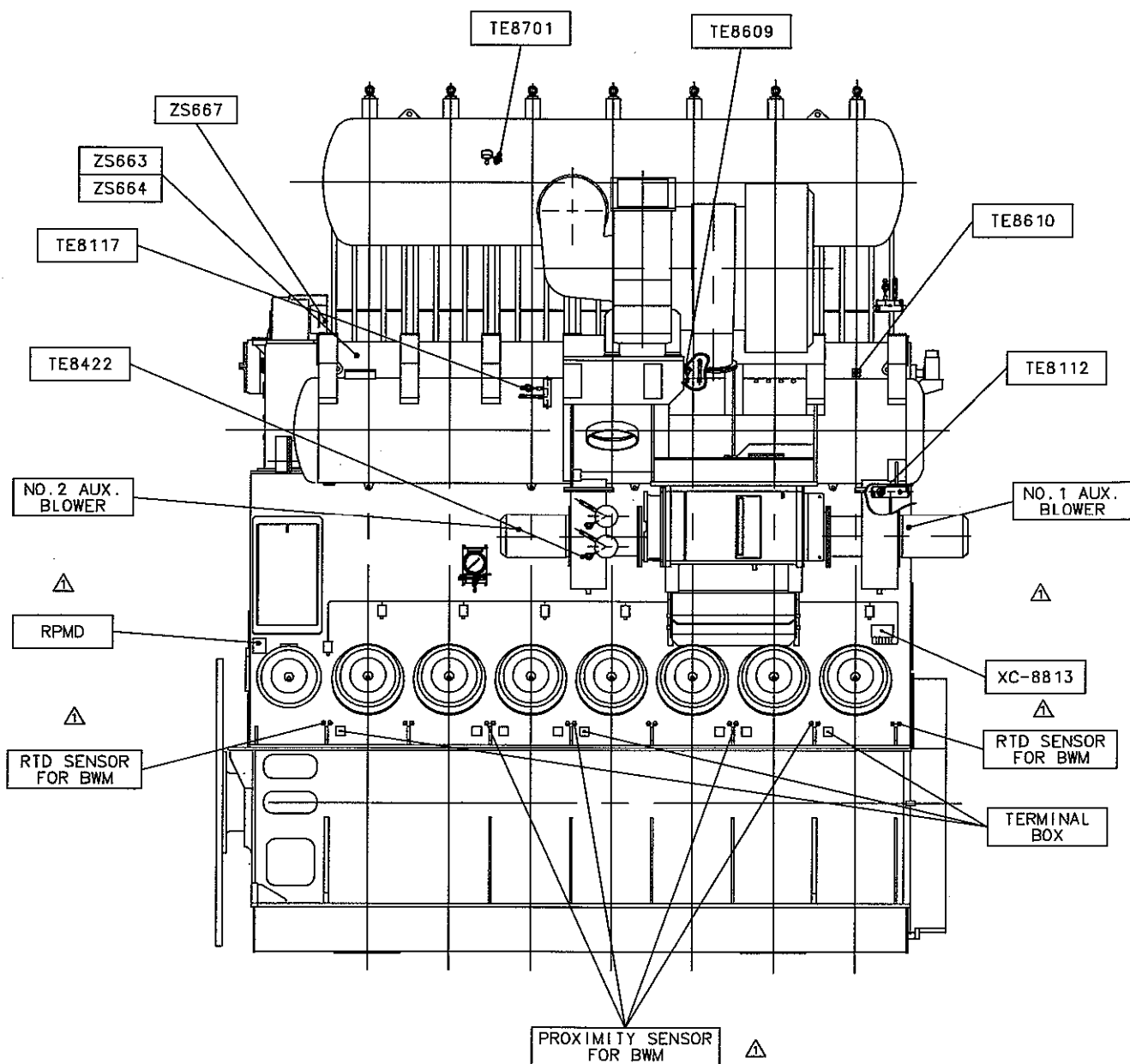
REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	090113	T.Y. JEON	SIZE	NAME
			CHKD	090113	M.H. CHOI	A4	INSTRUMENTATION
		WEIGHT	kg	APPD	090113		
SCALE		MODEL	 STX Engine Co., Ltd.			DRAWING NO.	
N/S		7S50MC-C				5154047270-01	

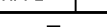
<PORT SIDE>



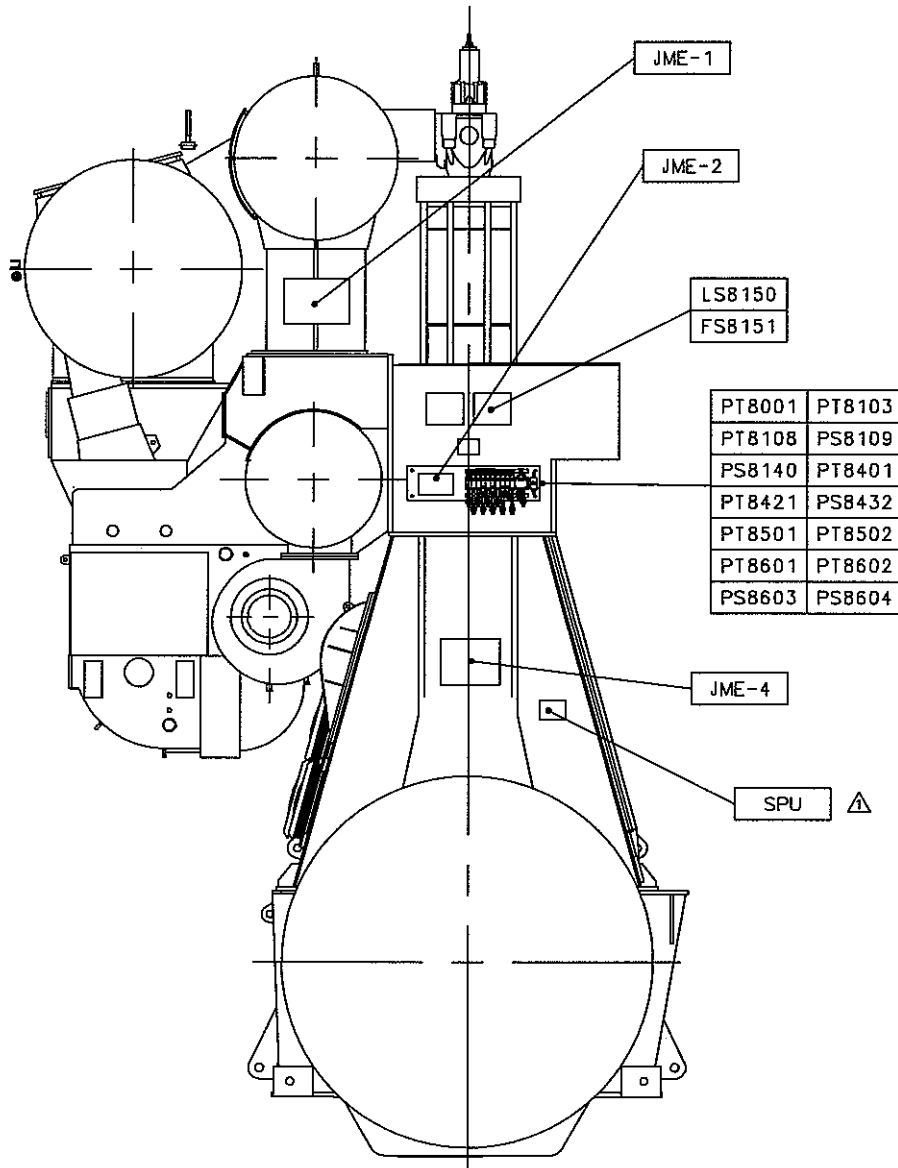
REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL		 CAD	DWN	090113	T. Y. JEON	SIZE NAME A4 INSTRUMENTATION
			CHKD	090113	M. H. CHO I	
			APPD	090113	T. B. KIM	
		WEIGHT		kg		
SCALE		MODEL	DRAWING NO.			
N/S		7S50MC-C	 STX Engine Co., Ltd.		5154047270-02	

<ST 'BD SIDE>



REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL		 CAD	DWN	100316	T.Y. JEON	SIZE A4 NAME INSTRUMENTATION
			CHKD	100316	M.H. CHOI	
			APPD	100316	T.B. KIM	
SCALE N/S		MODEL 7S50MC-C	 STX Engine Co., Ltd.			DRAWING NO. 5154047270-03

<FORE END SIDE>



REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	100316	T.Y. JEON	SIZE	NAME INSTRUMENTATION
			CHKD	100316	M.H. CHOI	A 4	
		WEIGHT kg	APPD	100316	T.B. KIM		
SCALE N/S		MODEL 7S50MC-C	 STX Engine Co., Ltd.			DRAWING NO. 5154047270-04	

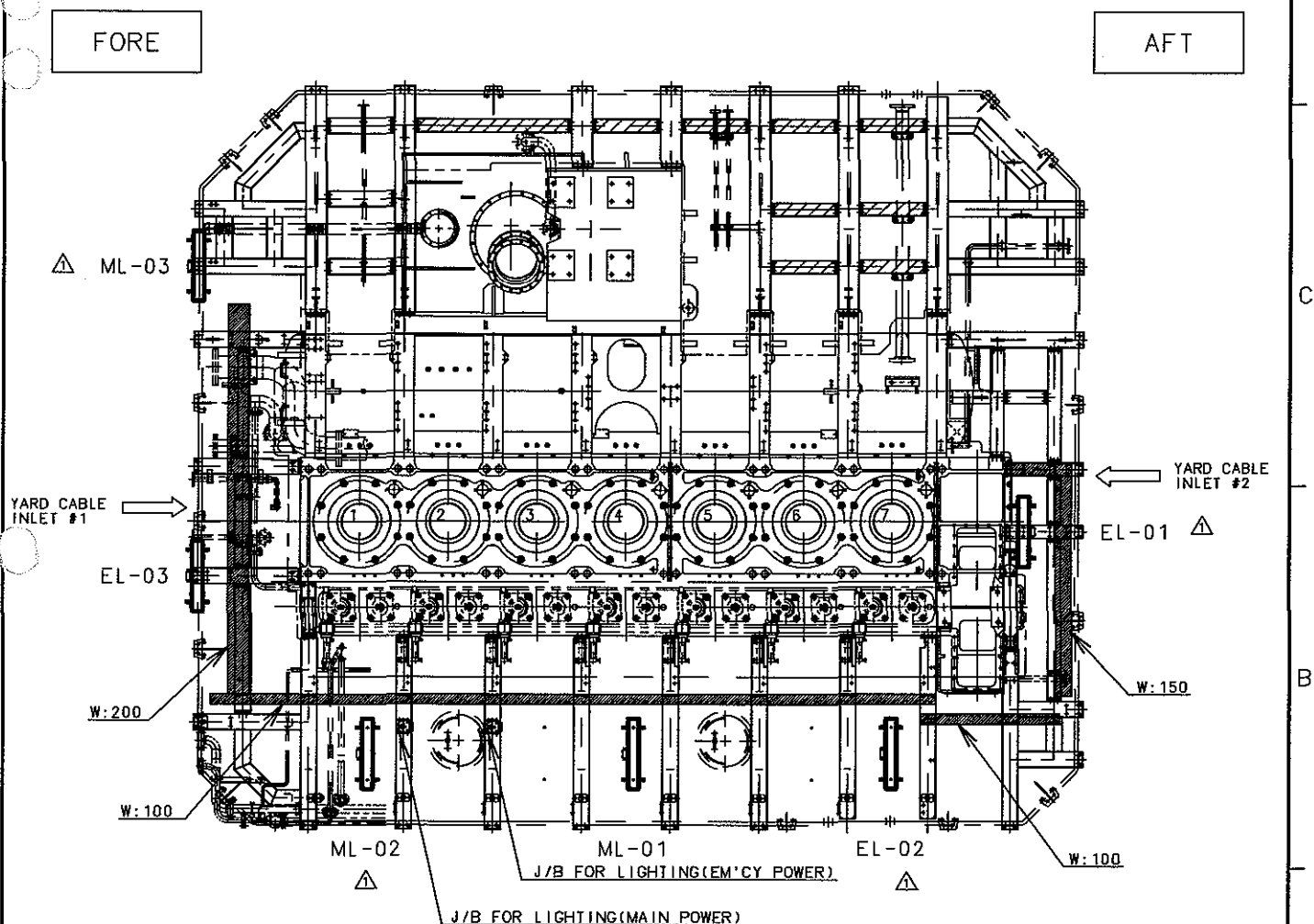
<UPPER PLATFORM>

FOR YARD CABLE INLET #1

- NO.1 AUX. BLOWER MOTOR
- JME-1
- JME-2
- JME-4
- J/B FOR OMD Δ
- SPU
- LIGHTING J/B Δ

FOR YARD CABLE INLET #2

- NO.2 AUX. BLOWER MOTOR
- JME-3
- Δ - RPMD
- COMBINED EL.BOX
- J/B FOR ACTUATOR

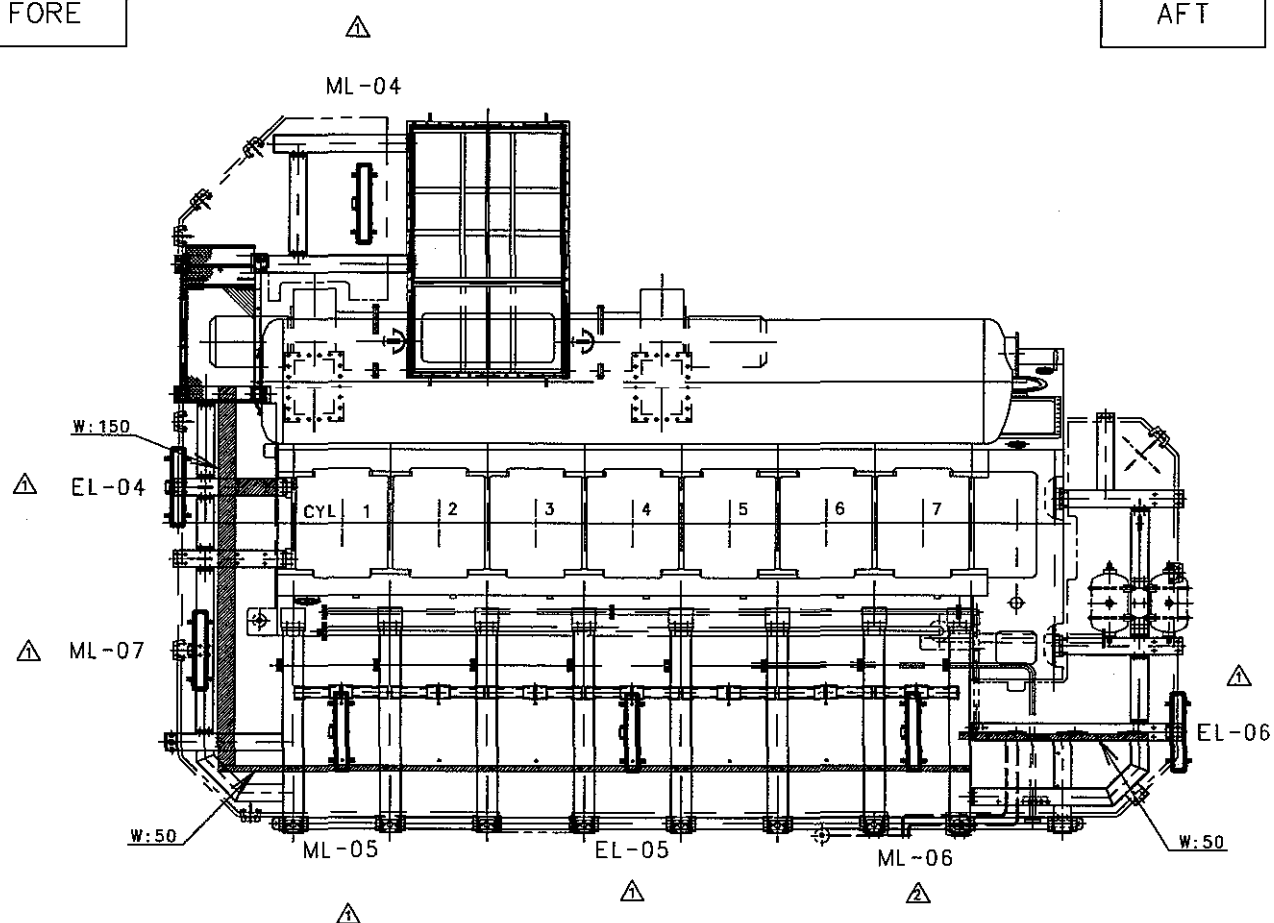


REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	091119	T.Y. JEON	SIZE A4	NAME LIGHTING ARRANGEMENT
			CHKD	091119	M.H. CHOI		
		WEIGHT	kg	APPD	091119		
SCALE N/S	MODEL 7S50MC-C	STX STX Engine Co., Ltd.			DRAWING NO. 5154047280-01		

<LOWER PLATFORM>

FORE

AFT



REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD WEIGHT kg	DWN	100624	T.Y. JEON	SIZE	NAME LIGHTING ARRANGEMENT
			CHKD	100624	M.H. CHO1	A4	
			APPD	100624	T.B. KIM		
SCALE N/S		MODEL 7S50MC-C	STX STX Engine Co., Ltd.			DRAWING NO. 5154047280-02	

12. 04 WIRING DIAGRAM

STX Heavy Industries Co., Ltd

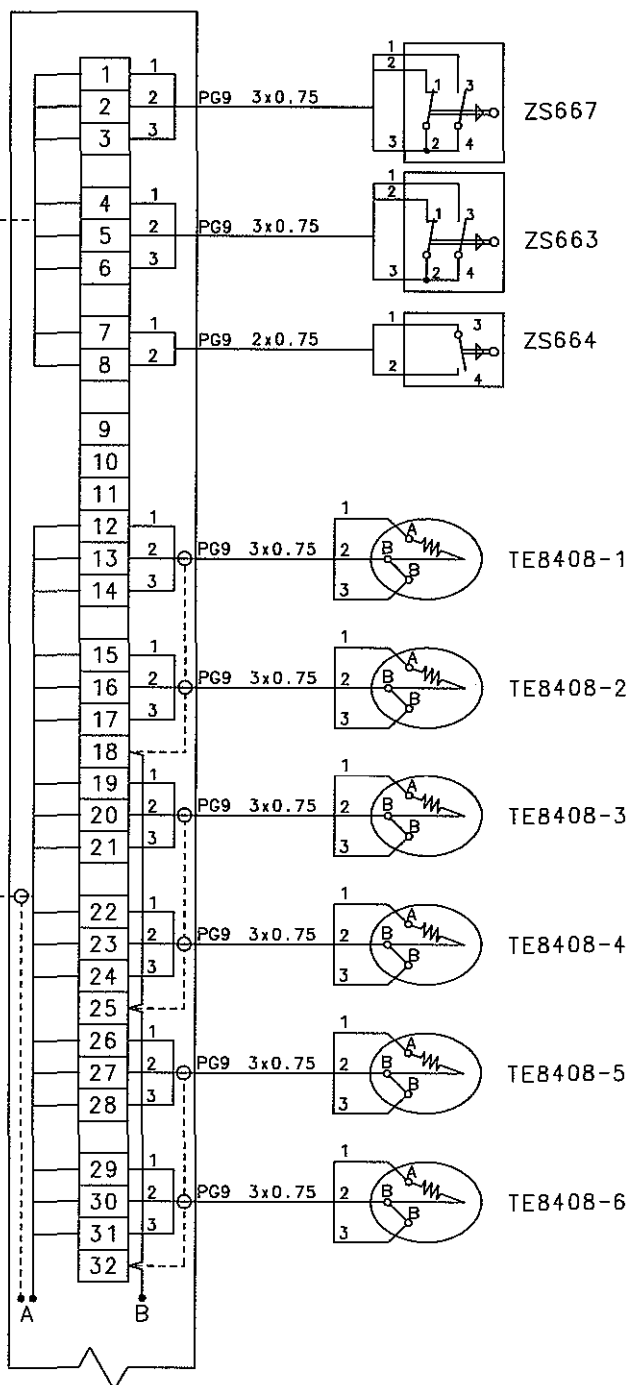
JME-1

TO E.C.C
(ME1-1)

MPYC-12

TO AMS

MPYCS-27



Starting air distributor
(Blocked pos.)(pos.119)
No.3,4 to be closed
in blocked position

Main starting v/v
(Service pos.)(pos.120)
No.3,4 to be closed
in serviced position

Main starting v/v
(Blocked pos.)(pos.121)
No.3,4 to be closed
in blocked position

No.1 Cyl. Jacket
C.W outlet temp.
(Ind./alarm/sld)

No.2 Cyl. Jacket
C.W outlet temp.
(Ind./alarm/sld)

No.3 Cyl. Jacket
C.W outlet temp.
(Ind./alarm/sld)

No.4 Cyl. Jacket
C.W outlet temp.
(Ind./alarm/sld)

No.5 Cyl. Jacket
C.W outlet temp.
(Ind./alarm/sld)

No.6 Cyl. Jacket
C.W outlet temp.
(Ind./alarm/sld)

NOTE

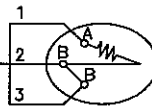
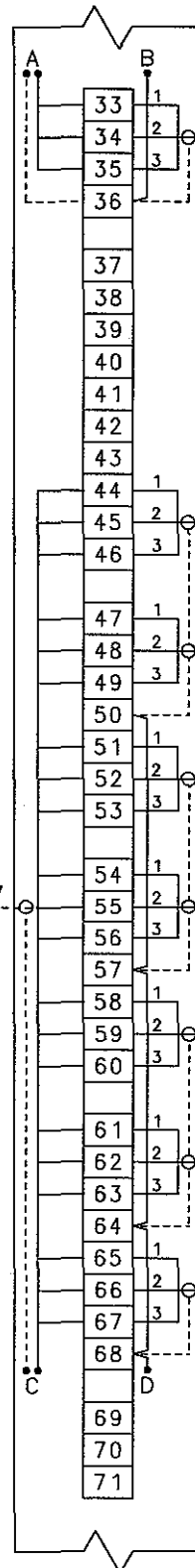
----- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL		 CAD WEIGHT kg	DWN	091119	T.Y. JEON	SIZE NAME A4 WIRING DIAGRAM FOR JME-1
			CHKD	091119	M.H. CHOI	
			APPD	091119	T.B. KIM	
SCALE N/S		MODEL 7S50MC-C	 STX Engine Co., Ltd.			DRAWING NO. 5154047290-01

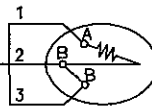
JME-1

TO AMS

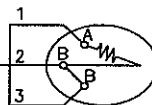
MPYCS-27



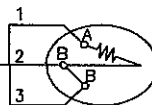
TE8408-7

No.7 Cyl. Jacket
C.W outlet temp.
(Ind./alarm/sld)

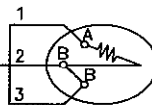
TE8610-1

No.1 Cyl. scav. air
Fire detection
(Ind./alarm/sld)

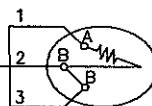
TE8610-2

No.2 Cyl. scav. air
Fire detection
(Ind./alarm/sld)

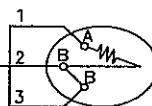
TE8610-3

No.3 Cyl. scav. air
Fire detection
(Ind./alarm/sld)

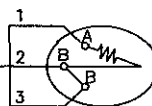
TE8610-4

No.4 Cyl. scav. air
Fire detection
(Ind./alarm/sld)

TE8610-5

No.5 Cyl. scav. air
Fire detection
(Ind./alarm/sld)

TE8610-6

No.6 Cyl. scav. air
Fire detection
(Ind./alarm/sld)

TE8610-7

No.7 Cyl. scav. air
Fire detection
(Ind./alarm/sld)

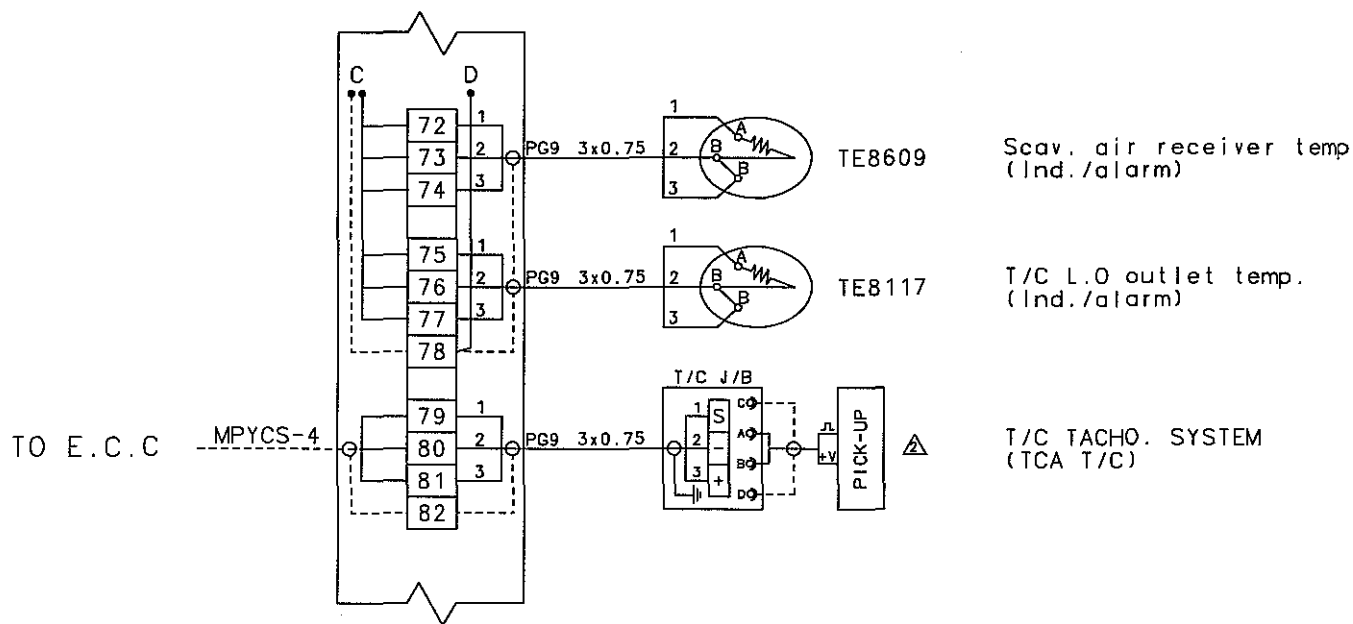
NOTE

----- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	091119	T.Y. JEON	SIZE A4	NAME WIRING DIAGRAM FOR JME-1
			CHKD	091119	M.H. CHOI		
		WEIGHT	kg	APPD	091119		
SCALE N/S	MODEL 7S50MC-C		 STX Engine Co., Ltd.			DRAWING NO. 5154047290-02	

STX STX Engine Co., Ltd.

JME-1

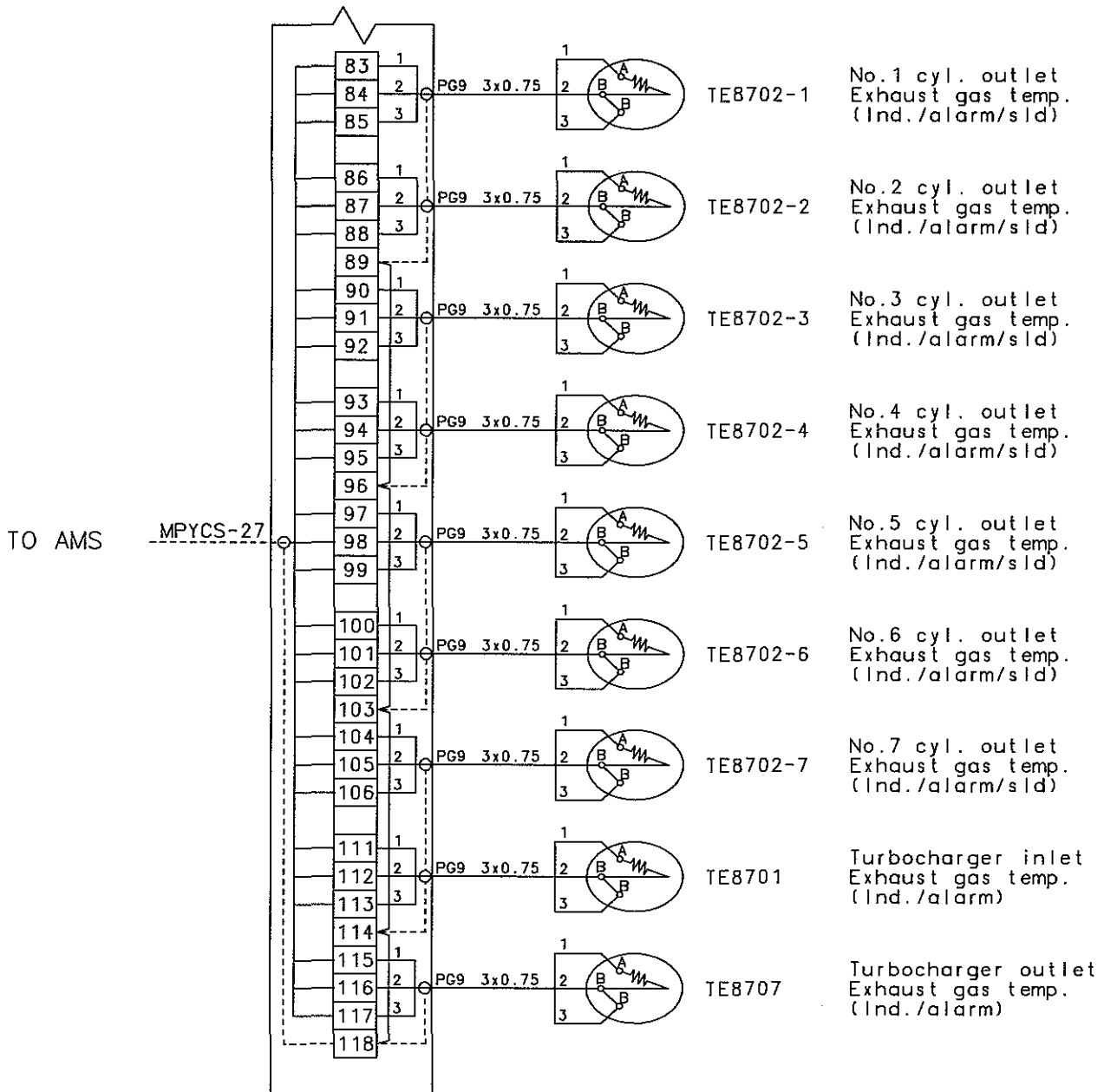


NOTE

----- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	100624	T.Y. JEON	SIZE A4	NAME WIRING DIAGRAM FOR JME-1
			CHKD	100624	M.H. CHOI		
		WEIGHT kg	APPD	100624	T.B. KIM		
SCALE N/S		MODEL 7S50MC-C	 STX Engine Co., Ltd.			DRAWING NO. 5154047290-03	

JME-1



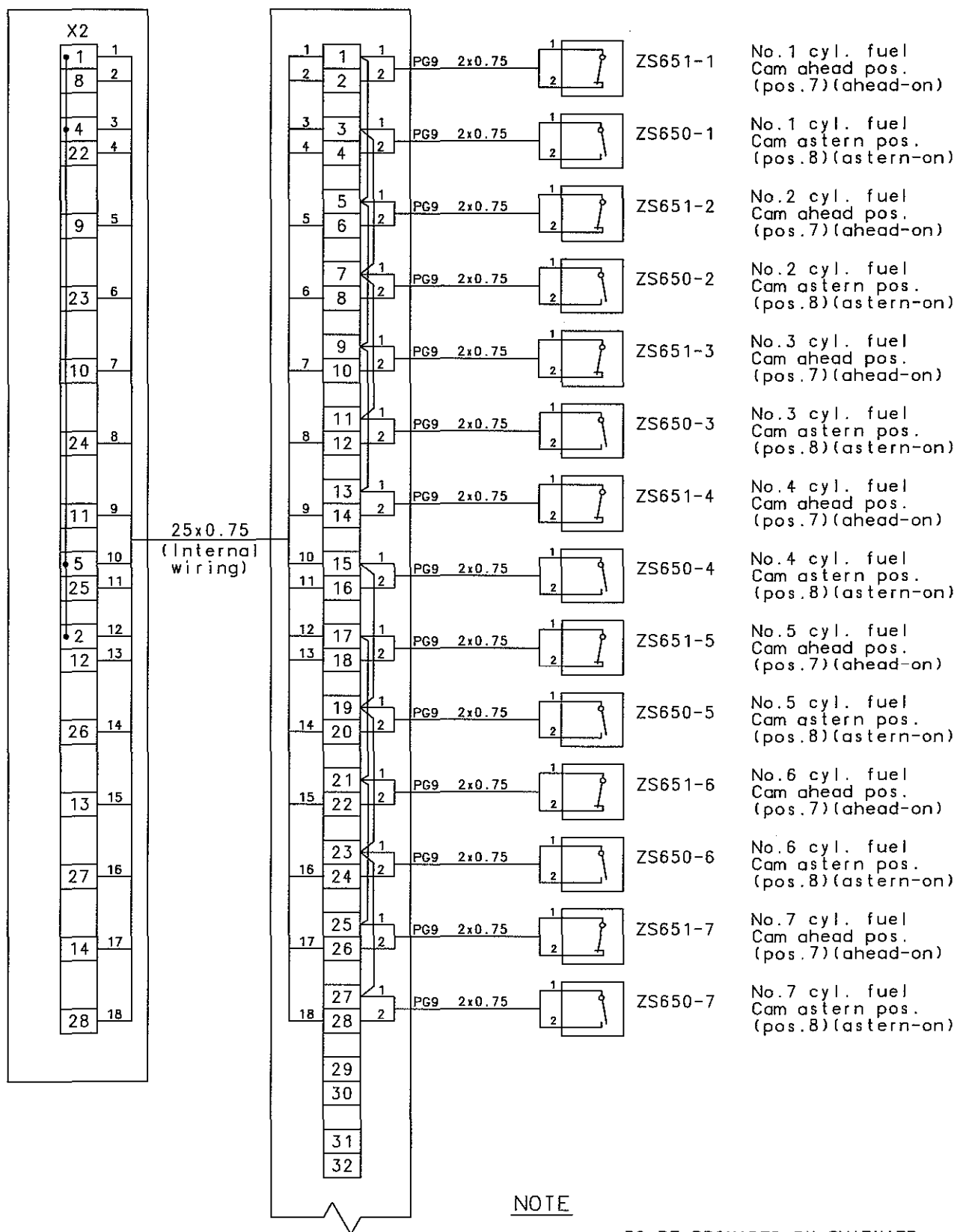
NOTE

----- : TO BE PROVIDED BY SHIPYARD

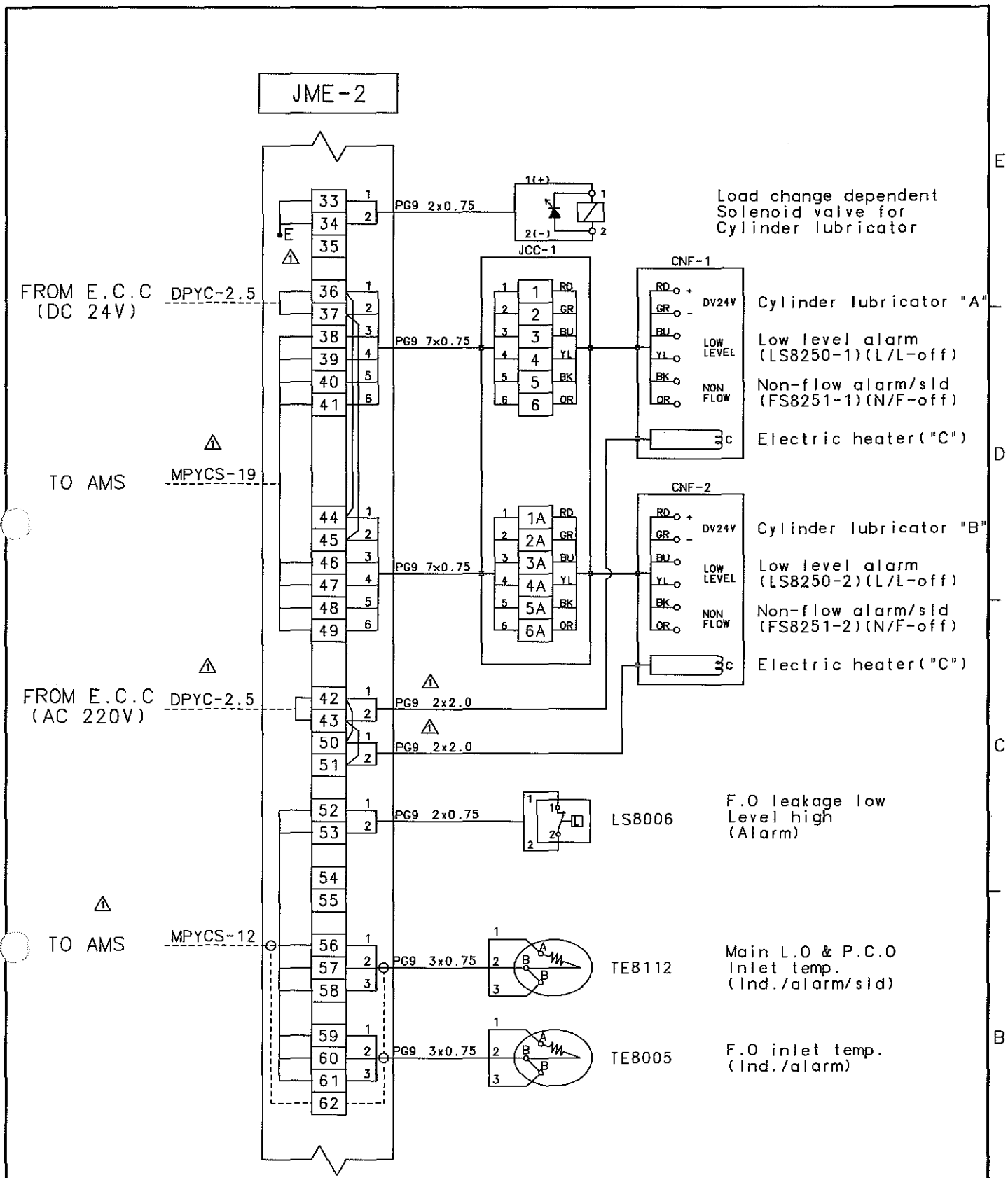
REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	091119	T.Y. JEON	SIZE	NAME WIRING DIAGRAM FOR JME-1
			CHKD	091119	M.H. CHOI	A4	
			APPD	091119	T.B. KIM		
		WEIGHT kg					
SCALE N/S		MODEL 7S50MC-C		 STX Engine Co., Ltd.		DRAWING NO. 5154047290-04	

COMBINED
EL. BOX

JME-2



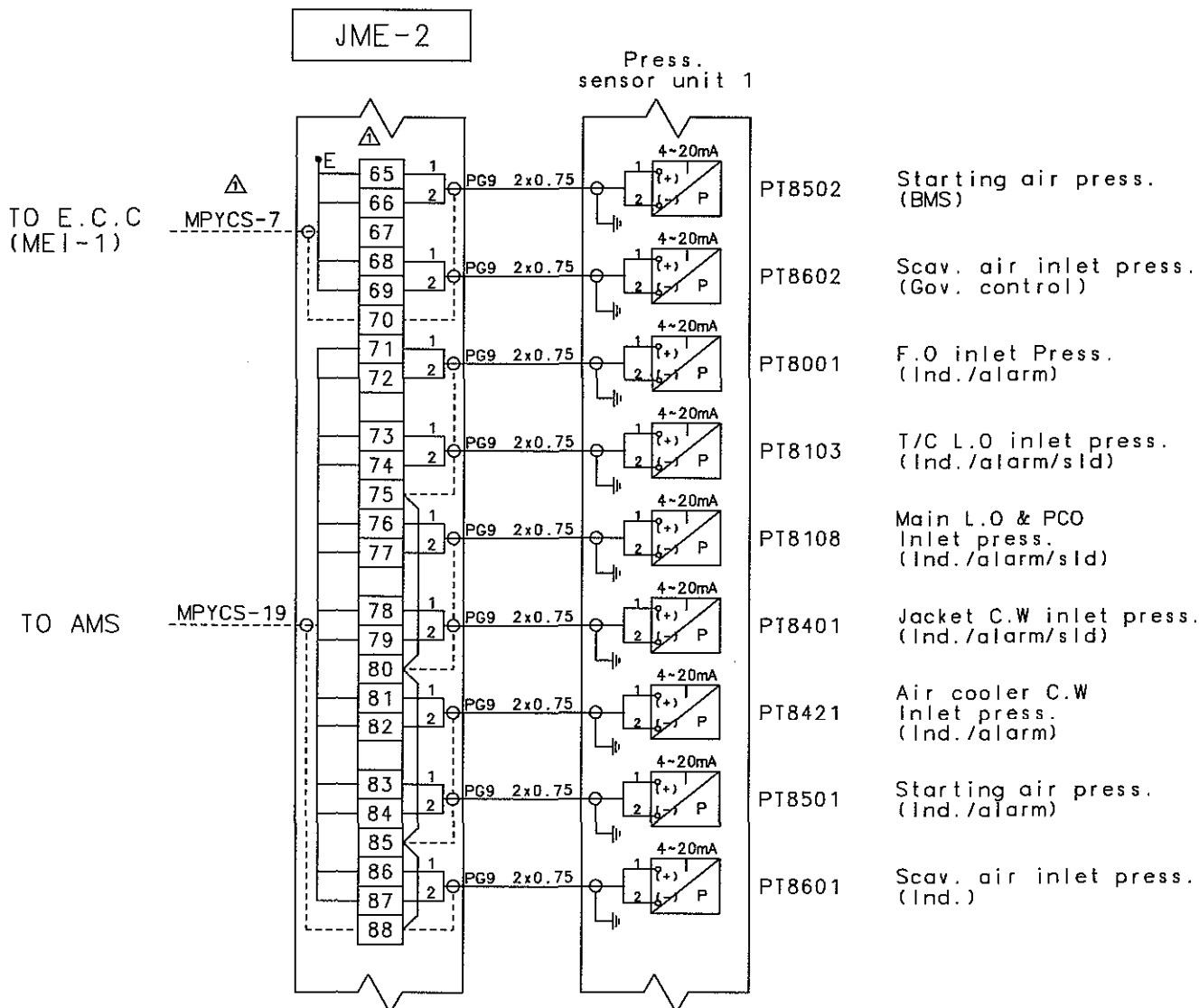
REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	090113	T.Y. JEON	SIZE A4	NAME WIRING DIAGRAM FOR JME-2
			CHKD	090113	M.H. CHOI		
		WEIGHT	kg	APPD	090113		
SCALE N/S		MODEL 7S50MC-C		 STX Engine Co., Ltd.		DRAWING NO. 5154047290-05	



NOTE

----- : TO BE PROVIDED BY SHIPYARD

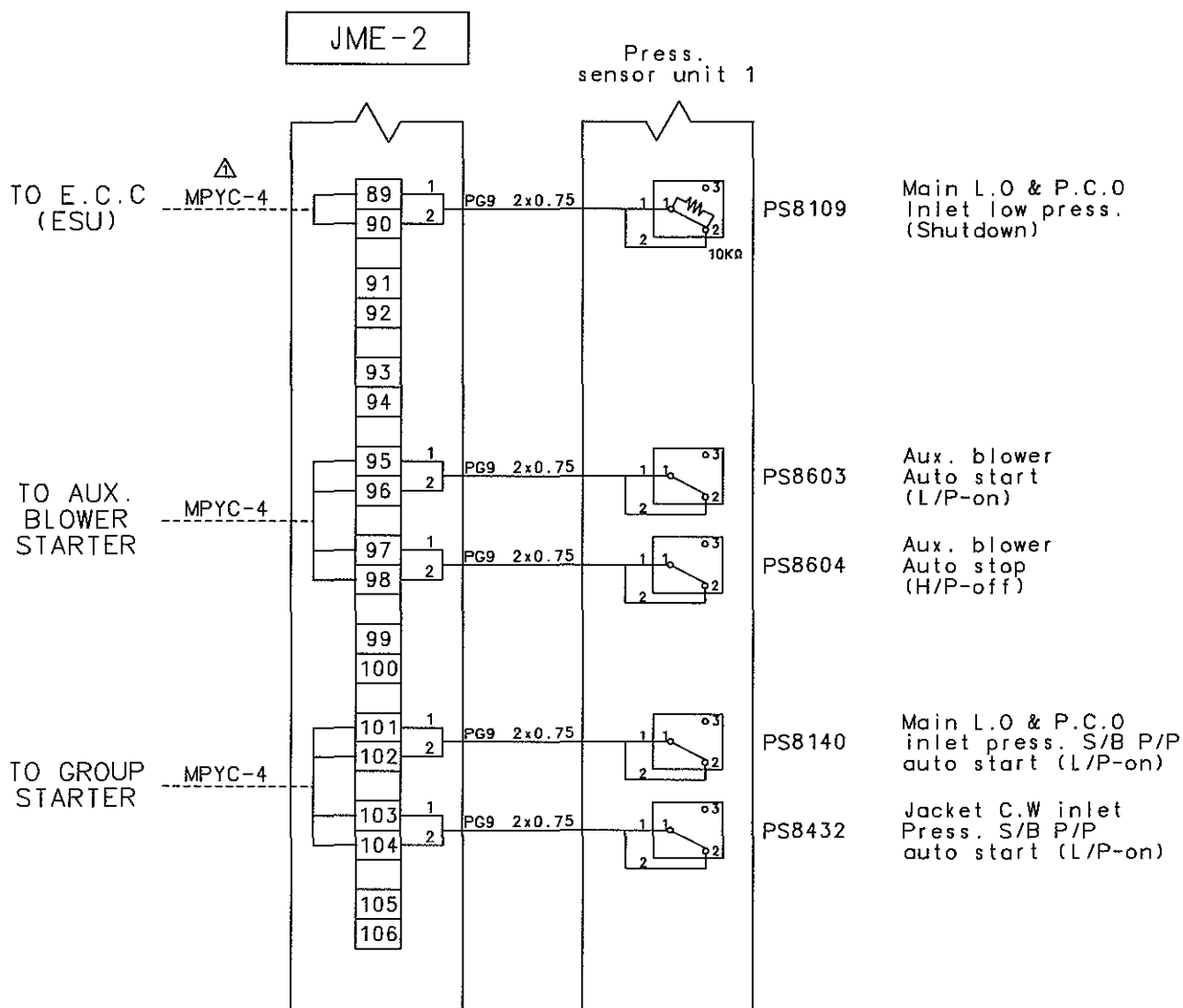
REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	091119	T.Y. JEON	SIZE	NAME WIRING DIAGRAM FOR JME-2
			CHKD	091119	M.H. CHOI	A4	
			APPD	091119	T.B. KIM		
SCALE N/S		MODEL 7S50MC-C	STX STX Engine Co., Ltd.			DRAWING NO. 5154047290-06	



NOTE

----- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL		 CAD WEIGHT kg	DWN	091119	T.Y. JEON	SIZE NAME A4 WIRING DIAGRAM FOR JME-2
			CHKD	091119	M.H. CHOI	
			APPD	091119	T.B. KIM	
SCALE N/S		MODEL 7S50MC-C	 STX Engine Co., Ltd.			DRAWING NO. 5154047290-07

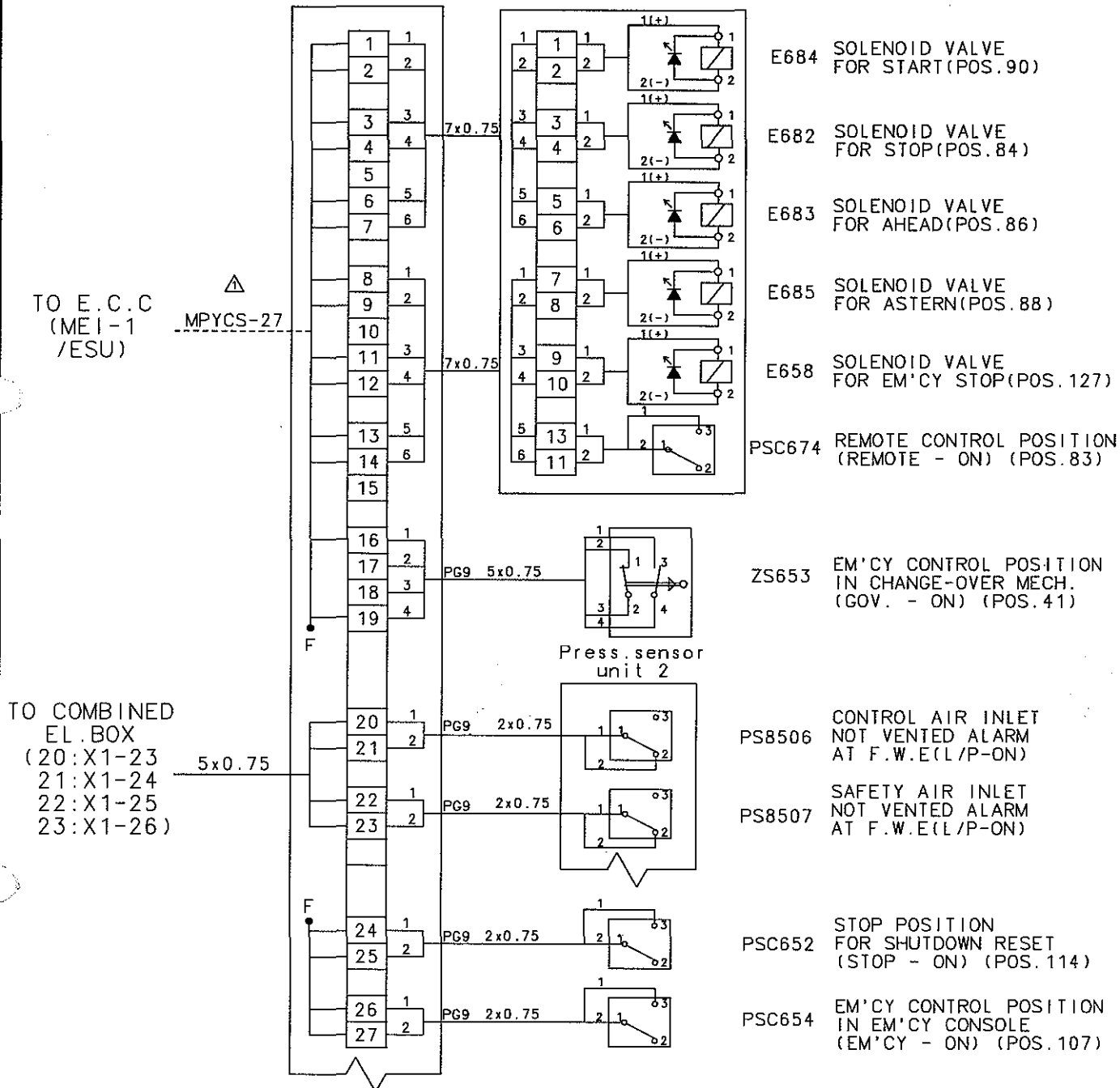


NOTE

----- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD WEIGHT kg	DWN	091119	T.Y. JEON	SIZE A4	NAME WIRING DIAGRAM FOR JME-2
			CHKD	091119	M.H. CHOI		
			APPD	091119	T.B. KIM		
SCALE N/S		MODEL 7S50MC-C	STX STX Engine Co., Ltd.			DRAWING NO. 5154047290-08	

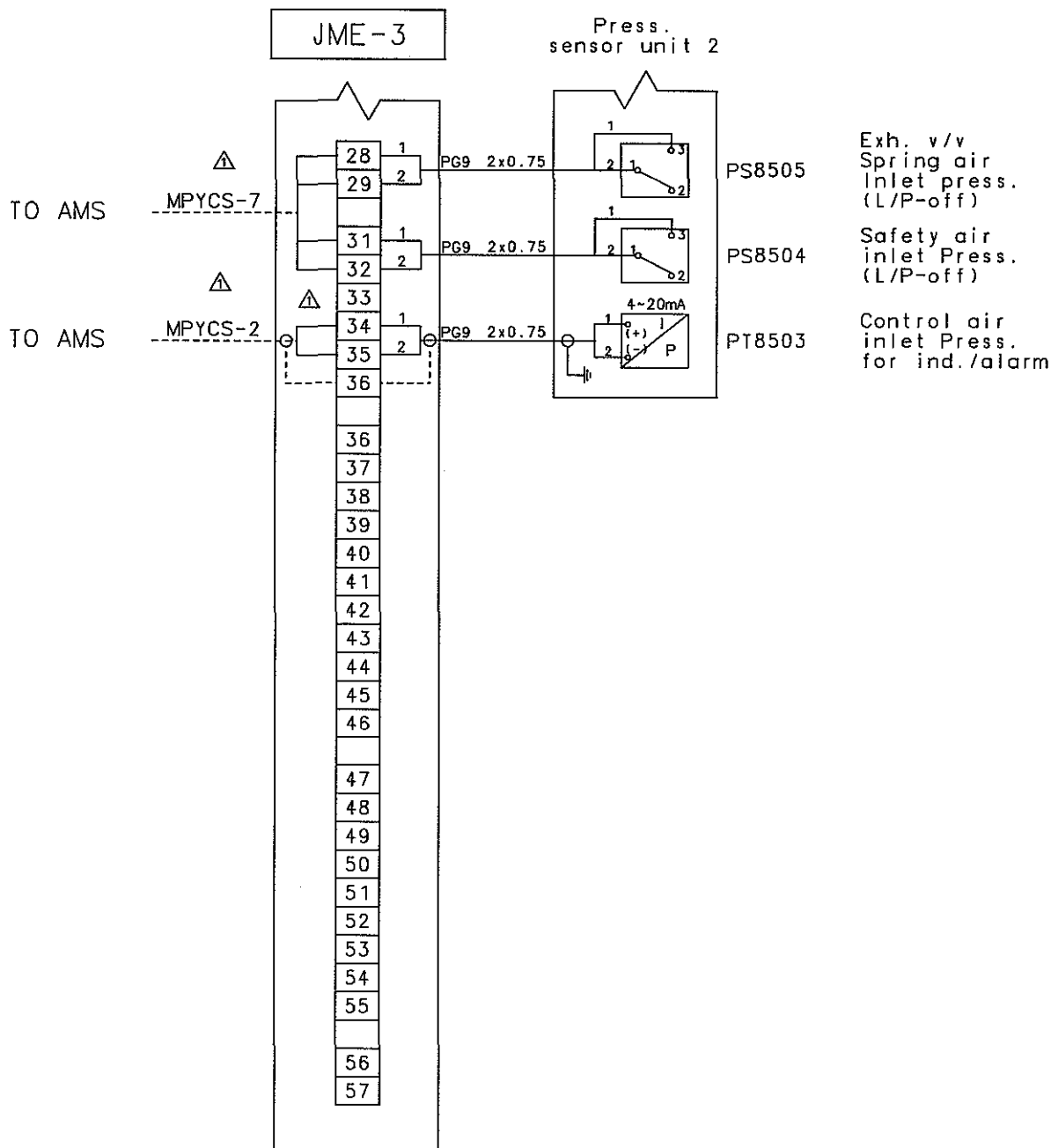
JME-3

PNEUMATIC
CONTROL CABINET

NOTE

----- : TO BE PROVIDED BY SHIPYARD

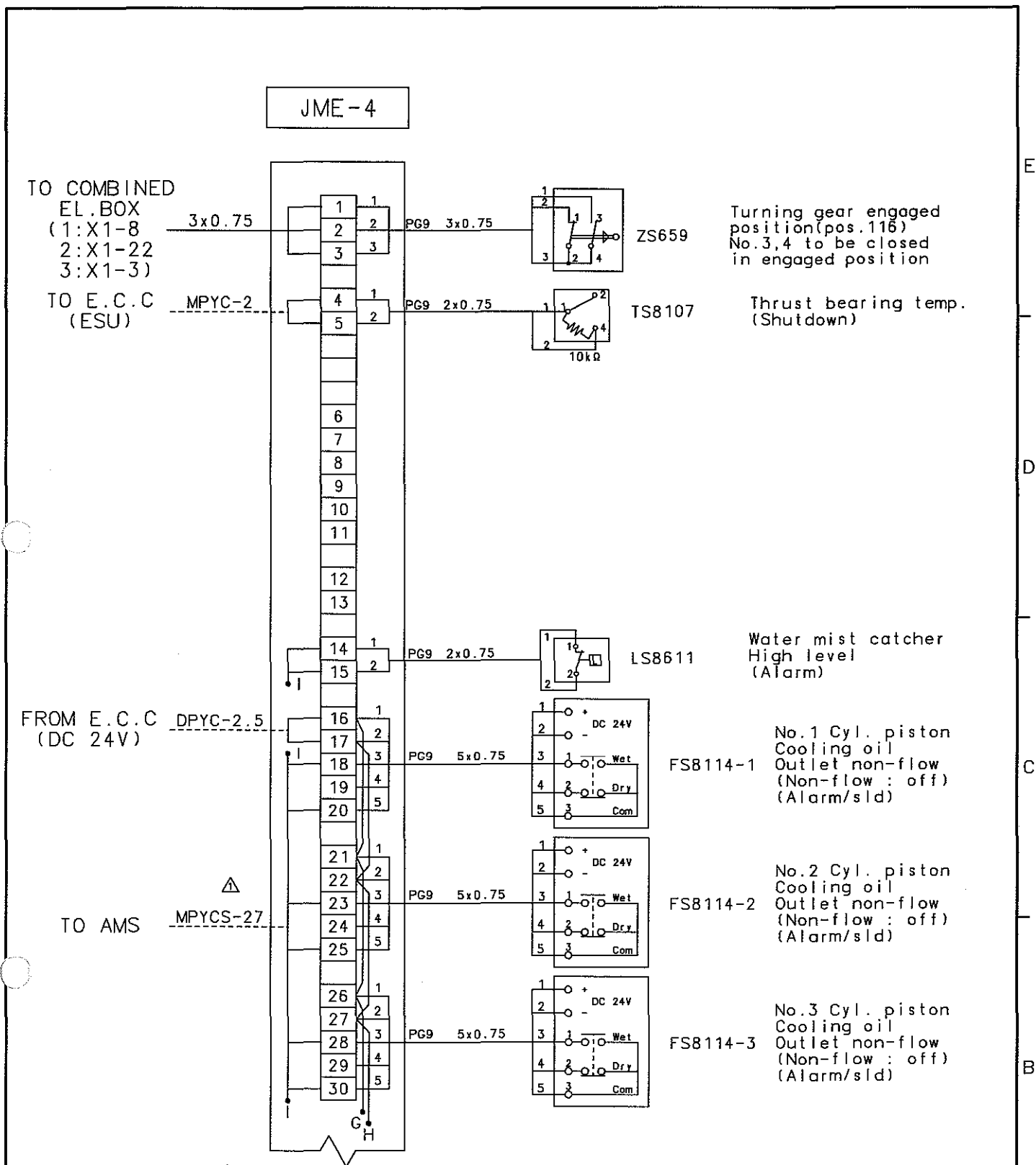
REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL		CAD	DWN	091119	T.Y. JEON	WIRING DIAGRAM FOR JME-3
WEIGHT		kg	CHKD	091119	M.H. CHOI	
SCALE		MODEL	APPD	091119	T.B. KIM	
N/S		7S50MC-C	STX STX Engine Co., Ltd.		DRAWING NO.	5154047290-09



NOTE

----- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL		DWN		091119	T.Y. JEON	SIZE
WEIGHT		CHKD		091119	M.H. CHOI	A4
SCALE		APPD		091119	T.B. KIM	NAME
N/S		MODEL		DRAWING NO.		
7S50MC-C		STX STX Engine Co., Ltd.			5154047290-10	

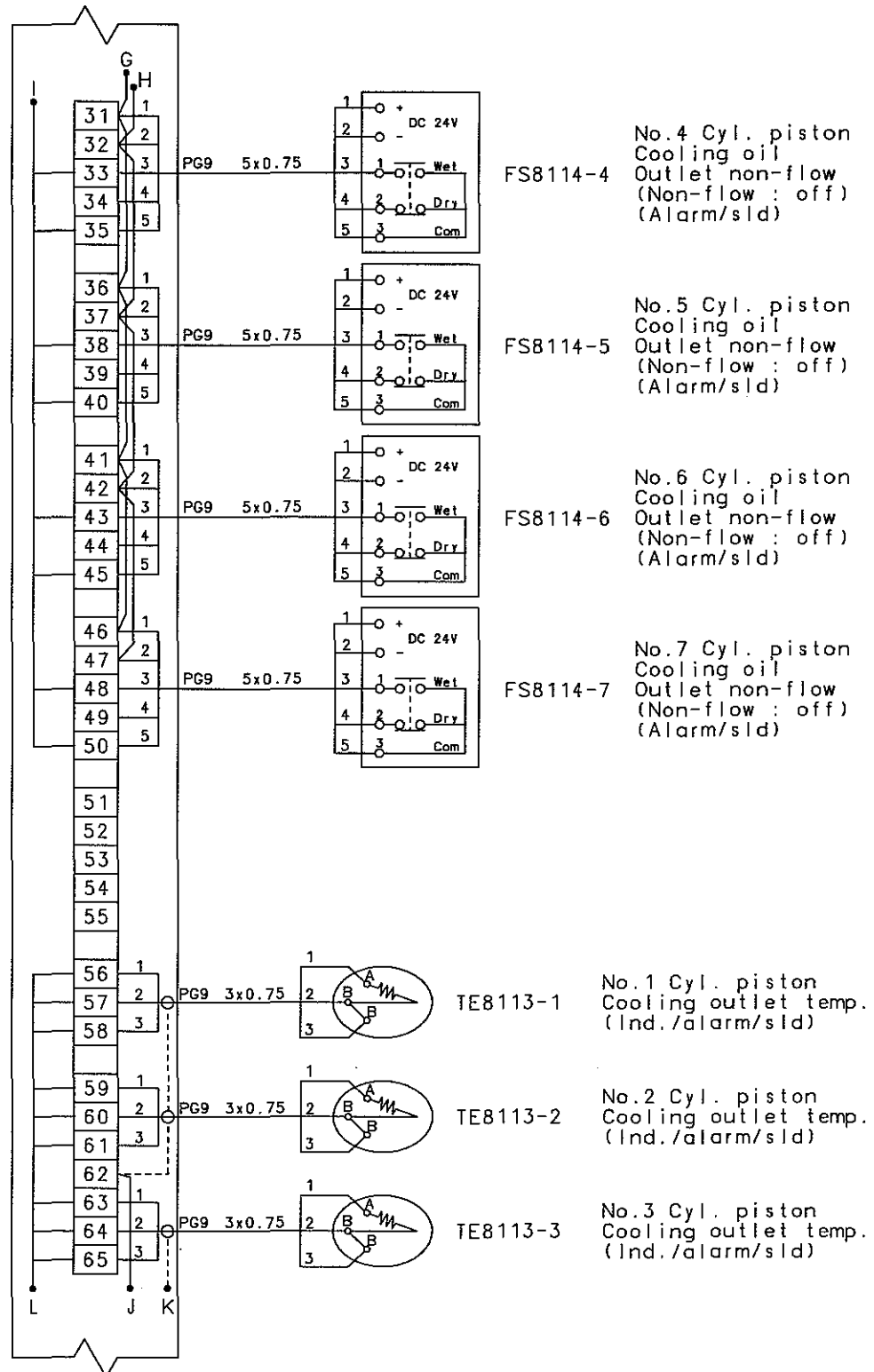


NOTE

----- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD WEIGHT kg	DWN	091119	T.Y. JEON	SIZE A4	NAME WIRING DIAGRAM FOR JME-4
			CHKD	091119	M.H. CHOI		
			APPD	091119	T.B. KIM		
SCALE N/S		MODEL 7S50MC-C	 STX Engine Co., Ltd.			DRAWING NO. 5154047290-11	

JME-4

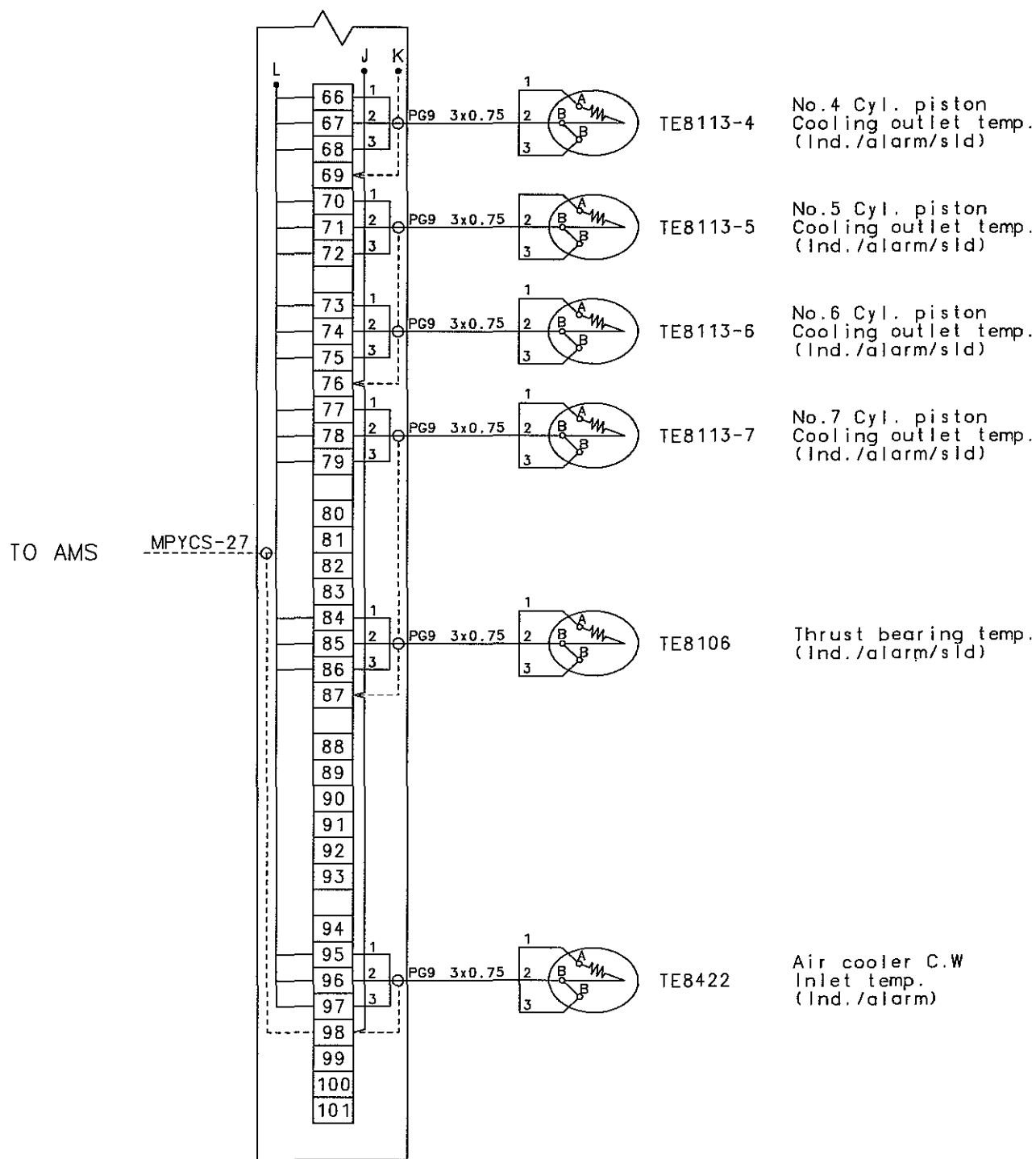


NOTE

----- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL		CAD	DWN	091119	T.Y. JEON	SIZE
WEIGHT		kg	CHKD	091119	M.H. CHOI	A4
SCALE		MODEL	APPD	091119	T.B. KIM	NAME
N/S		7S50MC-C	STX STX Engine Co., Ltd.		DRAWING NO.	
					5154047290-12	

JME-4



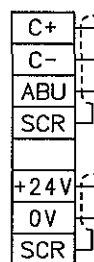
NOTE

----- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD WEIGHT kg	DWN	091119	T.Y. JEON	SIZE NAME A4 WIRING DIAGRAM FOR JME-4	
			CHKD	091119	M.H. CHOI		
			APPD	091119	T.B. KIM		
SCALE N/S		MODEL 7S50MC-C	 STX Engine Co., Ltd.			DRAWING NO. 5154047290-13	

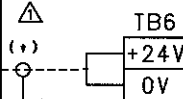
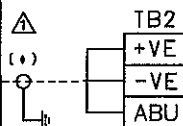
J/B FOR OMD

MONITORING PANEL FOR OMD ON E.C.C

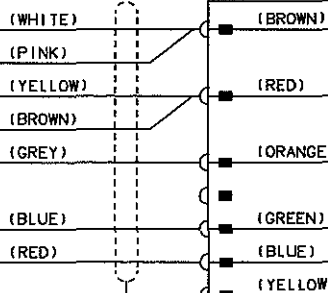
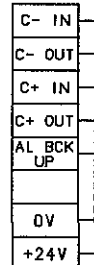


TIYCS-2

DPYCS-2.5

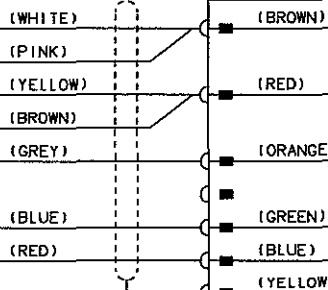
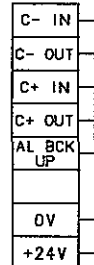


TB1 (SENSOR 1)



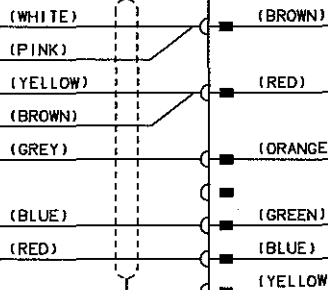
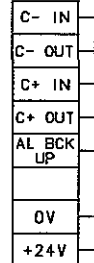
Oil mist detector 1

TB1 (SENSOR 2)



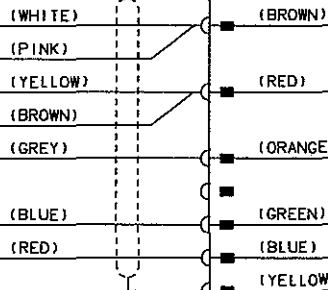
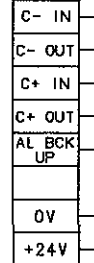
Oil mist detector 2

TB2 (SENSOR 3)



Oil mist detector 3

TB2 (SENSOR 4)



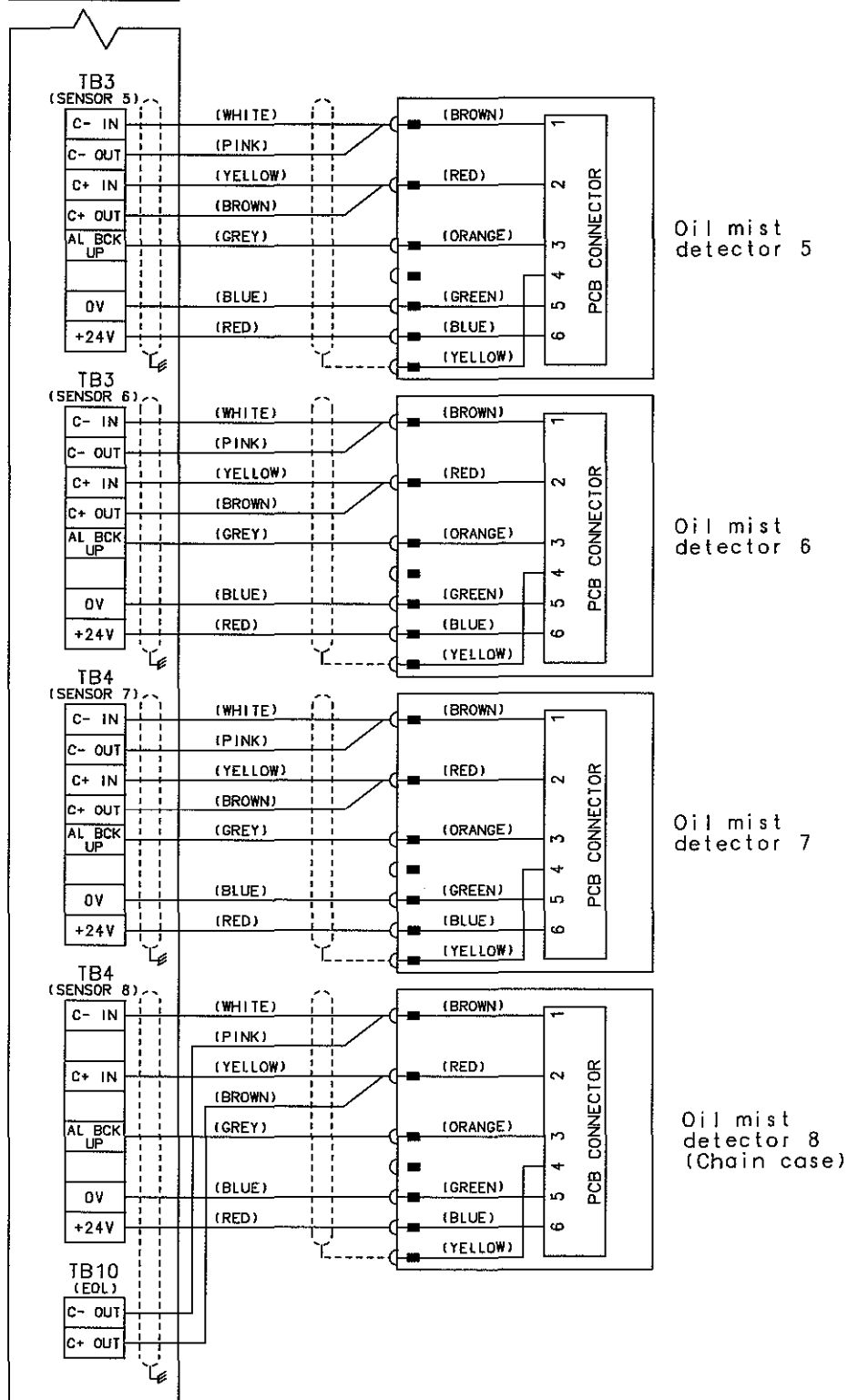
Oil mist detector 4

NOTE

----- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE
MATERIAL	CAD		DWN	091119	T.Y. JEON
	WEIGHT		CHKD	091119	M.H. CHOI
	kg		APPD	091119	T.B. KIM
SCALE	N/S	MODEL	STX STX Engine Co., Ltd.		
		7S50MC-C	DRAWING NO.		
			5154047290-14		

J/B FOR OMD

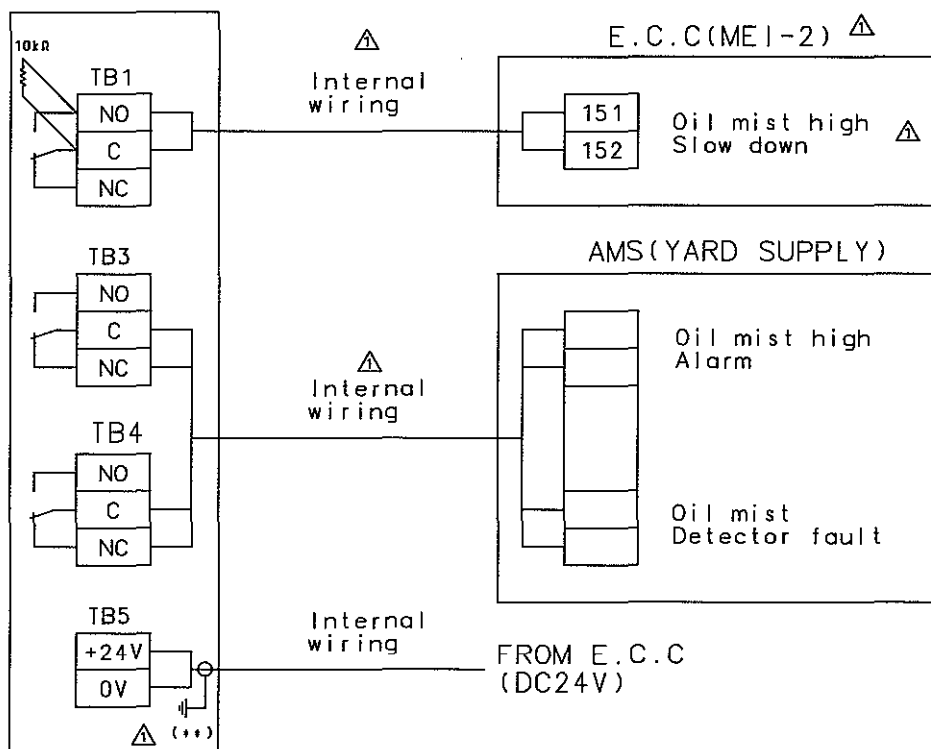


NOTE

----- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE
MATERIAL	 CAD		DWN	091119	T.Y. JEON
	WEIGHT		CHKD	091119	M.H. CHOI
	kg		APPD	091119	T.B. KIM
SCALE	N/S	MODEL	STX STX Engine Co., Ltd.		
		7S50MC-C	DRAWING NO.		
			5154047290-15		

MONITORING PANEL FOR OMD ON E.C.C



(+):

- △ 1. Shield screen of power and communication cable should be connected to IP65 metal gland in monitoring panel.
- 2. Power and communication with outer coating should come to control panel from J/B without using of terminal or disconnection.

(**):

- △ 1. DC24V input cable should be connected on earth.
- 2. Earth of panel should be made on earth stud in control panel and earth of the opposite side should be made on 24V earth rail.

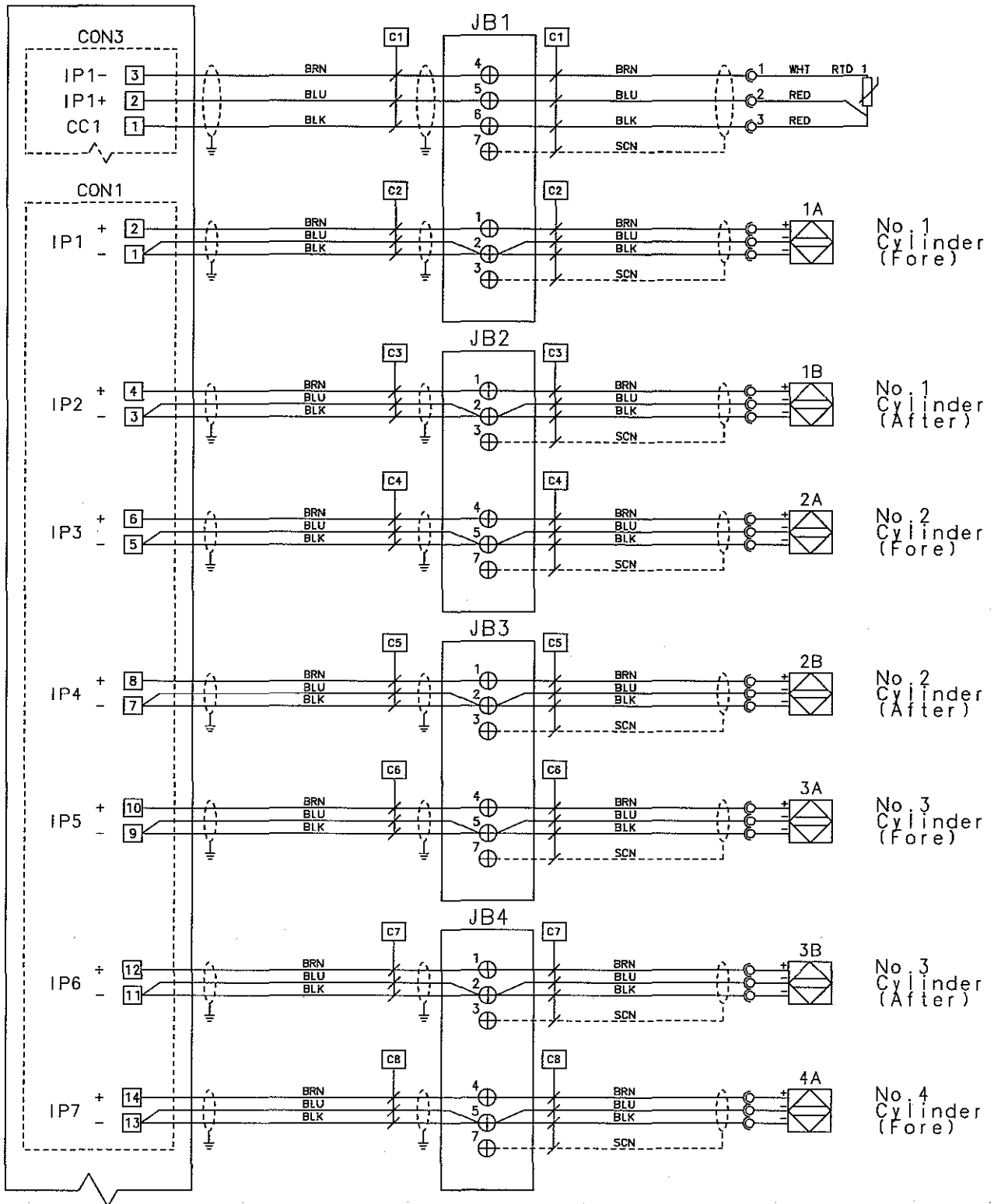
NOTE

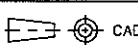
----- : TO BE PROVIDED BY SHIPYARD

REV	LOC	DATE	MADE	APPD	NOTE
MATERIAL	CAD	DWN	091119	T.Y.JEON	SIZE
WEIGHT	kg	CHKD	091119	M.H.CHOI	A4
SCALE	MODEL	APPD	091119	T.B.KIM	NAME
N/S	7S50MC-C				WIRING DIAGRAM FOR OMD
STX STX Engine Co., Ltd.			DRAWING NO. 5154047290-16		

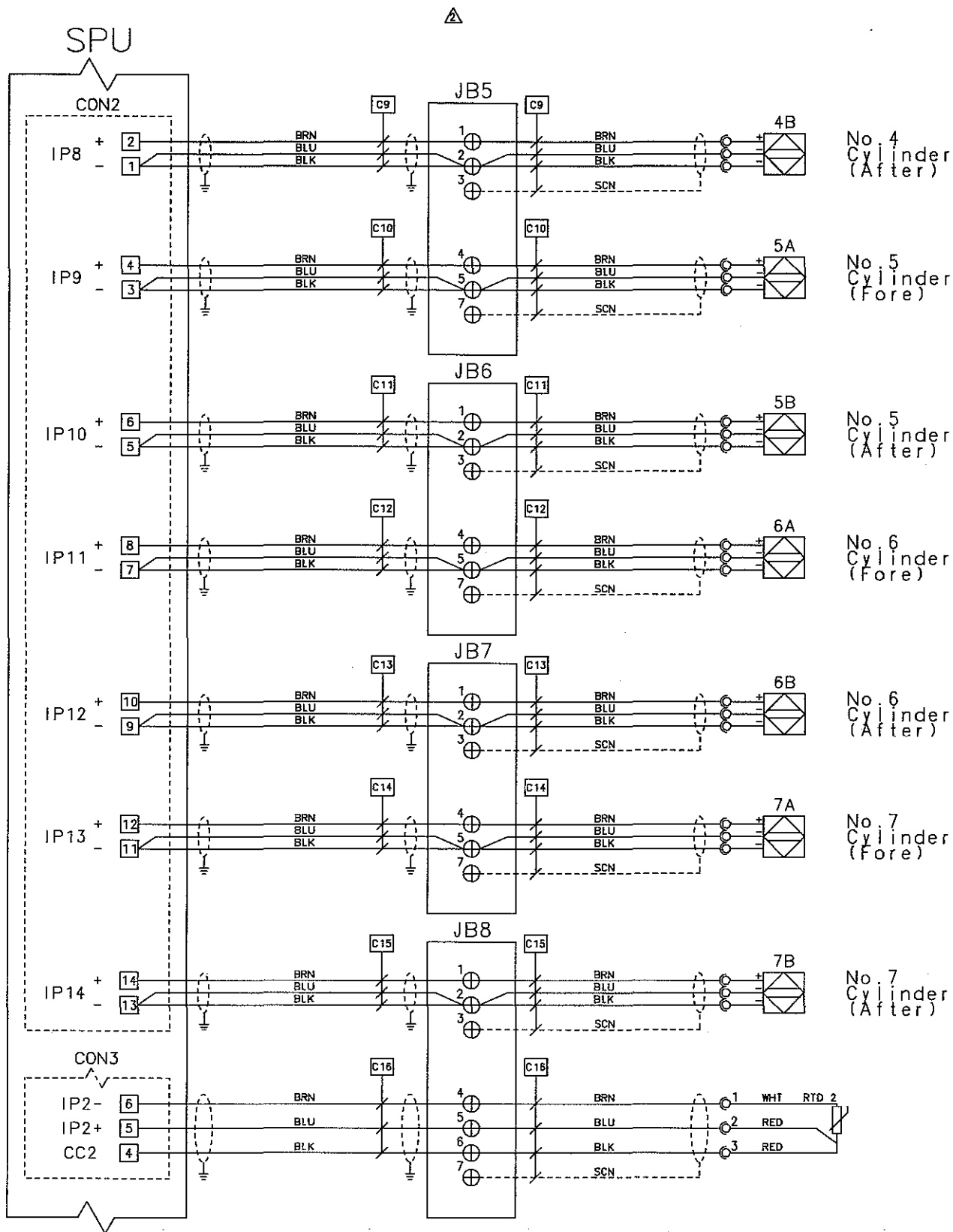


SPU



REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	101111	T.Y. JEON	SIZE A4	NAME WIRING DIAGRAM FOR BMW
			CHKD	101111	D.H. LEE		
			APPD	101111	T.B. KIM		
SCALE N/S		MODEL 7S50MC-C	STX STX Engine Co., Ltd.			DRAWING NO. 5154047290-17	

STX STX Engine Co., Ltd.



* All cables are supplied by AMOT

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD WEIGHT kg	DWN	101111	T.Y. JEON	SIZE A4	NAME WIRING DIAGRAM FOR BMW
			CHKD	101111	D.H. LEE		
			APPD	101111	T.B. KIM		
SCALE N/S		MODEL 7S50MC-C	 STX Engine Co., Ltd.			DRAWING NO. 5154047290-18	

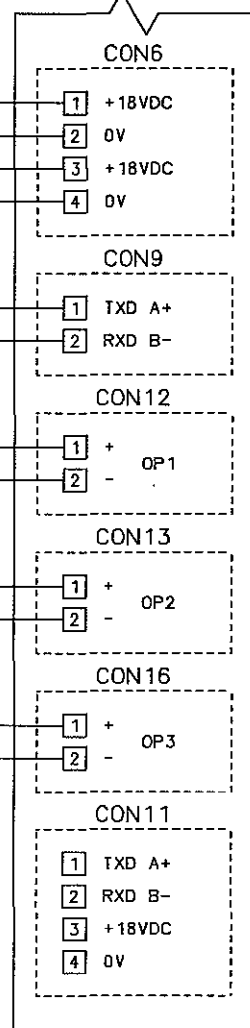
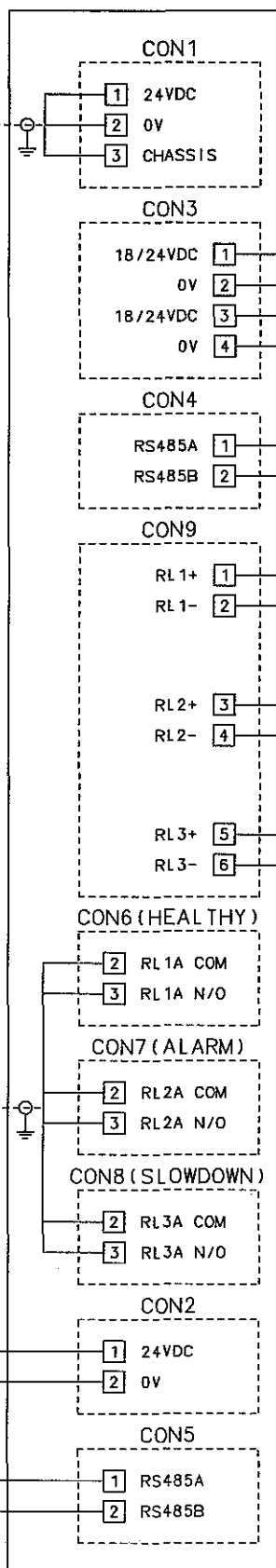
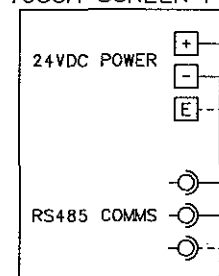
FROM E.C.C
(DC24V)

INTERNAL

IF +

SPU

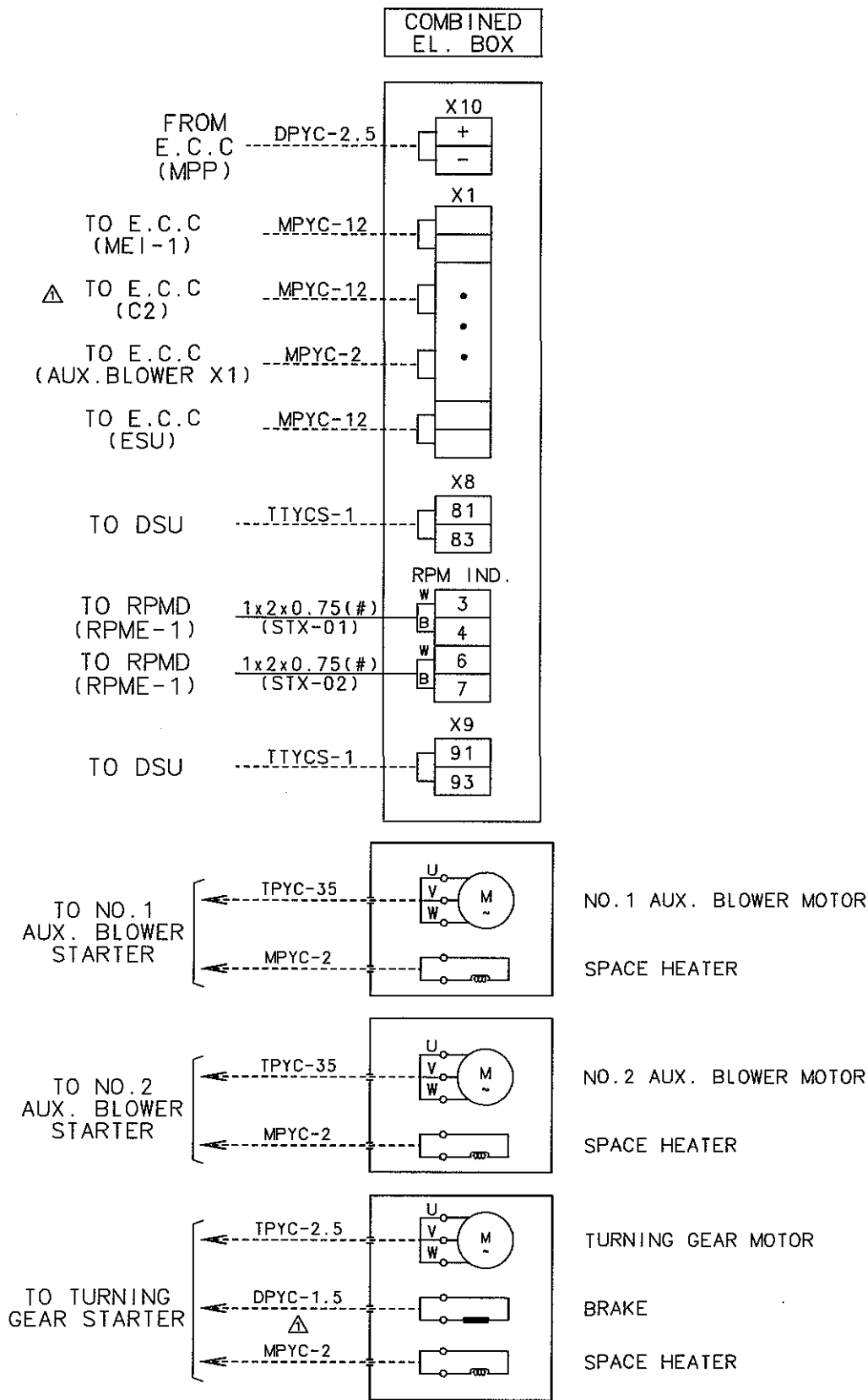
TOUCH SCREEN PC



NOTE

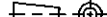
----- : TO BE PROVIDED BY SHIPYARD
----- : TO BE PROVIDED BY AMOT

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	101111	T.Y. JEON	SIZE A4	NAME WIRING DIAGRAM FOR BMW
			CHKD	101111	D.H. LEE		
		WEIGHT kg	APPD	101111	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C	STX STX Engine Co., Ltd.			DRAWING NO. 5154047290-19		

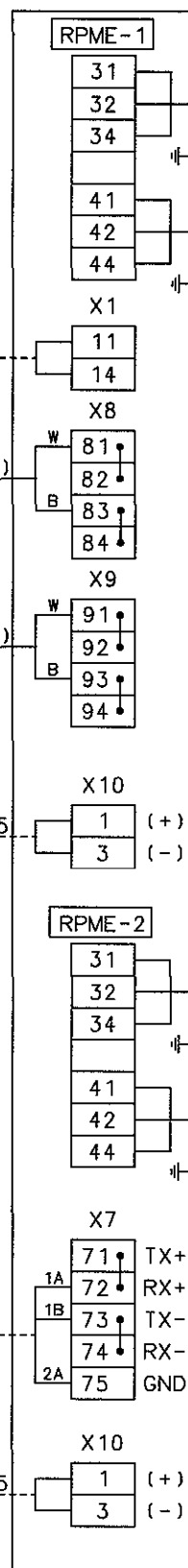


NOTE

----- : TO BE PROVIDED BY SHIPYARD
 (#) : TO BE CONNECTED BY COMMUNICATION CABLE

REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL		 CAD	DWN	091119	T.Y. JEON	SIZE NAME WIRING DIAGRAM FOR COMBINED EL. BOX & MOTOR STARTER
			CHKD	091119	M.H. CHOI	
		WEIGHT kg	APPD	091119	T.B. KIM	
SCALE N/S	MODEL 7S50MC-C	 STX Engine Co., Ltd.			DRAWING NO. 5154047290-20	

RPMD



Speed pick-up 1

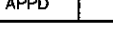
Speed pick-up 2

Speed pick-up 1

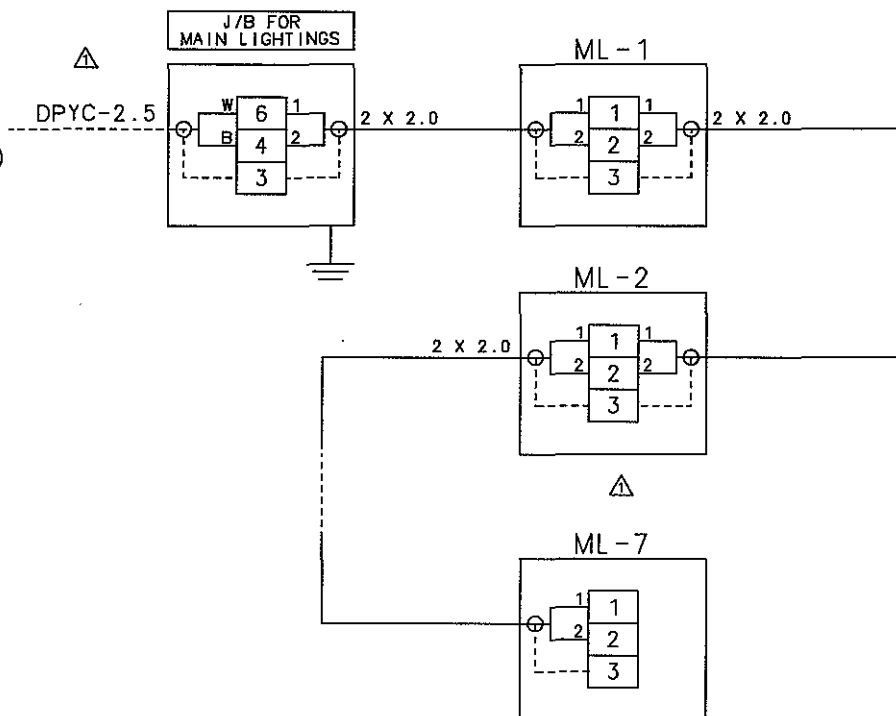
Speed pick-up 2

NOTE

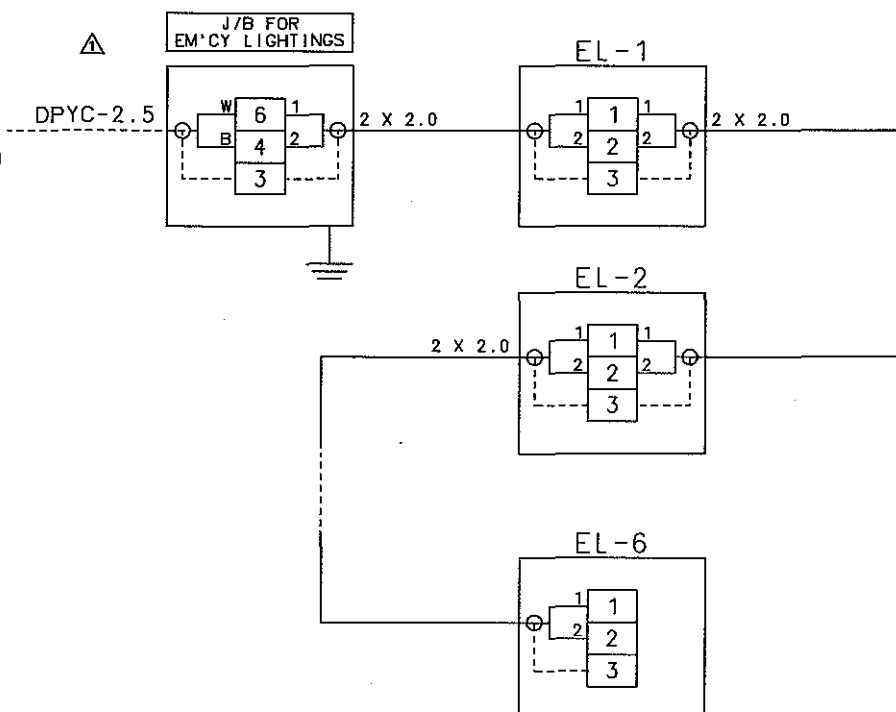
- : TO BE PROVIDED BY SHIPYARD
- : TO BE PROVIDED BY ENGINE MAKER
- (#) : TO BE CONNECTED BY COMMUNICATION CABLE

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	091119	T.Y. JEON	SIZE A4	NAME WIRING DIAGRAM FOR RPMD
			CHKD	091119	M.H. CHOI		
		WEIGHT kg	APPD	091119	T.B. KIM		
SCALE N/S	MODEL 7S50MC-C	 STX Engine Co., Ltd.			DRAWING NO. 5154047290-21		

FROM MSBD
MAIN POWER
(AC220V 1Ø 60Hz)



FROM MSBD
EM'CY POWER
(AC220V 1Ø 60Hz)



NOTE

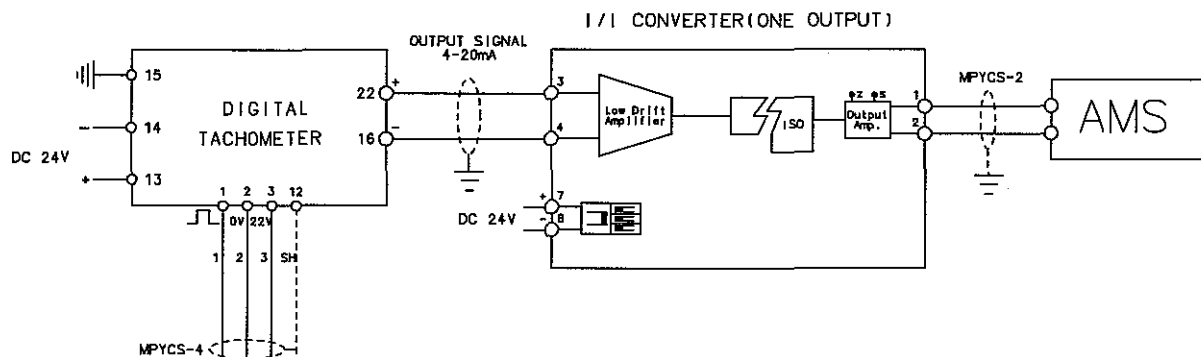
1. CABLE CONNECTIONS CAN BE CHANGED ACCORDING TO LIGHTING POSTION.
2. MAIN & EM'CY LIGHTING QUANTITIES AND POSITIONS ARE BASED ON LIGHTING ARRANGEMENT DRAWING.
3. LIGHTINGS OF 'L', 'EL' ARE CONSISTED OF 2PCS OF 20W.

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD WEIGHT kg	DWN	091119	T.Y. JEON	SIZE	NAME WIRING DIAGRAM FOR LIGHTING
			CHKD	091119	M.H. CHOI	A4	
			APPD	091119	T.B. KIM		
SCALE N/S		MODEL 7S50MC-C	 STX Engine Co., Ltd.			DRAWING NO.	5154047290-22

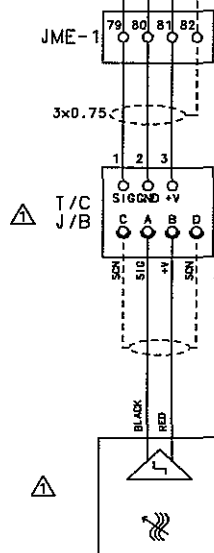
12. 05 T/C TACHO SYSTEM

STX Heavy Industries Co., Ltd

E.C.R

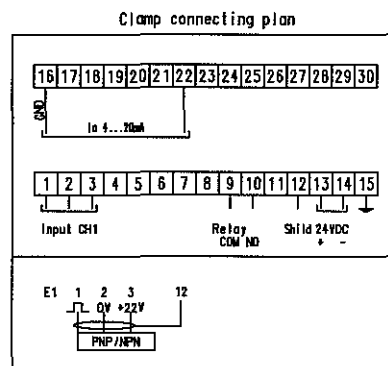
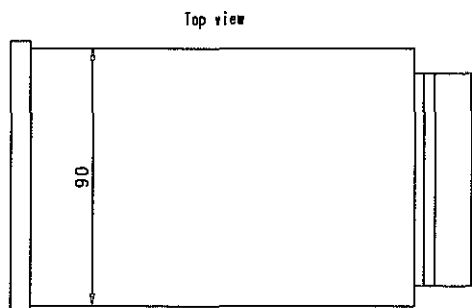
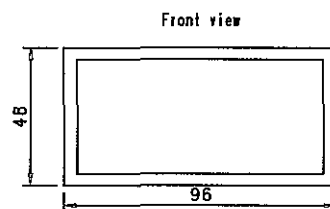
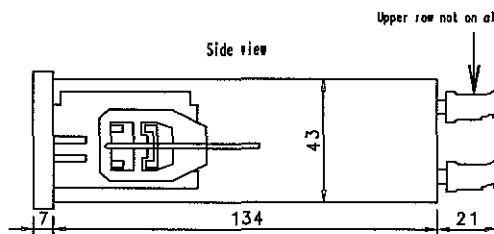


E/R

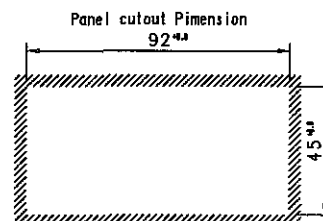
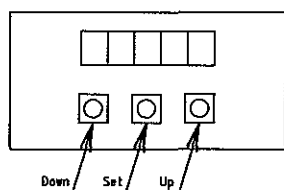


PICK UP FOR
TCA TURBO CHARGER

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD WEIGHT kg	DWN	100624	T.Y. JEON	SIZE A4	NAME TCA T/C TACHO SYSTEM (OUTPUT SIGNAL : AMS)
			CHKD	100624	M.H. CHOI		
			APPD	100624	T.B. KIM		
SCALE N/S		MODEL ALL	 STX Engine Co., Ltd.			DRAWING NO. 5154044810	



Instrument shown without front panel



Technical Data

Standard data :

Power supply
Power consumption
Overload safety switch
Operating temperature
Storage temperature
Relative humidity
Protection acc. to DIN 40050

Connection
Cross section for connection wires
Housing dimensions
Weight
EMC Requirements

Display :
Indicators
Segment height
Colour
Number of digits

Speed probe input
Input frequency range
Connectable probes
Failure : Display
Relay switch point

Relay output
Relay type
Max. switch load
Max. switch voltage
Max. switch current
Max. operating current
Min. switch cycles
Switching indication

Analogous output :
current output
Range
Max. burden
Resolution
Linearity failure
Max. reaction time

Digital tachometer

24VDC+30%/-25%
max. 5W
1AT
0°C...80°C
-10°C...80°C
< 90%, non condensing
IP64 Frontcover against housing
IP00 Housing and Connection
pluggable screw connection row with 15 poles
0,2-2,5mm² (IEC 947-7-1)
96x48mm, L=134mm
ca. 400g
EMV : EN 50 081, EN 50 082-2
NSR : EN 60 010

7-segment, LED
14,2mm
red
5

0...50kHz, lowest measurable frequency : 1Hz
-3-wire speed probe (NPN/PNP) with auxiliary voltage +22VDC, max. 30mA
≤ ± 0,05% of measured value ± 1 digit
≤ ± 0,3% of final value

NC or NO function free programmable

60W, 125VA
220VDC 250VAC
2A DC/AC
3A DC/AC
10
through red 3mm LED

4...20mA
5000SU
10 Bit D/A transformation
≤ ± 20μA
300ms

REV	LOC	DATE	MADE	APPD	NOTE
MATERIAL	CAD		DWN	080918	T.Y.JEON
	WEIGHT kg		CHKD	080918	D.H.LEE
			APPD	080918	T.B.KIM
SCALE	N/S	MODEL	ALL	DRAWING NO. 5154042440-01	

STX STX Engine Co., Ltd.

Pxx No.	Setting of MAN B&W Diesel AG	choice *2)	Designation/Information
P00	09876	09876	Password
P03	see table	00001-00255	Pulses per revolution
P16	see table	00000-99999	Upper switch point relay
P17	see table	00000-99999	Lower switch point relay
P24	00001	00000 00001	Effect direction relay : NO Effect direction relay : NC
P32	00000	00000-99999	Indication respectively process value at Input frequency = 0 Hz
P33	see table	00000-99999	Indication respectively process value at max. Input frequency see P34
P34	see table	00000-99999	Max. Speed Attention : Value must be identical with P33
P38	00000	00000	Selection of current output at 4~20mA
P51	00500	0100-9999	Display refreshment in ms

Use : Connection on transmitter with amplifier DSH1840.01 SHV

or transmitter with amplifier in terminal box DSH1840.03.SHZ

Signal output : Square voltage signal out of push-pull circuitry,

DC coupled with supply ; (0V = signal common),

Load current max : 25mA

Output voltage HI : > Supply voltage -2, 5V at current of 15mA

Output voltage LO : < 1, 5V at current of 15mA

*1) Programming of the relay

Entry possible at stillstanding or running turbine press by

SET for 5s minimum, display changes from operational modes to

P16 upper set point Relay : setting possible between 0...99999

Press SET 2s it shows :

P17 lower set point Relay : setting possible between 0...99999

(Select value min. 150 1/min lower than P16)

Press SET 2s it shows :

P24 Relay action set 0 = NO function : Set to 1 = NC function.

Press SET indicator returns to normal operational mode.

It display shows at normal operation E05 the pick-up delivers no Signal

Parameter setting at site

*2) Standard adjustment

Entry possible at stillstanding by simultaneous pressing Down and Up

Attention : unauthorized changing of the parameter is allowed.

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD WEIGHT kg	DWN	080918	T.Y. JEON	SIZE A4	NAME TCA T/C DIGITAL TACHOMETER FOR ECR
			CHKD	080918	D.H. LEE		
			APPD	080918	T.B. KIM		
SCALE N/S		MODEL ALL	 STX Engine Co., Ltd.			DRAWING NO.	5154042440-02

Code No. see data plate	Assignment				Analogous output (2)			Relay adjustable (1)		
	Compressor wheel		Turbocharger		Speed range final value (1=20mA) (1/min)/(RPM)	P03 pulses per revolution	P33 P34 max. speed	P16 Upper set point 0.97x n 11 max. (1/min)/(RPM)	P17 Lower set point 0.90x n 11 max. (1/min)/(RPM)	P24 Action set
	Geometry of blades	Number of main/ intermediate blades	type	Speed range (1/min)/(RPM)						
X11.56254-0137	7.2	11/11	NA70/T.9...	12200	15000	00011	15000	11830	10980	NC
X11.56254-0138	7.2	11/11	NA57/T.9...	15000	20000	00011	20000	14550	13500	NC
X11.56254-0139	7.2	11/11	NA48/S....	18600	20000	00011	20000	18040	16740	NC
X11.56254-0140	7.2	11/11	NA40/S....	22400	25000	00011	25000	21720	20160	NC
X11.56254-0141	7.2	11/11	NA34/S....	26300	30000	00011	30000	25510	23670	NC
X11.56254-0142	8.1	17	NA34/S....	26300	30000	00017	30000	25510	23670	NC
X11.56254-0143	8.3	9/9	NA70/T.9...	12200	15000	00009	15000	11830	10980	NC
X11.56254-0144	8.3	9/9	NA57/T.9...	15000	20000	00009	20000	14550	13500	NC
X11.56254-0145	8.3	9/9	NA48/S....	18600	20000	00009	20000	18040	16740	NC
X11.56254-0147	8.3	9/9	NA40/S....	22400	25000	00009	25000	21720	20160	NC
X11.56254-0148	8.3	9/9	NA34/S....	26300	30000	00009	30000	25510	23670	NC
X11.56254-0074	A1000	11/11	TCA44-4	23800	30000	00011	30000	23080	21420	NC
X11.56254-0075	A1000	11/11	TCA55-4	21000	25000	00011	25000	20370	18900	NC
X11.56254-0076	A1000	11/11	TCA66-4	18000	20000	00011	20000	17460	16200	NC
X11.56254-0077	A1000	11/11	TCA77-4	14900	20000	00011	20000	14450	13410	NC
X11.56254-0078	A1000	11/11	TCA88-4	12500	15000	00011	15000	12125	11250	NC
X11.56254-0150	A1045/A3045	11/11	TCA44-4	23800	30000	00011	30000	23080	21420	NC
X11.56254-0151	A1045/A3045	11/11	TCA55-4	19400	25000	00011	25000	19250	17860	NC
X11.56254-0152	A1045/A3045	11/11	TCA66-4	16900	20000	00011	20000	16200	15030	NC
X11.56254-0153	A1045/A3045	11/11	TCA77-4	14200	15000	00011	15000	14480	12640	NC
X11.56254-0154	A1045/A3045	11/11	TCA88-4	12000	15000	00011	15000	11440	10620	NC
X11.56254-0177	A2045	8/8	TCA44-4	23800	30000	00008	30000	23080	21420	NC
X11.56254-0183	A2045	8/8	TCA55-4	19400	25000	00008	25000	19250	17860	NC
X11.56254-0188	A2045	8/8	TCA66-4	16900	20000	00008	20000	16200	15030	NC
X11.56254-0193	A2045	8/8	TCA77-4	14200	15000	00008	15000	14480	12640	NC
X11.56254-0198	A2045	8/8	TCA88-4	12000	15000	00008	15000	11440	10620	NC
X11.56254-0079	A1000	11/11	TCA44-4	23000	30000	00011	30000	22310	20700	NC
X11.56254-0080	A1000	11/11	TCA55-4	19800	25000	00011	25000	19700	17820	NC
X11.56254-0081	A1000	11/11	TCA66-4	16700	20000	00011	20000	16600	15030	NC
X11.56254-0082	A1000	11/11	TCA77-4	14100	20000	00011	20000	14000	12690	NC
X11.56254-0083	A1000	11/11	TCA88-4	11900	15000	00011	15000	11540	10710	NC
X11.56254-0174	A1045/A3045	11/11	TCA44-4	23000	30000	00011	30000	22310	20700	NC
X11.56254-0180	A1045/A3045	11/11	TCA55-4	19400	25000	00011	25000	18820	17460	NC
X11.56254-0186	A1045/A3045	11/11	TCA66-4	16300	20000	00011	20000	15810	14670	NC
X11.56254-0191	A1045/A3045	11/11	TCA77-4	13700	20000	00011	20000	13290	12330	NC
X11.56254-0196	A1045/A3045	11/11	TCA88-4	11600	15000	00011	15000	11250	10440	NC
X11.56254-0178	A2045	8/8	TCA44-4	23000	30000	00008	30000	22310	20700	NC
X11.56254-0184	A2045	8/8	TCA55-4	19400	25000	00008	25000	18820	17460	NC
X11.56254-0189	A2045	8/8	TCA66-4	16300	20000	00008	20000	15810	14670	NC
X11.56254-0194	A2045	8/8	TCA77-4	13700	20000	00008	20000	13290	12330	NC
X11.56254-0199	A2045	8/8	TCA88-4	11600	15000	00008	15000	11250	10440	NC
X11.56254-0173	A1000	11/11	TCA44-2	23500	30000	00011	30000	22790	21150	NC
X11.56254-0179	A1000	11/11	TCA55-2	19850	25000	00011	25000	19250	17860	NC
X11.56254-0185	A1000	11/11	TCA66-2	16700	20000	00011	20000	16200	15030	NC
X11.56254-0190	A1000	11/11	TCA77-2	14050	20000	00011	20000	13630	12640	NC
X11.56254-0195	A1000	11/11	TCA88-2	11800	15000	00011	15000	11440	10620	NC
X11.56254-0175	A1045/A3045	11/11	TCA44-2	22500	25000	00011	25000	21820	20250	NC
X11.56254-0155	A1045/A3045	11/11	TCA55-2	19000	25000	00011	25000	18430	17100	NC
X11.56254-0156	A1045/A3045	11/11	TCA66-2	16000	20000	00011	20000	15520	14400	NC
X11.56254-0157	A1045/A3045	11/11	TCA77-2	13500	15000	00011	15000	13090	12150	NC
X11.56254-0158	A1045/A3045	11/11	TCA88-2	11350	15000	00011	15000	11010	10210	NC
X11.56254-0176	A2045	8/8	TCA44-2	22500	25000	00008	25000	21820	20500	NC
X11.56254-0181	A2045	8/8	TCA55-2	19000	25000	00008	25000	18430	17100	NC
X11.56254-0187	A2045	8/8	TCA66-2	16000	20000	00008	20000	15520	14400	NC
X11.56254-0192	A2045	8/8	TCA77-2	13500	15000	00008	15000	13090	12150	NC
X11.56254-0197	A2045	8/8	TCA88-2	11350	15000	00008	15000	10960	10170	NC
X11.56254-0172	-		NR29/S	31300	35000	00001	35000	30360	28170	NC

→

REV	LOC	DATE	MADE	APPD	NOTE
MATERIAL					
SCALE	N/S	MODEL	ALL		
		WEIGHT	kg		
		CAD			
		CHKD	080918	T. Y. JEON	SIZE
		APPD	080918	D. H. LEE	A4
				T. B. KIM	NAME
					TCA T/C DIGITAL TACHOMETER FOR ECR
					DRAWING NO. 5154042440-03

A

B

C

D

E

4

1

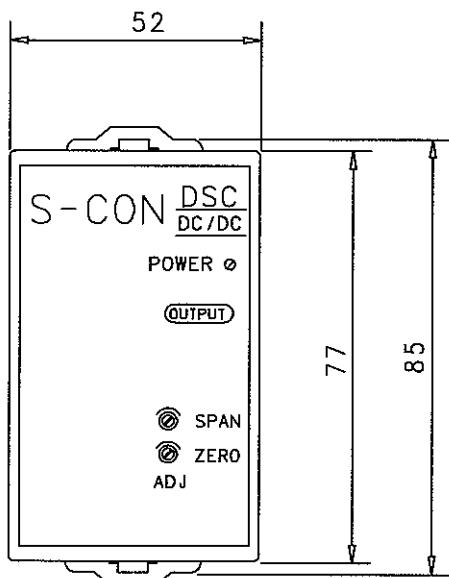
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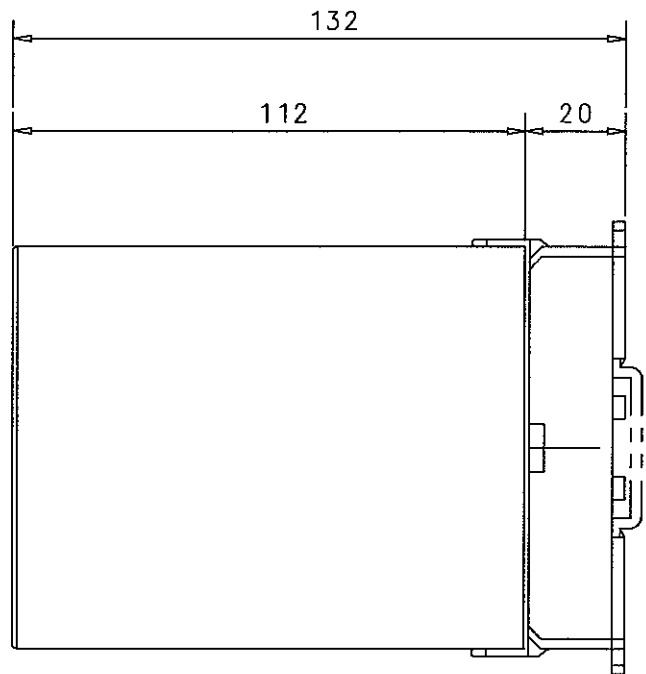
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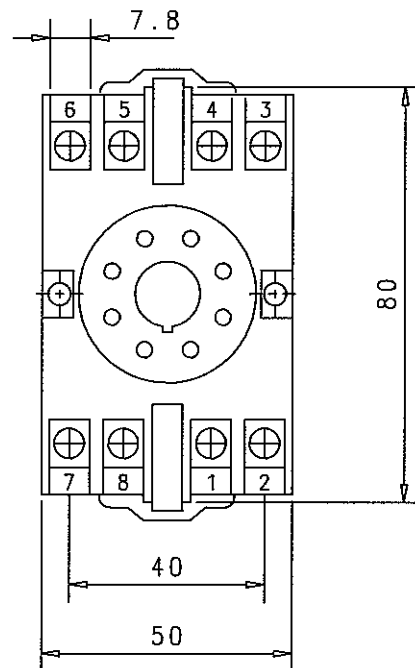
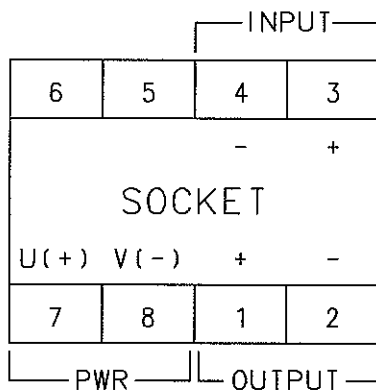
STX Engine Co., Ltd.



FRONT VIEW



SIDE VIEW



SPECIFICATIONS

1. INPUT SIGNAL : 4 ~ 20mA DC
(IMPEDANCE : 250Ω FIXED)
2. OUTPUT : 4 ~ 20mA DCx1EA
(LOAD RESISTANCE : 0~500Ω)
3. POWER SUPPLY : DC 24V (3Watt.)
4. I/O ISOLATION : INPUT : ISOLATION
OUTPUT : GALVANIC ISOLATION
5. ACCURACY : ±0.1% I/O
6. MAKER : S-CON
7. TYPE : DSC-9H8NY-GAL

SOCKET

REV	LOC	DATE	MADE	APPD	NOTE
MATERIAL		DWN		081105	T.Y. JEON
		CHKD		081105	D.H. LEE
		APPD		081105	T.B. KIM
SCALE		MODEL		DRAWING NO.	
N/S		ALL		5154044760	

STX STX Engine Co., Ltd.

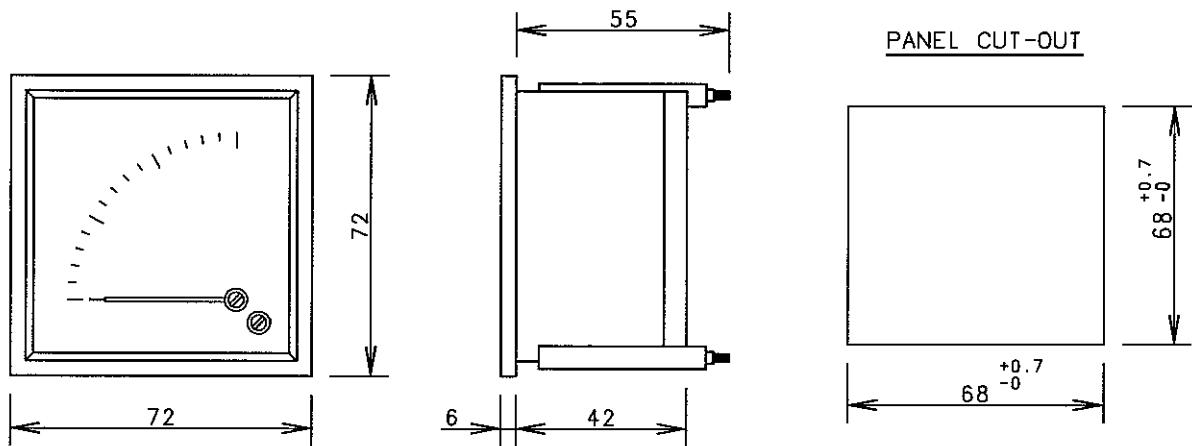
**12. 06 E.C.R PRESSURE
INDICATORS**

STX Heavy Industries Co., Ltd

E.C.R PRESS. INDICATORS

FOR STX/NNC
(S-4006/12/14/29)

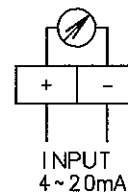
NO.	DESCRIPTION	Q'TY	RANGE	UNIT	WORKING RANGE (GREEN MARK)	PART NO.
1	SCAVENGE AIR INLET PRESS.	1	0 ~ 4	Kg/cm ² +Mpa	-	2902004370G-2A
2	T/C L.O INLET PRESS.	1	0 ~ 6	Kg/cm ² +Mpa	1.5 ~ 2.2	2902004370G-3A
3	MAIN L.O & P.C.O INLET PRESS.	1	0 ~ 6	Kg/cm ² +Mpa	2.0 ~ 2.3	2902004370G-3D
4	JACKET C.W INLET PRESS.	1	0 ~ 6	Kg/cm ² +Mpa	3.5 ~ 4.5	2902004370G-3G
5	AIR COOLER C.W INLET PRESS.	1	0 ~ 6	Kg/cm ² +Mpa	2.0 ~ 4.5	2902004370G-3H
6	CONTROL AIR INLET PRESS.	1	0 ~ 10	Kg/cm ² +Mpa	6.5 ~ 7.5	2902004370G-4A
7	F.O INLET PRESS.	1	0 ~ 16	Kg/cm ² +Mpa	7.0 ~ 8.0	2902004370G-5A
8	STARTING AIR INLET PRESS.	1	0 ~ 40	Kg/cm ² +Mpa	18.0 ~ 30.0	2902004370G-6A



CHARACTERISTICS

- ELECTRICAL : AS PER NFC 42100 (CLASS 1.5)
- DEFLECTION : STANDARD 90 DEGREES
- INPUT : 4~20mA
- CUT-OUT : 68 x 68mm
- WEIGHT : 0.28kg
- MAKER : CMR

CONNECTION

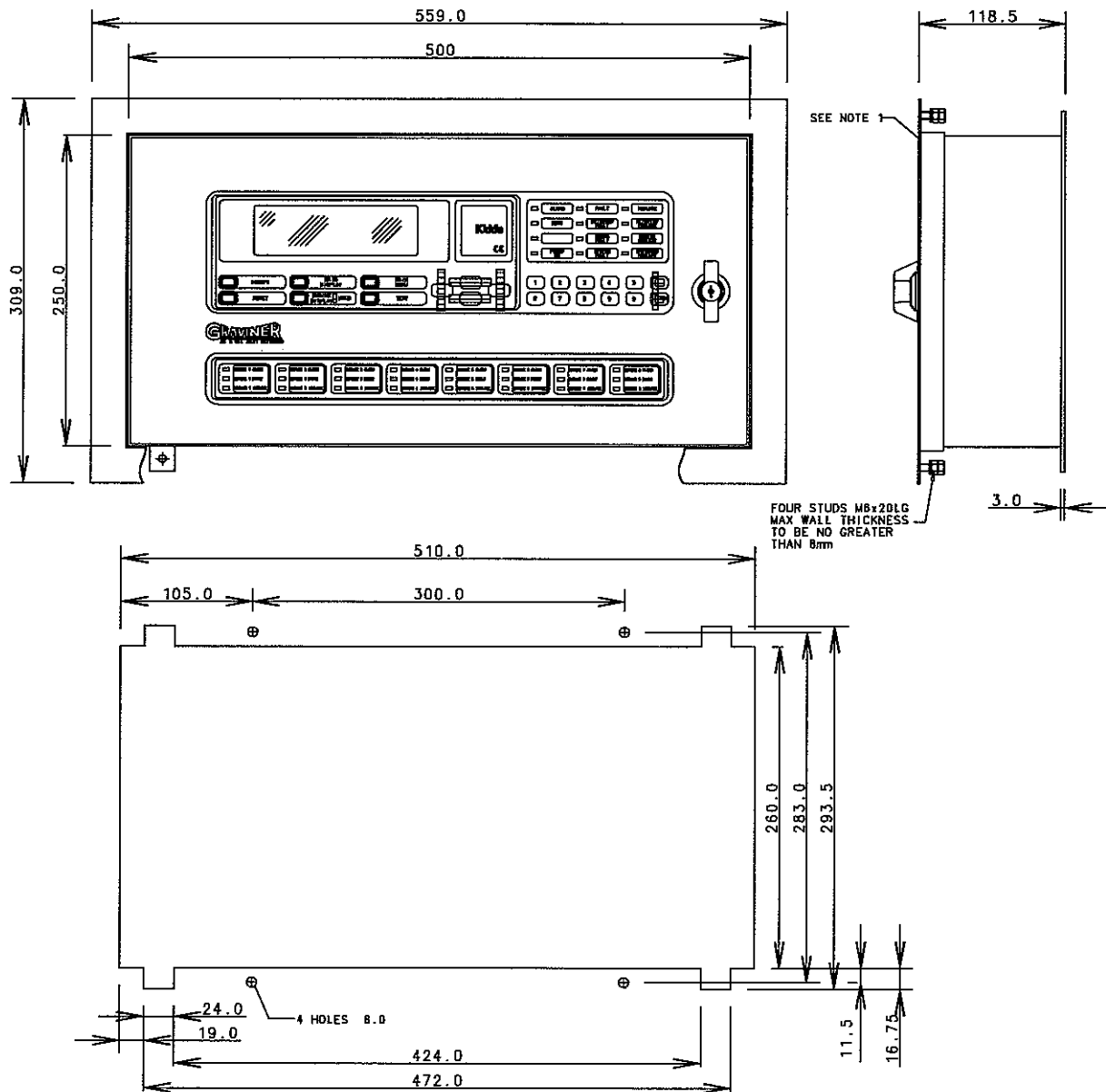


NO.	DESCRIPTION	PART NO.	TYPE	RANGE (Kg/Cm ² + Mpa)	GREEN MARK
1	SCAV. AIR	2902004370G-2A	Q72-90-4P-4-20	0~4Kg/Cm ² + 0~0.4Mpa	GREEN MARK 미표기
2	T/C L.O	2902004370G-3A	Q72-90-6P-4-20	0~6Kg/Cm ² + 0~0.6Mpa	1.5~2.2Kg/Cm ²
3	MAIN L.O & P.C.O	2902004370G-3B	Q72-90-6P-4-20	0~6Kg/Cm ² + 0~0.6Mpa	1.8~2.1Kg/Cm ²
4	MAIN L.O & P.C.O	2902004370G-3C	Q72-90-6P-4-20	0~6Kg/Cm ² + 0~0.6Mpa	1.9~2.2Kg/Cm ²
5	MAIN L.O & P.C.O	2902004370G-3D	Q72-90-6P-4-20	0~6Kg/Cm ² + 0~0.6Mpa	2.0~2.3Kg/Cm ²
6	MAIN L.O & P.C.O	2902004370G-3E	Q72-90-6P-4-20	0~6Kg/Cm ² + 0~0.6Mpa	2.1~2.4Kg/Cm ²
7	MAIN L.O & P.C.O	2902004370G-3F	Q72-90-6P-4-20	0~6Kg/Cm ² + 0~0.6Mpa	2.2~2.5Kg/Cm ²
8	JACKET C.W	2902004370G-3G	Q72-90-6P-4-20	0~6Kg/Cm ² + 0~0.6Mpa	3.5~4.5Kg/Cm ²
9	AIR COOLER C.W	2902004370G-3H	Q72-90-6P-4-20	0~6Kg/Cm ² + 0~0.6Mpa	2.0~4.5Kg/Cm ²
10	AIR COOLER C.W	2902004370G-3I	Q72-90-6P-4-20	0~6Kg/Cm ² + 0~0.6Mpa	2.0~2.5Kg/Cm ²
11	CONTROL AIR	2902004370G-4A	Q72-90-10P-4-20	0~10Kg/Cm ² + 0~1.0Mpa	6.5~7.5Kg/Cm ²
12	SAFETY AIR	2902004370G-4B	Q72-90-10P-4-20	0~10Kg/Cm ² + 0~1.0Mpa	6.5~7.5Kg/Cm ²
13	EXH. SPRING AIR	2902004370G-4C	Q72-90-10P-4-20	0~10Kg/Cm ² + 0~1.0Mpa	6.5~7.5Kg/Cm ²
14	SPEED SET AIR	2902004370G-4D	Q72-90-10P-4-20	0~10Kg/Cm ² + 0~1.0Mpa	0.5~5.0Kg/Cm ²
15	F.O	2902004370G-5A	Q72-90-16P-4-20	0~16Kg/Cm ² + 0~1.6Mpa	7~8Kg/Cm ²
16	STARTING AIR	2902004370G-6A	Q72-90-40P-4-20	0~40Kg/Cm ² + 0~4.0Mpa	18~30Kg/Cm ²

REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL		 CAD	DWN	080416	T.Y. JEON	SIZE NAME A4 PRESSURE INDICATOR (TYPE:DUAL RANGE Q72)
			CHKD	080416	D.H. LEE	
			APPD	080416	T.B.KIM	
		WEIGHT		kg		
SCALE		MODEL	DRAWING NO.			
N/S		ALL	STX STX Engine Co., Ltd.			2902004370G

**12. 07 OMD REMOTE
CONTROL PANEL**

STX Heavy Industries Co., Ltd



NOTES

1. IN ORDER FOR THE DOOR TO OPEN FULLY, AN ADDITIONAL 500mm MUST BE ALLOWED BEYOND THIS POINT.
2. SEE TABLE FOR THE NUMBER OF JUNCTION BOXES, RELAY OUTPUTS (ALARM, FAULT AND ENGINE SLOWDOWN) AND CABLE GLANDS/LOCKNUTS.
3. ON A ONE ENGINE SYSTEM ONE ALARM, ONE FAULT AND ONE ENGINE SLOW DOWN RELAY IS REQUIRED. IN CASES WHERE THERE ARE MORE ENGINES AN ADDITIONAL ENGINE SLOWDOWN RELAY WILL BE NEEDED FOR EACH ADDITIONAL ENGINE.
4. PROVISION HAS BEEN MADE FOR 15 CABLE GLAND ENTRIES AT THE TOP OF THE ENCLOSURE AND 15 AT THE BOTTOM. THESE ARE TO BE USED FOR POWER AND COMMUNICATIONS WIRING. THE SUGGESTED MANUFACTURER OF THE CABLE GLAND AND LOCKNUT IS JACOB. THE JACOB PART NUMBERS FOR THESE PARTS ARE 50620MREMYL AND 50220MPOT RESPECTIVELY. THESE EXTERNAL REFERENCE PART NUMBERS ARE FOR REFERENCE ONLY AND NEED NOT BE PURCHASED FROM THE SUGGESTED MANUFACTURER.

SYSTEM INSTALLATION REQUIREMENTS TABLE

NUMBER OF ENGINES	NUMBER OF JUNCTION BOXES	NUMBER OF RELAY OUTPUTS (SEE NOTE 3)	TOTAL NUMBER OF M20 CABLE GLANDS/LOCKNUTS REQUIRED.
1	1	3	6
2	2	4	9
3	3	5	12
4	4	6	15
5	5	7	18
6	6	8	21
7	7	9	24
8	8	10	27

REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL		 CAD WEIGHT kg	DWN	080401	T.Y. JEON	SIZE NAME MONITORING PANEL FOR O.M.D
			CHKD	080401	D.H. LEE	
			APPD	080401	T.B.KIM	
SCALE N/S		MODEL ALL	 STX Engine Co.,Ltd.			DRAWING NO. 5154012090

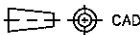
12. 08 BEARING CONDITION MONITORING SYSTEM

STX Heavy Industries Co., Ltd



AMOT XTS-W+ Bearing Condition Monitoring System

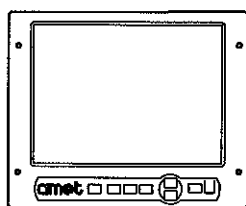
	STOCK CODE	STOCK CODE	QUANTITY	REMARKS
1	CYMAZ0009	CABLE 6PR 22AWG	100M	CONNECTION SPU TO IF
2	S4600-BK1A-001	BRACKET ASSEMBLY(2 PROX)	6	MOUNTED ON M/E
3	S4600-BK1B-001	BRACKET ASSEMBLY(1 RTD, 1 PROX)	2	MOUNTED ON M/E
4	S4600-SPU+003	SIGNAL PROCESSING UNIT ASSEMBLY	1	MOUNTED ON M/E
5	S4600-TE+002	XTSW+ TERMINAL ENCLOSURE ASSEMBLY	8	MOUNTED ON M/E
6	S4600-IF+002	INTERFACE UNIT ASSEMBLY	1	INSTALLED IN E.C.C
7	S4600-PC+002(DC)	TOUCHSCREEN PC ASSEMBLY & LEADS & USB MEMORY STICK	1	FITTED ON E.C.C
8		XTSW+ MANUAL	1	SUPPLY TO SHIPYARD
9	S4600-CK+003	XTSW+ COMMISSIONING KIT		SUPPLY TO SHIPYARD
	90423L020	FUSE 20 MM GLASS 2A	10	
	90423L050	FUSE 20 MM GLASS 5A	10	
	BTXRS0002	BATTERY 9V PP3	1	
	CEXSZ0003+	10M 3C+SCN PLG-FREE SENSOR CBL	1	
	IKXFA0004	PENDANT CHAIN 710MM	1	
	IMXCP0002	XTSW SENSOR CALIBRATOR	1	
	LSXHZ0159	XTSW SENSOR SETTING SHIM 3.0MM	1	
10	S4600-EIK6CYL+002	XTSW+ ELECTRICAL INSTALLATION KIT		MOUNTED ON M/E
	901B5	CABLE SLEEVNG 20MM LONG	100	
	CAXCZ0007	BOOTLACE FERRULE RED 1MM	500	
	CAXPZ0001	CABLE TIE 12" BLACK	200	
	CAXPZ0002	CABLE TIE 8" BLACK	200	
	CAXRS0031	2MM GREEN SILICON SLEEVE	10M	
	CAXSA0001	CABLE SADDLE	80	
	CYMSZ0001+	0.5MM 3C SCRN CABLE	300M	
	LSXEE0001	XTSW C1-C38 PRE-PRINTED LABELS	4	
11	S4600-SP+004	XTSW+ SPARES KIT		SUPPLY TO SHIPYARD
	90423L020	FUSE 20 MM GLASS 2A	20	
	90423L050	FUSE 20 MM GLASS 5A	20	
	CEXSZ0003+	10M 3C+SCN PLG-FREE SENSOR CBL	1	
	GLBFA0001	M16 x 1 GLAND NCK PLT	1	
	GLXLP0001	M20 x 1.5 N/P GLAND	1	
	GLXLP0003	M20 x 1.5 LONG GLAND	1	
	GLXLP0004	2 WAY GLAND INSERT	2	
	S4600-RTD-003	XTSW RTD ASSEMBLY 10CM	1	
	S4600-TE+002	XTSW+ TERMINAL BOX ASSEMBLY	1	
	SPXSZ0003	PROXIMITY SENSOR(REMOVE LABELS/NUTS & WASHERS)	2	

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL	 CAD		DWN	101111	T.Y.JEON	SIZE	NAME
	WEIGHT kg		CHKD	101111	D.H.LEE	A4	SCOPE OF SUPPLY FOR BEARING WEAR MONITORING SYSTEM
			APPD	101111	T.B.KIM		
SCALE	N/S	MODEL	7S50MC-C			DRAWING NO. 5154035070-01	

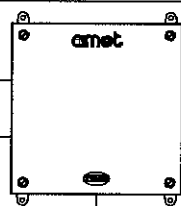
stx

ENGINE CONTROL ROOM

TOUCH SCREEN PC



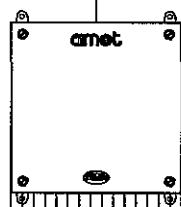
INTERFACE UNIT



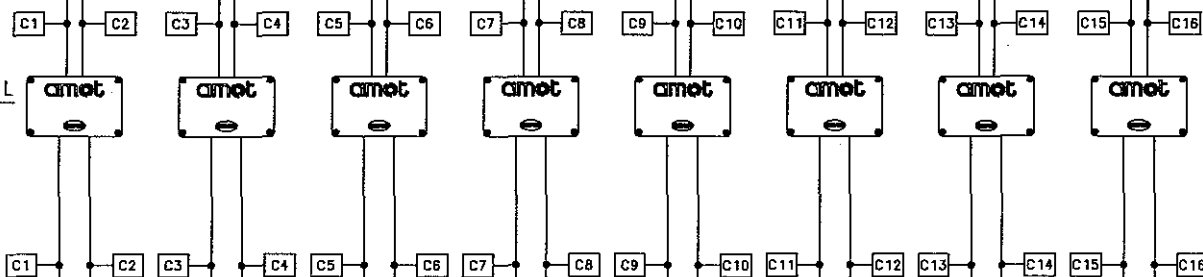
INTERNAL FROM E.C.C (DC24V)
INTERNAL TO AMS

ON ENGINE

SPU



TERMINAL BOX



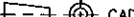
RTD1 1A 1B 2A 2B 3A 3B 4A 4B 5A 5B 6A 6B 7A 7B RTD2

PROXIMITY & TEMPERATURE SENSOR

NOTE

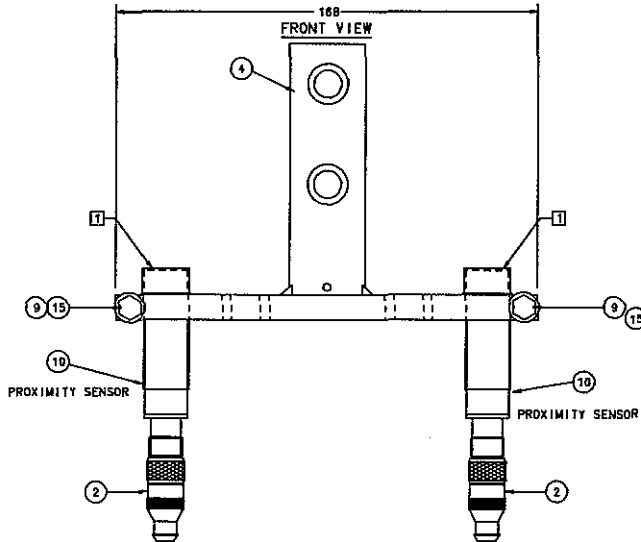
----- : TO BE PROVIDED BY SHIPYARD
----- : TO BE PROVIDED BY AMOT

IN ENGINE

REV	LOC	DATE	MADE	APPD	NOTE				
MATERIAL		 CAD	DWN	101111	T.Y. JEON	SIZE A4	NAME CONFIGURATION FOR BEARING WEAR MONITORING SYSTEM		
			CHKD	101111	D.H. LEE				
		WEIGHT		kg	APPD			101111	T.B. KIM
SCALE N/S		MODEL 7S50MC-C				DRAWING NO. 5154035070-02			

ⓘ ENSURE PROXIMITY SENSOR IS FITTED WITH PLASTIC CAP FOR SHIPPING.
MUST BE REMOVED WHEN INSTALLED IN ENGINE.

BRACKET SHOWN IS WITH 2 PROXIMITY SENSORS FITTED (S4600-BK-1A-001)



S4600-BK-1A-001

ITEM	QTY	DESCRIPTION	PART NUMBER
1			
2	2	10m, 3C+SC PLUG	CEXSZ0003
3			
4	1	PAINTED BRACKET	MBXAN0001
5			
6			
7			
8			
9	2	M6x25 HEX HEAD BOLT	SWXZZ0013
10	2	PROXIMITY SENSOR	SPXSZ0003
11			
12			
13			
14			
15	☆2	M6 PLAIN WASHER	90900W001
16	2	M10x45 HEX HEAD BOLT	BAXSR0005
17	2	M10 PLAIN WASHER	BAXSR0002
18	2	M10 SPLIT WASHER	BAXSR0003
19			

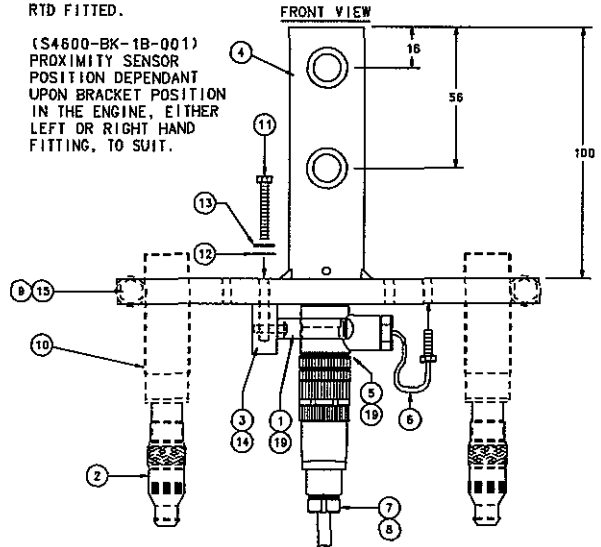
☆ KANBAN ITEMS

□ PARTS TO BE SUPPLIED LOOSE IN KIT

PROXIMITY SENSOR NOT TO BE FITTED
UNTIL AFTER INSTALLATION OF BRACKET

BRACKET SHOWN IS WITH 1
PROXIMITY SENSOR & 1
RTD FITTED.

(S4600-BK-1B-001)
PROXIMITY SENSOR
POSITION DEPENDANT
UPON BRACKET POSITION
IN THE ENGINE, EITHER
LEFT OR RIGHT HAND
FITTING, TO SUIT.



S4600-BK-1B-001

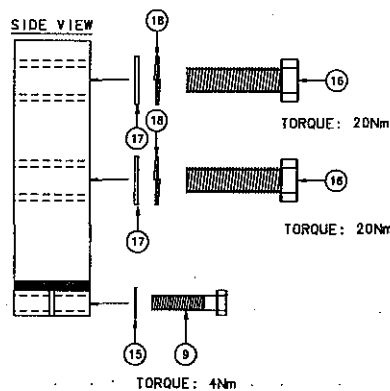
ITEM	QTY	DESCRIPTION	PART NUMBER
1	1	5/8" SPRING CLIP (BLK)	CAXRS0054
2	1	10m, 3C+SC PLUG	CEXSZ0003
3	1	RTD RETAINER BLOCK	MBXAZ0001
4	1	PAINTED BRACKET	MBXAN0001
5	1	4 WAY M12 PLUG	CSPRS0059
6	1	3W RTD 100, 10cm LEAD	RTXTC0022
7	1	4 PIN STRAIGHT SOCKET	CSPRS0069
8	10m	RTD EXTENSION CABLE	CYMSZ0001
9	1	M6x25 HEX HEAD BOLT	SWXZZ0013
10	1	PROXIMITY SENSOR	SPXSZ0003
11	☆2	M4x25 PAN HEAD(POS) SCREW	90910P100
12	☆2	M4 PLAIN WASHER	90910W001
13	☆2	M4 SHAKEPROOF WASHER	90910W002
14	☆1	M3x12 PAN HEAD(POS) SCREW	90920P050
15	☆1	M6 PLAIN WASHER	90900W001
16	2	M10x45 HEX HEAD BOLT	BAXSR0005
17	2	M10 PLAIN WASHER	BAXSR0002
18	2	M10 SPLIT WASHER	BAXSR0003
19	1	8" TIE WRAP	CAXPZ0002

☆ KANBAN ITEMS

□ PARTS TO BE SUPPLIED LOOSE IN KIT

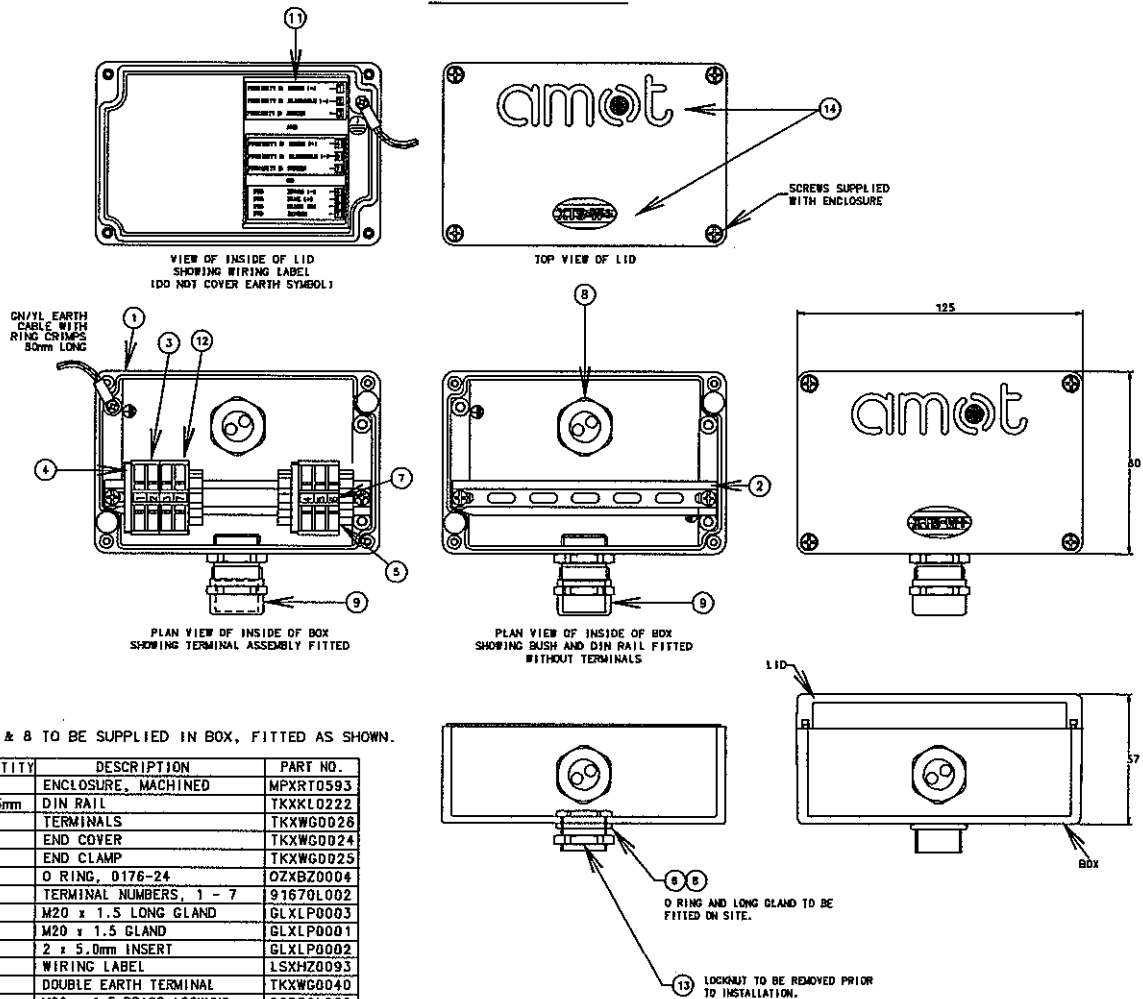
○ ITEMS 1 & 5 TO BE TIE WRAPPED TOGETHER

BRACKET SHOWN WITHOUT SENSORS FITTED

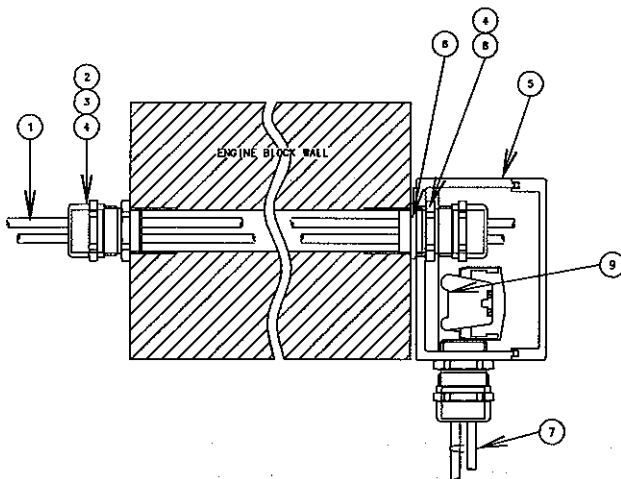


REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD WEIGHT kg	DWN	101111	T.Y. JEON	SIZE A4	NAME BRACKET FOR BEARING WEAR MONITORING SYSTEM
			CHKD	101111	D.H. LEE		
			APPD	101111	T.B.KIM		
SCALE N/S		MODEL 7S50MC-C				DRAWING NO. 5154035070-03	

ENCLOSURE



SCHEMATIC VIEW OF

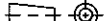


ITEM NO.	QUANTITY	DESCRIPTION	PART NO.
1	A/R	CABLE PLUG 3C+SCN	CEXSZ0003
2	1	M20 x 1.5 GLAND	GLXLP0001
3	1	O RING	PART OF ITEM 2
4	2	2 x 5.0mm INSERT	GLXLP0002
5	1	ENCLOSURE ASSEMBLY	S4600-TE+002
6	1	O RING, D176-24	OZXBZ0004
7	A/R	CABLE 3C+SCN	CYMSZ0001
8	1	M20 x 1.5 LONG GLAND	GLXLP0003

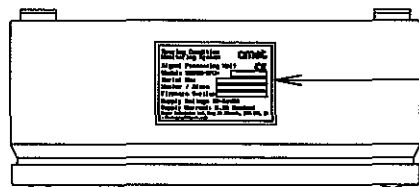
NOTE: 1 - CABLE CUT AND TERMINATED TO SUIT.

DIRECTIONS FOR INSTALLATION:

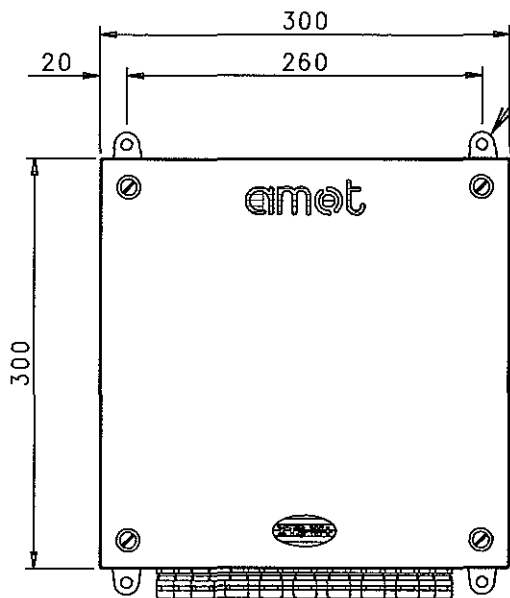
- 1) FIT ENCLOSURE TO ENGINE USING BUSH.
- 2) FIT DIN RAIL INTO ENCLOSURE WITH SCREWS PROVIDED.
- 3) FIT TERMINAL BLOCK ASSEMBLY ONTO DIN RAIL WITH END CLAMPS
- 4) CONNECT WIRES
- 5) FIT EARTH WIRE, 1.5mm² +100 TO LID & BOX
- 6) FIT ENCLOSED LID.

REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL		 CAD WEIGHT kg	DWN	101111	T.Y. JEON	SIZE NAME A4 JUNCTION BOX FOR BEARING WEAR MONITORING SYSTEM
			CHKD	101111	D.H. LEE	
			APPD	101111	T.B. KIM	
SCALE N/S		MODEL 7S50MC-C				DRAWING NO. 5154035070-04

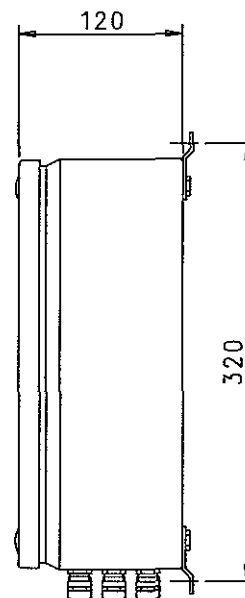
2



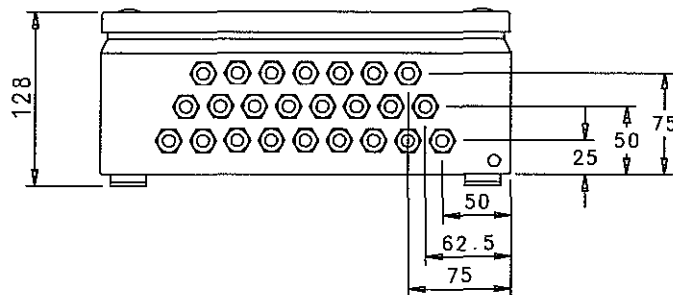
Bearing Condition Monitoring System	amet
Signal Processing Unit	CE
Model: S4800-SPU	
Serial No:	
Master / Slave	
Firmware Version	
Supply Voltage: 15-24VDC	
Supply Current: 0.5A Nominal	
Power Electronics Ltd, Bury St Edmunds, IP23 9EZ, UK	
salesupport@amet.com	



MOUNTING BRACKETS X4.
TYPE KL1594.000
Ø8mm FIXING DOWN
HOLE.



24 OFF CABLE ENTRIES,
EITHER M16x1.5 CABLE GLAND



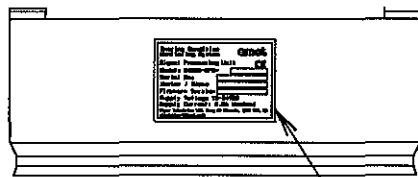
DESCRIPTION

THE SIGNAL PROCESSING UNIT IS MOUNTED DIRECTLY TO THE ENGINE. THE SENSORS ARE CONNECTED FROM THE ENGINE MOUNTED TERMINAL BOXES AND ROUTED TO THE SIGNAL PROCESSING UNIT. THE SIGNAL PROCESSING UNIT SHOULD BE MOUNTED WITH THE CABLE GLANDS FACING DOWN. THE IP RATING AND EMC COMPLIANCE ARE DEPENDANT ON THE LID BEING CORRECTLY FITTED.

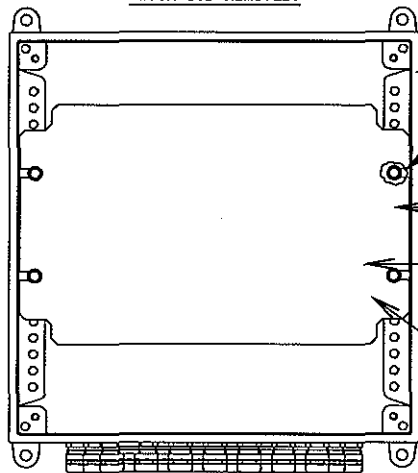
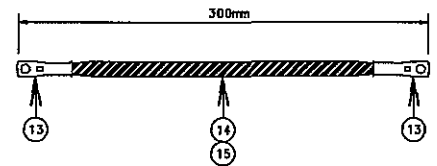
SPECIFICATION

MATERIAL:	FABRICATED STEEL BOX 1.25mm THK. INTERNAL MOUNTING PLATE 2.5mm THK.
FINISH:	EXTERNAL BOX - RAL 7035 POWDER COATED. INTERNAL BOX - ALUMINUM ZINC COATING. MOUNTING PLATE - ZINC PLATED
IP RATING:	IP55
VIBRATION:	MEETS GL & LLOYDS ON ENGINE SPEC 5-25Hz ±1.6mm DISPLACEMENT 25-100Hz ±4g ACCELERATION
OPERATING TEMPERATURE:	0°C TO +70°C
HUMIDITY:	20% TO 95% NON CONDENSING
STORAGE TEMPERATURE:	-20°C TO +70°C
INPUT VOLTAGE:	15 - 24VDC
CURRENT CONSUMPTION:	⊙ 18V 500mA NOMINAL
PROXIMITY SENSORS:	CONNECTIONS 14 (MAX PER SPU)
ANALOGUE LOOPS:	SELF POWERED 4-20mA
RTD CONNECTIONS:	4 X 3-WIRE PT100
BOX WEIGHT:	5Kgs

REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL	CAD WEIGHT kg	DWN	101111	T.Y. JEON	SIZE A4	NAME SPU FOR BEARING WEAR MONITORING SYSTEM (1/2)
		CHKD	101111	D.H. LEE		
		APPD	101111	T.B. KIM		
SCALE N/S	MODEL 7S50MC-C				DRAWING NO.	5154035070-05

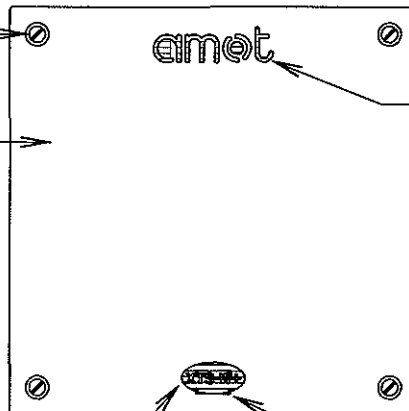


TOP VIEW
(WITH LID REMOVED)

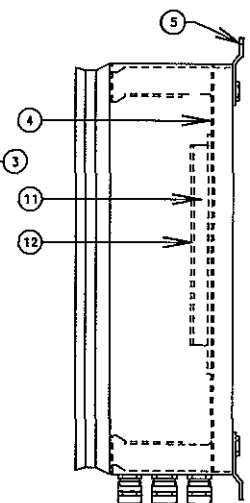


FRONT VIEW
(WITH LID REMOVED)

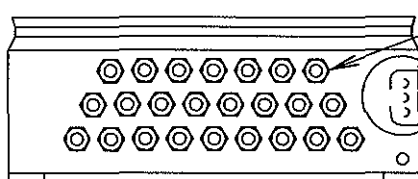
PLATE FIXING LOCATION
PLATE TO BE FIXED TO
6th HOLE DOWN FROM TOP OF
ENCLOSURE.



VIEW OF LID

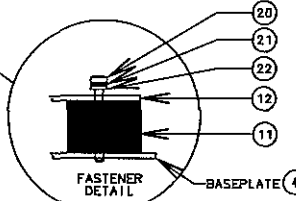


SIDE VIEW
(WITH LID REMOVED)



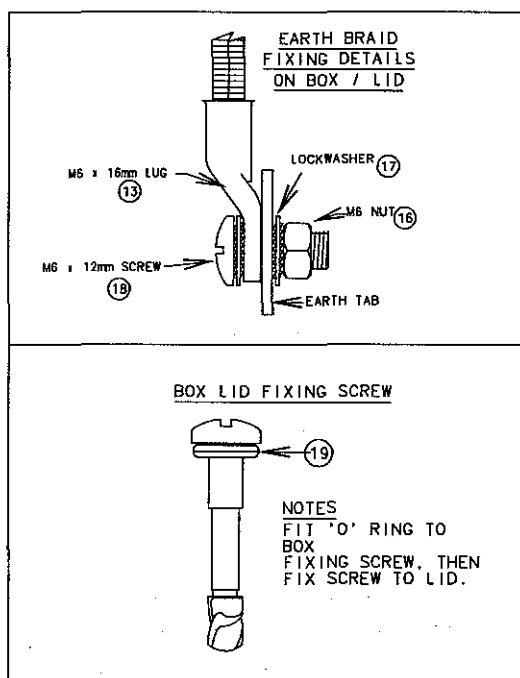
BASE VIEW
(WITH LID REMOVED)

SCRAP VIEW SHOWING EARTH
TAB LOCATED AT THE BOTTOM
OF THE BOX.

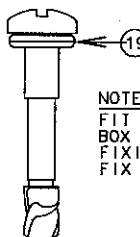


FASTENER DETAIL

SCRAP VIEW SHOWING TOP VIEW
OF EARTH TAB LOCATED AT THE
BOTTOM OF BOX LID.



BOX LID FIXING SCREW



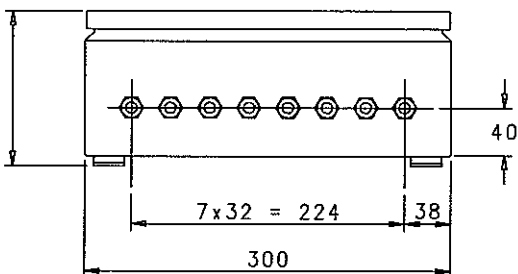
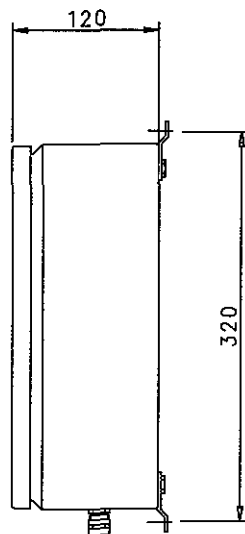
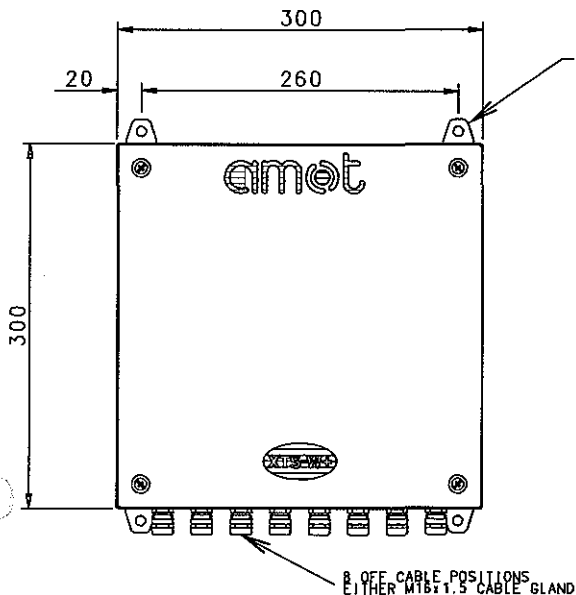
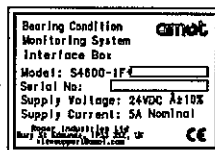
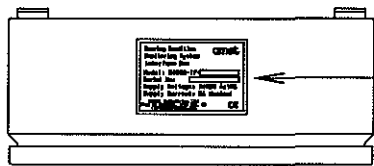
NOTES
FIT 'O' RING TO
BOX
FIXING SCREW, THEN
FIX SCREW TO LID.

NOTES

1. FIT EARTH BRAID BETWEEN ENCLOSURE AND ENCLOSURE LID.
2. APPLY LOCTITE TO ALL SCREWS APART FROM MEMORY CARD MOUNTING.
3. ITEM 9 MOUNTED ON UNDERSIDE OF ENCLOSURE LID, ITEM 10.
4. ITEM 6 OPERATING LEVERS ARE LOCATED IN A BAG WITHIN THE ENCLOSURE. THESE SHOULD BE USED FOR TERMINATING ALL CONNECTIONS.

ITEM NO.	QUANTITY	PART NO.	DESCRIPTION
1	24	GLBFA0001	M16x1. CABLE GLAND, NICKEL
2	24	GLBFA0002	M16. LOCKNUT, NICKEL
3	1	LSXHZ0112	VINYL AMOT LOGO
4	1	MPXDA0184	SPU BACKPLATE
5	4	MPXRT0180	WALL BRACKETS, STAINLESS STEEL
6	3	TKXWG0039	OPERATING LEVER
7	1	LSXHZ0195	VINYL XTS-W* LOGO LABEL
8	1	LSXHZ0223	XTS-W* SPU NAMEPLATE LABEL
9	1	LSXHZ0224	XTS-W* SPU INTERCONNECTION LABEL
10	1	MPXDA0241	SPU ENCLOSURE
11	1	PAAPZ0003	DRILLED PLATE, BE-4 FOR PCB
12	1	PBXGZ0007	XTS-W* SPU PCB ASSEMBLY
13	2	CAXLW0004	M6x16 EARTH LUG
14	300mm	EAXRS0001	EARTH BRAID
15	A/R	CAXRS0004	HEATSHRINK, GREEN/YELLOW
16	2	90900ND01	NUT, M6
17	4	90900W001	LOCKWASHER, M6
18	2	90900P050	SCREW, M6x12, STAINLESS STEEL
19	4	1392	'O' RING 3/16" x 5/16" x 1/16" Buna N
20	13	90920P100	SCREW, M3x25, POSI, STAINLESS STEEL
21	13	90920W002	M3 SHAKEPROOF WASHER, STAINLESS STEEL
22	13	90920W001	M3 FLAT WASHER, STAINLESS STEEL

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	101111	T.Y. JEON	SIZE	NAME SPU FOR BEARING WEAR MONITORING SYSTEM(2/2)
			CHKD	101111	D.H. LEE	A4	
			APPD	101111	T.B. KIM		
SCALE N/S		MODEL 7S50MC-C				DRAWING NO.	5154035070-06



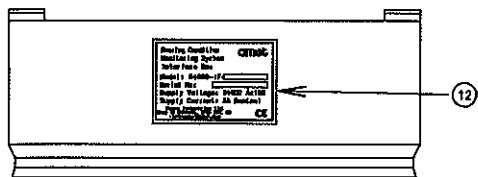
SPECIFICATION

MATERIAL: FABRICATED STEEL BOX 1.25mm THK.
FINISH: INTERNAL MOUNTING PLATE 2.5mm THK.
 EXTERNAL BOX - RAL 7035 POWDER COATED.
 INTERNAL BOX - ALUMINIUM ZINC COATED.
 MOUNTING PLATE - ZINC PLATED.
IP RATING: IP55
VIBRATION: MEETS GL & LLOYDS OFF ENGINE SPEC
 5-13.2Hz ±1.0mm DISPLACEMENT
 13.2-100Hz ±0.7g ACCELERATION
OPERATING TEMPERATURE: 0°C TO +40°C
HUMIDITY: 20% TO 95% NON CONDENSING
STORAGE TEMPERATURE: -5°C TO +45°C
INPUT VOLTAGE RANGE: 24VDC ±10%
CURRENT CONSUMPTION: 5A NOMINAL
DIGITAL OUTPUTS: HEALTHY = 2off CHANGEOVER CONTACTS NORMALLY ENERGISED.
 ALARM = 2off CHANGEOVER CONTACTS NORMALLY ENERGISED.
 SLOWDOWN = 2off CHANGEOVER CONTACTS NORMALLY DE-ENERGISED.
DIGITAL OUTPUT RATING: 30VDC, 5A RESISTIVE
COMMUNICATIONS: RS485, 2 WIRE
CONNECTIONS: RS485-A:D9S, PIN 2(PC RS485/USB CONVERTER)
 RS485-B:D9S, PIN 1(PC RS485/USB CONVERTER)
 38400,N,8,1
RS485 DATA FORMAT: 24VDC(OR 18VDC)
SPU BOX SUPPLY: 24VDC ONLY
PC SUPPLY OUTPUT: 20mm GLASS,
 F1=T2A PCB SUPPLIES
 F2=T5A PC SUPPLY
 F3=T2A SPU SUPPLY
 F4=T2A SPU SUPPLY 0V
FUSES: 5Kg
BOX WEIGHT: 5Kg

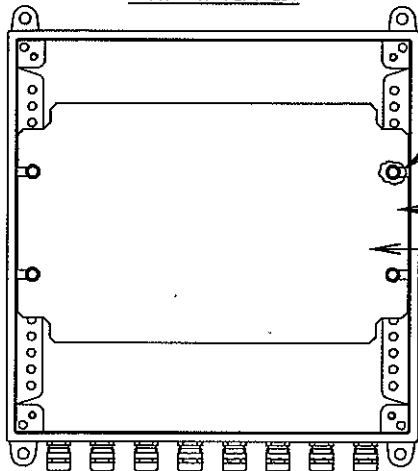
DESCRIPTION

THE INTERFACE UNIT IS DESIGNED TO BE MOUNTED IN CLOSE PROXIMITY TO THE XTS-W PC IN THE SHIPS CONTROL ROOM. THE PC SHOULD BE INSTALLED WITHIN 3M OF THE INTERFACE UNIT - CONSTRAINED BY THE MAXIMUM LENGTH OF PC CABLES. THE BOX SHOULD BE MOUNTED WITH THE CABLE GLANDS FACING DOWN. THE IP RATING AND THE EMC COMPLIANCE ARE DEPENDANT ON THE LID BEING CORRECTLY FITTED.

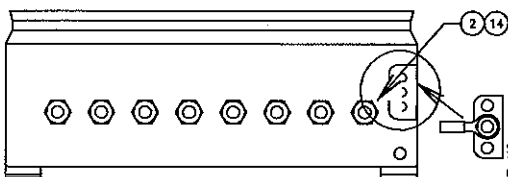
REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD WEIGHT kg	DWN	101111	T.Y. JEON	SIZE A4	NAME INTERFACE UNIT FOR BEARING WEAR MONITORING SYSTEM(1/2)
			CHKD	101111	D.H. LEE		
			APPD	101111	T.B.KIM		
SCALE N/S		MODEL 7S50MC-C				DRAWING NO. 5154035070-07	



TOP VIEW
(WITH LID REMOVED)

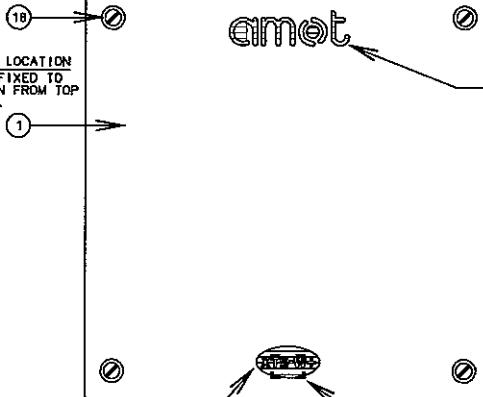


FRONT VIEW
(WITH LID REMOVED)

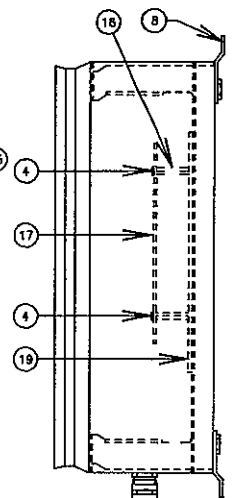


BASE VIEW
(WITH LID REMOVED)

PLATE FIXING LOCATION
PLATE TO BE FIXED TO
6TH HOLE DOWN FROM TOP
OF ENCLOSURE.

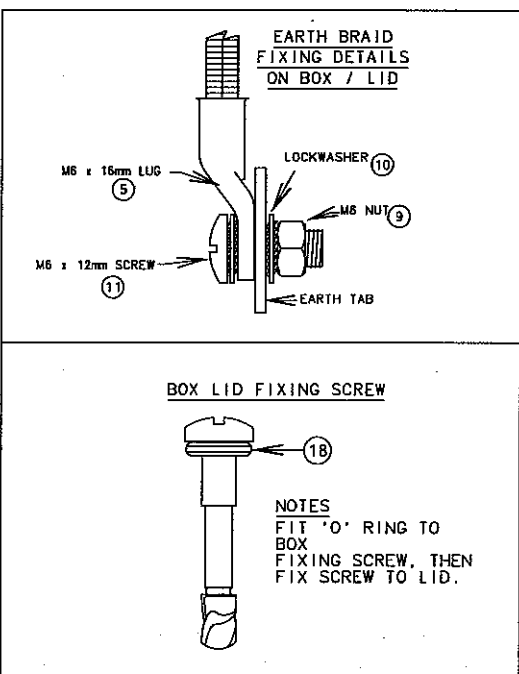
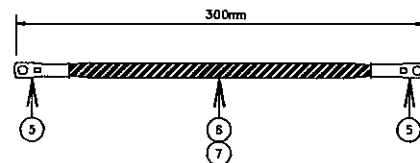


VIEW OF LID



SIDE VIEW
(WITH LID REMOVED)

SCRAP VIEW SHOWING TOP VIEW
OF EARTH TAB LOCATED AT THE
BOTTOM OF BOX LID.



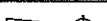
BOX LID FIXING SCREW

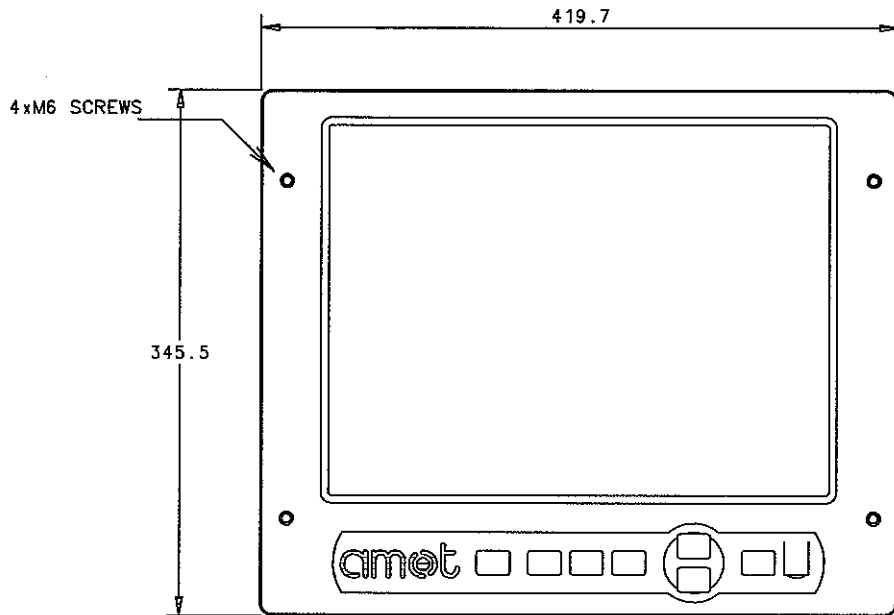
NOTES
FIT 'O' RING TO
BOX
FIXING SCREW, THEN
FIX SCREW TO LID.

NOTES

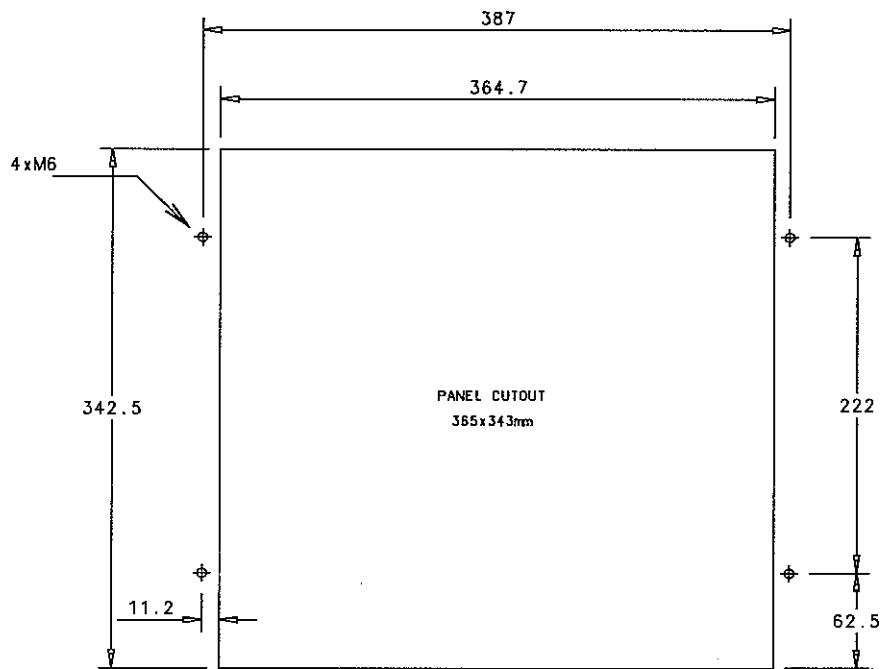
1. FIT EARTH BRAID BETWEEN ENCLOSURE AND ENCLOSURE LID.
2. APPLY LOCTITE TO ALL SCREWS.
3. ITEM 13 MOUNTED ON UNDERSIDE OF ENCLOSURE LID, ITEM 1.

ITEM NO.	QUANTITY	PART NO.	DESCRIPTION
1	1	MPXDA0242	MACHINED ENCLOSURE
2	8	GLBFA0001	M16 CABLE GLAND
3	1	LSXHZ0195	VINYL XTS-W+ LOGO LABEL
4	12	90920P025	M3x6 SCREW
5	2	CAXUW0004	M16 x M6 LUG
6	300mm	EAXRS0001	EARTH BRAID
7	A/R	CAXRS0004	HEATSHRINK, GREEN/YELLOW
8	4	MPXRT0180	WALL BRACKETS
9	2	90900N001	NUT, M6
10	4	90900W002	LOCKWASHER, M6
11	2	90900P050	SCREW, M6x12
12	1	LSXHZ0227	NAMEPLATE
13	1	LSXHZ0228	WIRING / CONNECTIONS LABEL
14	8	GLBFA0002	M16 LOCKNUT FOR CABLE GLAND
15	1	LSXHZ0112	VINYL AMOT LOGO
16	12	48419L015	PILLAR, M3x15, MALE/FEMALE, BRASS
17	1	PBXPR0004	IF BOX PCB REV.2
18	4	1392	'O' RING 3/16" x 5/16" x 1/16" Buna N
19	1	MPXDA0184	BACK PLATE

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	101111	T.Y. JEON	SIZE A4	NAME INTERFACE UNIT FOR BEARING WEAR MONITORING SYSTEM(2/2)
			CHKD	101111	D.H. LEE		
		WEIGHT kg	APPD	101111	T.B. KIM		
SCALE N/S	MODEL 7S50MC-C				DRAWING NO. 5154035070-08		



FRONT VIEW



PANEL CUTOUT
365x343mm

NOTES

THE PC IS DESIGNED FOR MOUNTING EITHER VERTICALLY OR AT A CONVENIENT ANGLE FOR VIEWING. IT SHOULD BE MOUNTED IN THE CONTROL ROOM OR SIMILAR.

MASS: 6kg

IP RATING: IP65 (FRONT PANEL)

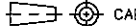
SHOCK: 10G PEAK ACCELERATION (11 M.SEC DURATION)

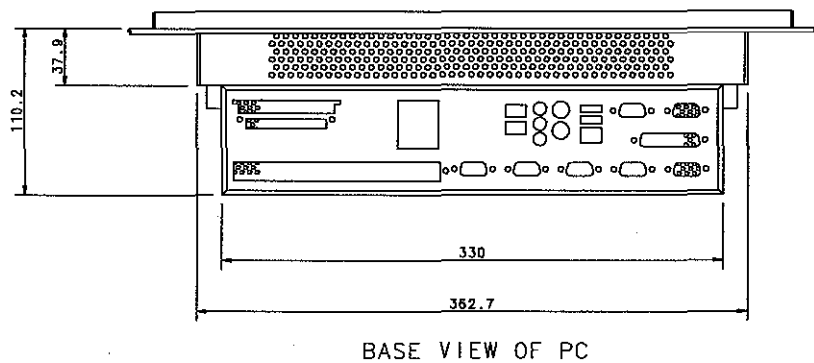
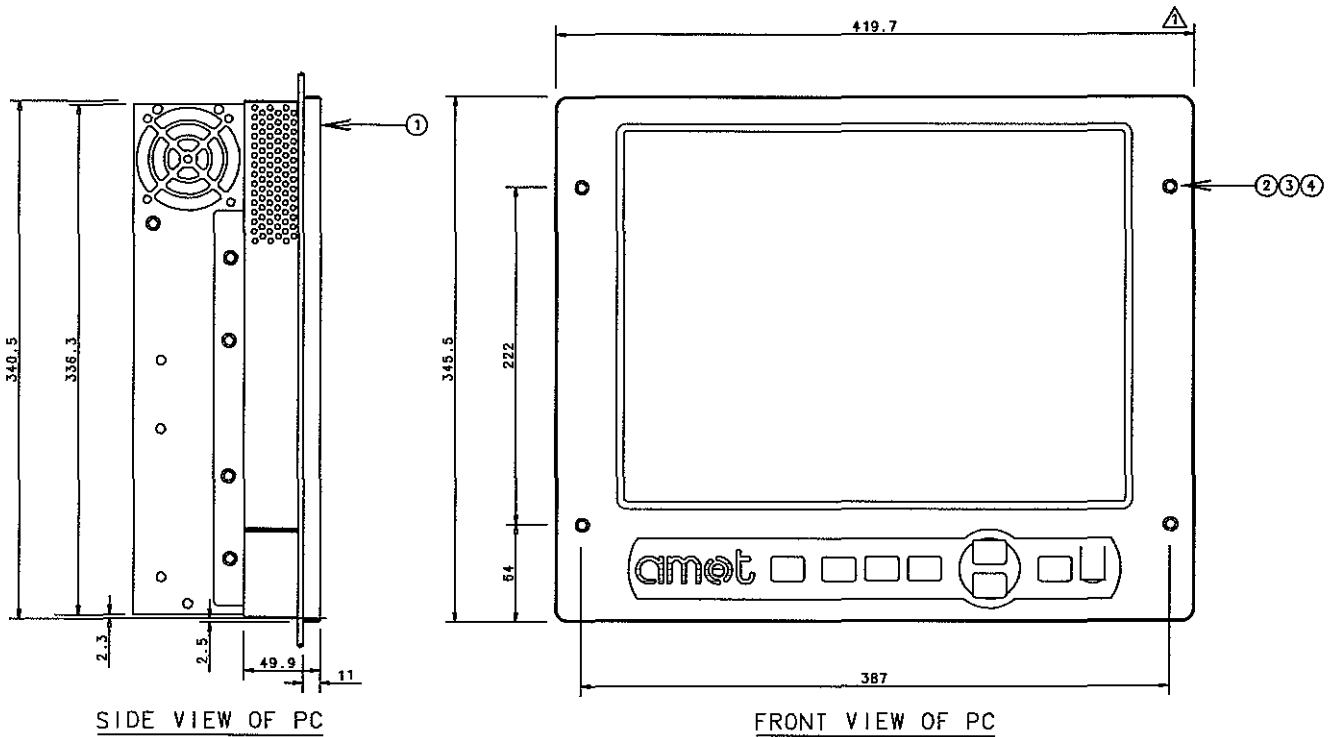
OPERATING TEMPERATURE: 0°C TO 45°C

INPUT VOLTAGE RANGE: 90VAC TO 264VAC

HUMIDITY: 10% TO 95% NON-CONDENSING AT 45°C.

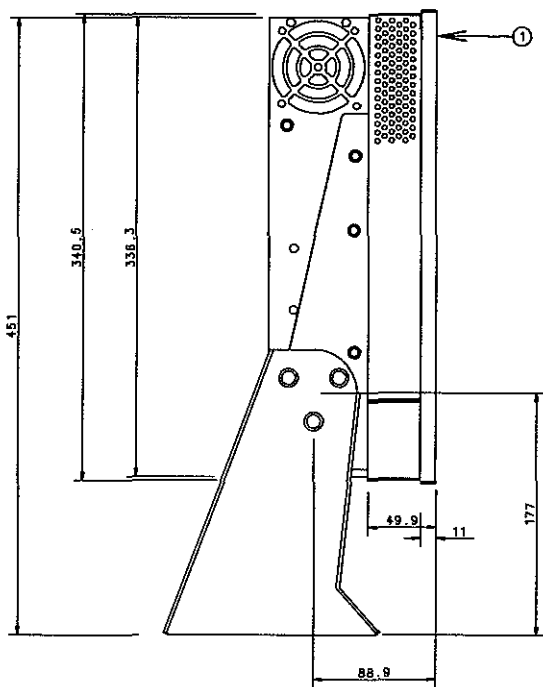
TOUCH SCREEN: CAPACITIVE, 15" 1024x768 SVGA.

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	101111	T.Y.JEON	SIZE A4	NAME TOUCH SCREEN PC FOR BEARING WEAR MONITORING SYSTEM(1/2)
			CHKD	101111	D.H.LEE		
		WEIGHT kg	APPD	101111	T.B.KIM		
SCALE N/S		MODEL 7S50MC-C				DRAWING NO. 5154035070-09	

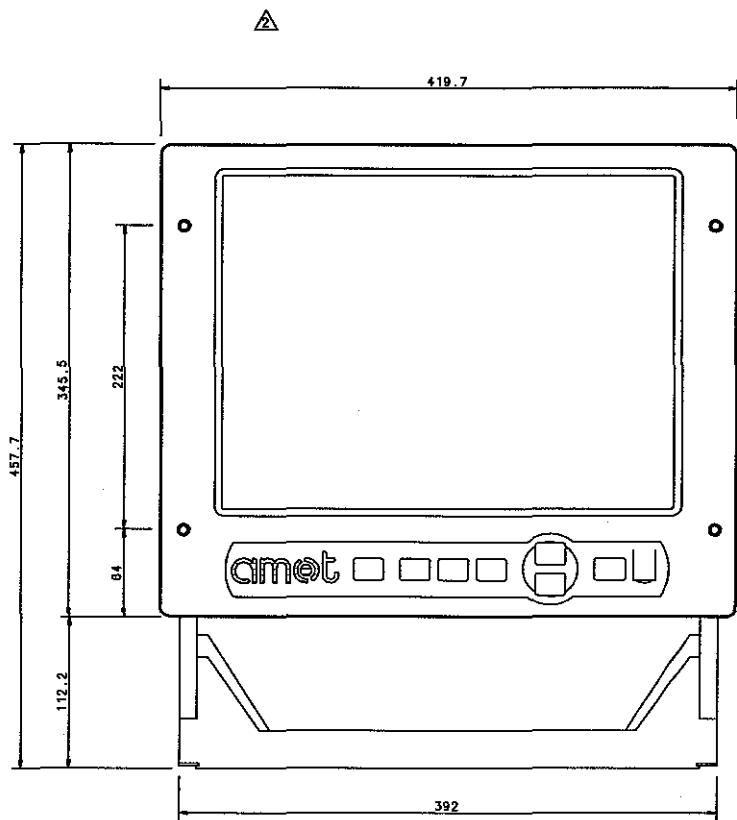


ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	PCXIS0001	TOUCH SCREEN PC KIT
2	4		M6 BOLT
3	4		M6 WASHER
4	4		M6 NUT

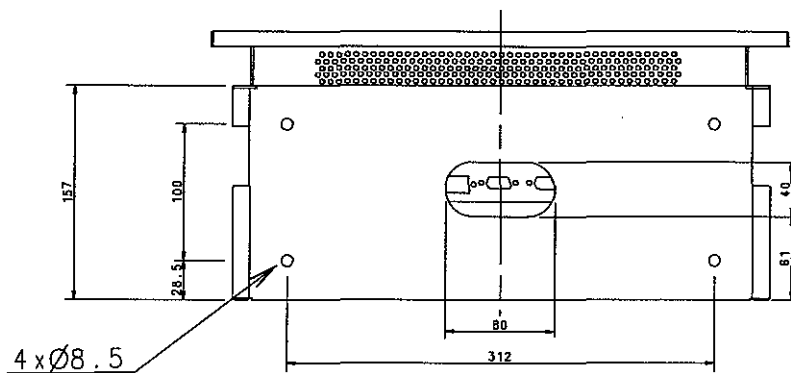
REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD WEIGHT kg	DWN	091119	T.Y. JEON	SIZE A4	NAME TOUCH SCREEN PC FOR BEARING WEAR MONITORING SYSTEM(2/2)
			CHKD	091119	M.H. CHOI		
			APPD	091119	T.B. KIM		
SCALE N/S		MODEL 7S50MC-C				DRAWING NO. 5154035070-10	



SIDE VIEW OF PC



FRONT VIEW OF PC



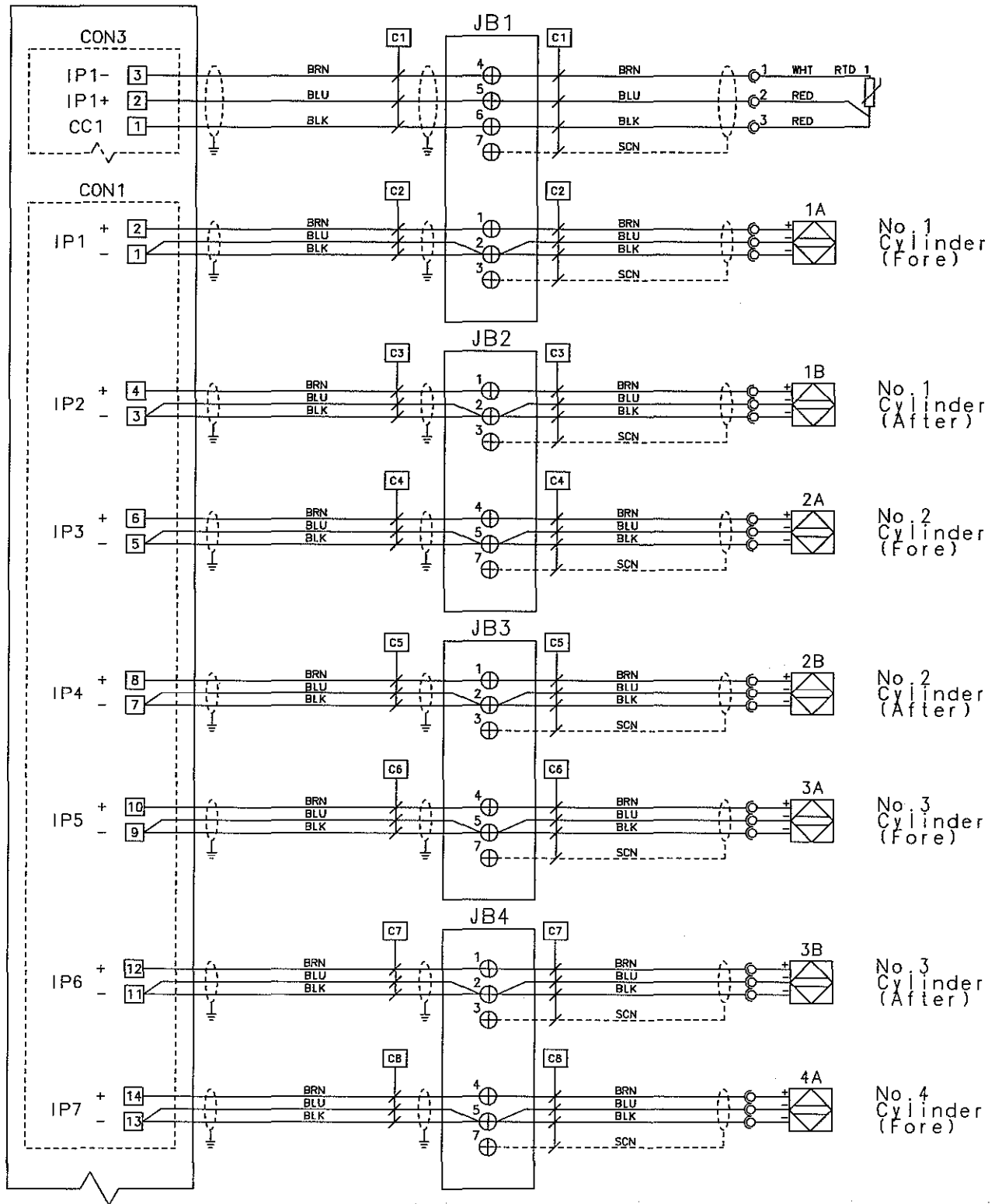
BASE VIEW OF PC

ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	PCXIS0001	TOUCH SCREEN PC KIT

REV	LOC	DATE	MADE	APPD	NOTE
MATERIAL	CAD		DWN	101111	T.Y. JEON
WEIGHT	kg		CHKD	101111	D.H. LEE
			APPD	101111	T.B. KIM
SCALE	N/S	MODEL	7S50MC-C		DRAWING NO.
				5154035070-11	

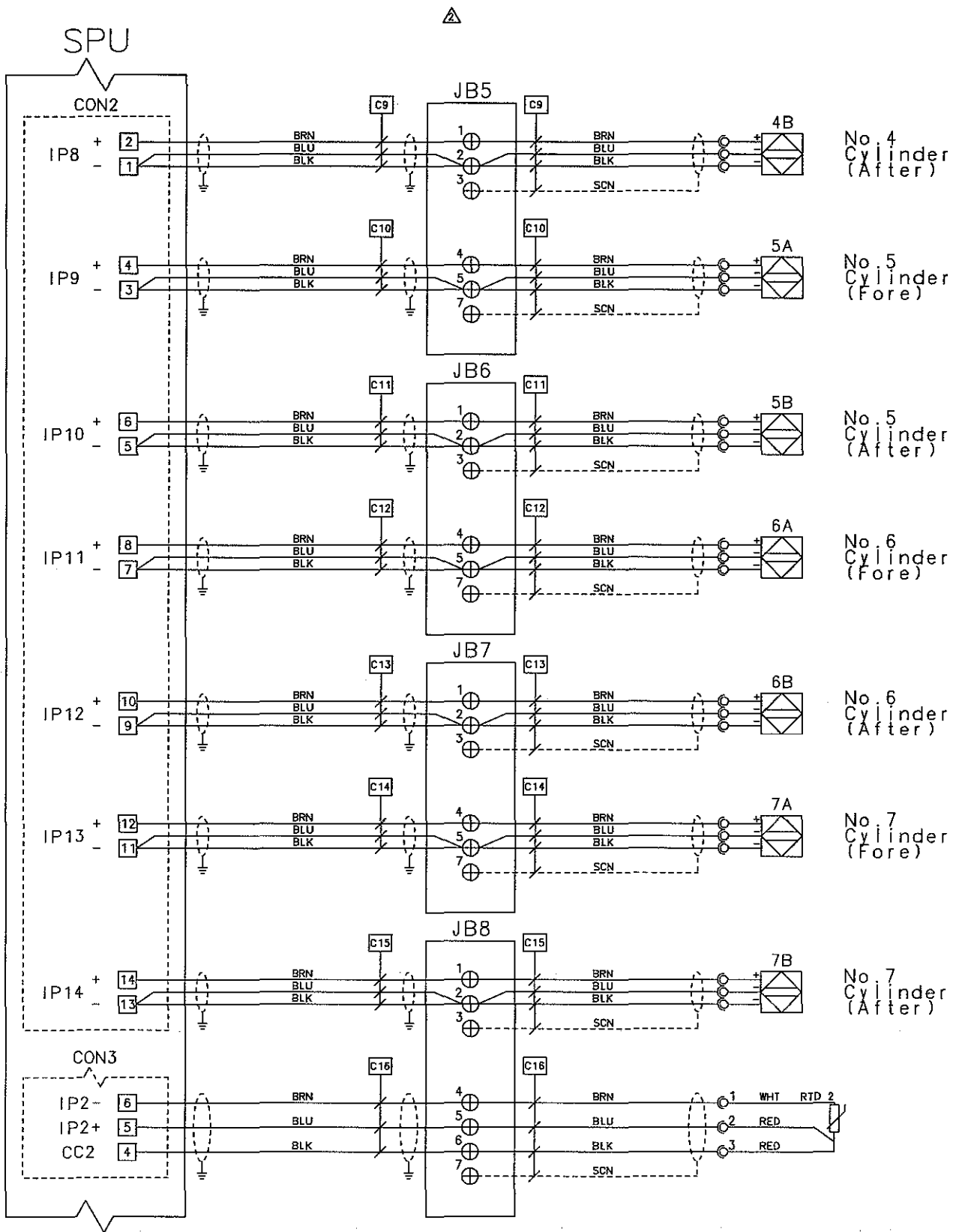
△

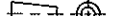
SPU



* All cables are supplied by AMOT

REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL	CAD WEIGHT kg	DWN	101111	T.Y. JEON	SIZE	NAME
			CHKD	101111		
			APPD	101111		
SCALE	MODEL				DRAWING NO.	
N/S	7S50MC-C				5154035070-12	



REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	101111	T.Y.JEON	SIZE A4	NAME WIRING DIAGRAM FOR BEARING WEAR MONITORING SYSTEM(2/3)
			CHKD	101111	D.H.LEE		
		WEIGHT kg	APPD	101111	T.B.KIM		
SCALE N/S	MODEL 7S50MC-C					DRAWING NO. 5154035070-13	

FROM E.C.C
(DC24V)

INTERNAL

IF +

SPU

TOUCH SCREEN PC

24VDC POWER

RS485 COMMS

CON1

1 24VDC
2 0V
3 CHASSIS

CON3

1 18/24VDC
2 0V
3 18/24VDC
4 0V

CON4

1 RS485A
2 RS485B

CON9

1 RL1+
2 RL1-

3 RL2+
4 RL2-

5 RL3+
6 RL3-

CON6 (HEALTHY)

2 RL1A COM
3 RL1A N/O

CON7 (ALARM)

2 RL2A COM
3 RL2A N/O

CON8 (SLOWDOWN)

2 RL3A COM
3 RL3A N/O

CON2

1 24VDC
2 0V

CON5

1 RS485A
2 RS485B

CON6

1 +18VDC
2 0V
3 +18VDC
4 0V

CON9

1 TXD A+
2 RXD B-

CON12

1 + OP1
2 - OP1

CON13

1 + OP2
2 - OP2

CON16

1 + OP3
2 - OP3

CON11

1 TXD A+
2 RXD B-
3 +18VDC
4 0V

BLK

WHT

BRN

WHT

GRN

WHT

RED

WHT

ORG

WHT

YEL

WHT

INTERNAL

TO AMS

NOTE

----- : TO BE PROVIDED BY SHIPYARD
----- : TO BE PROVIDED BY AMOT

REV	LOC	DATE	MADE	APPD	SIZE	NAME
					A4	WIRING DIAGRAM FOR BEARING WEAR MONITORING SYSTEM(3/3)
MATERIAL		CAD	DWN	101111	T.Y.JEON	
		WEIGHT	CHKD	101111	D.H.LEE	
		kg	APPD	101111	T.B.KIM	
SCALE	N/S	MODEL	stx			DRAWING NO.
		7S50MC-C				5154035070-14

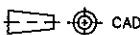
**12.09 AUX. BLOWER STARTER
&T/G MOTOR STARTER**

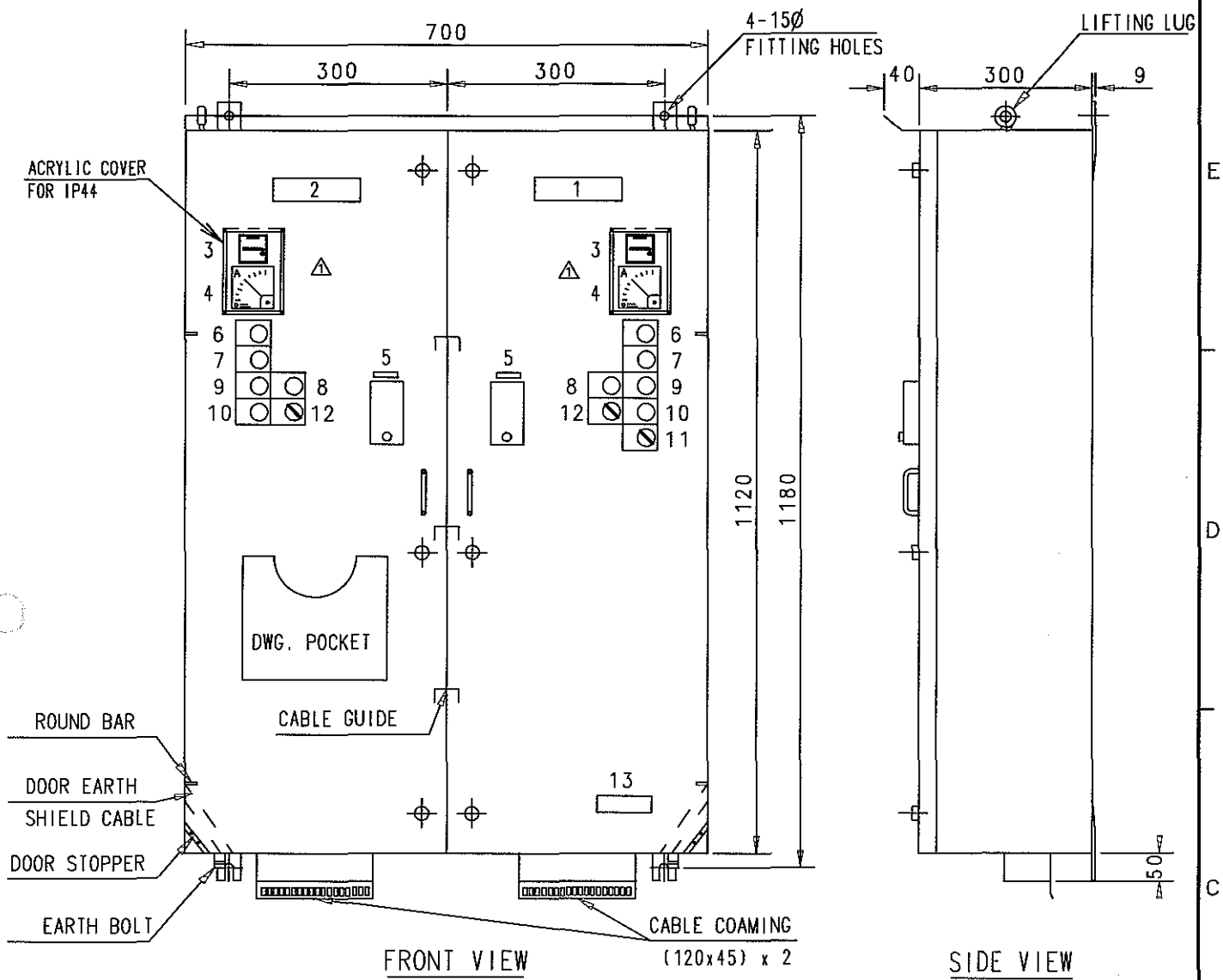
STX Heavy Industries Co., Ltd

1. GENERAL SPEC

NO	DESCRIPTION	AUX. BLOWER MOTOR	TURNING GEAR MOTOR
1	POWER SOURCE	440V 3Ø 60Hz	440V 3Ø 60Hz
2	MOTOR CAPACITY	55 kW	2.2 kW
3	RATED CURRENT	△ 86.6 A	4.9 A
4	STARTING METHOD	D.O.L	D.O.L
5	IP GARDE	IP 44	
6	PAINT COLOR	7.5 BG 7/2	

NO	NAME	TYPE	SPEC.	Q'TY		REMARK
				AUX. BLOWER	TURNING GEAR	
1	MCCB	ABS203b	225AF/100A	2		
		ABS33b	30AF/15A		1	
2	MAGNETIC CONTACTOR	GMC-125 △	AC440V	2		
		GMC-32	AC220V		2	
3	TRANSFORMER	IC-200	1Ø 440/220, 20V	2	1	200VA
		IC-200	1Ø 440/110V		1	200VA
4	ELECTRONIC O.C.R	EOCR-SS-05R-220	0.5 ~ 6A	2		AC220V
		EOCR-SS-30R-220	3 ~ 30A		1	AC220V
5	CURRENT TR	2CT-100	150:5, 5VA, ACC1%	2		
6	△ A.C AMMETER	BE-72, 0~150A, F3, CT; 150/5A RED MARK ; (86.6)A, MOVING IRON		2		CLASS 1.5
		BE-72, 0~10A, F3 RED MARK ; (4.9)A, MOVING IRON			1	CLASS 1.5
7	△ PILOT LAMP	YSPL22-024ARFW	Ø22, 24V, 1W	2	1	WHITE
		YSPL22-024ARFR	Ø22, 24V, 1W	2	1	RED
		YSPL22-024ARFO	Ø22, 6.3V, 1W	2	1	ORANGE
8	△ P.B W/LAMP	YSNPBL3-024ARFG	1a1b, 24V, 1W	2	2	GREEN
9	△ PUSHBUTTON	YSAP122-11RFR	Ø22, 1a1b	2	1	RED
10	RELAY	GMR-4	AC440V, 4a	2		
		GMR-4	AC220V, 3a1b	2	2	
11	RELAY W/BASE	SZR-LY2-N1	AC220V, 2c	4	2	
		SZR-LY4-N1	AC220V, 4c	4		
12	TIMER W/BASE	H3BA-N	AC220V, 2c	3		
13	HOUR METER	TH148S	AC220V, 60Hz	2	1	
14	RECTIFIER	KBPC-3510	35A, 1000V		1	
15	SURGE ABSORB	TNR23G471K			1	
16	△ SELECTOR S.W	YSAR3-222KB	Ø22, 2a2b	3	1	
		YSAR3-211KB	Ø22, 1a1b		1	
17	FUSE & HOLDER	SB-C1	1A	12	6	
		SB-C1	2A	4		
		SB-C1	3A		2	
		SB-C1	6A		1	

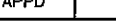
REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD WEIGHT kg	DWN	100107	T.Y. JEON	SIZE A4	NAME GENERAL SPEC & PARTS LIST FOR AUX.BLOWER & T/G STARTER
			CHKD	100107	M.H. CHOI		
			APPD	100107	T.B. KIM		
SCALE N/S		MODEL 7S50MC-C	 STX Engine Co., Ltd.			DRAWING NO. 5154047300-01	

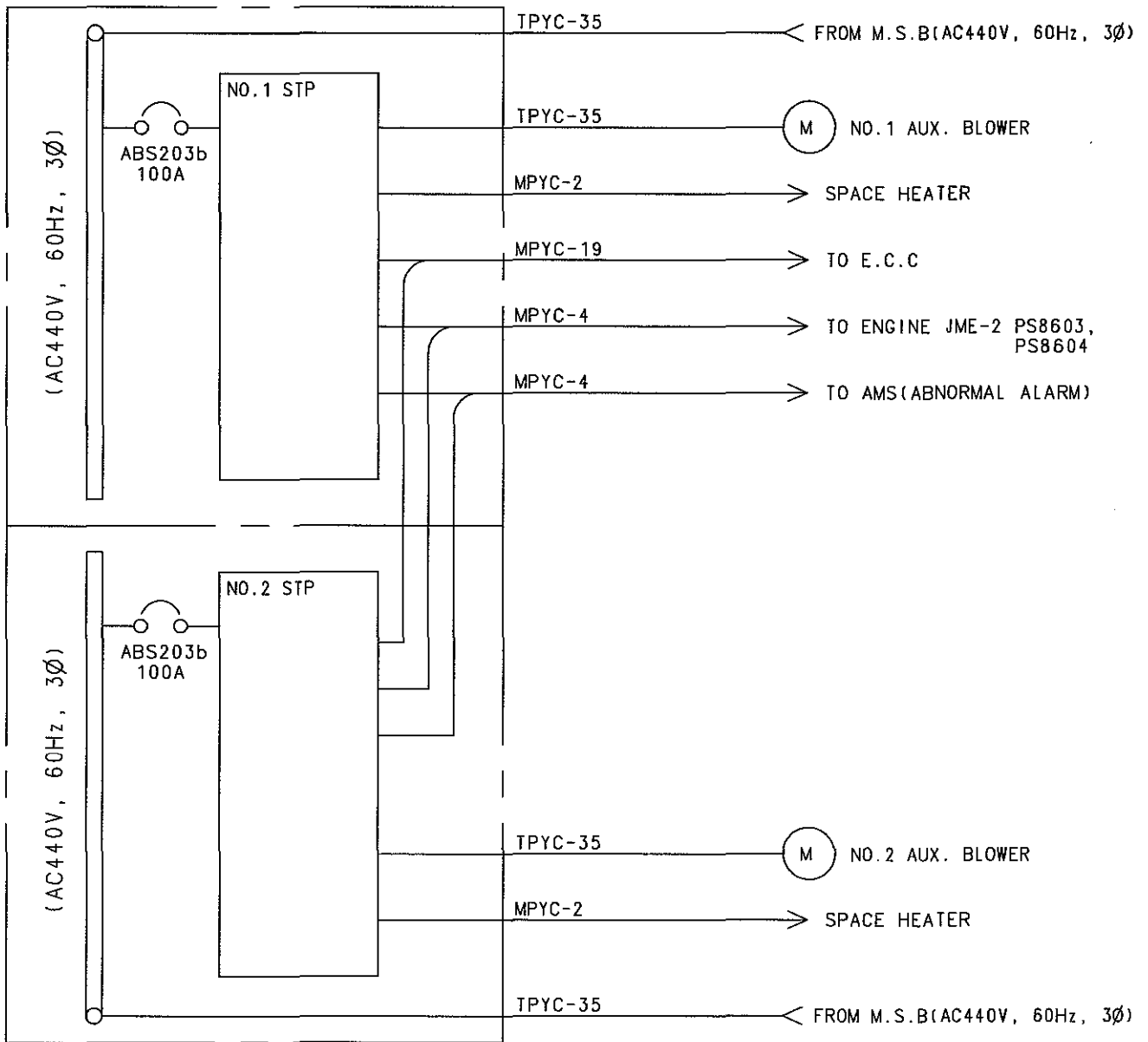


NO.	SYMBOL	DESCRIPTION	NAME PLATE		Q'TY	REMARKS
			LETTER	TYPE		
1		NAME PLATE	NO.1 M/E AUX. BLOWER STARTER	8A	1	
2		NAME PLATE	NO.2 M/E AUX. BLOWER STARTER	8A	1	
3	HM1,2	HOURLMETER			2	TH148S
4	A1,2	AMMETER			2	BE-72
5	52-1,2	MCCB	SOURCE	2A	2	ABS203b
6	WL1,2	IND. LAMP	SOURCE	SQ1	2	WHITE
7	RL1,2	IND. LAMP	OVERLOAD	SQ1	2	RED
8	OL1,2	IND. LAMP	HEATER ON	SQ1	2	ORANGE
9	3-C1,2 GL1,2	P.B W/LAMP	START/RUN	SQ1	2	GREEN
10	3-O1,2	P.B S/W	STOP	SQ1	2	RED
11	43	SELECTOR S/W	LOCAL - REMOTE	SQ1	1	
12	43SH1,2	SELECTOR S/W	HEATER OFF - ON	SQ1	2	
13		NAME PLATE	MAKER NAME		1	

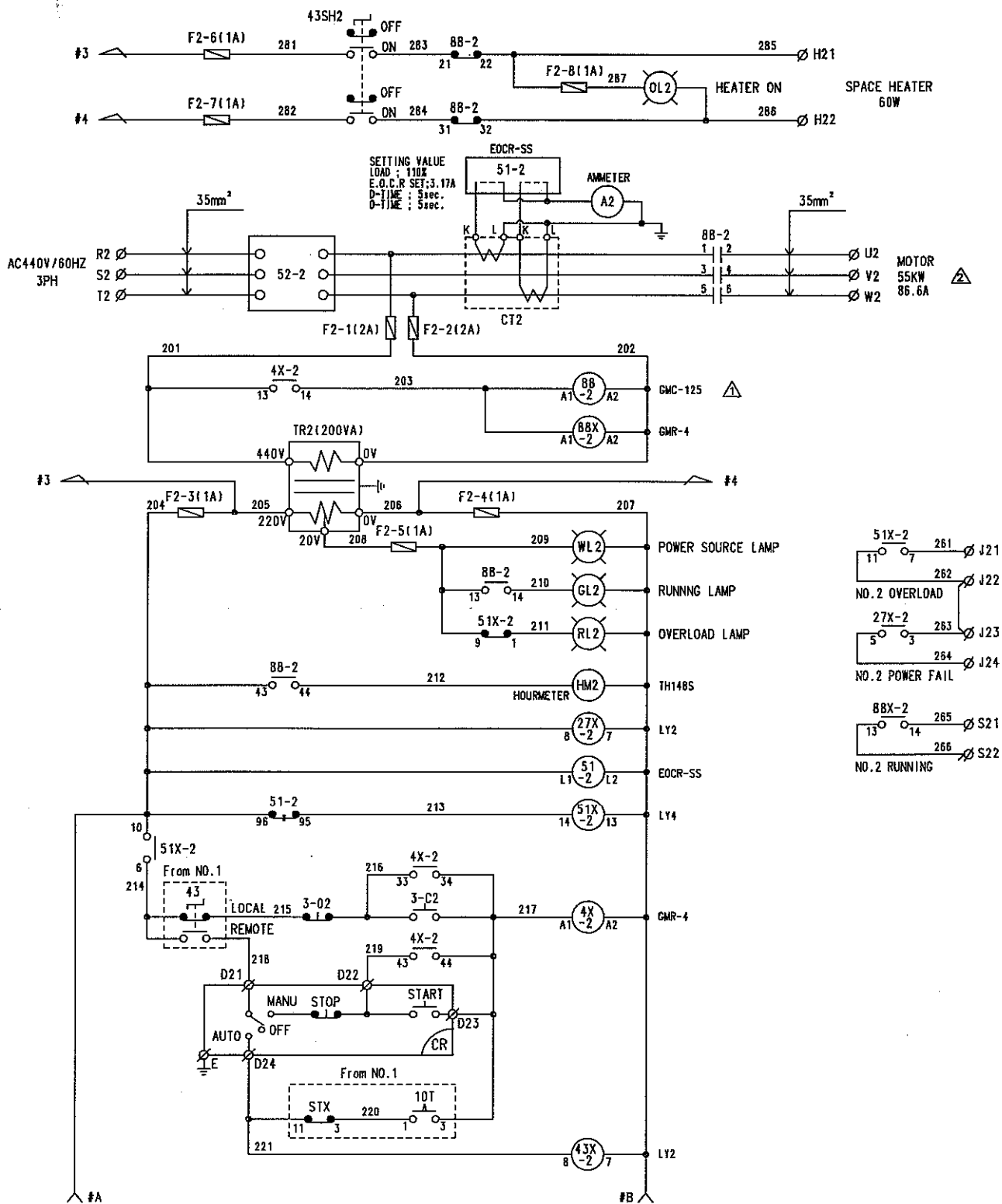
NOTE

1. PLATE : SHEET STEEL 2.3t, SCP1
2. PAINTING COLOR : MUNSELL NO. 7.5 BG 7/2
3. MOUNTING METHOD : WALL MOUNTING TYPE
4. PROTECTION : IP44
5. Q'TY : 1SET/SHIP
6. WEIGHT : abt 150Kg

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	091119	T.Y. JEON	SIZE	NAME OUTLINE OF AUX. BLOWER STARTER
			CHKD	091119	M.H. CHOI	A4	
			APPD	091119	T.B. KIM		
SCALE N/S		MODEL 7S50MC-C	 STX Engine Co., Ltd.			DRAWING NO. 5154047310-01	



REV	LOC	DATE	MADE	APPD	NOTE
MATERIAL	 CAD	DWN	090113	T.Y. JEON	SIZE NAME A4 OUTLINE DIAGRAM OF AUX. BLOWER STARTER
		CHKD	090113	M.H. CHOI	
		APPD	090113	T.B. KIM	
	WEIGHT kg				
SCALE N/S	MODEL 7S50MC-C	 STX Engine Co., Ltd.			DRAWING NO. 5154047310-02



LOCATION

1. CONTROL ROOM

REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL		 CAD	DWN	100107	T.Y. JEON	SIZE NAME CONNECTION DIAGRAM OF NO.2 AUX. BLOWER STARTER
			CHKD	100107	M.H. CHOI	
			APPD	100107	T.B. KIM	
SCALE N/S		MODEL 7S50MC-C	 STX Engine Co., Ltd.			DRAWING NO. 5154047310-04

NO.1 AUX. BLOWER STARTER

NO.	R1	S1	T1	U1	V1	W1	H11	H12
CAPACITY	150A						15A	
USE REMARK	NO1. SOURCE			MOTOR			HEATER	
CABLE SIZE	TPYC-35			TPYC-35			MPYC-2	

D11	D12	D13	D14	E	D21	D22	D23	D24	C11	C12	S11	S12	S21	S22	S13	S14	S15
15A																	
NO.1 REMOTE CONTROL				NO.2 REMOTE CONTROL					FWD & SHD		NO.1 RUNNING		NO.2 RUNNING		NO.1,2 RUNNING		
MPYC-19															NOT USED		

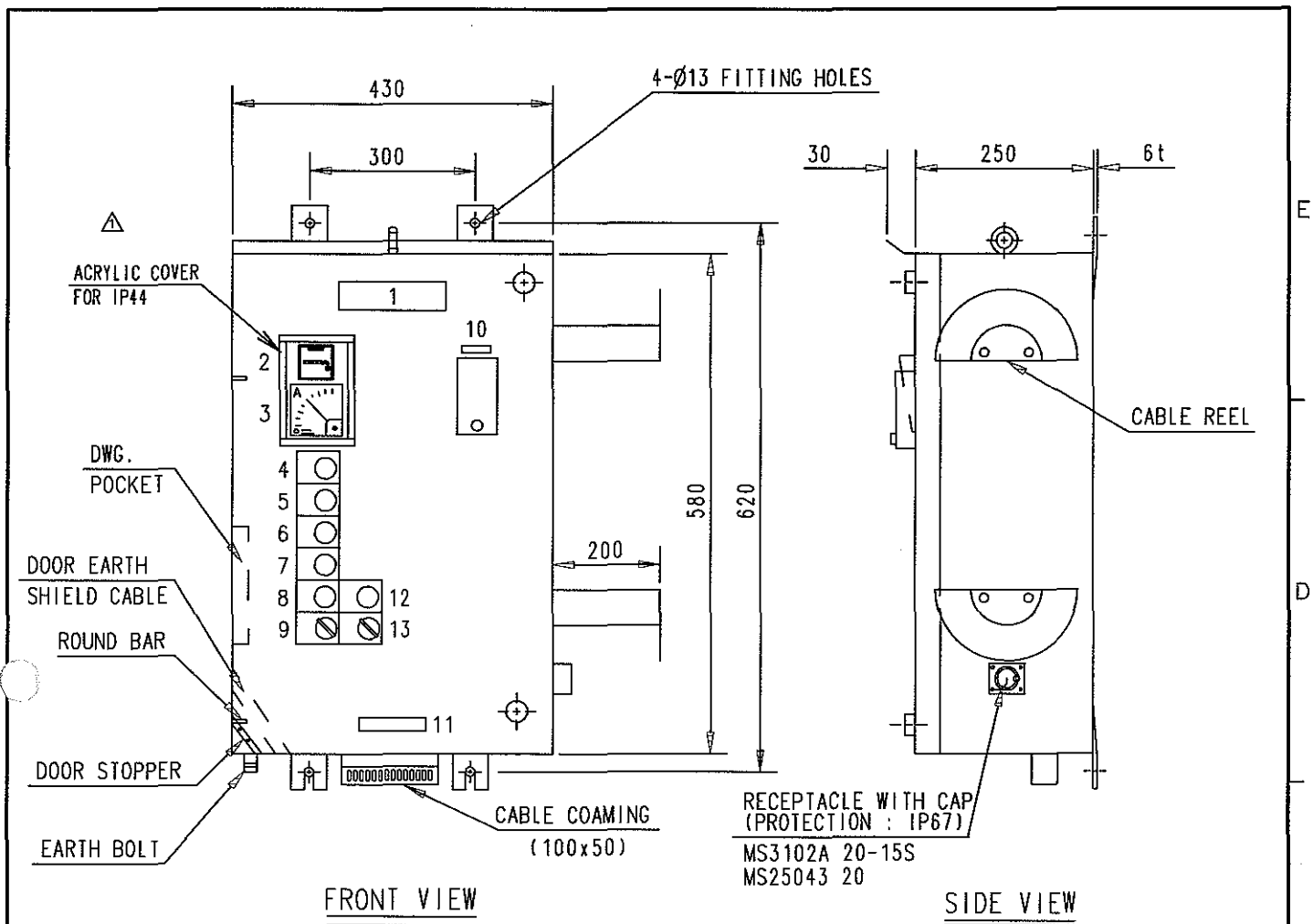


J11	J14	J21	J24	P11	P12	P13	P14
15A							
NO.1 ABNORMAL		NO.2 ABNORMAL		H/P-OFF		L/P-OFF	
MPYC-4				MPYC-4			

NO.2 AUX. BLOWER STARTER

NO.	R2	S2	T2	U2	V2	W2	H21	H22
CAPACITY	150A						15A	
USE REMARK	NO2. SOURCE			MOTOR			HEATER	
CABLE SIZE	TPYC-35			TPYC-35			MPYC-2	

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	091119	T. Y. JEON	SIZE A4	NAME TERMINAL ARR'T FOR AUX. BLOWER STARTER
			CHKD	091119	M. H. CHOI		
		WEIGHT	kg	APPD	091119		
SCALE N/S		MODEL 7S50MC-C		 STX Engine Co., Ltd.		DRAWING NO. 5154047310-05	

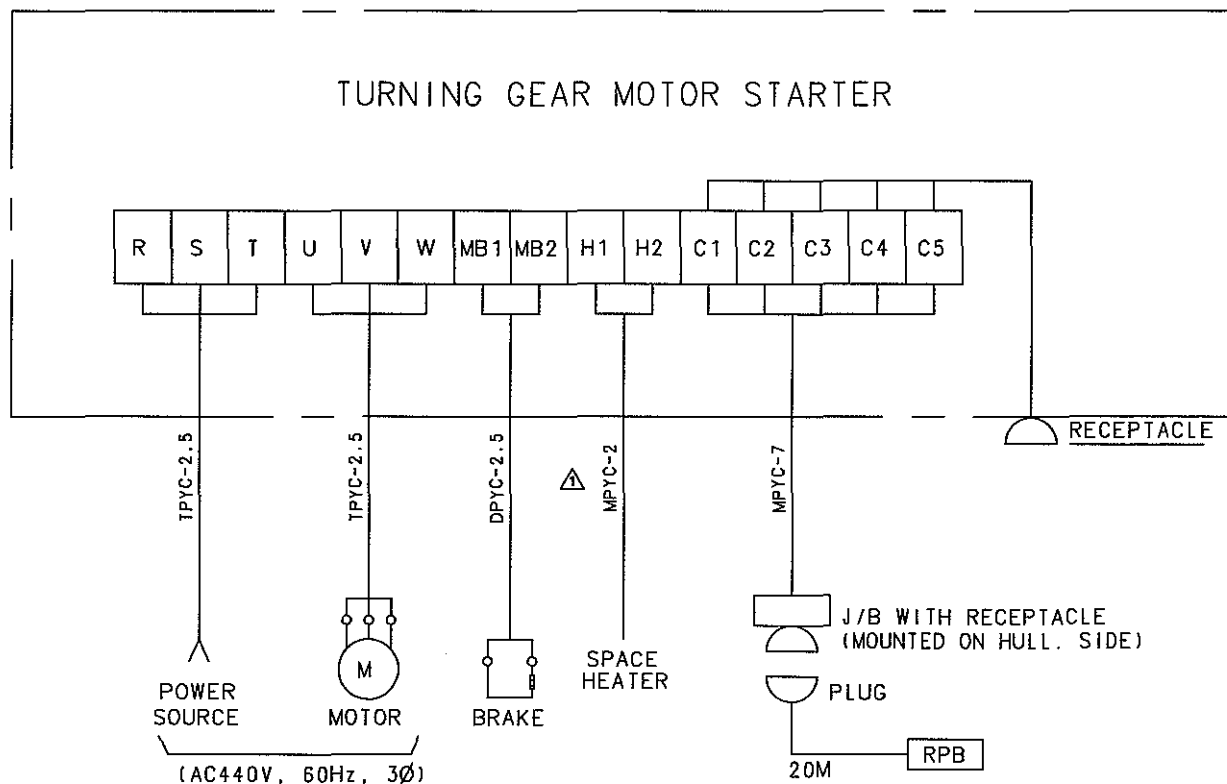


NO.	SYMBOL	DESCRIPTION	NAME PLATE		Q'TY	REMARKS
			LETTER	TYPE		
1		NAME PLATE	M/E TURNING GEAR STARTER	8A	1	
2	HM	HOURLMETER			1	TH148S
3	A	AMMETER			1	BE-72
4	WL	IND. LAMP	SOURCE	SQ1	1	WHITE
5	RL	IND. LAMP	OVERLOAD	SQ1	1	RED
6	3-CF, GLF	P.B W/LAMP	FWD. START/RUN	SQ1	1	GREEN
7	3-CR, GLR	P.B W/LAMP	REV. START/RUN	SQ1	1	GREEN
8	3-O	P.B S/W	STOP	SQ1	1	RED
9	43	SELECTOR S/W	LOCAL - REMOTE	SQ1	1	
10	52	MCCB	SOURCE	2A	1	ABS33b
11		NAME PLATE	MAKER NAME		1	
12	OL	IND. LAMP	HEATER ON	SQ1	1	ORANGE
13	43SH	SELECTOR S/W	HEATER OFF - ON	SQ1	1	

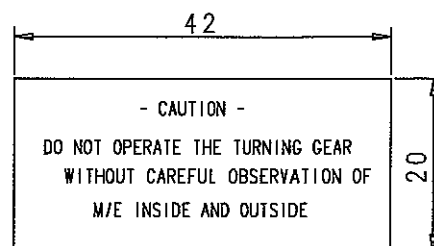
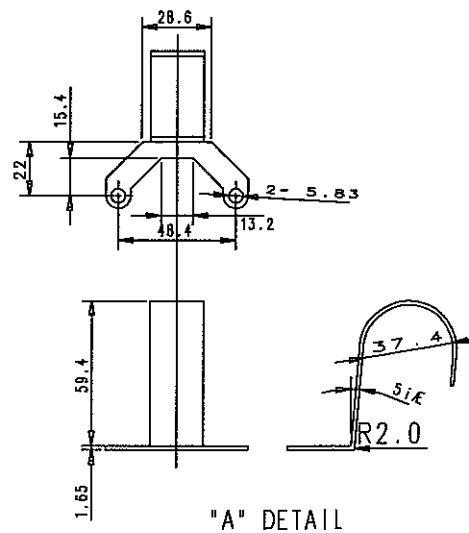
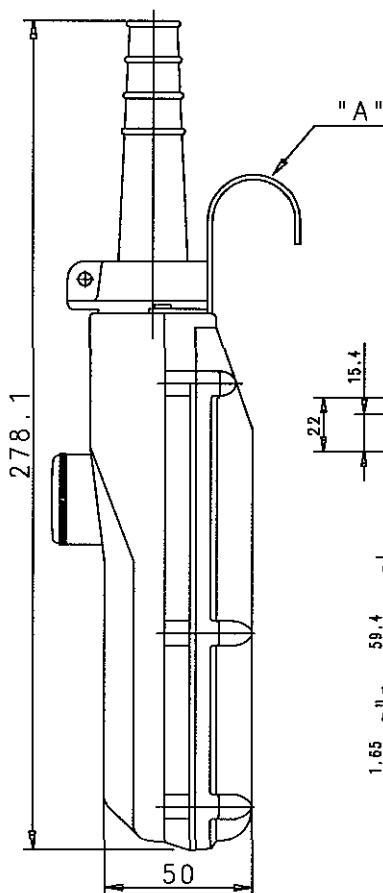
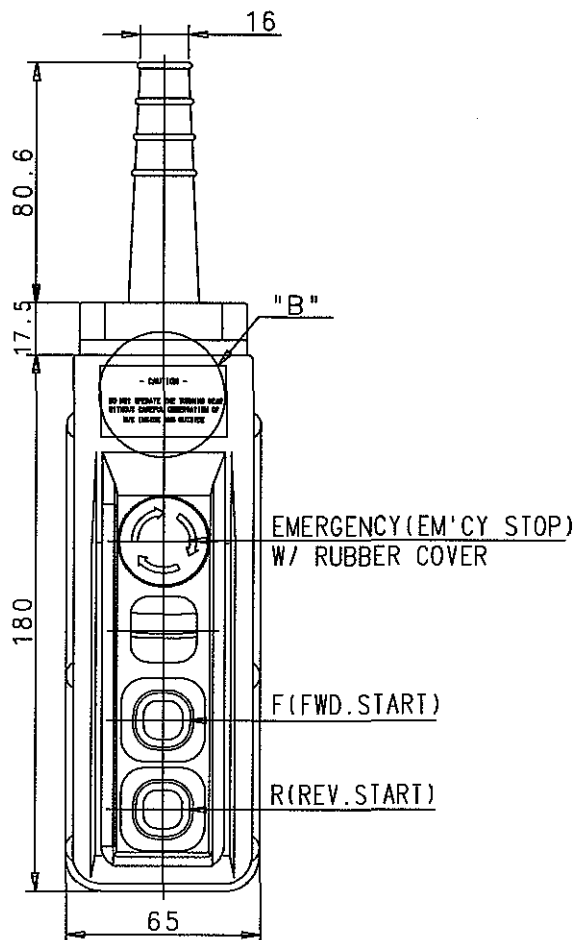
NOTE

1. PLATE : SHEET STEEL 2.3t, SCP1
2. PAINTING COLOR : MUNSELL NO. 7.5 BG 7/2
3. MOUNTING METHOD : WALL MOUNTING TYPE
4. PROTECTION : IP44
5. Q'TY : 1SET/SHIP
6. WEIGHT : obt 35Kg

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD WEIGHT kg	DWN	091119	T.Y. JEON	SIZE	NAME OUTLINE FOR TURNING GEAR STARTER
			CHKD	091119	M.H. CHOI	A4	
			APPD	091119	T.B. KIM		
SCALE N/S		MODEL 7S50MC-C	 STX Engine Co., Ltd.			DRAWING NO. 5154047320-01	

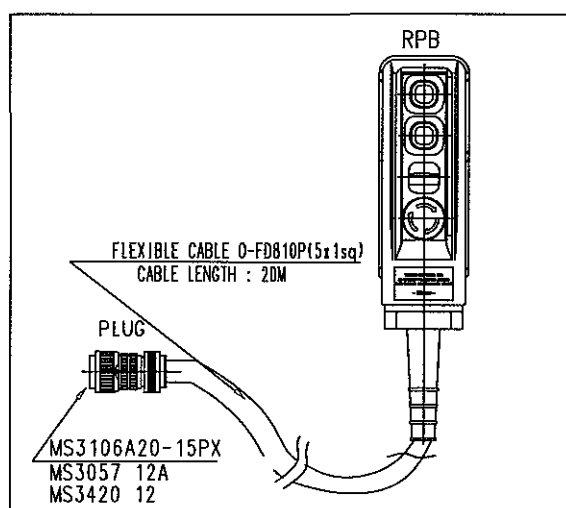


REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD WEIGHT kg	DWN	091119	T.Y.JEON	SIZE A4	NAME WIRING DIAGRAM FOR T/G MOTOR STARTER
			CHKD	091119	M.H.CHOI		
			APPD	091119	T.B.KIM		
SCALE N/S		MODEL 7S50MC-C	 STX Engine Co.,Ltd.			DRAWING NO. 5154047320-03	

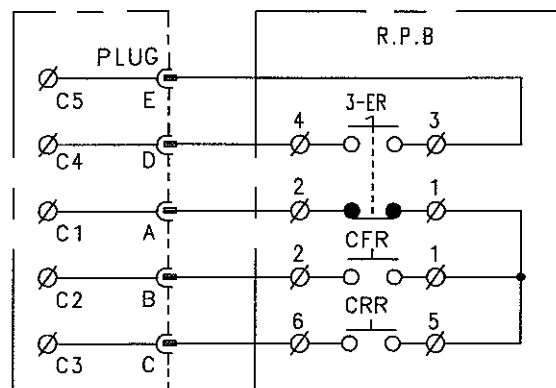


NOTE

1. IP GRADE : IP55
2. PAINT COLOR : MAKER STANDARD

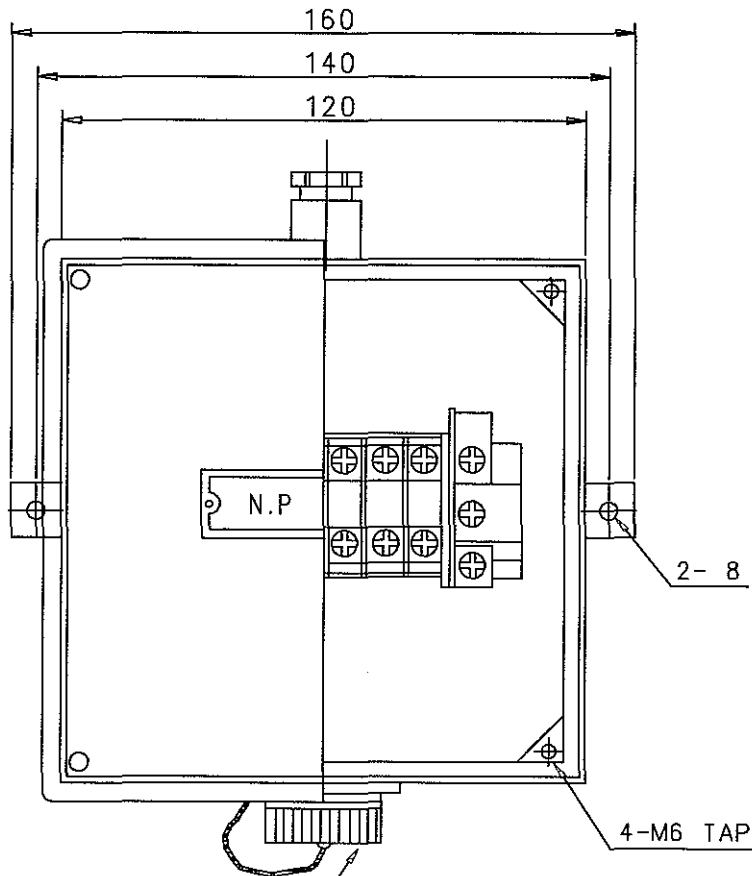


STARTER OR RECEPTACLE



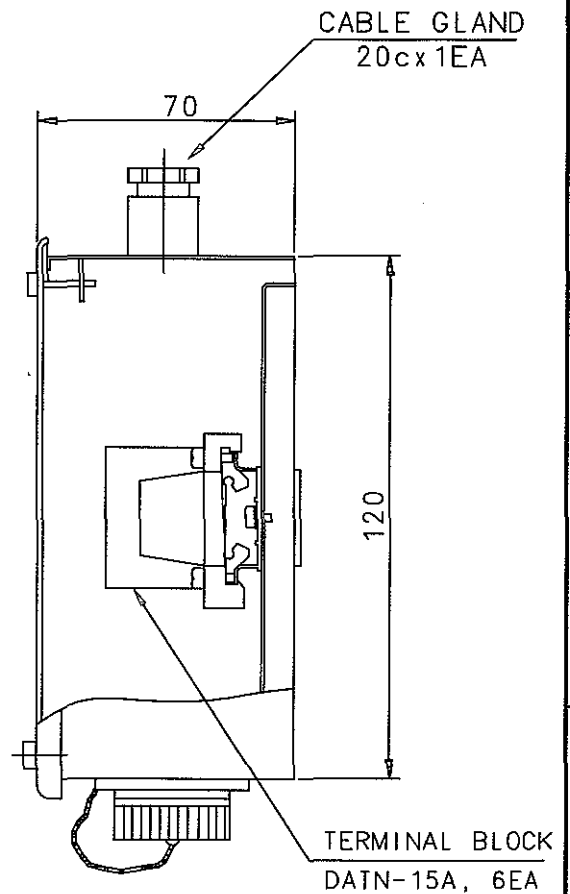
WIRING DIAGRAM

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	090113	T.Y. JEON	SIZE A4	NAME REMOTE PUSHBUTTON S.W BOX FOR TURNING GEAR STARTER (HY-1024)
			CHKD	090113	M.H. CHOI		
			APPD	090113	T.B. KIM		
SCALE N/S		MODEL 7S50MC-C	 STX Engine Co., Ltd.			DRAWING NO. 5154047320-04	

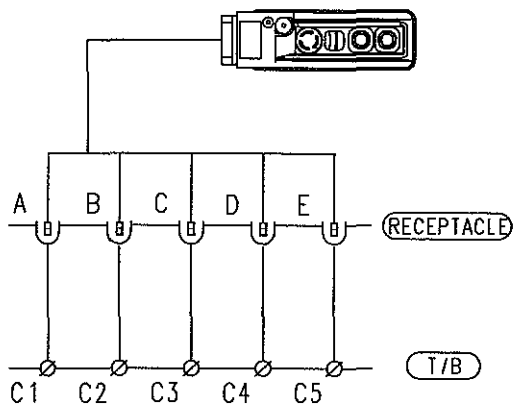


RECEPTACLE W/CAP
MS3102A 20-15S
MS25043, 20

FRONT VIEW



SIDE VIEW



WIRING DIAGRAM

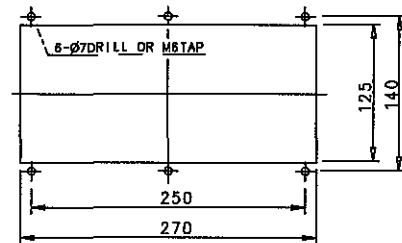
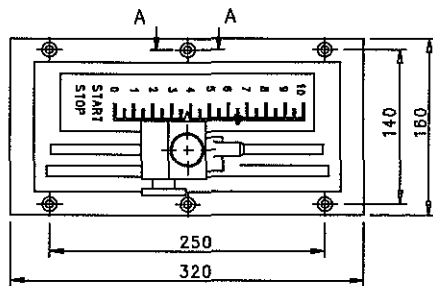
NOTE

1. MATERIAL : SCP1, 1.6t
2. PAINTING COLOR : MUNSELL NO. 7.5 BG 7/2
3. NAME PLATE : TYPE - 2A
LETTER : RECEPTACLE FOR M/E TURNING GEAR
4. Q'TY : 1SET/SHIP
5. PROTECTION : IP44

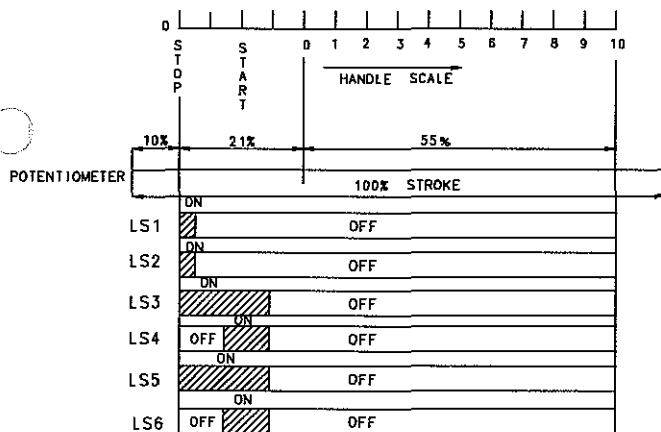
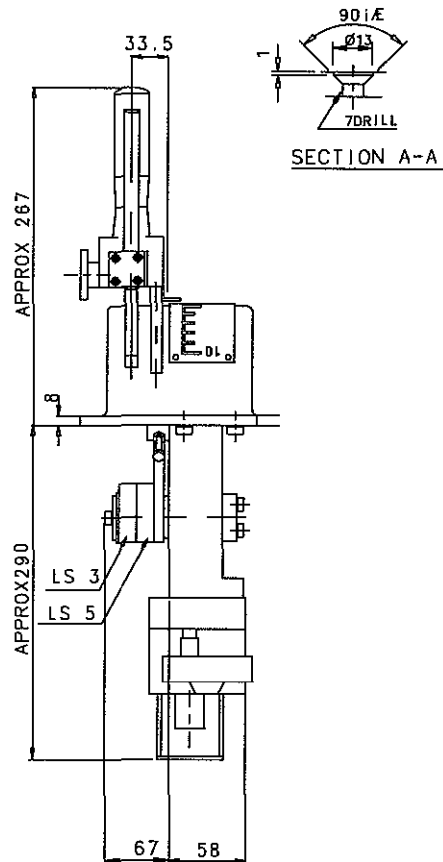
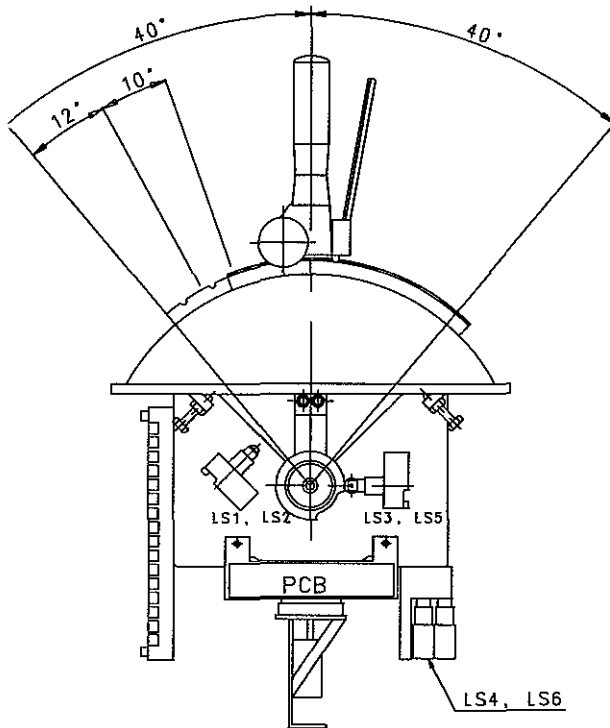
REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD WEIGHT kg	DWN	090113	T. Y. JEON	SIZE A4	NAME RECEPTACLE J/B FOR T/G MOTOR STARTER
			CHKD	090113	M. H. CHOI		
			APPD	090113	T. B. KIM		
SCALE N/S		MODEL 7S50MC-C	STX STX Engine Co., Ltd.			DRAWING NO. 5154047320-05	

12. 10 ECR MANEUVRING HANDLE

STX Heavy Industries Co., Ltd



PANEL CUTTING

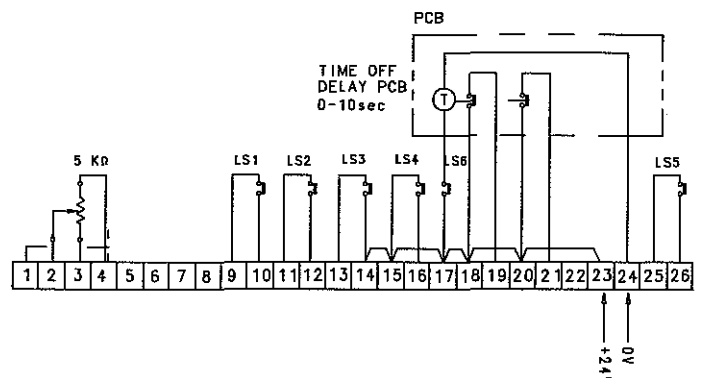


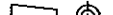
LIMIT SWITCH OPERATION POSITION

NOTE

1. SURROUNDING TEMP. : -20°C ~ +70 °C
2. L/S WORKING VOLTAGE : 15A 125,250 OR 480VAC 1/2A 125VDC

<INSTALLED ON E.C.C>

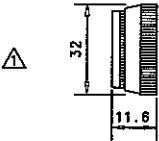
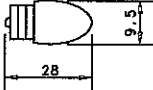
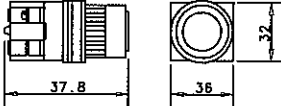
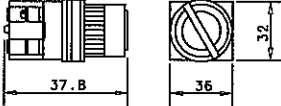
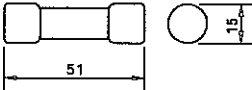
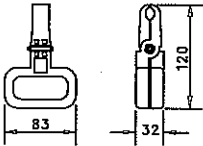
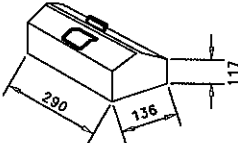


REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		 CAD	DWN	080401	T.Y. JEON	SIZE	NAME ELECTRIC MANEUVERING HANDLE (5KΩ, ONE-WAY TYPE)
			CHKD	080401	D.H. LEE		
			APPD	080401	T.B. KIM		
WEIGHT		kg					
SCALE	MODEL	DRAWING NO.					
N/S	ALL	STX STX Engine Co.,Ltd.				5154009200	

STX STX Engine Co., Ltd.

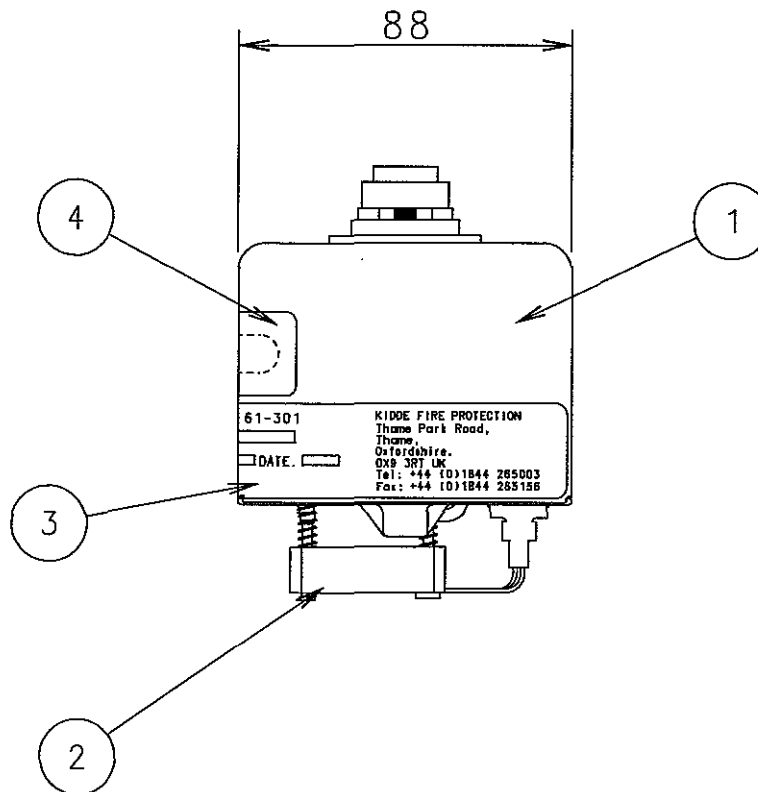
12. 11 SPARE PARTS

STX Heavy Industries Co., Ltd

NO.	NAME	SKETCH	SUPPLY (Q'TY)		UNIT	MAKER	REMARK
			WORK'G	SPARE			
1	GLOBE		3	1	EA	YONGSUNG	22Ø, WHITE
			3	1			22Ø, ORANGE
			3	1			22Ø, RED
2	BULB		10	10	EA	YONGSUNG	△ E-10 24V, 1W
3	PUSH BUTTON		3	1	EA	YONGSUNG	△ YSAP122 -11RFR
4	SELECTOR SWITCH		4	1	EA	YONGSUNG	YSAR22 -222KB
5	FUSE		1	1	EA	SB-FUSE	6A
			2	2			3A
			4	4			2A
			18	18			1A
6	FUSE PULLER			1	EA	SB-FUSE	FP-36
7	SPARE PART BOX WITH SUITABLE LOCKING DEVICE			1	EA		STEEL

REV	LOC	DATE	MADE	APPD	NOTE		
MATERIAL		CAD	DWN	091119	T.Y. JEON	SIZE	NAME
WEIGHT		kg	CHKD	091119	M.H. CHOI	A4	SPARE PARTS FOR AUX. BLOWER & T/G MOTOR STARTER
SCALE		MODEL	APPD	091119	T.B. KIM		
N/S		7S50MC-C	STX STX Engine Co., Ltd.			DRAWING NO.	5154047300-02

SPARE PART FOR OMD

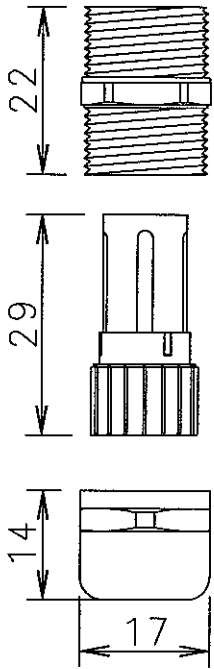
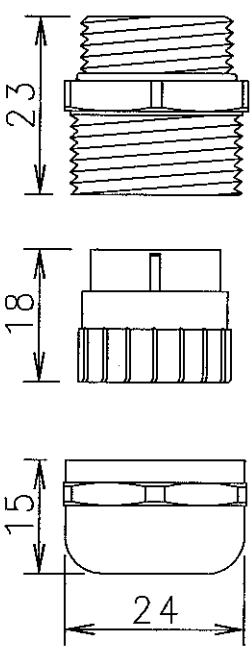


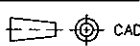
ITEM	DESCRIPTION	PART NO.
1	DETECTOR HEAD ASSEMBLY	D5622-001
2	FAN ASSEMBLY(PAPST)	D5622-005
2a (ALTERNATIVE NOT SHOW)	FAN ASSEMBLY(MICRONEL)	D5622-005-02
3	LABEL INVALIDATE GUARANTEE	C9175-803
4	LABEL SWITCH WINDOW	C9189-801

REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL		 CAD	DWN	100409	T.Y. JEON	SIZE NAME SPARE PART FOR OIL MIST DETECTOR
			CHKD	100409	M.H. CHOI	
			APPD	100409	T.B. KIM	
SCALE N/S		MODEL ALL	DRAWING NO. 5154048440			
			STX STX Engine Co., Ltd.			

△

SPARE PART FOR BWM

No	Part Description	Sketch	Material & Weight	Part no	Q'ty
1	M16x1 Cable gland		Nickle Plated Steel/Plastic & 0.021Kg	GLBFA0001	1
2	M20x1.5 Cable gland		Nickle Plated Steel/Plastic & 0.037Kg	GLXLP0001	1

REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL		 CAD	DWN	101111	T.Y. JEON	SIZE A4 NAME SPARE PART FOR BEARING WEAR MONITORING SYSTEM(1/5)
			CHKD	101111	D.H. LEE	
			APPD	101111	T.B.KIM	
SCALE		MODEL				DRAWING NO.
N/S		7S50MC-C				5154035070-15

△

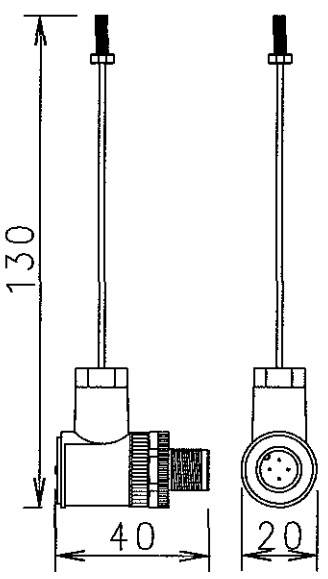
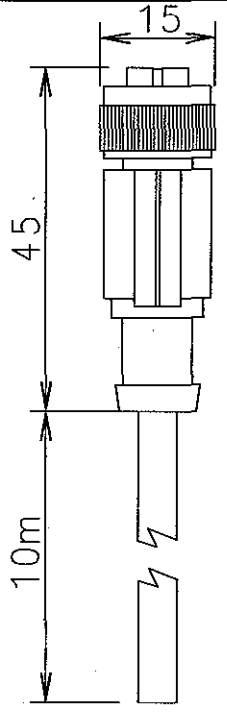
SPARE PART FOR BWM

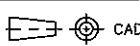
No	Part Description	Sketch	Material & Weight	Part no	Q'ty
3	Proximity Sensor		Nickle Plated & 0.021Kg	SPXSZ003	2
4	20mm Glass Fuse, 2A		Nickle Plated Steel/Plastic & 0.037Kg	90423L020	20

REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL		CAD WEIGHT kg	DWN	101111	T.Y. JEON	SIZE A4 NAME SPARE PART FOR BEARING WEAR MONITORING SYSTEM(2/5)
			CHKD	101111	D.H. LEE	
			APPD	101111	T.B. KIM	
SCALE N/S		MODEL 7S50MC-C				DRAWING NO. 5154035070-16



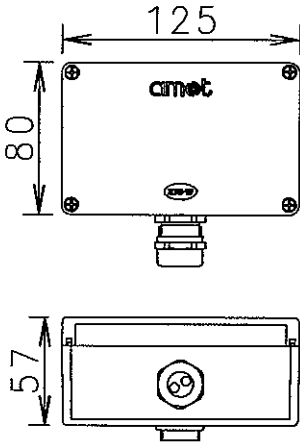
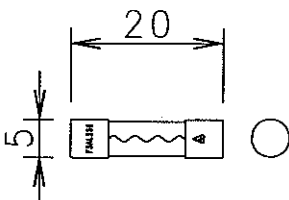
SPARE PART FOR BWM

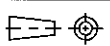
No	Part Description	Sketch	Material & Weight	Part no	Q'ty
5	RTD Assembly		Multiple Material Present & 0.025Kg	S4600-RTD-003	1
6	10m Sensor cable		Multiple Material Present & 0.462Kg	CEXSZ003+	1

REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL			DWN	101111	T.Y. JEON	SIZE A4 NAME SPARE PART FOR BEARING WEAR MONITORING SYSTEM(3/5)
			CHKD	101111	D.H. LEE	
			APPD	101111	T.B.KIM	
SCALE N/S		MODEL 7S50MC-C				DRAWING NO. 5154035070-17



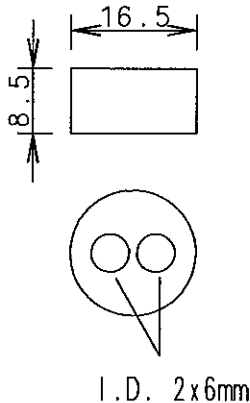
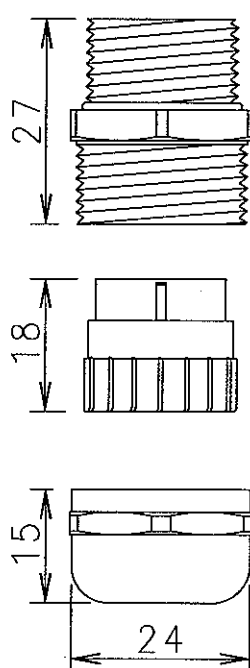
SPARE PART FOR BWM

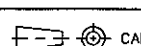
No	Part Description	Sketch	Material & Weight	Part no	Q'ty
7	Terminal Box Assembly		Multiple Materials Present & 0.61Kg	S4600-TE+002	1
8	20mm Glass Fuse, 5A		Nickle Plated Steel/Plastic & 0.001Kg	90423L050	20

REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL		 CAD	DWN	101111	T.Y. JEON	SIZE A4 NAME SPARE PART FOR BEARING WEAR MONITORING SYSTEM(4/5)
		WEIGHT	CHKD	101111	D.H. LEE	
		kg	APPD	101111	T.B.KIM	
SCALE		MODEL				DRAWING NO.
N/S		7S50MC-C				5154035070-18



SPARE PART FOR BWM

No	Part Description	Sketch	Material & Weight	Part no	Q'ty
9	Gland Insert		Silicone & 0.002Kg	GLXLP004	2
10	M20x1.5 Cable Gland		Nickle Plated Steel/Plastic & 0.041Kg	GLXLP003	1

REV	LOC	DATE	MADE	APPD	NOTE	
MATERIAL		 CAD	DWN	101111	T.Y. JEON	SIZE A4 NAME SPARE PART FOR BEARING WEAR MONITORING SYSTEM(5/5)
			CHKD	101111	D.H. LEE	
			APPD	101111	T.B. KIM	
SCALE N/S		MODEL 7S50MC-C				DRAWING NO. 5154035070-19

LIST OF SPARE PARTS FOR MAIN ENGINE

NOTE

The items marked with "S" in this list are to be arranged at the hull side and the steel box for strong them are not supplied

Steel Box : MS

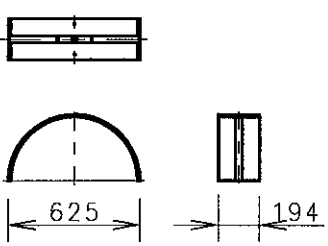
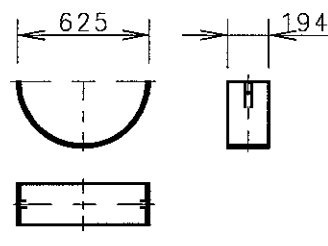
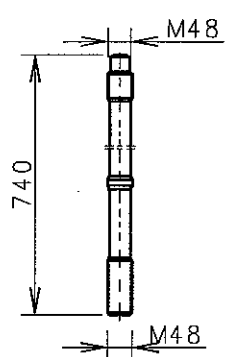
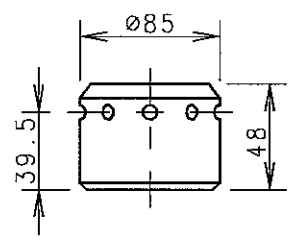
Eng. room : S

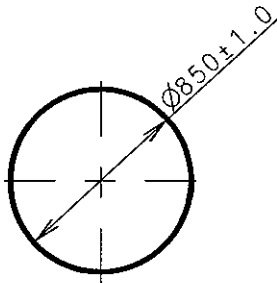
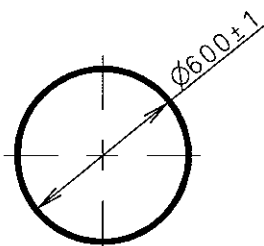
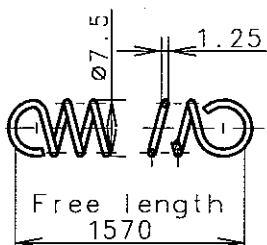
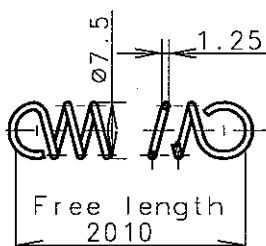
Date	Des.	Chk.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20091218	SJJ	SML	TBK	TBK	*	Pos No. 1,8,16 revised	1
20090317	KCJ	SML	TBK	TBK	*	First issue	0
Scale: 1:1	Size: A4	Type: S50MC-C				Page No.: 15-1-1	 Engine Co., Ltd.
Info No.: 7S50MC-CSP001		Description: LIST OF SPARE PART FOR MAIN ENGINE					Drawing no.: 2805030851(1/2)

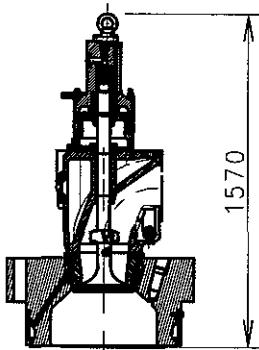
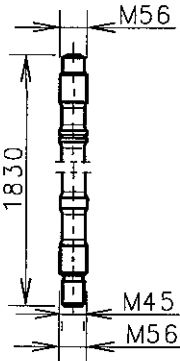
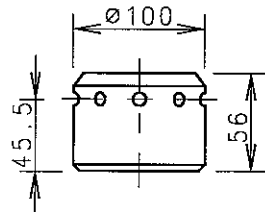
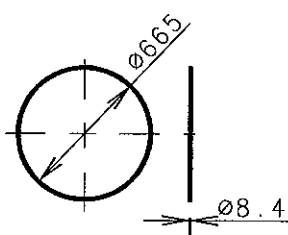
INDEX

NO.	DESCRIPTION	PAGE NO.	DRAWING NO.	REV.
1	MAIN BEARING	001-1~2	2805016411	△1
2	CYLINDER COVER	002-1	2805013680	
3	CYLINDER LINER	003-1~4	2805013690	
4	PISTON	004-1~3	2805013700	
5	CRANKPIN BEARING	005-1	2805013770	
6	CROSSHEAD BEARING	006-1~2	2805030060	
7	THRUST BEARING	007-1	2805021990	
8	VALVES ON CYLINDER COVERS	008-1	2805030071	△1
9	FUEL VALVE	009-1	2805020340	
10	EXHAUST VALVE	010-1~5	2805019260	
11	CAMSHAFT DRIVING GEAR	011-1~2	2805013840	
12	FUEL PUMP & HIGH PRESS PIPE	012-1~4	2805013850	
13	LUBRICATOR	013-1~4	2805022000	
14	FUEL & EXHAUST GEAR	014-1~4	2805013980	
15	INDICATOR VALVE	015-1	2805014000	
16	OTHERS SPARE	016-1~10	2805014011	△1
17	SPARE PARTS FOR THERMOMETER	017-1~3	2805020350	

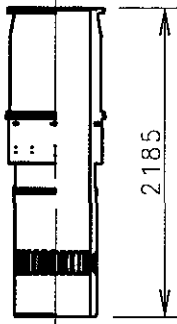
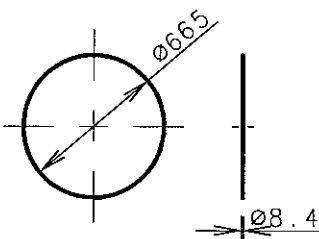
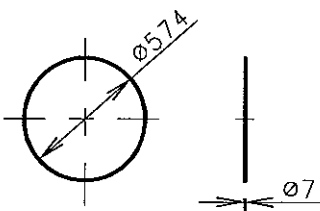
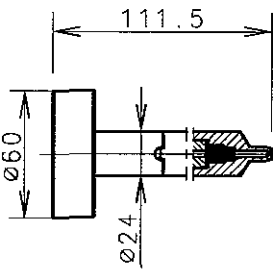
Date	Des.	Chk.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20091218	SJJ	SML	TBK	TBK	*	Pos No. 1,8,16 revised	1
20090317	KCJ	SML	TBK	TBK	*	First issue	0
Scale:	Size:	Type:				Page No.:	
1:1	A4	S50MC-C				15-1-2	
Info. No.:		Description:					Drawing no.:
7S50MC-CSP001		LIST OF SPARE PART FOR MAIN ENGINE					2805030851(2/2)

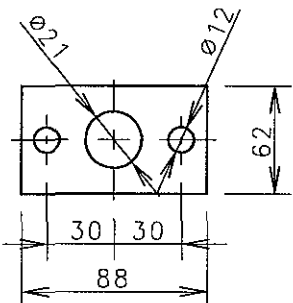
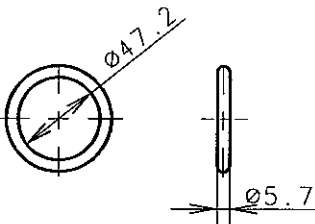
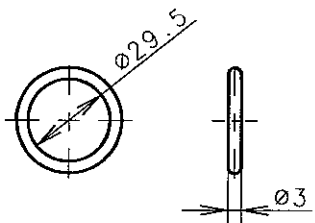
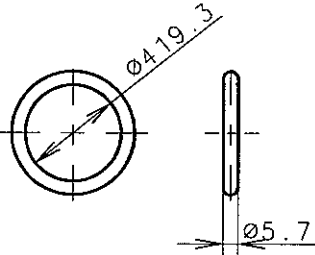
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE		PLACE	
				STD	ADD			
1	MAIN BRG. UPPER SHELL		WHITE METAL 16.5	N+2	1	P91210-0171 -108	OLD: 91210-0151 -108 MS	
2	MAIN BRG. LOWER SHELL		TIN- ALUMINIUM 18.3	N+2	1	P91210-0171 -121	OLD: 91210-0151 -121 MS	
3	MAIN BRG. STUD		Cr-Ni-Mo STEEL 8.0	4N+8	4	P91210-0171 -025	OLD: 91210-0151 -037 MS	
4	NUTS FOR HYDRAULIC TIGHTENING		CARBON STEEL 1.48	4N+8	4	P91210-0171 -013	OLD: 91210-0151 -013 MS	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
20090227		KCJ	SML	TBK	TBK	*	Plate No. revised	1
20070102		YWH	HCY	CSP	JSP	*	First issue	0
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1:1		A4		S50MC-C			001-1	
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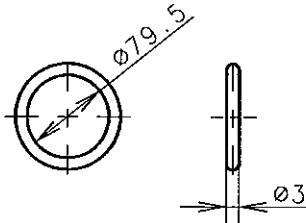
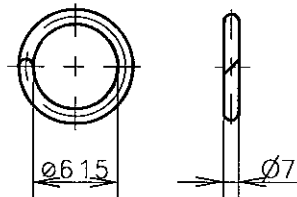
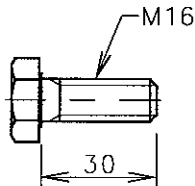
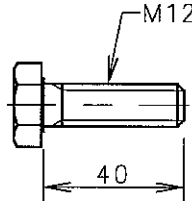
NO.	NAME	SKETCH	MATERIAL	Q' TY/ENG.			PLATE NO.	REMARKS
			WEIGHT (kg)	WORKING	SPARE	PLACE		
				STD	ADD			
5	OIL SEAL A.V DAMPER		TYPE : GHHD 850±1.0 ×874×12 -10500000	1	1	P91211-0065 -139		
			0.85				S	
6	OIL SEAL A.V DAMPER		TYPE : GHHD 600±1.0 ×624×12 -10500000	2	1	P91211-0065 -127		
			0.6				S	
7	TENSION SPRING		CARBON STEEL VALVE SPRING WIRE	2	1	P91211-0065 -020		
			0.18				MS	
8	TENSION SPRING		CARBON STEEL VALVE SPRING WIRE	1	1	P91211-0065 -019		
			0.21				MS	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	
						*		
						*		
20090227		KCJ	SML	TBK	TBK	*	Plate No. revised	
20070102		YWH	HCY	CSP	JSP	*	First issue	
Scale:		Size:	Type:			Page No.:		STX Engine Co., Ltd.
1 : 1		A4	S50MC-C			001-2		
Info. No.:		Description:					Drawing no.:	
1-S50MC-C-1		1. MAIN BEARING					2805016411(2/2)	

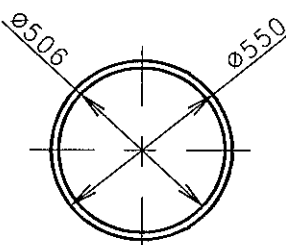
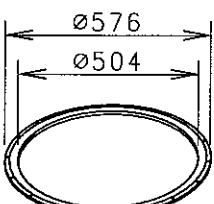
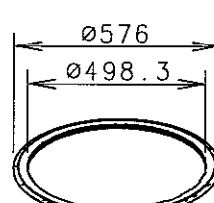
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.			PLATE NO.	REMARKS
			WEIGHT(kg)	WORKING	SPARE		PART NO.	PLACE
1	CYL. COVER COMPLETE WITH FUEL V/V(2EA) EXH. V/V(1EA) START. V/V(1EA) AND EACH STUD (TOTAL 10EA)		Cr-Mo STEEL	N	1		P90101-0104 + P90801-0213 & P90803-0049	REFER TO 15-2-3 PAGE
			1699					S
2	STUD FOR CYL. COVER		Cr-Ni-Mo STEEL	8N	4		P90301-0167 -040	
			26.6					S
3	NUTS FOR HYDRAULIC TIGHTENING		CARBON STEEL	8N	4		P90301-0167 -039	
			1.38					MS
4	O-RING		VITON	2N	4		90101-0104 -114	
			0.35					MS
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	
						*		
						*		
						*		
20061102		KCJ	JTS	CSP	JSP	*	First issue	
Scale:		Size:	Type:			Page No.:		SUX Engine Co., Ltd.
1:1		A4	S50MC-C			002-1		
Info. No.:			Description:				Drawing no.:	
2-S50MC-CA			2. CYLINDER COVER				2805013680	

STX Engine Co., Ltd.

NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.			PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE	PLACE			
					STD	ADD			
1	CYLINDER LINER		TARKALL-C	N	1		P90302-0184-046	REFER TO 15-2-4 PAGE	
			1630					S	
2	O-RING		VITON	2N	2		P90302-0184-117		
			0.35					MS	
3	O-RING		VITON	N	1		P90302-0184-129		
			0.16					MS	
4	NON-RETURN VALVE ASSY		HOT ROLLED CARBON STEEL	6N	6		P90304-0037-011		
			0.571					MS	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement		C.No
						*			3
						*			2
						*			1
20061102		KCJ	JTSC	CSP	JSP	*	First issue		0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 003-1		SUX Engine Co., Ltd.
Info. No.: 3-S50MC-C		Description: 3. CYLINDER LINER			Drawing no.: 2805013690(1/4)				

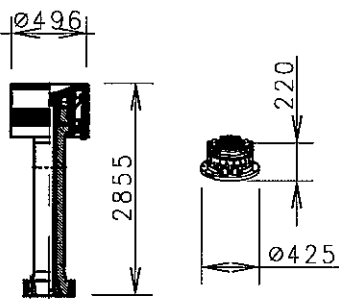
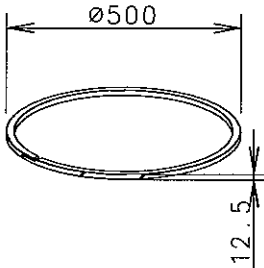
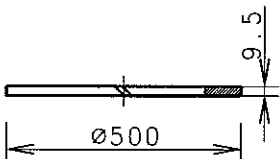
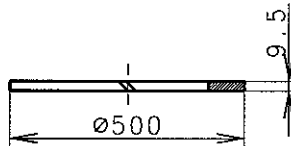
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS		
			WEIGHT(kg)	WORKING	SPARE		PLACE		
					STD	ADD			
5	PACKINGS		VITON	4N	6		P90302-0184 -178		
			0.004				MS		
6	O-RING		VITON	8N	4		P90301-0167 -052		
			0.01				MS		
7	O-RING		VITON	8N	12		P90302-0184 -201		
			0.002				MS		
8	SEALING RING		NBR	N	2		P90301-0167 -363		
			0.043				MS		
Date			Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
							*		3
							*		2
							*		1
20061102			KCJ	JTSC	SP	JSP	*	First issue	0
Scale:		Size:	Type:			Page No.:		STX Engine Co., Ltd.	
1:1		A4	S50MC-C			003-2			
Info. No.:			Description:					Drawing no.:	
3-S50MC-C			3. CYLINDER LINER					2805013690(2/4)	

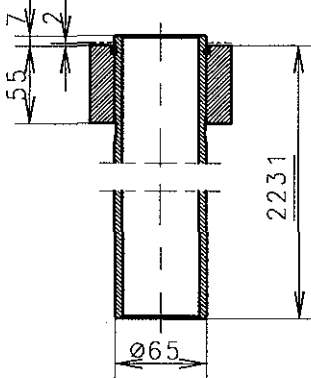
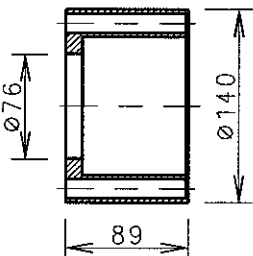
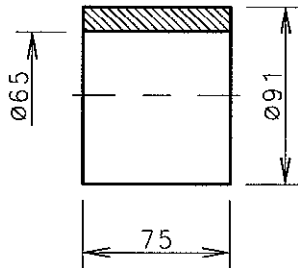
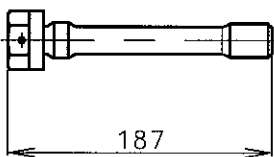
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE		PLACE	
					STD	ADD		
9	SEALING RING		NBR	2N	3		P90301-0167-267	
			0.002				MS	
10	O-RING		NBR	N	2		P90301-0167-326	
			0.095				MS	
11	HEX. BOLT		10.9T	-	10		P90301-0167-409	
			0.09				MS	
12	HEX. BOLT		10.9T	-	12		P90301-0167-351	
			0.09				MS	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20061102		KCJ	JTS	CSP	JSP	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C		Page No.: 003-3		SUX Engine Co., Ltd.
Info. No.: 3-S50MC-C		Description: 3. CYLINDER LINER					Drawing no.: 2805013690(3/4)	

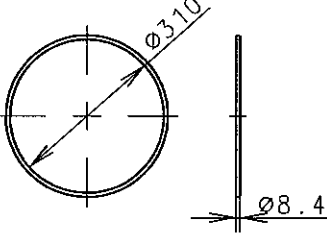
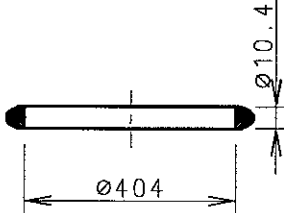
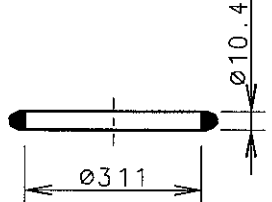
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS
				WORKING	SPARE		
			WEIGHT(kg)	STD	ADD	PLACE	
13	GASKET	 t=1	HOT ROLLED CARBON STEEL	N	2	P90302-0184-022	S
14	PISTON CLEANING RING (DUMMY)		Cr-Si V/V STEEL	-	1	P90302-0184-034	S
15	PISTON CLEANING RING		Cr-Si V/V STEEL	N	1	P90302-0184-034	S

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTSC	CSP	JSP	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 003-4	STX Engine Co., Ltd.
Info. No.: 3-S50MC-C		Description: 3. CYLINDER LINER		Drawing no.: 2805013690(4/4)

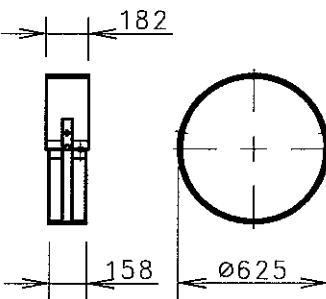
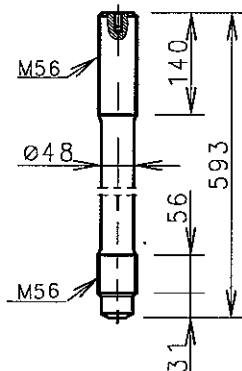
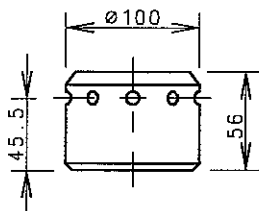
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE		PLACE	
					STD	ADD		
1	PISTON+ROD PISTON RING & STUFF. BOX		Cr-Mo STEEL (PISTON CROWN) GREY CAST IRON (PISTON SKIRT) CARBON STEEL (PISTON ROD) GREY CAST IRON (STUFF. BOX) 822&63.2	N	1		P90201-0204 & P90205-0115 REFER TO 15-2-5 PAGE	
							S	
2	PISTON RING -ALU-COAT		TARKLLOY-G 2.4	N	1		P90201-0204 -084	
							S	
3	PISTON RING -ALU-COAT		UBALLOY-S 1.8	2N	2		P90201-0204 -096	
							S	
4	PISTON RING -ALU-COAT		UBALLOY-S 1.8	N	1		P90201-0204 -106	
							S	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20061102		KCJ	JTSC	CSP	JSP	*	First issue	0
Scale:	Size:	Type:			Page No.:		STX Engine Co., Ltd.	
1:1	A4	S50MC-C			004-1			
Info. No.:		Description:					Drawing no.:	
4-S50MC-CA		4. PISTON					2805013700(1/3)	

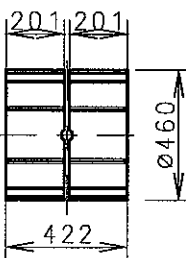
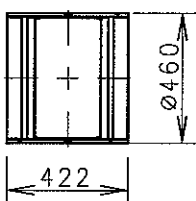
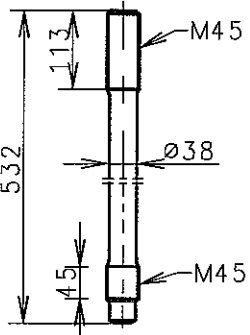
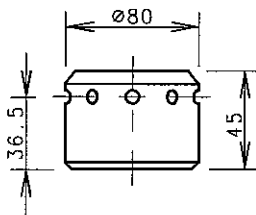
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE		PLACE	
					STD	ADD		
5	TELESCOPE PIPE		CARBON STEEL	N	1		P90401-0154-343	
			21.4				S	
6	HOUSING-STUFFING BOX		HOT ROLLED CARBON STEEL	N	1		P90401-0154-331	
			37.9				MS	
7	BUSHING		TIN-BRONZE	N	1		P90401-0154-318	
			2.1				MS	
8	SCREW, HEX. HEAD		Cr-Ni-Mo STEEL	4N	4		P90201-0204-060	
			0.7				MS	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20061102		KCJ	JTS	CSP	JSP	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C		Page No.: 004-2		SUX Engine Co., Ltd.
Info. No.: 4-S50MC-CA		Description: 4. PISTON					Drawing no.: 2805013700(2/3)	

NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS
			WEIGHT(kg)	WORKING	SPARE STD ADD		PLACE
9	O-RING		NBR	N	3	P90205-0115 -145	
			0.14				MS
10	D-RING		VITON	N	2	P90201-0204 -179	
			0.224				MS
11	D-RING		VITON	N	2	P90201-0204 -167	
			0.173				MS

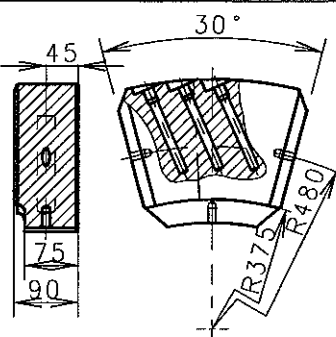
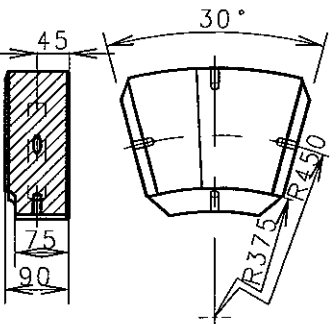
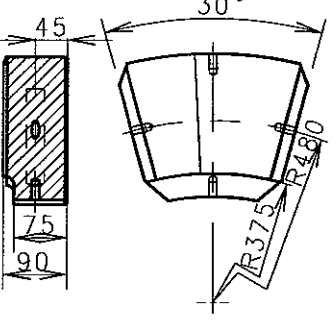
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTSC	CSP	JSP	*	First issue	0

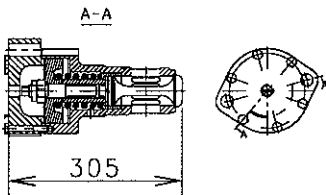
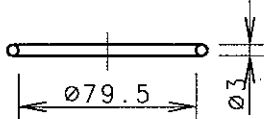
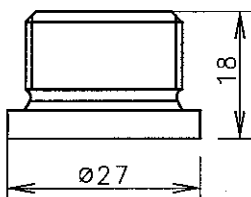
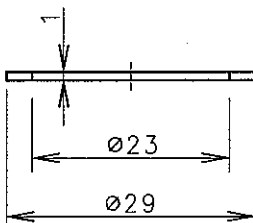
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Info. No.: 4-S50MC-CA		Description: 4. PISTON	Drawing no.: 2805013700(3/3)	

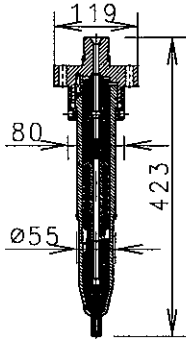
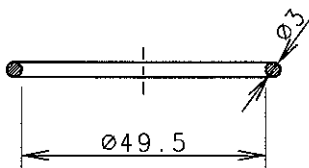
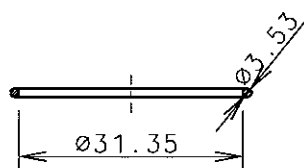
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS
			WEIGHT(kg)	WORKING	SPARE		PLACE
					STD	ADD	
1	CON-ROD BEARING COMPLETE		TIN-ALUMINIUM	N	1		P90401-0154-068
			30.3				MS
2	CON-ROD STUD		Cr-Ni-Mo STEEL	2N	2		P90401-0154-463
			9.3				MS
3	NUTS FOR HYDRAULIC TIGHTENING		CARBON STEEL	2N	2		P90401-0154-451
			2.42				MS
				</			

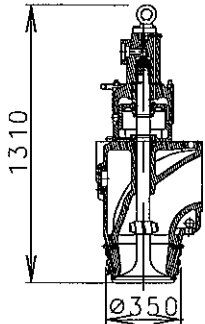
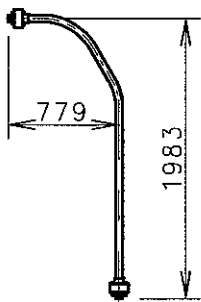
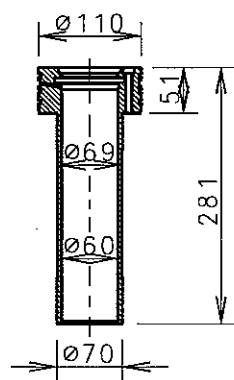
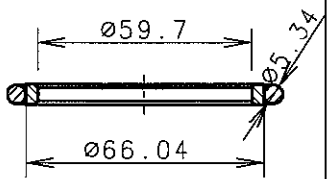
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE		PLACE	
				STD	ADD			
1	BRG. SHELL - CROSSHEAD (LOWER)		TIN- ALUMINIUM	N	1	P90401-0154 -140		
			28.0				S	
2	BRG. SHELL - CROSSHEAD (UPPER)		WHITE METAL	N	1	P90401-0154 -223		
			15.0				S	
3	CON-ROD STUD		Cr-Ni-Mo STEEL	4N	4	P90401-0154 -379		
			5.2				MS	
4	NUTS FOR HYDRAULIC TIGHTENING		CARBON STEEL	4N	4	P90401-0154 -380		
			1.22				MS	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090212		YWH	HCY	TBK	TBK	*	First issue	0
Scale: 1:1		Size: A4	Type: S50MC-C			Page No.: 006-1		SUX Engine Co., Ltd.
Info. No.: 6-S50MC-C		Description: 6. CROSSHEAD BEARING					Drawing no.: 2805030060(1/2)	

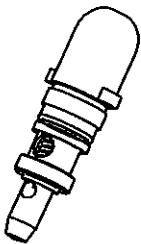
[illegible]

NO.	NAME	SKETCH	MATERIAL WEIGHT (kg)	Q'TY/ENG.		PLATE NO.	REMARKS	
				WOR- KING	SPARE		PLACE	
				STD	ADD			
1	SEGMENT		CARBON STEEL WITH WHITE METAL 26.35	1	1	P90505-0129 -063	S	
2	SEGMENT		CARBON STEEL WITH WHITE METAL 27.1	8	8	P90505-0129 -291	S	
3	SEGMENT		CARBON STEEL WITH WHITE METAL 27.1	7	7	P90505-0129 -075	S	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20080110		KCJ	JTSC	CSP	SKN	*	First issue	0
Scale:		Size:		Type:		Page No.:		STX Engine Co., Ltd.
1:1		A4		S50MC-C		007-1		
Info. No.:		Description:					Drawing no.:	
7-7S50MC-C		7. THRUST BEARING					2805021990	

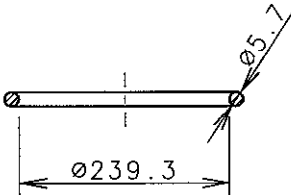
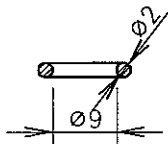
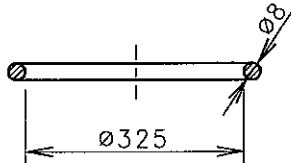
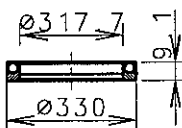
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.			PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE	STD		ADD	PLACE
1	STARTING V/V		CARBON STEEL + 13%Cr STEEL	N	1		P90704-0058		
			17.968					MS	
2	SEALING RING		RUBBER	N	2		P90704-0058 -124		
			0.004					MS	
3	PLUG SCREW		CARBON STEEL	N	2		EN47AR22 (PART NO.)		
								MS	
4	PACKING -ROUND		CARBON STEEL	N	2		1776190-7.1 (PART NO)		
			0.005					MS	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement		C.No
						*			3
						*			2
20091218		SJJ	SML	TBK	TBK	*	Pos No. 5 deleted		1
20090212		YWH	HCY	TBK	TBK	*	First issue		0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 008-1		STX Engine Co., Ltd.
Info. No.: 8-S50MC-CA		Description: 8. VALVES ON CYL. COVER						Drawing no.: 2805030071	

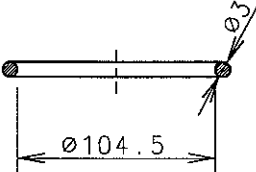
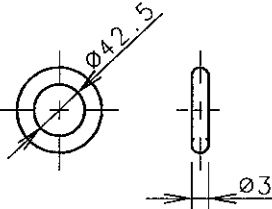
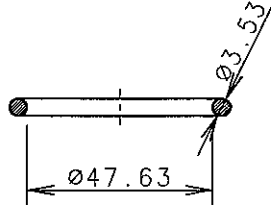
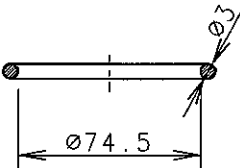
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS
			WEIGHT(kg)	WORKING	SPARE		PLACE
					STD	ADD	
1	FUEL VALVE		Cr-Mo-Al NITRIDING STEEL (SPINDLE GUIDE)	2N	14	P90910-0145	
			7.366				MS
2	SEALING RING		VITON	2N	2	P90910-0145 -124	
			0.003				MS
3	O-RING		VITON	2N	2	P90910-0145 -148	
			0.002				MS

NO.	NAME	SKETCH	MATERIAL WEIGHT(kg)	Q'TY/ENG.			PLATE NO.	REMARKS	
				WORKING	SPARE			PLACE	
				STD	ADD				
1	EXHAUST VALVE		NIMONIC 449.289	N	2		P90801-0213	REFER TO 15-2-6 PAGE S	
2	HYD. PIPE COMPLETE		CARBON STEEL 31.552	N	1		P90806-0105 -010	S	
3	SPINDLE GUIDE LOWER		GREY CAST IRON 3.8	N	1		P90801-0213 -133	MS	
4	SEALING RING		VITON 0.02	N	2		P90801-0213 -121	MS	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement		C.No
						*			3
						*			2
						*			1
20070711		KCJ	JT	SC	SP	SKN	* First issue		0
Scale:		Size:		Type:		Page No.:		STX Engine Co., Ltd.	
1 : 1		A4		S50MC-C		010-1			
Info. No.:		Description:					Drawing no.:		
10-S50MC-C-NA-1		10. EXHAUST VALVE					2805019260(1/5)		

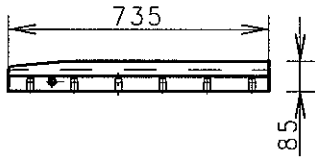
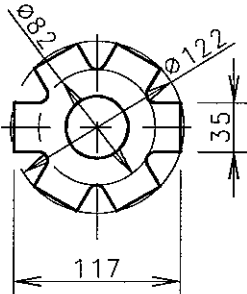
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS
			WEIGHT(kg)	WORKING	SPARE		PLACE
					STD	ADD	
5	SAFETY VALVE		TYPE : HY/VA 2/15. 50A(1/1) 0 532 002 065 0.212	N	2		P90804-0007 -033 <

NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS		
			WEIGHT(kg)	WORKING	SPARE		PLACE		
					STD	ADD			
9	PACKING-RECTANGLE	T=1	EN14R	N	2		P90803-0049-075	MS	
10	NON-RETURN VALVE, MALE	Ø19 G1/4A Ø10 65	FREE CUTTING STEEL	N	2		P90803-0049-482	MS	
11	PACKING, OVAL	R12 R24 Ø34 Ø11 30 30 T=2	EN14R	2N	2		P90801-0213-216	MS	
12	O-RING	Ø59.3 Ø5.7	VITON	2N	2		P90803-0049-408	MS	
			0.01					MS	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement		C.No
						*			3
						*			2
						*			1
20070711		KCJ	JTSC	SP	SKN	*	First issue		0
Scale:		Size:		Type:			Page No.:		STX Engine Co., Ltd.
1:1		A4		S50MC-C			010-3		
Info. No.:		Description:					Drawing no.:		
10-S50MC-C-NA-1		10. EXHAUST VALVE					2805019260(3/5)		

NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE		PLACE	
				STD	ADD			
13	SEALING RING		NBR	N	2	P90803-0049 -457		
			0.025				MS	
14	O-RING		NBR	N	2	P90803-0049 -278		
			0.001				MS	
15	O-RING		VITON	N	2	P90801-0213 -086		
			0.248				MS	
16	SEALING RING		POLYTETRA FLUORO ETHYLENE	N	2	P90801-0213 -098		
			0.083				MS	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20070711		KCJ	JT	SC	PS	SKN	* First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C		Page No.: 010-4		STX Engine Co., Ltd.
Info. No.: 10-S50MC-C-NA-1		Description: 10. EXHAUST VALVE					Drawing no.: 2805019260(4/5)	

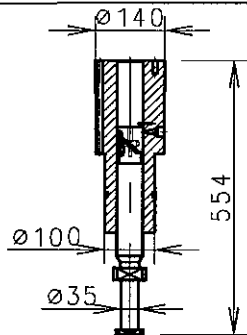
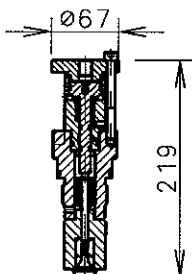
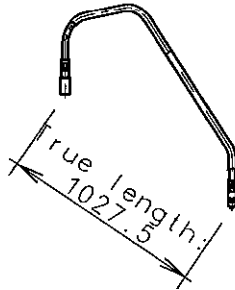
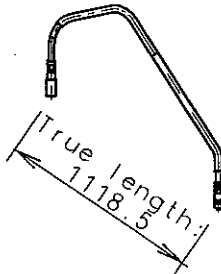
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE		PLACE	
				STD	ADD			
17	O-RING		VITON	3N	6	P90801-0213 -145		
			0.006				MS	
18	SEALING RING		VITON	2N	2	P90806-0105 -142		
			0.002				MS	
19	O-RING		VITON	2N	2	P90806-0105 -225		
			0.001				MS	
20	SEALING RING		NBR	2N	2	P90806-0105 -166		
			0.002				MS	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20070711		KCJ	JTSC	SP	SKN	*	First issue	0
Scale:		Size:		Type:		Page No.:		STX Engine Co., Ltd.
1:1		A4		S50MC-C		010-5		
Info. No.:		Description:					Drawing no.:	
10-S50MC-C-NA-1		10. EXHAUST VALVE					2805019260(5/5)	

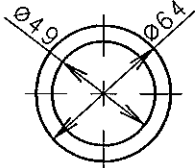
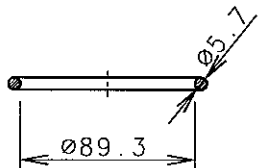
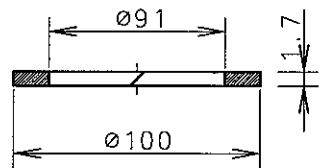
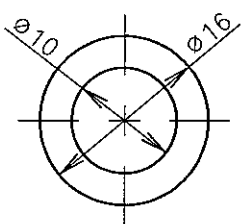
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE		PLACE	
					STD	ADD		
1	INNER&OUTER LINKS		STEEL	174	6		P90601-0099-327 (not applied 2nd order moment comp.) or P90601-0073-148 (applied 2nd order moment comp.)	
			6.0				MS	
2	THRUST RING		LEADED Sn-Zn BRONZE	4	4		P90610-0113-172	
			0.75				MS	
3	BEARING SHELL		CARBON STEEL & WHITE METAL	2	2		P90610-0113-039	
			2.8				MS	
4	GUIDE BAR		HOT ROLLED CARBON STEEL	2	1		P90601-0099-089 (not applied 2nd order moment comp.) or P90601-0073-100 (applied 2nd order moment comp.)	
			14.95				S	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20061102		KCJ	JTS	CSP	JSP	*	First issue	0
Scale:		Size:	Type:			Page No.:		STX Engine Co., Ltd.
1:1		A4	S50MC-C			011-1		
Info. No.:		Description:					Drawing no.:	
11-S50MC-C		11.CAMSHAFT DRIVING GEAR					2805013840(1/2)	

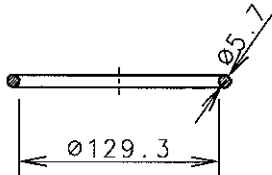
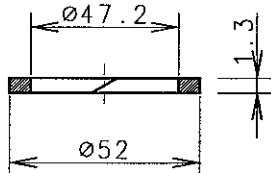
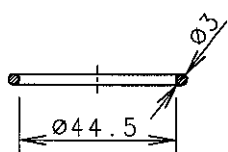
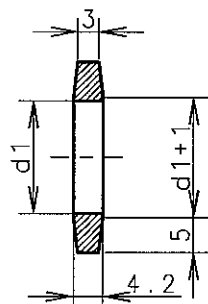
NO.	NAME	SKETCH	MATERIAL WEIGHT (kg)	Q'TY/ENG.		PLATE NO.	REMARKS PLACE
				WORKING	SPARE STD ADD		
5	GUIDE BAR		HOT ROLLED CARBON STEEL 11.5	4	1	P90601-0099-016 (not applied 2nd order moment comp.) or P90601-0073-041 (applied 2nd order moment comp.)	S
6	LOCKING PLATE		HOT ROLLED STEEL SHEET 0.1	2	2	P90603-0114-029	S

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 011-2	SIX Engine Co., Ltd.
Info. No.: 11-S50MC-C		Description: 11.CAMSHAFT DRIVING GEAR		Drawing no.: 2805013840(2/2)

NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE		PLACE	
				STD	ADD			
1	PUMP BARREL		Cr-Mo-Al NITRIDING STEEL	N	1	P90901-0203 -711		
			30.77				MS	
2	COMB. PUNCTURE + SUCT. V/V		Ni-Cr CASE HARDENING STEEL + HARDENED & TEMPERED Cr-Mo STEEL	N	1	P90901-0203 -772		
			1.594				MS	
3	HIGH PRESS FUEL OIL PIPE		Carbon Steel	N	1	P90913-0138 -012		
			7.462				S	
4	HIGH PRESS FUEL OIL PIPE		Carbon Steel	N	1	P90913-0138 -120		
			7.462				S	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20061102		KCJ	JTSCSP	JSP	*	First issue		0
Scale:		Size:		Type:		Page No.:		STX Engine Co., Ltd.
1:1		A4		S50MC-C		012-1		
Info. No.:		Description:					Drawing no.:	
12-S50MC-C		12. FUEL PUMP & HIGH PRESS PIPE					2805013850(1/4)	

NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE		PLACE	
					STD	ADD		
5	PACKING ROUND	 T=2	HOT ROLLED CARBON STEEL 0.021	N	2		P90901-0203-054 MS	
6	SEALING RING		VITON 0.018	N	1		P90901-0203-257 MS	
7	BACK-UP RING		PTFE 0.002	N	2		P90901-0203-269 MS	
8	PACKING ROUND	 T=1	HOT ROLLED CARBON STEEL 0.001	N	2		P90901-0203-198 MS	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20061102		KCJ	JT	SCSP	JSP	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C		Page No.: 012-2		STX Engine Co., Ltd.
Info. No.: 12-S50MC-C		Description: 12. FUEL PUMP & HIGH PRESS PIPE					Drawing no.: 2805013850(2/4)	

NO.	NAME	SKETCH	MATERIAL WEIGHT (kg)	Q'TY/ENG.			PLATE NO.	REMARKS	
				WOK- KING	SPARE			PLACE	
STD	ADD								
9	SEALING RING		VITON 0.026	N	2		P90901-0203 -365	MS	
10	BACK-UP RING		PTFE 0.002	2N	2		P90901-0203 -377	MS	
11	SEALING RING		VITON 0.002	N	2		P90901-0203 -389	MS	
12	PACKING FELT	 d1=16~22	TYPE : FI 110 0.001	N	2		P90901-0203 -316	MS	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement		C.No
						*			3
						*			2
						*			1
20061102		KCJ	JTSC	CSP	JSP	*	First issue		0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 012-3		STX Engine Co., Ltd.
Info. No.: 12-S50MC-C		Description: 12. FUEL PUMP & HIGH PRESS PIPE					Drawing no.: 2805013850(3/4)		

NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS
				WOR-KING	SPARE		
			WEIGHT(kg)	STD	ADD	PLACE	
13	SHIM		HOT ROLLED CARBON STEEL	16N	48	P90901-0203-149	MS
			0.04				
14	SEALING RING		VITON	N	2	P90901-0203-627	MS
			0.003				
15	O-RING		VITON	4N	12	P90913-0138-048	MS
			0.001				
16	SEALING RING		VITON	2N	6	P90913-0138-107	MS
			0.002				

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

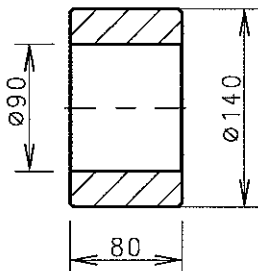
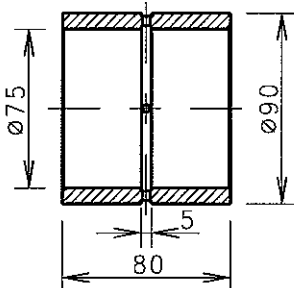
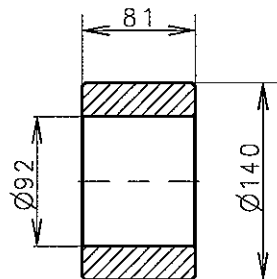
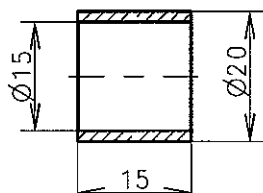
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1:1	A4	S50MC-C	012-4	

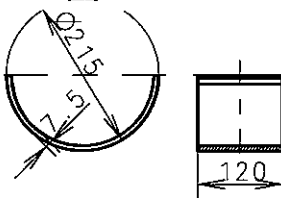
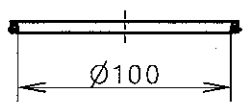
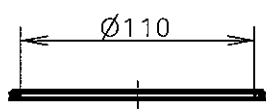
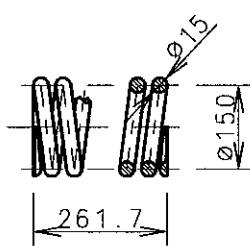
Info. No.:	Description:	Drawing no.:
12-S50MC-C	12. FUEL PUMP & HIGH PRESS PIPE	2805013850(4/4)

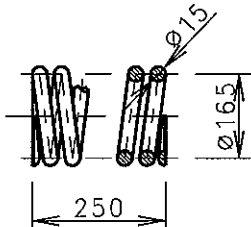
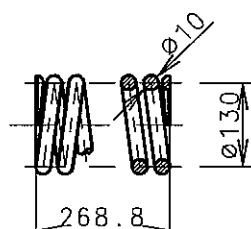
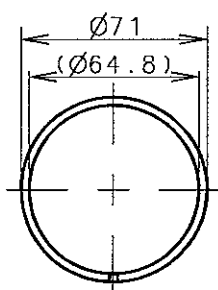
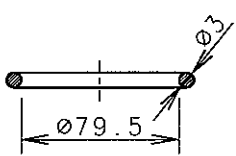
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.			PLATE NO.	REMARKS
				WEIGHT(kg)	WOKING	SPARE		
			STD				ADD	PLACE
1	CYLINDER LUBRICATOR		HOT ROLLED CARBON STEEL	2	1		90305-0083-017	
			42.0					S
2	COUPLING		TYPE : SPECIAL 1-28	3	1		90305-0083-030	
			0.809					MS

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20080110	KCJ	JTS	CSP	SKN	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 013-1	
Info. No.: 13-7S50MC-C-LCD	Description: 13.CYLINDER LUBRICATOR	Drawing no.: 2805022000		

NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE		PLACE	
				STD	ADD			
1	ROLLER		Ni-Cr CAST HARDENING STEEL	N	1	P90805-0141 -302		
			5.7					
2	BUSHING		TIN- BRONZE	N	1	P90805-0141 -135		
			1.3					
3	ROLLER		Ni-Cr CAST HARDENING STEEL	N	1	P90902-0131 -073		
			5.7					
4	BUSHING		ALLOY STEEL	N	1	P90902-0131 -181		
			0.023					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20061102		KCJ	JTSCSP	JSP	*	First issue		0
Scale: 1:1		Size: A4	Type: S50MC-C			Page No.: 014-1		STX Engine Co., Ltd.
Info. No.: 14-S50MC-C		Description: 14. FUEL & EXHAUST GEAR					Drawing no.: 2805013980(1/4)	

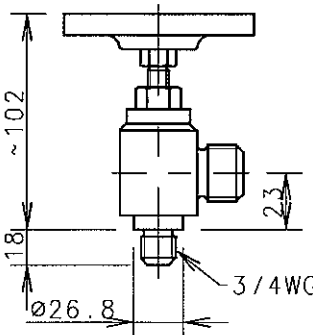
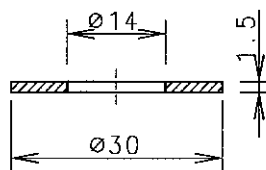
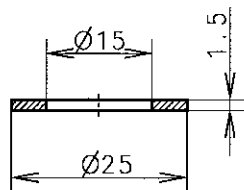
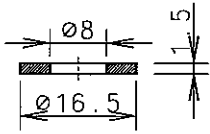
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE		PLACE	
					STD	ADD		
5	BEARING SHELL		CARBON STEEL WHITE METAL	N	1		P90613-0113 -118	
			2.57					
6	SCRAPER RING		TYPE : KL-099 -03935 -1000	N	2		P90902-0131 -431	
			0.01					
7	SCRAPER RING-AXIAL		TYPE : KL-099 -86055 -1100	N	2		P90902-0131 -418	
			0.012					
8	PRESS. SPRING RIGHT-HAND		Cr-V VALVE SPRING STEEL	N	1		P90805-0141 -279	
			4.57					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20061102		KCJ	JTSCSP	JSP	*	First issue		0
Scale: 1:1		Size: A4	Type: S50MC-C			Page No.: 014-2		SIX Engine Co., Ltd.
Info. No.: 14-S50MC-C		Description: 14. FUEL & EXHAUST GEAR					Drawing no.: 2805013980(2/4)	

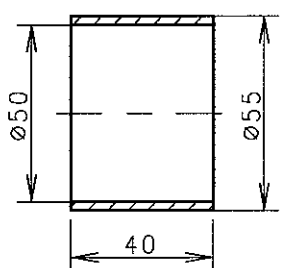
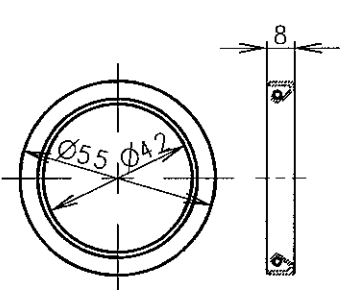
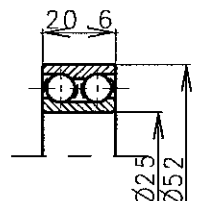
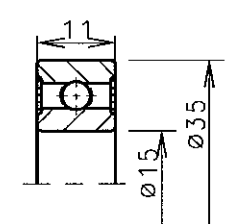
NO.	NAME	SKETCH	MATERIAL WEIGHT(kg)	Q'TY/ENG.		PLATE NO.	REMARKS
				WORKING	SPARE		PLACE
				STD	ADD		
9	PRESS. SPRING RIGHT-HAND		Cr-V VALVE SPRING STEEL 5.0	N	1	P90902-0131 -622	
10	PRESS. SPRING LEFT-HAND		Cr-V VALVE SPRING STEEL 2.25	N	1	P90902-0131 -610	
11	PISTON RING STEEL +C. IRON	 T=2.5	VITON 0.05	2N	4	P90805-0141 -231	
12	SEALING RING		VITON 0.004	N	2	P90902-0131 -671	

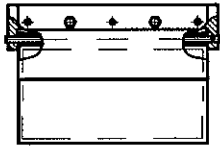
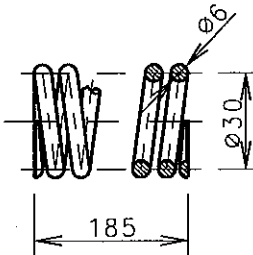
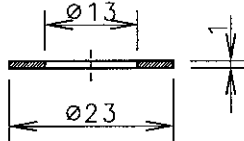
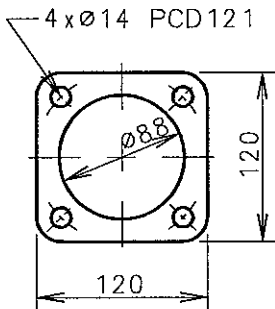
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement		C.No
					*			3
					*			2
					*			1
20061102	KCJ	JT	SCSP	JSP	*	First issue		0

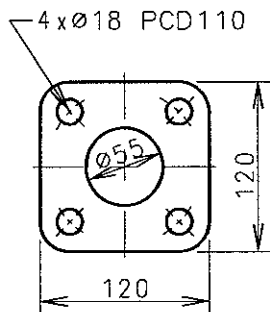
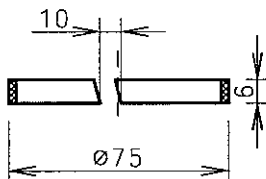
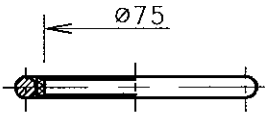
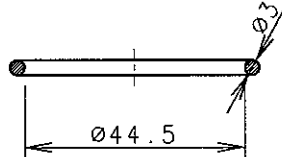
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Info. No.: 14-S50MC-C		Description: 14. FUEL & EXHAUST GEAR		Drawing no.: 2805013980(3/4)

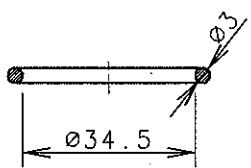
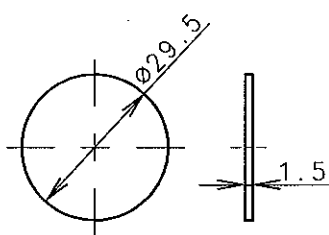
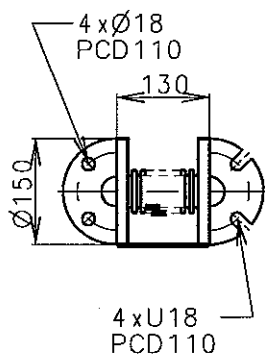
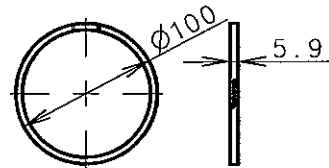
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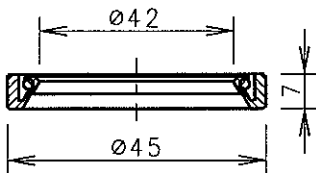
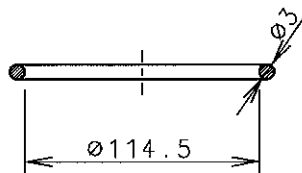
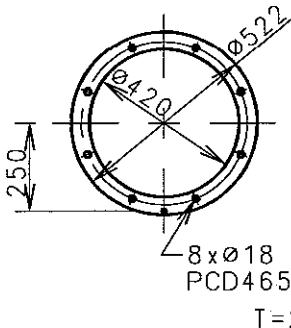
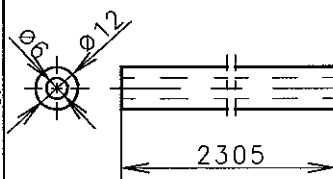
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			WEIGHT(kg)	WOR- KING	SPARE STD		ADD	PLACE
1	INDICATOR COCK-DETAIL		CARBON STEEL SPRING STEEL STAINLESS STEEL 1.0	N	2	P90612-0051 -303		
2	PACKING ROUND		COPPER BASE ALLOY 0.1	N	2	P90612-0051 -112		
3	PACKING ROUND		COPPER BASE ALLOY 0.1	N	2	P90612-0051 -268		
4	PACKING ROUND		COPPER BASE ALLOY 0.002	N	2	P90612-0051 -281		
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20060419		YWH	HCY	CSP	JSP	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 015-1	SDX Engine Co., Ltd.
Info. No.: 15-S50MC-C		Description: 15. INDICATOR VALVE					Drawing no.: 2805014000	

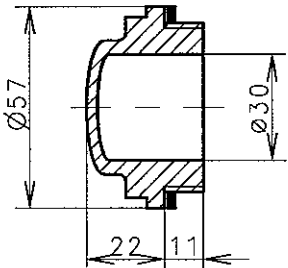
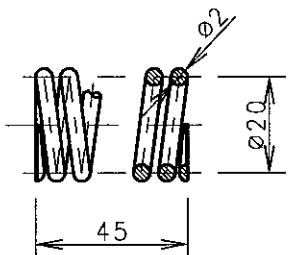
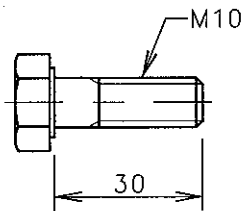
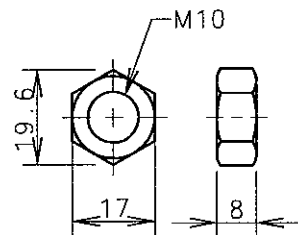
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			WEIGHT(kg)	WORKING	SPARE		PLACE	
				STD	ADD			
1	BUSHING		TYPE : PCM657070 BK 0.13	4	1	P90622-0072 -154		
2	SEALING RING		RUBBER SPRING STEEL 0.016	2	1	90620-0103 -714	OLD: 90620-0077 -643	
3	BALL BEARING		STEEL 0.19	1	1	90620-0103 -680	OLD: 90620-0077 -714	
4	BALL BEARING		STEEL 0.045	1	1	90620-0103 -631	OLD: 90620-0077 -476	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
20090123		KCJ	SML	JTS	TBK	*	Plate No. revised	1
20061101		KCJ	JTS	CSP	JSP	*	First issue	0
Scale:		Size:	Type:			Page No.:		STX Engine Co., Ltd.
1:1		A4	S50MC-C			016-1		
Info. No.:		Description:					Drawing no.:	
16-S50MC-C		16. OTHERS					2805014011(1/10)	

NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE		PLACE	
					STD	ADD		
5	NON-RETURN VALVE		CARBON STEEL	N	1		P91006-0026-122	
			13.566					
6	PRESS. SPRING RIGHT-HAND		CARBON STEEL VALVE SPRING WIRE	N	1		P90915-0053-407	
			0.42					
7	PACKING RING		EN14AA	N	1		P90915-0053	
			0.537					
8	PACKING RECTANGLE		EN14AA	N	1		P90915-0053-289	
			0.01					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
20090123		KCJ	SML	JTS	TBK	*	Plate No. revised	1
20061102		KCJ	JTS	CSP	JSP	*	First issue	0
Scale:		Size:		Type:		Page No.:		STX Engine Co., Ltd.
1:1		A4		S50MC-C		016-2		
Info. No.:		Description:					Drawing no.:	
16-S50MC-C		16. OTHERS					2805014011(2/10)	

NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE		PLACE	
					STD	ADD		
9	PACKING RECTANGLE		EN14AA	N	1		P90915-0053-241	
			0.01					
10	WEARING RING		TYPE : KL-99 -202-0750	N	1		P90915-0053-265	
			0.01					
11	SEALING RING		PTFE & VITON	N	1		P90915-0053-277	
			0.01					
12	SEALING RING		VITON	2	2		P90915-0053-013	
			0.002					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
20090123		KCJ	SML	JT	STBK	*	Plate No. revised	1
20061102		KCJ	JT	SCSP	JSP	*	First issue	0
Scale:		Size:		Type:		Page No.:		STX Engine Co., Ltd.
1:1		A4		S50MC-C		016-3		
Info. No.:		Description:					Drawing no.:	
16-S50MC-C		16. OTHERS					2805014011(3/10)	

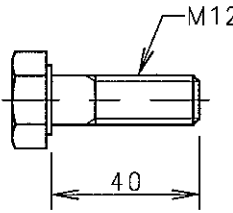
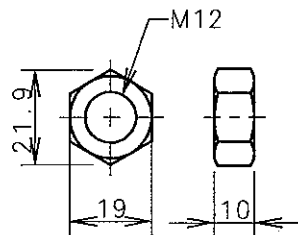
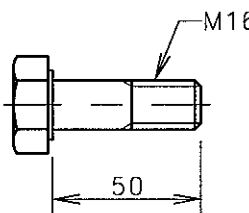
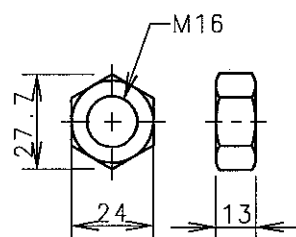
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.			PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE	PLACE			
				STD	ADD				
13	SEALING RING		VITON	1	1		P90915-0053 -086		
			0.002						
14	BURSTING DISC		COPPER BASE ALLOY	N	2		P91104-0080 -186		
			0.001						
15	COMPENSATOR (PCD110)		STAINLESS STEEL & MILA STEEL	N	1		P90810-0063 -359	OLD: P90810-0093 -036	
			4.1						
16	PISTON RING		PTFE		1		P90703-0122 -394		
			0.01						
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement		C.No
						*			3
						*			2
20090123		KCJ	SML	JTS	TBK	*	Plate No. revised		1
20061102		KCJ	JTS	CSP	JSP	*	First issue		0
Scale:		Size:		Type:			Page No.:		SIX Engine Co., Ltd.
1:1		A4		S50MC-C			016-4		
Info. No.:		Description:					Drawing no.:		
16-S50MC-C		16. OTHERS					2805014011(4/10)		

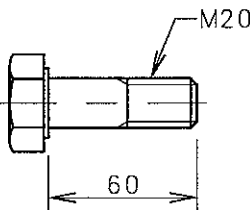
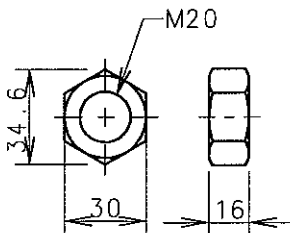
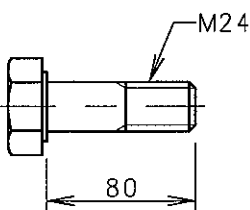
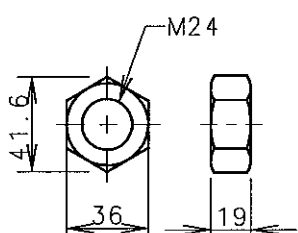
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE		PLACE	
					STD	ADD		
17	SEALING RING W. BACK-UP		TYPE : SM 32457 0.025		1		P90703-0122 -274	
18	SEALING RING		SILICONE RUBBER 0.006		1		P90703-0122 -370	
19	PACKING ROUND FOR RELIEF V/V		EN14R 0.3	7	2		P91205-0146 -148	
20	SEALING RING		NBR 0.29	N	2		P91205-0146 -028	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
20090123		KCJ	SML	JTS	TBK	*	Plate No. revised	1
20061102		KCJ	JTS	CSP	JSP	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C		Page No.: 016-5		STX Engine Co., Ltd. Drawing no.: 2805014011(5/10)
Info. No.: 16-S50MC-C		Description: 16. OTHERS						

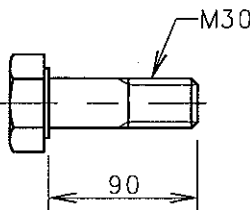
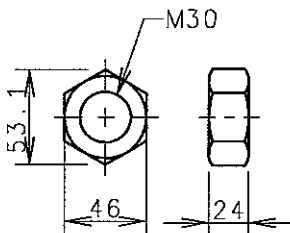
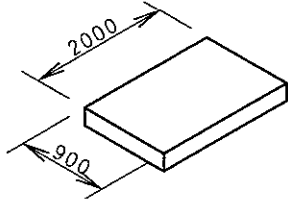
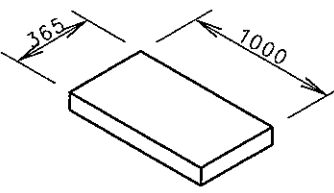
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				WORKING	SPARE STD ADD		
21	SIGHT GLASS		TRANS- PARENT ACRYLIC PLASTIC 0.1	N	1	P91207-0081 -113	
22	PRESS. SPRING RIGHT-HAND		CARBON STEEL VALVE SPRING WIRE 0.009	9N	9	P91205-0146 -435	
23	HEX. BOLT		10.9T 0.74		20	P000F0010030	
24	HEX. NUT		8T 0.011		20	P040H0010000	

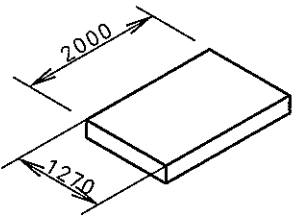
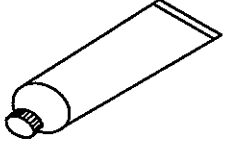
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20090123	KCJ	SML	JTS	TBK	*	Plate No. revised	1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale:	Size:	Type:	Page No.:	
1:1	A4	S50MC-C	016-6	
Info. No.: 16-S50MC-C				
Description:			Drawing no.:	
16. OTHERS			2805014011(6/10)	

NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS		
			WEIGHT(kg)	WORKING	SPARE			STD	ADD
25	HEX. BOLT		10.9T		20				
			0.047			P000F0012040			
26	HEX. NUT		8T		20				
			0.016			P040H0012000			
27	HEX. BOLT		10.9T		20				
			0.104			P000E0016050			
28	HEX. NUT		8T		20				
			0.03			P040H0016000			
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No	
						*		3	
						*		2	
20090123		KCJ	SML	JT	STBK	*	Plate No. revised	1	
20061102		KCJ	JT	CSP	JSP	*	First issue	0	
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Info. No.: 16-S50MC-C		Description: 16. OTHERS						Drawing no.: 2805014011(7/10)	

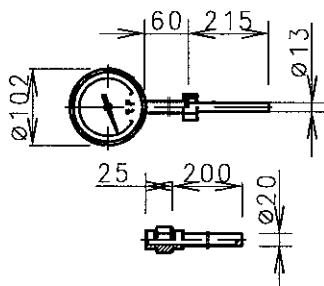
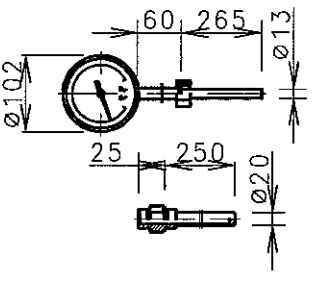
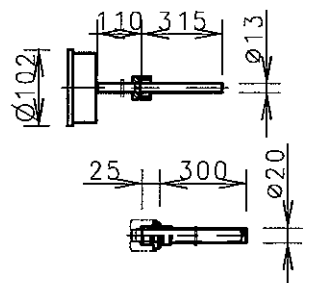
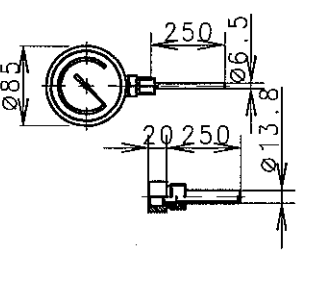
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE			STD
29	HEX. BOLT		10.9T		20			
			0.201			P000E0020060		
30	HEX. NUT		8T		20			
			0.055			P040H0020000		
31	HEX. BOLT		10.9T		20			
			0.374			P000E0024080		
32	HEX. NUT		8T		20			
			0.099			P040H0024000		
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
20090123		KCJ	SML	JTS	TBK	*	Plate No. revised	1
20061102		KCJ	JTS	CSP	JSP	*	First issue	0
Scale:		Size:		Type:			Page No.:	
1:1		A4		S50MC-C			016-8	
Info. No.:		Description:					Drawing no.:	
16-S50MC-C		16. OTHERS					2805014011(8/10)	

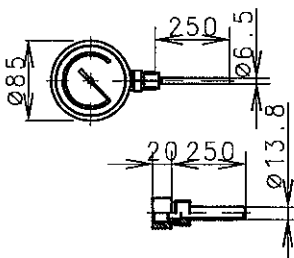
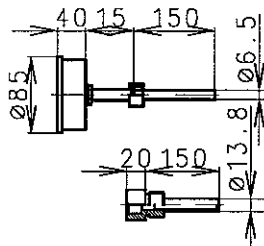
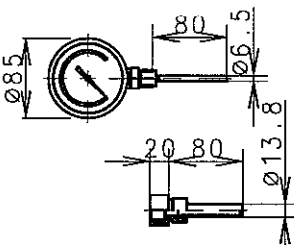
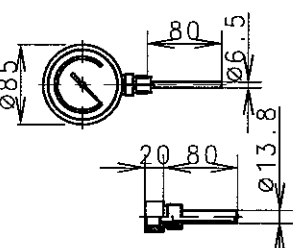
NO.	NAME	SKETCH	MATERIAL	Q'TY/ENG.		PLATE NO.	REMARKS	
			WEIGHT(kg)	WORKING	SPARE STD ADD	PART NO.	PLACE	
33	HEX. BOLT		10.9T		20			
			0.689			P000E0030090		
34	HEX. NUT		8T		20			
			0.202			P040H0030000		
35	RUBBER SHEET	 T=3.0	RUBBER		1			
			7.8			34500101		
36	COPPER SHEET	 T=0.5	COPPER		1			
			1.72			34500099		
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
20090123		KCJ	SML	JTS	TBK	*	Plate No. revised	1
20061102		KCJ	JTS	CSP	JSP	*	First issue	0
Scale:		Size:		Type:			Page No.:	
1:1		A4		S50MC-C			016-9	
Info. No.:		Description:					Drawing no.:	
16-S50MC-C		16. OTHERS					2805014011(9/10)	
STX Engine Co., Ltd.								

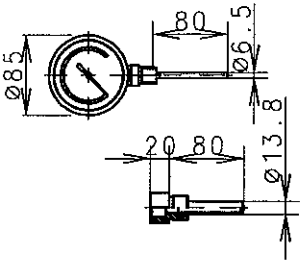
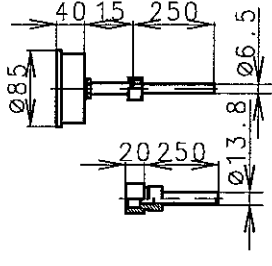
NO.	NAME	SKETCH	MATERIAL WEIGHT(kg)	Q'TY/ENG.		PLATE NO. PART NO.	REMARKS PLACE
				WORKING	SPARE STD ADD		
37	NON-ASBESTOS SHEET	 T=1.5	NON-ASBESTOS		1		
			5.98			34500100	
38	LIQUID PACKING		SILICON		3		
			0.12			34500102	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
20090123	KCJ	SML	JTS	TBK	*	Plate No. revised	1
20061102	KCJ	JTS	CSP	JSP	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 016-10	
Info. No.: 16-S50MC-C		Description: 16. OTHERS		
				Drawing no.: 2805014011(10/10)

NO.	NAME	SKETCH	MATERIAL WEIGHT (kg)	Q' TY/ENG.		RANGES PART NO.	WORKING POSITION PLACE		
				WOR- KING	SPARE STD ADD				
1	THERMOMETER		MAKER	N	1	50~650°C 5154022720-05	EXH. GAS OUTLET		
2	THERMOMETER		MAKER	1	1	50~650°C 5154022720-07	EXH. GAS T/C INLET		
3	THERMOMETER		MAKER	1	1	50~650°C 5154022730-03	EXH. GAS T/C OUTLET		
4	THERMOMETER		MAKER	1	1	0~300°C 5154022740-04	SCA. AIR BEFORE AIR COOLER		
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement		C.No
						*			3
						*			2
						*			1
20071002		KCJ	JTSC	SP	SKN	*	First issue		0
Scale:		Size:		Type:			Page No.:		STX Engine Co., Ltd.
1:1		A4		S50MC-C			017-1		
Info. No.:		Description:					Drawing no.:		
7S50MC-CSPARE2		17. SPARE FOR THERMOMETER					2805020350(1/3)		

NO.	NAME	SKETCH	MATERIAL WEIGHT (kg)	Q'TY/ENG.			RANGES PART NO.	WORKING POSITION PLACE	
				WOR- KING	SPARE				
			STD		ADD				
5	THERMOMETER		MAKER	1	1		0~120°C	SCA. AIR RECEIVER	
							5154022750-04		
6	THERMOMETER		MAKER	N+3	1		0~100°C	P.C.O OUTLET MLO & PCO INLET C.W INLET C.W OUTLET	
							5154022760-02		
7	THERMOMETER		MAKER	3	1		0~100°C	T/C L.O OUTLET J.C.W INLET J.C.W OUTLET -COMMON	
							5154022770-01		
8	THERMOMETER		MAKER	1	1		0~200°C	F.O INLET	
							5154022780-1		
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement		C.No
						*			3
						*			2
						*			1
20071002		KCJ	JTSC	CSP	SKN	*	First issue		0
Scale:		Size:		Type:			Page No.:		STX Engine Co., Ltd.
1:1		A4		S50MC-C			017-2		
Info. No.:		Description:					Drawing no.:		
7S50MC-CSPARE2		17. SPARE FOR THERMOMETER					2805020350(2/3)		

NO.	NAME	SKETCH	MATERIAL WEIGHT(kg)	Q'TY/ENG.		RANGES	WPRKING POSITION
				WOR-KING	SPARE		
				STD	ADD	PART NO.	PLACE
9	THERMOMETER		MAKER	N	1	0~150 °C	J.C.W OUTLET
						5154022790-01	
10	THERMOMETER		MAKER	1	1	0~100 °C	SCA. AIR AFTER AIR COOLER
						5154022760-04	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20071002	KCJ	JTS	CSP	SKN	*	First issue	0

Scale: 1 : 1 Size: A4 Type: S50MC-C

Page No.: 017-3

STX Engine Co., Ltd.

Info. No.: 7S50MC-CSPARE2

Description: 17. SPARE FOR THERMOMETER

Drawing no.: 2805020350(3/3)

LIST OF LARGE SPARE PARTS

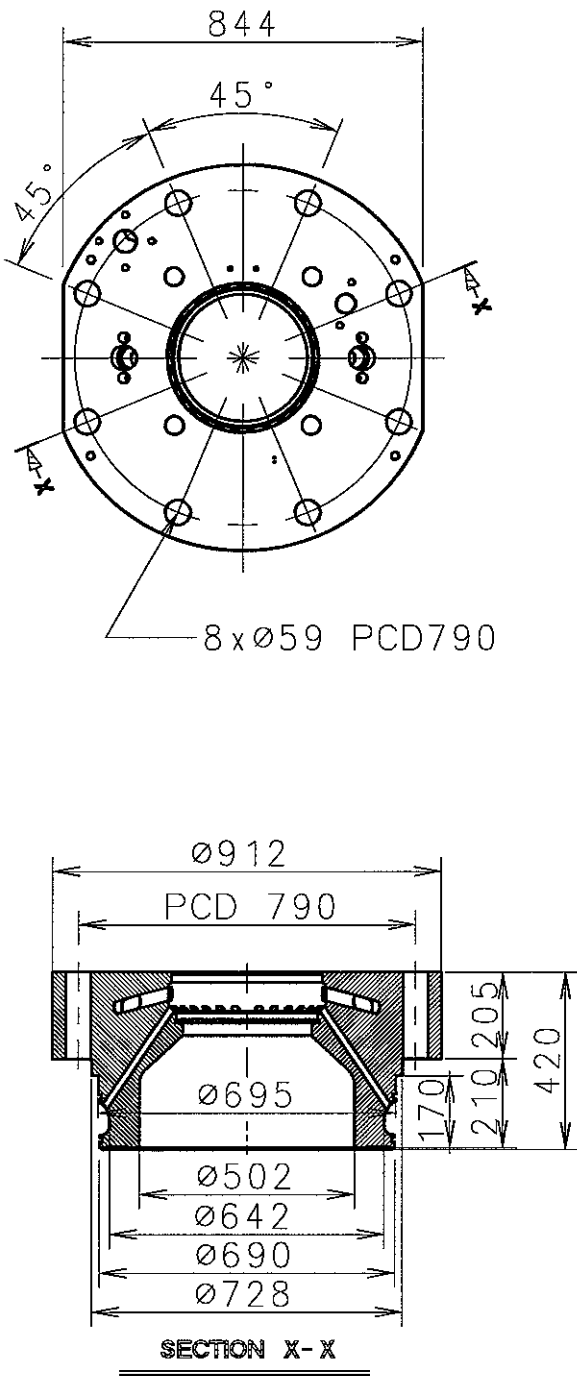
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Date	Des.	Chk.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070416	YWH	HCY	CSP	SKN	*	First issue	0
Scale: 1:1	Size: A4	Type: S50MC-C				Page No.: 15-2-1	 SUX Engine Co., Ltd.
Info No.:		Description: LIST OF LARGE SPARE PARTS				Drawing no.: 2805018180(1/2)	

INDEX

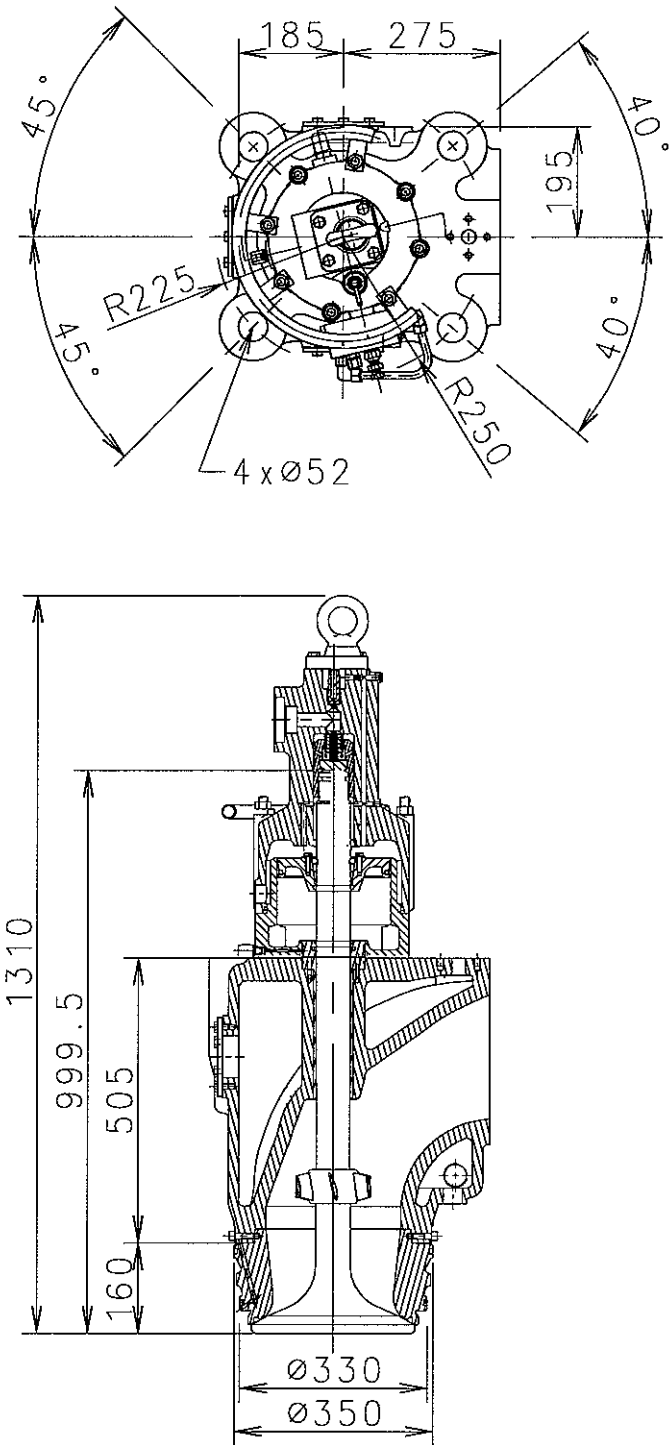
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Date	Des.	Chk.	Chk.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070416	YWH	HCY	CSP	SKN	*	First issue	0
Scale: 1:1	Size: A4	Type: S50MC-C				Page No.: 15-2-2	 STX Engine Co., Ltd.
Info. No.:		Description: LIST OF LARGE SPARE PARTS				Drawing no.: 2805018180(2/2)	

NO.	NAME	SKETCH	REMARKS				
			WEIGHT (kg)				
1	CYL. COVER		1089				
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061103	KCJ	JTS	CSP	JSP	*	First issue	0
Scale:	Size:	Type:			Page No.:		 STX Engine Co., Ltd.
1:1	A4	S50MC-C			15-2-3		
Info. No.:		Description:					Drawing no.:
		OUTLINE DWG. OF LARGE SPARE PARTS					2805014150

NO.	NAME	SKETCH	REMARKS				
			WEIGHT (kg)				
2	CYL. LINER		1630				
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061103	KCJ	JTS	CSP	JSP	*	First issue	0
Scale:	Size:	Type:		Page No.:		STX Engine Co., Ltd.	
1:1	A4	S50MC-C		15-2-4			
Info. No.:	Description:					Drawing no.:	
	OUTLINE DWG. OF LARGE SPARE PARTS					2805014160	

NO.	NAME	SKETCH	REMARKS				
			WEIGHT (kg)				
3	PISTON CROWN/ROD /SKIRT/RING & STUFF. BOX	Technical drawing of a piston assembly. The side view shows a piston crown with a width of $\varnothing 496$, a skirt diameter of $\varnothing 195$, and a total height of 2855. The skirt has a thickness of 98. The top view shows a rectangular shape with a width of $\varnothing 425$ and a height of 220. The PCD (Pitch Circle Diameter) of the skirt is 286.					
			839&63				
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20061103	KCJ	JTSCSP	JSP	*	First issue		0
Scale: 1:1	Size: A4	Type: S50MC-C				Page No.: 15-2-5	STX Engine Co., Ltd.
Info. No.:		Description: OUTLINE DWG. OF LARGE SPARE PARTS				Drawing no.: 2805014170	

NO.	NAME	SKETCH	REMARKS				
			WEIGHT (kg)				
4	EXH. VALVE		449.3				
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070416	YWH	HCY	CSP	SKN	*	First issue	0
Scale:	Size:	Type:			Page No.:		STX Engine Co., Ltd.
1:1	A4	S50MC-C			15-2-6		
Info. No.:		Description:					Drawing no.:
		OUTLINE DWG. OF LARGE SPARE PARTS					2805018190

LIST OF TOOL PARTS FOR MAIN ENGINE

NOTE

The items marked with "T" in this list are to be arranged at the hull side and the steel box for strong them are not supplied

Eng. room : T

Steel box : MT (Tool to be not mounted on panel)

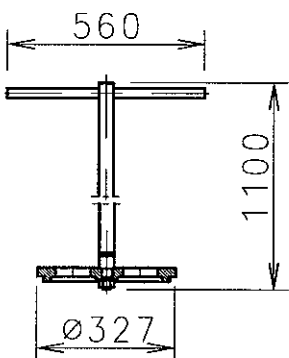
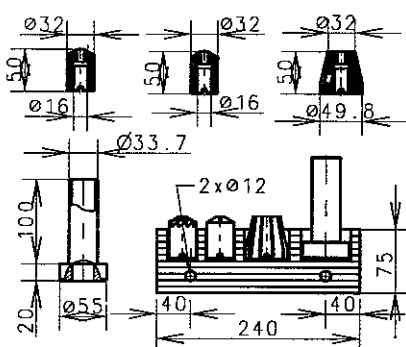
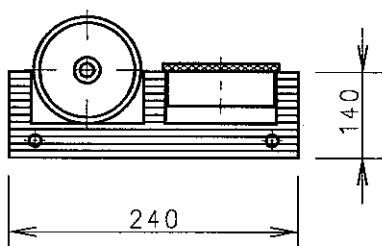
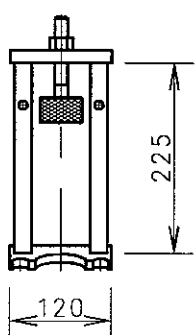
Type A - 800*400*300 = 19EA / TYPE B - 600*600*150 = 1EA

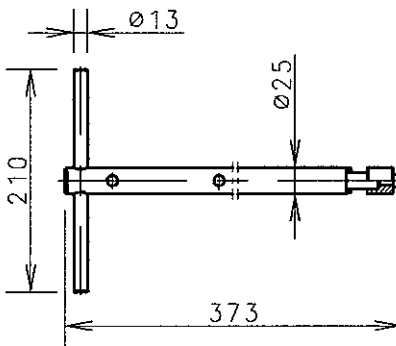
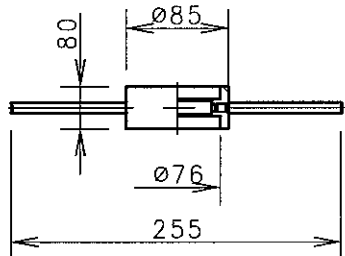
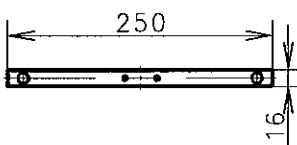
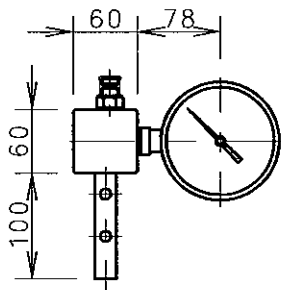
Type C - 650*550*500 = 1EA / TYPE D - 1350*350*300 = 1EA

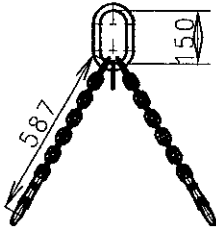
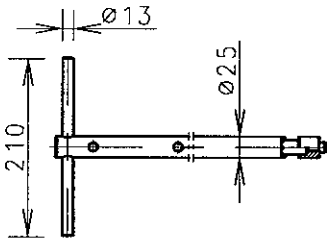
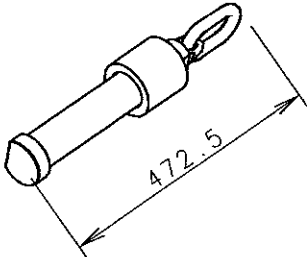
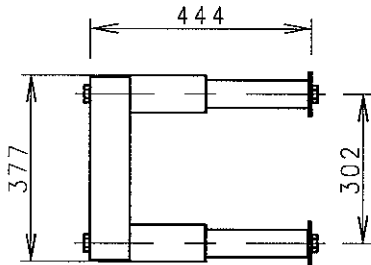
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					*		2
20100614	SJJ	DHE	JTS	TBK	*	Pos No. 1, 4 revised	1
20090317	KCJ	SML	TBK	TBK	*	First issue	0
Scale: 1:1	Size: A4	Type: S50MC-C				Page No.: 15-3-1	STX Engine Co., Ltd.
Info No.: S50MC -C7-01-22		Description: LIST OF TOOL PARTS FOR MAIN ENGINE					Drawing no.: 2805030861(1/2)

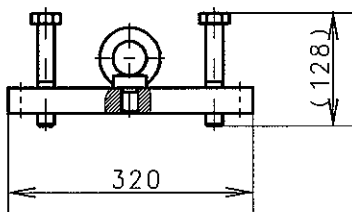
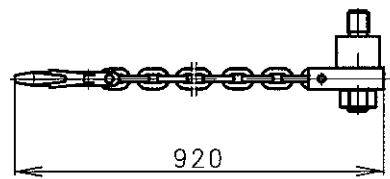
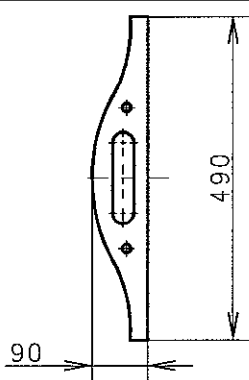
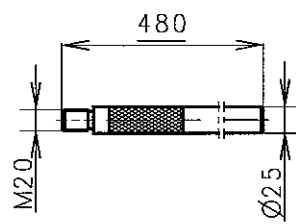
INDEX

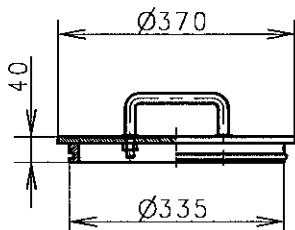
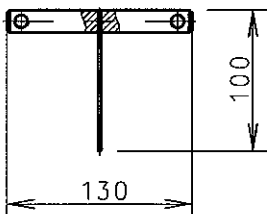
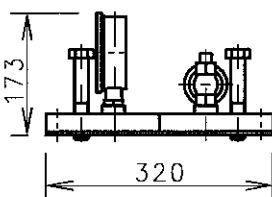
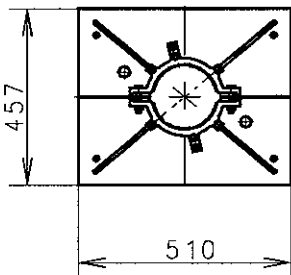
NO.	DESCRIPTION	PAGE NO.	DRAWING NO.	REV.
1	CYLINDER COVER	101-1~3	2805031870	△1
2	PISTON ROD, STUFFING BOX	102-1~4	2805030870	
3	CROSSHEAD + CON-ROD	103-1~3	2805029610	
4	CRANKSHAFT + THRUST BEARING	104-1~4	2805033620	△1
5	MECH. CONTROL GEAR	105-1~2	2805029630	
6	EXHAUST VALVE	106-1~3	2805029640	
7	FUEL OIL SYSTEM	107-1~9	2805030810	
8	TURBOCHARGER SYSTEM	108-1~3	2805029660	
9	GENERAL TOOLS	109-1~26	2805029670	
10	STDANDARD HYDR. JACK	110-1~18	2805029680	
11	HYDR. JACK-H/D & E/C BOLT	111-1~5	2805029690	
12	PLANIMETER	112-1	2805030840	
13	HYD. JACK PROPELLER	113-1~2	9005000360	

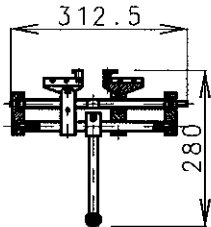
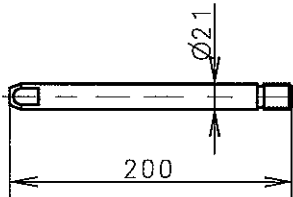
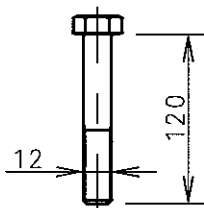
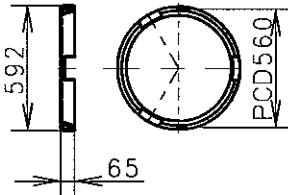
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT(kg)	STD	ADD		REMARKS
1	GRIND. TOOL- EXH. V/V SEAT		CAST IRON	1		P90151-0199 -046	MS901(046)
			18.96				
2	MILL+GRIND. WHEEL, FUEL VALVE		HIGH ALLOY STEEL ALLOY STEEL BEECH WOOD	1		P90151-0199 -058	MS901(058)
			1.84				
3	MILL+GRIND. WHEEL, STARTING V/V		HIGH ALLOY STEEL BEECH WOOD	1		P90151-0199 -060	MS901(060)
			6.04				
4	DISM. LEVER FOR FUEL VALVE		CARBON STEEL	1		P90151-0199 -071	MS901(071)
			4.39				
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement
						*	
						*	
						*	
20090617		YWH	SMK	HCY	TBK	*	First issue
Scale:		Size:	Type:	S50MC-C		Page No.:	C.No.
1:1		A4				101-1	3
Info. No.:		Description:	1. CYLINDER COVER		SUX Engine Co., Ltd.		2
S50MC -C7/C8STP901A					Drawing no.:		1
					2805031870(1/3)		0

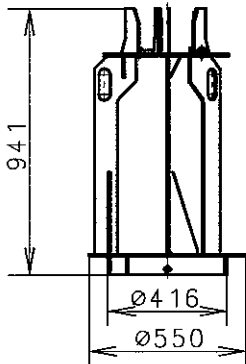
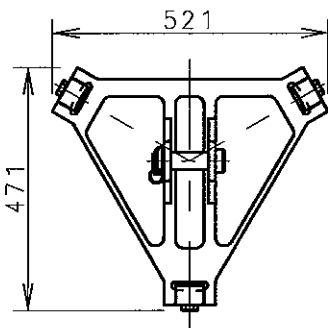
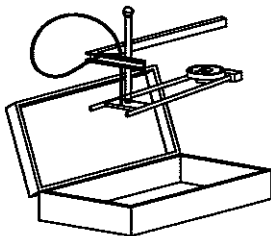
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
5	HANDLE SUNDRY TYPES		CARBON STEEL	1		P90151-0199 -083	MS901(083)	
			1.51					
6	GRINDING RING		CAST IRON	1		P90151-0199 -095	MS901(095)	
			1.30					
7	PIN WRENCH STARTING V/V		CARBON STEEL	1		P90151-0199 -105	MS901(105)	
			0.19					
8	TEST EQMT. COMB. SAFETY VALVE		CARBON STEEL	1		P90151-0199 -117	MS901(117)	
			2.21					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090617		YWH	SMK	HCY	TBK	*	First issue	0
Scale:		Size:		Type:		Page No.:		STX Engine Co., Ltd.
1:1		A4		S50MC-C		101-2		
Info. No.:		Description:					Drawing no.:	
S50MC -C7/C8STP901A		1. CYLINDER COVER					2805031870(2/3)	

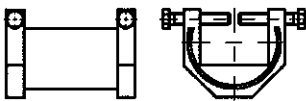
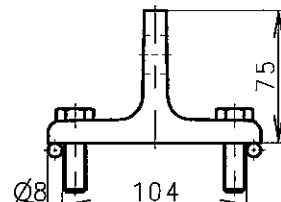
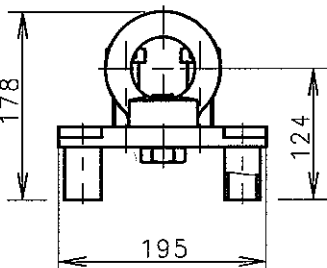
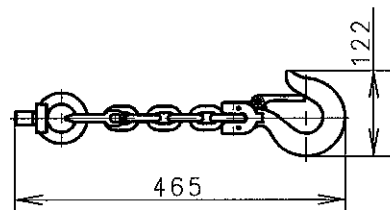
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.			
			WEIGHT (kg)	STD	ADD		REMARKS			
9	LIFTING CHAINS CYL. COVER		STEEL	2		P90151-0199 -130	MS901(130)			
			6.14							
10	HANDLE SUNDRY TYPES		CARBON STEEL	1		P90151-0199 -201	MS901(201)			
			2.361							
11	LIFTING TOOL FOR CYL. COVER		Cr-Ni-Mo STEEL	2		P90165-0003 -017				
12	LIFTING TOOL FOR CYL. COVER		Cr-Ni-Mo STEEL	1		P90165-0003 -029				
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No		
						*		3		
						*		2		
						*		1		
20090617		YWH	SMK	HCY	TBK	*	First issue	0		
Scale:		Size:	Type:			Page No.:		S50MC-C	101-3	SUX Engine Co., Ltd.
1:1		A4								
Info. No.:		Description:					Drawing no.:			
S50MC -C7/C8STP901A		1. CYLINDER COVER					2805031870(3/3)			

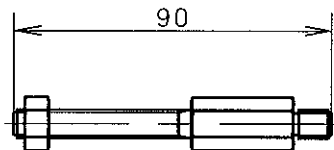
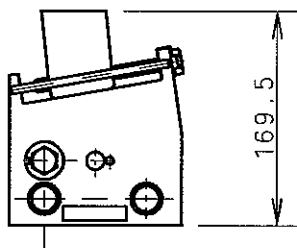
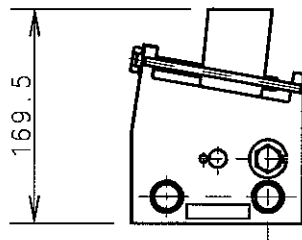
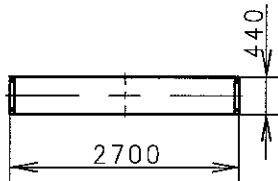
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
1	LIFT FOR PISTON ROD FOOT		CARBON STEEL	1		P90251-0237-040	MS902(040)	
			6.218					
2	LIFT TOOL FOR CYL. LINER		Cr-Ni-Mo STEEL	2		P90251-0237-051	MS902(051)	
			2.977					
3	TEMPLATER PISTON TOP		Cr-Ni STEEL	1		P90251-0237-075	MS902(075)	
			0.595					
4	DISTANCE PIECE		CARBON STEEL	4		P90251-0237-087	MS902(087)	
			2.20					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090317		KCJ	SML	TBK	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 102-1	
Info. No.: S50MC-C7STP902B		Description: 2. CYLINDER UNITS					Drawing no.: 2805030870(1/4)	
							STX Engine Co., Ltd.	

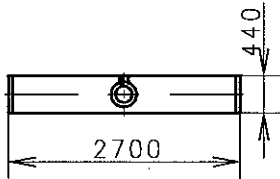
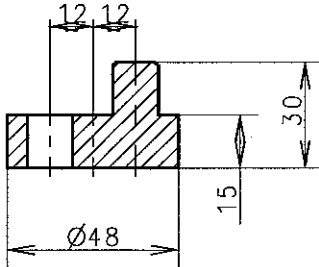
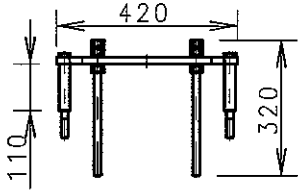
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT (kg)	STD	ADD		REMARKS
5	COVER FOR STUFF BOX HOLE		CARBON STEEL	1		P90251-0237-099	MS902(099)
6	MOUNT TOOL STUFF BOX SPRING		CARBON STEEL	2		P90251-0237-109	MS902(109)
			0.195				
7	PRESS. TEST TOOL PISTON		CARBON STEEL	1		P90251-0237-110	MS902(110)
			6.561				
8	WORK TABLE STUFFING BOX		CARBON STEEL	1		P90251-0237-122	MS902(122)
			12.298				
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement
						*	3
						*	2
						*	1
20090317		KCJ	SML	TBK	TBK	*	First issue
Scale: 1:1		Size: A4	Type: S50MC-C			Page No.: 102-2	SDX Engine Co., Ltd.
Info. No.: S50MC-C7STP902B		Description: 2. CYLINDER UNITS				Drawing no.: 2805030870(2/4)	

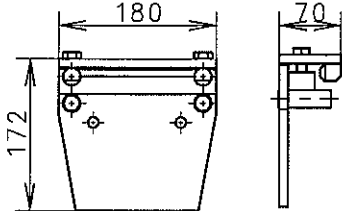
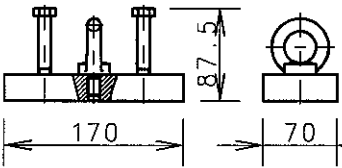
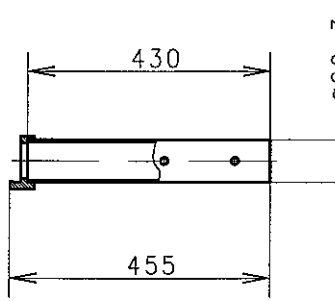
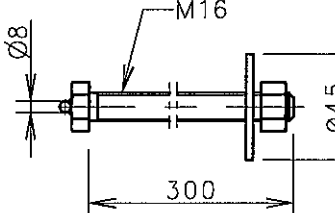
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.		
			WEIGHT (kg)	STD	ADD		REMARKS		
9	PISTON RING EXPANDER		CARBON STEEL	1		P90251-0237-134	MS902(134)		
			3.111						
10	GUIDE SCREW PISTON CROWN		CARBON STEEL	1		P90251-0237-158	MS902(158)		
			0.5						
11	HEX. BOLT		STEEL	2		P90251-0237-160	MS902(160)		
			0.09						
12	GUIDE RING FOR PISTON		CARBON STEEL	1		P90261-0108-019			
			23.7						
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No	
						*		3	
						*		2	
						*		1	
20090317		KCJ	SML	TBK	TBK	*	First issue	0	
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 102-3		SIX Engine Co., Ltd.
Info. No.: S50MC-C7STP902B		Description: 2. CYLINDER UNITS						Drawing no.: 2805030870(3/4)	

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
13	SUPPORT IRON FOR PISTON		CARBON STEEL	1		P90265-0001-010		
			44.652					
14	LIFTING TOOL FOR PISTON		CARBON STEEL	1		P90266-0002-016		
			24.918					
15	CROSS BAR FOR CYL. LINER		CARBON STEEL	1		P90366-0007-021		
			29.268					
16	MEASURING TOOL FOR CYL. LINER		WOODEN BOX	1		P90361-0084-036		
			4.00					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090317		KCJ	SML	TBK	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 102-4	SUX Engine Co., Ltd.
Info. No.: S50MC-C7STP902B		Description: 2. CYLINDER UNITS					Drawing no.: 2805030870(4/4)	

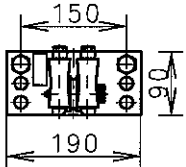
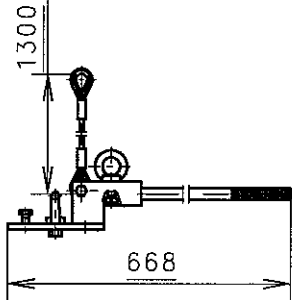
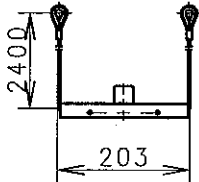
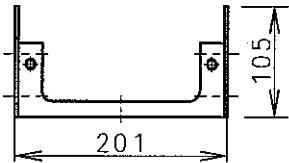
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT(kg)	STD	ADD		REMARKS	
1	WIRE GUIDE		CARBON STEEL	1		P90451-0133-047	MS904(047)	
			2.16					
2	LIFT ATTACHMENT CONN. ROD		CARBON STEEL	4		P90451-0133-059	MS904(059)	
			4.21					
3	LIFTINGS TOOL CROSSHEAD		CARBON-Mn STEEL	1		P90451-0133-060	MS904(060)	
			4.88					
4	CHAIN FOR SUSPEN. PISTON		STEEL	2		P90451-0133-072	MS904(072)	
			2.74					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 103-1	
Info. No.: S80MC -C7STP904A		Description: 3. CROSSHEAD + CONN-ROD					Drawing no.: 2805029610(1/3)	
STX Engine Co., Ltd.								

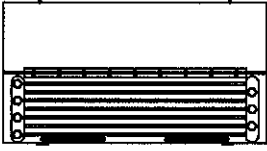
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
5	RET. TOOL TELESCOPIC PIPE		CARBON STEEL	1		P90451-0133-084	MS904(084)	
			0.055					
6	BRACKET SUPPORT OF X-HEAD		CARBON STEEL	2		P90451-0133-096	MS904(096)	
			3.9					
7	BRACKET SUPPORT OF X-HEAD		CARBON STEEL	2		P90451-0133-106	MS904(106)	
			3.9					
8	RUBBER COVER FOR CROSSHEAD		RUBBER	1		P90451-0133-118	MS904(118)	
			1.8					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 103-2	
Info. No.: S80MC -C7STP904A		Description: 3. CROSSHEAD + CONN-ROD					Drawing no.: 2805029610(2/3)	
SIX Engine Co., Ltd.								

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
9	RUBBER COVER FOR CROSSHEAD		RUBBER	1		P90451-0133-120	MS904(120)	
			1.8					
10	TORQUE WRENCH OFFSET TOOL		TOOL STEEL	1		P90451-0133-214	MS904(214)	
			0.21					
11	ALIGNMENT TOOL		CARBON STEEL	1		P90451-0133-226	MS904(226)	
			1.824					
12	GUIDE SHOE EXTRACTOR		CARBON STEEL	1		P90462-0020-011		
			8.546					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4	Type: S50MC-C			Page No.: 103-3		STX Engine Co., Ltd.
Info. No.: S80MC-C7STP904A		Description: 3. CROSSHEAD + CONN-ROD					Drawing no.: 2805029610(3/3)	

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT (kg)	STD	ADD		REMARKS
5	TOOL FOR TURN OUT SEGMENTS		CARBON STEEL	1		P90551-0227-100	MS905 (100)
6	LIFTING ATTACHMENT		CARBON STEEL	1		P90551-0227-111	MS905 (111)
7	RETAINING TOOL MAIN BRG. SHELL		STEEL	2		P90551-0227-123	MS905 (123)
8	LIFT TOOL FOR RELIEF VALVE		CARBON STEEL	1		P90551-0227-135	MS905 (135)
			0.7				
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement
						*	
						*	
						*	
20100513		SJJ	DHE	JT	STBK	*	First issue
Scale:		Size:	Type:		Page No.:		C.No
1:1		A4	S50MC-C		104-2		3
Info. No.:		Description:				Drawing no.:	
S50MC		4. CRANKSHAFT + THRUST BRG.				2805033620 (2/4)	
-C7STP905B							

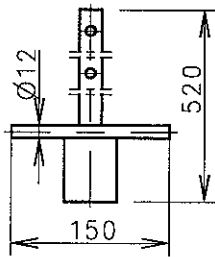
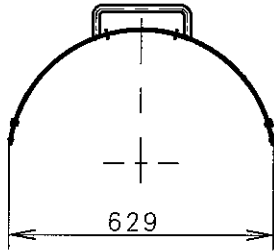
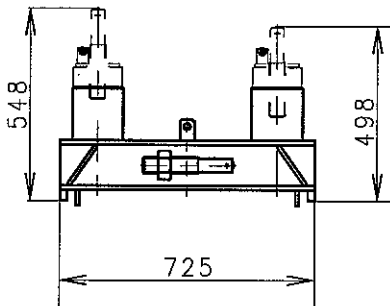
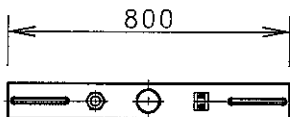
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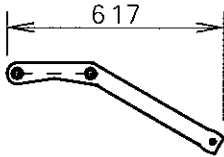
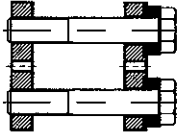
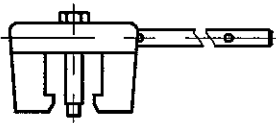
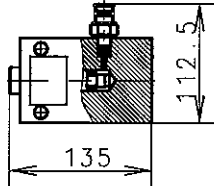
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT(kg)	STD	ADD		REMARKS	
1	PULLEY FOR WIRE FOR MAIN BRG.		CARBON STEEL	1		P90551-0227-040	MS905(040)	
			4.44					
2	LIFT TOOL MAIN BRG. CAP		CARBON STEEL	1		P90551-0227-076	MS905(076)	
			5.63					
3	DISM. TOOL MAIN BRG. SHELL		COPPER BASE ALLOY	1		P90551-0227-088	MS905(088)	
			0.495					
4	MOUNT. TOOL THIN BRG. SHELL		CARBON STEEL	1		P90551-0227-090	MS905(090)	
			1.642					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20100513		SJJ	DHE	JTS	TBK	*	First issue	0
Scale: 1:1		Size: A4	Type: S50MC-C			Page No.: 104-1		stx
Info. No.: S50MC-C7STP905B		Description: 4. CRANKSHAFT + THRUST BRG.					Drawing no.: 2805033620(1/4)	

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT (kg)	STD	ADD		REMARKS
13	FEELER GAUGE SET	 0.05 - 0.35 mm		1		P90572-0005-010	
14	FEELER GAUGE SET	 0.40 - 0.70 mm		1		P90572-0005-021	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20100513	SJJ	DHE	JTS	TBK	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 104-4	
Info. No.: S50MC-C7STP905B	Description: 4. CRANKSHAFT + THRUST BRG.	Drawing no.: 2805033620(4/4)		

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
9	MOUNT. TOOL THIN BRG. SHELL		CARBON STEEL	1		P90551-0227 -172	MS905 (172)	
			1.20					
10	LIFT TOOL MAIN BRG. SHELL		CARBON STEEL	1		P90551-0227 -184	MS905 (184)	
			1.348					
11	LIFT. TOOL FOR CRANKSHAFT		CARBON STEEL	1		P90562-0125 -015		
			58.1					
12	LIFT TOOL FOR THRUST SHAFT		CARBON STEEL	1		P90570-0004 -019		
			40.3					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20100513		SJJ	DHE	JT	STBK	*	First issue	0
Scale: 1:1		Size: A4	Type: S50MC-C			Page No.: 104-3		stx
Info. No.: S50MC -C7STP905B		Description: 4. CRANKSHAFT + THRUST BRG.					Drawing no.: 2805033620(3/4)	

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT (kg)	STD	ADD		REMARKS
1	FACE IMPACT WRENCH FUEL CAM		CARBON STEEL	1		P90651-0173 -044	MS906(044)
			8.11				
2	CHAIN ASSEMBLING		CARBON STEEL	1		P90651-0173 -056	MS906(056)
			9.79				
3	CHAIN DISASSEMBLING		CARBON STEEL	1		P90651-0173 -068	MS906(068)
			11.9				
4	HYDRAULIC JACK FOR LIFT OF CAMSHAFT		CARBON STEEL	1		P90651-0173 -152	MS906(152)
			3.99				

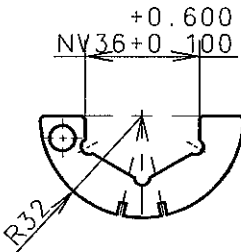
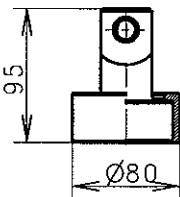
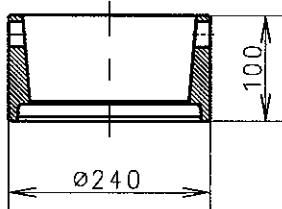
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1	Size: A4	Type: S50MC-C			Page No.: 105-1		STX Engine Co., Ltd.
Info. No.: S50MC -C7STP906A		Description: 5. MECH. CONTROL GEAR				Drawing no.: 2805029630(1/2)	

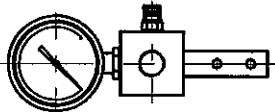
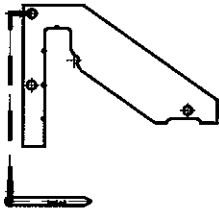
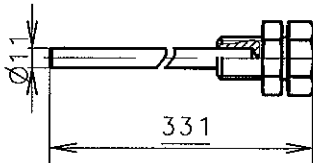
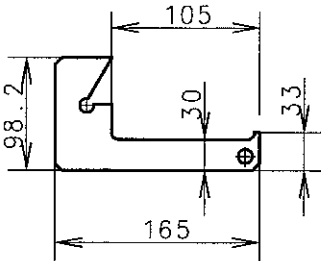
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT(kg)	STD	ADD		REMARKS
5	PIN GAUGE CAMSHAFT		CARBON STEEL	1		P90668-0003 -011	
			0.74				
6	PIN GAUGE CRANKSHAFT		CARBON STEEL	1		P90664-0005 -021	
			1.16				

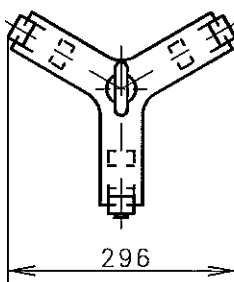
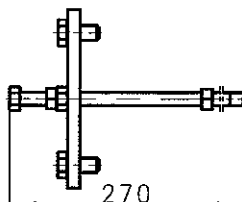
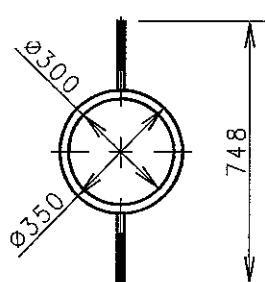
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

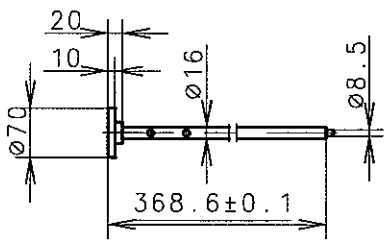
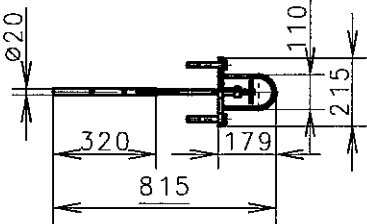
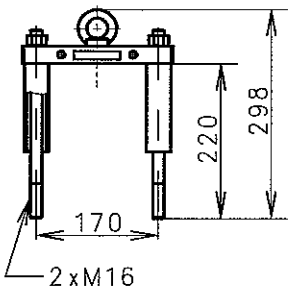
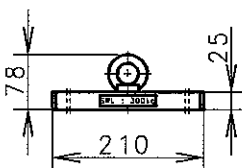
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1 : 1	A4	S50MC-C	105-2	

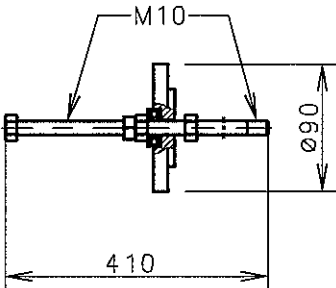
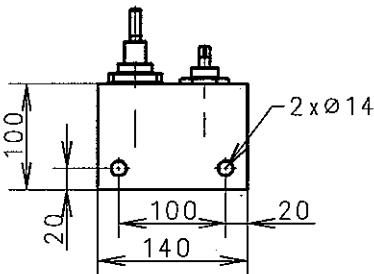
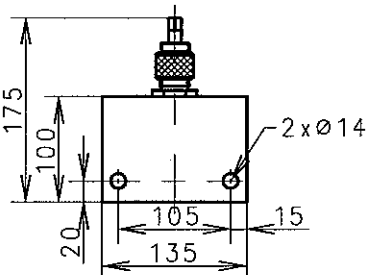
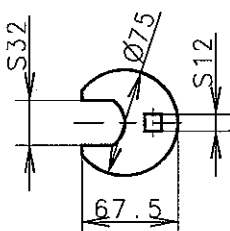
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S50MC -C7STP906A	5. MECH. CONTROL GEAR	2805029630(2/2)

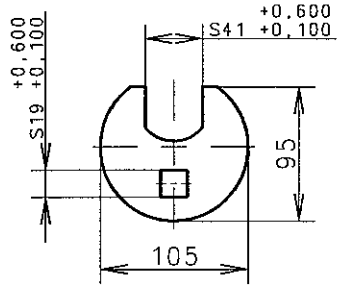
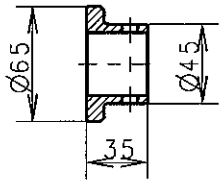
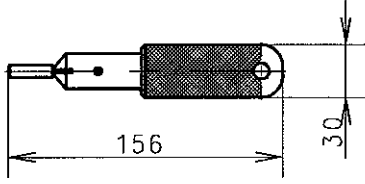
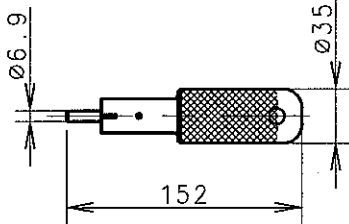
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
1	TIGHT. TEMP. ACTUATOR		Cr-Ni STEEL	1		P90851-0200-041	MS908(041)	
			0.024					
2	LIFTING TOOL		CARBON STEEL	1		P90851-0200-053	MS908(053)	
			0.78					
3	LIFTING TOOL FOR EXH. VALVE		CARBON STEEL	1		P90851-0200-065	MS908(065)	
			1.61					
4	CONE RING PNEUM PISTON		CARBON STEEL	1		P90851-0200-077	MS908(077)	
			8.48					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4	Type: S50MC-C			Page No.: 106-1		STX Engine Co., Ltd.
Info. No.: S50MC-C7STP908A		Description: 6. EXHAUST VALVE					Drawing no.: 2805029640(1/3)	

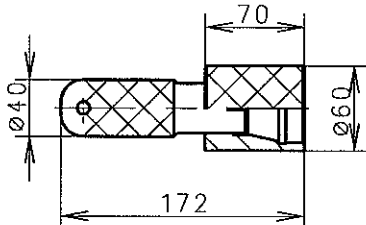
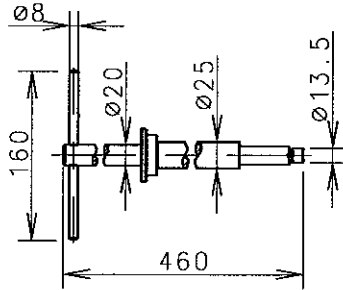
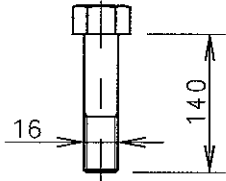
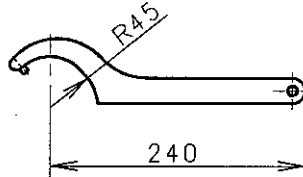
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
5	TEST EQUIPMENT EXH. VALVE		CARBON STEEL COPPER	1		P90851-0200 -089	MS908(089)	
			4.91					
6	GAUGE EXH. VALVE SPINDLE		Cr-Ni STEEL	1		P90851-0200 -090	MS908(090)	
			0.572					
7	TOOL EMERG. OPEN EXH. VALVE		STEEL	1		P90851-0200 -124	MS908(124)	
			0.592					
8	GAUGE EXH. VALVE BOTTOM PIECE		Cr-Ni STEEL	1		P90851-0200 -150	MS908(150)	
			0.176					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	H CY	TBK	*	First issue	0
Scale:		Size:		Type:		Page No.:		SUX Engine Co., Ltd.
1:1		A4		S50MC-C		106-2		
Info. No.:		Description:					Drawing no.:	
S50MC -C7STP908A		6. EXHAUST VALVE					2805029640(2/3)	

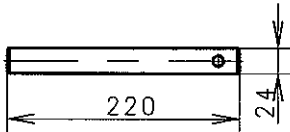
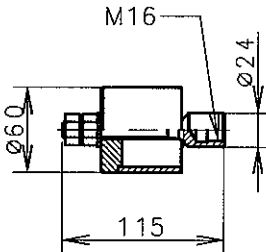
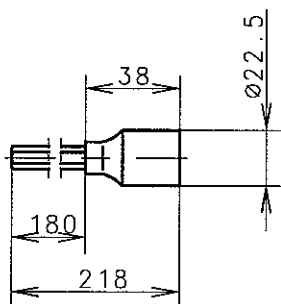
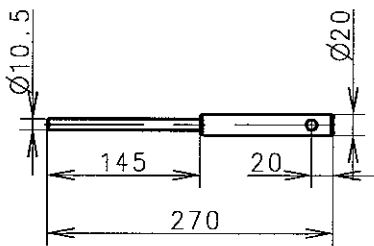
NO.	NAME	SKETCH	MATERIAL	Q'TY / ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT(kg)	STD	ADD		REMARKS	
9	LIFTING TOOL EXH. VALVE		CARBON STEEL	1		P90851-0200 -161	MS908(161)	
			2.844					
10	LIFT. TOOL ROLL.G. -EXH. VALVE		STEEL	1		P90851-0200 -173	MS908(173)	
			01.1476					
11	GRIND. RING FOR EXH. VALVE BOTTOM PIECE		CAST IRON CARBON STEEL	1		P90851-0200 -185	MS908(185)	
12	EXH. VALVE GRANDING MACHINE	REFER TO WORKING DWG. 15-6 PAGE	MAKER	1		2802008560 (PART NO.)		
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale:		Size:		Type:			Page No.:	
1:1		A4		S50MC-C			106-3	
Info. No.:		Description:					Drawing no.:	
S50MC -C7STP908A		6. EXHAUST VALVE					2805029640(3/3)	

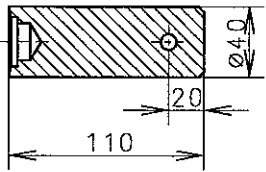
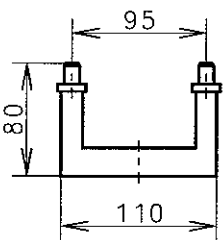
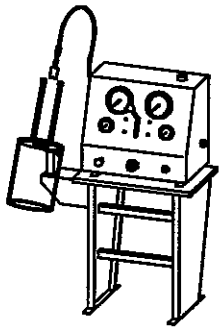
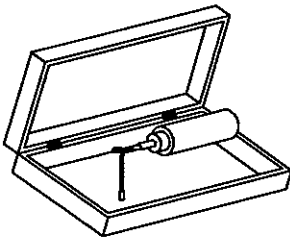
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT(kg)	STD	ADD		REMARKS	
1	MEAS. TOOL FUEL PUMP LEAD		CARBON STEEL	1		P90951-0444 -045	MS909(045)	
			0.95					
2	LIFTING TOOL FOR FUEL PUMP		CARBON STEEL	1		P90951-0444 -057	MS909(057)	
			3.29					
3	LIFTING TOOL FOR FUEL PUMP		CARBON STEEL	1		P90951-0444 -069	MS909(069)	
			3.21					
4	LIFTING TOOL FOR FUEL PUMP HOUSING		CARBON STEEL	1		P90951-0444 -070	MS909(070)	
			1.6					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090317		KCJ	SML	TBK	TBK	*	First issue	0
Scale:		Size:		Type:			Page No.:	
1:1		A4		S50MC-C			107-1	
Info. No.:		Description:					Drawing no.:	
S50MC -C7STP909C		7. FUEL OIL SYSTEM					2805030810(1/9)	
STX Engine Co., Ltd.								

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT(kg)	STD	ADD		REMARKS	
5	LIFTING TOOL ROLLER GUIDE FUEL PUMP		Cr-Ni-Mo STEEL	1		P90951-0444 -082	MS909(082)	
			0.89					
6	MILL TOOL SEAT F.O PIPE		CARBON STEEL BEECH WOOD	1		P90951-0444 -094	MS909(094)	
			2.089					
7	MILL TOOL FUEL OIL PIPE-HOSE		TOOL STEEL BEECH WOOD	1		P90951-0444 -104	MS909(104)	
			1.897					
8	CROWFOOT WRENCH HEAD		TOOL STEEL	1		P90951-0444 -116	MS909(116)	
			0.372					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090317		KCJ	SML	TBK	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 107-2	
Info. No.: S50MC -C7STP909C		Description: 7. FUEL OIL SYSTEM					STX Engine Co., Ltd. Drawing no.: 2805030810(2/9)	

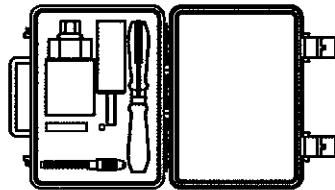
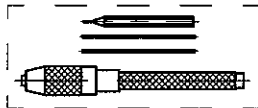
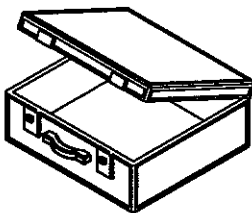
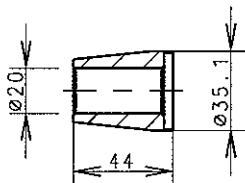
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT(kg)	STD	ADD	PART NO.	REMARKS	
9	CROWFOOT WRENCH HEAD		TOOL STEEL	1		P90951-0444 -128	MS909(128)	
			1.1			1395241-4.0		
10	FLANGE DISM NON RETURN V/V		CARBON STEEL	1		P90951-0444 -153	MS909(153)	
			0.29			1175068-6.1		
11	GRIND MANDREL VALVE HEAD		TOOL STEEL	1		P90951-0444 -160	MS909(160)	
			0.53			3101350-1.1		
12	GRIND MANDREL FOR THR. SPAIN		TOOL STEEL	1		P90951-0444 -177	MS909(177)	
			0.704			0980183-6.1		
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090317		KCJ	SML	TBK	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 107-3	
Info. No.: S50MC -C7STP909C		Description: 7. FUEL OIL SYSTEM					STX Engine Co., Ltd. Drawing no.: 2805030810(3/9)	

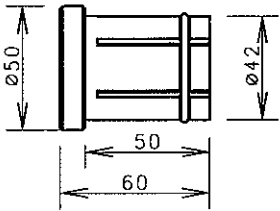
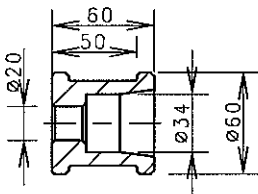
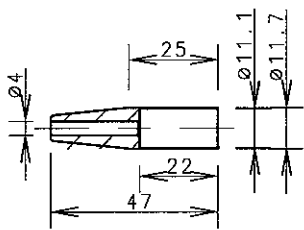
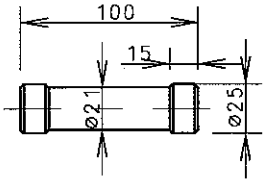
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
13	GRIND MANDREL HOLDER OUTSIDE		CAST IRON	1		P90951-0444-189	MS909(189)	
			1.64					
14	GRIND MANDREL HOLDER INSIDE		ALLOY STEEL	1		P90951-0444-190	MS909(190)	
			1.746					
15	HEX. BOLT		STEEL	2		P90951-0444-200	MS909(200)	
			0.33					
16	HOOK WRENCH		TOOL STEEL	1		P90951-0444-248	MS909(248)	
			0.2					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090317		KCJ	SML	TBK	TBK	*	First issue	0
Scale: 1:1		Size: A4	Type: S50MC-C			Page No.: 107-4		STX Engine Co., Ltd.
Info. No.: S50MC-C7STP909C		Description: 7. FUEL OIL SYSTEM					Drawing no.: 2805030810(4/9)	

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
17	DRIFT SPINDLE GUIDE		COPPER BASE ALLOY	1		P90951-0444 -250	MS909(250)	
			0.84					
18	EXTRACT TOOL SUCTION VALVE		CARBON STEEL	1		P90951-0444 -212	MS909(212)	
			0.73					
19	SOCKET WRENCH OUTSIDE HEX.		SPRING STEEL	1		P90951-0444 -344	MS909(344)	
			0.15					
20	DRIFT FOR PUNCTURE V/V		COOPER BASE ALLOY	1		P90951-0444 -261	MS909(261)	
			0.44					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090317		KCJ	SML	TBK	TBK	*	First issue	0
Scale:		Size:		Type:			Page No.:	
1:1		A4		S50MC-C			107-5	
Info. No.:		Description:					Drawing no.:	
S50MC -C7STP909C		7. FUEL OIL SYSTEM					2805030810(5/9)	
STX Engine Co., Ltd.								

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT (kg)	STD	ADD		REMARKS
21	DRIFT FOR NON-RETURN VALVE		COPPER BASE ALLOY	1		P90951-0444-273	MS909(273)
			1.0				
22	ASSEMBLING TOOLS		CARBON STEEL	1		P90951-0444-297	MS909(297)
			0.444				
23	TEST RIG & BENCH FUEL VALVE		MAKER	1		P90961-0059-014	
			105.83				
24	PROBELIGHT		PLASTIC BOX	1		P90968-0001-012	
			1.1				

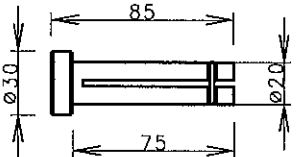
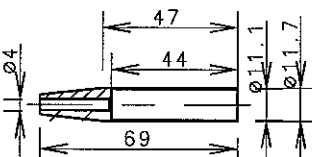
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090317	KCJ	SML	TBK	TBK	*	First issue	0
Scale: 1:1	Size: A4	Type: S50MC-C			Page No.: 107-6		SIX Engine Co., Ltd.
Info. No.: S50MC-C7STP909C		Description: 7. FUEL OIL SYSTEM				Drawing no.: 2805030810(6/9)	

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.		
			WEIGHT(kg)	STD	ADD		REMARKS		
25	CLEANING TOOL FUEL NOZZLE		CARBON STEEL	1		P90966-0008 -202			
			1.659						
26	CLEANING TOOL SLIDE NOZZLE		TOOL STEEL	1		P90974-0001 -206			
			0.001						
27	TOOLS MOUNT SEALS, FUEL VALVE		WOODEN BOX	1		P90980-0001			
			2.0						
28	CONE, FOR SEALING RING		POM	1		P90980-0001 -018			
			0.027						
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No	
						*		3	
						*		2	
						*		1	
20090317		KCJ	SML	TBK	TBK	*	First issue	0	
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 107-7		
Info. No.: S50MC -C7STP909C		Description: 7. FUEL OIL SYSTEM						Drawing no.: 2805030810(7/9)	

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT(kg)	STD	ADD		REMARKS
29	PUSHING TOOL, SEALS		POM	1		P90980-0001-020	
			0.05				
30	COMPRESSION TOOL, SEALS		POM	1		P90980-0001-031	
			0.124				
31	CONE, FOR SEALING RING		POM	1		P90980-0001-043	
			0.003				
32	COMPRESSION TOOL, SEALS		POM	1		P90980-0001-067	
			0.039				

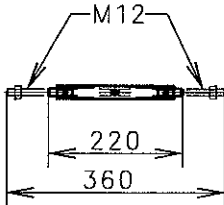
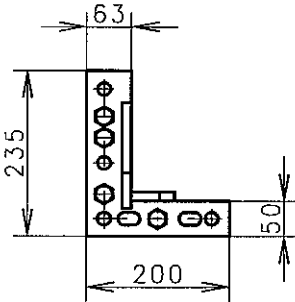
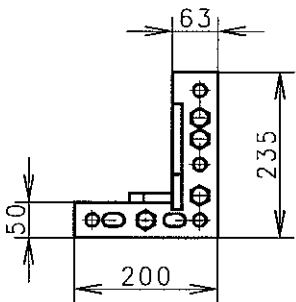
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

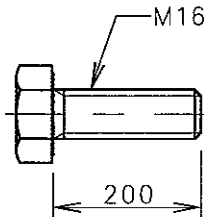
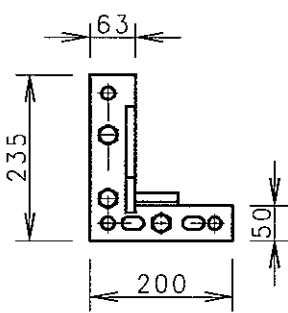
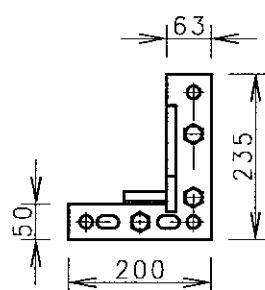
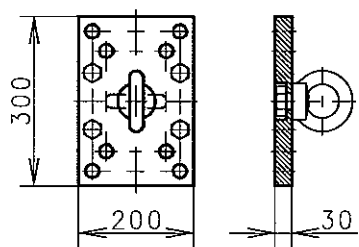
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Info. No.: S50MC-C7STP909C	Description: 7. FUEL OIL SYSTEM	Drawing no.: 2805030810(8/9)		

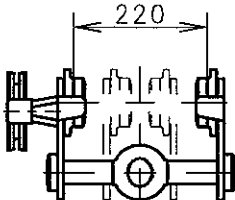
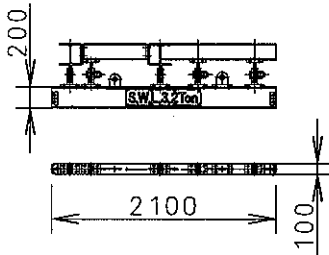
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT (kg)	STD	ADD		REMARKS
33	PUSHING TOOL, SEALS		POM	1		P90980-0001 -055	
			0.026				
34	CONE, FOR SEALING RING		POM	1		P90980-0001 -079	
			0.003				
				1			

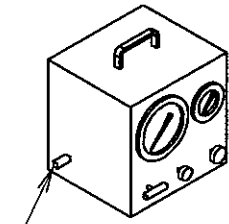
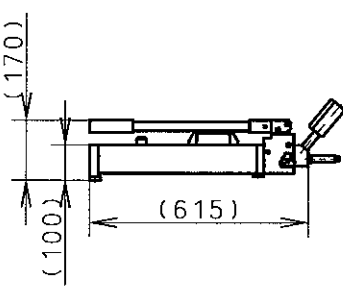
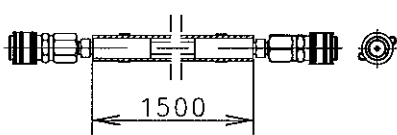
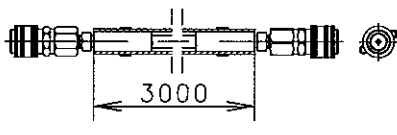
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090317	KCJ	SML	TBK	TBK	*	First issue	0

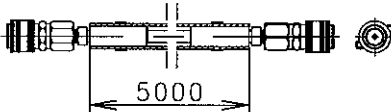
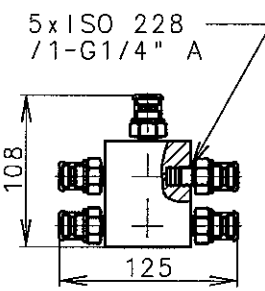
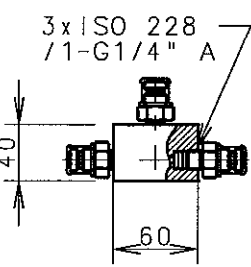
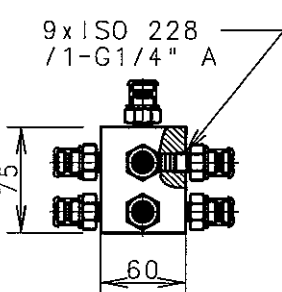
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Info. No.: S50MC -C7STP909C		Description: 7. FUEL OIL SYSTEM		Drawing no.: 2805030810(9/9)

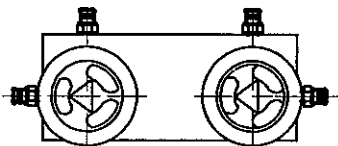
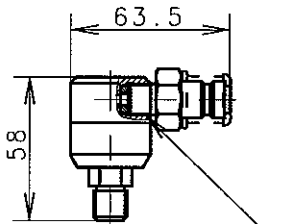
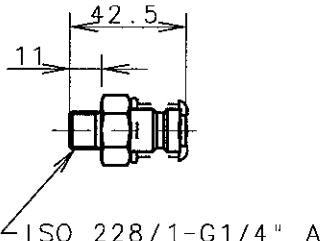
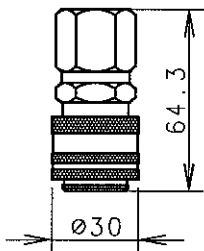
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT(kg)	STD	ADD		REMARKS	
1	PULLING TOOL		CARBON STEEL	3		P91061-0071-014		
			0.42					
2	LIFTING EYE BOLTS		STEEL	2		P91061-0071-026		
			0.2					
3	GUIDE RAIL A.C FORE END		CARBON STEEL	1		P91063-0017-015		
			6.97					
4	GUIDE RAIL A.C FORE END		CARBON STEEL	1		P91063-0017-015		
			6.97					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C		Page No.: 108-1		STX Engine Co., Ltd.
Info. No.: S50MC-C7STP910B		Description: 8. TURBOCHARGER SYSTEM PART					Drawing no.: 2805029660(1/3)	

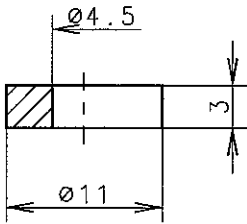
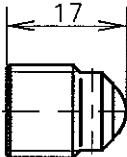
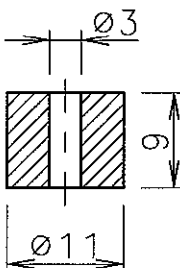
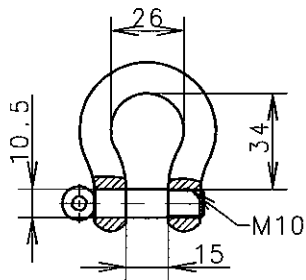
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			WEIGHT (kg)	STD	ADD		REMARKS
5	HEXAGON BOLT		STEEL	2		P91066-0001-023	
			0.302				
6	GUIDE RAIL A.C REAR END		CARBON STEEL	1		P91063-0017-027	
			5.83				
7	GUIDE RAIL A.C REAR END		CARBON STEEL	1		P91063-0017-027	
			5.83				
8	LIFT ATTACH A.C FORE END		CARBON STEEL	1		P91066-0001-011	
			14.9				
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement
						*	3
						*	2
						*	1
20090209		YWH	SMK	HCY	TBK	*	First issue
Scale:		Size:	Type:			Page No.:	STX Engine Co., Ltd.
1:1		A4	S50MC-C			108-2	
Info. No.:		Description:				Drawing no.:	
S50MC -C7STP910B		8. TURBOCHARGER SYSTEM PART				2805029660(2/3)	

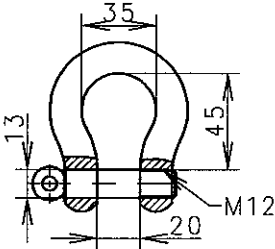
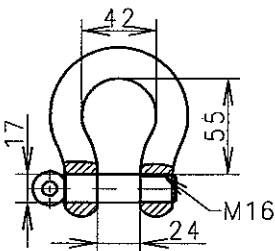
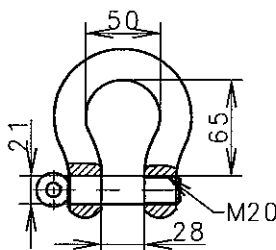
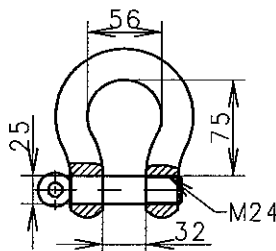
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT(kg)	STD	ADD		REMARKS
9	TRAVELLING TROLLEY	 Capacity:min. 2000kg	TYPE : YALE HTG-A 2000	2		P91061-0071 -010	
			18.0				
10	BRACKET		STEEL	1		280201916A (PART NO.)	
			90.8				
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement
						*	3
						*	2
						*	1
20090209		YWH	SMK	HCY	TBK	*	First issue
Scale: 1:1		Size: A4	Type: S50MC-C			Page No.: 108-3	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP910B		Description: 8. TURBOCHARGER SYSTEM PART				Drawing no.: 2805029660(3/3)	

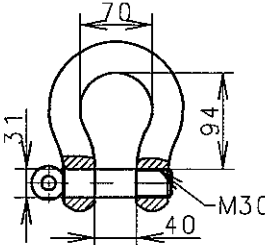
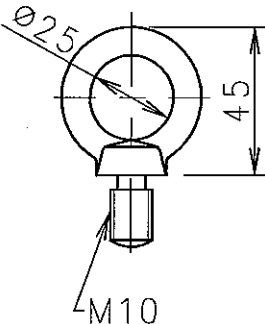
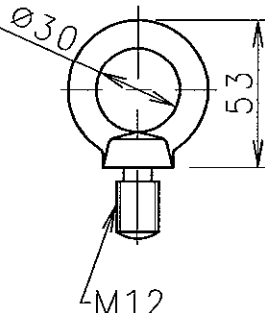
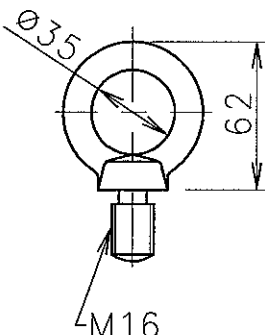
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT(kg)	STD	ADD		REMARKS
1	PUMP-PNEUM. AND HYDR.	Air inlet: ISO 228/1-G1/4  H.P outlet: ISO 228/1-G1/4	TYPE: HPU 2250	1		P91351-0037-010	
			30				
2	HYDRAULIC PUMP HAND-OPERATED		TYPE:EURO PRESS PML 16228BW	1		P91351-0037-022	
			8				
3	HOSE WITH 2 UNION		HOSE TYPE: ISO 228 /1-G1/4" L1.5m	8		P91351-0037-046	
			1.62				
4	HOSE WITH UNIONS		HOSE TYPE: ISO 228 /1-G1/4" L3.0m	1		P91351-0037-058	
			2.66				
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement
						*	
						*	
						*	
20090209		YWH	SMK	HCY	TBK	*	First issue
Scale: 1:1		Size: A4		Type: S50MC-C		Page No.: 109-1	
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS				Drawing no.: 2805029670(01/26)	
						SDX Engine Co., Ltd.	

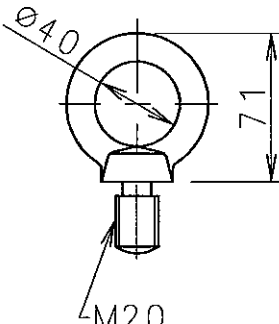
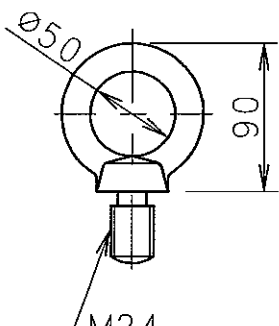
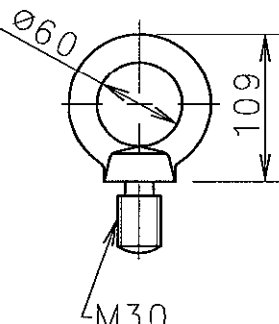
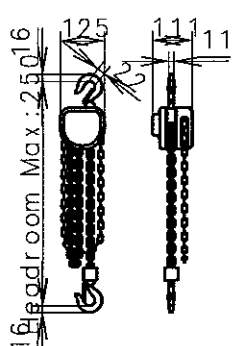
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT(kg)	STD	ADD		REMARKS
5	HOSE WITH 2 UNION		HOSE TYPE: ISO 228 /1-G1/4" L5.0m	1		P91351-0037 -060	
			4.02				
6	DISTRIBUTOR BLOCK		5x ISO 228 /1-G1/4" A	2		P91351-0037 -117	
			1.165				
7	DISTRIBUTOR BLOCK		3x ISO 228 /1-G1/4" A	1		P91351-0037 -105	
			0.657				
8	DISTRIBUTOR BLOCK		9x ISO 228 /1-G1/4" A	1		P91351-0037 -130	
			2.594				
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement
						*	3
						*	2
						*	1
20090209		YWH	SMK	HCY	TBK	*	First issue
Scale:		Size:	Type:			Page No.:	STX Engine Co., Ltd.
1:1		A4	S50MC-C			109-2	
Info. No.:		Description:				Drawing no.:	
S50MC -C7STP913A		9. GENERAL TOOLS				2805029670(02/26)	

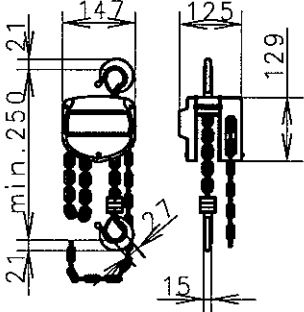
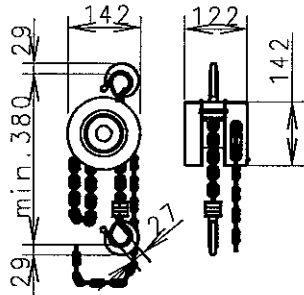
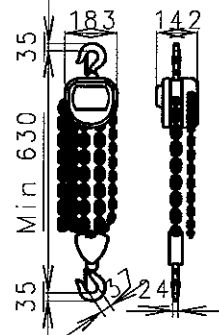
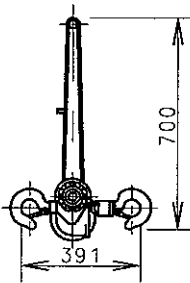
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT(kg)	STD	ADD		REMARKS
9	CONICAL VALVE	 WORKING PRESSURE : MAX. 1500 bar	KEY STEEL	1		P91351-0037-154	
			3.46				
10	ANGLE UNION	 ISO 228/1-G1/4" A	Cr-Ni-Mo STEEL	2		P91351-0037-166	
			0.306				
11	QUICK COUPLING MALE	 ISO 228/1-G1/4" A	TPYE: 10.125.5252	3		P91351-0037-201	
			0.06				
12	QUICK COUPLING FEMALE		TPYE: 10.125.1247	2		P91351-0037-213	
			0.21				
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement
						*	3
						*	2
						*	1
20090209		YWH	SMK	HCY	TBK	*	First issue
Scale: 1:1		Size: A4	Type: S50MC-C			Page No.: 109-3	STX Engine Co., Ltd.
Info. No.: S50MC-C7STP913A		Description: 9. GENERAL TOOLS				Drawing no.: 2805029670(03/26)	

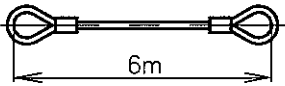
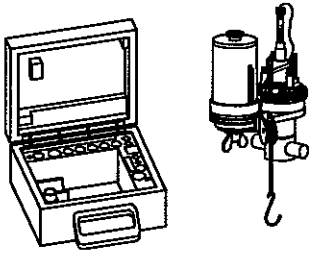
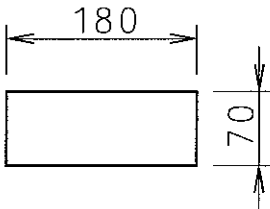
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT(kg)	STD	ADD		REMARKS	
13	DISC		COPPER	10		P91351-0037-225		
			0.003					
14	VENTING SCREW	 ISO 228/1-G1/4" A	STEEL	5		P91351-0037-237		
			0.02					
15	DISC		CARBON STEEL	10		P91351-0037-249		
			0.06					
16	FORGED-BENT SCREW SHACKLE		SIS STEEL 2108-01	4		P91356-0058-020		
			0.14					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale:		Size:		Type:			Page No.:	
1:1		A4		S50MC-C			109-4	
Info. No.:		Description:					Drawing no.:	
S50MC -C7STP913A		9. GENERAL TOOLS					2805029670(04/26)	
STX Engine Co., Ltd.								

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT(kg)	STD	ADD		REMARKS
17	FORGED-BENT SCREW SHACKLE		SIS STEEL 2108-01	4		P91356-0058 -031	
			0.26				
18	FORGED-BENT SCREW SHACKLE		SIS STEEL 2108-01	4		P91356-0058 -043	
			0.52				
19	FORGED-BENT SCREW SHACKLE		SIS STEEL 2108-01	4		P91356-0058 -055	
			0.91				
20	FORGED-BENT SCREW SHACKLE		SIS STEEL 2108-01	4		P91356-0058 -067	
			1.56				
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement
						*	
						*	
						*	
20090209		YWH	SMK	HCY	TBK	*	First issue
Scale:		Size:	Type:		S50MC-C		Page No.:
1:1		A4					109-5
Info. No.:		Description:					Drawing no.:
S50MC -C7STP913A		9. GENERAL TOOLS					2805029670(05/26)
STX Engine Co., Ltd.							

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
21	FORGED-BENT SCREW SHACKLE		SIS STEEL 2108-01	4		P91356-0058 -079		
			3.25					
22	LIFTING EYE BOLT		STEEL	4		P91356-0058 -114		
			0.1					
23	LIFTING EYE BOLT		STEEL	4		P91356-0058 -126		
			0.2					
24	LIFTING EYE BOLT		STEEL	4		P91356-0058 -138		
			0.3					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 109-6	STX Engine Co., Ltd.
Info. No.: S50MC-C7STP913A		Description: 9. GENERAL TOOLS					Drawing no.: 2805029670(06/26)	

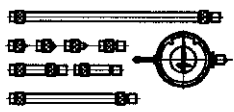
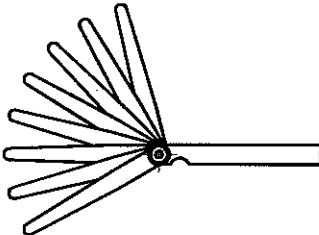
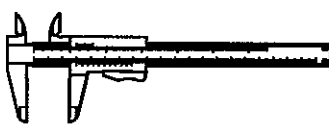
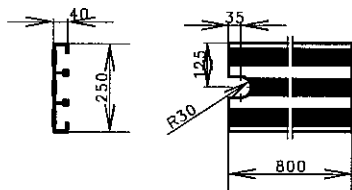
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT(kg)	STD	ADD		REMARKS
25	LIFTING EYE BOLT		STEEL	4		P91356-0058-140	
			0.5				
26	LIFTING EYE BOLT		STEEL	4		P91356-0058-151	
			0.7				
27	LIFTING EYE BOLT		STEEL	4		P91356-0058-163	
			1.7				
28	CHAIN TACKLE		STEEL	1		P91356-0058-210	
			9.0				
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement
						*	
						*	
						*	
20090209		YWH	SMK	HCY	TBK	*	First issue
Scale: 1:1		Size: A4	Type: S50MC-C			Page No.: 109-7	SDX Engine Co., Ltd.
Info. No.: S50MC-C7STP913A		Description: 9. GENERAL TOOLS				Drawing no.: 2805029670(07/26)	

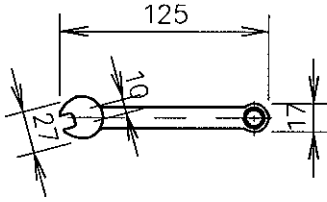
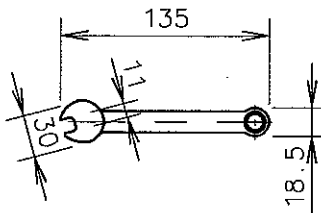
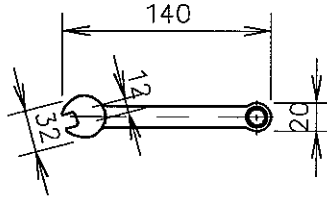
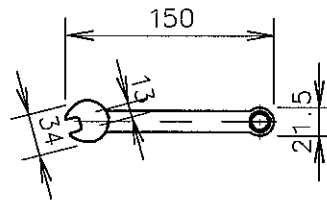
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
29	CHAIN TACKLE		STEEL	1		P91356-0058-222		
			11.2					
30	CHAIN TACKLE		STEEL	2		P91356-0058-234		
			14.4					
31	CHAIN TACKLE		STEEL	1		P91356-0058-258		
			32.2					
32	PULL LIFT		STEEL	2		P91356-0058-295		
			26.0					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C		Page No.: 109-8		STX Engine Co., Ltd.
Info. No.: S50MC-C7STP913A		Description: 9. GENERAL TOOLS				Drawing no.: 2805029670(08/26)		

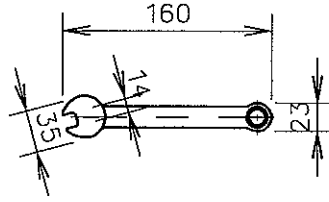
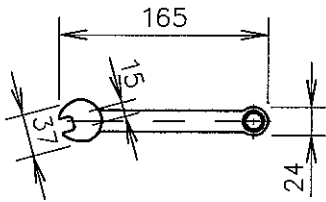
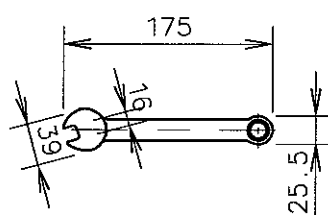
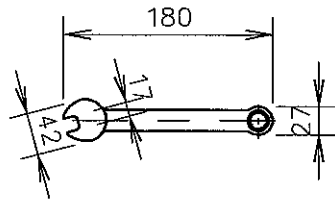
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT(kg)	STD	ADD		REMARKS
33	WIRE ROPE		STEEL	1		P91356-0058 -305	
			12.0				
34	INDICATOR		WOODEN BOX	1		P91366-0066 -012	
			4.4				
35	INDICATOR PAPER	 5pads, each of 50sheets	PAPER	1		P91366-0066 -024	
			0.26				
36	BLANK						

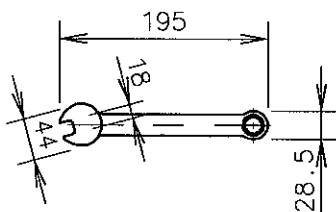
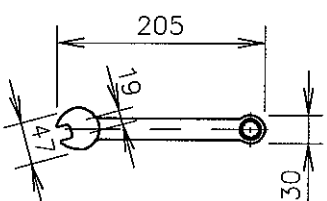
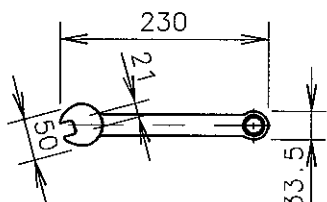
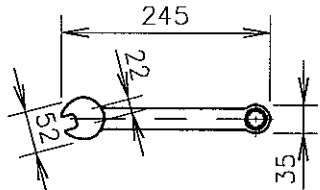
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 109-9	STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(09/26)

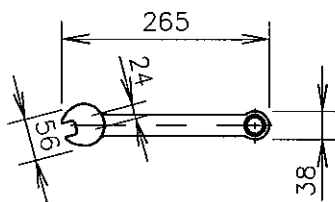
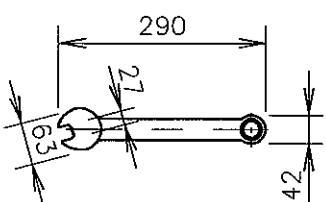
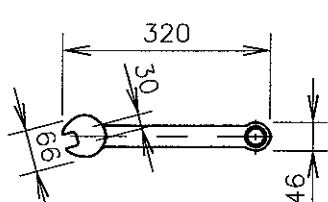
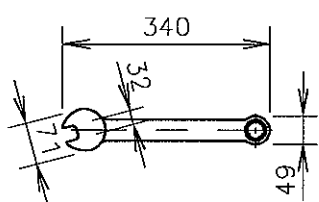
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
37	DEFLECTION GAUGE		STEEL	1		P91366-0066-048		
			0.457					
38	FEELER GAUGE SET		STEEL	1		P91366-0066-050		
			0.001					
39	SLIDE CALIPER		TYPE : 561-122	1		P91366-0066-061		
			0.153					
40	FOOT GRATING		ALUMINIUM	4		P91368-0004-013		
			3.44					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4	Type: S50MC-C			Page No.: 109-10		STX Engine Co., Ltd.
Info. No.: S50MC-C7STP913A		Description: 9. GENERAL TOOLS					Drawing no.: 2805029670(10/26)	

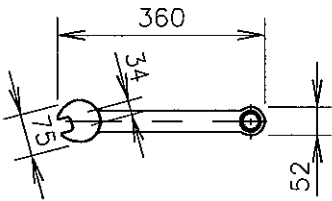
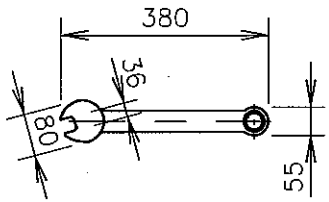
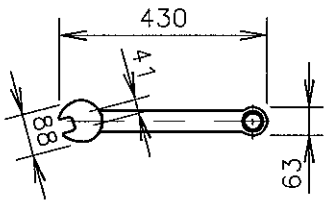
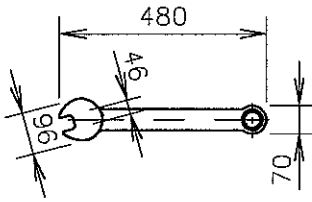
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
41	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052-015		
			0.06					
42	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052-027		
			0.08					
43	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052-039		
			0.09					
44	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052-040		
			0.1					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C		Page No.: 109-11		STX Engine Co., Ltd.
Info. No.: S50MC-C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(11/26)				

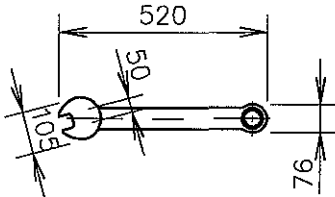
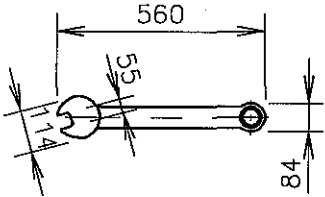
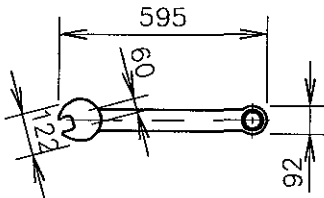
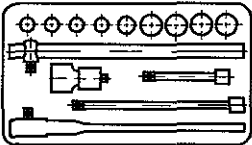
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
45	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052-052		
			0.13					
46	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052-064		
			0.15					
47	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052-076		
			0.185					
48	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052-088		
			0.2					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale:		Size:		Type:		Page No.:		STX Engine Co., Ltd.
1:1		A4		S50MC-C		109-12		
Info. No.:		Description:					Drawing no.:	
S50MC -C7STP913A		9. GENERAL TOOLS					2805029670(12/26)	

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT (kg)	STD	ADD		REMARKS
49	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052-090	
			0.23				
50	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052-100	
			0.25				
51	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052-111	
			0.3				
52	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052-123	
			0.4				

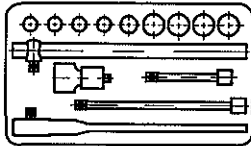
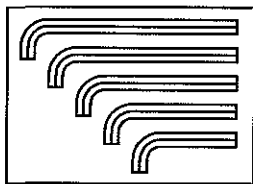
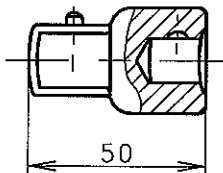
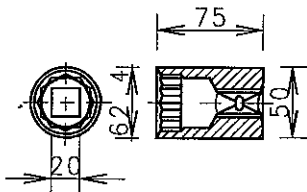
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1 : 1	Size: A4	Type: S50MC-C				Page No.: 109-13	
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS				Drawing no.: 2805029670(13/26)	

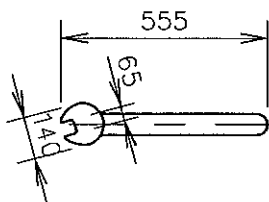
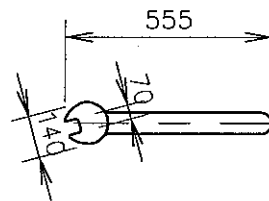
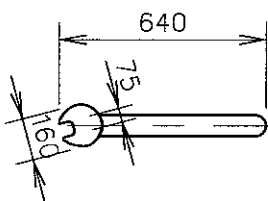
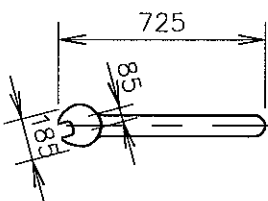
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
53	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052-135		
			0.5					
54	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052-147		
			0.65					
55	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052-159		
			0.95					
56	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052-160		
			1.15					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 109-14	STX Engine Co., Ltd.
Info. No.: S50MC-C7STP913A		Description: 9. GENERAL TOOLS					Drawing no.: 2805029670(14/26)	

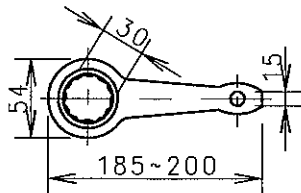
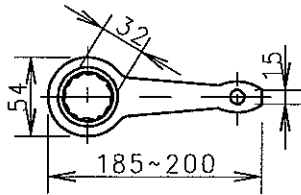
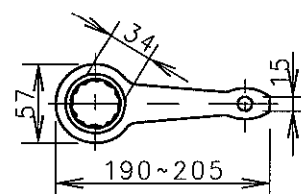
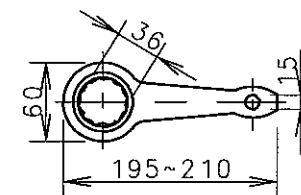
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
57	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052 -172		
			1.35					
58	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052 -184		
			1.5					
59	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052 -196		
			1.7					
60	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052 -206		
			2.0					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C		Page No.: 109-15		STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS		Drawing no.: 2805029670(15/26)				

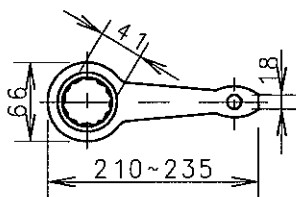
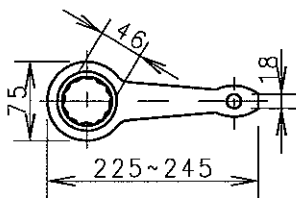
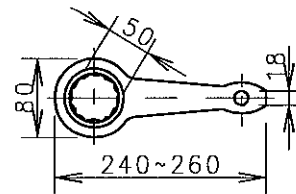
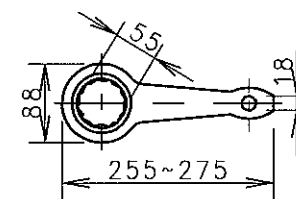
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT (kg)	STD	ADD		REMARKS
61	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052 -218	3102526-9.2 로 입고
			2.8				
62	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052 -220	3102526-9.2 로 입고
			3.6				
63	COMBINATION SPANNER		TOOL STEEL	2		P91361-0052 -231	3102526-9.2 로 입고
			4.4				
64	TOOL SETS, COMPLETE		STEEL	1		P91363-0040 -016	OLD: P91363-0035 -016
			5.7				

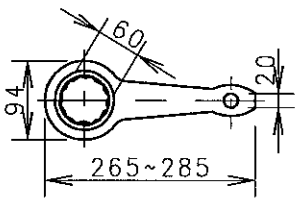
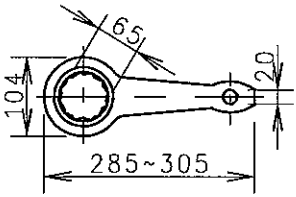
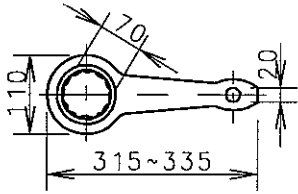
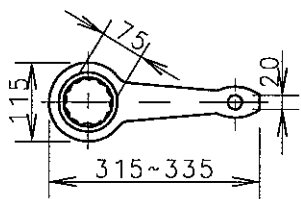
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1	Size: A4	Type: S50MC-C			Page No.: 109-16		STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS				Drawing no.: 2805029670(16/26)	

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.			
			WEIGHT(kg)	STD	ADD		REMARKS			
65	TOOL SETS, COMPLETE		STEEL	1		P91363-0040-028	OLD: P91363-0035-028			
			14.3							
66	BOX AND KEY SET, COMPLETE		STEEL	1		P91363-0040-030	OLD: P91363-0035-030			
			2.0							
67	ADAPTOR FOR SOCKET WRENCH		TOOL STEEL	1		P91363-0040-041	OLD: P91363-0035-041			
			0.2							
68	SOCKET SPANNERS, SQUARE DRIVE		TOOL STEEL	1		P91363-0040-100	OLD: P91363-0035-100			
			0.57							
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No		
						*		3		
						*		2		
						*		1		
20090209		YWH	SMK	HCY	TBK	*	First issue	0		
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 109-17		STX Engine Co., Ltd.	
Info. No.: S50MC-C7STP913A		Description: 9. GENERAL TOOLS					Drawing no.: 2805029670(17/26)			

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.		
			WEIGHT (kg)	STD	ADD		REMARKS		
69	ENGINE SPAN'S HEAD		TOOL STEEL	1		P91357-0005-013			
			1.95						
70	ENGINE SPAN'S HEAD		TOOL STEEL	1		P91357-0005-025			
			2.26						
71	ENGINE SPAN'S HEAD		TOOL STEEL	1		P91357-0005-037			
			2.802						
72	ENGINE SPAN'S HEAD		TOOL STEEL	1		P91357-0005-050			
			3.61						
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No	
						*		3	
						*		2	
						*		1	
20090209		YWH	SMK	HCY	TBK	*	First issue	0	
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 109-18		STX Engine Co., Ltd.
Info. No.: S50MC-C7STP913A		Description: 9. GENERAL TOOLS						Drawing no.: 2805029670(18/26)	

NO.	NAME	SKETCH	MATERIAL	Q'TY / ENG.		PLATE NO.	PANEL NO. & BOX NO.		
			WEIGHT (kg)	STD	ADD		REMARKS		
73	SLUGGING SPANNERS, RING		TOOL STEEL	1		P91362-0064-022			
			0.5						
74	SLUGGING SPANNERS, RING		TOOL STEEL	1		P91362-0064-034			
			0.5						
75	SLUGGING SPANNERS, RING		TOOL STEEL	1		P91362-0064-046			
			0.55						
76	SLUGGING SPANNERS, RING		TOOL STEEL	1		P91362-0064-058			
			0.6						
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No	
						*		3	
						*		2	
						*		1	
20090209		YWH	SMK	HCY	TBK	*	First issue	0	
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 109-19		SUX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS			Drawing no.: 2805029670(19/26)				

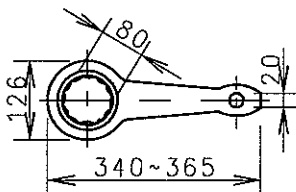
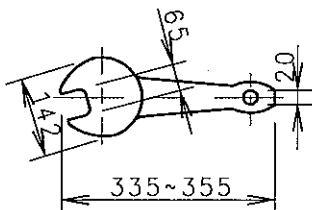
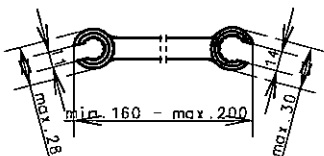
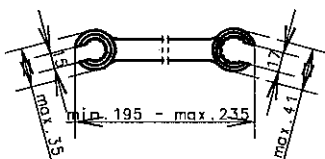
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
77	SLUGGING SPANNERS, RING		TOOL STEEL	1		P91362-0064-060		
			0.7					
78	SLUGGING SPANNERS, RING		TOOL STEEL	1		P91362-0064-071		
			0.9					
79	SLUGGING SPANNERS, RING		TOOL STEEL	1		P91362-0064-083		
			1.1					
80	SLUGGING SPANNERS, RING		TOOL STEEL	1		P91362-0064-095		
			1.5					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C		Page No.: 109-20		SIX Engine Co., Ltd.
Info. No.: S50MC-C7STP913A		Description: 9. GENERAL TOOLS					Drawing no.: 2805029670(20/26)	

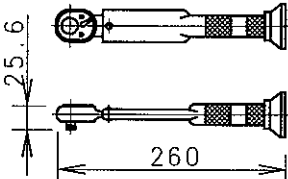
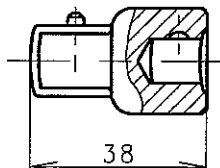
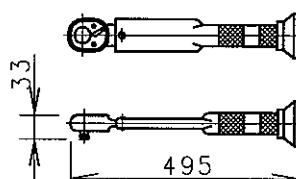
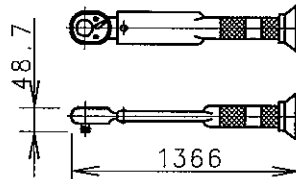
NO.	NAME	SKETCH	MATERIAL	Q'TY / ENG.		PLATE NO.	PANEL NO. & BOX NO.
				STD	ADD		REMARKS
			WEIGHT (kg)				
81	SLUGGING SPANNERS, RING		TOOL STEEL	1		P91362-0064-105	
			1.8				
82	SLUGGING SPANNERS, RING		TOOL STEEL	1		P91362-0064-117	
			2.2				
83	SLUGGING SPANNERS, RING		TOOL STEEL	1		P91362-0064-129	
			3.4				
84	SLUGGING SPANNERS, RING		TOOL STEEL	1		P91362-0064-130	
			3.4				

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0

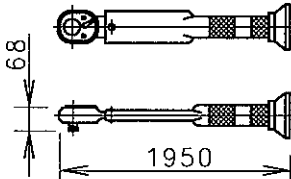
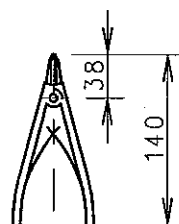
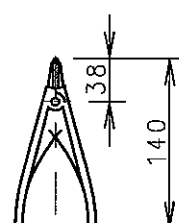
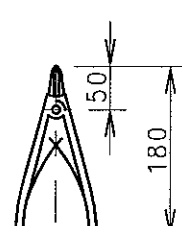
Scale:	Size:	Type:	Page No.:	STX Engine Co., Ltd.
1:1	A4	S50MC-C	109-21	

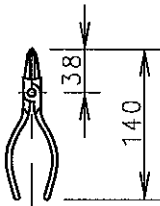
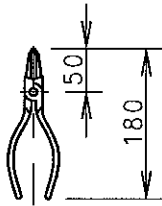
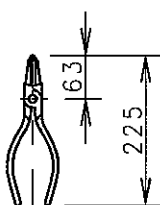
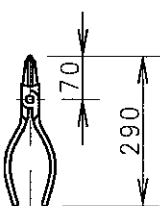
Info. No.:	Description:	Drawing no.:
S50MC -C7STP913A	9. GENERAL TOOLS	2805029670(21/26)

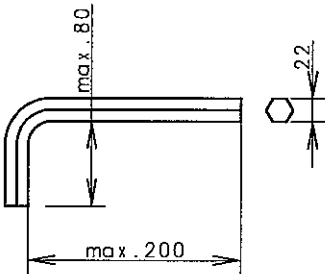
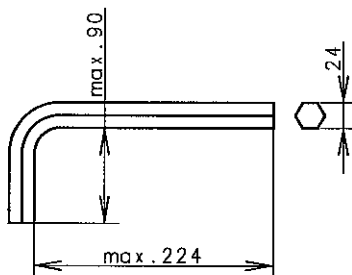
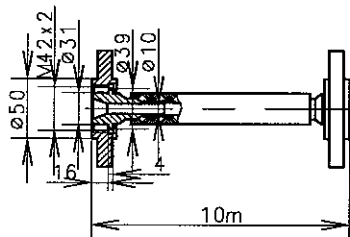
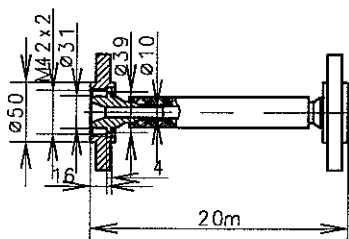
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
85	SLUGGING SPANNERS, RING		TOOL STEEL	1		P91362-0064 -142		
			4.5					
86	SLUGGING SPANNERS, OPEN		TOOL STEEL	1		P91358-0013 -127		
			3.4					
87	OPEN RING WRENCH		TOOL STEEL	1		P91364-0059 -011		
			0.13					
88	OPEN RING WRENCH		TOOL STEEL	1		P91364-0059 -023		
			0.25					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C		Page No.: 109-22		STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS					Drawing no.: 2805029670(22/26)	

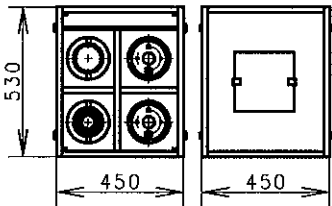
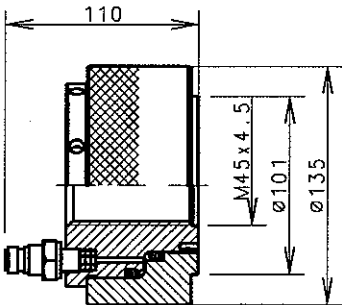
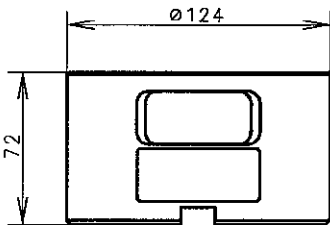
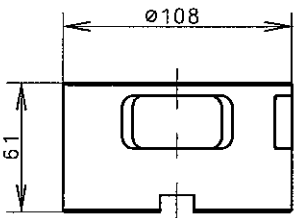
NO.	NAME	SKETCH	MATERIAL	Q'TY / ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT (kg)	STD	ADD		REMARKS
89	TORQUE WRENCH		TOOL STEEL	1		P91359-0003-014	ADJUST RANGES : 10-50Nm
90	ADAPTER FOR SOCKET WRENCH		TOOL STEEL	1		P91359-0003-026	
			0.1				
91	TORQUE WRENCH		TOOL STEEL	1		P91359-0003-038	ADJUST RANGES : 40-200Nm
			1-4				
92	TORQUE WRENCH		TOOL STEEL	1		P91359-0003-040	ADJUST RANGES : 100-750Nm
			5.0				

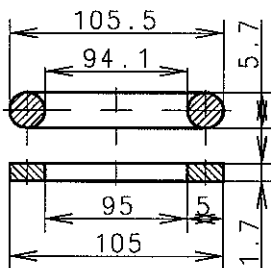
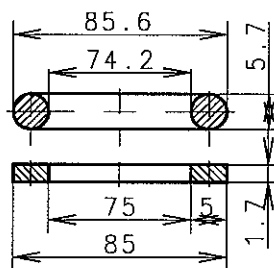
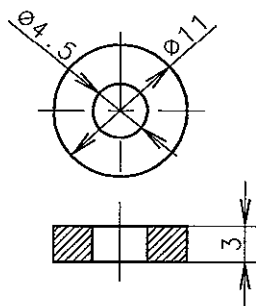
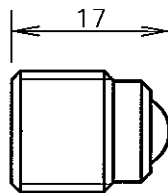
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1	Size: A4	Type: S50MC-C			Page No.: 109-23		STX Engine Co., Ltd.
Info. No.: S50MC-C7STP913A		Description: 9. GENERAL TOOLS				Drawing no.: 2805029670(23/26)	

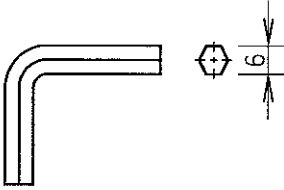
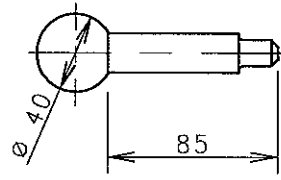
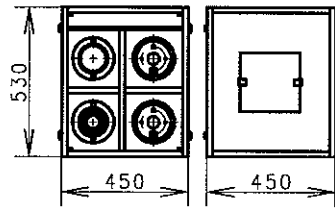
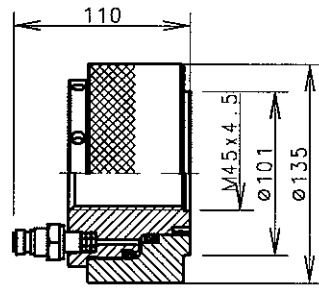
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
93	TORQUE WRENCH		TOOL STEEL	1		P91359-0003-051	ADJUST RANGES : 500-2100Nm	
			11.1					
94	PLIERS FOR RETAINING RING		TOOL STEEL	1		P91360-0003-010		
			0.085					
95	PLIERS FOR RETAINING RING		TOOL STEEL	1		P91360-0003-021		
			0.9					
96	PLIERS FOR RETAINING RING		TOOL STEEL	1		P91360-0003-033		
			0.16					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4	Type: S50MC-C			Page No.: 109-24		STX Engine Co., Ltd.
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS					Drawing no.: 2805029670(24/26)	

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.			
			WEIGHT (kg)	STD	ADD		REMARKS			
97	PLIERS FOR RETAINING RING		TOOL STEEL	1		P91360-0003 -128				
			0.09							
98	PLIERS FOR RETAINING RING		TOOL STEEL	1		P91360-0003 -130				
			0.16							
99	PLIERS FOR RETAINING RING		TOOL STEEL	1		P91360-0003 -141				
			0.23							
100	PLIERS FOR RETAINING RING		TOOL STEEL	1		P91360-0003 -153				
			0.4							
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No		
						*		3		
						*		2		
						*		1		
20090209		YWH	SMK	HCY	TBK	*	First issue	0		
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 109-25		SUX Engine Co., Ltd.	
Info. No.: S50MC -C7STP913A		Description: 9. GENERAL TOOLS						Drawing no.: 2805029670(25/26)		

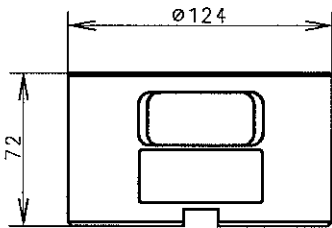
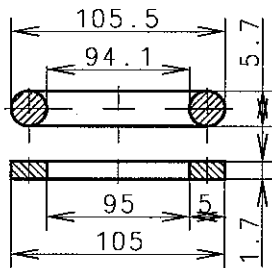
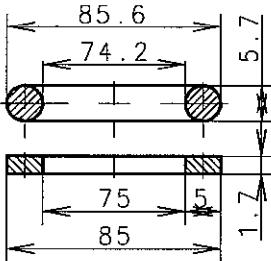
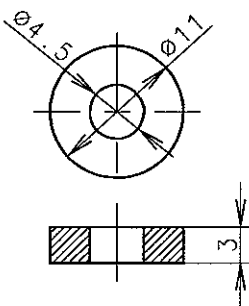
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT(kg)	STD	ADD		REMARKS
101	KEY, HEX. SOCKET SCREW		Cr-V VALVE SPRING STEEL	1		P91363-0040-150	
102	KEY, HEX. SOCKET SCREW		Cr-V VALVE SPRING STEEL	1		P91363-0040-161	
103	AIR HOSE FOR PUMP UNIT-10M		MAKER	1		2902001430-10 (PART NO.)	
104	AIR HOSE FOR PUMP UNIT-20M		MAKER	1		2902001430-20 (PART NO.)	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement
						*	3
						*	2
						*	1
20090209		YWH	SMK	H CY	TBK	*	First issue
Scale: 1:1		Size: A4	Type: S50MC-C			Page No.: 109-26	STX Engine Co., Ltd.
Info. No.: S50MC-C7STP913A		Description: 9. GENERAL TOOLS				Drawing no.: 2805029670(26/26)	

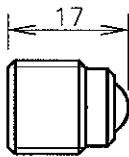
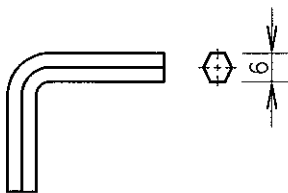
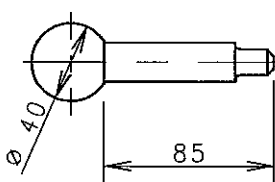
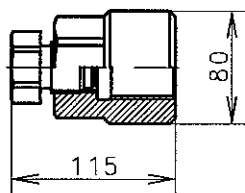
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
1	HYDR. JACK FOR CYL. COVER		PINE WOOD	2		P90161-0105-206		
2	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL	4		P90161-0105-027		
3	JACK-HYD. (SUPPORT)		Cr-Ni-Mo STEEL	4		P90161-0105-039		
			1.911					
4	JACK-HYD. (SUPPORT)		Cr-Ni-Mo STEEL	4		P90161-0105-040		
			1.501					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	H CY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 110-1	SUX Engine Co., Ltd.
Info. No.: S50MC-C7SHYD001		Description: 10. STANDARD HYDR. JACK					Drawing no.: 2805029680(01/18)	

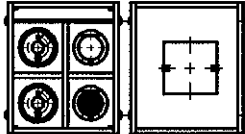
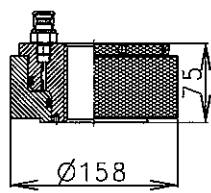
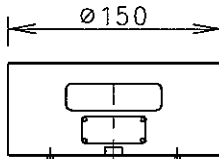
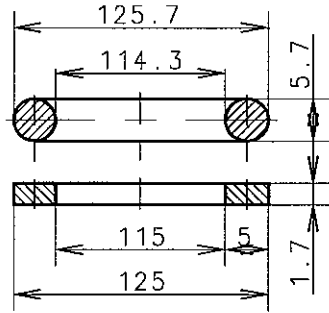
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT(kg)	STD	ADD		REMARKS	
5	SEALING RING W.BACK-UP		VITON PTFE	8		P90161-0105 -064		
			0.0323					
6	SEALING RING W.BACK-UP		VITON PTFE	8		P90161-0105 -052		
			0.0269					
7	DISC, ROUND PLAIN		COPPER	12		P91351-0037 -225		
			0.003					
8	VENTING SCREW		STEEL	4		P91351-0037 -237		
			0.028					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale:		Size:		Type:		Page No.:		S50 Engine Co., Ltd.
1:1		A4		S50MC-C		110-2		
Info. No.:		Description:				Drawing no.:		
S50MC -C7SHYD001		10. STANDARD HYDR. JACK				2805029680(02/18)		

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT (kg)	STD	ADD		REMARKS
9	KEY, HEX. SOCKET SCREWS		SPRING STEEL	2		P90161-0105 -090	
			0.03				
10	BALL HANDLE		TOOL STEEL	2		P90161-0105 -100	
			0.095				
11	HYDR. JACK FOR CYL. COVER		PINE WOOD	2		P90161-0105 -206	
12	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL	4		P90161-0105 -027	

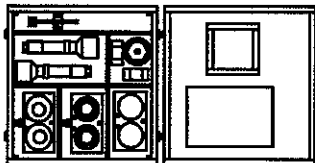
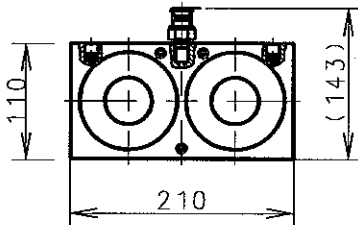
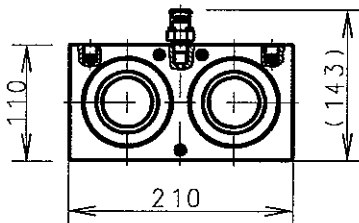
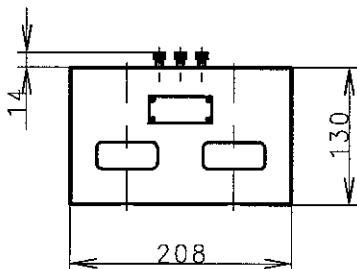
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	H CY	TBK	*	First issue	0
Scale: 1:1	Size: A4	Type: S50MC-C			Page No.: 110-3		STX Engine Co., Ltd.
Info. No.: S50MC -C7SHYD001		Description: 10. STANDARD HYDR. JACK				Drawing no.: 2805029680(03/18)	

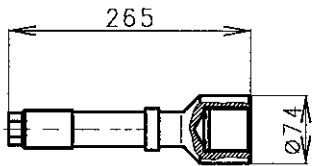
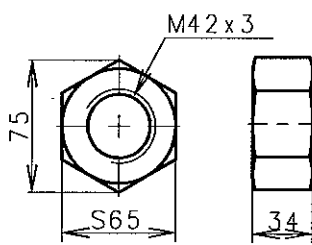
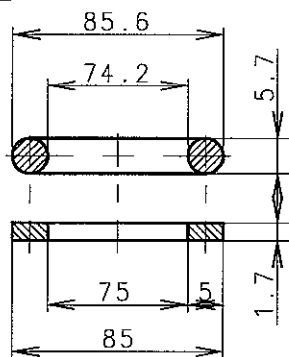
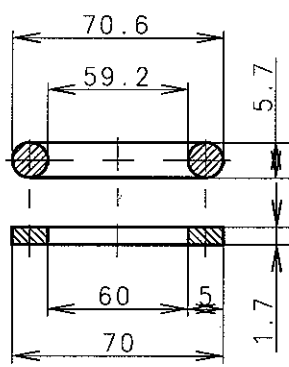
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
13	JACK-HYD. (SUPPORT)		Cr-Ni-Mo STEEL	4		P90161-0105-039		
			1.911					
14	SEALING RING W. BACK-UP		VITON PTFE	8		P90161-0105-064		
			0.0323					
15	SEALING RING W. BACK-UP		VITON PTFE	8		P90161-0105-052		
			0.0269					
16	DISC, ROUND PLAIN		COPPER	6		P91351-0037-225		
			0.003					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 110-4	SUX Engine Co., Ltd.
Info. No.: S50MC-C7SHYD001		Description: 10. STANDARD HYDR. JACK					Drawing no.: 2805029680(04/18)	

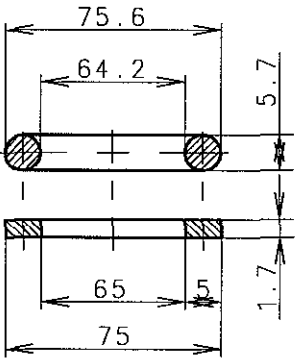
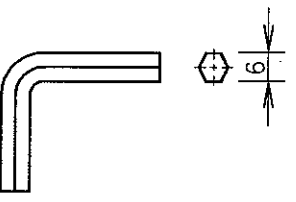
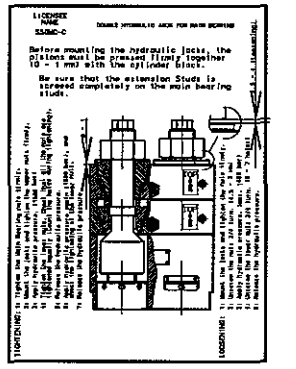
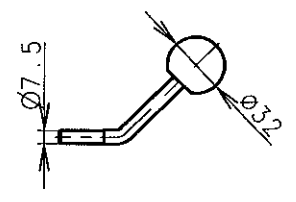
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT(kg)	STD	ADD		REMARKS	
17	VENTING SCREW		STEEL	2		P91351-0037-237		
			0.02					
18	KEY, HEX. SOCKET SCREWS		SPRING STEEL	4		P90161-0105-090		
			0.03					
19	BALL HANDLE		TOOL STEEL	2		P90161-0105-100		
			0.095					
20	STUD SETTER		CARBON STEEL	2		P90161-0105-111		
			2.61					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 110-5	STX Engine Co., Ltd.
Info. No.: S50MC-C7SHYD001		Description: 10. STANDARD HYDR. JACK					Drawing no.: 2805029680(05/18)	

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
21	HYD. JACK FOR CONN. ROD		PINE WOOD	1		P90461-0069-207		
			52.2					
22	JACK HYD. (COMPLETE)		Cr-Ni-Mo STEEL	2		P90461-0069-028		
			9.428					
23	JACK HYD. (SUPPORT)		Cr-Ni-Mo STEEL	2		P90461-0069-030		
			5.108					
24	SEALING RING W. BACK-UP		VITON PTFE	4		P90461-0069-053		
			0.02					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 110-6	STX Engine Co., Ltd.
Info. No.: S50MC-C7SHYD001		Description: 10. STANDARD HYDR. JACK					Drawing no.: 2805029680(06/18)	

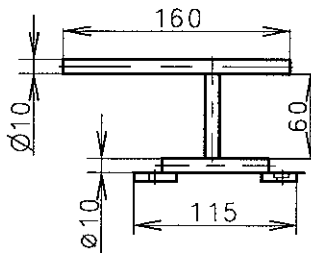
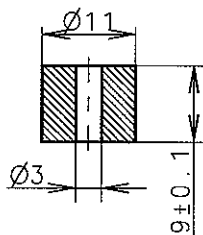
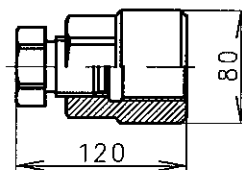
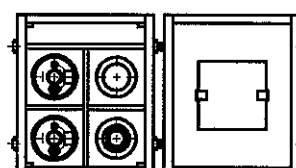
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			WEIGHT (kg)	STD	ADD		REMARKS
25	SEALING RING W. BACK-UP		VITON PTFE	4		P90461-0069 -065	
			0.028				
26	KEY, HEX. SOCKET SCREWS		SPRING STEEL	1		P90461-0069 -089	
			0.03				
27	BALL HANDLE		TOOL STEEL	1		P90461-0069 -090	
			0.095				
28	STUD SETTER		CARBON STEEL	1		P90461-0069 -100	
			5.29				
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement
						*	3
						*	2
						*	1
20090209		YWH	SMK	H CY	TBK	*	First issue
							0
Scale:	Size:	Type:			Page No.:		SUX Engine Co., Ltd.
1:1	A4	S50MC-C			110-7		
Info. No.:		Description:				Drawing no.:	
S50MC -C7SHYD001		10. STANDARD HYDR. JACK				2805029680(07/18)	

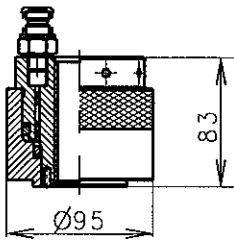
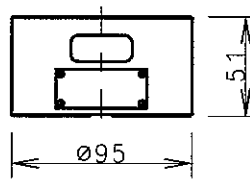
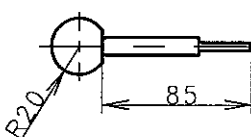
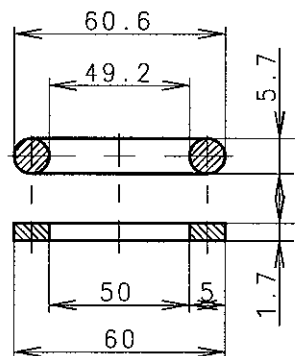
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
29	HYD. JACK FOR MAIN BEARING		PINE WOOD	2		P90561-0093-200		
			59.0					
30	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL	2		P90561-0093-021		
			10.013					
31	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL	2		P90561-0093-033		
			8.492					
32	JACK-HYD. (SUPPORT)		Cr-Ni-Mo STEEL	2		P90561-0093-045		
			11.78					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	H CY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 110-8	SUX Engine Co., Ltd.
Info. No.: S50MC-C7SHYD001		Description: 10. STANDARD HYDR. JACK					Drawing no.: 2805029680(08/18)	

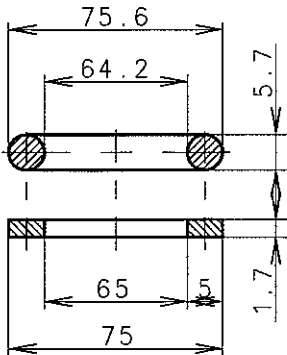
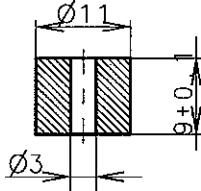
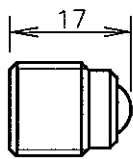
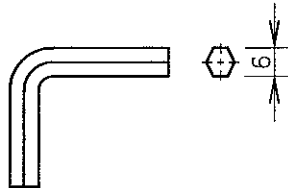
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			WEIGHT (kg)	STD	ADD		REMARKS
33	STUD		Cr-Ni-Mo STEEL	4		P90561-0093-057	
			3.37				
34	NUT		STEEL	4		P90561-0093-069	
			0.607				
35	SEALING RING W. BACK-UP		VITON PTFE	16		P90561-0093-070	
			0.027				
36	SEALING RING W. BACK-UP		VITON PTFE	8		P90561-0093-082	
			0.021				
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement
						*	3
						*	2
						*	1
20090209		YWH	SMK	HCY	TBK	*	First issue
Scale: 1:1		Size: A4	Type: S50MC-C			Page No.: 110-9	STX Engine Co., Ltd.
Info. No.: S50MC-C7SHYD001		Description: 10. STANDARD HYDR. JACK				Drawing no.: 2805029680(09/18)	

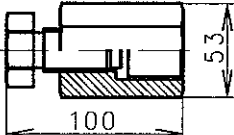
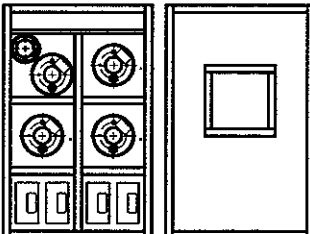
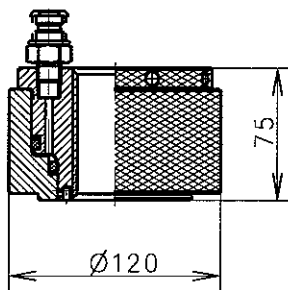
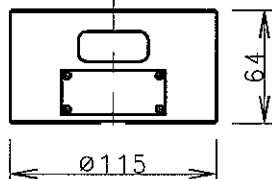
NO.	NAME	SKETCH	MATERIAL	Q'TY / ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT (kg)	STD	ADD		REMARKS
37	SEALING RING W. BACK-UP		VITON PTFE 0.022	8		P90561-0093 -094	
38	KEY, HEX. SOCKET SCREW		SPRING STEEL 0.03	2		P90561-0093 -104	
39	NAME PLATE		PLASTIC 0.1	2		3041056-8.0 (PART NO.)	
40	BALL HANDLE		TOOL STEEL 0.057	2		P90561-0093 -130	

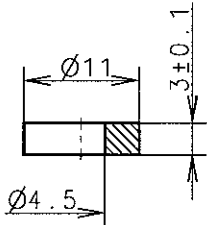
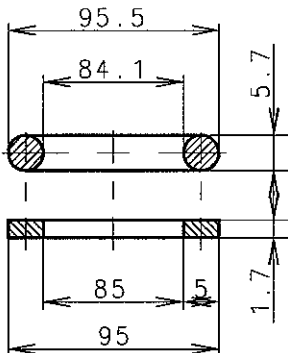
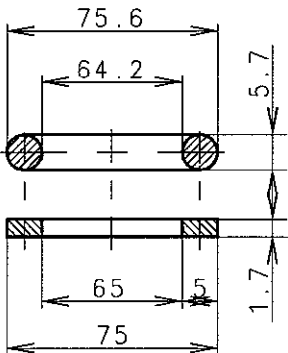
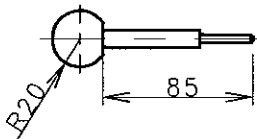
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	H CY	TBK	*	First issue	0
Scale: 1 : 1	Size: A4	Type: S50MC-C				Page No.: 110-10	 SUX Engine Co., Ltd.
Info. No.: S50MC -C7SHYD001		Description: 10. STANDARD HYDR. JACK				Drawing no.: 2805029680(10/18)	

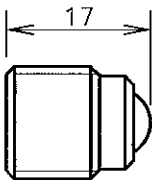
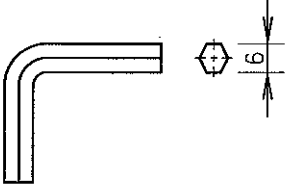
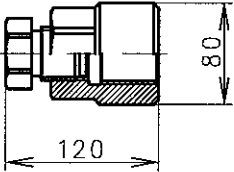
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
41	LIFTING TOOLS		CARBON STEEL	2		P90561-0093 -141		
			0.298					
42	DISC		CARBON STEEL	4		P91351-0037 -249		
			0.06					
43	STUD SETTER		CARBON STEEL	2		P91351-0037 -165		
			2.661					
44	HYDR. JACK FOR CAMSHAFT BRG		BEECH WOOD	1		P90662-0046 -211		
			24.065					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C		Page No.: 110-11		STX Engine Co., Ltd.
Info. No.: S50MC -C7SHYD001		Description: 10. STANDARD HYDR. JACK						

NO.	NAME	SKETCH	MATERIAL	Q'TY / ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
45	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL	2		P90662-0046 -020		
			3.74					
46	JACK-HYD. (SUPPORT)		Cr-Ni-Mo STEEL	2		P90662-0046 -032		
			1.156					
47	BALL HANDLE		TOOL STEEL	2		P90662-0046 -044		
			0.09					
48	SEALING RING W. BACK-UP		VITON PTFE	4		P90662-0046 -056		
			0.018					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 110-12	STX Engine Co., Ltd.
Info. No.: S50MC-C7SHYD001		Description: 10. STANDARD HYDR. JACK					Drawing no.: 2805029680(12/18)	

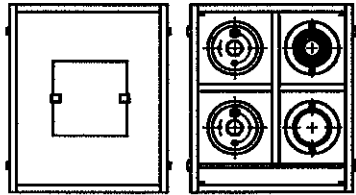
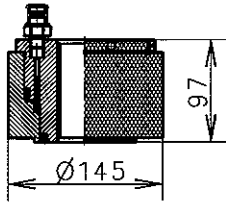
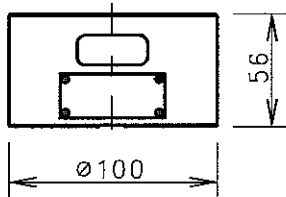
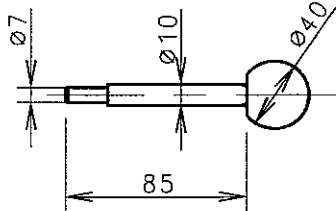
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.		
			WEIGHT (kg)	STD	ADD		REMARKS		
49	SEALING RING W. BACK-UP		VITON PTFE	4		P90662-0046 -068			
			0.022						
50	DISC		CARBON STEEL	2		P91351-0037 -249			
			0.06						
51	VENTING SCREW		STEEL	2		P91351-0037 -249			
			0.02						
52	KEY, HEX. SOCKET SCREW		SPRING STEEL	2		P90662-0046 -093			
			0.03						
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No	
						*		3	
						*		2	
						*		1	
20090209		YWH	SMK	HCY	TBK	*	First issue	0	
Scale:		Size:	Type:			S50MC-C		Page No.:	STX Engine Co., Ltd.
1:1		A4						110-13	
Info. No.:		Description:					Drawing no.:		
S50MC -C7SHYD001		10. STANDARD HYDR. JACK					2805029680(13/18)		

NO.	NAME	SKETCH	MATERIAL	Q'TY / ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
53	STUD SETTER		CARBON STEEL	1		P90862-0046-103		
			1.094					
54	HYD. JACK FOR EXH. VALVE		PINE WOOD	1		P90862-0043-207		
			45.374					
55	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL	4		P90862-0043-028		
			4.84					
56	JACK-HYD. (SUPPORT)		Cr-Ni-Mo STEEL	4		P90862-0043-030		
			1.902					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 110-14	
Info. No.: S50MC-C7SHYD001		Description: 10. STANDARD HYDR. JACK					Drawing no.: 2805029680(14/18)	
							STX Engine Co., Ltd.	

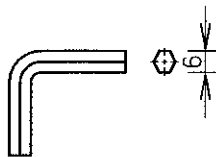
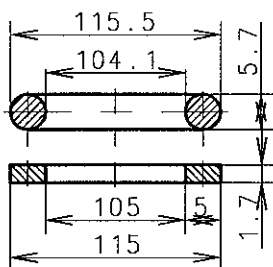
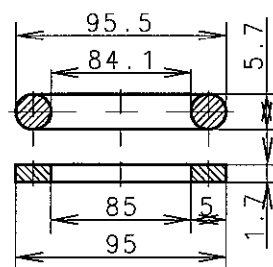
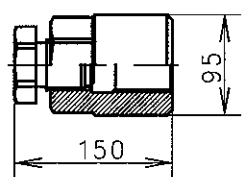
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT(kg)	STD	ADD		REMARKS	
57	DISC		COPPER	12		P91351-0037-225		
			0.003					
58	SEALING RING W.BACK-UP		VITON PTFE	8		P91351-0037-053		
			0.029					
59	SEALING RING W.BACK-UP		VITON PTFE	8		P91351-0037-065		
			0.022					
60	BALL HANDLE		TOOL STEEL	1		P90862-0043-077		
			0.09					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale:		Size:		Type:			Page No.:	
1:1		A4		S50MC-C			110-15	
Info. No.:		Description:					Drawing no.:	
S50MC -C7SHYD001		10. STANDARD HYDR. JACK					2805029680(15/18)	
STX Engine Co., Ltd.								

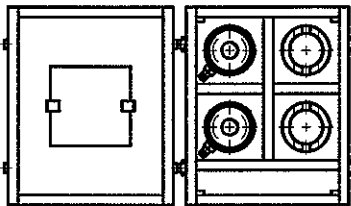
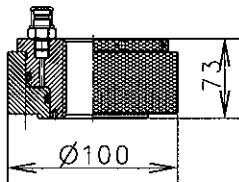
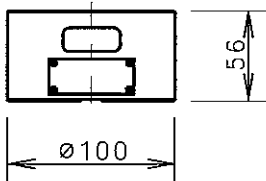
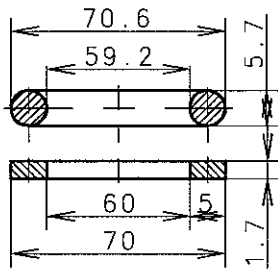
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT (kg)	STD	ADD		REMARKS
61	VENTING SCREW		STEEL	4		P91351-0037 -237	
			0.02				
62	KEY. HEX. SOCKET SCREW		SPRING STEEL	4		P90862-0043 -090	
			0.03				
63	STUD SETTER		CARBON STEEL	1		P90862-0043 -100	
			2.661				

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1	Size: A4	Type: S50MC-C			Page No.: 110-16		STX Engine Co., Ltd.
Info. No.: S50MC -C7SHYD001		Description: 10. STANDARD HYDR. JACK				Drawing no.: 2805029680(16/18)	

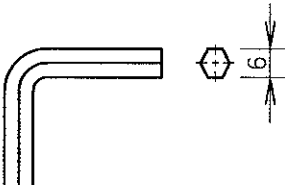
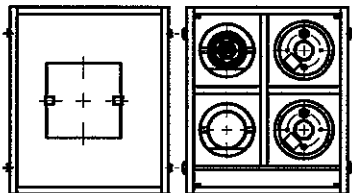
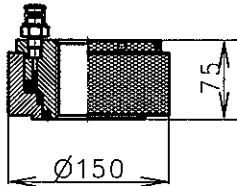
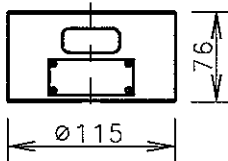
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT (kg)	STD	ADD		REMARKS
64	HYDRAULIC JACK FOR STAY BOLT		PINE WOOD	1		P91261-0073-072	
			53.624				
65	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL	2		P91261-0073-023	
			10.519				
66	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL	2		P91261-0073-035	
			4.306				
67	BALL HANDLE		TOOL STEEL	2		P91261-0073-047	
			0.09				

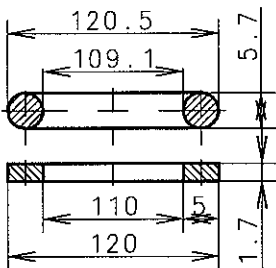
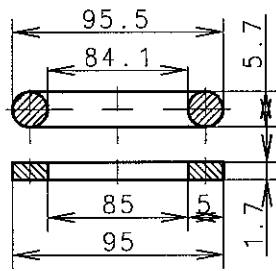
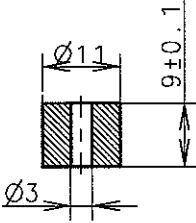
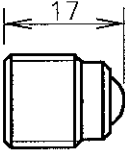
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1	Size: A4	Type: S50MC-C			Page No.: 110-17		STX Engine Co., Ltd.
Info. No.: S50MC-C7SHYD001		Description: 10. STANDARD HYDR. JACK				Drawing no.: 2805029680(17/18)	

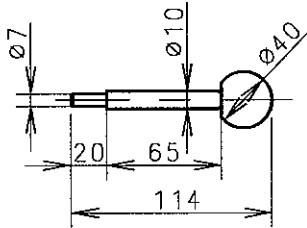
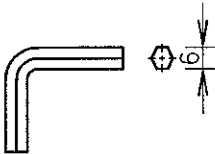
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
68	KEY, HEX. SCOKET SCREWS		SPRING STEEL	2		P91261-0073-084		
			0.03					
69	SEALING RING W. BACK-UP		VITON PTFE	4		P91261-0073-059		
			0.037					
70	SEALING RING W. BACK-UP		VITON PTFE	4		P91261-0073-060		
			0.029					
71	STUD SETTER		CARBON STEEL	1		P91261-0073-106		
			4.6					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 110-18	STX Engine Co., Ltd.
Info. No.: S50MC-C7SHYD001		Description: 10. STANDARD HYDR. JACK					Drawing no.: 2805029680(18/18)	

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT(kg)	STD	ADD		REMARKS
1	HYD. JACK FOR HOLDING DOWN BOLT		BEECH WOOD	1		P91265-0003-204	
			23.804				
2	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL	2		P91265-0003-025	
			3.442				
3	JACK-HYD. (SUPPORT)		Cr-Ni-Mo STEEL	2		P91265-0003-037	
			1.606				
4	SEALING RING W. BACK-UP		VITON PTFE	4		P91265-0003-049	
			0.021				
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement
						*	3
						*	2
						*	1
20090209		YWH	SMK	HCY	TBK	*	First issue
Scale: 1:1		Size: A4	Type: S50MC-C			Page No.: 111-1	STX Engine Co., Ltd.
Info. No.: S50MC-C70HYD001		Description: 11. HYDRAULIC JACK				Drawing no.: 2805029690(1/5)	

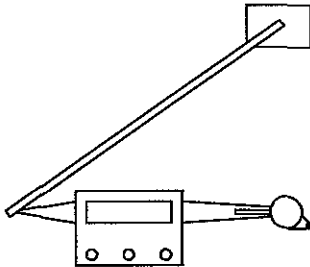
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT(kg)	STD	ADD		REMARKS	
5	SEALING RING W. BACK-UP		VITON PTFE	4		P91265-0003 -050		
			0.02					
6	DISC		COPPER	6		P91351-0037 225		
			0.003					
7	BALL HANDLE		TOOL STEEL	1		P91265-0003 -074		
			0.09					
8	SCREW WITH INSIDE HEXAGON		STEEL	2		P91351-0037 237		
			0.02					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale:		Size:		Type:		Page No.:		STX Engine Co., Ltd.
1:1		A4		S50MC-C		111-2		
Info. No.:		Description:					Drawing no.:	
S50MC -C70HYD001		11. HYDRAULIC JACK					2805029690(2/5)	

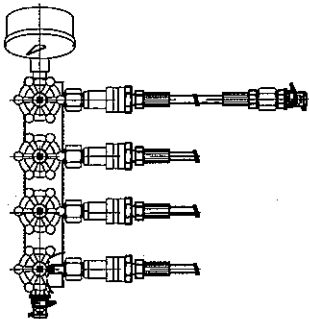
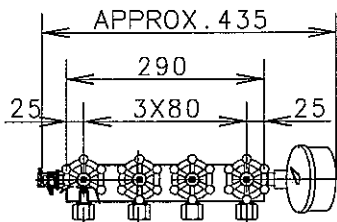
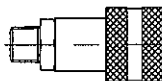
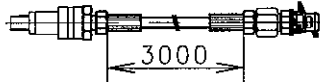
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD		REMARKS	
9	KEY, HEX. SOCKET SCREWS		SPRING STEEL	1		P91265-0003-098		
			0.03					
10	HYD. JACK FOR END CHOCK BOLT		BEECH WOOD	1		P91268-0001-200		
			32.124					
11	JACK-HYD. (COMPLETE)		Cr-Ni-Mo STEEL	2		P91268-0001-021		
			8.457					
12	JACK-HYD. (SUPPORT)		Cr-Ni-Mo STEEL	2		P91268-0001-033		
			4.466					
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20090209		YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 111-3	
Info. No.: S50MC-C70HYD001		Description: 11. HYDRAULIC JACK					Drawing no.: 2805029690(3/5)	
							STX Engine Co., Ltd.	

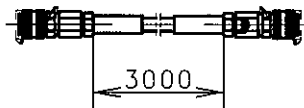
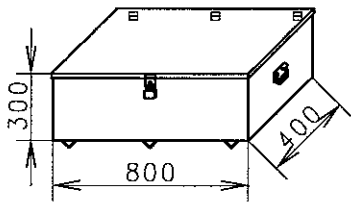
NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.		
			WEIGHT(kg)	STD	ADD		REMARKS		
13	SEALING RING W. BACK-UP		VITON PTFE	4		P91268-0001 -057			
			0.02						
14	SEALING RING W. BACK-UP		VITON PTFE	4		P91268-0001 -069			
			0.02						
15	DISC		CARBON STEEL	2		P91351-0037 -249			
			0.06						
16	SCREW WITH INSIDE HEXAGON		STEEL	2		P91351-0037 -237			
			0.02						
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No	
						*		3	
						*		2	
						*		1	
20090209		YWH	SMK	H CY	TBK	*	First issue	0	
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 111-4		STX Engine Co., Ltd.
Info. No.: S50MC -C70HYD001		Description: 11. HYDRAULIC JACK					Drawing no.: 2805029690(4/5)		

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT (kg)	STD	ADD		REMARKS
17	BALL HANDLE		TOOL STEEL	2		P91268-0001-057	
			0.09				
18	KEY, HEX. SOCKET SCREWS		SPRING STEEL	2		P91268-0001-069	
			0.03				

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090209	YWH	SMK	HCY	TBK	*	First issue	0
Scale: 1 : 1	Size: A4	Type: S50MC-C				Page No.: 111-5	 STX Engine Co., Ltd.
Info. No.: S50MC -C70HYD001		Description: 11. HYDRAULIC JACK				Drawing no.: 2805029690(5/5)	

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT(kg)	STD	ADD		REMARKS
1	PLANIMETER (DIGITAL)		PLASTIC	1		P91366-0066-073	
			0.8				

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.	
			WEIGHT (kg)	STD	ADD	PART NO.	REMARKS	
1	CONICAL V/V ASS'Y			1		9001000540		
2	DISTRIBUTOR BLOCK ASS'Y			1		TO BE INCLUDED ITEM NO. 1	-INLET : G1/4" -OUTLET : NPT3/8"	
3	QUICK COUPLING FE-MALE			4		TO BE INCLUDED ITEM NO. 1	NPT3/8"	
4	HYD HOSE WITH QUICK COUPLING FE-MALE			4		TO BE INCLUDED ITEM NO. 1	-NPT3/8" -1000bar	
Date		Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
						*		3
						*		2
						*		1
20070614		YWH	SJK	CSP	SKN	*	First issue	0
Scale: 1:1		Size: A4		Type: ALL		Page No.: 113-1		STX Engine Co., Ltd.
Info. No.: S35MC-CTP106		Description: 13. HYD. TOOL FOR PROPELLER					Drawing no.: 9005000320(1/2)	

NO.	NAME	SKETCH	MATERIAL	Q'TY /ENG.		PLATE NO.	PANEL NO. & BOX NO.
			WEIGHT(kg)	STD	ADD	PART NO.	REMARKS
5	HYD HOSE WITH UNION			1			G1/4"
						3101484-3.0	
6	TOOL BOX			1			400X800X300
						5D-44748R	

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20090317	KCJ	SML	TBK	TBK	*	First issue	0
Scale: 1:1	Size: A4	Type: ALL				Page No.: 113-2	STX Engine Co., Ltd.
Info. No.: S50MC -C70HYD002		Description: 13. HYD. TOOL FOR PROPELLER					Drawing no.: 9005000360(2/2)

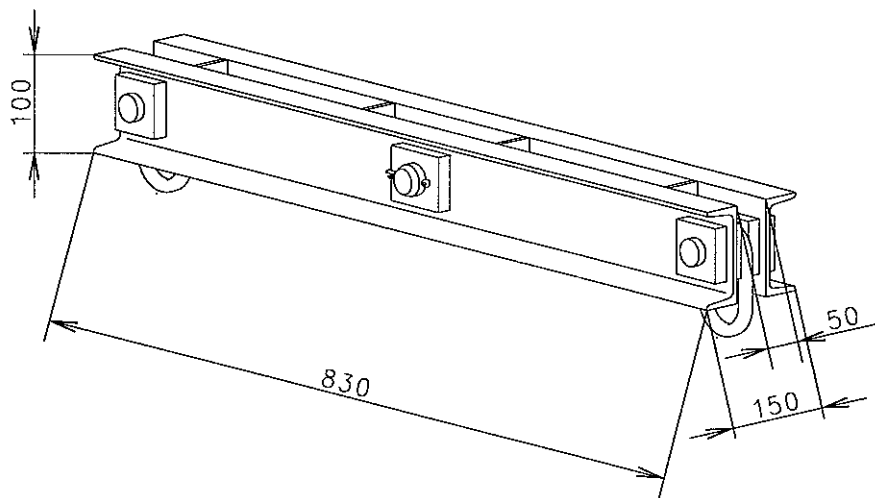
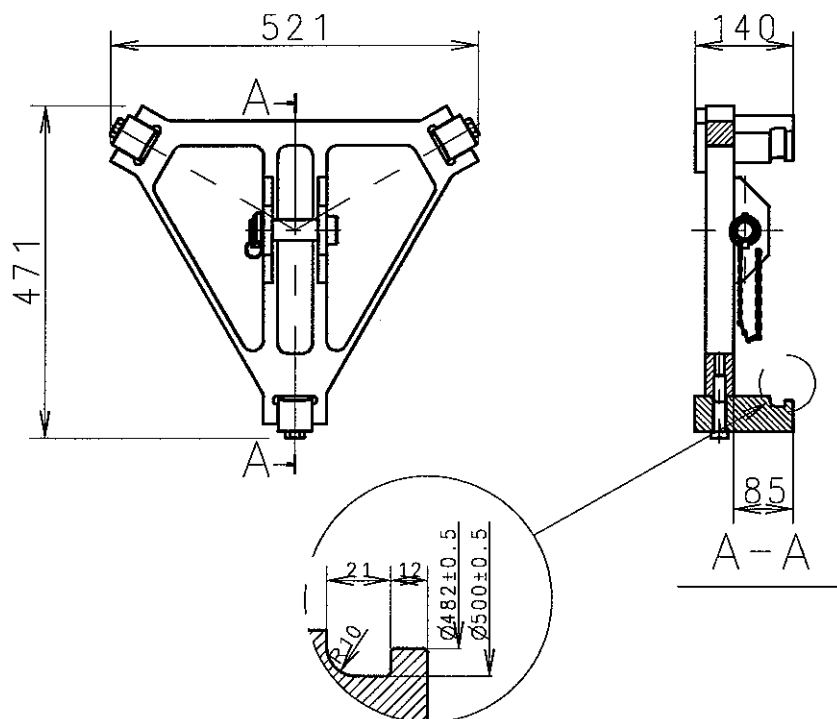
LIST OF LARGE TOOL PARTS

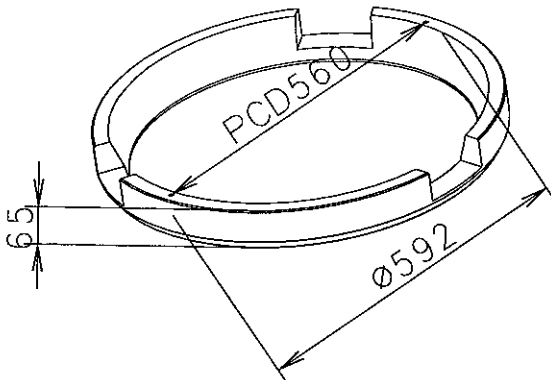
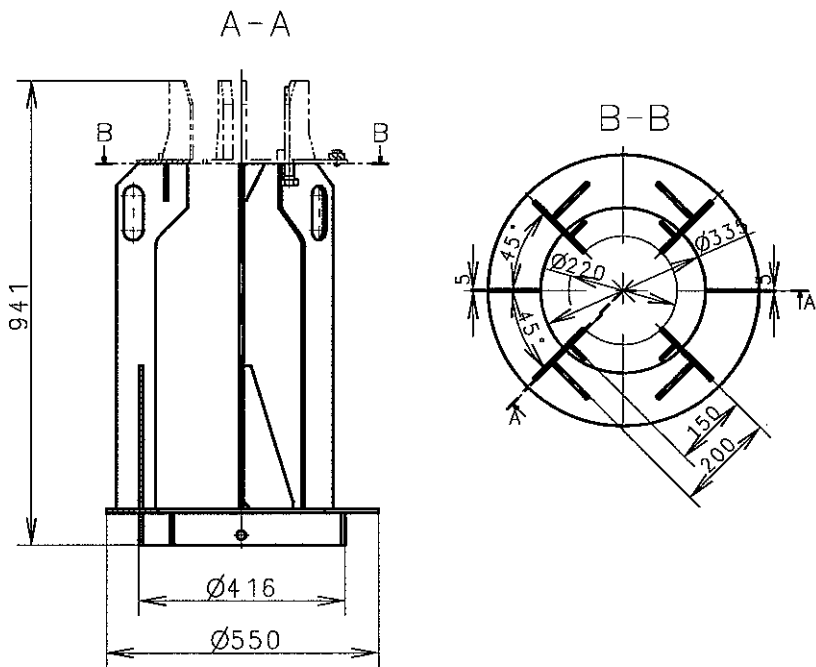
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Date	Des.	Chk.	Chk.	Appd.	A.C.	Change / Replacement	C.No
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					*		2
					*		1
20070103	KCJ	H CY	CSP	JSP	*	First issue	0
Scale: 1:1	Size: A4	Type: S50MC-C				Page No.: 15-4-1	 SUX Engine Co., Ltd.
Info No.:		Description: LIST OF LARGE TOOL PARTS					Drawing no.: 2805014190(1/2)

INDEX

[illegible]

NO.	NAME	SKETCH	REMARKS						
			WEIGHT (kg)						
1	CROSS BAR FOR CYL. LINER		29.268						
2	LIFTING TOOL FOR PISTON		24.918						
Date			Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
							*		3
							*		2
							*		1
20070103			YWH	HCY	CSP	JSP	*	First issue	0
Scale: 1:1		Size: A4		Type: S50MC-C			Page No.: 15-4-3		STX Engine Co., Ltd.
Info. No.:		Description: OUTLINE DWG OF LARGE TOOL PARTS						Drawing no.: 2805016490	

NO.	NAME	SKETCH	REMARKS
			WEIGHT (kg)
1	GUIDE RING FOR PISTON		26.15
2	SUPPORT IRON FOR PISTON		44.652

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070103	YWH	HCH	CSP	JSP	*	First issue	0

Scale:	Size:	Type:	Page No.:	STX Engine Co., Ltd.
1:1	A4	S50MC-C	15-4-4	
Info. No.:		Description:	Drawing no.:	
		OUTLINE DWG OF LARGE TOOL PARTS	2805016500	

NO.	NAME	SKETCH	REMARKS
			WEIGHT (kg)
1	LIFTING TOOL FOR CRANKSHAFT		58.1
2	SUPPORT IRON FOR PISTON		40.3

Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070103	YWH	HCY	CSP	JSP	*	First issue	0

Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 15-4-5	STX Engine Co., Ltd.
Info. No.:	Description: OUTLINE DWG OF LARGE TOOL PARTS	Drawing no.: 2805016510		

NO.	NAME	SKETCH	REMARKS				
			WEIGHT (kg)				
1	TEST RIG FOR FUEL VALVE		105.83				
Date	Des.	Chk.	Appd.	Appd.	A.C.	Change / Replacement	C.No
					*		3
					*		2
					*		1
20070103	YWH	HCY	CSP	JSP	*	First issue	0
Scale: 1:1	Size: A4	Type: S50MC-C	Page No.: 15-4-6		STX Engine Co., Ltd.		
Info. No.:		Description: OUTLINE DWG OF LARGE TOOL PARTS				Drawing no.: 2805016520	

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3

4

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A

F

B

C

D

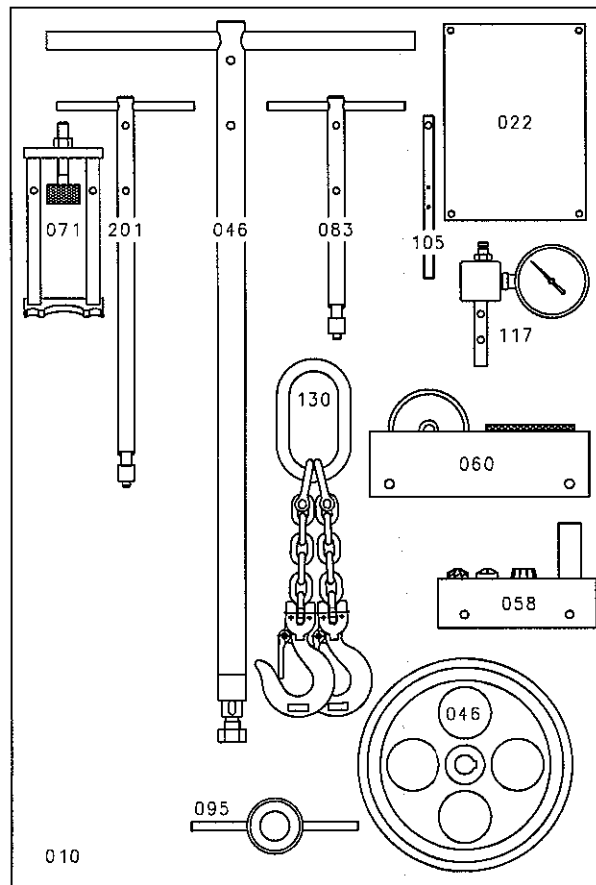
15-5-1

LICENSEE
NAME

CYLINDER COVER PANEL

PLATE 90151-199

S50MC-C



SIZE : 900x1350x25

Basic Standards (MBC 50) & Suppl. Drawing No.: 1175472-3.0		EN21C Surf. roughness		Projection:		Material / Blank:	
EN21F-m Tolerances		Max (g):		Final User Material:			
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement		
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							6
							5
							4
							3
							2
							1
							0
20010925							
Similar Drawing No.:		Replacement for Ident. No.:					
Scale:	1:1	Size:	A2	Type:	S50MC-C	Page No.:	01 (09)
Info. No.:	589901	Description:	PANEL FOR TOOLS(MS901)			Ident. No.:	2805028540
Final User Info. No.:	Final User Description:			Final User Ident. No.:			

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15-5-1



1

2

3

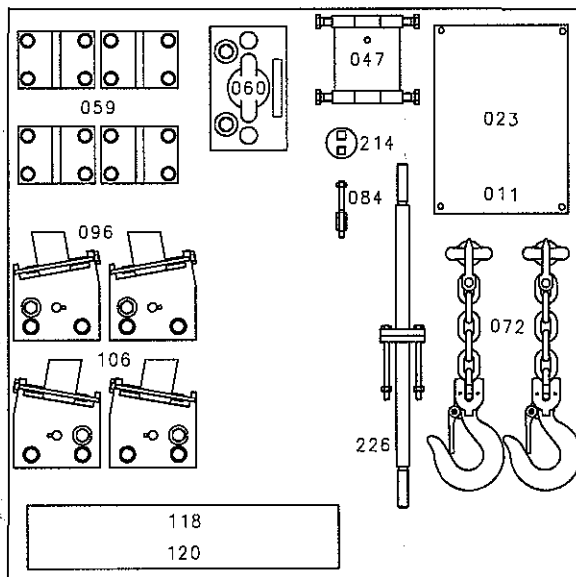
4

5

6

LICENCEE
NAME

CONNECTING ROD AND CROSSHEAD PANEL PLATE 90451-133
S50MC-C



SIZE : 900x900x25

Basic Standards (ISO 9001 & Suppl. Drawing No.: EN21C Surf. roughness		Projection:	Material / Blank:
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Date	Des.	Chg.	Appd.
Change / Replacement			C.No
Replaced by Ident. No.:			9
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			5
			4
			3
			2
			1
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20001012		Replacement for Ident. No.:	
Scale: 1:1		Page No.: 03 (09)	
Size: A2		Type: S50MC-C	
Info. No.: 589904		Description: PANEL FOR TOOLS(MS904)	
Final User Info. No.:		Final User Description:	
Ident. No.: 2805028540		Final User Ident. No.:	

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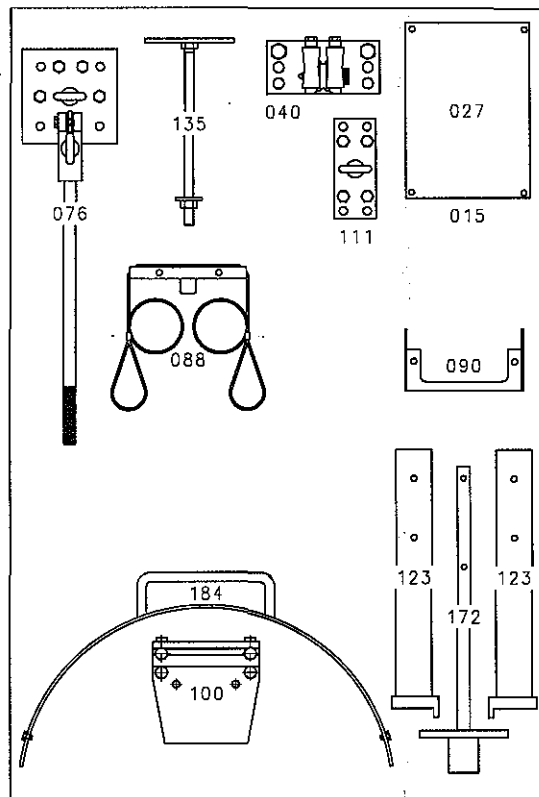
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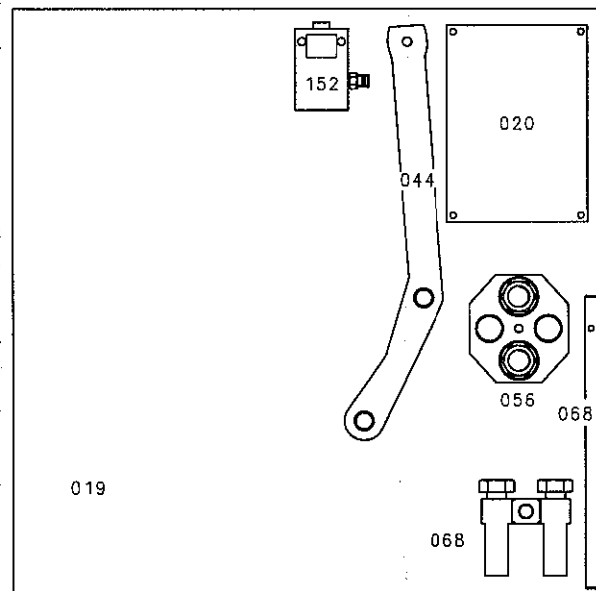
S50MC-C



SIZE : 900x1350x25

Basic Standards (M&B SBI & Suppl. Drawing No.:				EN21C Surf. roughness		Projection: 		Material / Blank:	
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Date	Des.	Chk.	Appd.	A.C.	Change / Replacement				C.No.
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				+					2
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20001012									
Similar Drawing No.:					Replacement for Ident. No.:				
Scale: 1:1		Size: A2		Type: S50MC-C			Page No. 1 04 (09)		SBX Engine Co. Ltd.
Info. No.:		Description:					Ident. No.:		
589905		PANEL FOR TOOLS(MS905)					2805028540		
Final User Info. No.:		Final User Description:					Final User Ident. No.:		

S50MC-C



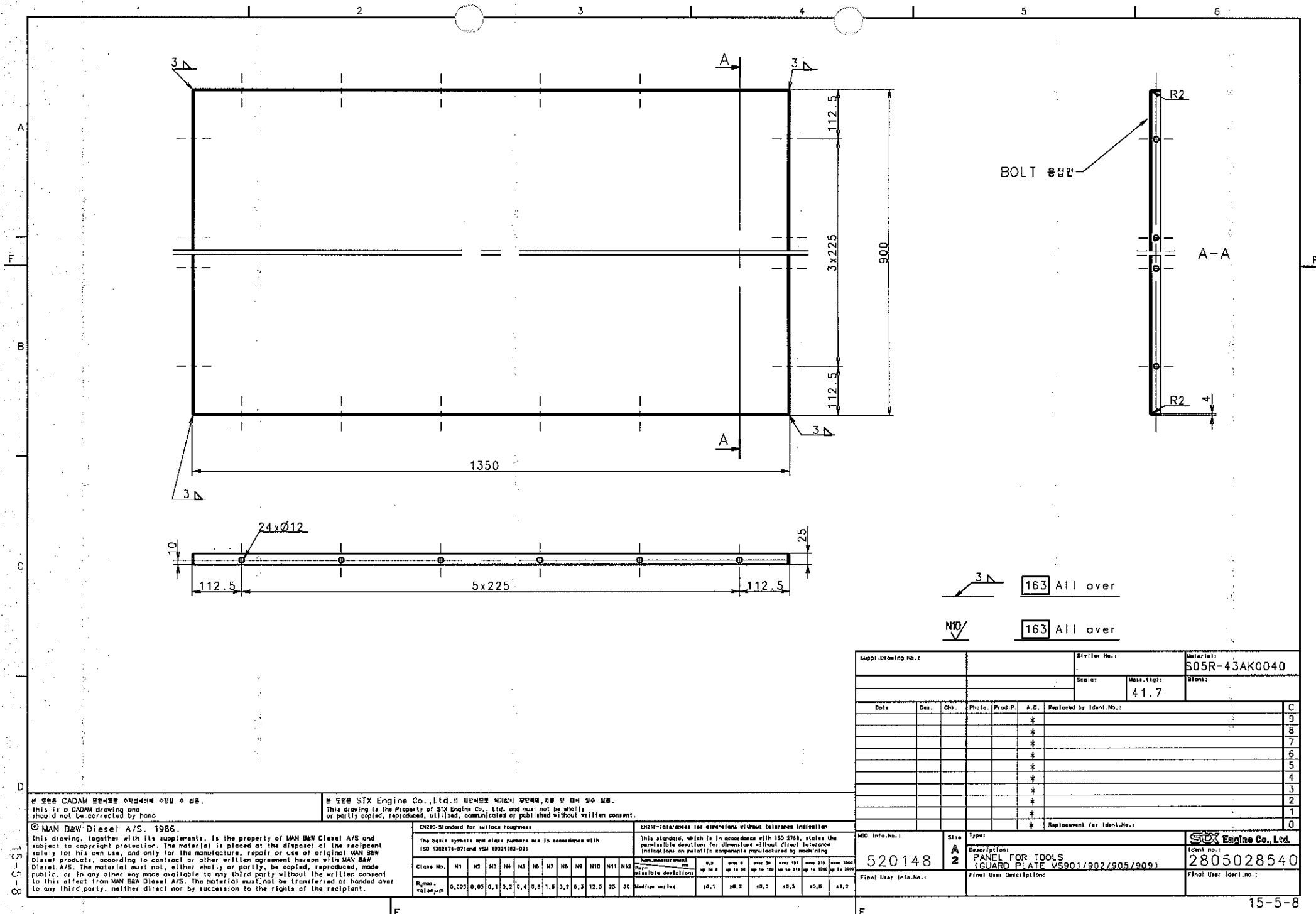
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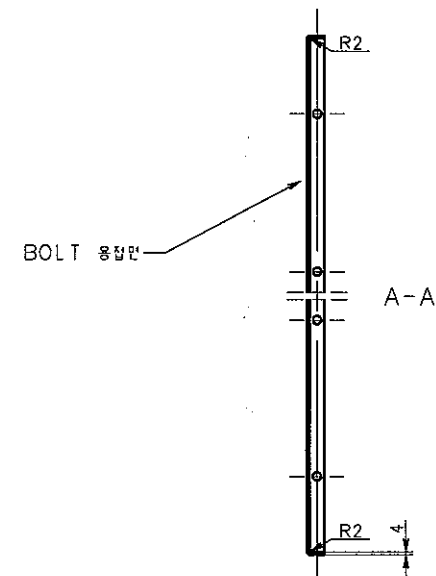
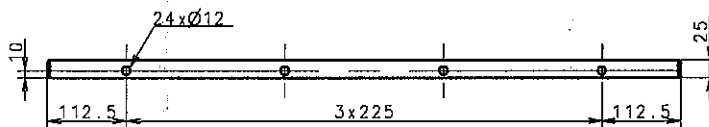
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1175454-4.0		EN21F-m Tolerances		Moss (sgt)		Final User Material:	
Date	Des.	Chg.	Appd.	A.C.	Change / Replacement		
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				†	8		
				†	7		
				†	6		
				†	5		
				†	4		
				†	3		
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Similar Drawing No.:		1175371-6		Replacement for Ident. No.:			
Scale:	Size:	Type:	S50MC-C		Page No.:	05 (09)	
Info. No.:	Description:		589906		STX Engine Co., Ltd.		
Final User Info. No.:	Final User Description:		PANEL FOR TOOLS(MS906)		Ident. No.:		
589906				2805028540			
Final User Ident. No.:				Final User Ident. No.:			

PLATE 90851-0200

[illegible]

Basic Standards (MSB) & Suppl. Drawing No.:						EN21C Surf. roughness		Projection: 		Material / Blank:	
1175386-1.2						EN21F-m Tolerances		Mass (kg):		Final User Material:	
Date	Des.	Chk.	Appd.	A.C.	Change / Replacement				C.N.		
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				#					7		
				#					6		
				#					5		
				#					4		
				#					3		
				Z3					2		
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19970206										0	
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Scale:		Size:		Type:				Page No.:		SOX Engine Co., Ltd.	
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Info. No.:				Description:				Ident. No.:			
589908				PANEL FOR TOOLS(MS908)				2805028540			
Final User Info. No.:				Final User Description:				Final User Ident. No.:			





163 All over

N10/ 163 All over

Suppl. Drawing No.:			Similar No.:			Material:		
						S05R-43AK0040		
			Scale:		Mass.(kg):		Blank:	
					28.3			
Date	Des.	Chf.	Photo.	Prod.P.	A.C.	Replaced by Ident.No.:		
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MFG Info.No.:			Size			Type:		
520148			A 2			Description: PANEL FOR TOOLS (GUIDE PLATE MS904/906/908)		
Final User Info.No.:			Final User Description:			Ident. No.:		
						SUX Engine Co., Ltd. 2805028540		

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EN210-Standard for surface roughness

The basic symbols and class numbers are in accordance with ISO 1502(74-07) and VDI 1032(87-09)

ENG21F-Tolerances for dimensions without tolerance indication

This standard, which is in accordance with ISO 2768, states the permissible deviations for dimensions without direct tolerance indications on metallic components manufactured by machining.

Instructions on how to fill in components of the form													Non-measurement						
Class No.	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12	per-	± 8	over 8	over 30	over 120	over 310	over 1000
													missible	mm	up to 8	up to 30	up to 120	up to 310	up to 1000
R _{max} , mm	0,025	0,03	0,1	0,2	0,4	0,6	1,0	3,2	6,3	12,5	25	50	Medium series	±0,1	±0,3	±0,5	±0,8	±0,8	±1,2

MED Info.No.: 520148

Size	Type:
A 2	Descr PA (C

Descriptions:	PANEL FOR TOOLS (GUIDE PLATE MS904/906/908)
---------------	--

Final User Info.No.:

Final User Description:	
-------------------------	--

Stux Engine Co., Ltd.

Ident no.: 2805028540

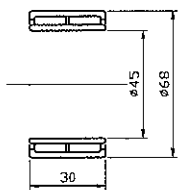
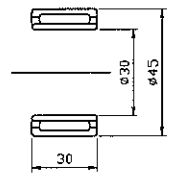
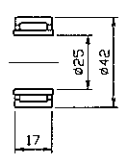
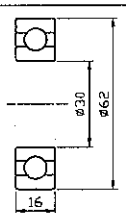
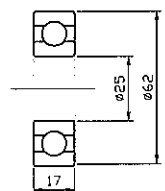
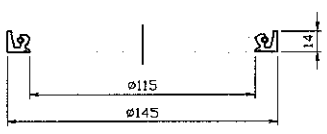
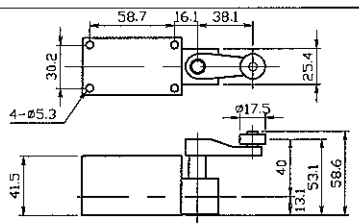
Final User Ident.no.:

LIST OF SPARE & TOOL PARTS FOR ACCESSORY COMPONENT

stx Heavy Industries

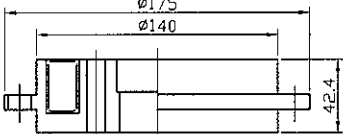
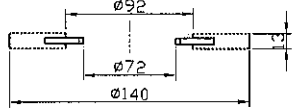
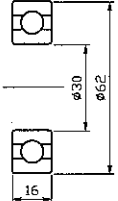
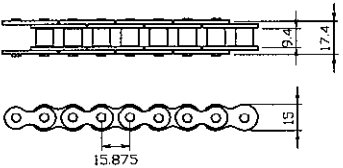
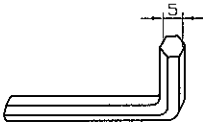
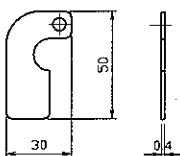
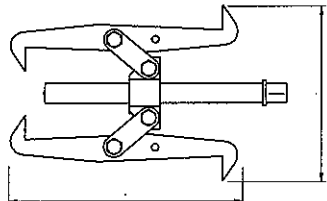
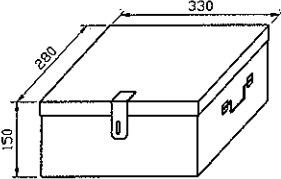
HULL. NO	S-4006/12/29/14	APPROVED BY	T. B. KIM
TYPE	7S50MC-C7	CHECKED BY	
DWG. NO.	2805030091 (1/2)	CHECKED BY	J. T. SEO
DATE	2010.06.22	DRAWN BY	S. M. LEE
REVISION	1	Aux. blower, T/C spare & tool list rev.	
	2		
	3		
			PAGE NO. 15-5-1

MAIN ENGINE TURNING GEAR WITH MOTOR SPARE PARTS LIST FOR REDUCTION GEAR

NO.	NAME	SKETCH	MATERIAL	SET	SPARE	PART NO	SIZE	REMARKS
				QUANTITY	QUANTITY			
1	NEEDLE BEARING		BEARING STEEL	8	1	(171)	NA #5909	PURCHASE
2	NEEDLE BEARING		BEARING STEEL	4	1	(151)	TAFI #304530	PURCHASE
3	NEEDLE BEARING		BEARING STEEL	3	1	(101)	NA #4905	PURCHASE
4	BALL BEARING		BEARING STEEL	1	1	(061)	#6206	PURCHASE
5	BALL BEARING		BEARING STEEL	1	1	(062)	#6305	PURCHASE
6	OIL SEAL		SYNTHETIC RUBBER	1	1	(042)	D TYPE 11514514	PURCHASE
7	OIL SEAL		SYNTHETIC RUBBER	1	1	(052)	D TYPE 30458	PURCHASE
8	LIMIT SWITCH			1	1	(54)	SZL-WL -A (HONEYWELL)	PURCHASE
15-5-3				YRTG 20-27	 YURIM PRECISION INDUSTRIAL CO.		YU-50MC-C-SPLST1	

MAIN ENGINE TURNING GEAR WITH MOTOR

SPARE PARTS LIST FOR MOTOR

NO.	NAME	SKETCH	MATERIAL	SET	SPARE	PART NO	SIZE	REMARKS
				QUANTITY	QUANTITY			
1	MAGNET HOUSING			1	1	YU-50S-56B (B4)	DB-8.0 (PART)	DAE KWANG ELECTRO MAGNETIC
2	FRICTION DISC		NON-ASBESTOS / SUS304	1	1	YU-50S-56B (B2)	DB-8.0 (PART)	DAE KWANG ELECTRO MAGNETIC
3	BALL BEARING		BEARING STEEL	2	2	YU-50S-56 (M7)	#6206	PURCHASE
4	ROLLER CHAIN		SK-5	2.1 M	0.5 M	(391)	HC #50	PURCHASE
5	WRENCH				1	YU-50S-56B (B11)		PURCHASE
6	THICKNESS GAUGE		SUS304		4	YU-50S-56B (B12)		YU RIM INDUSTRIAL
7	PULLER				1			PURCHASE
8	TOOL BOX				1			
15-5-4			YRTG 20-27	 YURIM PRECISION INDUSTRIAL CO.		YU-50MC-C-SPLST2		

Messrs. STX Heavy Industries Co., Ltd.

for Installation

排気弁研磨機 EXHAUST VALVE GRINDER

type: SEG-50MC/50MC-C-3ph440V

for

Main Diesel Engine 50MC/MC-C

CHIEF DEP.	CHIEF DEG.	DEG.	MFG. NO.	HULL NO.
			DATE	30-Nov-04
WAKEFIELD CORPORATION 株式会社 ウェイクフィールド			DWG. NO.	

Specifications 仕様

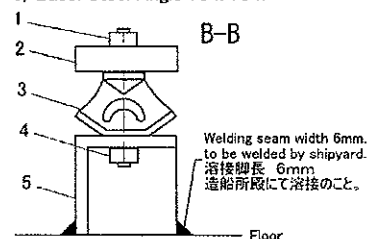
Grinding Angle/Seat	30.0 0/0.1°
Grinding Angle/Spindle	30.4 0.1/0°
ELECTRIC POWER SOURCE	440VAC, 3ph, 60Hz
GRINDING MOTOR	1400W
GRINDING MOTOR SPEED	4,200 min ⁻¹
GRINDING WHEEL SIZE	150 x 15 x 32 mm
ROTATING SPEED MOTOR 90W	
ROTATING SPEED/DIRECTION	0~20 min-1, Reversible Direction
MACHINE WEIGHT	175 kgs
MACHINE SIZE (H x W x D)	1250 x 970 x 750 mm

The Direction of Shock Absorbers

The machine has been assembled to be installed in a FORE-AFT direction at the time of delivery from factory.

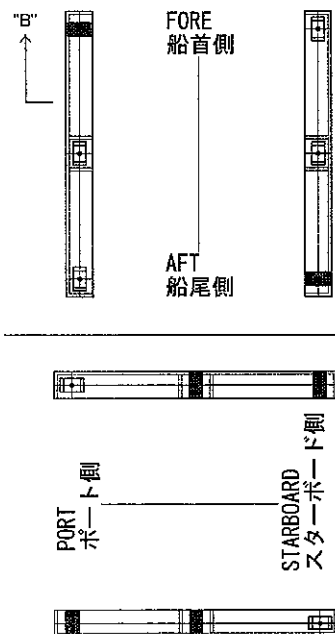
In case the machine is installed in a PORT-STARBOARD direction the machine's Shock Absorbers direction needs to be changed as per below sketch.

- 1) M8 Nut & Spring Washer
- 2) Machine Rest
- 3) Shock Absorber type EC4001
- 4) M8 Nut & Spring Washer
- 5) Base: Steel Angle 75 x 75 x 9mm

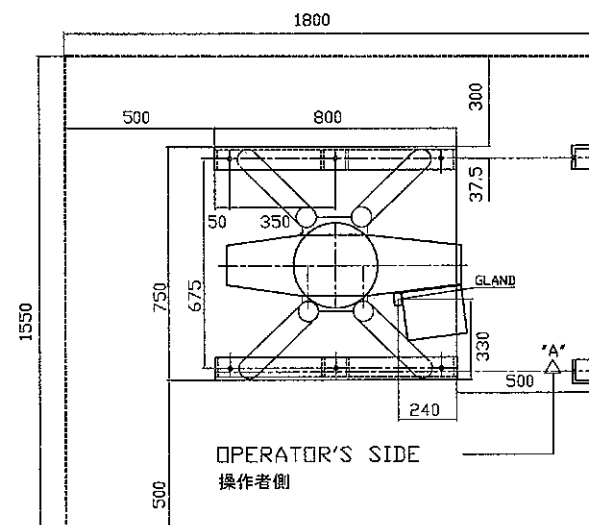
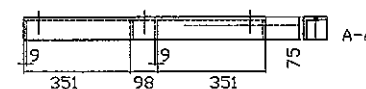
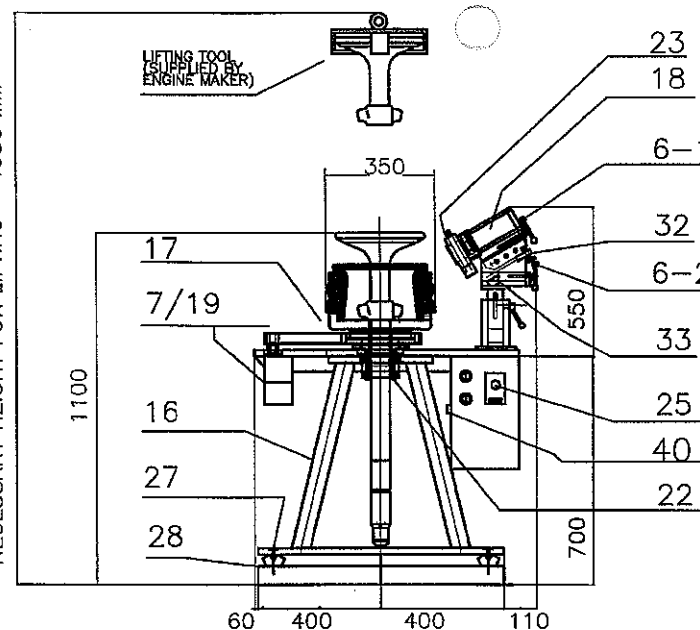


The Base to be welded to Floor.
ベースは床に溶接のこと。

Maker's standard paint; not changeable
Standard Paint ; Munsell 10R4/13

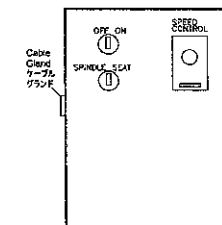


NECESSARY HEIGHT FOR LIFTING = 1950 MM



NECESSARY WORKING AREA
必要作業範囲

Control Box



6-1	Grinding Feed Handle
6-2	Cutting Feed Handle
7/19	Gear Box/ Rotation Motor (electric driven)
16	Machine Rest
17	Centering Chuck (Centering Bolts)
18	Grinding Motor (electric driven)
22	Securing Chuck
23	Grinding Wheel with wheel cover
25	Control Box
27	Shock Absorber
28	Base
32	Angle Adjustment Device
33	Angle Indication Plate
40	Cable Gland 20c

Engine type	50MC/MC-C
Valve Grinder	Type SEG-50MC/MC-C-440V3PH
Date:	Mar. 7, 2005
DRWG No:	505001440
WAKEFIELD CORPORATION	

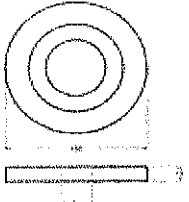
SPARE PARTS

No. 1

予備品リスト NO.SEG-SPL-3

28-Jun-04

SPARE PARTS LIST FOR EXHAUST VALVE GRINDER TYPE "SEG"

No.	NAME	SKETCH	WORKING	SPARE	WEIGHT	REMARKS
1	Grinding Wheel 砥石		1	0	0.4 kgs	P/N. KWM-709 for Valve Seat & Spindle 150 x 15 x 32mm
2	Timing Belt for Rotation Motor タイミングベルト 回転モーター用		1	1	0.1 kgs	P/N. 510L100
3	Timing Belt for Grinding Motor タイミングベルト 研磨モーター用		1	1	0.02 kgs	P/N. XL170
4	Fuse ヒューズ		1	1	0.005 kgs	15A
5						



Wakefield Corporation
 株式会社 ウェイクフィールド
 CALL: +81-(0)45-780-6120
 FAX: +81-(0)45-780-3392

1-8-24, Fukuura, Kanazawa-ku,
 Yokohama 236-0004 Japan
 e-mail : info@wakefield.co.jp

TOOL LIST

No. 1

要具リスト NO.SEG-TLL-5

17-Feb-05

TOOL LIST FOR EXHAUST VALVE GRINDER TYPE "SEG"

No.	NAME	SKETCH	Q'ty	weight	REMARKS
1	Hexagon Wrench 六角レンチ		Total 4 pcs	0.12 kgs	4mm x 1pc 5mm ball end long x 1pc 6mm x 1pc
2	Grinding Wheel Dresser 砥石ドレッサー		1	0.05 kgs	ND-1A6A Φ 11x30
3	Wheel Hub Tightening Bar ウィールハブ締結バー		2	0.15 kgs	φ 6mmx100mm
5	Dial Gauge with Magnetic Base ダイヤルゲージ (マグネットベース付 き)		1	1.5 kgs	unit; 0.01mm



Wakefield Corporation
株式会社 ウェイクフィールド

1-8-24, Fukuura, Kanazawa-ku,
Yokohama 236-0004 Japan

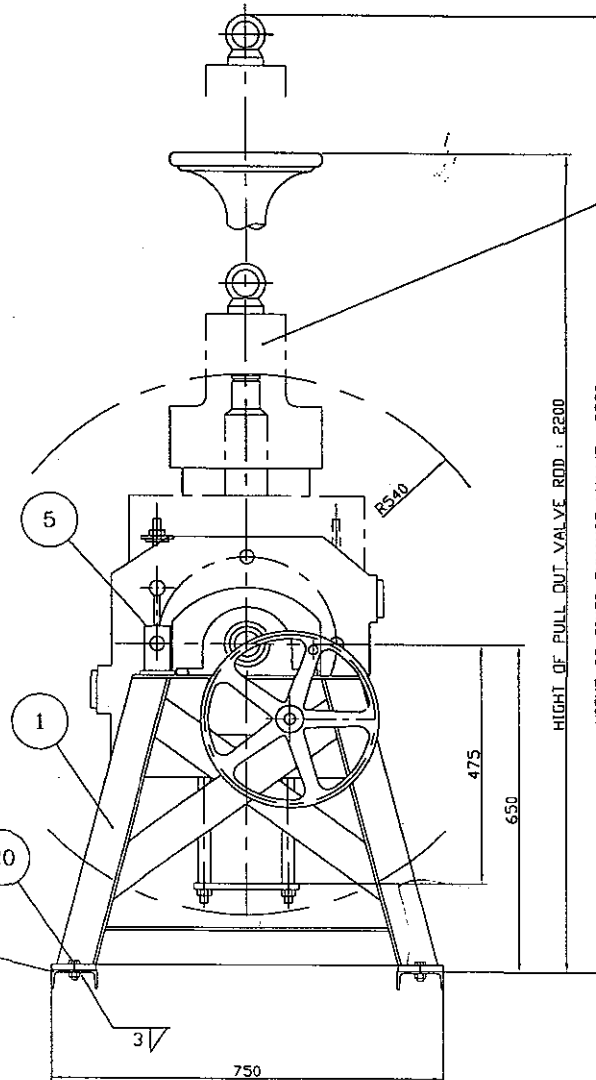
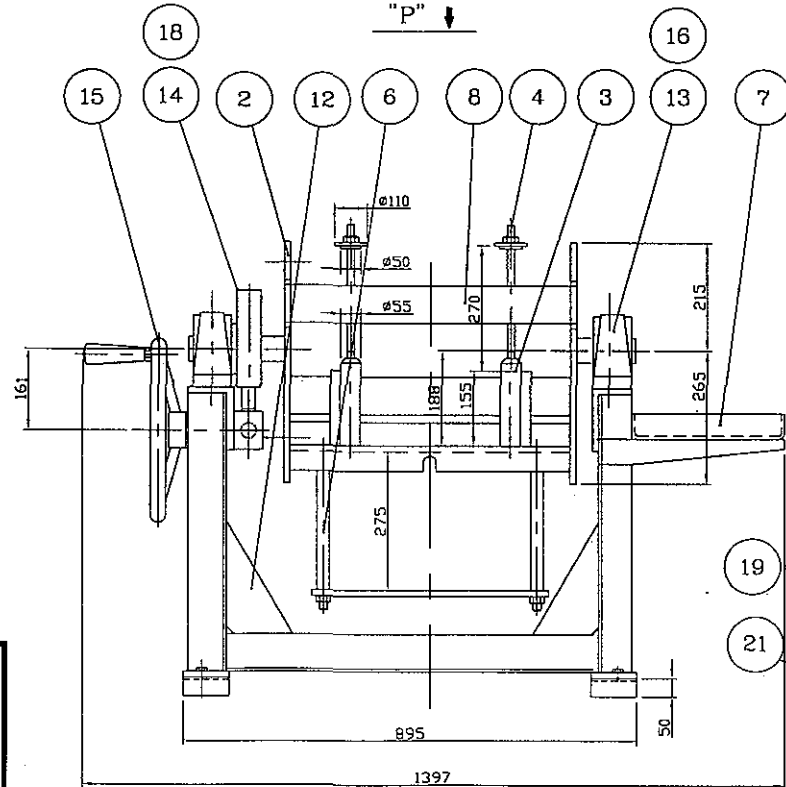
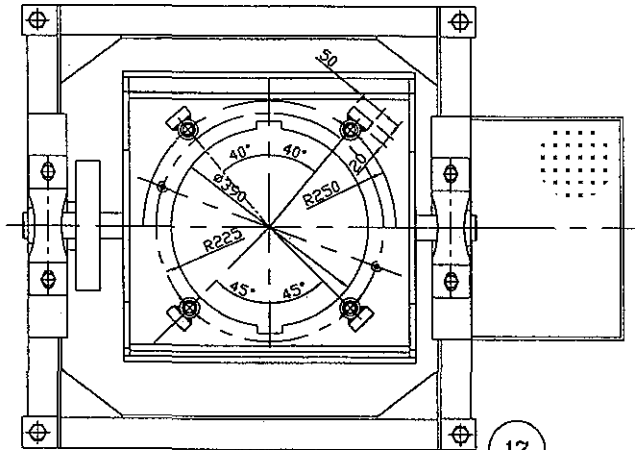
CALL: +81-(0)45-780-6120
FAX: +81-(0)45-780-3392

e-mail : info@wakefield.co.jp

REPAIR KIT FOR PNEUMATIC CONTROL SYSTEM (S50MC-C)

NO.	DESCRIPTION	REPAIR KIT	POS. NO.	REMARK
1	AIR CYLINDER	RAC-SP-50100	13	
2	MAGNET SWITCH	894.041.060.2	7,8	
3	3/2 WAY VALVE	371.076.000.2	10,11	
4	3/2 WAY VALVE	04904-310-04	26,117	
5	5/2 WAY VALVE	371.075.000.2	27	
6	CHECK CHOKE VALVE	534.106.000.2	32	
7	3/2 WAY VALVE	371.020.000.2	25,33	
8	3/2 WAY VALVE	363.003.002.2	55,56	
9	AIR CYLINDER	04903-943-03	57	
10	3/2 WAY SOLENOID V/V	563.102.000.2	84,86,88,90,127	
11	5/2 WAY VALVE	363.129.000.2	100, 105	
12	3/2 WAY MANUAL VALVE	577.202.000.2	101	

VIEW "P"



DATE	DRA	REVISION DESCRIPTION	
020701	KJA		0
021125	KJA	DIM ADDED.	1
021208	KJA	DIM MODIFY, ADDED.	2
030104	KJA	ADDED.	3
			4

PANINTING 地: LT315-RAL2004

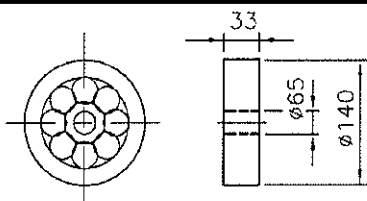
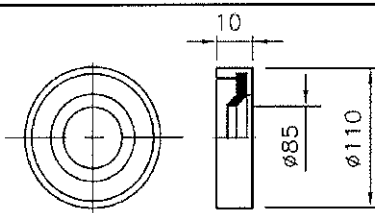
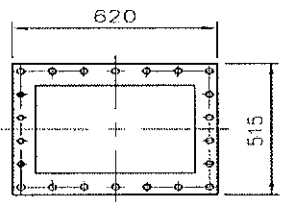
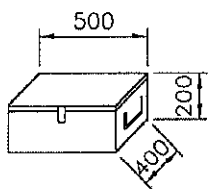
21	CHANEL	4	100x100x50	
20	NUT	4	M20	
19	B.8T BOLT	4	M20x65	
18	B.8T BOLT	4	M10x25	
17	SCM NUT	4	M20	
16	B.8T BOLT	4	M20x65	
15	HANDWHEEL	1	AC2B-F	
14	COVER	1	SS41	
13	BEARING	2	MAKER	
12	BRACKET	4	SS41	
8	SIDE BOARD	2	SS41	
7	TOOL SPACE	1	SS41	
6	V/V STOPPER	1	SS41	
5	STOPPER	1	SS41	
4	CLAMPING	4	SS41	
3	SETTING BASE	4	SS41	
2	TURN TABLE	1	SS41	
1	FRAME	1	SS41	
NO.	PART NAME	Q.T.Y	MATERIAL	REMARK



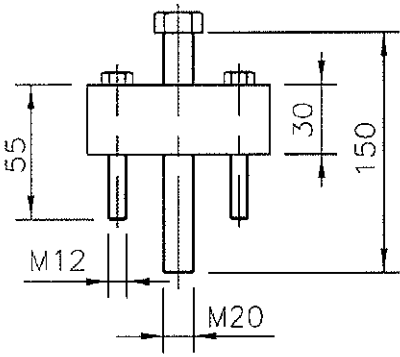
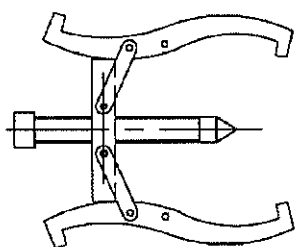
SANG KI CO., LTD.

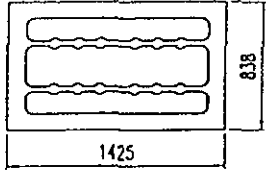
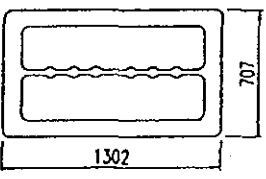
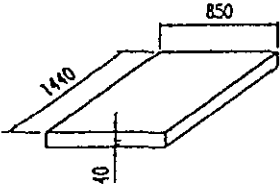
DATE	02.07.01	PRODUCT LINE	
APPROVAL	K.C.CHOI		I/S 50MC/-C
CHECK		PART NAME	
DRAWING	J.A.KIM	WORKING RIG FOR EXH.V/V	
SCALE	UNIT	WEIGHT (kg)	PART NO.
N/A		ab.220	MST-020750MC-3

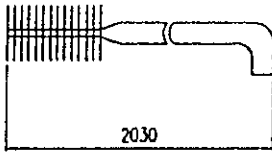
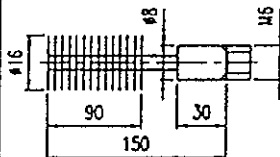
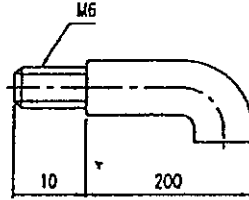
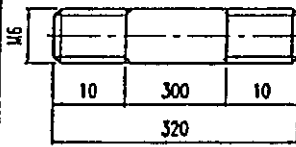
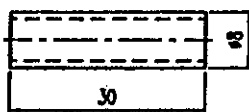
SPARE PART LIST

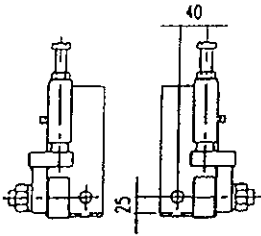
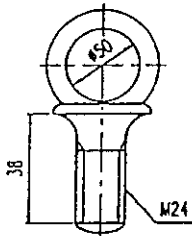
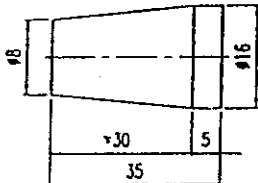
SHIP NO.		S-4006/12/29/14		A M E N D	REV.	DATE.	NOTE
DWG. NO.		H-0143P-A					
NO.	NAME OF PARTS	OUT LINE	QUANTITY			REMARK	
			WORKING		SPARE		
			PER BL.SET	PER ENG.			
1	BEARING		2	4	1	NO. 6313ZZ C3 (ABB 55Kw-2P)	
2	BEARING		2	4	2	THEROMOELASTIC SEAL	
3	PACKING		1	2	1	NON-ASBESTOS (1.5T)	
6	SPARE PART BOX				1	WITH KEY AND LOCK	

TOOL LIST

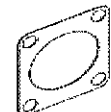
SHIP NO.		S-4006/12/29/14		A M E N D	REV.	DATE.	NOTE
DWG. NO.		H-0005T					
NO.	NAME OF PARTS	OUT LINE			Q'TY	REMARK	
1	IMPELLER TOOL				1		
2	BEARING TOOL				1	10"	

		SPARE PARTS LIST FOR AIR COOLER				PAGE			
						USED FOR			
						BOX NO.			
NO.	NAME	SKETCH	MATERIAL	SUPPLY per COOLER		DRAWING		REMARK	
				WORK	SPARE	NO.	PART NO.		
1	GASKET FOR CHANNEL COVER	 TH'K 1.5t	ASBESTOS-FREE	1	1	DHCA-0771T4Y4	8		
2	GASKET FOR REVERSING COVER	 TH'K 1.5t	ASBESTOS-FREE	1	1	DHCA-0771T4Y4	9		
3	SPARE GASKET BOX FOR SPARE PARTS NO. 1 - 2		WOOD	1				per SHIP	
4									
MFR'S NAME		DongHwa Entec							

STD.NO.		A00-TC-01		PAGE		A - 1		
TOOLS LIST FOR AIR COOLER						USED FOR		LKM-C/E/K
						BOX NO.		
NO.	NAME	SKETCH	MATERIAL	SUPPLY per ship		DRAWING		REMARK
				WORK	SPARE	NO.	PART NO.	
1	NYLON BRUSH		NYLON & SS400	1				
-a	NYLON BRUSH		NYLON & SS400	1		THIS TOOL TO BE USED TOGETHER WITH OTHER H/EX AND ASSEMBLED WITH ITEM a.b.c.d		
-b	HANDLE		SS400	1				
-c	BAR		SS400	5				
-d	JOINT		SS400	6				
MFR'S NAME		DongHwa Entec						

STD.NO. A00-TA-01D						PAGE		A - 2	
TOOLS LIST FOR AIR COOLER						USED FOR			
						BOX NO.			
NO.	NAME	SKETCH	MATERIAL	SUPPLY per ship		DRAWING		REMARK	
				WORK	SPARE	NO.	PART NO.		
2	ROLLER FOR FIXED SIDE		STEEL	1set					
3	EYE BOLT		STEEL	4					
4	PLUG		BRASS	60				PER COOLER	
5									
MFR'S NAME		DongHwa Entec							

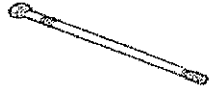
List of Spare Parts

Order No.	Number	Designation	Weight [kg]	Design
506.027	1	Gasket	0.06	 $d_i = 93$ $d_a = 190$
506.066	2	Hexagon bolt	0.1	 M16 x 40 SM
506.106	6	Hexagon bolt	0.2	 M16 x 80 SM
506.107	27	Lock washer pair	0.01	 $d_i = 17$
506.108	1	Hexagon bolt	0.1	 M16 x 40 SM
509.012	4	Hexagon bolt	0.1	 M12 x 70 SM
509.014	12	Lock washer pair	0.01	 $d_i = 13$
513.013	10	Hexagon bolt	0.06	 M12 x 35 SM
513.014	21	Lock washer pair	0.01	 $d_i = 13$
517.026	1	Gasket	0.1	 $d_i = 91$

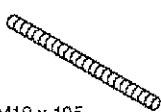
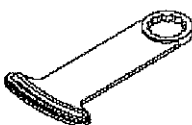
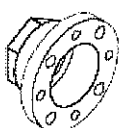
List of Spare Parts

Order No.	Number	Designation	Weight [kg]	Design
517.063	1	O-ring seal	0.2	 $d_i = 1005$
517.070	1	Gasket, turbine side	0.1	 $d_i = 187$
517.075	1	Bearing disk	1.4	
517.085	1	O-ring seal	0.1	 $d_i = 350$
517.095	14	Lock washer pair	0.01	 $d_i = 13$
517.102	2	Hexagon nut	0.02	 M12
517.109	1	Gasket	0.1	 $d_i = 53$ $d_o = 130$
517.154	1	Counter-thrust bearing	2.0	
517.156	12	Lock washer pair	0.01	 $d_i = 8.7$
517.157	3	Hexagon bolt	0.01	 M8 x 20

List of Spare Parts

Order No.	Number	Designation	Weight [kg]	Design
517.161	2	Bearing bush	2.7	
517.172	2	Countersunk bolt	0.01	 M8 x 30
517.173	2	Lock nut	0.01	 VM8
517.185	1	Gasket, compressor side	0.1	 $d_i = 178$ $d_a = 233$
517.211	1	Seal ring	0.02	 $d_i = 45$ $d_a = 100$
540.015	2	O-ring seal	0.15	 $d_i = 625$
546.073	1	Undercut bolt	0.3	 M16 x 250
554.013	1	O-ring seal	0.1	 $d_i = 388$
544.150	1	Fleece mat	1.4	 5220 x 710 x 16

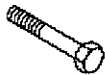
List of Tools

TOOL			Part No.	11.59700-0183		
TURBOCHARGER : TCA66			DATE	2010.05.18		
			SHEET	1/8	REV.	0
Order No	Designation	Representation	Weight in kg		Amount	
596.001	Puller ring, compressor wheel		2		1	
596.002	Assembly ring, compressor wheel		2		1	
596.003	Jack bolt	 M20 x 2 x 100	0.3		1	
596.004	Threaded bar	 M10 x 195	0.1		4	
596.005	Hexagon nut	 M10	0.01		4	
596.018	Locking key (emergency operation)		3.3		1	
596.022	Hub, (emergency operation)		1.6		1	
						

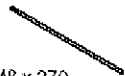
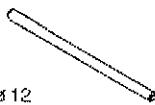
List of Tools

TOOL			Part No.	11.59700-0183	
TURBOCHARGER : TCA66			DATE	2010.05.18	
			SHEET	2/8	REV.
Order No	Designation	Representation	Weight in kg	Amount	
596.023	Cylindrical screw	 M10 x 35	0.1	4	
596.024	Lock washer pair	 $d_i = 10,7$	0.01	4	
596.027	Closing cover (compressor side), emergency operation	 $\varnothing=508$	19.1	1	
596.028	Closing cover (turbine side), emergency operation	 $\varnothing=301$	1.9	1	
596.029	Gasket (compressor side)	 $d_i = 380$ $d_a = 508$	0.5	1	
596.030	Gasket (turbine side)	 $d_i = 187$ $d_a = 290$	0.2	1	
596.031	Stud screw	 M12 x 658	0.6	1	
					

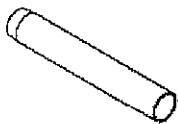
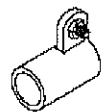
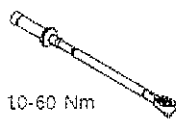
List of Tools

TOOL			Part No.	11.59700-0183	
TURBOCHARGER : TCA66			DATE	2010.05.18	
			SHEET	3/8	REV.
Order No	Designation	Representation	Weight in kg	Amount	
596.032	Hexagon nut	 M12	0.02	3	
596.033	Hexagon bolt	 M12 x 120	0.1	2	
596.040	Extension shaft		0.4	1	
596.041	Hexagon bolt	 M12 x 90	0.3	3	
596.044	Guide bar	 M12 x 600	0.5	2	
596.045	Guide bar	 M16 x 700	1.1	2	
596.046	Shackle	 A 1	0.4	1	
					

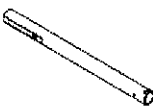
List of Tools

TOOL			Part No.	11.59700-0183	
TURBOCHARGER : TCA66			DATE	2010.05.18	
			SHEET	4/8	REV.
Order No	Designation	Representation	Weight in kg	Amount	
596.050	Guide bar	 M12 x 170	1.5	2	
596.051	Eye bolt	 M12	0.2	2	
596.052	Eye bolt	 M20 x 2	0,3	1	
596.055	Suspension plate, gas-admission casing		2.6	1	
596.058	Eye bolt	 M20 x 2	0.5	2	
596.070	Threaded bar	 M8 x 370	0.1	2	
596.071	Turn spin	 ø 12	0.2	1	
					

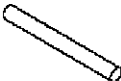
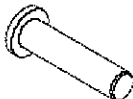
List of Tools

TOOL			Part No.	11.59700-0183		
TURBOCHARGER : TCA66			DATE	2010.05.18		
			SHEET	5/8	REV.	0
Order No	Designation	Representation	Weight in kg	Amount		
596.072	Hexagon nut	 M8	0.01	2		
596.073	Puller ring	 ø120	1.1	1		
596.075	Protection sleeve, turbine rotor		6.6	1		
596.087	Hexagon bolt	 M10 x 130	0.1	3		
596.504 596.092 596.124 596.125	Clamping sleeve Clamping sleeve Hexagon bolt Hexagon nut		0.9	1		
596.100	Torque spanner	 10-60 Nm	1.5	1		
596.101	Surface drive	 6 x 10		1		
						

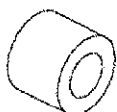
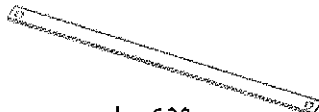
List of Tools

TOOL			Part No.	11.59700-0183	
TURBOCHARGER : TCA66			DATE	2010.05.18	
			SHEET	6/8	REV.
Order No	Designation	Representation	Weight in kg	Amount	
596.160	Pipe, Suspention device for comperssor wheel		5.3	1	
596.167	Sleeve	 $d_a = 46$	1.2	1	
596.168	Hexagon bolt, Suspention device for compessor wheel	 BM12 x 60	0.1	1	
596.170	Support		7.5	1	
596.197	Hexagon bolt	 M16 x 80	0.2	1	
596.198	Hexagon nut	 M16	0.03	1	
596.208	Shackle	 A 0.6	0.2	1	
					

List of Tools

TOOL			Part No.	11.59700-0183		
TURBOCHARGER : TCA66			DATE	2010.05.18		
			SHEET	7/8	REV.	0
Order No	Designation	Representation	Weight in kg		Amount	
596.267	Lever carrier		1.6		1	
596.270	Pipe	 30 x 3 x 900	1.8		1	
596.271	Slide		3.5		1	
596.272	Cylindrical screw	 M10 x 35	0.03		1	
596.273	Bolt	 ø6	0.01		1	
596.274	Cylindrical screw	 M10 x 45	0.03		3	
596.280	Hexagon bolt	 M16 x 50	0.1		3	
						

List of Tools

TOOL			Part No.	11.59700-0183		
TURBOCHARGER : TCA66			DATE	2010.05.18		
			SHEET	8/8	REV.	0
Order No	Designation	Representation	Weight in kg		Amount	
596.301	Fixation strap		0.767		1	
596.302	Cylindrical screw	 M8 x 20	0.015		2	
596.310	Sleeve	 d _a = 25	0.065		2	
596.311	Brace	 l = 620	0.597		1	
						